



Moneris Gateway API - Integration Guide - PHP – PHP

Version: 1.7.1

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## Security and Compliance

Your solution may be required to demonstrate compliance with the card associations' PCI/CISP/PABP requirements. For more information on how to make your application PCI-DSS compliant, contact the Moneris Sales Center and visit <https://developer.moneris.com> to download the PCI\_DSS Implementation Guide.

All Merchants and Service Providers that store, process, or transmit cardholder data must comply with PCI DSS and the Card Association Compliance Programs. However, certification requirements vary by business and are contingent upon your "Merchant Level" or "Service Provider Level".

The card association has some data security standards that define specific requirements for all organizations that store, process, or transmit cardholder data. As a Moneris client or partner using this method of integration, your solution must demonstrate compliance to the Payment Card Industry Data Security Standard (PCI DSS) and/or the Payment Application Data Security Standard (PA DSS). These standards are designed to help the cardholders and merchants in such ways as they ensure credit card numbers are encrypted when transmitted/stored in a database and that merchants have strong access control measures.

Non-compliant solutions may prevent merchant boarding with Moneris. A non-compliant merchant can also be subject to fines, fees, assessments or termination of processing services.

For further information on PCI DSS & PA DSS requirements, visit <http://www.pcisecuritystandards.org>.

## Confidentiality

You have a responsibility to protect cardholder and merchant related confidential account information. Under no circumstances should ANY confidential information be sent via email while attempting to diagnose integration or production issues. When sending sample files or code for analysis by Moneris staff, all references to valid card numbers, merchant accounts and transaction tokens should be removed and or obscured. Under no circumstances should live cardholder accounts be used in the test environment.

## Changes in v1.7.1

- Changed PriorAuthenticationInfo from optional to required within "MPI 3DS Authentication Request - 3RI with recurring" on page 260
- Changed RiIndicator note on Visa's support for particular values. Removed 07 and added 08 and 09 for Payment Transactions. Added 08 and 09 for Non-Payment Transactions.
- Added a note to PriorAuthenticationInfo on strict MasterCard formatting.
- Removed notes forbidding Interac pre-auths from the CAAV pre-authorization for ApplePay.

## Changes in v1.7.0

- Added Pay By Bank chapter with topics covering the new feature, the Konek authentication flow, and how to build and test an integration
- Added Pay By Bank optional object to Purchase, Preauthorization, Completion, and Refund basic transactions. This object contains new fields for linking a financial transaction to a Pay By Bank authentication, such as consent\_id, cryptogram, cryptogram\_expiry, payment\_method
- Added Pay By Bank transactions Create Consent, Get Consent Data, , Get Payment Transaction Data, and Cancel Consent
- Added the Pay By Bank response objects and their fields to the Definition of Response Fields in the Appendix. Each object has its own sub-topic to avoid confusion.
- Added the Pay By Bank request fields to the Definition of Request Fields in the Appendix.
- Added PBB to the list of possible values for wallet\_indicator

## Changes in v1.6.9

- Added Surcharge Lookup to the Basic Transaction set. Surcharge Lookup enables merchants to confirm card eligibility before applying surcharges to transactions.
- Added Vault Surcharge Lookup to the Vault Transaction set. Similar to the basic version, this lookup accepts a Moneris Vault data\_key for confirming eligibility of the underlying card.

- Added `surcharge_info` object to the transactions Purchase, Pre-Authorization, Vault Purchase, Vault Pre-Authorization, 3DS Purchase, 3DS Pre-Authorization, 3DS & Vault Purchase, and 3DS & Vault Pre-Authorization
- Added `surcharge_amount` as a child to the `surcharge_info` object.
- Added additional Response Codes for the Moneris Surcharge Eligibility service to "Response Codes" on page 1

### Changes in v1.6.8

- Added Vault Tokenize Credit Card request to the Vault transaction set. This transaction allows for tokenizing a card used in a previous financial transaction without resubmitting card data
- Added `return_issuer_id` to the Definition of Request Fields for Vault.
- Added new values to `request_type` in 3DS requests.
- Updated Visa Secure (3DS) support for the field `RIndicator` to support 01, 02, 06, 07, and 11 for Payment Transactions and 03, 04, 05, and 10 for Non-Payment Transactions

### Changes in v1.6.7

- Removed **Interac Online** chapter and related appendix topics for certification with Interac Online. Support for Interac Online has terminated effective October 31st, 2024.

### Changes in v1.6.6

- Added **Increment Preauthorization** in Basic transactions
- Added field `is_incremental` to the Basic Pre-Authorization and Vault Pre-Authorization transactions

### Changes in v1.6.5

- Added new request object and fields to support Visa Account Name Verification as an option- within the basic Card Verification transaction. See `account_name_verification` and its



subfields `first_name`, `middle_name`, and `last_name`

- Added response field `AccountNameVerificationResultCode` to the core response field definitions as part of the new Visa Account Name Verification

### Changes in v1.6.4

- Added `BrowserIP` to MPI 3DS Authentication Request - Browser Channel
- Added `WorkPhone`, `home_phone` and `mobile_phone` to MPI 3DS Authentication Request - Browser Channel
- Added 3 new transaction types `GooglePay Token Temp Add`, `GooglePay Token Purchase`, and `GooglePay Token Preauth`. These transactions allow for sending an encrypted GooglePay payload and receiving a Moneris temporary token in exchange for processing 3DS authentication.
- Added new request objects and fields to support GooglePay Token transactions such as `PaymentToken` and its subfields `signature`, `protocol version`, and `signed message`
- Added new response field `GooglePayPaymentMethod`

### Changes in v1.6.3

- Added Merchant Advice Code in Purchase, Pre-Authorization, CAVV Purchase; CAVV Pre-Authorization responses samples
- Updated DS trans ID Type and Limits, and Description in B.1 Definition of Response Fields – 3-D Secure
- Removed the `RequestType` field from MPI 3DS Authentication Request - 3RI with recurring
- Updated the note for the fields `RecurringFrequency` and `RecurringExpiry`
- Added Set Methods in MPI 3DS Authentication Request
- Removed status check in MPI 3DS Authentication Request

- Removed ThreeDSCompletionInd in MPI 3DS Authentication Request - 3RI with recurring/no recurring
- Moved PriorAuthenticationInfo field to optional section for MPI 3DS Prior Authentication Info for MPI 3DS Authentication Request - 3RI with recurring
- Updated the note related to the Field Email
- Added Merchant Advice Code field in the Appendix B - Definitions of Response Fields

### **Changes in v1.6.2**

- Updated the 3-D version to 2.2
- Added the 3RI flow chart Transaction Flow for 3-D Secure - 3RI channel
- Updated the field Email moving from optional to required
- Added a note to the field Email
- Added the status “D” in the TransStatus Code
- Updated transaction status reason Decline Codes
- Removed the Visa Checkout section. Visa Checkout has been decommissioned in June 1st 2023
- Updated the note related to the Field DSTransId
- Updated the note related to the Field ThreeDSSTransId
- Added a comment to Visa Secure
- Updated the note related to the Field RiIndicator
- Added note to ds\_trans\_id on only submitting it in financial transactions if using a 3rd party 3DS Secure service
- Added new 3DS fields to 3DS Authentication to support 3RI such as MessageCategory, DeviceChannel, RiIndicator
- Added new 3DS fields to 3DS Authentication to support 3RI Decoupled Authentication such as DecoupledRequestIndicator, DecoupledRequestMaxTime, DecoupledRequestAsyncUrl

- Added new 3DS object `prior_request_auth_data` to 3DS Authentication to support 3RI, including its fields `PriorAuthenticationInfo`, `prior_request_auth_method`, `prior_request_auth_ref`, `prior_request_auth_timestamp`
- Added additional fields to 3DS responses `ThreeDSVersion` and `AuthenticationType`
- Added additional topics on 3DS Authentication for the 3RI scenarios with and without recurring features
- Retitled the existing 3DS Authentication scenario to specify it is intended for the browser channel only
- Added topic on 3RI channel authentication flow and retitled previous 3DS flow to specify it is intended for browser channel only
- Added topic on Handling 3RI Decoupled Authentication flow to explain the asynchronous response handling
- Added topic on Server To Server endpoints to cover the separate URL for 3DS Authentication

### **Changes in v1.6.1**

- Added note to `ds_trans_id` on only submitting it in financial transactions if using a 3rd party 3DS Secure service

### **Changes in v1.6.0**

- Added new `foreign_indicator` field to Basic Transaction set: purchase and preauth
- Added new `foreign_indicator` field to 3-D Secure Transaction set: `cavv_purchase` and `cavvPreauth`
- Added new `foreign_indicator` values in Appendix A.2 Definition of Request Fields - Core Fields
- Updated sample code with new `foreign_indicator` field
- Updated AVS Response Codes table

## Changes in v1.5.0

- Added new definition for MPI 3DS Authentication Request transaction type
- Added new value V for request field **payment indicator** to reflect support for merchants who bill in recurring variable payments
- Amended allowable values for **e-commerce indicator** to reflect new **payment indicator** value and amended corresponding topic in Credential on File section
- Added note to the limit for request field **cardholder name** to indicate that accented characters are not allowable
- Added new section and topics about Installments by Visa
- Added topic about Vault and Installments in the Vault section

## Changes in v1.4.4

- Added new topic to Vault section listing Vault transactions that support temporary tokens
- Convenience Fee response field **convenience fee status** limit has been corrected to 3-character alphanumeric
- Information added to About Convenience Fee and response field definitions to indicate convenience fee transactions do not support MCP or digital wallets
- Transaction type descriptions have been added or amended in Vault Add Token, Vault Delete, Vault Is Corporate Card, Vault Add Credit Card,

## Changes in v1.4.3

- Added new MCP transaction topics MCP Purchase with 3-D Secure, MCP Purchase with 3-D Secure and Vault, MCP Pre-Authorization with 3-D Secure, and MCP Pre-Authorization with 3-D Secure
- Added transaction type topics for Purchase with 3-D Secure and Vault, Pre-Authorization with 3-D Secure and Vault, and Purchase with 3-D Secure and Recurring Billing in 3-D Secure 2.0 section

- Added new request field **DS transaction ID**
- Made billing-related request fields mandatory for 3DS Authentication Request transaction

### Changes in v1.4.2

- Corrected the limits for the request field **start date**
- Corrected presentation of set methods for the request fields **browser language**, **browser java enabled**, **browser screen height**, **browser screen width**,

### Changes in v1.4.1

- Added information about the request fields 3DS version and 3DS server indicator in the transaction topics for Purchase with 3-D Secure and Pre-Authorization with 3-D Secure
- corrected the variable name for the request field 3DS server indicator

### Changes in v1.4.0

- 3-D Secure 2.0 section has replaced MPI section (which related to 3-D Secure 1.0)
- Added new fields for 3-D Secure 2.0 to the Purchase with 3-D Secure and Pre-Authorization with 3-D Secure transactions
- References related to 3-D Secure 2.0 have been added, including CAVV Result Codes for Visa, Mastercard and American Express card brands, and 3-D Secure 2.0 TransStatus Codes
- Added Definition of Response Fields for 3-D Secure
- Added Definition of Request Fields – 3-D Secure 2.0 and removed the previous Definition of Request Fields – MPI

### Changes in v1.3.1

- Amended response code 959 under "Other Response Codes" to 599

## Changes in v1.3.0

- Removed wording about testing only with Visa cards in About Credential on File topic
- Removed differentiation between supported card brands in regards to AVS and CVD in Card Verification
- Added information about American Express support in Card Verification with AVS and CVD and Card Verification with Vault
- Removed the Re-Authorization transaction type and information related to it
- Removed the Mag Swipe transaction set section
- Added Response Codes reference
- Added new supported currencies to MCP Currency Codes
- Added transaction type definitions to Visa Checkout topics
- Added information about dynamic descriptor behaviour in Pre-Authorization transactions to all Pre-Authorization and Pre-Authorization Completion topics
- Added missing dynamic descriptor field information to Purchase with 3-D Secure transaction topic
- Added information about payment indicator and e-commerce indicator field values, including new topic Payment Indicator and E-Commerce Indicator Values
- Reorganized information for Convenience Fee, including new topics Supported Transactions for Convenience Fee and Convenience Fee Info Object
- Renamed Convenience Fee transaction type topics to emphasize that Purchase is the base transactions and Convenience Fee is an additional feature
- Reorganized Definition of Request Fields by feature sets; Mag Swipe request fields removed
- New request field definition topics for connection fields, core fields, Vault, MPI, and Convenience Fee

## Previous version changes

### **Changes in v1.2.12**

- Added warnings about 3-D Secure implementations using frames to MPI section and 3-D Secure related transaction type topics

### **Changes in v1.2.11**

- Added information about Google Pay™ and removed references to Android Pay
- Separated out transaction process flows into two separate topics for Apple Pay and Google Pay™, Apple Pay Transaction Process Overview and Google Pay™ Transaction Process Overview

### **Changes in v1.2.10**

- Added missing request fields in MCP Purchase with Vault

### **Changes in v1.2.9**

- Corrected sample code for some financial transactions (to remove old method of MCP processing)

### **Changes in v1.2.8**

- Added section about Multi-Currency Pricing (MCP) transactions
- Added new conceptual topics in Credential on File:
  - Merchant- vs. Cardholder-Initiated COF Transactions
  - COF With Previously Stored Credentials
- Discover test card number has been changed to 6011000992927602
- Amended electronic commerce indicator description in Definition of Response Fields to remove deprecated allowable values (8 and 9)

### **Changes in v1.2.7**

Missing limits for request variables amount, completion amount and transaction amount were replaced

### **Changes in v1.2.6**

Electronic commerce Indicator (crypt\_type) request variable's value of '7' amended to reflect that it also represents an American ExpressSafeKey non-authenticated transaction

### **Changes in v1.2.5**

Purchase transaction amended to include Customer ID variable

### **Changes in v1.2.4**

Changes limits in Amount, Transaction Amount, Completion Amount request variables to reflect 10 decimals.

### **Changes in v1.2.3**

This version adds information about passing Offlinx™ data for the Card Match pixel tag via Unified API transactions.



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## Getting Help

Moneris has help for you at every stage of the integration process.

| Getting Started                                                                   | During Development                                                                                                                                                                             | Production                                                                                                                                                                           |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Contact our Client Integration Specialists:<br><br>clientintegrations@moneris.com | If you are already working with an integration specialist and need technical development assistance, contact our eProducts Technical Consultants:<br><br>1-866-319-7450<br><br>api@moneris.com | If your application is already live and you need production support, contact Moneris Customer Service:<br><br>onlinepayments@moneris.com<br><br>1-866-319-7450<br><br>Available 24/7 |

For additional support resources, you can also make use of our community forums at

<http://community.moneris.com/product-forums/>

# 1 About This Documentation

## 1.1 Purpose

This document describes the transaction information for using the Moneris PHP API for sending credit card transactions. In particular, it describes the format for sending transactions and the corresponding responses you will receive.

## 1.2 Who Is This Guide For?

The Moneris Gateway API - Integration Guide - PHP is intended for developers integrating with the Moneris Gateway.

This guide assumes that the system you are trying to integrate meets the requirements outlined below and that you have some familiarity with the PHP programming language.

### System Requirements

- PHP
- Port 443 open for bi-directional communication
- Web server with a SSL certificate
- cURL - PHP interface – see Adding cURL CA Root Certificate to PHP API

## 1.3 Adding cURL CA Root Certificate to PHP API

### cURL CA Root Certificate File:

The default installation of PHP/cURL does not include the cURL CA root certificate file. In order for the Moneris Gateway PHP API to connect to the Moneris Gateway during transaction processing, the 'mpg-classes.php' file that's included with the PHP API package needs to be modified to include a path to the CA root certificate file.

To add the cURL CA root certificate file to the PHP API package, do the following:

1. If cURL was not installed separately from your PHP installation, libcurl is included in your PHP installation. You need to download the 'cacert.pem' file from <http://curl.haxx.se/docs/caextract.html> and save it to the necessary directory.



2. Once downloaded, rename the file to 'curl-ca-bundle.crt' (e.g., 'C:\path\to\curl-ca-bundle.crt'). If cURL was installed separately from PHP, you may need to determine the path to the cURL CA root certificate bundle on your system (e.g., 'C:\path\to\curl-ca-bundle.crt').
3. Insert the code below into the 'mpgclasses.php' file as part of the cURL option setting, at approximately line 73 below the line 'curl\_setopt(\$ch, CURLOPT\_SSL\_VERIFYPEER, TRUE);'

```
curl_setopt($ch, CURLOPT_CAINFO, 'C:\path\to\curl-ca-bundle.crt');
```

For more information regarding the `CURLOPT_SSL_VERIFYPEER` option, please refer to your PHP documentation.

## 2 Basic Transaction Set

- 2.1 Purchase
- 2.2 Pre-Authorization
- 2.4 Pre-Authorization Completion
- 2.5 Force Post
- 2.6 Purchase Correction
- 2.7 Refund
- 2.8 Independent Refund
- 2.9 Card Verification with AVS and CVD
- 2.11 Batch Close

## 2.1 Purchase

Verifies funds on the customer's card, removes the funds and prepares them for deposit into the merchant's account.

```
$txnArray = array('type'=>'purchase', ...);

$mpgTxn = new mpgTransaction($txnArray);

$mpgRequest = new mpgRequest($mpgTxn);

$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

### Purchase transaction request fields – Required

| Variable Name      | Type and Limits                                                                                                                                             |                        |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| order ID           | String<br>50-character alphanumeric                                                                                                                         | 'order_id'=>\$order_id |
| amount             | String<br>10-character decimal<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><div>EXAMPLE: 1234567.89</div> | 'amount'=>\$amount     |
| credit card number | String<br>20-character alphanumeric                                                                                                                         | 'pan'=>\$pan           |

| Variable Name                 | Type and Limits                                     |                              |
|-------------------------------|-----------------------------------------------------|------------------------------|
| expiry date                   | String<br>4-character alphanumeric<br>(YYMM format) | 'expiry_date'=>\$expiry_date |
| electronic commerce indicator | String<br>1-character alphanumeric                  | 'crypt_type'=>\$crypt        |

### Purchase transaction request fields – Optional

| Variable Name                                                                                                                                                                                                                                                                         | Type and Limits                     |                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|--------------------------------------------------------------|
| customer ID                                                                                                                                                                                                                                                                           | String<br>50-character alphanumeric | 'cust_id'=>\$cust_id                                         |
| CVD Information                                                                                                                                                                                                                                                                       | Object<br><br>N/A                   | \$mpgTxn->setCvdInfo<br>(\$mpgCvdInfo);                      |
| <div> <p><b>NOTE:</b> When storing credentials on the initial transaction, the CVD object must be sent; for subsequent transactions using stored credentials, CVD can be sent with cardholder-initiated transactions only—<b>merchants must not store CVD information.</b></p> </div> |                                     |                                                              |
| Surcharge Information                                                                                                                                                                                                                                                                 | Object<br><br>N/A                   | \$mpgTxn->setSurchargeInfo<br>(\$surchargeInfo);             |
| <div> <p><b>NOTE:</b> This object requires use of the Surcharge Lookup transaction prior to confirm card eligibility.</p> </div>                                                                                                                                                      |                                     |                                                              |
| Convenience Fee Information                                                                                                                                                                                                                                                           | Object<br><br>N/A                   | \$mpgConvFee = new<br>mpgConvFeeInfo<br>(\$convFeeTemplate); |
| <div> <p><b>NOTE:</b> This variable does not apply to Credential on File transactions.</p> </div>                                                                                                                                                                                     |                                     |                                                              |

| Variable Name                                                                                                                                                                             | Type and Limits                            |                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dynamic descriptor                                                                                                                                                                        | <i>String</i><br>20-character alphanumeric | 'dynamic_descriptor'=>\$dynamic_descriptor                                                                                                                                    |
| foreign indicator                                                                                                                                                                         | <i>Boolean</i><br>true or false            | 'foreign_indicator'=>\$foreign_indicator                                                                                                                                      |
| Pay By Bank Info<br>pbb_info                                                                                                                                                              | <i>Object</i><br>N/A                       | placeholder<br><br>Required when performing a financial transaction after obtaining an Interac Konek (Pay By Bank) authentication. Links the PBB consent to this transaction. |
| <div>For information on request fields for this object, see "Definition of Request Fields – PBB Info Object" on page 581</div>                                                            |                                            |                                                                                                                                                                               |
| Surcharge Information                                                                                                                                                                     | <i>Object</i><br>N/A                       | purchase<br><br>\$mpgTxn->setSurchargeInfo(\$surchargeInfo);<br><br>Contains fields related to Surcharge feature.                                                             |
| <div>NOTE: This object requires use of the Vault Surcharge Lookup transaction prior to confirm card eligibility.</div>                                                                    |                                            |                                                                                                                                                                               |
| Installment Info<br>For fields in this object, see 9.6 Installment Info Object                                                                                                            | <i>Object</i><br>N/A                       | \$mpgTxn->setInstallmentInfo(\$installmentInfo);                                                                                                                              |
| <div>NOTE: Do not send the Installment Info object on any transaction that is not intended to offer Installments by Visa functionality; doing so may cause the transaction to fail.</div> |                                            |                                                                                                                                                                               |

## Sample Purchase

```
<?php
require "../mpgClasses.php";

/***** Request Variables *****/
$store_id='monca03650';
$api_token='7Yw0MPTlhjBRcZiE6837';

/***** Transactional Variables *****/
$type='purchase';
$order_id='Test'.date("dmy-G:i:s");
$amount='6000.00';
```

```

$pan='4622943127023886';
$expdate='2212';
$crypt='7';
$dynamic_descriptor='123';
$status_check = 'false';
$foreign_indicator='true';

// TrId and TokenCryptogram are optional, refer documentation for more details.
$str_id = '50189815682';
$token_cryptogram = 'APmbM/411e0uAAH+s6xMAAADFA==';

/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'amount'=>$amount,
'pan'=>$pan,
'expdate'=>$expdate,
'crypt_type'=>$crypt
// 'dynamic_descriptor'=>$dynamic_descriptor
//, 'wallet_indicator' => '' //Refer to documentation for details
//, 'cm_id' => '8nAK8712sGaAkls56' //set only for usage with Offlinx - Unique max 50
alphanumeric characters transaction id generated by merchant
//, 'tr_id' => $str_id
'foreign_indicator'=>$foreign_indicator
//, 'token_cryptogram' => $token_cryptogram
);

/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);

/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("U");
$cof->setPaymentInformation("2");
$cof->setIssuerId("139X3130ASCXAS9");
$mpgTxn->setCofInfo($cof);
//$mpgTxn->setForeignIndicator($foreign_indicator);

/***** Installment Info *OPTIONAL* *****/
$installmentInfo = new InstallmentInfo();
$installmentInfo->setPlanId("ae859ef1-eb91-b708-8b80-1dd481746401");
$installmentInfo->setPlanIdRef("0000000065");
$installmentInfo->setTacVersion("2");
//$mpgTxn->setInstallmentInfo($installmentInfo);

/***** Surcharge Info *OPTIONAL* *****/
$surchargeInfo = new SurchargeInfo();
$surchargeInfo->setSurchargeAmount("1.00");
$mpgTxn->setSurchargeInfo($surchargeInfo);

/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions

/***** HTTPS Post Object *****/
/* Status Check Example
$mpgHttpPost =new mpgHttpPostStatus($store_id,$api_token,$status_check,$mpgRequest);
*/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);

/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());

```

```
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nStatusCode = " . $mpgResponse->getStatusCode());
print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
print("\nHostId = " . $mpgResponse->getHostId());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\nSourcePanLast4 = " . $mpgResponse->getSourcePanLast4());
print("\nAdviceCode = " . $mpgResponse->getAdviceCode());
// $installmentResults = $mpgResponse->getInstallmentResults();
// print("\nPlanId = " . $installmentResults->getPlanId());
// print("\nPlanIDRef = " . $installmentResults->getPlanIDRef());
// print("\nTacVersion = " . $installmentResults->getTacVersion());
// print("\nPlanAcceptanceId = " . $installmentResults->getPlanAcceptanceId());
// print("\nPlanStatus = " . $installmentResults->getPlanStatus());
// print("\nPlanResponse = " . $installmentResults->getPlanResponse());
?>
```

## 2.2 Pre-Authorization

Verifies and locks funds on the customer's credit card. The funds are locked for a specified amount of time based on the card issuer.

To retrieve the funds that have been locked by a Pre-Authorization transaction so that they may be settled in the merchant's account, a Pre-Authorization Completion transaction must be performed. A Pre-Authorization transaction may only be "completed" once.

### Things to Consider:

- If a Pre-Authorization transaction is not followed by a Pre-Authorization Completion transaction, it must be reversed via a Pre-Authorization Completion transaction for 0.00. See 2.4 Pre-Authorization Completion
- A Pre-Authorization transaction may only be "completed" once

```
$txnArray = array('type'=>'preauth', ...);

$mpgTxn = new mpgTransaction($txnArray);

$mpgRequest = new mpgRequest($mpgTxn);

$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

**Core connection object fields (all API transactions)**

| Variable Name | Type and Limits      |                          |
|---------------|----------------------|--------------------------|
| store ID      | <i>String</i><br>N/A | 'store_id'=>\$store_id   |
| API token     | <i>String</i><br>N/A | 'api_token'=>\$api_token |

**Pre-Authorization transaction request fields – Required**

| Variable Name                 | Type and Limits                                                                                                                                                    |                              |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| order ID                      | <i>String</i><br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                                                                                         | 'order_id'=>\$order_id       |
| amount                        | <i>String</i><br>10-character decimal<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><div>EXAMPLE: 1234567.89</div> | 'amount'=>\$amount           |
| credit card number            | <i>String</i><br>max 20-character alphanumeric                                                                                                                     | 'pan'=>\$pan                 |
| expiry date                   | <i>String</i><br>4-character alphanumeric<br>YYMM                                                                                                                  | 'expiry_date'=>\$expiry_date |
| electronic commerce indicator | <i>String</i><br>1-character alphanumeric                                                                                                                          | 'crypt_type'=>\$crypt        |



## Pre-Authorization transaction request fields – Optional

| Variable Name      | Type and Limits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dynamic descriptor | <i>String</i><br>20-character alphanumeric<br>total of 22 characters including your merchant name and separator<br><div> <b>NOTE:</b> For Pre-Authorization transactions: the value in the dynamic descriptor field will only be carried over to a Pre-Authorization Completion when executing the latter via the Merchant Resource Center; otherwise, the value for dynamic descriptor must be sent again in the Pre-Authorization Completion </div> <div> <b>NOTE:</b><br/>           Some special characters are not allowed:<br/>           &lt; &gt; \$ % = ? ^ { } [ ] \ </div> | 'dynamic_descriptor'=>\$dynamic_descriptor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| is_incremental     | <i>Boolean</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 'is_incremental'=>\$is_incremental                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| is_incremental     | true/false                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Indicates if this preauthorization is using an estimated amount. Estimations allow for incrementing the amount held via subsequent incrementalAuth requests. Defaults to false.<br><div> <b>NOTE:</b> Please note that if this field is true, the preauthorization is only eligible for a single Preauthorization Completion. Any completion sent for partial completion is treated as a full completion (ship_indicator= P is treated as = F when is_incremental= true on the original preauth). </div> <div> <b>NOTE:</b> For preauthorizations using Account-Based Payments, as part of Pay By Bank, is_incremental = true indicates the transaction may be completed for up to 15% more than the original preauthorization amount. </div> |
| foreign indicator  | <i>Boolean</i><br>true or false                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 'foreign_indicator'=>\$foreign_indicator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Variable Name                                                                                                                                                                                                                                                                  | Type and Limits                                                                                                                                                                  |                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Customer Information                                                                                                                                                                                                                                                           | <i>Object</i><br>N/A                                                                                                                                                             | <code>\$mpgTxn-&gt;setCustInfo<br/>(\$mpgCustInfo);</code>                                                             |
| AVS Information                                                                                                                                                                                                                                                                | <i>Object</i><br>N/A                                                                                                                                                             | <code>\$mpgTxn-&gt;setAvsInfo<br/>(\$mpgAvsInfo);</code>                                                               |
| CVD Information                                                                                                                                                                                                                                                                | <i>Object</i><br>N/A                                                                                                                                                             | <code>\$mpgTxn-&gt;setCvdInfo<br/>(\$mpgCvdInfo);</code>                                                               |
| <div> <b>NOTE:</b> When storing credentials on the initial transaction, the CVD object must be sent; for subsequent transactions using stored credentials, CVD can be sent with cardholder-initiated transactions only—<b>merchants must not store CVD information.</b> </div> |                                                                                                                                                                                  |                                                                                                                        |
| customer ID                                                                                                                                                                                                                                                                    | <i>String</i><br>50-character alphanumeric<br><div> <b>NOTE:</b><br/>           Some special characters are not allowed:<br/> <code>&lt; &gt; \$ % = ? ^ { } [ ] \</code> </div> | <code>'cust_id'=&gt;\$cust_id</code>                                                                                   |
| Surcharge Information                                                                                                                                                                                                                                                          | <i>Object</i><br>N/A                                                                                                                                                             | <code>\$mpgTxn-&gt;setSurchargeInfo<br/>(\$surchargeInfo);</code><br><br>Contains fields related to Surcharge feature. |
| <div> <b>NOTE:</b> This object requires use of the Surcharge Lookup transaction prior to confirm card eligibility.         </div>                                                                                                                                              |                                                                                                                                                                                  |                                                                                                                        |
| wallet indicator                                                                                                                                                                                                                                                               | <i>String</i><br>3-character alphanumeric                                                                                                                                        | <code>'wallet_indicator'=&gt;\$wallet_indicator</code>                                                                 |
| <div> <b>NOTE:</b> For basic Purchase and Preauthorization, the wallet indicator applies to Visa Checkout and MasterCard MasterPass only. For more, see Definition of Request Fields.         </div>                                                                           |                                                                                                                                                                                  |                                                                                                                        |

| Variable Name                                                                                                                  | Type and Limits |                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pay By Bank Info                                                                                                               | <i>Object</i>   | placeholder                                                                                                                                                |
| pbb_info                                                                                                                       | N/A             | Required when performing a financial transaction after obtaining an Interac Konek (Pay By Bank) authentication. Links the PBB consent to this transaction. |
| <div>For information on request fields for this object, see "Definition of Request Fields – PBB Info Object" on page 581</div> |                 |                                                                                                                                                            |
| Surcharge Information                                                                                                          | <i>Object</i>   | preauth                                                                                                                                                    |
| <div>NOTE: This object requires use of the Vault Surcharge Lookup transaction prior to confirm card eligibility.</div>         | N/A             | \$mpgTxn->setSurchargeInfo(\$surchargeInfo);<br><br>Contains fields related to Surcharge feature.                                                          |

### Sample Pre-Authorization

```

<?php
require "../mpgClasses.php";
$store_id='monca03650';
$api_token='7Yw0MPTlhjBRcZiE6837';
/***** Transactional Variables *****/
$type='preauth';
$cust_id='cust id';
$order_id='ord-' . date("dmy-G:i:s");
$amount='6000.00';
$pan='4622943127023886';
$expdate='2212';
$crypt='7';
$dynamic_descriptor='123';
$status_check = 'false';
$foreign_indicator='true';
$is_incremental= 'true';

// TrId and TokenCryptogram are optional, refer documentation for more details.
$str_id = '50189815682';
$token_cryptogram = 'APmbM/411e0uAAH+s6xMAAADFA==';
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'cust_id'=>$cust_id,
'amount'=>$amount,
'pan'=>$pan,
'expdate'=>$expdate,
'crypt_type'=>$crypt,
'dynamic_descriptor'=>$dynamic_descriptor
//,'cm_id' => '8nAK8712sGaAkls56' //set only for usage with Offlinx - Unique max 50
alphanumeric characters transaction id generated by merchant
//,'tr_id' => $str_id
//,'token_cryptogram' => $token_cryptogram
'foreign_indicator'=>$foreign_indicator
);
$mpgTxn = new mpgTransaction($txnArray);
/***** Credential on File *****/
$cof = new CofInfo();

```

```

$cof->setPaymentIndicator("U");
$cof->setPaymentInformation("2");
$cof->setIssuerId("139X3130ASCXAS9");
$mpgTxn->setCofInfo($cof);
/***** Installment Info *OPTIONAL* *****/
$installmentInfo = new InstallmentInfo();
$installmentInfo->setPlanId("ae859ef1-eb91-b708-8b80-1dd481746401");
$installmentInfo->setPlanIdRef("0000000065");
$installmentInfo->setTacVersion("2");
// $mpgTxn->setInstallmentInfo($installmentInfo);

/***** Surcharge Info *OPTIONAL* *****/
$surchargeInfo = new SurchargeInfo();
$surchargeInfo->setSurchargeAmount("1.00");
$mpgTxn->setSurchargeInfo($surchargeInfo);

$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
$mpgResponse = $mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\nSourcePanLast4 = " . $mpgResponse->getSourcePanLast4());
print("\nAdviceCode = " . $mpgResponse->getAdviceCode());

// $installmentResults = $mpgResponse->getInstallmentResults();
// print("\nPlanId = " . $installmentResults->getPlanId());
// print("\nPlanIDRef = " . $installmentResults->getPlanIDRef());
// print("\nTacVersion = " . $installmentResults->getTacVersion());
// print("\nPlanAcceptanceId = " . $installmentResults->getPlanAcceptanceId());
// print("\nPlanStatus = " . $installmentResults->getPlanStatus());
// print("\nPlanResponse = " . $installmentResults->getPlanResponse());
?>

```

## 2.3 Incremental Pre-Authorization

Increases the locked amount of funds in an existing pre-authorization for later settle by a single pre-authorization completion. There is no limit to the number of incremental pre-authorization transactions on the original estimated auth and each new incremental pre-authorization increases the hold on the customer's credit card.

Incremental Pre-authorizations require an estimated amount in the initial Pre-Authorization. This is set using the `<is_incremental>` field set to true.

For Mastercard only, an Incremental Pre-Authorization can be submitted with a \$0 value for the amount to request extending the allowable timeframe for completion (e.g, 30 days).

For additional details on using estimated amounts in Pre-Authorizations and using Incremental Pre-Authorizations to increase the locked amount of funds, see 1 Incremental Authorization Rules

### Incremental Pre-Authorization transaction object definition

```
$txnArray = array('type'=>'incremental_preauth', ...);

$mpgTxn = new mpgTransaction($txnArray);

$mpgRequest = new mpgRequest($mpgTxn);

$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

### Incremental Preauthorization transaction request fields – Required

| Variable Name | Type and Limits                                                                                                                                             |                        |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| order ID      | String<br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                                                                                         | 'order_id'=>\$order_id |
| amount        | String<br>10-character decimal<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><div>EXAMPLE: 1234567.89</div> | 'amount'=>\$amount     |

| Variable Name      | Type and Limits                                                                                   |                            |
|--------------------|---------------------------------------------------------------------------------------------------|----------------------------|
| transaction number | <i>String</i><br><br>255-character, alpha-numeric, hyphens or under-scores<br><br>variable length | 'txn_number'=>\$txn_number |

### Sample Incremental Pre-Authorization

```

<?php
require "../mpgClasses.php";
$store_id='monca03650';
$api_token='7Yw0MPTlhjBRcZiE6837';
$orderid='ord-290824-5:27:32';
$txnnumber='16204-0_879';
$amount='20.20';
## step 1) create transaction array ###
$txnArray=array('type'=>'incremental_preauth',
'order_id'=>$orderid,
'txn_number'=>$txnnumber,
'amount'=>$amount
// 'ship_indicator'=>$ship_indicator, //optional
);
## step 2) create a transaction object passing the hash created in
## step 1.
$mpgTxn = new mpgTransaction($txnArray);
## step 3) create a mpgRequest object passing the transaction object created
## in step 2
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false or comment out this line for production transactions
## step 4) create mpgHttpPost object which does an https post ##
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
## step 5) get an mpgResponse object ##
$mpgResponse=$mpgHttpPost->getMpgResponse();
## step 6) retrieve data using get methods
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nSourcePanLast4 = " . $mpgResponse->getSourcePanLast4());
?>

```

## 2.4 Pre-Authorization Completion

Retrieves funds that have been locked (by a Pre-Authorization transaction), and prepares them for settlement into the merchant's account.

### Things to Consider:

- A Pre-Authorization transaction can only be completed once
- To reverse the full amount of a Pre-Authorization transaction, use the Pre-Authorization Completion transaction with the amount set to 0.00
- To process this transaction, you need the order ID and transaction number from the original Pre-Authorization transaction

```
$txnArray = array('type'=>'completion', ...);
$mpgTxn = new mpgTransaction($txnArray);
$mpgRequest = new mpgRequest($mpgTxn);
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

### Pre-Authorization Completion transaction request fields – Required

| Variable Name           | Type and Limits                                                     |                              |
|-------------------------|---------------------------------------------------------------------|------------------------------|
| order ID                | String<br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces | 'order_id'=>\$order_id       |
| completion amountamount | String<br>10-character decimal                                      | 'comp_amount'=>\$comp_amount |

| Variable Name                 | Type and Limits                                                                         |                            |
|-------------------------------|-----------------------------------------------------------------------------------------|----------------------------|
|                               | Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point |                            |
|                               | <b>EXAMPLE:</b> 1234567.89                                                              |                            |
| transaction number            | String<br>255-character, alpha-numeric, hyphens or under-scores<br>variable length      | 'txn_number'=>\$txn_number |
| electronic commerce indicator | String<br>1-character alphanumeric                                                      | 'crypt_type'=>\$crypt      |

### Pre-Authorization Completion transaction request fields – Optional

| Variable Name      | Type and Limits                                                                                                                                                                       | Description                                |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| customer ID        | String<br>50-character alphanumeric<br><b>NOTE:</b><br>Some special characters are not allowed:<br><>\$%=?^{}[]\                                                                      | 'cust_id'=>\$cust_id                       |
| dynamic descriptor | String<br>20-character alphanumeric<br>total of 22 characters including your merchant name and separator<br><b>NOTE:</b><br>Some special characters are not allowed:<br><>\$%=?^{}[]\ | 'dynamic_descriptor'=>\$dynamic_descriptor |

**NOTE: For Pre-Authorization transactions:** the value in the dynamic descriptor field will only be carried over to a Pre-Authorization Completion when executing the latter via the Merchant Resource Center; otherwise, the value for dynamic descriptor must be sent again in the Pre-



| Variable Name                                                                                                                                      | Type and Limits                               | Description                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authorization Completion                                                                                                                           |                                               |                                                                                                                                                                                                    |
| Surcharge Information<br><div> <b>NOTE:</b> This object requires use of the Surcharge Lookup transaction prior to confirm card eligibility. </div> | <i>Object</i><br><br>N/A                      | <code>\$mpgTxn-&gt;setSurchargeInfo(\$surchargeInfo);</code><br><br>Contains fields related to Surcharge feature.                                                                                  |
| Pay By Bank Info                                                                                                                                   | <i>Object</i>                                 | placeholder                                                                                                                                                                                        |
| pbb_info                                                                                                                                           | N/A                                           | Required performing a completion transaction for an Account Based Payment (ABP) for Pay By Bank. Utilize Get Transaction Data to obtain the cryptogram. Links the PBB consent to this transaction. |
|                                                                                                                                                    |                                               | <div>           For information on request fields for this object, see "Definition of Request Fields – PBB Info Object" on page 581 </div>                                                         |
| shipping indicator                                                                                                                                 | <i>String</i><br><br>1-character alphanumeric | <code>'ship_indicator'=&gt;\$ship_indicator</code>                                                                                                                                                 |

### Sample Pre-Authorization Completion

```

<?php
require "../mpgClasses.php";
$store_id='monca03650';
$api_token='7Yw0MPTlhjBRcZiE6837';
$orderid='ord-290824-5:27:32';
$txnnumber='16204-0_879';
$compamount='200.00';
$dynamic_descriptor='123';
$ship_indicator = "F"; //optional
## step 1) create transaction array ###
$txnArray=array('type'=>'completion',
'txn_number'=>$txnnumber,
'order_id'=>$orderid,
'comp_amount'=>$compamount,
'crypt_type'=>'7',
'cust_id'=>'customer ID',
// 'ship_indicator'=>$ship_indicator, //optional
'dynamic_descriptor'=>$dynamic_descriptor
);
## step 2) create a transaction object passing the hash created in
## step 1.
$mpgTxn = new mpgTransaction($txnArray);
## step 3) create a mpgRequest object passing the transaction object created
## in step 2
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false or comment out this line for production transactions

```

```

## step 4) create mpgHttpPost object which does an https post ##
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
## step 5) get an mpgResponse object ##
$mpgResponse=$mpgHttpPost->getMpgResponse();
## step 6) retrieve data using get methods
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nSourcePanLast4 = " . $mpgResponse->getSourcePanLast4());
?>

```

## 2.5 Force Post

Retrieves the locked funds and prepares them for settlement into the merchant's account.

Used when a merchant obtains the authorization number directly from the issuer by a third-party authorization method (such as by phone).

### Things to Consider:

- This transaction is an independent completion where the original Pre-Authorization transaction was not processed via the same Moneris Gateway merchant account.
- It is not required for the transaction that you are submitting to have been processed via the Moneris Gateway. However, a credit card number, expiry date and original authorization number are required.
- Force Post transactions are not supported for UnionPay

```

$txnArray = array('type'=>'forcepost', ...);
$mpgTxn = new mpgTransaction($txnArray);
$mpgRequest = new mpgRequest($mpgTxn);
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);

```

**Core connection object fields (all API transactions)**

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

**Force Post transaction request fields – Required**

| Variable Name      | Type and Limits                                                                                                                                             |                              |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| order ID           | String<br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                                                                                         | 'order_id'=>\$order_id       |
| amount             | String<br>10-character decimal<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><div>EXAMPLE: 1234567.89</div> | 'amount'=>\$amount           |
| credit card number | String<br>max 20-character alphanumeric                                                                                                                     | 'pan'=>\$pan                 |
| expiry date        | String<br>4-character alphanumeric<br>YYMM                                                                                                                  | 'expiry_date'=>\$expiry_date |
| authorization code | String<br>8-character alphanumeric                                                                                                                          | 'auth_code'=>\$auth_code     |

| Variable Name                 | Type and Limits                    |                       |
|-------------------------------|------------------------------------|-----------------------|
| electronic commerce indicator | String<br>1-character alphanumeric | 'crypt_type'=>\$crypt |

### Force Post transaction request fields – Optional

| Variable Name      | Type and Limits                                                                                                                                                                               |                                            |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| customer ID        | String<br>50-character alphanumeric<br><br><b>NOTE:</b><br>Some special characters are not allowed:<br><>\$%=?^{}[]\                                                                          | 'cust_id'=>\$cust_id                       |
| dynamic descriptor | String<br>20-character alphanumeric<br><br>total of 22 characters including your merchant name and separator<br><br><b>NOTE:</b><br>Some special characters are not allowed:<br><>\$%=?^{}[]\ | 'dynamic_descriptor'=>\$dynamic_descriptor |

### Sample Force Post

```
<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
// $status = 'false';
/***** Transactional Variables *****/
$type='forcepost';
$cust_id='CUST13343';
$order_id='ord-' . date("dmy-G:i:s");
$amount='10.00';
$span='4242424242424242';
$expiry_date='0812';
$auth_code='123456';
$crypt='7';
$dynamic_descriptor='123456';
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
```

```

'cust_id'=>$cust_id,
'amount'=>$amount,
'pan'=>$pan,
'expdate'=>$expiry_date,
'auth_code'=>$auth_code,
'crypt_type'=>$crypt,
'dynamic_descriptor'=>$dynamic_descriptor
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
// Status check example
// $mpgHttpPost = new mpgHttpPostStatus($store_id, $api_token, $status, $mpgRequest);
/***** Response *****/
$mpgResponse = $mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
// print("\nStatusCode = " . $mpgResponse->getStatusCode());
// print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>

```

## 2.6 Purchase Correction

Restores the full amount of a previous Purchase, Pre-Authorization Completion or Force Post transaction to the cardholder's card, and removes any record of it from the cardholder's statement.

This transaction can be used against a Purchase or Pre-Authorization Completion transaction that occurred same day provided that the batch containing the original transaction remains open.

### Things to Consider:

- To process this transaction, you need the order ID and the transaction number from the original Completion, Purchase or Force Post transaction.

```
$txnArray = array('type'=>'purchasecorrection', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits      |                          |
|---------------|----------------------|--------------------------|
| store ID      | <i>String</i><br>N/A | 'store_id'=>\$store_id   |
| API token     | <i>String</i><br>N/A | 'api_token'=>\$api_token |

### Purchase Correction transaction request fields – Required

| Variable Name                 | Type and Limits                                                                           |                            |
|-------------------------------|-------------------------------------------------------------------------------------------|----------------------------|
| order ID                      | <i>String</i><br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                | 'order_id'=>\$order_id     |
| transaction number            | <i>String</i><br>255-character, alpha-numeric, hyphens or under-scores<br>variable length | 'txn_number'=>\$txn_number |
| electronic commerce indicator | <i>String</i><br>1-character alphanumeric                                                 | 'crypt_type'=>\$crypt      |

### Purchase Correction transaction request fields – Optional

| Variable Name | Type and Limits                                                                                                    |                      |
|---------------|--------------------------------------------------------------------------------------------------------------------|----------------------|
| customer ID   | <i>String</i><br>50-character alphanumeric<br><div><b>NOTE:</b><br/>Some special characters are not allowed:</div> | 'cust_id'=>\$cust_id |

| Variable Name                                                                                                                                          | Type and Limits                                                                                                                                                                                                                                     |                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                        | <>\$%=?^{}[]\                                                                                                                                                                                                                                       |                                                                                                                   |
| Surcharge Information<br><br><div> <b>NOTE:</b> This object requires use of the Surcharge Lookup transaction prior to confirm card eligibility. </div> | <i>Object</i><br><br>N/A                                                                                                                                                                                                                            | <code>\$mpgTxn-&gt;setSurchargeInfo(\$surchargeInfo);</code><br><br>Contains fields related to Surcharge feature. |
| dynamic descriptor                                                                                                                                     | <i>String</i><br><br>20-character alphanumeric<br><br>total of 22 characters including your merchant name and separator<br><br><div> <b>NOTE:</b><br/>           Some special characters are not allowed:<br/>           &lt;&gt;\$%=?^{}[]\ </div> | <code>'dynamic_descriptor'=&gt;\$dynamic_descriptor</code>                                                        |

### Sample Purchase Correction

```

<?php
require "../mpgClasses.php";
$store_id='store5';
$api_token='yesguy';
$orderid='ord-150816-12:36:20';
$txnnumber='117816-0_10';
$dynamic_descriptor='1234';
## step 1) create transaction hash ###
$txnArray=array('type'=>'purchasecorrection',
'txn_number'=>$txnnumber,
'order_id'=>$orderid,
'crypt_type'=>'7',
'cust_id'=>'customer ID',
'dynamic_descriptor'=>$dynamic_descriptor
);
## step 2) create a transaction object passing the array created in
## step 1.
$mpgTxn = new mpgTransaction($txnArray);
/***** Installment Info *OPTIONAL* *****/
$installmentInfo = new InstallmentInfo();
$installmentInfo->setPlanId("ae859ef1-eb91-b708-8b80-1dd481746401");
$installmentInfo->setPlanIdRef("00000000065");
$installmentInfo->setTacVersion("2");
// $mpgTxn->setInstallmentInfo($installmentInfo);
## step 3) create a mpgRequest object passing the transaction object created
## in step 2
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions

```

```

## step 4) create mpgHttpPost object which does an https post ##
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
## step 5) get an mpgResponse object ##
$mpgResponse=$mpgHttpPost->getMpgResponse();
## step 6) retrieve data using get methods
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
// $installmentResults = $mpgResponse->getInstallmentResults();
// print("\nPlanId = " . $installmentResults->getPlanId());
// print("\nPlanIDRef = " . $installmentResults->getPlanIDRef());
// print("\nTacVersion = " . $installmentResults->getTacVersion());
// print("\nPlanAcceptanceId = " . $installmentResults->getPlanAcceptanceId());
// print("\nPlanStatus = " . $installmentResults->getPlanStatus());
// print("\nPlanResponse = " . $installmentResults->getPlanResponse());
?>

```

## 2.7 Refund

Restores all or part of the funds from a Purchase, Pre-Authorization Completion or Force Post transaction to the cardholder's card.

Unlike a Purchase Correction, there is a record of both the initial charge and the refund on the cardholder's statement.

To process this transaction, you need the order ID and transaction number from the original Completion, Purchase or Force Post transaction.

```

$txnArray = array('type'=>'refund', ...);

$mpgTxn = new mpgTransaction($txnArray);

$mpgRequest = new mpgRequest($mpgTxn);

$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);

```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                        |
|---------------|-----------------|------------------------|
| store ID      | <i>String</i>   | 'store_id'=>\$store_id |
|               | N/A             |                        |



| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| API token     | <i>String</i>   | 'api_token'=>\$api_token |
|               | N/A             |                          |

**Refund transaction request fields – Required**

| Variable Name                 | Type and Limits                                                                                                                                                                |                            |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| order ID                      | <i>String</i><br><br>50-character alpha-numeric<br><br>a-Z A-Z 0-9 _ - : . @<br>spaces                                                                                         | 'order_id'=>\$order_id     |
| amount                        | <i>String</i><br><br>10-character decimal<br><br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><br><div>EXAMPLE: 1234567.89</div> | 'amount'=>\$amount         |
| transaction number            | <i>String</i><br><br>255-character, alpha-numeric, hyphens or underscores<br><br>variable length                                                                               | 'txn_number'=>\$txn_number |
| electronic commerce indicator | <i>String</i><br><br>1-character alphanumeric                                                                                                                                  | 'crypt_type'=>\$crypt      |

**Pre-Authorization Completion transaction request fields – Optional**

| Variable Name         | Type and Limits | Set Method                 |
|-----------------------|-----------------|----------------------------|
| Surcharge Information | <i>Object</i>   | \$mpgTxn->setSurchargeInfo |

| Variable Name                                                                                                                    | Type and Limits | Set Method                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div> <b>NOTE:</b> This object requires use of the Surcharge Lookup transaction prior to confirm card eligibility. </div>        | N/A             | (\$surchargeInfo);<br><br>Contains fields related to Surcharge feature.                                                                                                                        |
| Pay By Bank Info                                                                                                                 | Object          | placeholder                                                                                                                                                                                    |
| pbb_info                                                                                                                         | N/A             | Required performing a refund transaction for an Account Based Payment (ABP) for Pay By Bank. Utilize Get Transaction Data to obtain the cryptogram. Links the PBB consent to this transaction. |
| <div> For information on request fields for this object, see "Definition of Request Fields – PBB Info Object" on page 581 </div> |                 |                                                                                                                                                                                                |

## Sample Refund

```

<?php
##
## This program takes 4 arguments from the command line:
## 1. Store id
## 2. api token
## 3. order id
## 4. trans number
##
## Example php -q TestRefund.php store1 yesguy my_order_id 45109-89-0
##
require "../mpgClasses.php";
$store_id='store5';
$api_token='yesguy';
$orderid='ord-150816-11:56:58';
$txnnumber='117743-0_10';
$amount = '1.00';
$script_type = '7';
$dynamic_descriptor='123';
## step 1) create transaction array ###
$txnArray=array('type'=>'refund',
'txn_number'=>$txnnumber,
'order_id'=>$orderid,
'amount'=>$amount,
'crypt_type'=>$script_type,
'cust_id'=> 'Customer ID',
'dynamic_descriptor'=>$dynamic_descriptor
);
## step 2) create a transaction object passing the array created in
## step 1.
$mpgTxn = new mpgTransaction($txnArray);
/***** Installment Info *OPTIONAL* *****/
$installmentInfo = new InstallmentInfo();
$installmentInfo->setPlanId("ae859ef1-eb91-b708-8b80-1dd481746401");
$installmentInfo->setPlanIdRef("0000000065");
$installmentInfo->setTacVersion("2");
//$mpgTxn->setInstallmentInfo($installmentInfo);
## step 3) create a mpgRequest object passing the transaction object created
## in step 2
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment

```

```
$mpgRequest->setTestMode(true); //false for production transactions
## step 4) create mpgHttpPost object which does an https post ##
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
## step 5) get an mpgResponse object ##
$mpgResponse=$mpgHttpPost->getMpgResponse();
## step 6) retrieve data using get methods
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
// $installmentResults = $mpgResponse->getInstallmentResults();
// print("\nPlanId = " . $installmentResults->getPlanId());
// print("\nPlanIDRef = " . $installmentResults->getPlanIDRef());
// print("\nTacVersion = " . $installmentResults->getTacVersion());
// print("\nPlanAcceptanceId = " . $installmentResults->getPlanAcceptanceId());
// print("\nPlanStatus = " . $installmentResults->getPlanStatus());
// print("\nPlanResponse = " . $installmentResults->getPlanResponse());
?>
```

## 2.8 Independent Refund

Credits a specified amount to the cardholder's credit card. The credit card number and expiry date are mandatory.

It is not necessary for the transaction that you are refunding to have been processed via the Moneris Gateway.

### Things to Consider:

- Because of the potential for fraud, permission for this transaction is not granted to all accounts by default. If it is required for your business, it must be requested via your account manager.

**NOTE:** Independent Refund transactions do not support Installments

```
$txnArray = array('type'=>'ind_refund', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

### Independent Refund transaction request fields – Required

| Variable Name              | Type and Limits                                                                                                                                                      |                              |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| order ID                   | String<br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                                                                                                  | 'order_id'=>\$order_id       |
| amount                     | String<br>10-character decimal<br>Up to 7 digits (dollars) +<br>decimal point (.) + 2 digits<br>(cents) after the decimal<br>point<br><div>EXAMPLE: 1234567.89</div> | 'amount'=>\$amount           |
| credit card number         | String<br>max 20-character alpha-<br>numeric                                                                                                                         | 'pan'=>\$pan                 |
| expiry date                | String<br>4-character alphanumeric<br>YYMM                                                                                                                           | 'expiry_date'=>\$expiry_date |
| electronic commerce indic- | String                                                                                                                                                               | 'crypt_type'=>\$crypt        |

| Variable Name | Type and Limits          |  |
|---------------|--------------------------|--|
| ator          | 1-character alphanumeric |  |

### Independent Refund transaction request fields – Optional

| Variable Name      | Type and Limits                                                                                                                                                                                                                                 |                                            |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| customer ID        | <i>String</i><br>50-character alphanumeric<br><div> <b>NOTE:</b><br/>           Some special characters are not allowed:<br/>           &lt;&gt;\$%=?^{}[]\         </div>                                                                      | 'cust_id'=>\$cust_id                       |
| dynamic descriptor | <i>String</i><br>20-character alphanumeric<br>total of 22 characters including your merchant name and separator<br><div> <b>NOTE:</b><br/>           Some special characters are not allowed:<br/>           &lt;&gt;\$%=?^{}[]\         </div> | 'dynamic_descriptor'=>\$dynamic_descriptor |

### Sample Independent Refund

```

<?php
##
## This program takes 3 arguments from the command line:
## 1. Store id
## 2. api token
## 3. order id
##
## Example php -q TestIndependentRefund.php store1 yesguy unique_order_id
##
require "../mpgClasses.php";
$store_id='store5';
$api_token='yesguy';
$orderid='ord-' . date("dmy-G:i:s");
$dynamic_descriptor='123456';
## step 1) create transaction array ###
$txnArray=array('type'=>'ind_refund',
'order_id'=>$orderid,
'cust_id'=>'my cust id',
'amount'=>'1.00',
'pan'=>'4242424242424242',
'expdate'=>'1103',
'crypt_type'=>'7',

```

```

'dynamic_descriptor'=>$dynamic_descriptor
);
## step 2) create a transaction object passing the array created in
## step 1.
$mpgTxn = new mpgTransaction($txnArray);
## step 3) create a mpgRequest object passing the transaction object created
## in step 2
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
## step 4) create mpgHttpPost object which does an https post ##
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
## step 5) get an mpgResponse object ##
$mpgResponse=$mpgHttpPost->getMpgResponse();
## step 6) retrieve data using get methods
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
?>

```

## 2.9 Card Verification with AVS and CVD

Verifies the validity of the credit card, expiry date and any additional details (such as the Card Verification Digits or Address Verification details). It does not verify the available amount or lock any funds on the credit card.

### Things to Consider:

- The Card Verification transaction is only supported by Visa, Mastercard, Discover and American Express
- For some Credential on File transactions, Card Verification with AVS and CVD is used as a prior step to get the Issuer ID used in the subsequent transaction
- When testing Card Verification, please use the Visa and MasterCard test card numbers provided in the MasterCard Card Verification and Visa Card Verification tables available in CVD & AVS (E-Fraud) Simulator.

- For a full list of possible AVS & CVD result codes refer to the CVD and AVS Result Code tables.

```
$txnArray = array('type'=>'card_verification', ...);
$mpgTxn = new mpgTransaction($txnArray);
$mpgRequest = new mpgRequest($mpgTxn);
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

### Card Verification transaction request fields – Required

| Variable Name                 | Type and Limits                                                     |                              |
|-------------------------------|---------------------------------------------------------------------|------------------------------|
| order ID                      | String<br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces | 'order_id'=>\$order_id       |
| credit card number            | String<br>max 20-character alpha-numeric                            | 'pan'=>\$pan                 |
| expiry date                   | String<br>4-character alphanumeric<br>YYMM                          | 'expiry_date'=>\$expiry_date |
| electronic commerce indicator | String<br>1-character alphanumeric                                  | 'crypt_type'=>\$crypt        |

| Variable Name                                                                                                                                                                                                                                                                  | Type and Limits      |                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------------------------------|
| AVS Information                                                                                                                                                                                                                                                                | <i>Object</i><br>N/A | \$mpgTxn->setAvsInfo<br>(\$mpgAvsInfo) ; |
| CVD Information                                                                                                                                                                                                                                                                | <i>Object</i><br>N/A | \$mpgTxn->setCvdInfo<br>(\$mpgCvdInfo) ; |
| <div> <b>NOTE:</b> When storing credentials on the initial transaction, the CVD object must be sent; for subsequent transactions using stored credentials, CVD can be sent with cardholder-initiated transactions only—<b>merchants must not store CVD information.</b> </div> |                      |                                          |

### Account Name Verification object request fields

Request fields within the Account Name Verification object. The object can only be included in Card Verification transactions. Account name verification is only applicable to Visa credit cards.

| Variable Name | Type and Limits                            |                        |
|---------------|--------------------------------------------|------------------------|
| First Name    | <i>String</i><br>32-character alphanumeric | Cardholder last name   |
| Middle Name   | <i>String</i><br>32-character alphanumeric | Cardholder middle name |
| Last Name     | <i>String</i><br>32-character alphanumeric | Cardholder last name   |

### Sample Card Verification

```
<?php
require "../mpgClasses.php";
$store_id='store5';
$api_token="yesguy";
// TrId and TokenCryptogram are optional, refer documentation for more details.
$tr_id = '50189815682';
```



```
$token_cryptogram = 'APmbM/411e0uAAH+s6xMAAADFA==';

$txnArray=array('type'=>'card_verification',
'order_id'=>'ord-'.date("dmy-G:i:s"),
'cust_id'=>'my cust id',
'pan'=>'4242424242424242',
'expdate'=>'1512',
'crypt_type'=>'7'
//,'tr_id' => $tr_id
//,'token_cryptogram' => $token_cryptogram
);
$mpgTxn = new mpgTransaction($txnArray);
/***** AVS Variables *****/
$avs_street_number = '201';
$avs_street_name = 'Michigan Ave';
$avs_zipcode = 'M1M1M1';
/***** CVD Variables *****/
$cvd_indicator = '1';
$cvd_value = '198';

/***** Account Name Variables *****/
$first_name = 'FIRST';
$middle_name = 'MIDDL';
$last_name = 'LAST';

/***** AVS Associative Array *****/
$avsTemplate = array(
'avs_street_number'=>$avs_street_number,
'avs_street_name' =>$avs_street_name,
'avs_zipcode' => $avs_zipcode
);
/***** CVD Associative Array *****/
$cvdTemplate = array(
'cvd_indicator' => $cvd_indicator,
'cvd_value' => $cvd_value
);

/***** AccountName Array *****/
$accountNameTemplate = array(
'first_name'=>$first_name,
'middle_name' =>$middle_name,
'last_name' => $last_name
);

/***** AVS Object *****/
$mpgAvsInfo = new mpgAvsInfo ($avsTemplate);
/***** CVD Object *****/
$mpgCvdInfo = new mpgCvdInfo ($cvdTemplate);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("U");
$cof->setPaymentInformation("2");
$cof->setIssuerId("168451306048014");

/***** AccountNameObject *****/
$mpgAccountNameInfo = new mpgAccountNameInfo ($accountNameTemplate);

/***** Set Objects *****/
$mpgTxn->setAccountNameVerification($mpgAccountNameInfo);
$mpgTxn->setAvsInfo($mpgAvsInfo);
$mpgTxn->setCvdInfo($mpgCvdInfo);
$mpgTxn->setCofInfo($cof);

$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
```

```

$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\nSourcePanLast4 = " . $mpgResponse->getSourcePanLast4());
print("\nAccountNameVerificationResult = " . $mpgResponse->getAccountNameResult());
?>

```

## 2.10 Surcharge Lookup

Confirm eligibility of a card for merchant surcharges on a transaction.

Credit Surcharge is feature where merchants may append an additional charge to the amount of a transaction processed on an eligible credit card. Only eligible credit cards can be surcharged, so merchants must confirm eligibility of the card before processing the payment via surcharge lookup.

The following transactions support including surcharge amounts:

- Purchase (Basic)
- Pre-Authorization (Basic)
- Vault Purchase
- Vault Pre-Authorization
- 3DS Purchase
- 3DS Pre-Authorization
- 3DS & Vault Purchase
- 3DS & Vault Pre-Authorization
- Purchase Correction
- Pre-Authorization Completion
- Refund

### Things to Consider:

- Card association rules require that cardholders must be informed of any surcharge by a merchant.

- The cardholder must be offered an option for canceling the transaction if they refuse the surcharge.

```
$txnArray = array('type'=>'surcharge_lookup', ...);
$mpgTxn = new mpgTransaction($txnArray);
$mpgRequest = new mpgRequest($mpgTxn);
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

### Surcharge Lookup transaction request fields – Required

| Variable Name      | Type and Limits                                                                                                                                             |                    |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| credit card number | String<br>max 20-character alpha-numeric                                                                                                                    | 'pan'=>\$pan       |
| amount             | String<br>10-character decimal<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><div>EXAMPLE: 1234567.89</div> | 'amount'=>\$amount |

### Sample Surcharge Lookup

```
<?php
require "../mpgClasses.php";
```

```

$store_id='monca03650';
$api_token='7Yw0MPTlhjBRcZiE6837';
$amount = '50.00';
$pan='4242424242424242';
$txnArray=array('type'=>'surcharge_lookup',
'amount'=>$amount,
'pan'=>$pan
);
$mpgTxn = new mpgTransaction($txnArray);
$mpgRequest = new mpgRequest($mpgTxn);
//print the mpgrequest
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false or comment out this line for production transactions
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nMessage = " . $mpgResponse->getMessage());
print("\nCardType = " . $mpgResponse->getCardType());
print("\nIsSurchargeEligible = " . $mpgResponse->getIsSurchargeEligible());
print("\nMaxSurchargeRate = " . $mpgResponse->getMaxSurchargeRate());
print("\nMaxSurchargeAmount = " . $mpgResponse->getMaxSurchargeAmount());
print("\nServiceType = " . $mpgResponse->getServiceType());
?>

```

## 2.11 Batch Close

Takes the funds from all Purchase, Completion, Refund and Force Post transactions so that they will be deposited or debited the following business day.

For funds to be deposited the following business day, the batch must close before 11 pm Eastern Time.

```

$txnArray = array('type'=>'batchclose', ...);

$mpgTxn = new mpgTransaction($txnArray);

$mpgRequest = new mpgRequest($mpgTxn);

$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);

```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

### Batch Close transaction request fields – Required

| Variable Name        | Type and Limits |                          |
|----------------------|-----------------|--------------------------|
| ECR (electronic cash | String          | ecr_number=>\$ecr_number |

| Variable Name    | Type and Limits                      |  |
|------------------|--------------------------------------|--|
| register) number | No limit (value provided by Moneris) |  |

## Sample Batch Close

```
<?php
##
## This program takes 3 arguments from the command line:
## 1. Store id
## 2. api token
## 3. ecr number
##
## Example php -q TestBatchClose.php store1 yesguy 66002173
##
require "../mpgClasses.php";
$store_id='store5';
$api_token='yesguy';
$ecr_number='66013455';
## step 1) create transaction array ###
$txnArray=array('type'=>'batchclose',
'ecr_number'=>$ecr_number
);
$mpgTxn = new mpgTransaction($txnArray);
## step 2) create mpgRequest object ###
$mpgReq=new mpgRequest($mpgTxn);
$mpgReq->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgReq->setTestMode(true); //false for production transactions
## step 3) create mpgHttpPost object which does an https post ##
$mpgHttpPost=new mpgHttpPost($store_id,$api_token,$mpgReq);
## step 4) get an mpgResponse object ##
$mpgResponse=$mpgHttpPost->getMpgResponse();
##step 5) get array of all credit cards
$creditCards = $mpgResponse->getCreditCards($ecr_number);
## step 6) loop through the array of credit cards and get information
for($i=0; $i < count($creditCards); $i++)
{
print "\nCard Type = $creditCards[$i]";
print "\nPurchase Count = "
. $mpgResponse->getPurchaseCount($ecr_number,$creditCards[$i]);
print "\nPurchase Amount = "
. $mpgResponse->getPurchaseAmount($ecr_number,$creditCards[$i]);
print "\nRefund Count = "
. $mpgResponse->getRefundCount($ecr_number,$creditCards[$i]);
print "\nRefund Amount = "
. $mpgResponse->getRefundAmount($ecr_number,$creditCards[$i]);
print "\nCorrection Count = "
. $mpgResponse->getCorrectionCount($ecr_number,$creditCards[$i]);
print "\nCorrection Amount = "
. $mpgResponse->getCorrectionAmount($ecr_number,$creditCards[$i]);
}
?>
```

## 2.12 Open Totals

Returns the details about the currently open batch.

Similar to the Batch Close; the difference is that it does not close the batch for settlement.

```

$txnArray = array('type'=>'opentotals', ...);

$mpgTxn = new mpgTransaction($txnArray);

$mpgRequest = new mpgRequest($mpgTxn);

$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);

```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

### Open Totals transaction request fields – Required

For a full description of mandatory and optional values, see Appendix A Definition of Request Fields.

| Variable Name                         | Type and Limits                      |                          |
|---------------------------------------|--------------------------------------|--------------------------|
| ECR (electronic cash register) number | No limit (value provided by Moneris) | ecr_number=>\$ecr_number |

### Sample Open Totals

```

<?php
##
## This program takes 3 arguments from the command line:
## 1. Store id
## 2. api token
## 3. ecr number
##
## Example php -q TestOpenTotals.php store1 yesguy 66002163
##
require "../mpgClasses.php";
$store_id='store5';
$api_token='yesguy';
$ecr_number='66013455';
## step 1) create transaction array ###
$txnArray=array('type'=>'opentotals',
'ecr_number'=>$ecr_number
);
$mpgTxn = new mpgTransaction($txnArray);
## step 2) create mpgRequest object ###
$mpgReq= new mpgRequest($mpgTxn);
$mpgReq->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgReq->setTestMode(true); //false for production transactions
## step 3) create mpgHttpPost object which does an https post ##
$mpgHttpPost=new mpgHttpPost($store_id,$api_token,$mpgReq);
## step 4) get an mpgResponse object ##
$mpgResponse=$mpgHttpPost->getMpgResponse();

```

```
##step 5) get array of all credit cards
$creditCards = $mpgResponse->getCreditCards($secr_number);
## step 6) loop through the array of credit cards and get information
for($i=0; $i < count($creditCards); $i++)
{
print "\nCard Type = $creditCards[$i]";
print "\nPurchase Count = "
. $mpgResponse->getPurchaseCount($secr_number,$creditCards[$i]);
print "\nPurchase Amount = "
. $mpgResponse->getPurchaseAmount($secr_number,$creditCards[$i]);
print "\nRefund Count = "
. $mpgResponse->getRefundCount($secr_number,$creditCards[$i]);
print "\nRefund Amount = "
. $mpgResponse->getRefundAmount($secr_number,$creditCards[$i]);
print "\nCorrection Count = "
. $mpgResponse->getCorrectionCount($secr_number,$creditCards[$i]);
print "\nCorrection Amount = "
. $mpgResponse->getCorrectionAmount($secr_number,$creditCards[$i]);
}
?>
```

## 3 Credential on File

- 3.1 About Credential on File
- 3.2 Credential on File Info Object and Variables
- 3.3 Credential on File Transaction Types
- 3.5 Initial Transactions in Credential on File
- 3.9 Credential on File and Converting Temporary Tokens
- 3.8 Vault Tokenize Credit Card and Credential on File
- 3.10 Card Verification and Credential on File Transactions

### 3.1 About Credential on File

When storing customers' credit card credentials for use in future authorizations, or when using these credentials in subsequent transactions, card brands now require merchants to indicate this in the transaction request.

In the Moneris API, this is handled by the Moneris Gateway via the inclusion of the Credential on File info object and its variables in the transaction request.

While the requirements for handling Credential on File transactions relate to Visa, Mastercard and Discover only, in order to avoid confusion and prevent error, please implement these changes for all card types and the Moneris system will then correctly flow the relevant card data values as appropriate.

**NOTE:** If either the first transaction or a Card Verification authorization is declined when attempting to store cardholder credentials, those credentials cannot be stored —therefore the merchant must not use the credential for any subsequent transactions.

### 3.2 Credential on File Info Object and Variables

The Credential on File Info object is nested within the request for the applicable transaction types.

Credential on File Info Object:

cof

Variables in the cof object:

Payment Indicator

Payment Information



Issuer ID

For more information, see Definition of Request Fields – Credential on File.

### 3.3 Credential on File Transaction Types

The Credential on File Info object applies to the following transaction types:

- Purchase
- Pre-Authorization
- Purchase with 3-D Secure – cavv\_purchase
- Purchase with 3-D Secure and Recurring Billing
- Pre-Authorization with 3-D Secure – cavv\_preauth
- Purchase with Vault – ResPurchaseCC
- Pre-Authorization with Vault – ResPreauthCC
- Card Verification with AVS and CVD
- Card Verification with Vault – ResCardVerificationCC
- Vault Add Credit Card – ResAddCC
- Vault Update Credit Card – ResUpdateCC
- Vault Add Token – ResAddToken
- Vault Tokenize Credit Card – ResTokenizeCC
- Recurring Billing
- MCP Purchase
- MCP Pre-Authorization
- MCP Pre-Authorization Completion

- MCP Purchase with Vault
- MCP Pre-Authorization with Vault

### 3.4 Merchant- vs. Cardholder-Initiated COF Transactions

Transactions defined as Credential on File (COF) can be initiated one of two ways: by a merchant or by a cardholder. The initiator of a transaction is important because it determines which Credential on File indicator fields need to be sent in the transaction request.

**Merchant-initiated Credential on File transactions:** transactions in which the merchant intends to store cardholder credentials or use credentials that have already been stored. This includes sending the Credential on File Info object in the transaction request, and including all three of its fields: **issuer ID**, **payment indicator** and **payment information**.

**Cardholder-initiated Credential on File transactions:** transactions which are triggered by some action by the cardholder. For cardholder-initiated transactions, only the **payment indicator** and **payment information** fields are required.

For simplicity in developing your integration, the Moneris Gateway also allows cardholder-initiated transactions to be processed according to the same Credential on File rules as apply to merchant-initiated transactions. Technically, the **issuer ID** indicator is not required for cardholder-initiated transactions, but for convenience, if it is included in the transaction request, the Moneris Gateway will just ignore it when forwarding the request to the host.

### 3.5 Initial Transactions in Credential on File

When sending an *initial* transaction with the Credential on File Info object, i.e., a transaction request where the cardholder's credentials are being stored for the *first* time, it is important to understand the following:

- You must send the cardholder's Card Verification Digits (CVD)
- **Issuer ID** will be sent without a value on the initial transaction, because it is received in the response to that initial transaction; for all *subsequent* merchant-initiated transactions and all administrative transactions you send this **Issuer ID**
- The **payment information** field should always be set to a value of 0 on the first transaction
- The **payment indicator** field should be set to the value that is appropriate for the transaction

### 3.6 COF With Previously Stored Credentials

When processing a transaction with cardholder information that was already stored **prior to** the implementation of Credential on File requirements, you must:

- Include the Credential on File Info object in the transaction request, and
- Send the **payment information** value as 2, and
- Store the **issuer ID** returned in the transaction and associate it with the cardholder credentials for future use

Once the **issuer ID** has been stored and associated with the cardholder credentials, send it in all subsequent transactions going forward. **Issuer ID** is only required when sending merchant-initiated transactions.

### 3.7 Payment Indicator and E-Commerce Indicator Values

When sending Credential on File information in transaction requests that also include the **e-commerce indicator** request field (in code, referred to as `crypt_type`), acceptable values for the e-commerce indicator are dependent upon the value being sent for the **payment indicator**.

| If Payment Indicator Value is:   | Allowable e-commerce indicator values are:                                                                                                                                                            |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R<br><br>Fixed-Rate Recurring    | 2 – Mail Order / Telephone Order—Recurring<br><br>5 – Authenticated e-commerce transaction (3-D Secure)<br><br>6 – Non-authenticated e-commerce transaction (3-D Secure)                              |
| V<br><br>Variable-Rate Recurring | 2 – Mail Order / Telephone Order—Recurring (Variable)<br><br>5 – Authenticated e-commerce transaction (3-D Secure)<br><br>6 – Non-authenticated e-commerce transaction (3-D Secure)                   |
| <div>NOTE: MasterCard Only</div> |                                                                                                                                                                                                       |
| C<br><br>Credentials On File     | 1 – Mail Order / Telephone Order—Single<br><br>5 – Authenticated e-commerce transaction (3-D Secure)<br><br>6 – Non-authenticated e-commerce transaction (3-D Secure)<br><br>7 – SSL-enabled merchant |
| U                                | 1 – Mail Order / Telephone Order—Single                                                                                                                                                               |

| If Payment Indicator Value is:             | Allowable e-commerce indicator values are:                |
|--------------------------------------------|-----------------------------------------------------------|
| Unscheduled Stored Credential Transaction  | 7 – SSL-enabled merchant                                  |
| Z                                          | 1 – Mail Order / Telephone Order—Single                   |
| Unscheduled Customer-Initiated Transaction | 5 – Authenticated e-commerce transaction (3-D Secure)     |
|                                            | 6 – Non-authenticated e-commerce transaction (3-D Secure) |
|                                            | 7 – SSL-enabled merchant                                  |

### 3.8 Vault Tokenize Credit Card and Credential on File

When you want to store cardholder credentials from previous transactions into the Vault, you use the Vault Tokenize Credit Card transaction request. Credential on File rules require that only previous transactions with the Credential on File Info object can be tokenized to the Vault.

For more information about this transaction, see 4.5.10 Vault Tokenize Credit Card – ResTokenizeCC.

### 3.9 Credential on File and Converting Temporary Tokens

In the event you decide to convert a temporary token representing cardholder credentials into a permanent token, these credentials become stored credentials, and therefore it is necessary to send Credential on File information.

For Vault Temporary Token Add transactions where you subsequently decide to convert the temporary token into a permanent token (stored credentials):

1. Send a transaction request that includes the Credential on File Info object to get the Issuer ID; this can be a Card Verification, Purchase or Pre-Authorization request
2. After completing the transaction, send the Vault Add Token request with the Credential on File object (Issuer ID only) in order to convert the temporary token to a permanent one.

### 3.10 Card Verification and Credential on File Transactions

In the absence of a Purchase or Pre-Authorization, a Card Verification transaction is used to get the unique issuer ID value (**issuerId**) that is used in subsequent Credential on File transactions. Issuer ID is a variable included in the nested Credential on File Info object.

For all first-time transactions, including Card Verification transactions, you must also request the cardholder's Card Verification Details (CVD). For more on CVD, see 10.2 Card Validation Digits (CVD).

For a complete list of these variables, see each transaction type or Definition of Request Fields – Credential on File

The Card Verification request, including the Credential on File Info object, must be sent immediately prior to storing cardholder credentials.

For information about Card Verification, see 2.9 Card Verification with AVS and CVD.

### 3.10.1 When to Use Card Verification With COF

If you are not sending a Purchase or Pre-Authorization transaction (i.e., you are not charging the customer immediately), you must use Card Verification (or in the case of Vault Add Token, Card Verification with Vault) first before running the transaction in order to get the Issuer ID.

Transactions this applies to:

Vault Add Credit Card – ResAddCC

Vault Update Credit Card – ResUpdateCC

Vault Add Token – ResAddToken

Recurring Billing transactions, if:

- the first transaction is set to start on a future date

### 3.10.2 Credential on File and Vault Add Token

For Vault Add Token transactions:

1. Send Card Verification with Vault transaction request including the Credential on File object to get the Issuer ID
2. Send the Vault Add Token request including the Credential on File object (with Issuer ID only; other fields are not applicable)

For more on this transaction type, see 4.5.9 Vault Add Token – ResAddToken.

### 3.10.3 Credential on File and Vault Update Credit Card

For Vault Update Credit Card transactions where you are updating the credit card number:

1. Send Card Verification transaction request including the Credential on File object to get the Issuer ID
2. Send the Vault Update Credit Card request including the Credential on File Info object (Issuer ID only).

For more on this transaction type, see 4.5.3 Vault Update Credit Card – ResUpdateCC.

### 3.10.4 Credential on File and Vault Add Credit Card

For Vault Add Credit Card transactions:

1. Send Card Verification transaction request including the Credential on File object to get the Issuer ID
2. Send the Vault Add Credit Card request including the Credential on File Info object (Issuer ID only)

For more on this transaction type, see 4.5.1 Vault Add Credit Card – ResAddCC.

### 3.10.5 Credential on File and Recurring Billing

**NOTE:** The value of the **payment indicator** field must be **R** when sending Recurring Billing transactions.

For Recurring Billing transactions which are set to start **immediately**:

1. Send a Purchase transaction request with both the Recurring Billing and Credential on File info objects (with Recurring Billing object field **start now** = true)

For Recurring Billing transactions which are set to start on a **future** date:

1. Send Card Verification transaction request including the Credential on File info object to get the Issuer ID
2. Send Purchase transaction request with the Recur and Credential on File info objects included

For updating a Recurring Billing series where you are updating the card number (does not apply if you are only modifying the schedule or amount in a recurring series):

1. Send Card Verification request including the Credential on File info object to get the Issuer ID
2. Send a Recurring Billing Update transaction

For more information about the Recurring Billing object, see [Definition of Request Fields – Recurring](#).

## 4 Vault

- 4.1 About the Vault Transaction Set
- 4.2 Vault Transaction Types
- 4.3 Vault Transactions That Support Temporary Tokens
- 4.5 Vault Administrative Transactions
- 4.6 Vault Financial Transactions
- 4.7 Hosted Tokenization

### 4.1 About the Vault Transaction Set

The Vault feature allows merchants to create customer profiles, edit those profiles, and use them to process transactions without having to enter financial information each time. Customer profiles store customer data essential to processing transactions, including credit and signature debit.

The Vault is a complement to the Recurring Billing module. It securely stores customer account information on Moneris secure servers. This allows merchants to bill customers for routine products or services when an invoice is due.

### 4.2 Vault Transaction Types

The Vault API supports both administrative and financial transactions.

#### 4.2.1 Administrative Vault Transaction types

##### **ResAddCC**

Creates a new credit card profile, and generates a unique data key which can be obtained from the Receipt object.

This data key is the profile identifier that all future financial Vault transactions will use to associate with the saved information.

##### **EncResAddCC**

Creates a new credit card profile, but requires the card data to be either swiped or manually keyed in via a Moneris-provided encrypted mag swipe reader.

##### **ResTempAdd**

Creates a new temporary token credit card profile. This transaction requires a duration to be set to indicate how long the temporary token is to be stored for.

During the lifetime of this temporary token, it may be used for any other vault transaction before it is permanently deleted from the system.

##### **ResUpdateCC**

Updates a Vault profile (based on the data key) to contain credit card information.



All information contained within a credit card profile is updated as indicated by the submitted fields.

#### **EncResUpdateCC**

Updates a profile (based on the data key) to contain credit card information. The encrypted version of this transaction requires the card data to either be swiped or manually keyed in via a Moneris-provided encrypted mag swipe reader.

#### **ResDelete**

Deletes an existing Vault profile of any type using the unique data key that was assigned when the profile was added.

It is important to note that after a profile is deleted, the information which was saved within can no longer be retrieved.

#### **ResLookupFull**

Verifies what is currently saved under the Vault profile associated with the given data key. The response to this transaction returns the latest active data for that profile.

Unlike ResLookupMasked (which returns the masked credit card number), this transaction returns both the masked and the unmasked credit card numbers.

#### **ResLookupMasked**

Verifies what is currently saved under the Vault profile associated with the given data key. The response to this transaction returns the latest active data for that profile.

Unlike ResLookupFull (which only returns both the masked and the unmasked credit card numbers), this transaction only returns the masked credit card number.

#### **ResGetExpiring**

Verifies which profiles have credit cards that are expiring during the current and next calendar month. For example, if you are processing this transaction on September 30, then it will return all cards that expire(d) in September and October of this year.

When generating a list of profiles with expiring credit cards, only the **masked** credit card numbers are returned.

This transaction can be performed no more than 2 times on any given calendar day, and it only applies to credit card profiles.

#### **ResIsCorporateCard**

Determines whether a profile has a corporate card registered within it.

After sending the transaction, the response field to the Receipt object's getCorporateCard method is either `true` or `false` depending on whether the associated card is a corporate card.

#### **ResAddToken**

Converts a Hosted Tokenization temporary token to a permanent Vault token.

A temporary token is valid for 15 minutes after it is created.

**ResTokenizeCC**

Creates a new credit card profile using the credit card number, expiry date and e-commerce indicator that were submitted in a previous financial transaction. A transaction that was previously done in Moneris Gateway is taken, and the card data from that transaction is stored in the Moneris Vault.

As with ResAddCC, a unique data key is generated and returned to the merchant via the Receipt object. This is the profile identifier that all future financial Vault transactions will use to associate with the saved information.

## 4.2.2 Financial Vault Transaction types

**ResPurchaseCC**

Uses the data key to identify a previously registered credit card profile. The details saved within the profile are then submitted to perform a Purchase transaction.

**ResPreauthCC**

Uses the data key to identify a previously registered credit card profile. The details within the profile are submitted to perform a Pre-Authorization transaction.

**ResIndRefundCC**

Uses the unique data key to identify a previously registered credit card profile, and credits a specified amount to that credit card.

**ResMpiTxn**

Uses the data key (as opposed to a credit card number) in a VBV/SecureCode Txn MPI transaction. The merchant uses the data key with ResMpiTxn request, and then reads the response fields to verify whether the card is enrolled in Verified by Visa or MasterCard SecureCode. Retrieves the vault transaction value to pass on to Visa or MasterCard.

After it has been validated that the data key is enrolled in 3-D Secure, a window appears in which the customer can enter the 3-D Secure password. The merchant may initiate the forming of the validation form `getMpiInlineForm()`.

For more information on integrating with MonerisMPI, refer to 1 MPI.

## 4.3 Vault Transactions That Support Temporary Tokens

The following transactions support temporary as well as permanent tokens:

- Purchase with Vault – ResPurchaseCC
- Pre-Authorization with Vault – ResPreauthCC
- Card Verification with Vault – ResCardVerificationCC
- Purchase with Vault and 3-D Secure

- Pre-Authorization with Vault & 3-D Secure
- Force Post with Vault – ResForcePostCC
- Vault Independent Refund – ResIndRefundCC
- Vault Is Corporate Card - ResIsCorporateCard

## 4.4 Vault and Installments

Installments functionality is also available on transactions using cardholder credentials stored in the Moneris Vault. To offer this feature to the customer, send the Vault Installment Plan Lookup transaction prior to running a Purchase with Vault or Pre-Authorization with Vault.

For more about Installments, see 9 Installments by Visa

## 4.5 Vault Administrative Transactions

Administrative transactions allow you to perform such tasks as creating new Vault profiles, deleting existing Vault profiles and updating profile information.

Some Vault Administrative Transactions require the Credential on File object to be sent with the **issuer ID** field only.

### 4.5.1 Vault Add Credit Card – ResAddCC

Creates a new credit card profile, and generates a unique data key which can be obtained from the Receipt object.

This data key is the profile identifier that all future financial Vault transactions will use to associate with the saved information.

#### **Vault Add Credit Card transaction object definition**

```
$txnArray = array('type'=>'res_add_cc', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

#### **HttpPostRequest object for Vault Add Credit Card transaction**

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

#### **Vault Add Credit Card transaction request fields – Required**

For a full description of mandatory and optional values, see Appendix A Definition of Request Fields.

| Variable Name                                                                                                                                                                                                                                            | Type and Limits                                   | Set Method                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|------------------------------|
| credit card number                                                                                                                                                                                                                                       | <i>String</i><br>max 20-character alpha-numeric   | 'pan'=>\$pan                 |
| expiry date                                                                                                                                                                                                                                              | <i>String</i><br>4-character alphanumeric<br>YYMM | 'expiry_date'=>\$expiry_date |
| electronic commerce indicator                                                                                                                                                                                                                            | <i>String</i><br>1-character alphanumeric         | 'crypt_type'=>\$crypt        |
| Credential on File Info                                                                                                                                                                                                                                  | <i>Object</i>                                     | \$mpgTxn->setCofInfo(\$cof); |
| cof                                                                                                                                                                                                                                                      | N/A                                               |                              |
| <b>NOTE:</b> This is a nested object within the transaction, and required when storing or using the customer's stored credentials. For information about fields in the Credential on File Info object, see Credential on File Info Object and Variables. |                                                   |                              |

### Vault Add Credit Card transaction request fields – Optional

| Variable Name | Type and Limits                                                                                                         | Set Method           |
|---------------|-------------------------------------------------------------------------------------------------------------------------|----------------------|
| customer ID   | <i>String</i><br>50-character alphanumeric<br><b>NOTE:</b><br>Some special characters are not allowed:<br><>\$%=?^{}[]\ | 'cust_id'=>\$cust_id |
| email address | <i>String</i><br>30-character alphanumeric                                                                              | 'email'=>\$email     |

| Variable Name   | Type and Limits                            | Set Method                           |
|-----------------|--------------------------------------------|--------------------------------------|
| phone number    | <i>String</i><br>30-character alphanumeric | 'phone'=>\$phone                     |
| note            | <i>String</i><br>30-character alphanumeric | 'note'=>\$note                       |
| data key format | <i>String</i><br>2-character alphanumeric  | 'data_key_format'=>\$data_key_format |
| AVS Information | <i>Object</i><br>N/A                       | \$mpgTxn->setAvsInfo(\$mpgAvsInfo);  |

### Credential on File Info object request fields

| Variable Name                                                                                                                                                                                                                                                                                   | Type and Limits                                                       | Set Method                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| issuer ID<br><br><div> <b>NOTE:</b> This variable is required for all merchant-initiated transactions following the first one; upon sending the first transaction, the issuer ID value is received in the transaction response and then used in subsequent transaction requests.         </div> | <i>String</i><br><br>15-character alphanumeric<br><br>variable length | \$cof->setIssuerId("VALUE_FOR_ISSUER_ID");<br><br><div> <b>NOTE:</b> For a list and explanation of the possible values to send for this variable, see Definition of Request Fields – Credential on File         </div> |

### Sample Vault Add Credit Card

```

<?php
##
## Example php -q TestResAddCC.php store3 yesguy
##
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
/***** Transactional Variables *****/
$type='res_add_cc';
$cust_id='customer1';

```

## Sample Vault Add Credit Card

```

$phone = '5555551234';
$email = 'bob@smith.com';
$note = 'this is my note';
$pan='5454545454545454';
$expiry_date='1412';
$crypt_type='1';
$data_key_format = "0";
$avs_street_number = '123';
$avs_street_name = 'lakeshore blvd';
$avs_zipcode = '90210';
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'cust_id'=>$cust_id,
'phone'=>$phone,
'email'=>$email,
'note'=>$note,
'pan'=>$pan,
'expdate'=>$expiry_date,
// 'data_key_format'=>$data_key_format, //optional
'crypt_type'=>$crypt_type
);
/***** AVS Associative Array *****/
$avsTemplate = array(
'avs_street_number' => $avs_street_number,
'avs_street_name' => $avs_street_name,
'avs_zipcode' => $avs_zipcode
);
/***** AVS Object *****/
$mpgAvsInfo = new mpgAvsInfo ($avsTemplate);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setIssuerId("139X3130ASCXAS9");
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
$mpgTxn->setAvsInfo($mpgAvsInfo);
$mpgTxn->setCofInfo($cof);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
//----- ResolveData -----
print("\n\nCust ID = " . $mpgResponse->getResDataCustId());
print("\n\nPhone = " . $mpgResponse->getResDataPhone());
print("\n\nEmail = " . $mpgResponse->getResDataEmail());
print("\n\nNote = " . $mpgResponse->getResDataNote());

```

### Sample Vault Add Credit Card

```
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\nCrypt Type = " . $mpgResponse->getResDataCryptType());
print("\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
print("\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
?>
```

## Vault response fields

For a list and explanation of (Receipt object) response fields that are available after sending this Vault transaction, see Definitions of Response Fields (page 593).

### 4.5.1.1 Vault Data Key

The ResAddCC sample code includes the following instruction from the Receipt object:

```
print("\nDataKey = " . $mpgResponse->getDataKey());
```

The data key response field is populated when you send a Vault Add Credit Card – ResAddCC (page 71), Vault Encrypted Add Credit Card – EncResAddCC (page 75), Vault Tokenize Credit Card – ResTokenizeCC (page 99), Vault Temporary Token Add – ResTempAdd (page 78) or Vault Add Token – ResAddToken (page 96) transaction. It is the profile identifier that all future financial Vault transactions will use to associate with the saved information.

The data key is a maximum 28-character alphanumeric string.

### 4.5.1.2 Vault Encrypted Add Credit Card – EncResAddCC

## Vault Encrypted Add Credit Card transaction object definition

```
$txnArray = array('type'=>'enc_res_add_cc', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

## HttpPostRequest object for Vault Encrypted Add Credit Card transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

## Vault Encrypted Add Credit Card transaction request fields – Required

For a full description of mandatory and optional values, see Appendix A Definition of Request Fields.

| Variable Name          | Type and Limits | Set Method                 |
|------------------------|-----------------|----------------------------|
| encrypted track 2 data | <i>String</i>   | 'enc_track2'=>\$enc_track2 |

| Variable Name                 | Type and Limits                                       | Set Method                   |
|-------------------------------|-------------------------------------------------------|------------------------------|
|                               | 40-character numeric                                  |                              |
| device type                   | String<br>30-character alphanumeric<br>case sensitive | 'device_type'=>\$device_type |
| electronic commerce indicator | String<br>1-character alphanumeric                    | 'crypt_type'=>\$crypt        |

### Vault Encrypted Add Credit Card transaction request fields – Optional

| Variable Name   | Type and Limits                                                                                                                     | Set Method                              |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| customer ID     | String<br>50-character alphanumeric<br><div><b>NOTE:</b><br/>Some special characters are not allowed:<br/>&lt;&gt;\$%=?^{}[]\</div> | 'cust_id'=>\$cust_id                    |
| AVS Information | Object<br>N/A                                                                                                                       | \$mpgTxn->setAvsInfo<br>(\$mpgAvsInfo); |
| email address   | String<br>30-character alphanumeric                                                                                                 | 'email'=>\$email                        |
| phone number    | String<br>30-character alphanumeric                                                                                                 | 'phone'=>\$phone                        |
| note            | String<br>30-character alphanumeric                                                                                                 | 'note'=>\$note                          |
| data key format | String<br>2-character alphanumeric                                                                                                  | 'data_key_format'=>\$data_key_format    |



## Sample Vault Encrypted Add Credit Card

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
/***** Transactional Variables *****/
$type='enc_res_add_cc';
$cust_id='cust1';
$phone = '6479996999';
$email = 'bob@smith.com';
$note = 'this is my note';
$enc_track2 = 'ENCRYPTEDTRACK2DATA';
$device_type='idtech_bdk';
$data_key_format="0";
$script_type='7';
$avs_street_number = '11';
$avs_street_name = 'lakeshore blvd';
$avs_zipcode = 'm8x2x2';
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'cust_id'=>$cust_id,
'phone'=>$phone,
'email'=>$email,
'note'=>$note,
'enc_track2'=>$enc_track2,
'device_type'=>$device_type,
// 'data_key_format'=>$data_key_format, //optional
'crypt_type'=>$script_type
);
/***** AVS Associative Array *****/
$avsTemplate = array(
'avs_street_number' => $avs_street_number,
'avs_street_name' => $avs_street_name,
'avs_zipcode' => $avs_zipcode
);
/***** AVS Object *****/
$mpgAvsInfo = new mpgAvsInfo ($avsTemplate);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
$mpgTxn->setAvsInfo($mpgAvsInfo);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
//----- ResolveData -----
print("\n\nCust ID = " . $mpgResponse->getResDataCustId());

```

### Sample Vault Encrypted Add Credit Card

```
print("\nPhone = " . $mpgResponse->getResDataPhone());
print("\nEmail = " . $mpgResponse->getResDataEmail());
print("\nNote = " . $mpgResponse->getResDataNote());
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\nCrypt Type = " . $mpgResponse->getResDataCryptType());
print("\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
print("\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
?>
```

### Vault response fields

For a list and explanation of (Receipt object) response fields that are available after sending this Vault transaction, see Definitions of Response Fields (page 593).

## 4.5.2 Vault Temporary Token Add – ResTempAdd

Creates a new temporary token credit card profile. This transaction requires a duration to be set to indicate how long the temporary token is to be stored for.

During the lifetime of this temporary token, it may be used for any other Vault transaction before it is permanently deleted from the system.

#### Things to Consider:

- The duration, or lifetime, of the temporary token can be set to be a maximum of 15 minutes.

### Vault Temporary Token Add transaction object definition

```
$txnArray = array('type'=>'res_temp_add', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for Vault Temporary Token Add transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Vault Temporary Token Add transaction request fields – Required

For a full description of mandatory and optional values, see Appendix A Definition of Request Fields.

| Variable Name                 | Type and Limits                                             | Set Method                   |
|-------------------------------|-------------------------------------------------------------|------------------------------|
| credit card number            | <i>String</i><br>max 20-character alpha-numeric             | 'pan'=>\$pan                 |
| expiry date                   | <i>String</i><br>4-character alphanumeric<br>YYMM           | 'expiry_date'=>\$expiry_date |
| duration                      | <i>String</i><br>3-character numeric<br>maximum 900 seconds | 'duration'=>\$duration       |
| electronic commerce indicator | <i>String</i><br>1-character alphanumeric                   | 'crypt_type'=>\$crypt        |

### Vault Temporary Token Add transaction request fields – Optional

| Variable Name   | Type and Limits                           | Set Method                           |
|-----------------|-------------------------------------------|--------------------------------------|
| data key format | <i>String</i><br>2-character alphanumeric | 'data_key_format'=>\$data_key_format |

#### Sample Vault Temporary Token Add

```
<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
/***** Transactional Variables *****/
$type='res_temp_add';
$pan='5454545454545454';
$expiry_date='1509';
$duration='900';
$data_key_format = "0";
$crypt_type='7';
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'pan'=>$pan,
'expdate'=>$expiry_date,
'duration'=>$duration,
// 'data_key_format'=>$data_key_format, //optional
'crypt_type'=>$crypt_type
```

### Sample Vault Temporary Token Add

```
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
//----- ResolveData -----
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
?>
'crypt_type'=>$crypt_type
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
//----- ResolveData -----
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
?>
```

### Vault response fields

For a list and explanation of (Receipt object) response fields that are available after sending this Vault transaction, see Definitions of Response Fields (page 593).

### 4.5.3 Vault Update Credit Card – ResUpdateCC

Updates an existing Vault profile (referencing the profile's unique **data key**) with cardholder information.

Information contained within a credit card profile is updated as indicated by the submitted fields; if any field representing an item of cardholder information is not sent in this request, that item will remain unchanged in the profile.

If the Vault profile is being updated with a new credit card number, then you first need to send a Purchase, Pre-Authorization or Card Verification transaction, with the Credential on File Info object included, before performing Vault Update Credit Card. If the credit card number is not one of the profile items being updated, this step is not required.

#### Things to Consider:

- To update a specific element in the profile, set that element using the corresponding set method
- When updating a credit card number, first send a Purchase, Pre-Authorization, or Card Verification with the Credential on File Info object before sending this transaction; send the issuer ID received in the response in the subsequent Vault Update Credit Card request
- If the credit card number is not one of the profile items being updated, the Credential on File info object is not required

#### Vault Update Credit Card transaction object definition

```
$txnArray = array('type'=>'res_update_cc', ...);  
$mpgTxn = new mpgTransaction($txnArray);
```

#### HttpPostRequest object for Vault Update Credit Card transaction

```
$mpgRequest = new mpgRequest($mpgTxn);  
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

#### Vault Update Credit Card transaction request fields – Required

For a full description of mandatory and optional values, see Appendix A Definition of Request Fields.

| Variable Name | Type and Limits | Set Method             |
|---------------|-----------------|------------------------|
| data key      | <i>String</i>   | 'data_key'=>\$data_key |

| Variable Name | Type and Limits           | Set Method |
|---------------|---------------------------|------------|
|               | 25-character alphanumeric |            |

Optional values that are submitted to the ResUpdateCC object are updated. Unsubmitted optional values (with one exception) remain unchanged. This allows you to change only the fields you want.

If a profile contains AVS information, but a Vault Update Credit Card transaction is submitted without an AVS Info object, the existing AVS Info details are deactivated and the new credit card information is registered without AVS.

### Vault Update Credit Card transaction request fields – Optional

| Variable Name                 | Type and Limits                                                                                                                            | Set Method                   |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| credit card number            | <i>String</i><br>20-character alphanumeric                                                                                                 | 'pan'=>\$pan                 |
| expiry date                   | <i>String</i><br>4-character alphanumeric<br>YYMM                                                                                          | 'expiry_date'=>\$expiry_date |
| electronic commerce indicator | <i>String</i><br>1-character alphanumeric                                                                                                  | 'crypt_type'=>\$crypt        |
| customer ID                   | <i>String</i><br>50-character alphanumeric<br><div><b>NOTE:</b><br/>Some special characters are not allowed:<br/>&lt;&gt;\$%=?^{}[]\</div> | 'cust_id'=>\$cust_id         |
| email address                 | <i>String</i><br>30-character alphanumeric                                                                                                 | 'email'=>\$email             |
| phone number                  | <i>String</i><br>30-character alphanumeric                                                                                                 | 'phone'=>\$phone             |
| note                          | <i>String</i><br>30-character alphanumeric                                                                                                 | 'note'=>\$note               |

| Variable Name                                                                                                                                                                                                                                                                | Type and Limits      | Set Method                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------------------------------|
| AVS Information                                                                                                                                                                                                                                                              | <i>Object</i><br>N/A | \$mpgTxn->setAvsInfo<br>(\$mpgAvsInfo) ; |
| Credential on File Info<br>cof                                                                                                                                                                                                                                               | <i>Object</i><br>N/A | \$mpgTxn->setCofInfo (\$cof) ;           |
| <div> <p><b>NOTE:</b> This is a nested object within the transaction, and required when storing or using the customer's stored credentials. For information about fields in the Credential on File Info object, see Credential on File Info Object and Variables.</p> </div> |                      |                                          |

### Credential on File Info object request fields

| Variable Name                                                                                                                                                                                                                                                                                                                                                                                                                                     | Type and Limits                                               | Set Method                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------|
| issuer ID                                                                                                                                                                                                                                                                                                                                                                                                                                         | <i>String</i><br>15-character alphanumeric<br>variable length | \$cof->setIssuerId ("VALUE_ FOR_ISSUER_ID") ; |
| <div> <p><b>NOTE:</b> This variable is required for all merchant-initiated transactions following the first one; upon sending the first transaction, the issuer ID value is received in the transaction response and then used in subsequent transaction requests.</p> </div> <div> <p><b>NOTE:</b> For a list and explanation of the possible values to send for this variable, see Definition of Request Fields – Credential on File</p> </div> |                                                               |                                               |

### Sample Vault Update Credit Card

```

<?php
##
## Example php -q TestResUpdateCC.php store3 yesguy
##
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
/***** Transactional Variables *****/
$type='res_update_cc';

```

## Sample Vault Update Credit Card

```

$data_key='D8cpd4r7REXoN8NIJPI512xPh';
$cust_id='customer1';
$phone = '5555555555';
$email = 'bob@smith.com';
$note = 'stuff';
$pan='5454545454545454';
$expiry_date='0909';
$crypt_type='7';
$savs_street_number = '123';
$savs_street_name = 'stuff dr';
$savs_zipcode = '90215';
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'data_key'=>$data_key,
'cust_id'=>$cust_id,
'phone'=>$phone,
'email'=>$email,
'note'=>$note,
'pan'=>$pan,
'expdate'=>$expiry_date,
'crypt_type'=>$crypt_type
);
/***** AVS Associative Array *****/
$savsTemplate = array(
'avs_street_number' => $savs_street_number,
'avs_street_name' => $savs_street_name,
'avs_zipcode' => $savs_zipcode
);
/***** AVS Object *****/
$mpgAvsInfo = new mpgAvsInfo ($savsTemplate);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setIssuerId("168451306048014");
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
$mpgTxn->setAvsInfo($mpgAvsInfo);
$mpgTxn->setCofInfo($cof);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
//----- ResolveData -----
print("\n\nCust ID = " . $mpgResponse->getResDataCustId());
print("\n\nPhone = " . $mpgResponse->getResDataPhone());
print("\n\nEmail = " . $mpgResponse->getResDataEmail());
print("\n\nNote = " . $mpgResponse->getResDataNote());

```



### Sample Vault Update Credit Card

```
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\nCrypt Type = " . $mpgResponse->getResDataCryptType());
print("\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
print("\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
?>
```

### Vault response fields

For a list and explanation of (Receipt object) response fields that are available after sending this Vault transaction, see Definitions of Response Fields (page 593).

#### 4.5.3.1 Vault Encrypted Update CC – EncResUpdateCC

### Vault Encrypted Update CC transaction object definition

```
$txnArray = array('type'=>'enc_res_update_cc', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for Vault Encrypted Update CC transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Vault Encrypted Update CC transaction request fields – Required

For a full description of mandatory and optional values, see Appendix A Definition of Request Fields.

| Variable Name          | Type and Limits                                              | Set Method                   |
|------------------------|--------------------------------------------------------------|------------------------------|
| data key               | <i>String</i><br>25-character alphanumeric                   | 'data_key'=>\$data_key       |
| encrypted track 2 data | <i>String</i><br>40-character numeric                        | 'enc_track2'=>\$enc_track2   |
| device type            | <i>String</i><br>30-character alphanumeric<br>case sensitive | 'device_type'=>\$device_type |

Optional values that are submitted to the ResUpdateCC object are updated, while unsubmitted optional values (with one exception) remain unchanged. This allows you to change only the fields you want.

The exception is that if you are making changes to the payment type, **all** of the variables in the optional values table below must be submitted.

If you update a profile to a different payment type, it is automatically deactivated and a new credit card profile is created and assigned to the data key. The only values from the prior profile that will remain unchanged are the customer ID, phone number, email address, and note.

**EXAMPLE:** If a profile contains AVS information, but a ResUpdateCC transaction is submitted without an AVSInfo object, the existing AVSInfo details are deactivated and the new credit card information is registered without AVS.

### Vault Encrypted Update CC transaction request fields – Optional

| Variable Name                                                                        | Type and Limits                            | Set Method            |
|--------------------------------------------------------------------------------------|--------------------------------------------|-----------------------|
| electronic commerce indicator                                                        | <i>String</i><br>1-character alphanumeric  | 'crypt_type'=>\$crypt |
| customer ID                                                                          | <i>String</i><br>50-character alphanumeric | 'cust_id'=>\$cust_id  |
| <b>NOTE:</b><br>Some special characters are not allowed:<br>< > \$ % = ? ^ { } [ ] \ |                                            |                       |
| electronic commerce indicator                                                        | <i>String</i><br>1-character alphanumeric  | 'crypt_type'=>\$crypt |
| email address                                                                        | <i>String</i><br>30-character alphanumeric | 'email'=>\$email      |
| phone number                                                                         | <i>String</i><br>30-character alphanumeric | 'phone'=>\$phone      |
| note                                                                                 | <i>String</i><br>30-character alphanumeric | 'note'=>\$note        |

## Sample Vault Encrypted Update CC

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
/***** Transactional Variables *****/
$type='enc_res_update_cc';
$data_key='F91LyeEJjv8OvpOdmXYWKH7dV';
$cust_id='cust2';
$phone = '4169996999';
$email = 'bob@email.com';
$note = 'note4';
$enc_track2 = 'ENCRYPTEDTRACK2DATA';
$device_type='idtech_bdk';
$script_type='7';
$avs_street_number = '3300';
$avs_street_name = 'bloor street west';
$avs_zipcode = 'm8x2x3';
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'data_key'=>$data_key,
'cust_id'=>$cust_id,
'phone'=>$phone,
'email'=>$email,
'note'=>$note,
'enc_track2'=>$enc_track2,
'device_type'=>$device_type,
'crypt_type'=>$crypt_type
);
/***** AVS Associative Array *****/
$avsTemplate = array(
'avs_street_number' => $avs_street_number,
'avs_street_name' => $avs_street_name,
'avs_zipcode' => $avs_zipcode
);
/***** AVS Object *****/
$mpgAvsInfo = new mpgAvsInfo ($avsTemplate);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
$mpgTxn->setAvsInfo($mpgAvsInfo);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
//----- ResolveData -----
print("\n\nCust ID = " . $mpgResponse->getResDataCustId());

```

### Sample Vault Encrypted Update CC

```
print("\nPhone = " . $mpgResponse->getResDataPhone());
print("\nEmail = " . $mpgResponse->getResDataEmail());
print("\nNote = " . $mpgResponse->getResDataNote());
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\nCryp Type = " . $mpgResponse->getResDataCryptType());
print("\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
print("\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
?>
```

### Vault response fields

For a list and explanation of (Receipt object) response fields that are available after sending this Vault transaction, see Definitions of Response Fields (page 593).

### 4.5.4 Vault Delete – ResDelete

Deletes an existing Vault profile of any type using the unique data key that was assigned when the profile was added.

**NOTE:** Once a profile is deleted, the information that was saved within can no longer be retrieved.

### Vault Delete transaction object definition

```
$txnArray = array('type'=>'res_delete', ...);
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for Vault Delete transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Vault Delete transaction request fields – Required

For a full description of mandatory and optional values, see Appendix A Definition of Request Fields.

| Variable Name | Type and Limits                     | Set Method             |
|---------------|-------------------------------------|------------------------|
| data key      | String<br>25-character alphanumeric | 'data_key'=>\$data_key |

## Sample Vault Delete

```

<?php
##
## Example php -q TestResDelete.php store3 yesguy
##
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
/***** Transactional Variables *****/
$type='res_delete';
$data_key='YjNEwYw6U2pPwquXOkOme3G7g';
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'data_key'=>$data_key
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
//----- ResolveData -----
print("\n\nCust ID = " . $mpgResponse->getResDataCustId());
print("\nPhone = " . $mpgResponse->getResDataPhone());
print("\nEmail = " . $mpgResponse->getResDataEmail());
print("\nNote = " . $mpgResponse->getResDataNote());
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\nCrypt Type = " . $mpgResponse->getResDataCryptType());
print("\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
print("\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
?>

```

## Vault response fields

For a list and explanation of (Receipt object) response fields that are available after sending this Vault transaction, see Definitions of Response Fields (page 593).

### 4.5.5 Vault Lookup Full – ResLookupFull

Verifies what is currently saved under the Vault profile associated with the given data key. The response to this transaction returns the latest active data for that profile.

Unlike Vault Lookup Masked (which returns a masked credit card number), this transaction returns both the masked and unmasked credit card number.

### Vault Lookup Full transaction object definition

```
$txnArray = array('type'=>'res_lookup_full', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for Vault Lookup Full transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Vault Lookup Full transaction request fields – Required

| Variable Name | Type and Limits                     | Set Method             |
|---------------|-------------------------------------|------------------------|
| data key      | String<br>25-character alphanumeric | 'data_key'=>\$data_key |

### Sample Vault Lookup Full

```
<?php
##
## Example php -q TestResLookupFull.php store3 yesguy
##
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
/***** Transactional Variables *****/
$type='res_lookup_full'; //will return both the full & masked card number
$data_key='t8RCndWBNFnt4Dx32CCnl2tlz';
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'data_key'=>$data_key
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
```

**Sample Vault Lookup Full**

```

print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
//----- ResolveData -----
print("\n\nCust ID = " . $mpgResponse->getResDataCustId());
print("\nPhone = " . $mpgResponse->getResDataPhone());
print("\nEmail = " . $mpgResponse->getResDataEmail());
print("\nNote = " . $mpgResponse->getResDataNote());
print("\nPan = " . $mpgResponse->getResDataPan());
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\nCrypt Type = " . $mpgResponse->getResDataCryptType());
print("\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
print("\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
?>

```

**Vault response fields**

For a list and explanation of (Receipt object) response fields that are available after sending this Vault transaction, see Definitions of Response Fields (page 593).

**4.5.6 Vault Lookup Masked – ResLookupMasked**

Verifies what is currently saved under the Vault profile associated with the given data key. The response to this transaction returns the latest active data for that profile.

Unlike Vault Lookup Full (which returns both the masked and the unmasked credit card numbers), this transaction only returns the masked credit card number.

**Vault Lookup Masked transaction object definition**

```

$txnArray = array('type'=>'res_lookup_masked', ...);

$mpgTxn = new mpgTransaction($txnArray);

```

**HttpPostRequest object for Vault Lookup Masked transaction**

```

$mpgRequest = new mpgRequest($mpgTxn);

$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);

```

**Vault Lookup Masked transaction request fields – Required**

For a full description of mandatory and optional values, see Appendix A Definition of Request Fields.

| Variable Name | Type and Limits                     | Set Method             |
|---------------|-------------------------------------|------------------------|
| data key      | String<br>25-character alphanumeric | 'data_key'=>\$data_key |

### Sample Vault Lookup Masked

```
<?php
##
## Example php -q TestResLookupMasked.php store3 yesguy
##
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
/***** Transactional Variables *****/
$type='res_lookup_masked'; //will only return the masked card number
$data_key='t8RCndWBNFnt4Dx32CCnl2tlz';
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'data_key'=>$data_key
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
//----- ResolveData -----
print("\n\nCust ID = " . $mpgResponse->getResDataCustId());
print("\nPhone = " . $mpgResponse->getResDataPhone());
print("\nEmail = " . $mpgResponse->getResDataEmail());
print("\nNote = " . $mpgResponse->getResDataNote());
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\nCrypt Type = " . $mpgResponse->getResDataCryptType());
print("\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
print("\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
?>
```

### Vault response fields

For a list and explanation of (Receipt object) response fields that are available after sending this Vault transaction, see Definitions of Response Fields (page 593).

### 4.5.7 Vault Get Expiring – ResGetExpiring

Verifies which profiles have credit cards that are expiring during the current and next calendar month.



**EXAMPLE:** if you are processing this transaction on September 30, then it will return all cards that expire(d) in September and October of this year.

When generating a list of profiles with expiring credit cards, only the masked credit card numbers are returned. Can be performed no more than 2 times on any given calendar day.

### Vault Get Expiring transaction object definition

```
$txnArray = array('type'=>'res_get_expiring', ...);

$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for Vault Get Expiring transaction

```
$mpgRequest = new mpgRequest($mpgTxn);

$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

### Vault Get Expiring transaction request fields – Required

This transaction has no required request fields.

#### Sample Vault Get Expiring

```
<?php
##
## Example php -q TestResGetExpiring.php store3 yesguy
##
//There is a max number of attempts set for this transaction per calendar day
//Can not surpass or will receive Invalid Transaction error
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
/***** Transactional Variables *****/
$type='res_get_expiring';
/***** Transactional Associative Array *****/
$txnArray = array( 'type'=>$type );
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nComplete = " . $mpgResponse->getComplete());
```

### Sample Vault Get Expiring

```
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
//----- ResolveData -----
$DataKeys = $mpgResponse->getDataKeys();
for($i=0; $i < count($DataKeys); $i++)
{
    $mpgResponse->setResolveData($DataKeys[$i]);
    print("\n\nData Key = " . $DataKeys[$i]);
    print("\nCust ID = " . $mpgResponse->getResDataCustId());
    print("\nPhone = " . $mpgResponse->getResDataPhone());
    print("\nEmail = " . $mpgResponse->getResDataEmail());
    print("\nNote = " . $mpgResponse->getResDataNote());
    print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
    print("\nExp Date = " . $mpgResponse->getResDataExpDate());
    print("\nCryp Type = " . $mpgResponse->getResDataCryptType());
    print("\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
    print("\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
    print("\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
}
?>
```

### Vault response fields

For a list and explanation of (Receipt object) response fields that are available after sending this Vault transaction, see Definitions of Response Fields (page 593).

### 4.5.8 Vault Is Corporate Card - ResIsCorporateCard

Determines whether a profile has a corporate card registered within it.

After sending the transaction, the response field to the Receipt object's `getCorporateCard` method is either true or false depending on whether the associated card is a corporate card.

**NOTE:** This transaction supports both temporary and permanent tokens.

### Vault Is Corporate Card transaction object definition

```
$txnArray = array('type'=>'res_iscorporategcard', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for Vault Is Corporate Card transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

**Vault Is Corporate Card transaction request fields – Required**

| Variable Name | Type and Limits                                | Set Method             |
|---------------|------------------------------------------------|------------------------|
| data key      | <i>String</i><br><br>25-character alphanumeric | 'data_key'=>\$data_key |

**Sample Vault Is Corporate Card**

```

<?php
##
## Example php -q TestResIsCorporateCard.php moneris hurgle
##
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
/***** Transactional Variables *****/
$type='res_iscorporatcard';
$data_key='t8RCndWBNFnt4Dx32CCnl2tlz';
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'data_key'=>$data_key
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nCorporateCard = " . $mpgResponse->getCorporateCard());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
?>

```

**Vault response fields**

For a list and explanation of (Receipt object) response fields that are available after sending this Vault transaction, see Definitions of Response Fields (page 593).

### 4.5.9 Vault Add Token – ResAddToken

Converts a Hosted Tokenization temporary token to a permanent Vault token.

A temporary token is valid for 15 minutes after it is created. This transaction must be performed within that time frame if the token is to be changed to a permanent one for future use.

Using the temporary token, send either a Purchase with Vault, Pre-Authorization with Vault or Card Verification with Vault transaction request including the Credential on File object to get the issuer ID.

#### Vault Add Token transaction object definition

```
$txnArray = array('type'=>'res_add_token', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

#### HttpPostRequest object for Vault Add Token transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

#### Vault Add Token transaction request fields – Required

For a full description of mandatory and optional values, see Appendix A Definition of Request Fields.

| Variable Name                                                                                                                                                                                                                                                         | Type and Limits                            | Set Method                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|------------------------------|
| data key                                                                                                                                                                                                                                                              | <i>String</i><br>25-character alphanumeric | 'data_key'=>\$data_key       |
| electronic commerce indicator                                                                                                                                                                                                                                         | <i>String</i><br>1-character alphanumeric  | 'crypt_type'=>\$crypt        |
| Credential on File Info<br>cof                                                                                                                                                                                                                                        | <i>Object</i><br>N/A                       | \$mpgTxn->setCofInfo(\$cof); |
| <div> <b>NOTE:</b> This is a nested object within the transaction, and required when storing or using the customer's stored credentials. For information about fields in the Credential on File Info object, see Credential on File Info Object and Variables. </div> |                                            |                              |

## Vault Add Token transaction request fields – Optional

Table 1: Vault Add Token transaction optional values

| Variable Name   | Type and Limits                                                                                                                                                                       | Set Method                           |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| customer ID     | <i>String</i><br>50-character alphanumeric<br><div> <b>NOTE:</b><br/>           Some special characters are not allowed:<br/>           &lt; &gt; \$ % = ? ^ { } [ ] \         </div> | 'cust_id'=>\$cust_id                 |
| AVS Information | <i>Object</i><br>N/A                                                                                                                                                                  | \$mpgTxn->setAvsInfo(\$mpgAvsInfo);  |
| email address   | <i>String</i><br>30-character alphanumeric                                                                                                                                            | 'email'=>\$email                     |
| phone number    | <i>String</i><br>30-character alphanumeric                                                                                                                                            | 'phone'=>\$phone                     |
| note            | <i>String</i><br>30-character alphanumeric                                                                                                                                            | 'note'=>\$note                       |
| data key format | <i>String</i><br>2-character alphanumeric                                                                                                                                             | 'data_key_format'=>\$data_key_format |

## Credential on File Info object request fields

| Variable Name                                                                                                                                                                                                                                                           | Type and Limits                                               | Set Method                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| issuer ID<br><div> <b>NOTE:</b> This variable is required for all merchant-initiated transactions following the first one; upon sending the first transaction, the issuer ID value is received in the transaction response and then used in subsequent           </div> | <i>String</i><br>15-character alphanumeric<br>variable length | \$cof->setIssuerId("VALUE_FOR_ISSUER_ID");<br><div> <b>NOTE:</b> For a list and explanation of the possible values to send for this variable, see Definition of Request Fields – Credential on File           </div> |

| Variable Name         | Type and Limits | Set Method |
|-----------------------|-----------------|------------|
| transaction requests. |                 |            |

### Sample Vault Add Token

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
/***** Transactional Variables *****/
$type='res_add_token';
$temp_data_key='ot-mtNKdu8NcxDoChqOJKZJZ1BOB';
$cust_id='customer1';
$phone = '5555551234';
$email = 'bob@smith.com';
$note = 'this is my note';
$expiry_date='1811';
$data_key_format = "0";
$script_type='1';
$avs_street_number = '123';
$avs_street_name = 'lakeshore blvd';
$avs_zipcode = '90210';
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'data_key'=>$temp_data_key,
'cust_id'=>$cust_id,
'phone'=>$phone,
'email'=>$email,
'note'=>$note,
'expdate'=>$expiry_date,
// 'data_key_format'=>$data_key_format, //optional
'crypt_type'=>$script_type
);
/***** AVS Associative Array *****/
$avsTemplate = array(
'avs_street_number' => $avs_street_number,
'avs_street_name' => $avs_street_name,
'avs_zipcode' => $avs_zipcode
);
/***** AVS Object *****/
$mpgAvsInfo = new mpgAvsInfo ($avsTemplate);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setIssuerId("168451306048014");
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
$mpgTxn->setAvsInfo($mpgAvsInfo);
$mpgTxn->setCofInfo($cof);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());

```

### Sample Vault Add Token

```
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
//----- ResolveData -----
print("\n\nCust ID = " . $mpgResponse->getResDataCustId());
print("\nPhone = " . $mpgResponse->getResDataPhone());
print("\nEmail = " . $mpgResponse->getResDataEmail());
print("\nNote = " . $mpgResponse->getResDataNote());
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\nCrypt Type = " . $mpgResponse->getResDataCryptType());
print("\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
print("\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
?>
```

### Vault response fields

For a list and explanation of (Receipt object) response fields that are available after sending this Vault transaction, see Definitions of Response Fields (page 593).

### 4.5.10 Vault Tokenize Credit Card – ResTokenizeCC

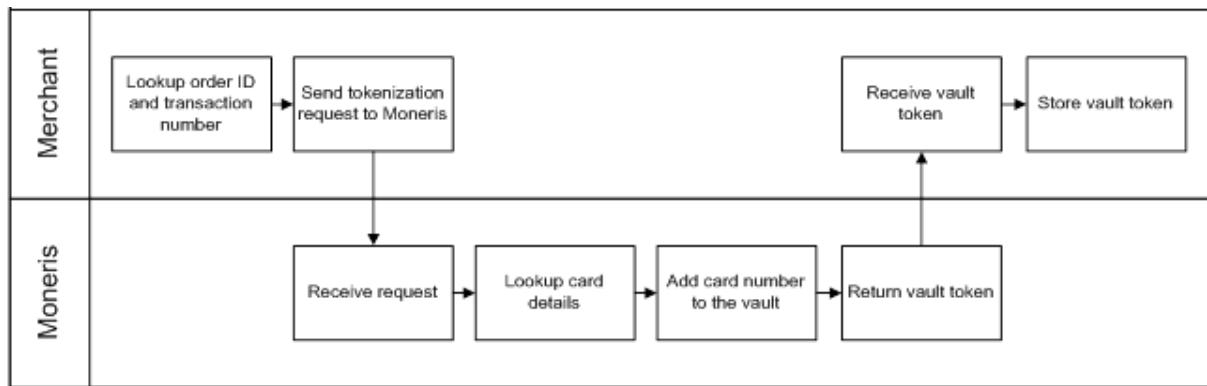
Creates a new credit card profile using the credit card number, expiry date and e-commerce indicator that were submitted in a previous financial transaction. Previous transactions to be tokenized must have included the Credential on File Info object.

The Issuer ID received in the previous transaction response is sent in the Vault Tokenize Credit Card request to reference that this is a stored credential. If you require this Issuer ID in the response to this request, include **return\_issuer\_ID** as **true**; this allows for retrieval of the Issuer ID from the previous financial transaction.

Basic transactions that can be tokenized are:

- Purchase
- Pre-Authorization
- Card Verification

The tokenization process is outlined below :



**Figure 1: Tokenize process diagram**

### Vault Tokenize Credit Card transaction object definition

```
$txnArray = array('type'=>'res_tokenize_cc', ...);
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for Vault Tokenize Credit Card transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Vault Tokenize Credit Card transaction request fields – Required

These mandatory values reference a previously processed credit card financial transaction. The credit card number, expiry date, and e-commerce indicator from the original transaction are registered in the Vault for future financial Vault transactions.

| Variable Name      | Type and Limits                                                                    | Set Method                 |
|--------------------|------------------------------------------------------------------------------------|----------------------------|
| order ID           | String<br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                | 'order_id'=>\$order_id     |
| transaction number | String<br>255-character, alpha-numeric, hyphens or under-scores<br>variable length | 'txn_number'=>\$txn_number |



## Vault Tokenize Credit Card transaction request fields – Optional

| Variable Name                                                                                                                                                                                                                        | Type and Limits                                                                                                                                                                       | Set Method                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| customer ID                                                                                                                                                                                                                          | <i>String</i><br>50-character alphanumeric<br><div> <b>NOTE:</b><br/>           Some special characters are not allowed:<br/>           &lt; &gt; \$ % = ? ^ { } [ ] \         </div> | 'cust_id'=>\$cust_id                                    |
| return issuer ID                                                                                                                                                                                                                     | <i>Boolean</i><br>true/false                                                                                                                                                          | resTokenizeCC<br>'return_issuer_id'=>\$return_issuer_id |
| email address                                                                                                                                                                                                                        | <i>String</i><br>30-character alphanumeric                                                                                                                                            | 'email'=>\$email                                        |
| phone number                                                                                                                                                                                                                         | <i>String</i><br>30-character alphanumeric                                                                                                                                            | 'phone'=>\$phone                                        |
| note                                                                                                                                                                                                                                 | <i>String</i><br>30-character alphanumeric                                                                                                                                            | 'note'=>\$note                                          |
| AVS Information                                                                                                                                                                                                                      | <i>Object</i><br>N/A                                                                                                                                                                  | \$mpgTxn->setAvsInfo (\$mpgAvsInfo) ;                   |
| data key format                                                                                                                                                                                                                      | <i>String</i><br>2-character alphanumeric                                                                                                                                             | 'data_key_format'=>\$data_key_format                    |
| Credential on File Info                                                                                                                                                                                                              | <i>Object</i>                                                                                                                                                                         | \$mpgTxn->setCofInfo (\$cof) ;                          |
| cof                                                                                                                                                                                                                                  | N/A                                                                                                                                                                                   |                                                         |
| <div> <b>NOTE:</b> This is a nested object within the transaction, and required when storing or using the customer's stored credentials. For information about fields in the Credential on File Info object, see Cre-         </div> |                                                                                                                                                                                       |                                                         |

| Variable Name                                                     | Type and Limits | Set Method |
|-------------------------------------------------------------------|-----------------|------------|
| <div> <p>credential on File Info Object and Variables.</p> </div> |                 |            |

### Credential on File Info object request fields

| Variable Name                                                                                                                                                                                                                                                                                  | Type and Limits                                                       | Set Method                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>issuer ID</p> <div> <p><b>NOTE:</b> This variable is required for all merchant-initiated transactions following the first one; upon sending the first transaction, the issuer ID value is received in the transaction response and then used in subsequent transaction requests.</p> </div> | <p><i>String</i></p> <p>15-character alphanumeric variable length</p> | <pre>\$cof-&gt;setIssuerId("VALUE_FOR_ISSUER_ID");</pre> <div> <p><b>NOTE:</b> For a list and explanation of the possible values to send for this variable, see Definition of Request Fields – Credential on File</p> </div> |

Any field that is not set in the tokenize request is not stored with the transaction. That is, Moneris Gateway does not automatically take the optional information that was part of the original transaction.

The ResolveData that is returned in the response fields indicates what values were registered for this profile.

| Sample Vault Tokenize Credit Card                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>&lt;?php require "../mpgClasses.php"; /***** Request Variables *****/ \$store_id='store5'; \$api_token='yesguy'; /***** Transactional Variables *****/ \$type='res_tokenize_cc'; \$order_id='res-purch-110515-12:56:49'; \$txn_number='31570-0_10'; \$return_issuer_id = 'false'; \$data_key_format = "0"; \$cust_id='customer1'; \$phone = '4165555555'; \$email = 'bob@smith.com'; \$note = 'this is my note'; \$savs_street_number = '123'; \$savs_street_name = 'lakeshore blvd'; \$savs_zipcode = '90210'; /***** Transactional Associative Array *****/ \$txnArray=array('type'=&gt;\$type,</pre> |

## Sample Vault Tokenize Credit Card

```

'order_id'=>$order_id,
'txn_number'=>$txn_number,
// 'data_key_format'=>$data_key_format, //optional
'cust_id'=>$cust_id,
'phone'=>$phone,
'email'=>$email,
'note'=>$note
);
/***** AVS Associative Array *****/
$avsTemplate = array(
'avs_street_number' => $avs_street_number,
'avs_street_name' => $avs_street_name,
'avs_zipcode' => $avs_zipcode
);
/***** AVS Object *****/
$mpgAvsInfo = new mpgAvsInfo ($avsTemplate);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setIssuerId("168451306048014");
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
$mpgTxn->setAvsInfo($mpgAvsInfo);
$mpgTxn->setCofInfo($cof);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
//----- ResolveData -----
print("\nCust ID = " . $mpgResponse->getResDataCustId());
print("\nPhone = " . $mpgResponse->getResDataPhone());
print("\nEmail = " . $mpgResponse->getResDataEmail());
print("\nNote = " . $mpgResponse->getResDataNote());
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\nCrypt Type = " . $mpgResponse->getResDataCryptType());
print("\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
print("\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
?>

```

## Vault response fields

For a list and explanation of (Receipt object) response fields that are available after sending this Vault transaction, see Definitions of Response Fields (page 593).

## 4.6 Vault Financial Transactions

After a financial transaction is complete, the response fields indicate all the values that are currently saved under the profile that was used.

### 4.6.1 Customer ID Changes

Some financial transactions take the customer ID as an optional value. The customer ID may or may not already be in the Vault profile when the transaction is sent. Therefore, it is possible to change the value of the customer ID by performing a financial transaction

The table below shows what the customer ID will be in the response field after a financial transaction is performed.

**Table 2: Customer ID use in response fields**

| Already in profile? | Passed in? | Version used in response            |
|---------------------|------------|-------------------------------------|
| No                  | No         | Customer ID not used in transaction |
| No                  | Yes        | Passed in                           |
| Yes                 | No         | Profile                             |
| Yes                 | Yes        | Passed in                           |

### 4.6.2 Purchase with Vault – ResPurchaseCC

**NOTE:** This transaction supports both temporary and permanent tokens.

#### Purchase with Vault transaction object definition

```
$txnArray = array('type'=>'res_purchase_cc', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

#### HttpPostRequest object for Purchase with Vault transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

#### Purchase with Vault transaction request fields – Required

For a full description of mandatory and optional values, see Appendix A Definition of Request Fields.

| Variable Name                  | Type and Limits                                                                                                                                                                                                                                                                      | Set Method                   |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| data key                       | <i>String</i><br>25-character alphanumeric                                                                                                                                                                                                                                           | 'data_key'=>\$data_key       |
| order ID                       | <i>String</i><br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                                                                                                                                                                                                           | 'order_id'=>\$order_id       |
| amount                         | <i>String</i><br>10-character decimal<br><br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><br><b>EXAMPLE:</b> 1234567.89                                                                                                               | 'amount'=>\$amount           |
| electronic commerce indicator  | <i>String</i><br>1-character alphanumeric                                                                                                                                                                                                                                            | 'crypt_type'=>\$crypt        |
| Credential on File Info<br>cof | <i>Object</i><br>N/A<br><br><b>NOTE:</b> This is a nested object within the transaction, and required when storing or using the customer's stored credentials. For information about fields in the Credential on File Info object, see Credential on File Info Object and Variables. | \$mpgTxn->setCofInfo(\$cof); |

**Purchase with Vault transaction request fields – Optional**

| Variable Name | Type and Limits | Set Method         |
|---------------|-----------------|--------------------|
| status check  | <i>Boolean</i>  | \$mpgHttpPost =new |

| Variable Name        | Type and Limits                                                                                                                                                                                         | Set Method                                                                    |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
|                      | true/false                                                                                                                                                                                              | <code>mpgHttpPostStatus(\$store_id,\$api_token,\$status,\$mpgRequest);</code> |
| expiry date          | <i>String</i><br>4-character numeric<br>YYMM format.<br>(Note that this is reversed from the date displayed on the card, which is MMY)                                                                  | <code>'expiry_date'=&gt;\$expiry_date</code>                                  |
|                      | <b>NOTE:</b> This field is required if referencing a temporary token in Installments by Visa                                                                                                            |                                                                               |
| customer ID          | <i>String</i><br>50-character alphanumeric<br><b>NOTE:</b><br>Some special characters are not allowed:<br>< > \$ % = ? ^ { } [ ] \                                                                      | <code>'cust_id'=&gt;\$cust_id</code>                                          |
| dynamic descriptor   | <i>String</i><br>20-character alphanumeric<br>total of 22 characters including your merchant name and separator<br><b>NOTE:</b><br>Some special characters are not allowed:<br>< > \$ % = ? ^ { } [ ] \ | <code>'dynamic_descriptor'=&gt;\$dynamic_descriptor</code>                    |
| Customer Information | <i>Object</i><br>N/A                                                                                                                                                                                    | <code>\$mpgTxn-&gt;setCustInfo(\$mpgCustInfo);</code>                         |
| AVS Information      | <i>Object</i><br>N/A                                                                                                                                                                                    | <code>\$mpgTxn-&gt;setAvsInfo(\$mpgAvsInfo);</code>                           |
| CVD Information      | <i>Object</i><br>N/A                                                                                                                                                                                    | <code>\$mpgTxn-&gt;setCvdInfo(\$mpgCvdInfo);</code>                           |

| Variable Name                                                                                                                                                                                                                                                            | Type and Limits          | Set Method                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------|
| <p><b>NOTE:</b> When storing credentials on the initial transaction, the CVD object must be sent; for subsequent transactions using stored credentials, CVD can be sent with cardholder-initiated transactions only—<b>merchants must not store CVD information.</b></p> |                          |                                                                                                                   |
| Recurring Billing                                                                                                                                                                                                                                                        | <i>Object</i><br><br>N/A | <code>\$mpgTxn-&gt;setRecur(\$mpgRecur);</code>                                                                   |
| Surcharge Information<br><br><p><b>NOTE:</b> This object requires use of the Vault Surcharge Lookup transaction prior to confirm card eligibility.</p>                                                                                                                   | <i>Object</i><br><br>N/A | <code>\$mpgTxn-&gt;setSurchargeInfo(\$surchargeInfo);</code><br><br>Contains fields related to Surcharge feature. |

### Credential on File Info object request fields

| Variable Name                                                                                                                                                                                                                                                                     | Type and Limits                                                       |                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| issuer ID<br><br><p><b>NOTE:</b> This variable is required for all merchant-initiated transactions following the first one; upon sending the first transaction, the issuer ID value is received in the transaction response and then used in subsequent transaction requests.</p> | <i>String</i><br><br>15-character alphanumeric<br><br>variable length | <code>\$cof-&gt;setIssuerId("VALUE_FOR_ISSUER_ID");</code><br><br><p><b>NOTE:</b> For a list and explanation of the possible values to send for this variable, see Definition of Request Fields – Credential on File</p> |
| payment indicator<br><br><p><b>NOTE:</b> In Credential on File transactions where the request field e-commerce indicator is also being sent,</p>                                                                                                                                  | <i>String</i><br><br>1-character alphabetic                           | <code>\$cof-&gt;setPaymentIndicator("PAYMENT_INDICATOR_VALUE");</code><br><br><p><b>NOTE:</b> For a list and explanation of the possible values to send for this variable, see</p>                                       |

| Variable Name                                                                                                                                             | Type and Limits               |                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the acceptable values for e-commerce indicator are dependent on the value sent for payment indicator; see request field definitions for more information. |                               | Definition of Request Fields – Credential on File                                                                                                                                                              |
| payment information                                                                                                                                       | String<br>1-character numeric | \$cof->setPaymentInformation<br>("PAYMENT_INFO_VALUE");<br><br><b>NOTE:</b> For a list and explanation of the possible values to send for this variable, see Definition of Request Fields – Credential on File |

### Sample Purchase with Vault

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='monca00597';
$api_token='027AbCbxQorPggMQe6hU';
/***** Transaction Variables *****/
$data_key='4HIme0ZGURXE3NRBXHUj6nSc4';
$orderid='res-purch-' . date("dmy-G:i:s");
$amount='18.00';
$custid='customer1';
$cript_type='7';
$expdate='2301'; //For Temp Tokens only

//NT Response Option
$get_nt_response = 'false'; //Optional - set it true only if you want to get network
tokenization response.
/***** Transaction Array *****/
$txnArray=array('type'=>'res_purchase_cc',
'data_key'=>$data_key,
'order_id'=>$orderid,
'cust_id'=>$custid,
'amount'=>$amount,
'crypt_type'=>$cript_type,
'expdate'=>$expdate,
'dynamic_descriptor'=>'12484',
'get_nt_response'=>$get_nt_response
);

/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);

/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("U");

```



## Sample Purchase with Vault

```

$cof->setPaymentInformation("2");
$cof->setIssuerId("139X3130ASCXAS9");
$mpgTxn->setCofInfo($cof);

/***** Installment Info *OPTIONAL* *****/
$installmentInfo = new InstallmentInfo();
$installmentInfo->setPlanId("ae859ef1-eb91-b708-8b80-1dd481746401");
$installmentInfo->setPlanIdRef("0000000065");
$installmentInfo->setTacVersion("2");
//$mpgTxn->setInstallmentInfo($installmentInfo);

/***** Surcharge Info *OPTIONAL* *****/
$surchargeInfo = new SurchargeInfo();
$surchargeInfo->setSurchargeAmount("1.00");
$mpgTxn->setSurchargeInfo($surchargeInfo);

/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions

/***** mpgHttpPost Object *****/
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);

/***** Response Object *****/
$mpgResponse = $mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nAVSResponse = " . $mpgResponse->getAvsResultCode());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\nSourcePanLast4 = " . $mpgResponse->getSourcePanLast4());

// $installmentResults = $mpgResponse->getInstallmentResults();
// print("\nPlanId = " . $installmentResults->getPlanId());
// print("\nPlanIDRef = " . $installmentResults->getPlanIDRef());
// print("\nTacVersion = " . $installmentResults->getTacVersion());
// print("\nPlanAcceptanceId = " . $installmentResults->getPlanAcceptanceId());
// print("\nPlanStatus = " . $installmentResults->getPlanStatus());
// print("\nPlanResponse = " . $installmentResults->getPlanResponse());

if($get_nt_response == 'true')
{
print("\nNTResponseCode = " . $mpgResponse->getNTResponseCode());
print("\nNTMessage = " . $mpgResponse->getNTMessage());
}

```

### Sample Purchase with Vault

```
print("\nNTUsed = " . $mpgResponse->getNTUsed());
print("\nNTTokenBin = " . $mpgResponse->getNTTokenBin());
print("\nNTTokenLast4 = " . $mpgResponse->getNTTokenLast4());
print("\nNTTokenExpDate = " . $mpgResponse->getNTTokenExpDate());
}
//----- ResolveData -----
print("\n\nCust ID = " . $mpgResponse->getResDataCustId());
print("\nPhone = " . $mpgResponse->getResDataPhone());
print("\nEmail = " . $mpgResponse->getResDataEmail());
print("\nNote = " . $mpgResponse->getResDataNote());
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\nCrypT Type = " . $mpgResponse->getResDataCryptType());
print("\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
print("\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
?>
```

### Vault response fields

For a list and explanation of (Receipt object) response fields that are available after sending this Vault transaction, see Definitions of Response Fields (page 593).

## 4.6.3 Pre-Authorization with Vault – ResPreauthCC

**NOTE:** This transaction supports both temporary and permanent tokens.

### Pre-Authorization with Vault transaction object definition

```
$txnArray = array('type'=>'res_preauth_cc', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for Pre-Authorization with Vault transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Pre-Authorization with Vault transaction request fields – Required

| Variable Name | Type and Limits                            | Set Method             |
|---------------|--------------------------------------------|------------------------|
| data key      | <i>String</i><br>25-character alphanumeric | 'data_key'=>\$data_key |
| order ID      | <i>String</i>                              | 'order_id'=>\$order_id |

| Variable Name                  | Type and Limits                                                                                                                                                                                                                                                                                       | Set Method                   |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
|                                | 50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                                                                                                                                                                                                                                             |                              |
| amount                         | <i>String</i><br>10-character decimal<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><div>EXAMPLE: 1234567.89</div>                                                                                                                                    | 'amount'=>\$amount           |
| electronic commerce indicator  | <i>String</i><br>1-character alphanumeric                                                                                                                                                                                                                                                             | 'crypt_type'=>\$crypt        |
| Credential on File Info<br>cof | <i>Object</i><br>N/A<br><div> <b>NOTE:</b> This is a nested object within the transaction, and required when storing or using the customer's stored credentials. For information about fields in the Credential on File Info object, see Credential on File Info Object and Variables.         </div> | \$mpgTxn->setCofInfo(\$cof); |

### Pre-Authorization with Vault transaction request fields – Optional

| Value              | Limits                                     | Set method                                                                                                   |
|--------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| status check       | <i>Boolean</i><br>true/false               | <pre>\$mpgHttpPost =new<br/>mpgHttpPostStatus(\$store_<br/>id,\$api_<br/>token,\$status,\$mpgRequest);</pre> |
| dynamic descriptor | <i>String</i><br>20-character alphanumeric | 'dynamic_descriptor'=>\$dynamic_descriptor                                                                   |

| Value                                                                                                                                                                                                                                                                                                                        | Limits                                                                                                                                                                       | Set method                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>NOTE: For Pre-Authorization transactions:</b> the value in the dynamic descriptor field will only be carried over to a Pre-Authorization Completion when executing the latter via the Merchant Resource Center; otherwise, the value for dynamic descriptor must be sent again in the Pre-Authorization Completion</p> | <p>total of 22 characters including your merchant name and separator</p> <p><b>NOTE:</b><br/>Some special characters are not allowed:<br/>&lt; &gt; \$ % = ? ^ { } [ ] \</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <p>is incremental</p> <p>is_incremental</p>                                                                                                                                                                                                                                                                                  | <p><i>Boolean</i></p> <p>true/false</p>                                                                                                                                      | <p>'is_incremental'=&gt;\$is_incremental</p> <p>Indicates if this preauthorization is using an estimated amount. Estimations allow for incrementing the amount held via subsequent incrementalAuth requests. Defaults to false.</p> <p><b>NOTE:</b> Please note that if this field is true, the preauthorization is only eligible for a single Preauthorization Completion. Any completion sent for partial completion is treated as a full completion (ship_indicator= P is treated as = F when is_incremental= true on the original preauth).</p> <p><b>NOTE:</b> For preauthorizations using Account-Based Payments, as part of Pay By Bank, is_incremental = true indicates the transaction may be completed for up to 15% more than the original preauthorization amount.</p> |
| <p>expiry date</p> <p><b>NOTE:</b> This field is required if referencing a temporary token in Installments by Visa</p>                                                                                                                                                                                                       | <p><i>String</i></p> <p>4-character alphanumeric</p> <p>YYMM</p>                                                                                                             | <p>'expiry_date'=&gt;\$expiry_date</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p>customer ID</p>                                                                                                                                                                                                                                                                                                           | <p><i>String</i></p> <p>50-character alphanumeric</p>                                                                                                                        | <p>'cust_id'=&gt;\$cust_id</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Value                                                                                                                                                                                                                                                                                     | Limits                                                                    | Set method                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                           | <b>NOTE:</b><br>Some special characters are not allowed:<br><>\$%=?^{}[]\ |                                                                                                   |
| final authorization                                                                                                                                                                                                                                                                       | String                                                                    | 'final_auth' => 'true'                                                                            |
| <b>NOTE:</b> Applies to Mastercard transactions only                                                                                                                                                                                                                                      | true/false                                                                |                                                                                                   |
| Customer Information                                                                                                                                                                                                                                                                      | Object<br>N/A                                                             | \$mpgTxn->setCustInfo(\$mpgCustInfo);                                                             |
| AVS Information                                                                                                                                                                                                                                                                           | Object<br>N/A                                                             | \$mpgTxn->setAvsInfo(\$mpgAvsInfo);                                                               |
| CVD Information<br><br><b>NOTE:</b> When storing credentials on the initial transaction, the CVD object must be sent; for subsequent transactions using stored credentials, CVD can be sent with cardholder-initiated transactions only— <b>merchants must not store CVD information.</b> | Object<br>N/A                                                             | \$mpgTxn->setCvdInfo(\$mpgCvdInfo);                                                               |
| Surcharge Information<br><br><b>NOTE:</b> This object requires use of the Surcharge Lookup transaction prior to confirm card eligibility.                                                                                                                                                 | Object<br>N/A                                                             | \$mpgTxn->setSurchargeInfo(\$surchargeInfo);<br><br>Contains fields related to Surcharge feature. |

### Credential on File Info object request fields

| Variable Name | Type and Limits |                                |
|---------------|-----------------|--------------------------------|
| issuer ID     | String          | \$cof->setIssuerId("VALUE_FOR_ |

| Variable Name                                                                                                                                                                                                                                             | Type and Limits                                  |                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NOTE:</b> This variable is required for all merchant-initiated transactions following the first one; upon sending the first transaction, the issuer ID value is received in the transaction response and then used in subsequent transaction requests. | 15-character alphanumeric<br><br>variable length | ISSUER_ID" ) ;<br><br><b>NOTE:</b> For a list and explanation of the possible values to send for this variable, see Definition of Request Fields – Credential on File |

payment indicator

*String*

1-character alphabetic

```
$cof->setPaymentIndicator
("PAYMENT_INDICATOR_VALUE") ;
```

**NOTE:** In Credential on File transactions where the request field e-commerce indicator is also being sent, the acceptable values for e-commerce indicator are dependent on the value sent for payment indicator; see request field definitions for more information.

**NOTE:** For a list and explanation of the possible values to send for this variable, see Definition of Request Fields – Credential on File

payment information

*String*

1-character numeric

```
$cof->setPaymentInformation
("PAYMENT_INFO_VALUE") ;
```

**NOTE:** For a list and explanation of the possible values to send for this variable, see Definition of Request Fields – Credential on File

### Sample Pre-Authorization with Vault

```
<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='monca00597';
$api_token='027AbCbxQorPggMQe6hU';
/***** Transaction Variables *****/
$data_key='4HIme0ZGURXE3NRBXHUj6nSc4';
$orderid='res-preauth-'.date("dmy-G:i:s");
$amount='1.00';
$custid='customer1'; //if sent will be submitted, otherwise cust_id from profile will be used
$crypt_type='1';
```

## Sample Pre-Authorization with Vault

```

$expdate='2301';
$is_incremental= 'true';

//NT Response Option
$get_nt_response = 'false';//Optional - set it true only if you want to get network
tokenization response.
/***** Transaction Array *****/
$txnArray =array('type'=>'res_preauth_cc',
'data_key'=>$data_key,
'order_id'=>$orderid,
'cust_id'=>$custid,
'amount'=>$amount,
'crypt_type'=>$crypt_type,
'expdate'=>$expdate,
'dynamic_descriptor'=>'12424',
'get_nt_response'=>$get_nt_response
);

/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);

/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("U");
$cof->setPaymentInformation("2");
$cof->setIssuerId("139X3130ASCXAS9");
$mpgTxn->setCofInfo($cof);

/***** Installment Info *OPTIONAL* *****/
$installmentInfo = new InstallmentInfo();
$installmentInfo->setPlanId("ae859ef1-eb91-b708-8b80-1dd481746401");
$installmentInfo->setPlanIdRef("0000000065");
$installmentInfo->setTacVersion("2");
//$mpgTxn->setInstallmentInfo($installmentInfo);

/***** Surcharge Info *OPTIONAL* *****/
$surchargeInfo = new SurchargeInfo();
$surchargeInfo->setSurchargeAmount("1.00");
$mpgTxn->setSurchargeInfo($surchargeInfo);

/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions

/***** mpgHttpPost Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);

/***** Response Object *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());

```

### Sample Pre-Authorization with Vault

```

print("\nTransType = " . $mpgResponse->getTransType());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nAVSResponse = " . $mpgResponse->getAvsResultCode());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\n\nSourcePanLast4 = " . $mpgResponse->getSourcePanLast4());

// $installmentResults = $mpgResponse->getInstallmentResults();
// print("\nPlanId = " . $installmentResults->getPlanId());
// print("\nPlanIDRef = " . $installmentResults->getPlanIDRef());
// print("\nTacVersion = " . $installmentResults->getTacVersion());
// print("\nPlanAcceptanceId = " . $installmentResults->getPlanAcceptanceId());
// print("\nPlanStatus = " . $installmentResults->getPlanStatus());
// print("\nPlanResponse = " . $installmentResults->getPlanResponse());

if($get_nt_response == 'true')
{
print("\n\nNTResponseCode = " . $mpgResponse->getNTResponseCode());
print("\nNTMessage = " . $mpgResponse->getNTMessage());
print("\nNTUsed = " . $mpgResponse->getNTUsed());
print("\nNTTokenBin = " . $mpgResponse->getNTTokenBin());
print("\nNTTokenLast4 = " . $mpgResponse->getNTTokenLast4());
print("\nNTTokenExpDate = " . $mpgResponse->getNTTokenExpDate());
}
//----- ResolveData -----
print("\n\nCust ID = " . $mpgResponse->getResDataCustId());
print("\nPhone = " . $mpgResponse->getResDataPhone());
print("\nEmail = " . $mpgResponse->getResDataEmail());
print("\nNote = " . $mpgResponse->getResDataNote());
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\nCryp Type = " . $mpgResponse->getResDataCryptType());
print("\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
print("\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
?>

```

### Vault response fields

For a list and explanation of (Receipt object) response fields that are available after sending this Vault transaction, see Definitions of Response Fields (page 593).

### 4.6.4 Vault Independent Refund – ResIndRefundCC

**NOTE:** This transaction supports both temporary and permanent tokens.



**Vault Independent Refund transaction object definition**

```
$txnArray = array('type'=>'res_ind_refund_cc', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

**HttpPostRequest object for Vault Independent Refund transaction**

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

**Vault Independent Refund transaction values**

For a full description of mandatory and optional values, see Appendix A Definition of Request Fields.

| Variable Name                 | Type and Limits                                                                                                                                                        | Set Method             |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| data key                      | <i>String</i><br>25-character alphanumeric                                                                                                                             | 'data_key'=>\$data_key |
| order ID                      | <i>String</i><br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                                                                                             | 'order_id'=>\$order_id |
| amount                        | <i>String</i><br>10-character decimal<br><br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><br><b>EXAMPLE:</b> 1234567.89 | 'amount'=>\$amount     |
| electronic commerce indicator | <i>String</i><br>1-character alphanumeric                                                                                                                              | 'crypt_type'=>\$crypt  |

**Vault Independent Refund transaction request fields – Optional**

| Value       | Limits                                     | Set method           |
|-------------|--------------------------------------------|----------------------|
| customer ID | <i>String</i><br>50-character alphanumeric | 'cust_id'=>\$cust_id |

| Value              | Limits                                                                                                                                                                                                         | Set method                                                                                        |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|                    | <b>NOTE:</b><br>Some special characters are not allowed:<br><>\$%=?^{}[]\                                                                                                                                      |                                                                                                   |
| expiry date        | <i>String</i><br><br>4-character alphanumeric<br><br>YYMM                                                                                                                                                      | 'expiry_date'=>\$expiry_date                                                                      |
| status check       | <i>Boolean</i><br><br>true/false                                                                                                                                                                               | <pre>\$mpgHttpPost =new mpgHttpsPostStatus(\$store_ id,\$api_ token,\$status,\$mpgRequest);</pre> |
| dynamic descriptor | <i>String</i><br><br>20-character alphanumeric<br><br>total of 22 characters<br>including your merchant<br>name and separator<br><br><b>NOTE:</b><br>Some special characters are not allowed:<br><>\$%=?^{}[]\ | 'dynamic_descriptor'=>\$dynamic_descriptor                                                        |

### Sample Vault Independent Refund

```
<?php
##
## This program takes 3 arguments from the command line:
## 1. Store id
## 2. api token
## 3. order id
##
## Example php -q TestResIndRefundCC.php store3 yesguy unique_order_id cust_id 15.00 1
##
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
/***** Transaction Variables *****/
$data_key='t8RCndWBNFNt4Dx32CCnl2tlz';
```

## Sample Vault Independent Refund

```

$orderid='res-ind-refund-' . date("dmy-G:i:s");
$amount='1.00';
$custid='';
$crypt_type='1';

//NT Response Option
$get_nt_response = 'true'; //Optional - set it true only if you want to get network
tokenization response.
/***** Transaction Array *****/
$txnArray = array('type'=>'res_ind_refund_cc',
'data_key'=>$data_key,
'order_id'=>$orderid,
'cust_id'=>$custid,
'amount'=>$amount,
'crypt_type'=>$crypt_type,
'dynamic_descriptor'=>'12346'
'get_nt_response'=>$get_nt_response
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** mpgHttpPost Object *****/
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
/***** Response Object *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
print("\n\nSourcePanLast4 = " . $mpgResponse->getSourcePanLast4());

if($get_nt_response == 'true')
{
print("\n\nNTResponseCode = " . $mpgResponse->getNTResponseCode());
print("\n\nNTMessage = " . $mpgResponse->getNTMessage());
print("\n\nNTUsed = " . $mpgResponse->getNTUsed());
print("\n\nNTTokenBin = " . $mpgResponse->getNTTokenBin());
print("\n\nNTTokenLast4 = " . $mpgResponse->getNTTokenLast4());
print("\n\nNTTokenExpDate = " . $mpgResponse->getNTTokenExpDate());
}
//----- ResolveData -----
print("\n\nCust ID = " . $mpgResponse->getResDataCustId());
print("\nPhone = " . $mpgResponse->getResDataPhone());

```

### Sample Vault Independent Refund

```
print("\nEmail = " . $mpgResponse->getResDataEmail());
print("\nNote = " . $mpgResponse->getResDataNote());
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\nCryt Type = " . $mpgResponse->getResDataCryptType());
print("\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
print("\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
?>
```

### Vault response fields

For a list and explanation of (Receipt object) response fields that are available after sending this Vault transaction, see Definitions of Response Fields (page 593).

### 4.6.5 Force Post with Vault – ResForcePostCC

**NOTE:** This transaction supports both temporary and permanent tokens.

### Force Post with Vault transaction object definition

```
$txnArray = array('type'=>'res_forcepost_cc', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for Force Post with Vault transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Force Post with Vault transaction request fields – Required

| Variable Name | Type and Limits                                                                                                                                                                | Set Method         |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| amount        | <i>String</i><br><br>10-character decimal<br><br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><br><div>EXAMPLE: 1234567.89</div> | 'amount'=>\$amount |

| Variable Name                 | Type and Limits                            | Set Method               |
|-------------------------------|--------------------------------------------|--------------------------|
| data key                      | <i>String</i><br>25-character alphanumeric | 'data_key'=>\$data_key   |
| authorization code            | <i>String</i><br>8-character alphanumeric  | 'auth_code'=>\$auth_code |
| electronic commerce indicator | <i>String</i><br>1-character alphanumeric  | 'crypt_type'=>\$crypt    |

### Force Post with Vault transaction object definition

| Variable Name      | Type and Limits                                                                                                                                                                                                 | Set Method                                                                              |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| customer ID        | <i>String</i><br>50-character alphanumeric<br><br><b>NOTE:</b><br>Some special characters are not allowed:<br>< > \$ % = ? ^ { } [ ] \                                                                          | 'cust_id'=>\$cust_id                                                                    |
| dynamic descriptor | <i>String</i><br>20-character alphanumeric<br><br>total of 22 characters including your merchant name and separator<br><br><b>NOTE:</b><br>Some special characters are not allowed:<br>< > \$ % = ? ^ { } [ ] \ | 'dynamic_descriptor'=>\$dynamic_descriptor                                              |
| status check       | <i>Boolean</i><br>true/false                                                                                                                                                                                    | \$mpgHttpPost =new<br>mpgHttpsPostStatus(\$store_id,\$api_token,\$status,\$mpgRequest); |

## Sample Force Post with Vault

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
/***** Transaction Variables *****/
$data_key='uroyVNSxzk5hHoT0kpQDBCw4';
$orderid='res-forcepost-'.date("dmy-G:i:s");
$amount='1.00';
$custid='cust';
$crypt_type='7';
$auth_code='256452';
$dynamic_descriptor='my descriptor';

//NT Response Option
$get_nt_response = 'true';//Optional - set it true only if you want to get network
tokenization response.
/***** Transaction Array *****/
$txnArray=array('type'=>'res_forcepost_cc',
'order_id'=>$orderid,
'cust_id'=>$custid,
'amount'=>$amount,
'data_key'=>$data_key,
'crypt_type'=>$crypt_type,
'auth_code'=>$auth_code,
'dynamic_descriptor'=>$dynamic_descriptor,
'get_nt_response'=>$get_nt_response
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** mpgHttpPost Object *****/
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response Object *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nAVSResponse = " . $mpgResponse->getAvsResultCode());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\n\nSourcePanLast4 = " . $mpgResponse->getSourcePanLast4());

```

**Sample Force Post with Vault**

```

if($get_nt_response == 'true')
{
print("\n\nNTResponseCode = " . $mpgResponse->getNTResponseCode());
print("\nNTMessage = " . $mpgResponse->getNTMessage());
print("\nNTUsed = " . $mpgResponse->getNTUsed());
print("\nNTTokenBin = " . $mpgResponse->getNTTokenBin());
print("\nNTTokenLast4 = " . $mpgResponse->getNTTokenLast4());
print("\nNTTokenExpDate = " . $mpgResponse->getNTTokenExpDate());
}
//----- ResolveData -----
print("\n\nCust ID = " . $mpgResponse->getResDataCustId());
print("\nPhone = " . $mpgResponse->getResDataPhone());
print("\nEmail = " . $mpgResponse->getResDataEmail());
print("\nNote = " . $mpgResponse->getResDataNote());
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\nCrypt Type = " . $mpgResponse->getResDataCryptType());
print("\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
print("\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
?>

```

**4.6.6 Card Verification with Vault – ResCardVerificationCC**

**NOTE:** This transaction supports both temporary and permanent tokens.

**Things to Consider:**

- This transaction type only applies to Visa, Mastercard, American Express and Discover transactions
- The card number and expiry date for this transaction are passed using a token, as represented by the data key value
- When using a temporary token (e.g., such as with Hosted Tokenization) **and** you intend to store the cardholder credentials, this transaction must be run prior to running the Vault Add Token transaction

**Card Verification with Vault object definition**

```

$txnArray = array('type'=>'res_card_verification_cc', ...);
$mpgTxn = new mpgTransaction($txnArray);

```

## HttpPostRequest object for Card Verification with Vault transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

### Card Verification with Vault transaction request fields – Required

For a full description of mandatory and optional values, see Appendix A Definition of Request Fields.

| Variable Name                                                                                                                                                                                                                                                                  | Type and Limits                                                            | Set Method                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------|
| order ID                                                                                                                                                                                                                                                                       | <i>String</i><br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces | 'order_id'=>\$order_id              |
| data key                                                                                                                                                                                                                                                                       | <i>String</i><br>25-character alphanumeric                                 | 'data_key'=>\$data_key              |
| electronic commerce indicator                                                                                                                                                                                                                                                  | <i>String</i><br>1-character alphanumeric                                  | 'crypt_type'=>\$crypt               |
| AVS Information                                                                                                                                                                                                                                                                | <i>Object</i><br>N/A                                                       | \$mpgTxn->setAvsInfo(\$mpgAvsInfo); |
| CVD Information                                                                                                                                                                                                                                                                | <i>Object</i><br>N/A                                                       | \$mpgTxn->setCvdInfo(\$mpgCvdInfo); |
| <div> <b>NOTE:</b> When storing credentials on the initial transaction, the CVD object must be sent; for subsequent transactions using stored credentials, CVD can be sent with cardholder-initiated transactions only—<b>merchants must not store CVD information.</b> </div> |                                                                            |                                     |
| Credential on File Info                                                                                                                                                                                                                                                        | <i>Object</i>                                                              |                                     |
| cof                                                                                                                                                                                                                                                                            | N/A                                                                        | \$mpgTxn->setCofInfo(\$cof);        |
| <div> <b>NOTE:</b> This is a nested object within the transaction, and required when storing or using the customer's stored </div>                                                                                                                                             |                                                                            |                                     |



| Variable Name                                                                                                                             | Type and Limits | Set Method |
|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------|
| <p>credentials. For information about fields in the Credential on File Info object, see Credential on File Info Object and Variables.</p> |                 |            |

### Credential on File Info object request fields

| Variable Name                                                                                                                                                                                                                                                                                              | Type and Limits                                                       |                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>issuer ID</p> <p><b>NOTE:</b> This variable is required for all merchant-initiated transactions following the first one; upon sending the first transaction, the issuer ID value is received in the transaction response and then used in subsequent transaction requests.</p>                          | <p><i>String</i></p> <p>15-character alphanumeric variable length</p> | <pre>\$cof-&gt;setIssuerId("VALUE_FOR_ISSUER_ID");</pre> <p><b>NOTE:</b> For a list and explanation of the possible values to send for this variable, see Definition of Request Fields – Credential on File</p>             |
| <p>payment indicator</p> <p><b>NOTE:</b> In Credential on File transactions where the request field e-commerce indicator is also being sent, the acceptable values for e-commerce indicator are dependent on the value sent for payment indicator; see request field definitions for more information.</p> | <p><i>String</i></p> <p>1-character alphabetic</p>                    | <pre>\$cof-&gt;setPaymentIndicator("PAYMENT_INDICATOR_VALUE");</pre> <p><b>NOTE:</b> For a list and explanation of the possible values to send for this variable, see Definition of Request Fields – Credential on File</p> |
| <p>payment information</p>                                                                                                                                                                                                                                                                                 | <p><i>String</i></p> <p>1-character numeric</p>                       | <pre>\$cof-&gt;setPaymentInformation("PAYMENT_INFO_VALUE");</pre> <p><b>NOTE:</b> For a list and explanation of the possible values to send for this variable, see Definition of Request Fields – Credential on File</p>    |

### Sample Card Verification with Vault

```
<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='monca00597';
$api_token='027AbCbxQorPggMQe6hU';
/***** Transaction Variables *****/
$data_key='4HIme0ZGURXE3NRBXHUj6nSc4';
$orderid='res-purch-'.date("dmy-G:i:s");
$crypt_type='1';
$expdate='2301'; //for temp token

//NT Response Option
$get_nt_response = 'false'; //Optional - set it true only if you want to get network
tokenization response.
/***** Transaction Array *****/
$txnArray=array('type'=>'res_card_verification_cc',
'data_key'=>$data_key,
'order_id'=>$orderid,
'crypt_type'=>$crypt_type,
'expdate'=>$expdate,
'get_nt_response'=>$get_nt_response
);
/***** CVD Variables *****/
$cvd_indicator = '1';
$cvd_value = '198';
/***** CVD Associative Array *****/
$cvdTemplate = array(
'cvd_indicator' => $cvd_indicator,
'cvd_value' => $cvd_value
);
$mpgCvdInfo = new mpgCvdInfo ($cvdTemplate);
/***** AVS Variables *****/
//The AVS portion is optional if AVS details are already stored in this profile
//If AVS details are resent in Purchase transaction, they will replace stored details
$avs_street_number = '';
$avs_street_name = 'bloor st';
$avs_zipcode = '111111';
/***** AVS Associative Array *****/
$avsTemplate = array(
'avs_street_number' => $avs_street_number,
'avs_street_name' => $avs_street_name,
'avs_zipcode' => $avs_zipcode
);
$mpgAvsInfo = new mpgAvsInfo ($avsTemplate);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
$mpgTxn->setCvdInfo($mpgCvdInfo);
$mpgTxn->setAvsInfo($mpgAvsInfo);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("U");
$cof->setPaymentInformation("2");
$cof->setIssuerId("168451306048014");
$mpgTxn->setCofInfo($cof);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** mpgHttpPost Object *****/
```

### Sample Card Verification with Vault

```
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response Object *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nCVDResponse = " . $mpgResponse->getCvdResultCode());
print("\nAVSResponse = " . $mpgResponse->getAvsResultCode());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\n\nSourcePanLast4 = " . $mpgResponse->getSourcePanLast4());

if($get_nt_response == 'true')
{
print("\n\nNTResponseCode = " . $mpgResponse->getNTResponseCode());
print("\n\nNTMessage = " . $mpgResponse->getNTMessage());
print("\n\nNTUsed = " . $mpgResponse->getNTUsed());
print("\n\nNTTokenBin = " . $mpgResponse->getNTTokenBin());
print("\n\nNTTokenLast4 = " . $mpgResponse->getNTTokenLast4());
print("\n\nNTTokenExpDate = " . $mpgResponse->getNTTokenExpDate());
}
//----- ResolveData -----
print("\n\nCust ID = " . $mpgResponse->getResDataCustId());
print("\n\nPhone = " . $mpgResponse->getResDataPhone());
print("\n\nEmail = " . $mpgResponse->getResDataEmail());
print("\n\nNote = " . $mpgResponse->getResDataNote());
print("\n\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\n\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\n\nCrypt Type = " . $mpgResponse->getResDataCryptType());
print("\n\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
print("\n\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\n\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
?>
```

#### 4.6.7 Surcharge Lookup with Vault

Confirm eligibility of a tokenized card for merchant surcharges on a transaction.

Credit Surcharge is feature where merchants may append an additional charge to the amount of a transaction processed on an eligible credit card. Only eligible credit cards can be surcharged, so merchants must confirm eligibility of the card before processing the payment via surcharge lookup.

The following transactions support including surcharge amounts:

- Purchase (Basic)
- Pre-Authorization (Basic)
- Vault Purchase
- Vault Pre-Authorization

**Things to Consider:**

- Card association rules require that cardholders must be informed of any surcharge by a merchant.
- The cardholder must be offered an option for canceling the transaction if they refuse the surcharge.

```
$txnArray = array('type'=>'res_surcharge_lookup', ...);

$mpgTxn = new mpgTransaction($txnArray);

$mpgRequest = new mpgRequest($mpgTxn);

$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

**Core connection object fields (all API transactions)**

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

**Surcharge Lookup transaction request fields – Required**

| Variable Name | Type and Limits                                                                              |                        |
|---------------|----------------------------------------------------------------------------------------------|------------------------|
| data key      | String<br>25-character alphanumeric                                                          | 'data_key'=>\$data_key |
| amount        | String<br>10-character decimal<br>Up to 7 digits (dollars) +<br>decimal point (.) + 2 digits | 'amount'=>\$amount     |

| Variable Name | Type and Limits                 |  |
|---------------|---------------------------------|--|
|               | (cents) after the decimal point |  |
|               | <b>EXAMPLE:</b> 1234567.89      |  |

### Sample Vault Surcharge Lookup

```
<?php
require "../mpgClasses.php";
$store_id='monca03650';
$api_token='7Yw0MPTlhjBRcZiE6837';
$amount = '50.00';
$pan='4242424242424242';
$txnArray=array('type'=>'surcharge_lookup',
'amount'=>$amount,
'pan'=>$pan
);
$mpgTxn = new mpgTransaction($txnArray);
$mpgRequest = new mpgRequest($mpgTxn);
//print the mpgrequest
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false or comment out this line for production transactions
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nMessage = " . $mpgResponse->getMessage());
print("\nCardType = " . $mpgResponse->getCardType());
print("\nIsSurchargeEligible = " . $mpgResponse->getIsSurchargeEligible());
print("\nMaxSurchargeRate = " . $mpgResponse->getMaxSurchargeRate());
print("\nMaxSurchargeAmount = " . $mpgResponse->getMaxSurchargeAmount());
print("\nServiceType = " . $mpgResponse->getServiceType());
?>
```

## 4.7 Hosted Tokenization

Moneris Hosted Tokenization is a solution for online e-commerce merchants who do not want to handle credit card numbers directly on their websites, yet want the ability to fully customize their check-out web page appearance.

When an hosted tokenization transaction is initiated, the Moneris Gateway displays (on the merchant's behalf) a single text box on the merchant's checkout page. The cardholder can then securely enter the credit card information into the text box. Upon submission of the payment information on the checkout page, Moneris Gateway returns a temporary token representing the credit card number to the merchant. This is then used in an API call to process a financial transaction directly with Moneris to charge the card. After receiving a response to the financial transaction, the merchant generates a receipt and allows the cardholder to continue with online shopping.

For more details on how to implement the Moneris Hosted Tokenization feature, see the Hosted Solutions Integration Guide. The guide can be downloaded from the Moneris Developer Portal at

[developer.moneris.com](http://developer.moneris.com)

## 5 Level 2/3 Transactions

- 5.1 About Level 2/3 Transactions
- 5.2 Level 2/3 Visa Transactions
- 5.3 Level 2/3 Mastercard Transactions
- 5.4 Level 2/3 American Express Transactions

### 5.1 About Level 2/3 Transactions

The Moneris Gateway API supports passing Level 2/3 purchasing card transaction data for Visa, MasterCard and American Express corporate cards.

All Level 2/3 transactions use the same Pre-Authorization transaction as described in the topic Pre-Authorization (page 27).

### 5.2 Level 2/3 Visa Transactions

- 5.2.1 Level 2/3 Transaction Types for Visa
- 5.2.2 Level 2/3 Transaction Flow for Visa
- 5.2.3 VS Completion
- 5.2.5 VS Force Post
- 5.2.4 VS Purchase Correction
- 5.2.6 VS Refund
- 5.2.7 VS Independent Refund
- 5.2.8 VS Corpais
- 1 VS Corpais Invoice
- 1 VS Corpais – Passenger Itinerary

#### 5.2.1 Level 2/3 Transaction Types for Visa

This transaction set includes a suite of corporate card financial transactions as well as a transaction that allows for the passing of Level 2/3 data. Please ensure that Visa Level 2/3 support is enabled on your

merchant account. Batch Close, Open Totals and Pre-authorization are identical to the transactions outlined in the section Basic Transaction Set (page 22).

- When the Pre-authorization response contains CorporateCard equal to true then you can submit the Visa transactions.
- If CorporateCard is false then the card does not support Level 2/3 data and non Level 2/3 transaction are to be used. If the card is not a corporate card, please refer to the section 2 Basic Transaction Set for the appropriate non-corporate card transactions.

**NOTE:** This transaction set is intended for transactions where Corporate Card is true and Level 2/3 data will be submitted. If the credit card is found to be a corporate card but you do not wish to send any Level 2/3 data then you may submit Visa transactions using the basic transaction set outlined in 2 Basic Transaction Set.

#### **Pre-authorization– (authorization/pre-authorization)**

Pre-authorization verifies and locks funds on the customer's credit card. The funds are locked for a specified amount of time, based on the card issuer. To retrieve the funds from a preauth so that they may be settled in the merchant account a capture must be performed. CorporateCard will return as true if the card supports Level 2/3.

#### **VS Completion – (Capture/Pre-authorization Completion)**

Once a Pre-authorization is obtained the funds that are locked need to be retrieved from the customer's credit card. The capture retrieves the locked funds and readies them for settlement into the merchant account. Prior to performing a VS Completion, a Pre-authorization must be performed. Once the transaction is completed, VS Corpais must be used to process the Level 2/3 data.

#### **VS Force Post – (Force Capture/Pre-authorization Completion)**

This transaction is an alternative to VS Completion to obtain the funds locked on Pre-auth obtained from IVR or equivalent terminal. The VS Force Post retrieves the locked funds and readies them for settlement in to the merchant account. Once the transaction is completed, VS Corpais must be used to process the Level 2/3 data.

#### **VS Purchase Correction (Void, Correction)**

VS Completion and VS Force Post can be voided the same day\* that they occur. A VS Purchase Correction must be for the full amount of the transaction and will remove any record of it from the cardholder statement.

#### **VS Refund – (Credit)**

A VS Refund can be performed against a VS Completion to refund any part or all of the transaction. Once the transaction is completed, VS Corpais must be used to process the Level 2/3 data.

**VS Independent Refund – (Credit)**

A VS Independent Refund can be performed against a purchase or a capture to refund any part, or all of the transaction. Independent refund is used when the originating transaction was not performed through Moneris Gateway. Once the transaction is completed, VS Corpais must be used to process the Level 2/3 data.

**NOTE:** the Independent Refund transaction may or may not be supported on your account. If you receive a transaction not allowed error when attempting an independent refund, it may mean the transaction is not supported on your account. If you wish to have the Independent Refund transaction type temporarily enabled (or re-enabled), please contact the Service Centre at 1-866-319-7450.

**VS Corpais – (Level 2/3 Data)**

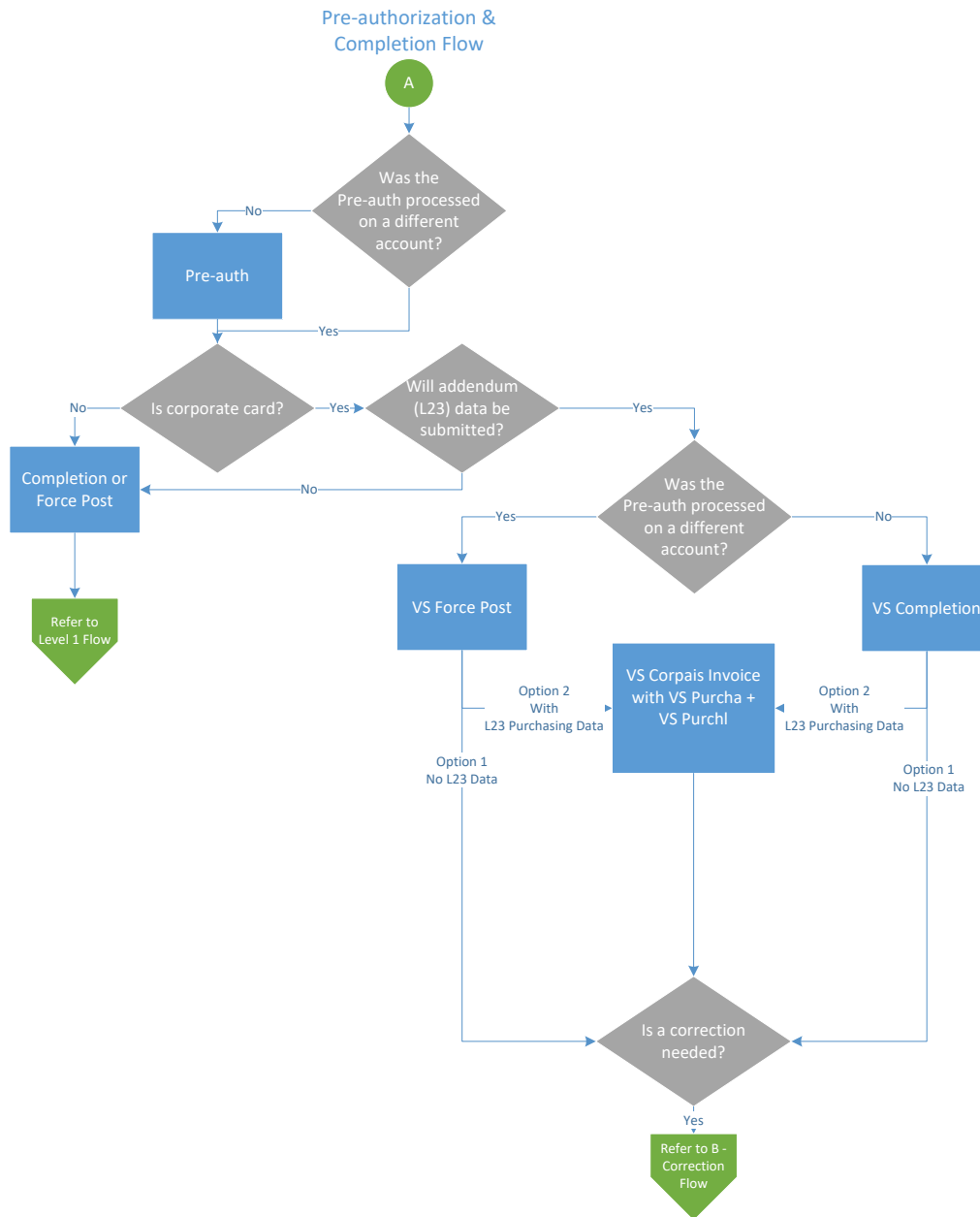
VS Corpais will contain all the required and optional data fields for Level 2/3 Business to Business data. VS Corpais data can be sent when the card has been identified in the Pre-authorization transaction request as being a corporate card.

\* A VS Purchase Correction can be performed against a transaction as long as the batch that contains the original transaction remains open. When using the automated closing feature, the batch close occurs daily between 10 – 11 pm EST.

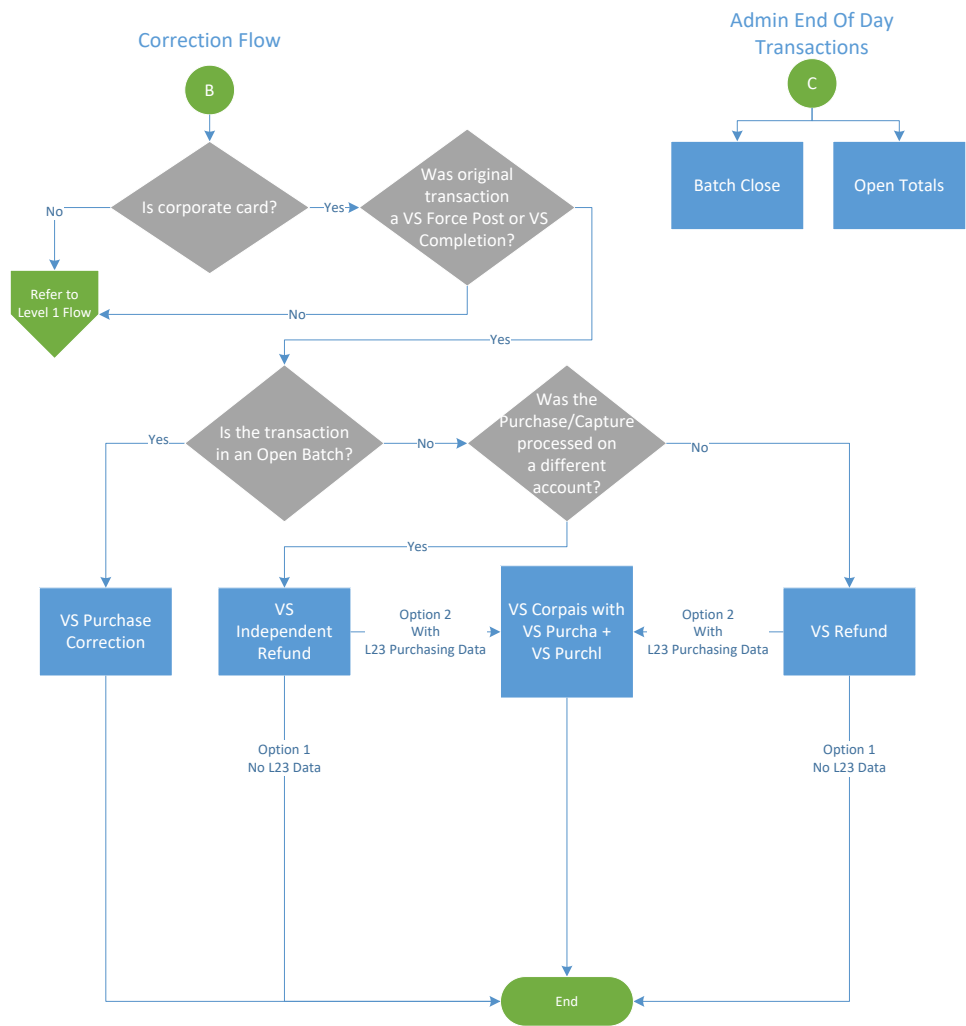


## 5.2.2 Level 2/3 Transaction Flow for Visa

### Pre-authorization/Completion Transaction Flow



Purchase Correction Transaction Flow



### 5.2.3 VS Completion

Once a Pre-authorization is obtained, the funds that are locked need to be retrieved from the customer's credit card. This VS Completion transaction is used to secure the funds locked by a pre-authorization transaction and readies them for settlement into the merchant account.

**NOTE:** Once you have completed this transaction successfully, to submit the complete supplemental level 2/3 data, please proceed to VS Corpais.

#### VS Completion transaction object definition

```
$txnArray = array('type'=>'vscompletion', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

#### HttpPostRequest object for VS Completion transaction object

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

#### VS Completion transaction request fields – Required

| Variable Name        | Limits                                                                                                                                                          | Set Method                   |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Order ID             | String<br>50-character alphanumeric                                                                                                                             | 'order_id'=>\$order_id       |
| Completion amount    | String<br>10-character decimal<br><br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><br><b>EXAMPLE: 1234567.89</b> | 'comp_amount'=>\$comp_amount |
| Transaction number   | String<br>255-character alphanumeric                                                                                                                            | 'txn_number'=>\$txn_number   |
| E-Commerce Indicator | String                                                                                                                                                          | 'crypt_type'=>\$crypt        |

| Variable Name | Limits | Set Method |
|---------------|--------|------------|
|---------------|--------|------------|

1-character alphanumeric

Table 1 Visa - Corporate Card Common Data - Level 2 Request Fields

| Req* | Value                                               | Limits                    | Set Method                           | Description                                                                                                                                                                                                                                 |
|------|-----------------------------------------------------|---------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Y    | National Tax                                        | 12-character decimal      | 'national_tax'=>\$national_tax       | <p>Must reflect the amount of National Tax (GST or HST) appearing on the invoice.</p> <p>Minimum - 0.01 Maximum - 999999.99. Must have 2 decimal places.</p>                                                                                |
| Y    | Merchant VAT Registration/Single Business Reference | 20-character alphanumeric | 'merchant_vat_no'=>\$merchant_vat_no | <p>Merchant's Tax Registration Number</p> <p>must be provided if tax is included on the invoice</p> <div style="border: 1px solid black; padding: 5px; margin-top: 10px;"> <b>NOTE:</b> Must not be all spaces or all zeroes         </div> |
| C    | Local Tax                                           | 12-character decimal      | 'local_tax'=>\$local_tax             | <p>Must reflect the amount of Local Tax (PST or QST) appearing on the invoice</p>                                                                                                                                                           |

| Req* | Value                                      | Limits                    | Set Method                           | Description                                                                                                                                                                                                                                    |
|------|--------------------------------------------|---------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                            |                           |                                      | <p>If Local Tax included then must not be all spaces or all zeroes; Must be provided if Local Tax (PST or QST) applies</p> <p>Minimum = 0.01</p> <p>Maximum = 999999.99</p> <p>Must have 2 decimal places</p>                                  |
| C    | Local Tax (PST or QST) Registration Number | 15-character alphanumeric | 'local_tax_no'=>\$local_tax_no       | <p>Merchant's Local Tax (PST/QST) Registration Number</p> <p>Must be provided if tax is included on the invoice; If Local Tax included then must not be all spaces or all zeroes</p> <p>Must be provided if Local Tax (PST or QST) applies</p> |
| C    | Customer VAT Registration Number           | 13-character alphanumeric | 'customer_vat_no'=>\$customer_vat_no | If the Customer's Tax Registration                                                                                                                                                                                                             |

| Req* | Value                                             | Limits                    | Set Method                         | Description                                                                                                                     |
|------|---------------------------------------------------|---------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
|      |                                                   |                           |                                    | Number appears on the invoice to support tax exempt transactions it must be provided here                                       |
| C    | Customer Code/Customer Reference Identifier (CRI) | 16-character alphanumeric | 'cri'=>\$cri                       | Value which the customer may choose to provide to the supplier at the point of sale – must be provided if given by the customer |
| N    | Customer Code                                     | 17-character alphanumeric | 'customer_code'=>\$customer_code   | Optional customer code field that will not be passed along to Visa, but will be included on Moneris reporting                   |
| N    | Invoice Number                                    | 17-character alphanumeric | 'invoice_number'=>\$invoice_number | Optional invoice number field that will not be passed along to Visa, but will be included on Moneris reporting                  |

\*Y = Required, N = Optional, C = Conditional

| Sample VS Completion |
|----------------------|
| <?php                |

## Sample VS Completion

```

require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='moneris';
$api_token='hurgle';
//$status = 'false';
/***** Transactional Variables *****/
$type='vscompletion';
$order_id='ord-210916-15:14:46';
$comp_amount='5.00';
$txn_number = '19002-0_11';
$crypt='7';
$national_tax = "1.23";
$merchant_vat_no = "gstno111";
$local_tax = "2.34";
$customer_vat_no = "gstno999";
$cri = "CUST-REF-002";
$customer_code="ccvsfp";
$invoice_number="invsfp";
$local_tax_no="ltaxno";
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'comp_amount'=>$comp_amount,
'txn_number'=>$txn_number,
'crypt_type'=>$crypt,
'national_tax'=>$national_tax,
'merchant_vat_no'=>$merchant_vat_no,
'local_tax'=>$local_tax,
'customer_vat_no'=>$customer_vat_no,
'cri'=>$cri,
'local_tax_no'=>$local_tax_no
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
//Status check example
//$mpgHttpPost = new mpgHttpPostStatus($store_id,$api_token,$status,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());

```

### Sample VS Completion

```
//print("\nStatusCode = " . $mpgResponse-&gtgetStatusCode());
//print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>
```

## 5.2.4 VS Purchase Correction

The VS Purchase Correction (also known as a "void") transaction is used to cancel a transaction that was performed in the current batch. No amount is required because a void is always for 100% of the original transaction. The only transaction that can be voided using VS Purchase Correction is a VS Completion or VS Force Post. To send a void the order\_id and txn\_number from the VS Completion/VS Force Post are required.

### VS Purchase Correction transaction object definition

```
$txnArray = array('type'=>'vspurchasecorrection', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for VS Purchase Correction transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### VS Purchase Correction transaction request fields – Required

| Variable Name        | Type and Limits                             | Set Method                 |
|----------------------|---------------------------------------------|----------------------------|
| Order ID             | <i>String</i><br>50-character alphanumeric  | 'order_id'=>\$order_id     |
| Transaction number   | <i>String</i><br>255-character alphanumeric | 'txn_number'=>\$txn_number |
| E-Commerce Indicator | <i>String</i><br>1-character alphanumeric   | 'crypt_type'=>\$crypt      |

### Sample VS Purchase Correction

```
<?php
```



## Sample VS Purchase Correction

```

require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='moneris';
$api_token='hurgle';
//$status = 'false';
/***** Transactional Variables *****/
$type='vspurchasecorrection';
$order_id='ord-210916-15:28:01';
$amount='5.00';
$txn_number = '19017-0_11';
$crypt='7';
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'amount'=>$amount,
'txn_number'=>$txn_number,
'crypt_type'=>$crypt
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
//Status check example
//$mpgHttpPost = new mpgHttpPostStatus($store_id,$api_token,$status,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
//print("\nStatusCode = " . $mpgResponse->getStatusCode());
//print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>

```

### 5.2.5 VS Force Post

The VS Force Post transaction is used to secure the funds locked by a pre-authorization transaction performed over IVR or equivalent terminal. When sending a force post request, you will need Order ID, Amount, Credit Card Number, Expiry Date, E-commerce Indicator and the Authorization Code received in the pre-authorization response.

**NOTE:** Once you have completed this transaction successfully, to submit the complete supplemental level 2/3 data, please proceed to VS Corpais.

### VS Force Post transaction object definition

```
$txnArray = array('type'=>'vsforcepost', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for VS Force Post transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### VS Force Post transaction request fields – Required

| Variable Name      | Type and Limits                                                                                                                                                        | Set Method                   |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Order ID           | <i>String</i><br>50-character alphanumeric                                                                                                                             | 'order_id'=>\$order_id       |
| Amount             | <i>String</i><br>10-character decimal<br><br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><br><b>EXAMPLE: 1234567.89</b> | 'amount'=>\$amount           |
| Credit card number | <i>String</i><br>20-character numeric                                                                                                                                  | 'pan'=>\$pan                 |
| Expiry Date        | <i>String</i><br>4-character numeric<br>YYMM format                                                                                                                    | 'expiry_date'=>\$expiry_date |
| Authorization code | <i>String</i><br>8-character alphanumeric                                                                                                                              | 'auth_code'=>\$auth_code     |

| Variable Name        | Type and Limits                           | Set Method            |
|----------------------|-------------------------------------------|-----------------------|
| E-commerce Indicator | <i>String</i><br>1-character alphanumeric | 'crypt_type'=>\$crypt |

**VS Force Post transaction request fields – Optional**

| Variable Name | Type and Limits                            | Set Method           |
|---------------|--------------------------------------------|----------------------|
| Customer ID   | <i>String</i><br>50-character alphanumeric | 'cust_id'=>\$cust_id |

**Table 1 Visa - Corporate Card Common Data - Level 2 Request Fields**

| Req* | Value                                               | Limits                    | Set Method                           | Description                                                                                                                                                  |
|------|-----------------------------------------------------|---------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Y    | National Tax                                        | 12-character decimal      | 'national_tax'=>\$national_tax       | <p>Must reflect the amount of National Tax (GST or HST) appearing on the invoice.</p> <p>Minimum - 0.01 Maximum - 999999.99. Must have 2 decimal places.</p> |
| Y    | Merchant VAT Registration/Single Business Reference | 20-character alphanumeric | 'merchant_vat_no'=>\$merchant_vat_no | <p>Merchant's Tax Registration Number</p> <p>must be provided if tax is included on the invoice</p> <p><b>NOTE:</b> Must</p>                                 |

| Req* | Value                                      | Limits                    | Set Method                     | Description                                                                                                                                                                                                                                                                                     |
|------|--------------------------------------------|---------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                            |                           |                                | <div>not be all spaces or all zeroes</div>                                                                                                                                                                                                                                                      |
| C    | Local Tax                                  | 12-character decimal      | 'local_tax'=>\$local_tax       | <p>Must reflect the amount of Local Tax (PST or QST) appearing on the invoice</p> <p>If Local Tax included then must not be all spaces or all zeroes; Must be provided if Local Tax (PST or QST) applies</p> <p>Minimum = 0.01</p> <p>Maximum = 999999.99</p> <p>Must have 2 decimal places</p> |
| C    | Local Tax (PST or QST) Registration Number | 15-character alphanumeric | 'local_tax_no'=>\$local_tax_no | <p>Merchant's Local Tax (PST/QST) Registration Number</p> <p>Must be provided if tax is included on the invoice; If Local Tax included then</p>                                                                                                                                                 |

| Req* | Value                                             | Limits                    | Set Method                           | Description                                                                                                                     |
|------|---------------------------------------------------|---------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
|      |                                                   |                           |                                      | <p>must not be all spaces or all zeroes</p> <p>Must be provided if Local Tax (PST or QST) applies</p>                           |
| C    | Customer VAT Registration Number                  | 13-character alphanumeric | 'customer_vat_no'=>\$customer_vat_no | If the Customer's Tax Registration Number appears on the invoice to support tax exempt transactions it must be provided here    |
| C    | Customer Code/Customer Reference Identifier (CRI) | 16-character alphanumeric | 'cri'=>\$cri                         | Value which the customer may choose to provide to the supplier at the point of sale – must be provided if given by the customer |
| N    | Customer Code                                     | 17-character alphanumeric | 'customer_code'=>\$customer_code     | Optional customer code field that will not be passed along to Visa, but will be included on Moneris reporting                   |
| N    | Invoice Number                                    | 17-character alphanumeric | 'invoice_number'=>\$invoice_number   | Optional invoice number field that                                                                                              |

| Req* | Value | Limits | Set Method | Description                                                                 |
|------|-------|--------|------------|-----------------------------------------------------------------------------|
|      |       |        |            | will not be passed along to Visa, but will be included on Moneris reporting |

\*Y = Required, N = Optional, C = Conditional

#### Sample VS Force Post

```
<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='moneris';
$api_token='hurgle';
// $status = 'false';
/***** Transactional Variables *****/
$type='vsforcepost';
$cust_id='CUST13343';
$order_id='ord-' . date("dmy-G:i:s");
$amount='5.00';
$pan='4242424254545454';
$expiry_date='2012';
$auth_code='123456';
$crypt='7';
$national_tax = "1.23";
$merchant_vat_no = "gstnol11";
$local_tax = "2.34";
$customer_vat_no = "gstno999";
$cri = "CUST-REF-002";
$customerCode="ccvsfp";
$invoiceNumber="invsfp";
$local_tax_no="ltaxno";
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'amount'=>$amount,
'pan'=>$pan,
'expdate'=>$expiry_date,
'auth_code'=>$auth_code,
'crypt_type'=>$crypt,
'national_tax'=>$national_tax,
'merchant_vat_no'=>$merchant_vat_no,
'local_tax'=>$local_tax,
'customer_vat_no'=>$customer_vat_no,
'cri'=>$cri,
'local_tax_no'=>$local_tax_no
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
```

### Sample VS Force Post

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
//Status check example
//$mpgHttpPost = new mpgHttpPostStatus($store_id,$api_token,$status,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
//print("\nStatusCode = " . $mpgResponse->getStatusCode());
//print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>
```

## 5.2.6 VS Refund

VS Refund will credit a specified amount to the cardholder's credit card. A refund can be sent up to the full value of the original VS Completion or VS Force Post. To send a VS Refund you will require the Order ID and Transaction Number from the original VS Completion or VS Force Post.

**NOTE:** Once you have completed this transaction successfully, to submit the complete supplemental level 2/3 data, please proceed to VS Corpais.

### VS Refund transaction object definition

```
$txnArray = array('type'=>'vsrefund', ...);

$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for VS Refund transaction

```
$mpgRequest = new mpgRequest($mpgTxn);

$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

## VS Refund transaction object values

| Variable Name        | Type and Limits                                                                                                                                             | Set Method                 |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| Order ID             | String<br>50-character alphanumeric                                                                                                                         | 'order_id'=>\$order_id     |
| Transaction number   | String<br>255-character alphanumeric                                                                                                                        | 'txn_number'=>\$txn_number |
| Amount               | String<br>10-character decimal<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><div>EXAMPLE: 1234567.89</div> | 'amount'=>\$amount         |
| E-Commerce Indicator | String<br>1-character alphanumeric                                                                                                                          | 'crypt_type'=>\$crypt      |

Table 1 Visa - Corporate Card Common Data - Level 2 Request Fields

| Req* | Value        | Limits               | Set Method                     | Description                                                                                                               |
|------|--------------|----------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Y    | National Tax | 12-character decimal | 'national_tax'=>\$national_tax | Must reflect the amount of National Tax (GST or HST) appearing on the invoice.<br><br>Minimum - 0.01 Maximum - 999999.99. |



| Req* | Value                                               | Limits                    | Set Method                           | Description                                                                                                                                                                                                                                                   |
|------|-----------------------------------------------------|---------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                                     |                           |                                      | Must have 2 decimal places.                                                                                                                                                                                                                                   |
| Y    | Merchant VAT Registration/Single Business Reference | 20-character alphanumeric | 'merchant_vat_no'=>\$merchant_vat_no | <p>Merchant's Tax Registration Number</p> <p>must be provided if tax is included on the invoice</p> <div> <b>NOTE:</b> Must not be all spaces or all zeroes </div>                                                                                            |
| C    | Local Tax                                           | 12-character decimal      | 'local_tax'=>\$local_tax             | <p>Must reflect the amount of Local Tax (PST or QST) appearing on the invoice</p> <p>If Local Tax included then must not be all spaces or all zeroes; Must be provided if Local Tax (PST or QST) applies</p> <p>Minimum = 0.01</p> <p>Maximum = 999999.99</p> |

| Req* | Value                                             | Limits                    | Set Method                           | Description                                                                                                                                                                                                                                    |
|------|---------------------------------------------------|---------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                                   |                           |                                      | Must have 2 decimal places                                                                                                                                                                                                                     |
| C    | Local Tax (PST or QST) Registration Number        | 15-character alphanumeric | 'local_tax_no'=>\$local_tax_no       | <p>Merchant's Local Tax (PST/QST) Registration Number</p> <p>Must be provided if tax is included on the invoice; If Local Tax included then must not be all spaces or all zeroes</p> <p>Must be provided if Local Tax (PST or QST) applies</p> |
| C    | Customer VAT Registration Number                  | 13-character alphanumeric | 'customer_vat_no'=>\$customer_vat_no | If the Customer's Tax Registration Number appears on the invoice to support tax exempt transactions it must be provided here                                                                                                                   |
| C    | Customer Code/Customer Reference Identifier (CRI) | 16-character alphanumeric | 'cri'=>\$cri                         | Value which the customer may choose to provide to the supplier at the point of sale – must be provided if                                                                                                                                      |

| Req* | Value          | Limits                    | Set Method                         | Description                                                                                                    |
|------|----------------|---------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------|
|      |                |                           |                                    | given by the customer                                                                                          |
| N    | Customer Code  | 17-character alphanumeric | 'customer_code'=>\$customer_code   | Optional customer code field that will not be passed along to Visa, but will be included on Moneris reporting  |
| N    | Invoice Number | 17-character alphanumeric | 'invoice_number'=>\$invoice_number | Optional invoice number field that will not be passed along to Visa, but will be included on Moneris reporting |

\*Y = Required, N = Optional, C = Conditional

#### Sample VS Refund

```
<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='moneris';
$api_token='hurgle';
//$status = 'false';
/***** Transactional Variables *****/
$type='vsrefund';
$order_id='ord-210916-15:14:46';
$amount='5.00';
$txn_number = '19003-1_11';
$crypt='7';
$national_tax = "1.23";
$merchant_vat_no = "gstno111";
$local_tax = "2.34";
$customer_vat_no = "gstno999";
$cri = "CUST-REF-002";
$customerCode="ccvsfp";
$invoiceNumber="invsfp";
$local_tax_no="ltaxno";
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
```

## Sample VS Refund

```

'amount'=>$amount,
'txn_number'=>$txn_number,
'crypt_type'=>$crypt,
'national_tax'=>$national_tax,
'merchant_vat_no'=>$merchant_vat_no,
'local_tax'=>$local_tax,
'customer_vat_no'=>$customer_vat_no,
'cri'=>$cri,
'local_tax_no'=>$local_tax_no
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
//Status check example
//$mpgHttpPost = new mpgHttpPostStatus($store_id,$api_token,$status,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
//print("\nStatusCode = " . $mpgResponse->getStatusCode());
//print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>

```

## 5.2.7 VS Independent Refund

VS Independent Refund will credit a specified amount to the cardholder's credit card. The independent refund does not require an existing order to be logged in the Moneris Gateway; however, the credit card number and expiry date will need to be passed. The transaction format is almost identical to a pre-authorization.

**NOTE:** Once you have completed this transaction successfully, to submit the complete supplemental level 2/3 data, please proceed to VS Corpais.

**VS Independent Refund transaction object definition**

```
$txnArray = array('type'=>'vsind_refund', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

**HttpPostRequest object for VS Independent Refund transaction**

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

**VS Independent Refund transaction request fields – Required**

| Variable Name        | Type and Limits                                                                                                                                                        | Set Method                   |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Order ID             | <i>String</i><br>50-character alphanumeric                                                                                                                             | 'order_id'=>\$order_id       |
| Amount               | <i>String</i><br>10-character decimal<br><br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><br><b>EXAMPLE: 1234567.89</b> | 'amount'=>\$amount           |
| Credit card number   | <i>String</i><br>20-character numeric                                                                                                                                  | 'pan'=>\$pan                 |
| Expiry date          | <i>String</i><br>4-character numeric<br>YYMM format                                                                                                                    | 'expiry_date'=>\$expiry_date |
| E-commerce indicator | <i>String</i><br>1-character alphanumeric                                                                                                                              | 'crypt_type'=>\$crypt        |

**VS Independent Refund transaction request fields – Optional**

| Variable Name | Type and Limits           | Set Method           |
|---------------|---------------------------|----------------------|
| Customer ID   | 50-character alphanumeric | 'cust_id'=>\$cust_id |

Table 1 Visa - Corporate Card Common Data - Level 2 Request Fields

| Req* | Value                                               | Limits                    | Set Method                           | Description                                                                                                                                                               |
|------|-----------------------------------------------------|---------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Y    | National Tax                                        | 12-character decimal      | 'national_tax'=>\$national_tax       | <p>Must reflect the amount of National Tax (GST or HST) appearing on the invoice.</p> <p>Minimum - 0.01 Maximum - 999999.99. Must have 2 decimal places.</p>              |
| Y    | Merchant VAT Registration/Single Business Reference | 20-character alphanumeric | 'merchant_vat_no'=>\$merchant_vat_no | <p>Merchant's Tax Registration Number</p> <p>must be provided if tax is included on the invoice</p> <div> <p><b>NOTE:</b> Must not be all spaces or all zeroes</p> </div> |
| C    | Local Tax                                           | 12-character decimal      | 'local_tax'=>\$local_tax             | <p>Must reflect the amount of Local Tax (PST or QST) appearing on the invoice</p> <p>If Local Tax</p>                                                                     |

| Req* | Value                                      | Limits                    | Set Method                           | Description                                                                                                                                                                                                                                    |
|------|--------------------------------------------|---------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                            |                           |                                      | <p>included then must not be all spaces or all zeroes; Must be provided if Local Tax (PST or QST) applies</p> <p>Minimum = 0.01</p> <p>Maximum = 999999.99</p> <p>Must have 2 decimal places</p>                                               |
| C    | Local Tax (PST or QST) Registration Number | 15-character alphanumeric | 'local_tax_no'=>\$local_tax_no       | <p>Merchant's Local Tax (PST/QST) Registration Number</p> <p>Must be provided if tax is included on the invoice; If Local Tax included then must not be all spaces or all zeroes</p> <p>Must be provided if Local Tax (PST or QST) applies</p> |
| C    | Customer VAT Registration Number           | 13-character alphanumeric | 'customer_vat_no'=>\$customer_vat_no | If the Customer's Tax Registration Number                                                                                                                                                                                                      |

| Req* | Value                                             | Limits                    | Set Method                         | Description                                                                                                                     |
|------|---------------------------------------------------|---------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
|      |                                                   |                           |                                    | appears on the invoice to support tax exempt transactions it must be provided here                                              |
| C    | Customer Code/Customer Reference Identifier (CRI) | 16-character alphanumeric | 'cri'=>\$cri                       | Value which the customer may choose to provide to the supplier at the point of sale – must be provided if given by the customer |
| N    | Customer Code                                     | 17-character alphanumeric | 'customer_code'=>\$customer_code   | Optional customer code field that will not be passed along to Visa, but will be included on Moneris reporting                   |
| N    | Invoice Number                                    | 17-character alphanumeric | 'invoice_number'=>\$invoice_number | Optional invoice number field that will not be passed along to Visa, but will be included on Moneris reporting                  |

\*Y = Required, N = Optional, C = Conditional

#### Sample VS Independent Refund

```
<?php
require "../mpgClasses.php";
/***** Request Variables *****/
```



## Sample VS Independent Refund

```

$store_id='moneris';
$api_token='hurgle';
// $status = 'false';
/***** Transactional Variables *****/
$type='vsind_refund';
$cust_id='CUST13343';
$order_id='ord-' . date("dmy-G:i:s");
$amount='5.00';
$pan='4242424254545454';
$expiry_date='2012';
$crypt='7';
$national_tax = "1.23";
$merchant_vat_no = "gstno111";
$local_tax = "2.34";
$customer_vat_no = "gstno999";
$cri = "CUST-REF-002";
$customerCode="ccvsfp";
$invoiceNumber="invsfp";
$local_tax_no="ltaxno";
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'cust_id'=>$cust_id,
'amount'=>$amount,
'pan'=>$pan,
'expdate'=>$expiry_date,
'crypt_type'=>$crypt,
'national_tax'=>$national_tax,
'merchant_vat_no'=>$merchant_vat_no,
'local_tax'=>$local_tax,
'customer_vat_no'=>$customer_vat_no,
'cri'=>$cri,
'local_tax_no'=>$local_tax_no
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
// Status check example
// $mpgHttpPost = new mpgHttpPostStatus($store_id,$api_token,$status,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());

```

### Sample VS Independent Refund

```
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
//print("\nStatusCode = " . $mpgResponse->getStatusCode());
//print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>
```

## 5.2.8 VS Corpais

VS Corpais will contain all the required and optional data fields for Level 2/3 Purchasing Card Addendum data. VS Corpais data can be sent when the card has been identified in the Pre-authorization transaction request as being a corporate card.

In addition to the order ID and transaction number, this transaction also contains two objects:

- VS Purcha – Corporate Card Common Data
- VS Purchl – Line Item Details

VS Corpais request must be preceded by a financial transaction (VS Completion, VS Force Post, VS Refund, VS Independent Refund) and the Corporate Card flag must be set to “true” in the Pre-authorization response.

### VS Corpais transaction object definition

```
$txnArray = array('type'=>'vscorpais', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for VS Corpais transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### VS Corpais transaction request fields – Required

| Variable Name      | Type and Limits                             | Set Method                   |
|--------------------|---------------------------------------------|------------------------------|
| Order ID           | <i>String</i><br>50-character alphanumeric  | 'order_id'=>\$order_id       |
| Transaction number | <i>String</i><br>255-character alphanumeric | 'txn_number'=>\$txn_number   |
| vsPurcha           | <i>String</i>                               | \$vsPurcha = new vsPurcha(); |

| Variable Name                                                               | Type and Limits | Set Method                                                                                                  |
|-----------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------|
| For a list of the variables that appear in this object, see the table below | n/a             | <pre>\$mpgVsLevel123 = new mpgVsLevel123();  \$mpgVsLevel123-&gt;setVsPurch (\$vsPurcha, \$vsPurchl);</pre> |

vsPurchl *String* `$vsPurchl = new vsPurchl();`

For a list of the variables that appear in this object, see the table below

n/a

```
$mpgVsLevel123 = new
mpgVsLevel123();

$mpgVsLevel123->setVsPurch
($vsPurcha, $vsPurchl);
```

\*Y = Required, N = Optional, C = Conditional

### 5.2.8.1 VS Purcha – Corporate Card Common Data

VS Corpais transactions use the VS Purcha object to contain Level 2 data.

| Variable Name  | Type and Limits                            | Description                                                                                                                                                                                                                                           |
|----------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Buyer Name     | <i>String</i><br>30-character alphanumeric | Buyer/Recipient Name<br><div> <b>NOTE:</b> Name required by CRA on transactions &gt;\$150 </div>                                                                                                                                                      |
| Local Tax Rate | <i>String</i><br>4-character decimal       | Indicates the detailed tax rate applied in relationship to a local tax amount<br><div> <b>EXAMPLE:</b> 8% PST should be 8.0 </div> Minimum = 0.01<br>Maximum = 99.99<br><div> <b>NOTE:</b> Must be provided if Local Tax (PST or QST) applies. </div> |
| Duty Amount    | <i>String</i><br>9-character decimal       | Duty on total purchase amount<br>A minus sign means 'amount is a credit', plus sign or no sign means                                                                                                                                                  |

| Variable Name                    | Type and Limits                            | Description                                                                                                                                                                                                                                                                   |
|----------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  |                                            | 'amount is a debit'<br>maximum without sign is 999999.99                                                                                                                                                                                                                      |
| Invoice Discount Treatment       | <i>String</i><br>1-character numeric       | Indicates how the merchant is managing discounts<br><br>Must be one of the following values:<br><br>0 - if no invoice level discounts apply for this invoice<br><br>1 - if Tax was calculated on Post-Discount totals<br><br>2 - if Tax was calculated on Pre-Discount totals |
| Invoice Level Discount Amount    | <i>String</i><br>9-character decimal       | Amount of discount (if provided at the invoice level according to the Invoice Discount Treatment)<br><br>Must be non-zero if Invoice Discount Treatment is 1 or 2<br><br>Minimum amount is 0.00 and maximum is 999999.99                                                      |
| Ship To Postal Code / Zip Code   | <i>String</i><br>10-character alphanumeric | The postal code or zip code for the destination where goods will be delivered<br><br><div><b>NOTE:</b> Required if shipment is involved</div><br>Full alpha postal code - Valid ANA<space>NAN format required if shipping to an address within Canada                         |
| Ship From Postal Code / Zip Code | <i>String</i><br>10-character alphanumeric | The postal code or zip code from which items were shipped<br><br>For Canadian addresses, requires full alpha postal code for the merchant with Valid ANA<space>NAN format                                                                                                     |
| Destination Country Code         | 2-character alphanumeric                   | Code of country where purchased goods will be delivered                                                                                                                                                                                                                       |

| Variable Name                         | Type and Limits                     | Description                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       |                                     | Use ISO 3166-1 alpha-2 format                                                                                                                                                                                                                                                                                                                                                                              |
|                                       |                                     | <b>NOTE:</b> Required if it appears on the invoice for an international transaction                                                                                                                                                                                                                                                                                                                        |
| Unique VAT Invoice Reference Number   | String<br>25-character alphanumeric | Unique Value Added Tax Invoice Reference Number<br><br>Must be populated with the invoice number and this cannot be all spaces or zeroes                                                                                                                                                                                                                                                                   |
| Tax Treatment                         | String<br>1-character alphanumeric  | Must be one of the following values:<br><br>0 = Net Prices with tax calculated at line item level;<br><br>1 = Net Prices with tax calculated at invoice level;<br><br>2 = Gross prices given with tax information provided at line item level;<br><br>3 = Gross prices given with tax information provided at invoice level;<br><br>4 = No tax applies (small merchant) on the invoice for the transaction |
| Freight/Shipping Amount (Ship Amount) | String<br>9-character decimal       | Freight charges on total purchase<br><br>If shipping is not provided as a line item it must be provided here, if applicable<br><br>Signed monetary amount:<br><br>Minus (-) sign means 'amount is a credit',<br><br>Plus (+) sign or no sign means 'amount is a debit'<br><br>Maximum without sign is 999999.99                                                                                            |
| GST HST Freight Rate                  | String<br>4-character decimal       | Rate of GST (excludes PST) or HST charged on the shipping amount (in accordance with the Tax Treatment)                                                                                                                                                                                                                                                                                                    |

| Variable Name          | Type and Limits                                 | Description                                                                                                                                                                                                                                                                     |
|------------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        |                                                 | <p>If Freight/Shipping Amount is provided then this (National GST or HST) tax rate must be provided.</p> <p>Monetary amount, maximum is 99.99. Such as 13% HST is 13.00</p>                                                                                                     |
| GST HST Freight Amount | <p><i>String</i></p> <p>9-character decimal</p> | <p>Amount of GST (excludes PST) or HST charged on the shipping amount</p> <p>If Freight/Shipping Amount is provided then this (National GST or HST) tax amount must be provided if taxTreatment is 0 or 2</p> <p>Signed monetary amount: maximum without sign is 999999.99.</p> |

### 5.2.8.2 VS Purchl – Line Item Details

VS Corpais transactions use the VS Purchl object to contain Level 3 data.

#### Line Item Details for VS Purchl

```

$item_com_code = array("X3101", "X84802");
$product_code = array("CHR123", "DDSK200");
$item_description = array("Office Chair", "Disk Drive");
$item_quantity = array("3", "1");
$item_uom = array("EA", "EA");
$unit_cost = array("0.20", "0.40");
$vat_tax_amt = array("0.00", "0.00");
$vat_tax_rate = array("13.00", "13.00");
$discount_treatmentL = array("0", "0");
$discount_amtL = array("0.00", "0.00");

```

#### Setting VS Purchl Line Item Details

```

$vsPurchl->setVsPurchl($item_com_code[0], $product_code[0], $item_description
[0], $item_quantity[0], $item_uom[0], $unit_cost[0], $vat_tax_amt[0], $vat_
tax_rate[0], $discount_treatmentL[0], $discount_amtL[0]);

```

**Table 1 Corporate Card Common Data - Level 3 Request Fields - VSPurchl**

| Req* | Value                | Limits                     | Variable/Field   | Description                                                                                                                                                                                                                                                                                     |
|------|----------------------|----------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C    | Item Commodity Code  | 12-character alpha-numeric | item_com_code    | Line item Comodity Code (if this field is not sent, then Product Code must be sent)                                                                                                                                                                                                             |
| Y    | Product Code         | 12-character alpha-numeric | product_code     | <p>Product code for this line item – merchant’s product code, manufacturer’s product code or buyer’s product code</p> <p>Typically this will be the SKU or identifier by which the merchant tracks and prices the item or service</p> <p>This should always be provided for every line item</p> |
| Y    | Item Description     | 35-character alpha-numeric | item_description | Line item description                                                                                                                                                                                                                                                                           |
| Y    | Item Quantity        | 12-character decimal       | item_quantity    | <p>Quantity invoiced for this line item</p> <p>Up to 4 decimal places supported, whole numbers are accepted</p> <p>Minimum = 0.0001</p> <p>Maximum = 999999999999</p>                                                                                                                           |
| Y    | Item Unit of Measure | 2-character alpha-numeric  | item_uom         | Unit of measure                                                                                                                                                                                                                                                                                 |

| Req* | Value              | Limits               | Variable/Field      | Description                                                                                                                                                                                                            |
|------|--------------------|----------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                    |                      |                     | Use ANSI X-12 EDI Allowable Units of Measure and Codes                                                                                                                                                                 |
| Y    | Item Unit Cost     | 12-character decimal | unit_cost           | <p>Line item cost per unit</p> <p>2-4 decimal places accepted</p> <p>Minimum = 0.0001</p> <p>Maximum = 999999.9999</p>                                                                                                 |
| N    | VAT Tax Amount     | 12-character decimal | vat_tax_amt         | <p>Any value-added tax or other sales tax amount</p> <p>Must have 2 decimal places</p> <p>Minimum = 0.01</p> <p>Maximum = 999999.99</p>                                                                                |
| N    | VAT Tax Rate       | 4-character decimal  | vat_tax_rate        | <p>Sales tax rate</p> <div> <p><b>EXAMPLE:</b> 8% PST should be 8.0</p> </div> <p>maximum 99.99</p>                                                                                                                    |
| Y    | Discount Treatment | 1-character numeric  | discount_treatmentL | <p>Must be one of the following values:</p> <p>0 if no invoice level discounts apply for this invoice</p> <p>1 if Tax was calculated on Post-Discount totals</p> <p>2 if Tax was calculated on Pre-Discount totals</p> |



| Req* | Value           | Limits               | Variable/Field | Description                                                                                                                                                                                                                                                  |
|------|-----------------|----------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C    | Discount Amount | 12-character decimal | discount_amtL  | <p>Amount of discount, if provided for this line item according to the Line Item Discount Treatment</p> <p>Must be non-zero if Line Item Discount Treatment is 1 or 2</p> <p>Must have 2 decimal places</p> <p>Minimum = 0.01</p> <p>Maximum = 999999.99</p> |

### 5.2.8.3 Sample Code for VS Corpais

| Sample VS Corpais                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> &lt;?php require "../mpgClasses.php"; /***** Request Variables *****/ \$store_id='moneris'; \$api_token='hurgle'; //\$status = 'false'; /***** Transactional Variables *****/ \$type='vscorpais'; \$cust_id='CUST13343'; \$order_id='ord-160916-15:31:39'; \$txn_number='18306-0_11'; \$buyer_name = "Buyer Manager"; \$local_tax_rate = "13.00"; \$duty_amount = "0.00"; \$discount_treatment = "0"; \$discount_amt = "0.00"; \$freight_amount = "0.20"; \$ship_to_pos_code = "M8X 2W8"; \$ship_from_pos_code = "M1K 2Y7"; \$des_cou_code = "CAN"; \$vat_ref_num = "VAT12345"; \$tax_treatment = "3";//3 = Gross prices given with tax information provided at invoice level \$gst_hst_freight_amount = "0.00"; \$gst_hst_freight_rate = "13.00"; \$item_com_code = array("X3101", "X84802"); \$product_code = array("CHR123", "DDSK200"); \$item_description = array("Office Chair", "Disk Drive"); </pre> |

## Sample VS Corpsais

```

$item_quantity = array("3", "1");
$item_uom = array("EA", "EA");
$unit_cost = array("0.20", "0.40");
$vat_tax_amt = array("0.00", "0.00");
$vat_tax_rate = array("13.00", "13.00");
$discount_treatmentL = array("0", "0");
$discount_amtL = array("0.00", "0.00");
//Create and set VsPurcha
$vsPurcha = new vsPurcha();
$vsPurcha->setBuyerName($buyer_name);
$vsPurcha->setLocalTaxRate($local_tax_rate);
$vsPurcha->setDutyAmount($duty_amount);
$vsPurcha->setDiscountTreatment($discount_treatment);
$vsPurcha->setDiscountAmt($discount_amt);
$vsPurcha->setFreightAmount($freight_amount);
$vsPurcha->setShipToPostalCode($ship_to_pos_code);
$vsPurcha->setShipFromPostalCode($ship_from_pos_code);
$vsPurcha->setDesCouCode($des_cou_code);
$vsPurcha->setVatRefNum($vat_ref_num);
$vsPurcha->setTaxTreatment($tax_treatment);
$vsPurcha->setGstHstFreightAmount($gst_hst_freight_amount);
$vsPurcha->setGstHstFreightRate($gst_hst_freight_rate);
//Create and set VsPurchl
$vsPurchl = new vsPurchl();
$vsPurchl->setVsPurchl($item_com_code[0], $product_code[0], $item_description[0], $item_
quantity[0], $item_uom[0], $unit_cost[0], $vat_tax_amt[0], $vat_tax_rate[0], $discount_
treatmentL[0], $discount_amtL[0]);
$vsPurchl->setVsPurchl($item_com_code[1], $product_code[1], $item_description[1], $item_
quantity[1], $item_uom[1], $unit_cost[1], $vat_tax_amt[1], $vat_tax_rate[1], $discount_
treatmentL[1], $discount_amtL[1]);
//Create and set VsLevel23
$mpgVsLevel23 = new mpgVsLevel23();
$mpgVsLevel23->setVsPurch($vsPurcha, $vsPurchl);
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'txn_number'=>$txn_number,
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
$mpgTxn->setLevel23Data($mpgVsLevel23);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
//Status check example
//$mpgHttpPost = new mpgHttpPostStatus($store_id,$api_token,$status,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());

```

**Sample VS Corpais**

```
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
//print("\nStatusCode = " . $mpgResponse->getStatusCode());
//print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>
```

## 5.3 Level 2/3 Mastercard Transactions

- 5.3.1 Level 2/3 Transaction Types for Mastercard
- 5.3.2 Level 2/3 Transaction Flow for Mastercard
- 5.3.3 MC Completion
- 5.3.4 MC Force Post
- 5.3.5 MC Purchase Correction
- 5.3.6 MC Refund
- 5.3.7 MC Independent Refund
- 1 MC Corpais – Level 2/3 Transactions

### 5.3.1 Level 2/3 Transaction Types for Mastercard

This transaction set includes a suite of corporate card financial transactions as well as a transaction that allows for the passing of Level 2/3 data. Please ensure MC Level 2/3 processing support is enabled on your merchant account. Batch Close, Open Totals and Pre-authorization are identical to the transactions outlined in the section Basic Transaction Set (page 22).

When the Preauth response contains CorporateCard equal to true then you can submit the MC transactions.

If CorporateCard is false then the card does not support Level 2/3 data and non Level 2/3 transaction are to be used. If the card is not a corporate card, please refer to section 4 for the appropriate non-corporate card transactions.

**NOTE:** This transaction set is intended for transactions where Corporate Card is true and Level 2/3 data will be submitted. If the credit card is found to be a corporate card but you do not wish to send any Level 2/3 data then you may submit MC transactions using the transaction set outlined in Basic Transaction Set (page 22).

#### **Pre-auth – (authorization/pre-authorization)**

The pre-auth verifies and locks funds on the customer's credit card. The funds are locked for a specified amount of time, based on the card issuer. To retrieve the funds from a pre-auth so that they may be settled in the merchant account a capture must be performed. Level 2/3 data submission is not supported as part of a pre-auth as a pre-auth is not settled. When CorporateCard is returned true then Level 2/3 data may be submitted.

#### **MC Completion – (Capture/Preauth Completion)**

Once a Pre-authorization is obtained the funds that are locked need to be retrieved from the customer's credit card. The capture retrieves the locked funds and readies them for settlement in to the merchant account. Prior to performing an MCCompletion a Pre-auth must be performed.

#### **MC Force Post – (Force Capture/Preauth Completion)**

This transaction is an alternative to MC Completion to obtain the funds locked on Preauth obtained from IVR or equivalent terminal. The MC Force Post requires that the original Pre-authorization's auth code is provided and it retrieves the locked funds and readies them for settlement in to the merchant account.

#### **MC Purchase Correction – (Void, Correction)**

MC Completions can be voided the same day\* that they occur. A void must be for the full amount of the transaction and will remove any record of it from the cardholder statement. \* An MC Purchase Correction can be performed against a transaction as long as the batch that contains the original transaction remains open. When using the automated closing feature batch close occurs daily between 10 – 11 pm EST.

#### **MC Refund – (Credit)**

A MC Refund can be performed against an MC Completion or MC Force Post to refund an amount less than or equal to the amount of the original transaction.

#### **MC Independent Refund – (Credit)**

A MC Independent Refund can be performed against an completion to refund any part, or all of the transaction. Independent refund is used when the originating transaction was not performed through Moneris Gateway. Please note, the MC Independent Refund transaction may or may not be supported on your account. If you receive a transaction not allowed error when attempting an MC Independent Refund, it may mean the transaction is not supported on your account. If you wish to have the MC Independent Refund transaction type temporarily enabled (or re-enabled), please contact the Service Centre at 1-866-319-7450.

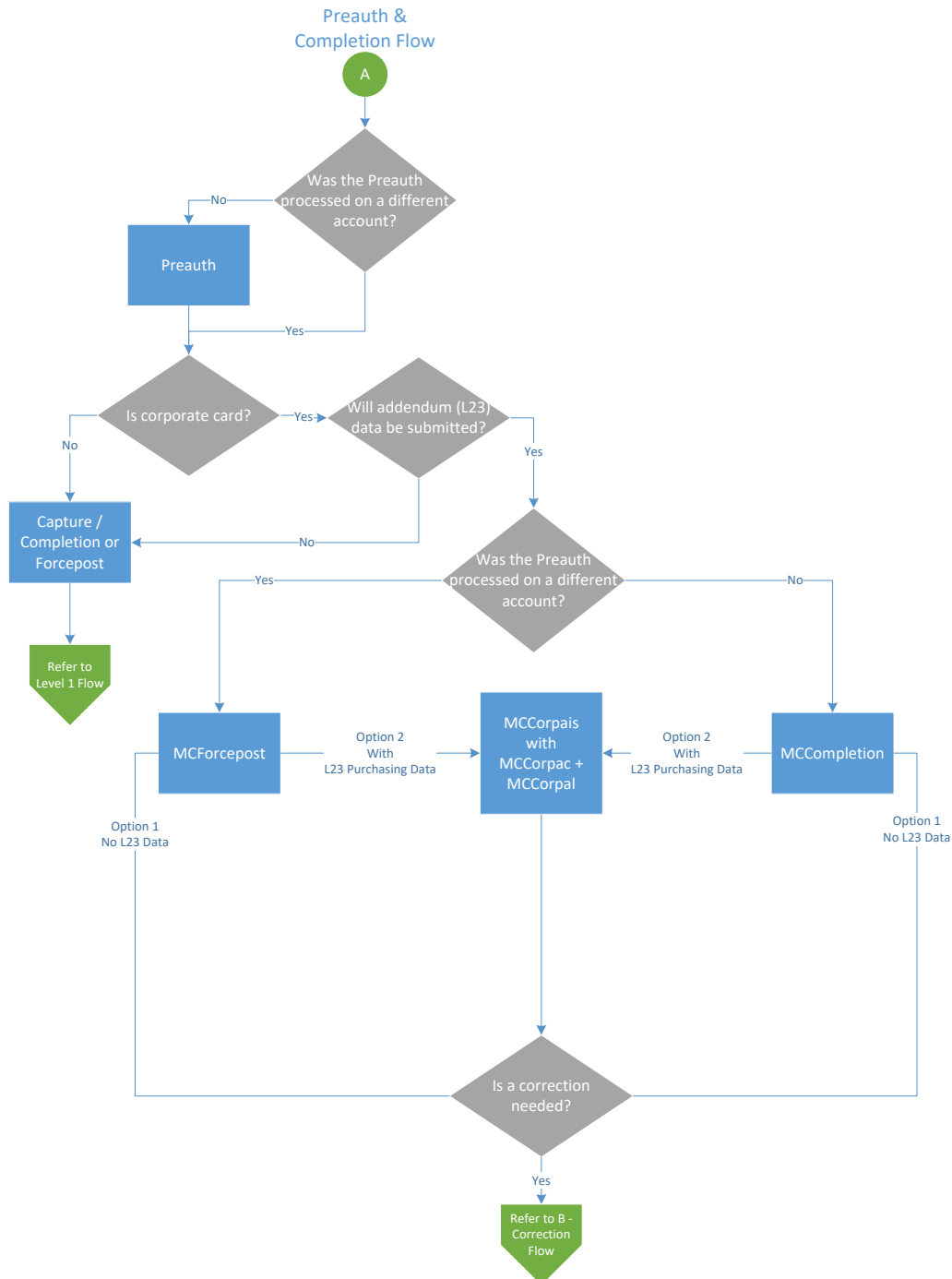
**MC Corpais Common Line Item – (Level 2/3 Data)**

MC Corpais Common Line Item will contain the entire required and optional data field for Level 2/3 data. MCCorpais Common Line Item data can be sent when the card has been identified in the transaction request as being a corporate card. This transaction supports multiple data types and combinations:

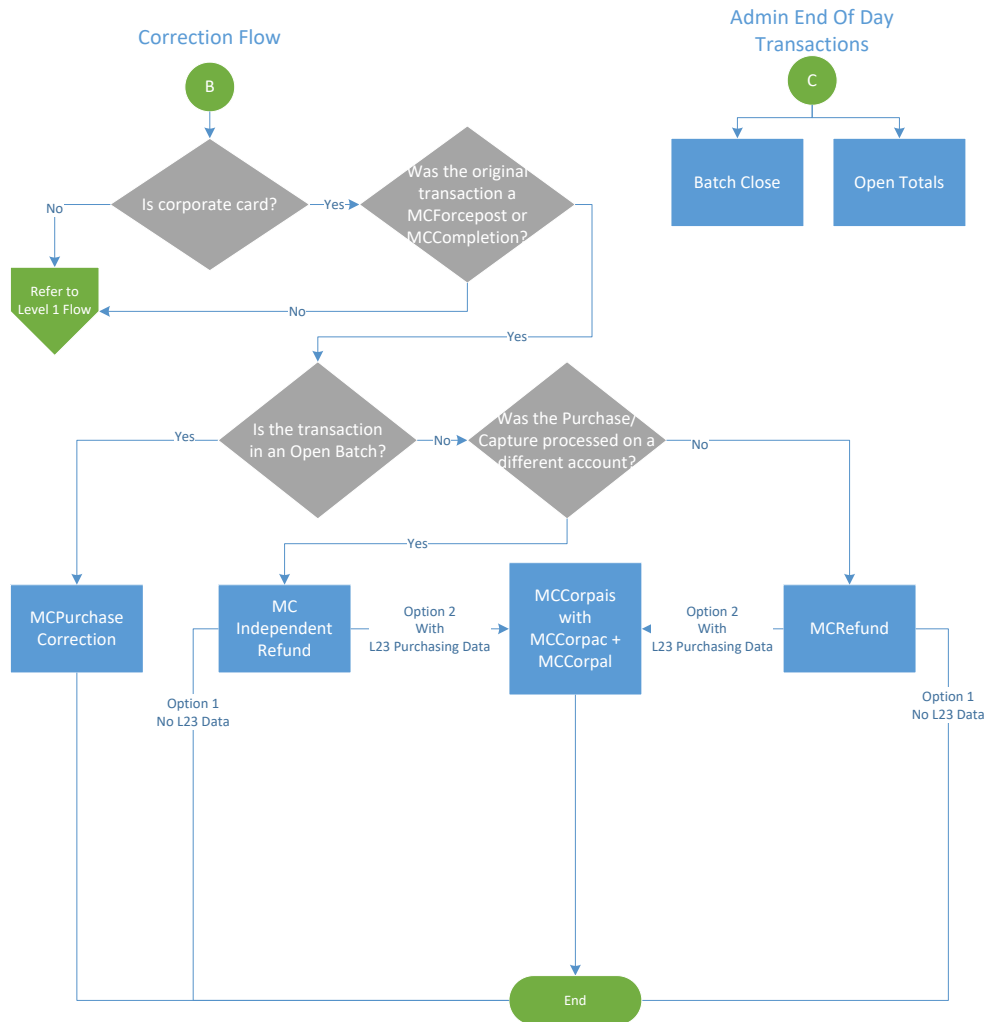
- Purchasing Card Data:
  - Corporate card common data with Line Item Details

### 5.3.2 Level 2/3 Transaction Flow for Mastercard

#### Pre-Authorization/Completion Transaction Flow



## Purchase Correction Transaction Flow



### 5.3.3 MC Completion

The MC Completion transaction is used to secure the funds locked by a Pre-Authorization transaction. When sending a capture request you will need two pieces of information from the original pre-authorization– the order ID and the transaction number from the returned response.

Once you have completed this transaction successfully, to submit the complete supplemental level 2/3 data, please proceed to MC Corpais.

#### MC Completion transaction object definition

```
$txnArray = array('type'=>'mccompletion', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

#### HttpPostRequest object for MC Completion transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

#### MC Completion transaction request fields – Required

| Variable Name           | Limits                                                                                                                                                      | Set Method                           |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| Order ID                | String<br>50-character alphanumeric                                                                                                                         | 'order_id'=>\$order_id               |
| Completion amount       | String<br>10-character decimal<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><div>EXAMPLE: 1234567.89</div> | 'comp_amount'=>\$comp_amount         |
| Transaction number      | String<br>255-character alphanumeric                                                                                                                        | 'txn_number'=>\$txn_number           |
| Merchant reference num- | String                                                                                                                                                      | 'merchant_ref_no'=>\$merchant_ref_no |



| Variable Name        | Limits                             | Set Method            |
|----------------------|------------------------------------|-----------------------|
| ber                  | 19-character alphanumeric          |                       |
| E-commerce indicator | String<br>1-character alphanumeric | 'crypt_type'=>\$crypt |

## Sample MC Completion

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='moneris';
$api_token='hurgle';
//$status = 'false';
/***** Transactional Variables *****/
$type='mccompletion';
$order_id='ord-210916-16:13:11';
$comp_amount='5.00';
$txn_number='19021-0_11';
$crypt='7';
$merchant_ref_no = "319038";
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'comp_amount'=>$comp_amount,
'txn_number'=>$txn_number,
'merchant_ref_no' => $merchant_ref_no,
'crypt_type'=>$crypt
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
//Status check example
//$mpgHttpPost = new mpgHttpPostStatus($store_id,$api_token,$status,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());

```

### Sample MC Completion

```
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
//print("\nStatusCode = " . $mpgResponse->getStatusCode());
//print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>
```

### 5.3.4 MC Force Post

MC Force Post transaction is used to secure the funds locked by a Pre-Authorization transaction performed over IVR or equivalent terminal. When sending a Force Post request, you will need order\_id, amount, pan (card number), expiry date, crypt type and the authorization code received in the Pre-Authorization response.

#### MC Force Post transaction object definition

```
$txnArray = array('type'=>'mcforcepost', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

#### HttpPostRequest object for MC Force Post transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

#### MC Force Post transaction request fields – Required

| Variable Name      | Type and Limits                                                                                                                                                 | Set Method             |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Order ID           | String<br>50-character alphanumeric                                                                                                                             | 'order_id'=>\$order_id |
| Amount             | String<br>10-character decimal<br><br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><br><b>EXAMPLE: 1234567.89</b> | 'amount'=>\$amount     |
| Credit card number | String                                                                                                                                                          | 'pan'=>\$pan           |

| Variable Name             | Type and Limits                                     | Set Method                           |
|---------------------------|-----------------------------------------------------|--------------------------------------|
|                           | 20-character alphanumeric                           |                                      |
| Expiry date               | String<br>4-character alphanumeric<br>(YYMM format) | 'expiry_date'=>\$expiry_date         |
| Authorization code        | String<br>8-character alphanumeric                  | 'auth_code'=>\$auth_code             |
| E-commerce indicator      | String<br>1-character alphanumeric                  | 'crypt_type'=>\$crypt                |
| Merchant reference number | String<br>19-character alphanumeric                 | 'merchant_ref_no'=>\$merchant_ref_no |

**MC Force Post transaction request fields – Optional**

| Variable Name | Type and Limits                     | Set Method           |
|---------------|-------------------------------------|----------------------|
| Customer ID   | String<br>50-character alphanumeric | 'cust_id'=>\$cust_id |

**Sample MC Force Post**

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='moneris';
$api_token='hurgle';
//$status = 'false';
/***** Transactional Variables *****/
$type='mcforcepost';
$cust_id='CUST13343';
$order_id='ord-' . date("dmy-G:i:s");
$amount='5.00';
$pan='5454545442424242';
$expiry_date='2012';
$auth_code='123456';
$crypt='7';
$merchant_ref_no = "319038";
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,

```

### Sample MC Force Post

```
'order_id'=>$order_id,
'cust_id'=>$cust_id,
'amount'=>$amount,
'pan'=>$pan,
'expdate'=>$expiry_date,
'auth_code'=>$auth_code,
'merchant_ref_no' => $merchant_ref_no,
'crypt_type'=>$crypt
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
//Status check example
//$mpgHttpPost = new mpgHttpPostStatus($store_id,$api_token,$status,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
//print("\nStatusCode = " . $mpgResponse->getStatusCode());
//print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>
```

### 5.3.5 MC Purchase Correction

The MC Purchase Correction (void) transaction is used to cancel a transaction that was performed in the current batch. No amount is required because a void is always for 100% of the original transaction. The only transaction that can be voided is completion. To send a void, the order ID and transaction number from the MC Completion or MC Force Post are required.

#### MC Purchase Correction transaction object definition

```
$txnArray = array('type'=>'mcpurchasecorrection', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

**HttpPostRequest object for MC Purchase Correction transaction**

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

**MC Purchase Correction transaction request fields – Required**

| Variable Name        | Type and Limits                              | Set Method                 |
|----------------------|----------------------------------------------|----------------------------|
| Order ID             | <i>String</i><br>50-character alphanumeric   | 'order_id'=>\$order_id     |
| Transaction number   | <i>String</i><br>255-character alpha-numeric | 'txn_number'=>\$txn_number |
| E-commerce indicator | <i>String</i><br>1-character alphanumeric    | 'crypt_type'=>\$crypt      |

**Sample MC Purchase Correction**

```
<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='moneris';
$api_token='hurgle';
//$status = 'false';
/***** Transactional Variables *****/
$type='mcpurchasecorrection';
$order_id='ord-210916-16:15:50';
$txn_number='66011731642016265161550929-0_11';
$crypt='7';
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'txn_number'=>$txn_number,
'crypt_type'=>$crypt
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
//Status check example
//$mpgHttpPost = new mpgHttpPostStatus($store_id,$api_token,$status,$mpgRequest);
/***** Response *****/
```

### Sample MC Purchase Correction

```
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
//print("\nStatusCode = " . $mpgResponse->getStatusCode());
//print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>
```

### 5.3.6 MC Refund

The MC Refund will credit a specified amount to the cardholder's credit card. A refund can be sent up to the full value of the original capture. To send a refund you will require the Order ID and Transaction Number from the original MC Completion or MC Force Post.

#### MC Refund transaction object definition

```
$txnArray = array('type'=>'mcrefund', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

#### HttpPostRequest object for MC Refund transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

#### MC Refund transaction request fields – Required

| Variable Name | Type and Limits                                                                                                              | Set Method             |
|---------------|------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Order ID      | 50-character alphanumeric                                                                                                    | 'order_id'=>\$order_id |
| Amount        | 10-character decimal<br><br>Up to 7 digits (dollars) +<br>decimal point (.) + 2 digits<br>(cents) after the decimal<br>point | 'amount'=>\$amount     |

| Variable Name                  | Type and Limits             | Set Method                           |
|--------------------------------|-----------------------------|--------------------------------------|
| <div>EXAMPLE: 1234567.89</div> |                             |                                      |
| Transaction number             | 255-character alpha-numeric | 'txn_number'=>\$txn_number           |
| E-commerce indicator           | 1-character alphanumeric    | 'crypt_type'=>\$crypt                |
| Merchant reference number      | 19-character alphanumeric   | 'merchant_ref_no'=>\$merchant_ref_no |

### Sample MC Refund

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='moneris';
$api_token='hurgle';
//$status = 'false';
/***** Transactional Variables *****/
$type='mcrefund';
$order_id='ord-210916-16:13:11';
$amount='5.00';
$txn_number='19021-1_11';
$crypt='7';
$merchant_ref_no = "319038";
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'amount'=>$amount,
'txn_number'=>$txn_number,
'merchant_ref_no' => $merchant_ref_no,
'crypt_type'=>$crypt
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost =new mpgHttpsPost($store_id,$api_token,$mpgRequest);
//Status check example
//$mpgHttpPost = new mpgHttpsPostStatus($store_id,$api_token,$status,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());

```

### Sample MC Refund

```
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
//print("\nStatusCode = " . $mpgResponse->getStatusCode());
//print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>
```

### 5.3.7 MC Independent Refund

MC Independent Refund is used when the originating transaction was not performed through Moneris Gateway and does not require an existing order to be logged in the Moneris Gateway; however, the credit card number and the expiry date will need to be passed. The transaction format is almost identical to a purchase or a pre-authorization.

**NOTE:** Independent refund transactions are not supported on all accounts. If you receive a transaction not allowed error when attempting an independent refund transaction, it may mean the feature is not supported on your account. To have Independent Refund transaction functionality temporarily enabled (or re-enabled), please contact the Moneris Customer Service Centre at 1-866-319-7450.

Once you have completed this transaction successfully, to submit the complete supplemental level 2/3 data, please proceed to MC Corpais.

#### MC Independent Refund transaction object definition

```
$txnArray = array('type'=>'mcind_refund', ...);
$mpgTxn = new mpgTransaction($txnArray);
```

#### HttpPostRequest object for MC Independent Refund transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```



**MC Independent Refund transaction request fields – Required**

| Variable Name             | Type and Limits                                                                                                                                       | Set Method                           |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| Order ID                  | 50-character alphanumeric                                                                                                                             | 'order_id'=>\$order_id               |
| Amount                    | 10-character decimal<br><br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><br><b>EXAMPLE: 1234567.89</b> | 'amount'=>\$amount                   |
| E-commerce indicator      | 1-character alphanumeric                                                                                                                              | 'crypt_type'=>\$crypt                |
| Credit card number        | 20-character numeric                                                                                                                                  | 'pan'=>\$pan                         |
| Expiry date               | 4-character numeric<br>(YYMM format)                                                                                                                  | 'expiry_date'=>\$expiry_date         |
| Merchant reference number | 19-character alphanumeric                                                                                                                             | 'merchant_ref_no'=>\$merchant_ref_no |

**MC Independent Refund transaction request fields – Optional****Table 1 MC Independent Refund transaction object optional values**

| Variable Name | Type and Limits                                | Set Method           |
|---------------|------------------------------------------------|----------------------|
| Customer ID   | <i>String</i><br><br>50-character alphanumeric | 'cust_id'=>\$cust_id |

**Sample MC Independent Refund**

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='moneris';
$api_token='hurgle';
// $status = 'false';
/***** Transactional Variables *****/
$type='mcind_refund';
$cust_id='CUST13343';

```

## Sample MC Independent Refund

```

$order_id='ord-' . date("dmy-G:i:s");
$amount='5.00';
$pan='5454545442424242';
$expiry_date='2012';
$crypt='7';
$merchant_ref_no = "319038";
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'cust_id'=>$cust_id,
'amount'=>$amount,
'pan'=>$pan,
'expdate'=>$expiry_date,
'merchant_ref_no' => $merchant_ref_no,
'crypt_type'=>$crypt
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
// Status check example
// $mpgHttpPost = new mpgHttpPostStatus($store_id,$api_token,$status,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
// print("\nStatusCode = " . $mpgResponse->getStatusCode());
// print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>

```

### 5.3.8 MC Corpais - Corporate Card Common Data with Line Item Details

This transaction example includes the following elements for Level 2 and 3 purchasing card corporate card data processing:

- Corporate Card Common Data (MC Corpac)
  - only 1 set of MC Corpac fields can be submitted
  - this data set includes data elements that apply to the overall order, e.g., the total overall taxes
- Line Item Details (MC Corpal)
  - 1-998 counts of MC Corpal line items can be submitted
  - This data set includes the details about each individual item or service purchased

The MC Corpais request must be preceded by a financial transaction (MC Completion, MC Force Post, MC Refund, MC Independent Refund) and the Corporate Card flag must be set to “true” in the Preauthorization response. The MC Corpais request will need to contain the Order ID of the financial transaction as well as the Transaction Number.

In addition, MC Corpais has a tax array object that can be sent via the Tax fields in MC Corpac and MC Corpal. For more about the tax array object, see 5.3.8.3 Tax Array Object - MC Corpais.

For descriptions of the Level 2/3 fields, please see Definition of Request Fields for Level 2/3 - Mastercard (page 553).

### **MC Corpais transaction object definition**

```
$txnArray = array('type'=>'mccorpais', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### **HttpPostRequest object for MC Corpais transaction**

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

## MC Corpais transaction object values

Table 1 MC Corpais transaction object mandatory values

| Value              | Type   | Limits                      | Set Method                                                                                      |
|--------------------|--------|-----------------------------|-------------------------------------------------------------------------------------------------|
| Order ID           | String | 50-character alpha-numeric  | 'order_id'=>\$order_id                                                                          |
| Transaction number | String | 255-character alpha-numeric | 'txn_number'=>\$txn_number                                                                      |
| MCCorpac           | Object | n/a                         | <pre>\$mpgMcLevel23 = new mpgMcLevel23();  \$mpgMcLevel23-&gt;setMcCorpac (\$mcCorpac);</pre>   |
| MC Corpal          | Object | n/a                         | <pre>\$mpgMcLevel23 = new mpgMcLevel23();  \$mpgMcLevel23-&gt;setMcCorpall (\$mcCorpall);</pre> |

\*Y = Required, N = Optional, C = Conditional

## 5.3.8.1 MC Corpac - Corporate Card Common Data

Table 1 Corporate Card Common Data - Level 2 Request Fields - MCCorpac

| Re-q* | Value               | Limits                     | Set Method                                                | Description                                                                                                                       |
|-------|---------------------|----------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| N     | Austin-Tetra Number | 15-character alpha-numeric | \$mcCorpac->setAustinTetraNumber (\$austin_tetra_number); | The Austin-Tetra Number assigned to the card acceptor                                                                             |
| N     | NAICS Code          | 15-character alpha-numeric | \$mcCorpac->setNaicsCode (\$naics_code);                  | North American Industry Classification System (NAICS) code assigned to the card acceptor                                          |
| N     | Customer Code       | 25-character alpha-numeric | \$mcCorpac->setCustomerCode1 (\$customer_code1_c);        | A control number, such as purchase order number, project number, department allocation number or name that the purchaser supplied |

| Re-q* | Value                  | Limits                                   | Set Method                                                                       | Description                                                                                                                              |
|-------|------------------------|------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
|       |                        |                                          |                                                                                  | the merchant<br><br>Left-justified; may be spaces                                                                                        |
| N     | Unique Invoice Number  | 17-character alpha-numeric               | <code>\$mcCorpac-&gt;setUniqueInvoiceNumber(\$unique_invoice_number_c);</code>   | Unique number associated with the individual transaction provided by the merchant                                                        |
| N     | Commodity Code         | 15-character alpha-numeric               | <code>\$mcCorpac-&gt;setCommodityCode(\$commodity_code);</code>                  | Code assigned by the merchant that best categorizes the item(s) being purchased                                                          |
| N     | Order Date             | 6-character numeric<br><br>YYMMDD format | <code>\$mcCorpac-&gt;setOrderDate(\$order_date_c);</code>                        | The date the item was ordered<br><br><div><b>NOTE:</b> If present, must contain a valid date</div>                                       |
| N     | Corporation VAT Number | 20-character alpha-numeric               | <code>\$mcCorpac-&gt;setCorporationVatNumber(\$corporation_vat_number_c);</code> | Contains a corporation's value added tax (VAT) number                                                                                    |
| N     | Customer VAT Number    | 20-character alpha-numeric               | <code>\$mcCorpac-&gt;setCustomerVatNumber(\$customer_vat_number_c);</code>       | Contains the VAT number for the customer / cardholder used to identify the customer when purchasing goods and services from the merchant |
| N     | Freight Amount         | 12-character decimal                     | <code>\$mcCorpac-&gt;setFreightAmount1(\$freight_amount_c);</code>               | The freight on the total purchase<br><br>Must have 2 decimals<br><br>Minimum = 0.00 Maximum = 999999.99                                  |

| Re-q* | Value                             | Limits                                                     | Set Method                                                                           | Description                                                                                                                                                        |
|-------|-----------------------------------|------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N     | Duty Amount                       | 12-character decimal                                       | <code>\$vsPurcha-&gt;setDutyAmount(\$duty_amount);</code>                            | <p>The duty on the total purchase</p> <p>Must have 2 decimals</p> <p>Minimum = 0.00</p> <p>Maximum = 999999.99</p>                                                 |
| N     | Destination State / Province Code | 3-character alpha-numeric                                  | <code>\$mcCorpac-&gt;setDestinationProvinceCode(\$destination_province_code);</code> | <p>State or Province of the country where the goods will be delivered</p> <p>Left justified with trailing spaces</p> <div>EXAMPLE: ONT = Ontario</div>             |
| N     | Destination Country Code          | 3-character alpha-numeric<br><br>ISO 3166-1 alpha-3 format | <code>\$mcCorpac-&gt;setDestinationCountryCode(\$destination_country_code);</code>   | <p>The country code where goods will be delivered</p> <p>Left justified with trailing spaces</p> <p>ISO 3166-1 alpha-3 format</p> <div>EXAMPLE: CAN = Canada</div> |
| N     | Ship From Postal Code             | 10-character alpha-numeric<br><br>ANA NAN format           | <code>\$mcCorpac-&gt;setShipFromPosCode(\$ship_from_pos_code);</code>                | <p>The postal code or zip code from which items were shipped</p> <p>Full alpha postal code - Valid ANA&lt;space&gt;NAN format</p>                                  |
| N     | Destination Postal Code           | 10-character alpha-numeric                                 | <code>\$mcCorpac-&gt;setShipToPosCode(\$ship_to_pos_code_c);</code>                  | The postal code or zip code where goods will be delivered                                                                                                          |

| Re-q* | Value                         | Limits                     | Set Method                                                                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------|-------------------------------|----------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |                               |                            |                                                                                             | Full alpha postal code - Valid ANA<space>NAN format if shipping to an address within Canada                                                                                                                                                                                                                                                                                                                                                                                              |
| N     | Authorized Contact Name       | 36-character alpha-numeric | <code>\$mcCorpac-&gt;setAuthorizedContactName(\$authorized_contact_name_c);</code>          | Name of an individual or company contacted for company authorized purchases                                                                                                                                                                                                                                                                                                                                                                                                              |
| N     | Authorized Contact Phone      | 17-character alpha-numeric | <code>\$mcCorpac-&gt;setAuthorizedContactPhone(\$authorized_contact_phone);</code>          | Phone number of an individual or company contacted for company authorized purchases                                                                                                                                                                                                                                                                                                                                                                                                      |
| N     | Additional Card Acceptor Data | 40-character alpha-numeric | <code>\$mcCorpac-&gt;setAdditionalCardAcceptorData(\$additional_card_acceptor_data);</code> | Information pertaining to the card acceptor                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| N     | Card Acceptor Type            | 8-character alpha-numeric  | <code>\$mcCorpac-&gt;setCardAcceptorType(\$card_acceptor_type);</code>                      | <p>Various classifications of business ownership characteristics</p> <p>This field takes 8 characters. Each character represents a different component, as follows:</p> <p>1st character represents 'Business Type' and contains a code to identify the specific classification or type of business:</p> <ul style="list-style-type: none"> <li>Corporation</li> <li>Not known</li> <li>Individual/Sole Proprietorship</li> <li>Partnership</li> <li>Association/Estate/Trust</li> </ul> |

| Re-q* | Value | Limits | Set Method | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------|-------|--------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |       |        |            | <p>Tax Exempt Organizations (501C)</p> <p>International Organization</p> <p>Limited Liability Company (LLC)</p> <p>Government Agency</p> <p>2nd character represents 'Business Owner Type'. Contains a code to identify specific characteristics about the business owner.</p> <p>1 - No application classification</p> <p>2 - Female business owner</p> <p>3 - Physically handicapped female business owner</p> <p>4 - Physically handicapped male business owner</p> <p>0 - Unknown</p> <p>3rd character represents 'Business Certification Type'. Contains a code to identify specific characteristics about the business certification type, such as small business, disadvantaged, or other certification type:</p> <p>1 - Not certified</p> <p>2 - Small Business Administration (SBA) certification small business</p> <p>3 - SBA certification</p> |



| Re-q* | Value | Limits | Set Method | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------|-------|--------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |       |        |            | <p>as small disadvantaged business</p> <p>4 - Other government or agency-recognized certification (such as Minority Supplier Development Council)</p> <p>5 - Self-certified small business</p> <p>6 - SBA certification as small and other government or agency-recognized certification</p> <p>7 - SBA certification as small disadvantaged business and other government or agency-recognized certification</p> <p>8 - Other government or agency-recognized certification and self-certified small business</p> <p>A - SBA certification as 8(a)</p> <p>B - Self-certified small disadvantaged business (SDB)</p> <p>C - SBA certification as HUBZone</p> <p>0 - Unknown</p> <p>4th character represents 'Business Racial/Ethnic Type'. Contains a code identifying</p> |

| Re-q* | Value | Limits | Set Method | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------|-------|--------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |       |        |            | <p>the racial or ethnic type of the majority owner of the business.</p> <p>1 - African American</p> <p>2 - Asian Pacific American</p> <p>3 - Subcontinent Asian American</p> <p>4 - Hispanic American</p> <p>5 - Native American Indian</p> <p>6 - Native Hawaiian</p> <p>7 - Native Alaskan</p> <p>8 - Caucasian</p> <p>9 - Other</p> <p>0 - Unknown</p> <p>5th character represents 'Business Type Provided Code'</p> <p>Y - Business type is provided.</p> <p>N - Business type was not provided.</p> <p>R - Card acceptor refused to provide business type</p> <p>6th character represents 'Business Owner Type Provided Code'</p> <p>Y - Business owner type is provided.</p> |

| Re-q* | Value                | Limits                     | Set Method                                                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------|----------------------|----------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |                      |                            |                                                                             | <p>N - Business owner type was not provided.</p> <p>R - Card acceptor refused to provide business type</p> <p>7th character represents 'Business Certification Type Provided Code'</p> <p>Y - Business certification type is provided.</p> <p>N - Business certification type was not provided.</p> <p>R - Card acceptor refused to provide business type</p> <p>8th character represents 'Business Racial/Ethnic Type'</p> <p>Y - Business racial/ethnic type is provided.</p> <p>N - Business racial/ethnic type was not provided.</p> <p>R - Card acceptor refused to provide business racial/ethnic type</p> |
| N     | Card Acceptor Tax ID | 20-character alpha-numeric | <code>\$mcCorpac-&gt;setCardAcceptorTaxTd(\$card_acceptor_tax_id_c);</code> | US federal tax ID number or value-added tax (VAT) ID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Req* | Value                          | Limits                     | Set Method                                                                                    | Description                                                                                                                                                          |
|------|--------------------------------|----------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N    | Card Acceptor Reference Number | 25-character alpha-numeric | <code>\$mcCorpac-&gt;setCardAcceptorReferenceNumber(\$card_acceptor_reference_number);</code> | Code that facilitates card acceptor/corporation communication and record keeping                                                                                     |
| N    | Card Acceptor VAT Number       | 20-character alpha-numeric | <code>\$mcCorpac-&gt;setCardAcceptorVatNumber(\$card_acceptor_vat_number_c);</code>           | Value added tax (VAT) number for the card acceptor location<br><br>Used to identify the card acceptor when collecting and reporting taxes                            |
| C    | Tax                            | Up to 6 arrays             | <code>\$mcCorpac-&gt;setTax(\$mcTax_c);</code>                                                | Can have up to 6 arrays containing different tax details<br><br><b>NOTE:</b> If you use this variable, you must fill in all the fields of tax array mentioned below. |

### 5.3.8.2 MC Corpal - Line Item Details

#### MC Corpal Object - Line Item Details

```
$mcCorpac->setMcCorpac($customer_code1_l[0], $line_item_date1_l[0], $ship_date1_l[0], $order_date1_l[0], $product_code1_l[0], $item_description1_l[0], $item_quantity1_l[0], $unit_cost1_l[0], $item_unit_measure1_l[0], $ext_item_amount1_l[0], $discount_amount1_l[0], $commodity_code1_l[0], $type_of_supply1_l[0], $vat_ref_num1_l[0], $mcTax1_l[0]);
```

**Table 1 Line Item Details - Level 3 Request Fields - MC Corpal**

| Req* | Value         | Limits                     | Variable                      | Description                                                                        |
|------|---------------|----------------------------|-------------------------------|------------------------------------------------------------------------------------|
| N    | Customer Code | 25-character alpha-numeric | <code>customer_code1_l</code> | A control number, such as purchase order number, project number, department alloc- |

| Req* | Value          | Limits                               | Variable         | Description                                                                                                                                           |
|------|----------------|--------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                |                                      |                  | ation number or name that the purchaser supplied the merchant                                                                                         |
| N    | Line Item Date | 6-character numeric<br>YYMMDD format | line_item_date_l | The purchase date of the line item referenced in the associated Corporate Card Line Item Detail<br><br>Fixed length 6<br>Numeric, in<br>YYMMDD format |
| N    | Ship Date      | 6-character numeric<br>YYMMDD format | ship_date_l      | The date the merchandise was shipped to the destination<br><br>Fixed length 6<br>Numeric, in<br>YYMMDD format                                         |
| N    | Order Date     | 6-character numeric<br>YYMMDD format | order_date1_l    | The date the item was ordered<br><br>Fixed length 6-character numeric, in<br>YYMMDD format                                                            |
| Y    | Product Code   | 12-character alpha-numeric           | product_code1_l  | Line item Product Code<br><br>Contains the non-fuel related product code of the individual item purchased                                             |

| Req* | Value                | Limits                     | Variable            | Description                                                                                                                                                                |
|------|----------------------|----------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Y    | Item Description     | 35-character alpha-numeric | item_description_l  | Line Item description<br><br>Contains the description of the individual item purchased                                                                                     |
| Y    | Item Quantity        | 12-character alpha-numeric | item_quantity_l     | Quantity of line item<br><br>Up to 5 decimal places supported<br><br>Minimum amount is 0.0 and maximum is 9999999.99999                                                    |
| Y    | Unit Cost            | 12-character decimal       | unit_cost_l         | Line item cost per unit.<br><br>Must contain a minimum of 2 decimal places, up to 5 decimal places supported.<br><br>Minimum amount is 0.00001 and maximum is 999999.99999 |
| Y    | Item Unit Measure    | 12-character alpha-numeric | item_unit_measure_l | The line item unit of measurement code<br><br>ANSI X-12 EDI Allowable Units of Measure and Codes                                                                           |
| Y    | Extended Item Amount | 9-character decimal        | ext_item_amount_l   | Contains the individual item amount that is normally cal-                                                                                                                  |

| Req* | Value           | Limits                    | Variable          | Description                                                                                                                                                                                                                                                   |
|------|-----------------|---------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                 |                           |                   | <p>culated as price multiplied by quantity</p> <p>Must contain 2 decimal places</p> <p>Minimum amount is 0.00 and maximum is 999999.99</p>                                                                                                                    |
| N    | Discount Amount | 9-character decimal       | discount_amount_l | <p>Contains the item discount amount</p> <p>Must contain 2 decimal places</p> <p>Minimum amount is 0.00 and maximum is 999999.99</p>                                                                                                                          |
| N    | Commodity Code  | 15-character alphanumeric | commodity_code_l  | Code assigned to the merchant that best categorizes the item(s) being purchased                                                                                                                                                                               |
| C    | Tax             | Up to 6 arrays            | tax_l             | <p>Can have up to 6 arrays containing different tax details –see Tax Array Request Fields table below for each field description</p> <div> <p><b>NOTE:</b> If you use this variable, you must fill in all the fields of tax array mentioned below.</p> </div> |

### 5.3.8.3 Tax Array Object - MC Corpais

The tax array object is used when you use the Tax field of both MC Corpac and MC Corpal. If you use the tax array object, all of the array fields must be sent.

Setting the tax array differs slightly between the two objects.

#### Setting tax array for MC Corpac

```
//Tax Details

$tax_amount_c = array("1.19", "1.29");

$tax_rate_c = array("6.0", "7.0");

$tax_type_c = array("GST", "PST");

$tax_id_c = array("gst1298", "pst1298");

$tax_included_in_sales_c = array("Y", "N");

//Create and set Tax for McCorpac

$mcTax_c = new mcTax();

$mcTax_c->setTax($tax_amount_c[0], $tax_rate_c[0], $tax_type_c[0], $tax_id_c
[0], $tax_included_in_sales_c[0]);

$mcTax_c->setTax($tax_amount_c[1], $tax_rate_c[1], $tax_type_c[1], $tax_id_c
[1], $tax_included_in_sales_c[1]);
```

#### Setting tax array for MC Corpal

```
//Tax Details for Items

$tax_amount_l = array("0.52", "1.48");

$tax_rate_l = array("13.0", "13.0");

$tax_type_l = array("HST", "HST");

$tax_id_l = array("hst1298", "hst1298");

$tax_included_in_sales_l = array("Y", "Y");

//Create and set Tax for McCorpall

$mcTax_l = array(new mcTax(), new mcTax());

$mcTax_l[0]->setTax($tax_amount_l[0], $tax_rate_l[0], $tax_type_l[0], $tax_id_
l[0], $tax_included_in_sales_l[0]);

$mcTax_l[1]->setTax($tax_amount_l[1], $tax_rate_l[1], $tax_type_l[1], $tax_id_
l[1], $tax_included_in_sales_l[1]);
```



Table 1 MC Corpaix Tax Array Request Fields

| Req* | Value                           | Limits                     | Variable                                        | Description                                                                                                                                                                                                                                                |
|------|---------------------------------|----------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Y    | Tax Amount                      | 12-character decimal       | tax_amount_c/tax_amount_l                       | <p>Contains detail tax amount for purchase of goods or services</p> <p>Must be 2 decimal places. Minimum amount is 0.00 and maximum is 999999.99</p>                                                                                                       |
| Y    | Tax Rate                        | 5-character decimal        | tax_rate_c/tax_rate_l                           | <p>Contains the detailed tax rate applied in relationship to a specific tax amount</p> <div> <p><b>EXAMPLE:</b> 5% GST should be '5.0' or 9.975% QST should be '9.975'</p> </div> <p>May contain up to 3 decimals, minimum 0.001, maximum up to 9999.9</p> |
| Y    | Tax Type                        | 4-character alpha-numeric  | tax_type_c/tax_type_l                           | Contains tax type, such as GST,QST,PST,HST                                                                                                                                                                                                                 |
| Y    | Tax ID                          | 20-character alpha-numeric | tax_id_c/tax_id_l                               | Provides an identification number used by the card acceptor with the tax authority in relationship to a specific tax amount, such as GST/HST number                                                                                                        |
| Y    | Tax included in sales indicator | 1-character alpha-numeric  | tax_included_in_sales_c/tax_included_in_sales_l | This is the indicator used to reflect addi-                                                                                                                                                                                                                |

| Req* | Value | Limits | Variable | Description                                                                                                                                                            |
|------|-------|--------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |       |        |          | <p>tional tax capture and reporting</p> <p>Valid values are:</p> <p>Y = Tax included in total purchase amount</p> <p>N = Tax not included in total purchase amount</p> |

#### 5.3.8.4 Sample Code for MC Corpais

| Sample MC Corpais - Corporate Card Common Data with Line Item Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> &lt;?php require "../mpgClasses.php"; /***** Request Variables *****/ \$store_id='moneris'; \$api_token='hurgle'; //\$status = 'false'; /***** Transactional Variables *****/ \$type='mccorpais'; \$cust_id='CUST13343'; \$order_id='ord-200916-13:29:27'; \$txn_number='66011731632016264132927986-0_11'; \$customer_code1_c ="CustomerCode123"; \$card_acceptor_tax_id_c ="UrTaxId";//Merchant tax id which is mandatory \$corporation_vat_number_c ="cvt123"; \$freight_amount_c ="1.23"; \$duty_amount_c ="2.34"; \$ship_to_pos_code_c ="M1R 1W5"; \$order_date_c ="141211"; \$customer_vat_number_c ="customervn231"; \$unique_invoice_number_c ="uin567"; \$authorized_contact_name_c ="John Walker"; //Tax Details \$tax_amount_c = array("1.19", "1.29"); \$tax_rate_c = array("6.0", "7.0"); \$tax_type_c = array("GST", "PST"); \$tax_id_c = array("gst1298", "pst1298"); \$tax_included_in_sales_c = array("Y", "N"); //Item Details \$customer_code1_l = array("customer code", "customer code2"); \$line_item_date_l = array("150114", "150114"); \$ship_date_l = array("150120", "150122"); \$order_date1_l = array("150114", "150114"); \$medical_services_ship_to_health_industry_number_l = array(null, null); \$contract_number_l = array(null, null); \$medical_services_adjustment_l = array(null, null); \$medical_services_product_number_qualifier_l = array(null, null); \$product_code1_l = array("pc11", "pc12"); \$item_description_l = array("Good item", "Better item"); \$item_quantity_l = array("4", "5"); \$unit_cost_l = array("1.25", "10.00"); </pre> |

## Sample MC Corps - Corporate Card Common Data with Line Item Details

```

$item_unit_measure_l = array("EA", "EA");
$ext_item_amount_l = array("5.00", "50.00");
$discount_amount_l = array("1.00", "50.00");
$commodity_code_l = array("cCode11", "cCode12");
$type_of_supply_l = array(null, null);
$vat_ref_num_l = array(null, null);
//Tax Details for Items
$tax_amount_l = array("0.52", "1.48");
$tax_rate_l = array("13.0", "13.0");
$tax_type_l = array("HST", "HST");
$tax_id_l = array("hst1298", "hst1298");
$tax_included_in_sales_l = array("Y", "Y");
//Create and set Tax for McCorpac
$mcTax_c = new mcTax();
$mcTax_c->setTax($tax_amount_c[0], $tax_rate_c[0], $tax_type_c[0], $tax_id_c[0], $tax_
included_in_sales_c[0]);
$mcTax_c->setTax($tax_amount_c[1], $tax_rate_c[1], $tax_type_c[1], $tax_id_c[1], $tax_
included_in_sales_c[1]);
//Create and set McCorpac for common data - only set values that you know
$mcCorpac = new mcCorpac();
$mcCorpac->setCustomerCode($customer_code_c);
$mcCorpac->setCardAcceptorTaxTd($card_acceptor_tax_id_c);
$mcCorpac->setCorporationVatNumber($corporation_vat_number_c);
$mcCorpac->setFreightAmount1($freight_amount_c);
$mcCorpac->setDutyAmount1($duty_amount_c);
$mcCorpac->setShipToPosCode($ship_to_pos_code_c);
$mcCorpac->setOrderDate($order_date_c);
$mcCorpac->setCustomerVatNumber($customer_vat_number_c);
$mcCorpac->setUniqueInvoiceNumber($unique_invoice_number_c);
$mcCorpac->setAuthorizedContactName($authorized_contact_name_c);
$mcCorpac->setTax($mcTax_c);
//Create and set Tax for McCorpall
$mcTax_l = array(new mcTax(), new mcTax());
$mcTax_l[0]->setTax($tax_amount_l[0], $tax_rate_l[0], $tax_type_l[0], $tax_id_l[0], $tax_
included_in_sales_l[0]);
$mcTax_l[1]->setTax($tax_amount_l[1], $tax_rate_l[1], $tax_type_l[1], $tax_id_l[1], $tax_
included_in_sales_l[1]);
//Create and set McCorpall for each item
$mcCorpall = new mcCorpall();
$mcCorpall->setMcCorpall($customer_code_l[0], $line_item_date_l[0], $ship_date_l[0], $order_
date_l[0], $medical_services_ship_to_health_industry_number_l[0], $contract_number_l[0],
$medical_services_adjustment_l[0], $medical_services_product_number_qualifier_l[0],
$product_code_l[0], $item_description_l[0], $item_quantity_l[0],
$unit_cost_l[0], $item_unit_measure_l[0], $ext_item_amount_l[0], $discount_amount_l[0],
$commodity_code_l[0], $type_of_supply_l[0], $vat_ref_num_l[0], $mcTax_l[0]);
$mcCorpall->setMcCorpall($customer_code_l[1], $line_item_date_l[1], $ship_date_l[1], $order_
date_l[1], $medical_services_ship_to_health_industry_number_l[1], $contract_number_l[1],
$medical_services_adjustment_l[1], $medical_services_product_number_qualifier_l[1],
$product_code_l[1], $item_description_l[1], $item_quantity_l[1],
$unit_cost_l[1], $item_unit_measure_l[1], $ext_item_amount_l[1], $discount_amount_l[1],
$commodity_code_l[1], $type_of_supply_l[1], $vat_ref_num_l[1], $mcTax_l[1]);
//Create and set McLevel23
$mpgMcLevel23 = new mpgMcLevel23();
$mpgMcLevel23->setMcCorpac($mcCorpac);
$mpgMcLevel23->setMcCorpall($mcCorpall);
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'txn_number'=>$txn_number,

```

### Sample MC Corpais - Corporate Card Common Data with Line Item Details

```
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
$mpgTxn->setLevel23Data($mpgMcLevel23);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
// Status check example
// $mpgHttpPost = new mpgHttpPostStatus($store_id, $api_token, $status, $mpgRequest);
/***** Response *****/
$mpgResponse = $mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
// print("\nStatusCode = " . $mpgResponse->getStatusCode());
// print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>
```

## 5.4 Level 2/3 American Express Transactions

- 1 American Express Level 2/3 Standard Transactions
- 1 American Express L23 Air and Rail Transactions

### 5.4.1 Level 2/3 Transaction Types for Amex

This transaction set includes a suite of corporate card financial transactions as well as a transaction that allows for the passing of Level 2/3 data. Please ensure American Express Level 2/3 processing support is enabled on your merchant account. Batch Close, Open Totals and Pre-authorization are identical to the transactions outlined in the section Basic Transaction Set (page 22).

- When the Pre-authorization response contains CorporateCard equal to true then you can submit the AX transactions.

- If CorporateCard is false then the card does not support Level 2/3 data and non Level 2/3 transaction are to be used. If the card is not a corporate card, please refer to 2 Basic Transaction Set for the appropriate non-corporate card transactions.

**NOTE:** This transaction set is intended for transactions where Corporate Card is true and Level 2/3 data will be submitted. If the credit card is found to be a corporate card but you do not wish to send any Level 2/3 data then you may submit AX transactions using the transaction set outlined in the section Basic Transaction Set (page 22).

**Pre-authorization – (authorization)**

The preauth verifies and locks funds on the customer's credit card. The funds are locked for a specified amount of time, based on the card issuer. To retrieve the funds from a pre-auth so that they may be settled in the merchant account a capture must be performed. CorporateCard will return as true if the card supports Level 2/3.

**AX Completion – (Capture/Pre-authorization Completion)**

Once a Pre-authorization is obtained the funds that are locked need to be retrieved from the customer's credit card. The capture retrieves the locked funds and readies them for settlement in to the merchant account. Prior to performing an AXCompletion a Preauth must be performed.

**AX Force Post – (Force Capture/Pre-authorization Completion)**

This transaction is an alternative to AX Completion to obtain the funds locked on a Pre-authorization obtained from IVR or equivalent terminal. The capture retrieves the locked funds and readies them for settlement in to the merchant account.

**AX Purchase Correction – (Void, Correction)**

AX Completion and AX Force Post can be voided the same day\* that they occur. A void must be for the full amount of the transaction and will remove any record of it from the cardholder statement. \* An AX Purchase Correction can be performed against a transaction as long as the batch that contains the original transaction remains open. When using the automated closing feature, the batch close occurs daily between 10 – 11 pm EST.

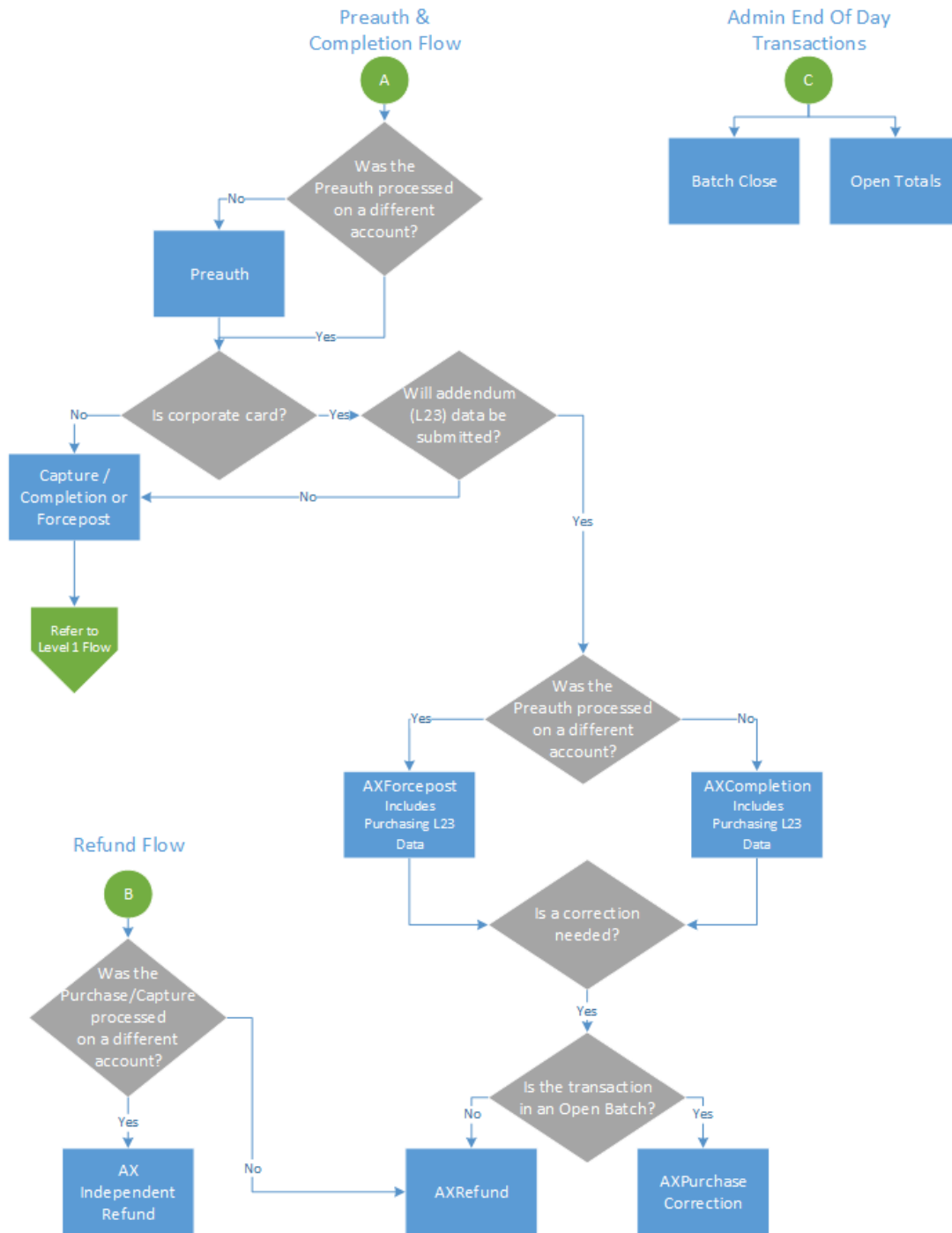
**AX Refund – (Credit)**

An AX Refund can be performed against an AX Completion and AX Force Post to refund any part, or all of the transaction.

**AX Independent Refund – (Credit)**

An AX Independent Refund can be performed against a purchase or a capture to refund any part, or all of the transaction. Independent refund is used when the originating transaction was not performed through Moneris Gateway. Please note, the Independent Refund transaction may or may not be supported on your account. If you receive a transaction not allowed error when attempting an independent refund, it may mean the transaction is not supported on your account. If you wish to have the AX Independent Refund transaction type temporarily enabled (or re-enabled), please contact the Service Centre at 1-866-319-7450.

## 5.4.2 Level 2/3 Transaction Flow for Amex



### 5.4.3 Level 2/3 Data Objects in Amex

- 5.4.3.1 About the Level 2/3 Data Objects for Amex
- 5.4.3.2 Definition of the AxLevel23 Object
- Table 1 Object
- Table 2 Object
- Table 3 Object

#### 5.4.3.1 About the Level 2/3 Data Objects for Amex

Many of the Level 2/3 transaction requests using American Express also include a mandatory data object called AxLevel23. AxLevel23 is also comprised of other objects, also described in this section.

The Level 2/3 data objects within this section apply to all of the following transactions and are passed as part of the transaction request for:

- AX Completion
- AX Force Post
- AX Refund
- AX Independent Refund

#### Things to Consider:

- Please ensure the addendum data below is complete and accurate.
- Please ensure the math on quantities calculations, amounts, discounts, taxes, etc. properly adds up to the overall transaction amount. Incorrect amounts will cause the transaction to be rejected.

#### 5.4.3.2 Definition of the AxLevel23 Object

##### AxLevel23 object definition

```
$mpgAxLevel23 = new mpgAxLevel23();
```

The AxLevel23 object itself has three objects, Table1, Table2 and Table3, all of which are mandatory.



**Table 1 AxLevel23 Object**

| Variable Name | Type and Limits | Description                                                | Set Method                                                                        |
|---------------|-----------------|------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Table1        | Object          | Refer below for further breakdown and definition of table1 | <code>\$mpgAxLevel23-&gt;setTable1(\$big04, \$big05, \$big10, \$axN1Loop);</code> |
| Table2        | Object          | Refer below for further breakdown and definition of table2 | <code>\$mpgAxLevel23-&gt;setTable2(\$axItLoop);</code>                            |
| Table3        | Object          | Refer below for further breakdown and definition of table3 | <code>\$mpgAxLevel23-&gt;setTable3(\$taxTbl3);</code>                             |

**Table 1 Object**

Table 1 contains the addendum data heading information. Contains information such as identification elements that uniquely identify an invoice (transaction), the customer name and shipping address.

**Table 1 object definition**

```
$mpgAxLevel23->setTable1($big04, $big05, $big10, $axN1Loop);
```

**Table 1 AxLevel23 object - Table 1 object fields**

| Req* | Value                 | Limits                     | Set Method       | Description                                                                                                                                                                                                                                                                                                                                |
|------|-----------------------|----------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C    | Purchase Order Number | 22-character alpha-numeric | 'big04'=>\$big04 | <p>The cardholder supplied Purchase Order Number, which is entered by the merchant at the point-of-sale</p> <p>This entry is used in the Statement/Reporting process and may include accounting information specific to the client</p> <div> <b>NOTE:</b> This element is mandatory, if the merchant's customer provides a Purchase </div> |

| Req* | Value          | Limits                     | Set Method         | Description                                                                                                              |
|------|----------------|----------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------|
|      |                |                            |                    | Order Number.                                                                                                            |
| N    | Release Number | 30-character alpha-numeric | 'big05'=>\$big05   | A number that identifies a release against a Purchase Order previously placed by the parties involved in the transaction |
| N    | Invoice Number | 8-character alpha-numeric  | 'big10'=>\$big10   | Contains the Amex invoice/reference number                                                                               |
| N    | N1Loop         | Object                     | 'n1Loop'=>\$n1Loop | Refer below for further breakdown and definition of N1Loop object                                                        |

\*Y = Required, N = Optional, C = Conditional

Table 1 also has its own objects:

- N1Loop object
- AxRef object

Table 1 - Setting the N1Loop Object

The N1Loop data set contains the Requester names. It can also optionally contain the buying group, ship from, ship to and receiver details.

A minimum of at least 1 n1Loop must be set. Up to 5 n1Loop can be set.

### N1Loop object definition

```
$axN1Loop = new axN1Loop();
```

```
$axN1Loop->setN1Loop($n101, $n102, $n301, $n401, $n402, $n403, $axRef1);
```

Table 1 AxLevel23 object - Table 1 object - N1Loop object fields

| Req* | Value                  | Limits                    | Variable or Set Method | Description                                    |
|------|------------------------|---------------------------|------------------------|------------------------------------------------|
| Y    | Entity Identifier Code | 2-character alpha-numeric | n101                   | Supported values:<br>R6 - Requester (required) |

| Req*      | Value             | Limits                     | Variable or Set Method               | Description                                                                                                                                                                                                                                                                            |           |              |    |                |    |                   |    |                |    |              |    |               |
|-----------|-------------------|----------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------|----|----------------|----|-------------------|----|----------------|----|--------------|----|---------------|
|           |                   |                            |                                      | BG - Buying Group (optional)<br><br>SF - Ship From (optional)<br><br>ST - Ship To (optional)<br><br>40 - Receiver (optional)                                                                                                                                                           |           |              |    |                |    |                   |    |                |    |              |    |               |
| Y         | Name              | 40-character alpha-numeric | n102                                 | <table><tr><th>n101 code</th><th>n102 meaning</th></tr><tr><td>R6</td><td>Requester Name</td></tr><tr><td>BG</td><td>Buying Group Name</td></tr><tr><td>SF</td><td>Ship From Name</td></tr><tr><td>ST</td><td>Ship To Name</td></tr><tr><td>40</td><td>Receiver Name</td></tr></table> | n101 code | n102 meaning | R6 | Requester Name | BG | Buying Group Name | SF | Ship From Name | ST | Ship To Name | 40 | Receiver Name |
| n101 code | n102 meaning      |                            |                                      |                                                                                                                                                                                                                                                                                        |           |              |    |                |    |                   |    |                |    |              |    |               |
| R6        | Requester Name    |                            |                                      |                                                                                                                                                                                                                                                                                        |           |              |    |                |    |                   |    |                |    |              |    |               |
| BG        | Buying Group Name |                            |                                      |                                                                                                                                                                                                                                                                                        |           |              |    |                |    |                   |    |                |    |              |    |               |
| SF        | Ship From Name    |                            |                                      |                                                                                                                                                                                                                                                                                        |           |              |    |                |    |                   |    |                |    |              |    |               |
| ST        | Ship To Name      |                            |                                      |                                                                                                                                                                                                                                                                                        |           |              |    |                |    |                   |    |                |    |              |    |               |
| 40        | Receiver Name     |                            |                                      |                                                                                                                                                                                                                                                                                        |           |              |    |                |    |                   |    |                |    |              |    |               |
| N         | Address           | 40-character alpha-numeric | n301                                 | Address                                                                                                                                                                                                                                                                                |           |              |    |                |    |                   |    |                |    |              |    |               |
| N         | City              | 30-character alpha-numeric | n401                                 | City                                                                                                                                                                                                                                                                                   |           |              |    |                |    |                   |    |                |    |              |    |               |
| N         | State or Province | 2-character alpha-numeric  | n402                                 | State or province                                                                                                                                                                                                                                                                      |           |              |    |                |    |                   |    |                |    |              |    |               |
| N         | Postal Code       | 15-character alpha-numeric | n403                                 | Postal Code                                                                                                                                                                                                                                                                            |           |              |    |                |    |                   |    |                |    |              |    |               |
| N         | AxRef             | Object                     | <code>\$axRef1 = new axRef();</code> | <p>Refer below for further breakdown and definition of AxRef object.</p> <p>This object contains the customer postal code (mandatory) and customer reference number (optional)</p> <p>A minimum of 1 axRef1 must be set; maximum of 2 axRef1's may be set</p>                          |           |              |    |                |    |                   |    |                |    |              |    |               |

\*Y = Required, N = Optional, C = Conditional

Table 1 - Setting the AxRef Object

**Setting AXRef object**

```

$axRef1 = new axRef();

$ref01 = array("4C", "CR"); //Reference ID Qualifier

$ref02 = array("M5T3A5", "16802309004"); //Reference ID

$axRef1->setRef($ref01[0], $ref02[0]);

$axRef1->setRef($ref01[1], $ref02[1]);

```

**Table 1 AxLevel23 object - Table 1 object - AxRef object fields**

| Req*       | Value                                                                                                                   | Limits                    | Variable | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |                  |    |                                                                                                                         |    |     |    |     |    |     |    |     |
|------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|----|-------------------------------------------------------------------------------------------------------------------------|----|-----|----|-----|----|-----|----|-----|
| Y          | Reference Identification Qualifier                                                                                      | 2-character alphanumeric  | ref01    | <div>This element may contain the following qualifiers for the corresponding occurrences of the N1Loop:</div> <table><tr><th>n101 value</th><th>ref01 denotation</th></tr><tr><td>R6</td><td>Supported values:<br/><br/>4C - Shipment Destination Code (mandatory)<br/><br/>CR - Customer Reference Number (conditional)</td></tr><tr><td>BG</td><td>n/a</td></tr><tr><td>SF</td><td>n/a</td></tr><tr><td>ST</td><td>n/a</td></tr><tr><td>40</td><td>n/a</td></tr></table> | n101 value | ref01 denotation | R6 | Supported values:<br><br>4C - Shipment Destination Code (mandatory)<br><br>CR - Customer Reference Number (conditional) | BG | n/a | SF | n/a | ST | n/a | 40 | n/a |
| n101 value | ref01 denotation                                                                                                        |                           |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                  |    |                                                                                                                         |    |     |    |     |    |     |    |     |
| R6         | Supported values:<br><br>4C - Shipment Destination Code (mandatory)<br><br>CR - Customer Reference Number (conditional) |                           |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                  |    |                                                                                                                         |    |     |    |     |    |     |    |     |
| BG         | n/a                                                                                                                     |                           |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                  |    |                                                                                                                         |    |     |    |     |    |     |    |     |
| SF         | n/a                                                                                                                     |                           |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                  |    |                                                                                                                         |    |     |    |     |    |     |    |     |
| ST         | n/a                                                                                                                     |                           |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                  |    |                                                                                                                         |    |     |    |     |    |     |    |     |
| 40         | n/a                                                                                                                     |                           |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                  |    |                                                                                                                         |    |     |    |     |    |     |    |     |
| Y          | Reference Identification                                                                                                | 15-character alphanumeric | ref02    | This field must be populated for each ref01 provided                                                                                                                                                                                                                                                                                                                                                                                                                       |            |                  |    |                                                                                                                         |    |     |    |     |    |     |    |     |

| Req* | Value | Limits | Variable | Description           |                                                                                                                                                                                                                                                                                                                                           |
|------|-------|--------|----------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |       |        |          | <b>ref01 value</b>    | <b>ref02 denotation</b>                                                                                                                                                                                                                                                                                                                   |
|      |       |        |          | 4C (n101 value = R6)  | This element must contain the Amex Ship-to Postal Code of the destination where the commodity was shipped. If the Ship-to Postal Code is unavailable, the postal code of the merchant location where the transaction took place may be substituted.                                                                                       |
|      |       |        |          | CR (n101 value = R6): | <p>This element must contain the Amex Card member Reference Number (e.g., purchase order, cost center, project number, etc.) that corresponds to this transaction, if provided by the Cardholder.</p> <p>This information may be displayed in the statement/reporting process and may include client-specific accounting information.</p> |

\*Y = Required, N = Optional, C = Conditional

### Table 2 Object

Table 2 includes the transaction's addendum detail. It contains transaction data including reference codes, debit or credit and tax amounts, line item detail descriptions, shipping information and much more. All transaction data in an invoice relate to a single transaction and cardholder account number.

### Table 2 object definition

```
$mpgAxLevel123->setTable2 ($axItLoop) ;
```

**Table 1 AxLevel23 object - Table 2 object fields**

| Req* | Value   | Limits | Set Method               | Description                                                          |
|------|---------|--------|--------------------------|----------------------------------------------------------------------|
| N    | It1loop | Object | 'axIt1Loop'=>\$axIt1Loop | Refer below for further break-down and definition of object details. |

\*Y = Required, N = Optional, C = Conditional

**Table 2 - Setting the AxIt1Loop Object**

The AxIt1Loop data defines the baseline item data for the invoice. This data is defined for each item/service purchased and included within this invoice. This data set contains basic transaction data, including quantity, unit of measure, unit price and goods/services reference information.

- A minimum of 1 it1Loop required
- A maximum of 999 it1Loop's supported

### **AxIt1Loop object definition**

```
$axItLoop = new axIt1Loop();
```

```
$axItLoop->setIt1Loop($it102[0], $it103[0], $it104[0], $it105[0], $it106s[0],  
$txi[0], $pam05[0], $pid05[0]);
```

```
$axItLoop->setIt1Loop($it102[1], $it103[1], $it104[1], $it105[1], $it106s[1],  
$txi[1], $pam05[1], $pid05[1]);
```

**Table 1 AxLevel23 object - Table 2 object - AxIt1Loop object fields**

| Req* | Value                       | Limits               | Variable | Description                                                                                                          |
|------|-----------------------------|----------------------|----------|----------------------------------------------------------------------------------------------------------------------|
| Y    | Line Item Quantity Invoiced | 10-character decimal | it102    | Quantity of line item<br><br>Up to 2 decimal places supported<br><br>Minimum amount is 0.0 and maximum is 9999999999 |

| Req* | Value                              | Limits                    | Variable | Description                                                                                                                                                                                                                                                                                                                           |
|------|------------------------------------|---------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Y    | Unit or Basis for Measurement Code | 2-character alpha-numeric | it103    | <p>The line item unit of measurement code</p> <p>Must contain a code that specifies the units in which the value is expressed or the manner in which a measurement is taken</p> <div> <b>EXAMPLE:</b> EA = each, E5=inches </div> <p>See <a href="#">ANSI X-12 EDI Allowable Units of Measure and Codes</a> for the list of codes</p> |
| Y    | Unit Price                         | 15-character decimal      | it104    | <p>Line item cost per unit</p> <p>Must contain 2 decimal places</p> <p>Minimum amount is 0.00 and maximum is 999999.99</p>                                                                                                                                                                                                            |
| N    | Basis or Unit Price Code           | 2-character alpha-numeric | it105    | <p>Code identifying the type of unit price for an item</p> <div> <b>EXAMPLE:</b> DR = dealer, AP = advise price </div> <p>See <a href="#">ASC X12 004010 Element</a></p>                                                                                                                                                              |

| Req* | Value                     | Limits              | Variable | Description                                                                                                                                                                                                                                                                                                                                  |
|------|---------------------------|---------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                           |                     |          | <a href="#">639</a> for list of codes                                                                                                                                                                                                                                                                                                        |
| N    | Axlt106s                  | object              | it106s   | Refer below for further break-down and definition of object details.                                                                                                                                                                                                                                                                         |
| N    | AxTxi                     | object              | txi      | <p>Refer below for further break-down and definition of object details</p> <p>A maximum of 12 AxTxi (tax information data sets) may be defined</p> <div> <p><b>NOTE:</b> that if line item level tax information is populated in AxTxi in Table2, then tax totals for the entire invoice (transaction) must be entered in Table3.</p> </div> |
| Y    | Line Item Extended Amount | 8-character decimal | pam05    | <p>Contains the individual item amount that is normally calculated as price multiplied by quantity</p> <p>Must contain 2 decimal places</p> <p>Minimum amount is 0.00</p>                                                                                                                                                                    |



| Req* | Value                 | Limits                    | Variable | Description                                                                                                                                                  |
|------|-----------------------|---------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                       |                           |          | and maximum is 99999.99                                                                                                                                      |
| Y    | Line Item Description | 80-character alphanumeric | pid05    | <p>Line Item description</p> <p>Contains the description of the individual item purchased</p> <p>This field pertain to each line item in the transaction</p> |

\*Y = Required, N = Optional, C = Conditional

Table 2 - Setting the AxIt106s Object

```

$it10618 = array("MG", "MG", "MG", "MG", "MG"); //Product/Service ID qualifier

$it10719 = array("DJFR4", "JFJ49", "FEF33", "FEE43", "DISCOUNT");
//Product/Service ID (corresponds to it10618)

$it106s = array();

$it106s[0] = new axIt106s($it10618[0], $it10719[0]);
$it106s[1] = new axIt106s($it10618[1], $it10719[1]);
$it106s[2] = new axIt106s($it10618[2], $it10719[2]);
$it106s[3] = new axIt106s($it10618[3], $it10719[3]);
$it106s[4] = new axIt106s($it10618[4], $it10719[4]);

```

Table 1 AxLevel23 object - Table 2 object - AxIt106s object fields

| Req* | Value                        | Limits                   | Set Method           | Description                                                                                                                    |
|------|------------------------------|--------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| N    | Product/Service ID Qualifier | 2-character alphanumeric | 'it10618'=>\$it10618 | <p>Supported values:</p> <p>MG - Manufacturer's Part Number</p> <p>VC - Supplier Catalog Number</p> <p>SK - Supplier Stock</p> |

| Req*    | Value                     | Limits                                                                                                                                                                                                                                    | Set Method | Description                                                                                                                                                   |    |                           |    |                           |       |                           |                      |                                                                                                                                                    |
|---------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---------------------------|----|---------------------------|-------|---------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
|         |                           |                                                                                                                                                                                                                                           |            | Keeping Unit Number<br><br>UP - Universal Product Code<br><br>VP – Vendor Part Number<br><br>PO – Purchase Order Number<br><br>AN – Client Defined Asset Code |    |                           |    |                           |       |                           |                      |                                                                                                                                                    |
| N       | Product/Service ID        | <table><tr><th>it10618</th><th>it10719 - size/type</th></tr><tr><td>VC</td><td>20-character alphanumeric</td></tr><tr><td>PO</td><td>22-character alphanumeric</td></tr><tr><td>Other</td><td>30-character alphanumeric</td></tr></table> | it10618    | it10719 - size/type                                                                                                                                           | VC | 20-character alphanumeric | PO | 22-character alphanumeric | Other | 30-character alphanumeric | 'it10719'=>\$it10719 | Product/Service ID corresponds to the preceding qualifier defined by it10618<br><br>The maximum length depends on the qualifier defined in it10618 |
| it10618 | it10719 - size/type       |                                                                                                                                                                                                                                           |            |                                                                                                                                                               |    |                           |    |                           |       |                           |                      |                                                                                                                                                    |
| VC      | 20-character alphanumeric |                                                                                                                                                                                                                                           |            |                                                                                                                                                               |    |                           |    |                           |       |                           |                      |                                                                                                                                                    |
| PO      | 22-character alphanumeric |                                                                                                                                                                                                                                           |            |                                                                                                                                                               |    |                           |    |                           |       |                           |                      |                                                                                                                                                    |
| Other   | 30-character alphanumeric |                                                                                                                                                                                                                                           |            |                                                                                                                                                               |    |                           |    |                           |       |                           |                      |                                                                                                                                                    |

\*Y = Required, N = Optional, C = Conditional

Table 2 - Setting the AxTxi Object

### Table 2 AxTxi object definition

```

$txi01_GST = array("GS", "GS", "GS", "GS", "GS"); //Tax type code
$txi02_GST = array("0.70", "1.75", "1.00", "0.80","0.00"); //Monetary amount
$txi03_GST = array("5.0", "5.0", "5.0", "5.0","5.0"); //Percent
$txi06_GST = array("", "", "", "", ""); //Tax exempt code
$txi01_PST = array("PG", "PG", "PG","PG","PG"); //Tax type code
$txi02_PST = array("0.80", "2.00", "1.00", "0.80","0.00"); //Monetary amount
$txi03_PST = array("7.0", "7.0", "7.0", "7.0","7.0"); //Percent
$txi06_PST = array("", "", "", "", ""); //Tax exempt code
$txi = array(new axTxi(), new axTxi(), new axTxi(), new axTxi(), new axTxi());
$txi[0]->setTxi($txi01_GST[0], $txi02_GST[0], $txi03_GST[0], $txi06_GST[0]);

```

```

$txi[0]->setTxi($txi01_PST[0], $txi02_PST[0], $txi03_PST[0], $txi06_PST[0]);
$txi[1]->setTxi($txi01_GST[1], $txi02_GST[1], $txi03_GST[1], $txi06_GST[1]);
$txi[1]->setTxi($txi01_PST[1], $txi02_PST[1], $txi03_PST[1], $txi06_PST[1]);
$txi[2]->setTxi($txi01_GST[2], $txi02_GST[2], $txi03_GST[2], $txi06_GST[2]);
$txi[2]->setTxi($txi01_PST[2], $txi02_PST[2], $txi03_PST[2], $txi06_PST[2]);
$txi[3]->setTxi($txi01_GST[3], $txi02_GST[3], $txi03_GST[3], $txi06_GST[3]);
$txi[3]->setTxi($txi01_PST[3], $txi02_PST[3], $txi03_PST[3], $txi06_PST[3]);
$txi[4]->setTxi($txi01_GST[4], $txi02_GST[4], $txi03_GST[4], $txi06_GST[4]);
$txi[4]->setTxi($txi01_PST[4], $txi02_PST[4], $txi03_PST[4], $txi06_PST[4]);

```

**Table 1 AxLevel23 object - Table 2 object - AxiTxi object fields**

| Req* | Value         | Limits | Variable                  | Description                                                                                                                                                                                                                                                                                                                                                                                               |
|------|---------------|--------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C    | Tax Type code | txi01  | 2-character alpha-numeric | <p>Tax type code applicable to Canada and US only</p> <p>For Canada, this field must contain a code that specifies the type of tax</p> <p>If txi01 is used, then txi02, txi03 or txi06 must be populated</p> <p>Valid codes include the following:</p> <p>CT – County/Tax (optional)</p> <p>CA – City Tax (optional)</p> <p>EV – Environmental Tax (optional)</p> <p>GS – Good and Services Tax (GST)</p> |

| Req* | Value           | Limits | Variable            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------|-----------------|--------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                 |        |                     | <p>(optional)</p> <p>LS – State and Local Sales Tax (optional)</p> <p>LT – Local Sales Tax (optional)</p> <p>PG – Provincial Sales Tax (PST) (optional)</p> <p>SP – State/Provincial Tax a.k.a. Quebec Sales Tax (QST) (optional)</p> <p>ST – State Sales Tax (optional)</p> <p>TX – All Taxes (required)</p> <p>VA – Value-Added Tax a.k.a. Canadian Harmonized Sales Tax (HST) (optional)</p>                                                                            |
| C    | Monetary Amount | txi02  | 6-character decimal | <p>This element may contain the monetary tax amount that corresponds to the Tax Type Code in txi01</p> <div data-bbox="1198 1339 1409 1743"> <p><b>NOTE:</b></p> <p>If txi02 is used in mandatory occurrence txi01=TX, txi02 must contain the total tax amount applicable to the entire invoice (transaction)</p> <p>If taxes are not applicable for the entire invoice (transaction), txi02 must be 0.00.</p> </div> <p>The maximum value that can be entered in this</p> |

| Req* | Value           | Limits | Variable                  | Description                                                                                                                                                                                                                                                                                                               |
|------|-----------------|--------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                 |        |                           | <p>field is "9999.99", which is \$9,999.99 (CAD)</p> <p>A debit is entered as: 9999.99</p> <p>A credit is entered as: – 9999.99</p>                                                                                                                                                                                       |
| C    | Percent         | txi03  | 10-character decimal      | <p>Contains the tax percentage (in decimal format) that corresponds to the tax type code defined in txi01</p> <p>Up to 2 decimal places supported</p>                                                                                                                                                                     |
| C    | Tax Exempt Code | txi06  | 1-character alpha-numeric | <p>This element may contain the Tax Exempt Code that identifies the exemption status from sales and tax that corresponds to the Tax Type Code in txi01</p> <p>Supported values:</p> <p>1 – Yes (Tax Exempt)</p> <p>2 – No (Not Tax Exempt)</p> <p>4 – Not Exempt/For Resale</p> <p>A – Labor Taxable, Material Exempt</p> |

| Req* | Value | Limits | Variable | Description                                                                                                                                                                                                                     |
|------|-------|--------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |       |        |          | B – Material Taxable, Labor Exempt<br><br>C – Not Taxable<br><br>F – Exempt (Goods / Services Tax)<br><br>G – Exempt (Provincial Sales Tax)<br><br>L – Exempt Local Service<br><br>R – Recurring Exempt<br><br>U – Usage Exempt |

\*Y = Required, N = Optional, C = Conditional

### Table 3 Object

Table 3 includes the transaction addendum summary. It contains the total invoice (transaction) amount, sales tax, freight and/or handling charges and invoice summary information, including total line items, number of segments in the invoice, and the transaction set control number (a.k.a., batch number).

### Table 3 object definition

```
$mpgAxLevel23->setTable3($taxTbl3);
```

**Table 1 AxLevel23 object - Table 3 object fields**

| Req* | Value | Limits | Set Method           | Description                                                                                                                                                                                                                                                       |
|------|-------|--------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C    | AxTxi | Object | 'taxTbl3'=>\$taxTbl3 | Refer below for further breakdown and definition of object details.<br><br><div style="border: 1px solid black; padding: 5px; margin-top: 10px;"> <b>NOTE:</b> if line item level tax information is populated in AxTxi in Table2, then tax totals         </div> |

| Req* | Value | Limits | Set Method | Description                                                                                                   |
|------|-------|--------|------------|---------------------------------------------------------------------------------------------------------------|
|      |       |        |            | for the entire invoice (transaction) must be entered in Table3. A maximum of 10 AxTxi's may be set in Table3. |

\*Y = Required, N = Optional, C = Conditional

**Table 3 - Setting the AxTxi Object**

The mandatory tax information data set must contain the total tax amount applicable to the entire invoice (transaction) which includes all line items identified in Table2. If taxes are not applicable for the entire invoice (transaction), then txi02 must be set to 0.00.

Tax totals must be entered in this mandatory tax information segment in Table 3, even if line item detail level tax data is reported in Table 2.

At least one occurrence of txi02, txi03 or txi06 is required.

### Table 3 AxTxi object definition

```
$taxTbl3 = new axTxi();

$taxTbl3->setTxi("GS", "4.25","5.0",""); //sum of GST taxes
$taxTbl3->setTxi("PG", "4.60","7.0",""); //sum of PST taxes
$taxTbl3->setTxi("TX", "8.85","13.0",""); //sum of all taxes

$mpgAxLevel23->setTable3($taxTbl3);
```

**Table 1 AxLevel23 object - Table 3 object - AxTxi object fields**

| Req* | Value         | Limits | Variable                  | Description                                                                                                                     |
|------|---------------|--------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| C    | Tax Type code | txi01  | 2-character alpha-numeric | Tax type code applicable to Canada and US only<br><br>For Canada, this field must contain a code that specifies the type of tax |

| Req* | Value           | Limits | Variable            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------|-----------------|--------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                 |        |                     | <p>If txi01 is used, then txi02, txi03 or txi06 must be populated</p> <p>Valid codes include the following:</p> <p>CT – County/Tax (optional)</p> <p>CA – City Tax (optional)</p> <p>EV – Environmental Tax (optional)</p> <p>GS – Good and Services Tax (GST) (optional)</p> <p>LS – State and Local Sales Tax (optional)</p> <p>LT – Local Sales Tax (optional)</p> <p>PG – Provincial Sales Tax (PST) (optional)</p> <p>SP – State/Provincial Tax a.k.a. Quebec Sales Tax (QST) (optional)</p> <p>ST – State Sales Tax (optional)</p> <p>TX – All Taxes (required)</p> <p>VA – Value-Added Tax a.k.a. Canadian Harmonized Sales Tax (HST) (optional)</p> |
| C    | Monetary Amount | txi02  | 6-character decimal | This element may contain the monetary tax amount that corresponds                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |



| Req* | Value           | Limits | Variable                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|-----------------|--------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                 |        |                           | <p>to the Tax Type Code in txi01</p> <div> <p><b>NOTE:</b><br/>If txi02 is used in mandatory occurrence txi01=TX, txi02 must contain the total tax amount applicable to the entire invoice (transaction)<br/>If taxes are not applicable for the entire invoice (transaction), txi02 must be 0.00.</p> </div> <p>The maximum value that can be entered in this field is "9999.99", which is \$9,999.99 (CAD)</p> <p>A debit is entered as: 9999.99</p> <p>A credit is entered as: – 9999.99</p> |
| C    | Percent         | txi03  | 10-character decimal      | <p>Contains the tax percentage (in decimal format) that corresponds to the tax type code defined in txi01</p> <p>Up to 2 decimal places supported</p>                                                                                                                                                                                                                                                                                                                                           |
| C    | Tax Exempt Code | txi06  | 1-character alpha-numeric | <p>This element may contain the Tax Exempt Code that identifies the</p>                                                                                                                                                                                                                                                                                                                                                                                                                         |

| Req* | Value | Limits | Variable | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|-------|--------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |       |        |          | <p>exemption status from sales and tax that corresponds to the Tax Type Code in txi01</p> <p>Supported values:</p> <p>1 – Yes (Tax Exempt)</p> <p>2 – No (Not Tax Exempt)</p> <p>4 – Not Exempt/For Resale</p> <p>A – Labor Taxable, Material Exempt</p> <p>B – Material Taxable, Labor Exempt</p> <p>C – Not Taxable</p> <p>F – Exempt (Goods / Services Tax)</p> <p>G – Exempt (Provincial Sales Tax)</p> <p>L – Exempt Local Service</p> <p>R – Recurring Exempt</p> <p>U – Usage Exempt</p> |

\*Y = Required, N = Optional, C = Conditional

#### 5.4.4 AX Completion

The AX Completion transaction is used to secure the funds locked by a pre-authorization transaction. When sending a capture request you will need two pieces of information from the original pre-authorization – the Order ID and the transaction number from the returned response.

**AX Completion transaction object definition**

```
$txnArray = array('type'=>'axcompletion', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

**HttpPostRequest object for AX Completion**

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

**AX Completion transaction request fields – Required**

| Variable Name        | Type and Limits                                                                                                                                                     | Set Method                                |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Order ID             | String<br>50-character alphanumeric                                                                                                                                 | 'order_id'=>\$order_id                    |
| Completion amount    | String<br>10-character decimal<br><br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><br><div>EXAMPLE: 1234567.89</div> | 'comp_amount'=>\$comp_amount              |
| Transaction number   | String<br>255-character alphanumeric                                                                                                                                | 'txn_number'=>\$txn_number                |
| E-commerce indicator | String<br>1-character alphanumeric                                                                                                                                  | 'crypt_type'=>\$crypt                     |
| Level 2/3 Data       | Object<br>n/a                                                                                                                                                       | \$mpgTxn->setLevel23Data(\$mpgAxLevel23); |

## Sample AX Completion

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='moneris';
$sapi_token='hurgle';
// $status = 'false';
/***** Transactional Variables *****/
$type='axcompletion';
$order_id='ord-210916-12:06:38';
$comp_amount='62.37';
$txn_number = '18924-0_11';
$script = '7';
// Create AxLevel23 Object
$mpgAxLevel23 = new mpgAxLevel23();
// Create Table 1 with details
$n101 = "R6"; //Entity ID Code
$n102 = "Retailing Inc. International"; //Name
$n301 = "919 Oriole Rd."; //Address Line 1
$n401 = "Toronto"; //City
$n402 = "On"; //State or Province
$n403 = "H1T6W3"; //Postal Code
$ref01 = array("4C", "CR"); //Reference ID Qualifier
$ref02 = array("M5T3A5", "16802309004"); //Reference ID
$big04 = "P07758545"; //Purchase Order Number
$big05 = "RN0049858"; //Release Number
$big10 = "INV99870E"; //Invoice Number
$saxRef1 = new axRef();
$saxRef1->setRef($ref01[0], $ref02[0]);
$saxRef1->setRef($ref01[1], $ref02[1]);
$saxN1Loop = new axN1Loop();
$saxN1Loop->setN1Loop($n101, $n102, $n301, $n401, $n402, $n403, $saxRef1);
$mpgAxLevel23->setTable1($big04, $big05, $big10, $saxN1Loop);
// Create Table 2 with details
// the sum of the extended amount field (pam05) must equal the level 1 amount field

$it102 = array("1", "1", "1", "1", "1"); //Line item quantity invoiced
$it103 = array("EA", "EA", "EA", "EA", "EA"); //Line item unit or basis of measurement code
$it104 = array("10.00", "25.00", "8.62", "10.00", "-10.00"); //Line item unit price
$it105 = array("", "", "", "", ""); //Line item basis of unit price code

$it10618 = array("MG", "MG", "MG", "MG", "MG"); //Product/Service ID qualifier
$it10719 = array("DJFR4", "JFJ49", "FEF33", "FEE43", "DISCOUNT"); //Product/Service ID
(corresponds to it10618)

$txi01_GST = array("GS", "GS", "GS", "GS", "GS"); //Tax type code
$txi02_GST = array("0.70", "1.75", "1.00", "0.80", "0.00"); //Monetary amount
$txi03_GST = array("", "", "", "", ""); //Percent
$txi06_GST = array("", "", "", "", ""); //Tax exempt code

$txi01_PST = array("PG", "PG", "PG", "PG", "PG"); //Tax type code
$txi02_PST = array("0.80", "2.00", "1.00", "0.80", "0.00"); //Monetary amount
$txi03_PST = array("", "", "", "", ""); //Percent
$txi06_PST = array("", "", "", "", ""); //Tax exempt code
$pam05 = array("11.50", "28.75", "10.62", "11.50", "-10.00"); //Extended line-item amount
$pid05 = array("Stapler", "Lamp", "Bottled Water", "Fountain Pen", "DISCOUNT"); //Line item
description
$it106s = array(new axIt106s(), new axIt106s(), new axIt106s(), new axIt106s(), new
axIt106s());
$it106s[0]->setIt10618($it10618[0]);

```

## Sample AX Completion

```

$it106s[0]->setIt10719($it10719[0]);
$it106s[1]->setIt10618($it10618[1]);
$it106s[1]->setIt10719($it10719[1]);
$it106s[2]->setIt10618($it10618[2]);
$it106s[2]->setIt10719($it10719[2]);
$it106s[3]->setIt10618($it10618[3]);
$it106s[3]->setIt10719($it10719[3]);
$it106s[4]->setIt10618($it10618[4]);
$it106s[4]->setIt10719($it10719[4]);
$txi = array(new axTxi(), new axTxi(), new axTxi(), new axTxi(), new axTxi());
$txi[0]->setTxi($txi01_GST[0], $txi02_GST[0], $txi03_GST[0], $txi06_GST[0]);
$txi[0]->setTxi($txi01_PST[0], $txi02_PST[0], $txi03_PST[0], $txi06_PST[0]);
$txi[1]->setTxi($txi01_GST[1], $txi02_GST[1], $txi03_GST[1], $txi06_GST[1]);
$txi[1]->setTxi($txi01_PST[1], $txi02_PST[1], $txi03_PST[1], $txi06_PST[1]);
$txi[2]->setTxi($txi01_GST[2], $txi02_GST[2], $txi03_GST[2], $txi06_GST[2]);
$txi[2]->setTxi($txi01_PST[2], $txi02_PST[2], $txi03_PST[2], $txi06_PST[2]);
$txi[3]->setTxi($txi01_GST[3], $txi02_GST[3], $txi03_GST[3], $txi06_GST[3]);
$txi[3]->setTxi($txi01_PST[3], $txi02_PST[3], $txi03_PST[3], $txi06_PST[3]);
$txi[4]->setTxi($txi01_GST[4], $txi02_GST[4], $txi03_GST[4], $txi06_GST[4]);
$txi[4]->setTxi($txi01_PST[4], $txi02_PST[4], $txi03_PST[4], $txi06_PST[4]);
$axItLoop = new axItLoop();
$axItLoop->setItLoop($it102[0], $it103[0], $it104[0], $it105[0], $it106s[0], $txi[0],
$spam05[0], $pid05[0]);
$axItLoop->setItLoop($it102[1], $it103[1], $it104[1], $it105[1], $it106s[1], $txi[1],
$spam05[1], $pid05[1]);
$axItLoop->setItLoop($it102[2], $it103[2], $it104[2], $it105[2], $it106s[2], $txi[2],
$spam05[2], $pid05[2]);
$axItLoop->setItLoop($it102[3], $it103[3], $it104[3], $it105[3], $it106s[3], $txi[3],
$spam05[3], $pid05[3]);
$axItLoop->setItLoop($it102[4], $it103[4], $it104[4], $it105[4], $it106s[4], $txi[4],
$spam05[4], $pid05[4]);
$mpgAxLevel23->setTable2($axItLoop);
//Create Table 3 with details
$taxTbl3 = new axTxi();
$taxTbl3->setTxi("GS", "4.25","", ""); //sum of GST taxes
$taxTbl3->setTxi("PG", "4.60","", ""); //sum of PST taxes
$taxTbl3->setTxi("TX", "8.85","", ""); //sum of all taxes
$mpgAxLevel23->setTable3($taxTbl3);
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'comp_amount'=>$comp_amount,
'txn_number'=> $txn_number,
'crypt_type'=>$crypt
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
$mpgTxn->setLevel23Data($mpgAxLevel23);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
//Status check example
//$mpgHttpPost = new mpgHttpPostStatus($store_id,$api_token,$status,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());

```

### Sample AX Completion

```
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
//print("\nStatusCode = " . $mpgResponse->getStatusCode());
//print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>
```

### 5.4.5 AX Force Post

The AX Force Post transaction is used to secure the funds locked by a pre-authorization transaction performed over IVR or equivalent terminal. When sending an AX Force Post request, you will need the order ID, amount, credit card number, expiry date, authorization code and e-commerce indicator.

#### AX Force Post transaction object definition

```
$txnArray = array('type'=>'axforcepost', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

#### HttpPostRequest object for AX Force Post transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

#### AX Force Post transaction request fields – Required

| Value    | Limits                                                                                                              | Set Method             |
|----------|---------------------------------------------------------------------------------------------------------------------|------------------------|
| Order ID | String<br>50-character alphanumeric                                                                                 | 'order_id'=>\$order_id |
| Amount   | String<br>10-character decimal<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal | 'amount'=>\$amount     |

| Value                | Limits                                              | Set Method                                    |
|----------------------|-----------------------------------------------------|-----------------------------------------------|
|                      | point<br><div>EXAMPLE: 1234567.89</div>             |                                               |
| Credit card number   | String<br>20-character alphanumeric                 | 'pan'=>\$pan                                  |
| Expiry date          | String<br>4-character alphanumeric<br>(YYMM format) | 'expiry_date'=>\$expiry_date                  |
| Authorization code   | String<br>8-character alphanumeric                  | 'auth_code'=>\$auth_code                      |
| E-commerce indicator | String<br>1-character alphanumeric                  | 'crypt_type'=>\$crypt                         |
| Level 2/3 Data       | Object<br>n/a                                       | \$mpgTxn->setLevel23Data<br>(\$mpgAxLevel23); |

**AX Force Post transaction request fields – Optional**

| Variable Name | Type and Limits                     | Set Method           |
|---------------|-------------------------------------|----------------------|
| Customer ID   | String<br>50-character alphanumeric | 'cust_id'=>\$cust_id |

**Sample AX Force Post**

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='moneris';
$sapi_token='hurgle';
//$status = 'false';
/***** Transactional Variables *****/
$type='axforcepost';

```

## Sample AX Force Post

```

$cust_id='CUST13343';
$order_id='ord-' . date("dmy-G:i:s");
$amount='62.37';
$pan='373269005095005';
$expiry_date='2012';
$auth_code='123456';
$script = '7';
//Create AxLevel23 Object
$mpgAxLevel23 = new mpgAxLevel23();
//Create Table 1 with details
$n101 = "R6"; //Entity ID Code
$n102 = "Retailing Inc. International"; //Name
$n301 = "919 Oriole Rd."; //Address Line 1
$n401 = "Toronto"; //City
$n402 = "On"; //State or Province
$n403 = "H1T6W3"; //Postal Code
$rref01 = array("4C", "CR"); //Reference ID Qualifier
$rref02 = array("M5T3A5", "16802309004"); //Reference ID
$b104 = "PO7758545"; //Purchase Order Number
$b105 = "RN0049858"; //Release Number
$b1010 = "INV99870E"; //Invoice Number
$axRef1 = new axRef();
$axRef1->setRef($rref01[0], $rref02[0]);
$axRef1->setRef($rref01[1], $rref02[1]);
$axN1Loop = new axN1Loop();
$axN1Loop->setN1Loop($n101, $n102, $n301, $n401, $n402, $n403, $axRef1);
$mpgAxLevel23->setTable1($b104, $b105, $b1010, $axN1Loop);
//Create Table 2 with details
//the sum of the extended amount field (pam05) must equal the level 1 amount field

$it102 = array("1", "1", "1", "1", "1"); //Line item quantity invoiced
$it103 = array("EA", "EA", "EA", "EA", "EA"); //Line item unit or basis of measurement code
$it104 = array("10.00", "25.00", "8.62", "10.00", "-10.00"); //Line item unit price
$it105 = array("", "", "", "", ""); //Line item basis of unit price code

$it10618 = array("MG", "MG", "MG", "MG", "MG"); //Product/Service ID qualifier
$it10719 = array("DJFR4", "JFJ49", "FEF33", "FEE43", "DISCOUNT"); //Product/Service ID
(corresponds to it10618)

$txi01_GST = array("GS", "GS", "GS", "GS", "GS"); //Tax type code
$txi02_GST = array("0.70", "1.75", "1.00", "0.80", "0.00"); //Monetary amount
$txi03_GST = array("", "", "", "", ""); //Percent
$txi06_GST = array("", "", "", "", ""); //Tax exempt code

$txi01_PST = array("PG", "PG", "PG", "PG", "PG"); //Tax type code
$txi02_PST = array("0.80", "2.00", "1.00", "0.80", "0.00"); //Monetary amount
$txi03_PST = array("", "", "", "", ""); //Percent
$txi06_PST = array("", "", "", "", ""); //Tax exempt code
$pam05 = array("11.50", "28.75", "10.62", "11.50", "-10.00"); //Extended line-item amount
$pid05 = array("Stapler", "Lamp", "Bottled Water", "Fountain Pen", "DISCOUNT"); //Line item
description
$it106s = array();
$it106s[0] = new axIt106s($it10618[0], $it10719[0]);
$it106s[1] = new axIt106s($it10618[1], $it10719[1]);
$it106s[2] = new axIt106s($it10618[2], $it10719[2]);
$it106s[3] = new axIt106s($it10618[3], $it10719[3]);
$it106s[4] = new axIt106s($it10618[4], $it10719[4]);
$txi = array(new axTxi(), new axTxi(), new axTxi(), new axTxi(), new axTxi());
$txi[0]->setTxi($txi01_GST[0], $txi02_GST[0], $txi03_GST[0], $txi06_GST[0]);

```



## Sample AX Force Post

```

$txi[0]->setTxi($txi01_PST[0], $txi02_PST[0], $txi03_PST[0], $txi06_PST[0]);
$txi[1]->setTxi($txi01_GST[1], $txi02_GST[1], $txi03_GST[1], $txi06_GST[1]);
$txi[1]->setTxi($txi01_PST[1], $txi02_PST[1], $txi03_PST[1], $txi06_PST[1]);
$txi[2]->setTxi($txi01_GST[2], $txi02_GST[2], $txi03_GST[2], $txi06_GST[2]);
$txi[2]->setTxi($txi01_PST[2], $txi02_PST[2], $txi03_PST[2], $txi06_PST[2]);
$txi[3]->setTxi($txi01_GST[3], $txi02_GST[3], $txi03_GST[3], $txi06_GST[3]);
$txi[3]->setTxi($txi01_PST[3], $txi02_PST[3], $txi03_PST[3], $txi06_PST[3]);
$txi[4]->setTxi($txi01_GST[4], $txi02_GST[4], $txi03_GST[4], $txi06_GST[4]);
$txi[4]->setTxi($txi01_PST[4], $txi02_PST[4], $txi03_PST[4], $txi06_PST[4]);
$axItLoop = new axItLoop();
$axItLoop->setItLoop($it102[0], $it103[0], $it104[0], $it105[0], $it106s[0], $txi[0],
$spam05[0], $pid05[0]);
$axItLoop->setItLoop($it102[1], $it103[1], $it104[1], $it105[1], $it106s[1], $txi[1],
$spam05[1], $pid05[1]);
$axItLoop->setItLoop($it102[2], $it103[2], $it104[2], $it105[2], $it106s[2], $txi[2],
$spam05[2], $pid05[2]);
$axItLoop->setItLoop($it102[3], $it103[3], $it104[3], $it105[3], $it106s[3], $txi[3],
$spam05[3], $pid05[3]);
// $axItLoop->setItLoop($it102[4], $it103[4], $it104[4], $it105[4], $it106s[4], $txi[4],
$spam05[4], $pid05[4]);
$mpgAxLevel23->setTable2($axItLoop);
// Create Table 3 with details
$taxTbl3 = new axTxi();
$taxTbl3->setTxi("GS", "4.25","", ""); //sum of GST taxes
$taxTbl3->setTxi("PG", "4.60","", ""); //sum of PST taxes
$taxTbl3->setTxi("TX", "8.85","", ""); //sum of all taxes
$mpgAxLevel23->setTable3($taxTbl3);
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'amount'=>$amount,
'pan'=>$pan,
'expdate'=>$expiry_date,
'auth_code'=>$auth_code,
'crypt_type'=>$crypt
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
$mpgTxn->setLevel23Data($mpgAxLevel23);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
// Status check example
// $mpgHttpPost = new mpgHttpPostStatus($store_id, $api_token, $status, $mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());

```

### Sample AX Force Post

```
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
//print("\nStatusCode = " . $mpgResponse->getStatusCode());
//print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>
```

## 5.4.6 AX Purchase Correction

The AX Purchase Correction (Void) transaction is used to cancel a transaction that was performed in the current batch. No amount is required because a void is always for 100% of the original transaction. The only transaction that can be voided using AX Purchase Correction is AX Completion and AX Force Post. To send an AX Purchase Correction the Order ID and transaction number from the AX Completion or AX Force Post are required.

### AX Purchase Correction transaction object definition

```
$txnArray = array('type'=>'axpurchasecorrection', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for AX Purchase Correction transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### AX Purchase Correction transaction request fields – Required

| Variable Name        | Type and Limits                             | Set Method                 |
|----------------------|---------------------------------------------|----------------------------|
| Order ID             | <i>String</i><br>50-character alphanumeric  | 'order_id'=>\$order_id     |
| Transaction number   | <i>String</i><br>255-character alphanumeric | 'txn_number'=>\$txn_number |
| E-commerce indicator | <i>String</i><br>1-character alphanumeric   | 'crypt_type'=>\$crypt      |

### AX Purchase Correction

```
<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='moneris';
$api_token='hurgle';
//$status = 'false';
/***** Transactional Variables *****/
$type='axpurchasecorrection';
$order_id='ord-210916-12:12:18';
$txn_number = '66011731632016265121219276-0_11';
$crypt_type = '7';
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'txn_number'=>$txn_number,
'crypt_type'=>$crypt_type
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
//Status check example
//$mpgHttpPost = new mpgHttpPostStatus($store_id,$api_token,$status,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
//print("\nStatusCode = " . $mpgResponse->getStatusCode());
//print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>
```

### 5.4.7 AX Refund

The AX Refund will credit a specified amount to the cardholder's credit card. A refund can be sent up to the full value of the original AX Completion or AX Force Post. To send an AX Refund you will require the order ID and transaction number from the original AX Completion or AX Force Post.

**AX Refund transaction object definition**

```
$txnArray = array('type'=>'axrefund', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

**HttpPostRequest object for AX Refund transaction**

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

**AX Refund transaction request fields – Required**

| Variable Name        | Type and Limits                                                                                                                                             | Set Method                                |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Order ID             | String<br>50-character alphanumeric                                                                                                                         | 'order_id'=>\$order_id                    |
| Transaction number   | String<br>255-character alphanumeric                                                                                                                        | 'txn_number'=>\$txn_number                |
| Amount               | String<br>10-character decimal<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><div>EXAMPLE: 1234567.89</div> | 'amount'=>\$amount                        |
| E-commerce indicator | String<br>1-character alphanumeric                                                                                                                          | 'crypt_type'=>\$crypt                     |
| Level 2/3 Data       | Object<br>n/a                                                                                                                                               | \$mpgTxn->setLevel23Data(\$mpgAxLevel23); |

## Sample AX Refund

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='moneris';
$api_token='hurgle';
//$status = 'false';
/***** Transactional Variables *****/
$type='axrefund';
$order_id='ord-210916-12:06:38';
$amount='62.37';
$txn_number = '18924-1_11';
$script = '7';
//Create AxLevel23 Object
$mpgAxLevel23 = new mpgAxLevel23();
//Create Table 1 with details
$n101 = "R6"; //Entity ID Code
$n102 = "Retailing Inc. International"; //Name
$n301 = "919 Oriole Rd."; //Address Line 1
$n401 = "Toronto"; //City
$n402 = "On"; //State or Province
$n403 = "H1T6W3"; //Postal Code
$ref01 = array("4C", "CR"); //Reference ID Qualifier
$ref02 = array("M5T3A5", "16802309004"); //Reference ID
$big04 = "P07758545"; //Purchase Order Number
$big05 = "RN0049858"; //Release Number
$big10 = "INV99870E"; //Invoice Number
$axRef1 = new axRef();
$axRef1->setRef($ref01[0], $ref02[0]);
$axRef1->setRef($ref01[1], $ref02[1]);
$axN1Loop = new axN1Loop();
$axN1Loop->setN1Loop($n101, $n102, $n301, $n401, $n402, $n403, $axRef1);
$mpgAxLevel23->setTable1($big04, $big05, $big10, $axN1Loop);
//Create Table 2 with details
//the sum of the extended amount field (pam05) must equal the level 1 amount field

$it102 = array("1", "1", "1", "1", "1"); //Line item quantity invoiced
$it103 = array("EA", "EA", "EA", "EA", "EA"); //Line item unit or basis of measurement code
$it104 = array("10.00", "25.00", "8.62", "10.00", "-10.00"); //Line item unit price
$it105 = array("", "", "", "", ""); //Line item basis of unit price code

$it10618 = array("MG", "MG", "MG", "MG", "MG"); //Product/Service ID qualifier
$it10719 = array("DJFR4", "JFJ49", "FEF33", "FEE43", "DISCOUNT"); //Product/Service ID
(corresponds to it10618)

$txi01_GST = array("GS", "GS", "GS", "GS", "GS"); //Tax type code
$txi02_GST = array("0.70", "1.75", "1.00", "0.80", "0.00"); //Monetary amount
$txi03_GST = array("", "", "", "", ""); //Percent
$txi06_GST = array("", "", "", "", ""); //Tax exempt code

$txi01_PST = array("PG", "PG", "PG", "PG", "PG"); //Tax type code
$txi02_PST = array("0.80", "2.00", "1.00", "0.80", "0.00"); //Monetary amount
$txi03_PST = array("", "", "", "", ""); //Percent
$txi06_PST = array("", "", "", "", ""); //Tax exempt code
$pam05 = array("11.50", "28.75", "10.62", "11.50", "-10.00"); //Extended line-item amount
$pid05 = array("Stapler", "Lamp", "Bottled Water", "Fountain Pen", "DISCOUNT"); //Line item
description
$it106s = array();
$it106s[0] = new axIt106s($it10618[0], $it10719[0]);
$it106s[1] = new axIt106s($it10618[1], $it10719[1]);

```

## Sample AX Refund

```

$it106s[2] = new axIt106s($it10618[2], $it10719[2]);
$it106s[3] = new axIt106s($it10618[3], $it10719[3]);
$it106s[4] = new axIt106s($it10618[4], $it10719[4]);
$txi = array(new axTxi(), new axTxi(), new axTxi(), new axTxi(), new axTxi());
$txi[0]->setTxi($txi01_GST[0], $txi02_GST[0], $txi03_GST[0], $txi06_GST[0]);
$txi[0]->setTxi($txi01_PST[0], $txi02_PST[0], $txi03_PST[0], $txi06_PST[0]);
$txi[1]->setTxi($txi01_GST[1], $txi02_GST[1], $txi03_GST[1], $txi06_GST[1]);
$txi[1]->setTxi($txi01_PST[1], $txi02_PST[1], $txi03_PST[1], $txi06_PST[1]);
$txi[2]->setTxi($txi01_GST[2], $txi02_GST[2], $txi03_GST[2], $txi06_GST[2]);
$txi[2]->setTxi($txi01_PST[2], $txi02_PST[2], $txi03_PST[2], $txi06_PST[2]);
$txi[3]->setTxi($txi01_GST[3], $txi02_GST[3], $txi03_GST[3], $txi06_GST[3]);
$txi[3]->setTxi($txi01_PST[3], $txi02_PST[3], $txi03_PST[3], $txi06_PST[3]);
$txi[4]->setTxi($txi01_GST[4], $txi02_GST[4], $txi03_GST[4], $txi06_GST[4]);
$txi[4]->setTxi($txi01_PST[4], $txi02_PST[4], $txi03_PST[4], $txi06_PST[4]);
$axItLoop = new axItLoop();
$axItLoop->setItLoop($it102[0], $it103[0], $it104[0], $it105[0], $it106s[0], $txi[0],
$spam05[0], $pid05[0]);
$axItLoop->setItLoop($it102[1], $it103[1], $it104[1], $it105[1], $it106s[1], $txi[1],
$spam05[1], $pid05[1]);
$axItLoop->setItLoop($it102[2], $it103[2], $it104[2], $it105[2], $it106s[2], $txi[2],
$spam05[2], $pid05[2]);
$axItLoop->setItLoop($it102[3], $it103[3], $it104[3], $it105[3], $it106s[3], $txi[3],
$spam05[3], $pid05[3]);
//$axItLoop->setItLoop($it102[4], $it103[4], $it104[4], $it105[4], $it106s[4], $txi[4],
$spam05[4], $pid05[4]);
$mpgAxLevel23->setTable2($axItLoop);
//Create Table 3 with details
$taxTbl3 = new axTxi();
$taxTbl3->setTxi("GS", "4.25","", ""); //sum of GST taxes
$taxTbl3->setTxi("PG", "4.60","", ""); //sum of PST taxes
$taxTbl3->setTxi("TX", "8.85","", ""); //sum of all taxes
$mpgAxLevel23->setTable3($taxTbl3);
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'amount'=>$amount,
'txn_number'=> $txn_number,
'crypt_type'=>$crypt
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
$mpgTxn->setLevel23Data($mpgAxLevel23);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
//Status check example
//$mpgHttpPost = new mpgHttpPostStatus($store_id,$api_token,$status,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());

```

**Sample AX Refund**

```

print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
//print("\nStatusCode = " . $mpgResponse->getStatusCode());
//print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>

```

### 5.4.8 AX Independent Refund

The AX Independent Refund will credit a specified amount to the cardholder's credit card. The independent refund does not require an existing order to be logged in the Moneris Gateway; however, the credit card number and expiry date will need to be passed.

**AX Independent Refund transaction object definition**

```
$txnArray = array('type'=>'axind_refund', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

**HttpPostRequest object for AX Independent Refund transaction**

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

**AX Independent Refund transaction request fields – Required**

| Variable Name | Type and Limits                                                                                                               | Set Method                 |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| Order ID      | String<br>50-character alphanumeric                                                                                           | 'order_id'=>\$order_id     |
| Amount        | String<br>10-character decimal<br><br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point | 'amount'=>\$amount         |
|               |                                                                                                                               | <b>EXAMPLE:</b> 1234567.89 |

| Variable Name        | Type and Limits                                            | Set Method                   |
|----------------------|------------------------------------------------------------|------------------------------|
| Credit card number   | <i>String</i><br>20-character alphanumeric                 | 'pan'=>\$pan                 |
| Expiry date          | <i>String</i><br>4-character alphanumeric<br>(YYMM format) | 'expiry_date'=>\$expiry_date |
| E-commerce indicator | <i>String</i><br>1-character alphanumeric                  | 'crypt_type'=>\$crypt        |

### AX Independent Refund transaction request fields – Optional

| Variable Name | Type and Limits                            | Set Method           |
|---------------|--------------------------------------------|----------------------|
| Customer ID   | <i>String</i><br>50-character alphanumeric | 'cust_id'=>\$cust_id |

#### Sample AX Independent Refund

```
<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='moneris';
$api_token='hurgle';
//$status = 'false';
/***** Transactional Variables *****/
$type='axind_refund';
$cust_id='CUST13343';
$order_id='ord-'.date("dmy-G:i:s");
$amount='62.37';
$pan='373269005095005';
$expiry_date='2012';
$crypt = '7';
//Create AxLevel23 Object
$mpgAxLevel23 = new mpgAxLevel23();
//Create Table 1 with details
$n101 = "R6"; //Entity ID Code
$n102 = "Retailing Inc. International"; //Name
$n301 = "919 Oriole Rd."; //Address Line 1
$n401 = "Toronto"; //City
$n402 = "On"; //State or Province
$n403 = "H1T6W3"; //Postal Code
$ref01 = array("4C", "CR"); //Reference ID Qualifier
$ref02 = array("M5T3A5", "16802309004"); //Reference ID
$big04 = "P07758545"; //Purchase Order Number
```



## Sample AX Independent Refund

```

$big05 = "RN0049858"; //Release Number
$big10 = "INV99870E"; //Invoice Number
$axRef1 = new axRef();
$axRef1->setRef($ref01[0], $ref02[0]);
$axRef1->setRef($ref01[1], $ref02[1]);
$axN1Loop = new axN1Loop();
$axN1Loop->setN1Loop($n101, $n102, $n301, $n401, $n402, $n403, $axRef1);
$mpgAxLevel23->setTable1($big04, $big05, $big10, $axN1Loop);
//Create Table 2 with details
//the sum of the extended amount field (pam05) must equal the level 1 amount field

$it102 = array("1", "1", "1", "1", "1"); //Line item quantity invoiced
$it103 = array("EA", "EA", "EA", "EA", "EA"); //Line item unit or basis of measurement code
$it104 = array("10.00", "25.00", "8.62", "10.00", "-10.00"); //Line item unit price
$it105 = array("", "", "", "", ""); //Line item basis of unit price code

$it10618 = array("MG", "MG", "MG", "MG", "MG"); //Product/Service ID qualifier
$it10719 = array("DJFR4", "JFJ49", "FEF33", "FEE43", "DISCOUNT"); //Product/Service ID
(corresponds to it10618)

$txi01_GST = array("GS", "GS", "GS", "GS", "GS"); //Tax type code
$txi02_GST = array("0.70", "1.75", "1.00", "0.80", "0.00"); //Monetary amount
$txi03_GST = array("", "", "", "", ""); //Percent
$txi06_GST = array("", "", "", "", ""); //Tax exempt code

$txi01_PST = array("PG", "PG", "PG", "PG", "PG"); //Tax type code
$txi02_PST = array("0.80", "2.00", "1.00", "0.80", "0.00"); //Monetary amount
$txi03_PST = array("", "", "", "", ""); //Percent
$txi06_PST = array("", "", "", "", ""); //Tax exempt code

$spam05 = array("11.50", "28.75", "10.62", "11.50", "-10.00"); //Extended line-item amount
$spid05 = array("Stapler", "Lamp", "Bottled Water", "Fountain Pen", "DISCOUNT"); //Line item
description
$it106s = array();
$it106s[0] = new axIt106s($it10618[0], $it10719[0]);
$it106s[1] = new axIt106s($it10618[1], $it10719[1]);
$it106s[2] = new axIt106s($it10618[2], $it10719[2]);
$it106s[3] = new axIt106s($it10618[3], $it10719[3]);
$it106s[4] = new axIt106s($it10618[4], $it10719[4]);
$txi = array(new axTxi(), new axTxi(), new axTxi(), new axTxi(), new axTxi());
$txi[0]->setTxi($txi01_GST[0], $txi02_GST[0], $txi03_GST[0], $txi06_GST[0]);
$txi[0]->setTxi($txi01_PST[0], $txi02_PST[0], $txi03_PST[0], $txi06_PST[0]);
$txi[1]->setTxi($txi01_GST[1], $txi02_GST[1], $txi03_GST[1], $txi06_GST[1]);
$txi[1]->setTxi($txi01_PST[1], $txi02_PST[1], $txi03_PST[1], $txi06_PST[1]);
$txi[2]->setTxi($txi01_GST[2], $txi02_GST[2], $txi03_GST[2], $txi06_GST[2]);
$txi[2]->setTxi($txi01_PST[2], $txi02_PST[2], $txi03_PST[2], $txi06_PST[2]);
$txi[3]->setTxi($txi01_GST[3], $txi02_GST[3], $txi03_GST[3], $txi06_GST[3]);
$txi[3]->setTxi($txi01_PST[3], $txi02_PST[3], $txi03_PST[3], $txi06_PST[3]);
$txi[4]->setTxi($txi01_GST[4], $txi02_GST[4], $txi03_GST[4], $txi06_GST[4]);
$txi[4]->setTxi($txi01_PST[4], $txi02_PST[4], $txi03_PST[4], $txi06_PST[4]);
$axItLoop = new axItLoop();
$axItLoop->setItLoop($it102[0], $it103[0], $it104[0], $it105[0], $it106s[0], $txi[0],
$spam05[0], $spid05[0]);
$axItLoop->setItLoop($it102[1], $it103[1], $it104[1], $it105[1], $it106s[1], $txi[1],
$spam05[1], $spid05[1]);
$axItLoop->setItLoop($it102[2], $it103[2], $it104[2], $it105[2], $it106s[2], $txi[2],
$spam05[2], $spid05[2]);
$axItLoop->setItLoop($it102[3], $it103[3], $it104[3], $it105[3], $it106s[3], $txi[3],
$spam05[3], $spid05[3]);
//$axItLoop->setItLoop($it102[4], $it103[4], $it104[4], $it105[4], $it106s[4], $txi[4],

```

## Sample AX Independent Refund

```

$spam05[4], $pid05[4]);
$mpgAxLevel23->setTable2($axItLoop);
//Create Table 3 with details
$taxTbl3 = new axTxi();
$taxTbl3->setTxi("GS", "4.25","", ""); //sum of GST taxes
$taxTbl3->setTxi("PG", "4.60","", ""); //sum of PST taxes
$taxTbl3->setTxi("TX", "8.85","", ""); //sum of all taxes
$mpgAxLevel23->setTable3($taxTbl3);
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'cust_id'=>$cust_id,
'amount'=>$amount,
'pan'=>$pan,
'expdate'=>$expiry_date,
'crypt_type'=>$crypt
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
$mpgTxn->setLevel23Data($mpgAxLevel23);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
//Status check example
//$mpgHttpPost = new mpgHttpPostStatus($store_id,$api_token,$status,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
//print("\nStatusCode = " . $mpgResponse->getStatusCode());
//print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>

```



## 6 3-D Secure 2.2

- 6.1 About 3-D Secure 2.2
- 6.2 Building Your 3-D Secure 2.2 Integration
- 6.3 Implementing Card Lookup Request
- 6.5 Implementing MPI 3DS Authentication Request
- 6.6 Handling the Challenge Flow
- 6.8 Performing the Authorization
- 6.9 Testing Your 3-D Secure 2.2 Integration
- 6.10 Moving to Production With 3-D Secure 2.2
- 6.11 3-D Secure 2.2 TransStatus Codes
- 6.12 3-D Secure 2.2 Commons TransStatusReason Decline Codes
- 6.13 CAVV Result Codes

### 6.1 About 3-D Secure 2.2

3-D Secure 2.2 is an EMVCo payment authentication protocol designed to reduce card not present fraud by making a risk assessment based on transaction and device data, while also supporting further risk minimization measures, such as a challenge to the cardholder. In some cases, a liability shift takes effect for certain card-not-present fraud-related chargebacks enabling the merchant to provide goods and services with confidence.

The Moneris Gateway can enable transactions using the 3-D Secure protocol via Moneris 3DS Server and Access Control Server (ACS).

Moneris Gateway supports the following 3-D Secure implementations:

- Visa Secure (please note: Visa Secure does not support all the RI Indicators available in the 3D Secure 2.2. Check the RI Indicators status field to confirm the status Visa Secure support.)
- Mastercard Identity Check
- American Express SafeKey (please note: American Express only supports authentication requests for merchants who have an Amex OFI merchant account)

### 6.1.1 3-D Secure Implementations

Visa Secure, Mastercard Identity Check and American Express SafeKey are programs based on the 3-D Secure Protocol to improve the security of online transactions.

These programs involve authentication of the cardholder during an online e-commerce transaction.

Authentication is based on the issuer's selected method of authentication.

The following are examples of authentication methods:

- Risk-based authentication
- Dynamic passwords
- Static passwords

Some benefits of these programs are reduced risk of fraudulent transactions and protection against chargebacks for certain fraudulent transactions.

The PHP 3DS 2.2 API supports two message categories and two device channels from the 3-D Secure authentication protocol:

1. Message Categories:

- **Payment Authentication** – Cardholder authentication prior to an eCommerce transaction. After a successful 3DS authentication, you proceed with a purchase or pre-authorization.
- **Non-Payment Authentication (NPA)**– Identity verification and account confirmation performed without an accompanying financial transaction. After a successful 3DS authentication, you might proceed with:
  - Tokenizing the card for future payments
  - Allowing log-in for client portals
  - Any other activity relying on identity or account confirmation

2. Device Channels:

- **Browser** – The transaction originates from a website utilized via a browser on the cardholder's device.

- For example, an eCommerce transaction originating on the merchant's website with a check-out process that the cardholder is using via their personal computer or mobile phone's web browser (Chrome, Edge, Safari, etc.).
- **3DS Requestor Initiated** – Account confirmations and cardholder authentication with no direct cardholder originating the transaction.
  - 3RI can be used for authenticating Mail-Telephone Order (MOTO) transactions.
  - 3RI can be used to authenticate follow-on transactions as part of a subscription, such as recurring transactions. The first cardholder payment might use a browser-based authentication, with subsequent payments utilizing a 3RI authentication linking to the previous.
  - In situations where a merchant business model accommodates waiting before processing their payment, they can utilize Decoupled Authentication to allow the cardholder to authenticate directly with their issuer via a non-3DS challenge, such as a push notification to a banking app.

### 6.1.2 Out of Scope/Not Supported Check

- In-app

### 6.1.3 Version Compatibility

All development to the Moneris API must be able to support the addition of new fields in the response and new error conditions in the response. Otherwise any changes that affect backwards compatibility will be communicated by Moneris Solutions with an appropriate period of notice. When developing to the solution it is recommended to validate for success state of the request and then handle errors states separately and ensure there is a final catch for any unexpected/undocumented errors that are returned.

### 6.1.4 Upgrading from 3-D Secure 2.0 to 3-D Secure 2.2 Check

The 3DS 2.2 API is different from the 3DS 2.0 API therefore developers will have to complete the steps described in the section 6.2 Building Your 3-D Secure 2.2 Integration.

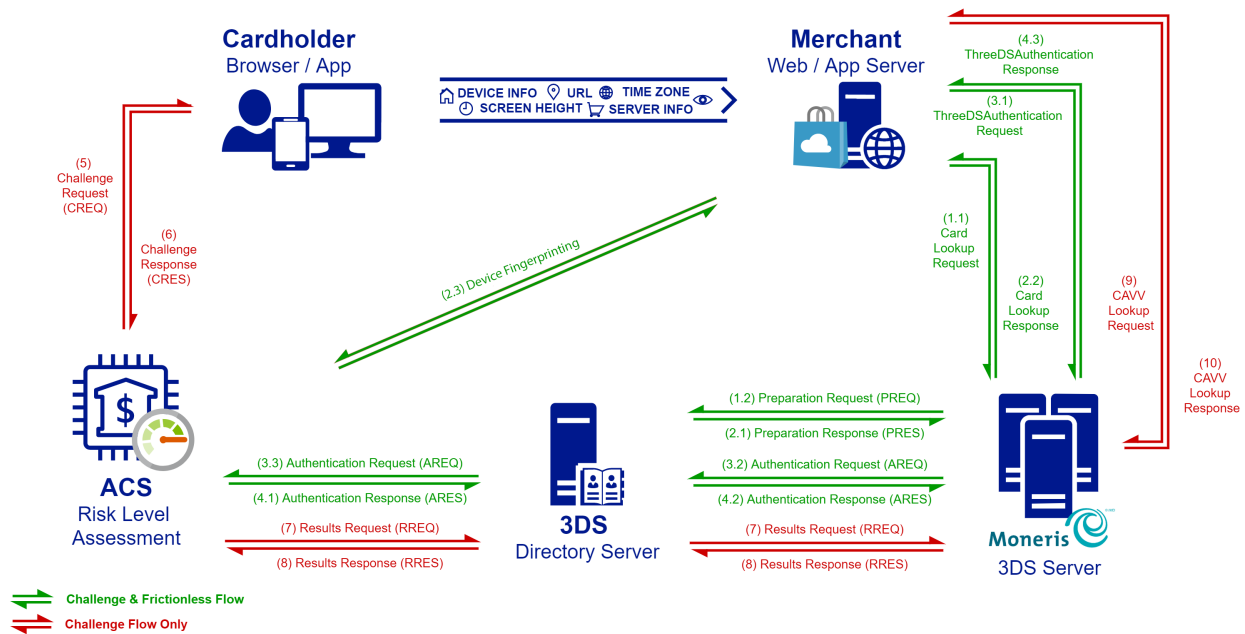
## 6.2 Building Your 3-D Secure 2.2 Integration

- 6.2.1 Activating 3-D Secure Functionality
- 6.2.2 Transaction Flow for 3-D Secure - Browser channel
- 6.2.3 Transaction Flow for 3-D Secure - 3RI channel

### 6.2.1 Activating 3-D Secure Functionality

To activate Visa Secure, Mastercard Identity Check and/or American Express SafeKey transaction functionality, call Moneris Sales Support at 1-855-465-4980 to have Moneris enroll you in the program(s) and enable the functionality on your account.

### 6.2.2 Transaction Flow for 3-D Secure - Browser channel



The 3DS 2.2 API is called when the customer wishes to checkout. An optional card lookup request can be performed to initiate cardholder browser fingerprinting. Once the fingerprint is complete, or as a first step if not performing a fingerprint, the transactional information can then be transmitted to the 3DS 2.2 service so a risk assessment may be initiated.

The flow can then proceed in one of two ways. The two different flows are referred to as “frictionless” and “challenge”.

The “frictionless” flow is invisible to a cardholder. If the issuing financial institution has enough information to make a risk assessment and assume liability, this will manifest itself as with an authentication attempt or success with an accompanying CAVV value. No cardholder challenge is presented.

In the “challenge” flow the issuing financial institution may wish to take a further step and issue a challenge to the cardholder. In this case the cardholder’s browser gets re-directed to the issuer’s 3DS platform for authentication. Once this challenge is complete, the cardholder browser is again re-directed back to the merchant’s site. The merchant’s server then issues a server-to-server request in order to obtain the CAVV value from Moneris.

### **Steps 1 – 2 (Optional)**

An optional card lookup request can be performed to initiate cardholder browser fingerprinting. The merchant website collects device information and provides them to Moneris via the `card_lookup` request (1.1). Moneris submits this data to the 3DS Directory Server and returns with the `card_lookup` response containing the card’s supported 3DS version, an ACS URL, and 3DS Method Data representing the fingerprint (2.2). The merchant browser then submits an HTTP POST to the ACS URL with the method data. (2.3)

Once the fingerprint is complete, or as a first step if not performing a fingerprint, the transactional information can then be transmitted to the 3DS 2.2 service so a risk assessment may be initiated.

### **Steps 3 – 4 (Required)**

The 3DS authentication request `threeDSAuthentication` is performed by the merchant website to initiate validating the cardholder identity. Moneris communicates with the 3DS Directory and the ACS system for that issuer to provide an initial risk assessment (3.2-4.2). Moneris returns a `threeDSAuthentication` response to the merchant with a `TransStatus` indicating the action for the website to perform:

- A `TransStatus` = “Y” or “A” means the website can proceed immediately to the financial transaction with the CAVV value provided. This is a frictionless transaction flow without presenting a challenge.
- A `TransStatus` = “C” indicates that the cardholder must be presented a challenge. To present the challenge, you must POST a `<form>` with a “creq” field, which contains the `ChallengeData`, to the URL defined in the `ChallengeURL` field.
- A `TransStatus` = “D” indicates that the cardholder must be presented a challenge via Decoupled Authentication. See Decoupled Authentication.

### **Steps 5 – 10 (Challenge Only)**

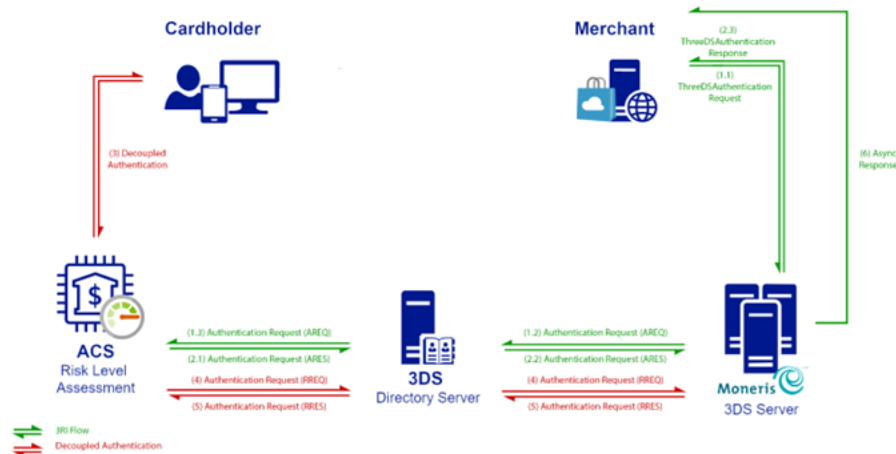
In scenarios where a challenge is required, the merchant website sends an HTTP POST to the Challenge URL with the `ChallengeData` sent as a “CREQ” value (5). The ACS system will present a challenge to the cardholder, who will supply whatever credentials their issuer requires. The merchant website receives a



“CRES” value from the ACS via the HTTP POST response (6). Meanwhile, the ACS supplies the results to the 3DS Directory, which then forwards it to Moneris (7-8).

The merchant’s website then sends a CAVV Lookup to Moneris via a `cavv_lookup` request and includes their “CRES” (9). Moneris responds with the `cavv_lookup` response with the necessary ECI and CAVV values. With the 3DS authentication complete, you can proceed to the financial transaction.

### 6.2.3 Transaction Flow for 3-D Secure - 3RI channel



In a 3DS Requestor Initiated flow, the cardholder is not directly triggering the transaction flow via a browser experience as above. It is possible they are initiating the transaction outside the 3DS protocol, such as mailing or phoning the merchant (Mail-Telephone Order, aka MOTO), or it is possible the merchant is processing a recurring or installment plan on behalf of the cardholder’s subscription. It is also possible the merchant requires a non-payment authentication as part of tokenizing the card for later use.

3RI flows do not have direct cardholder interaction. The merchant sends their `<threeDSAuthentication>` request per steps 3-4 above but include additional fields to describe their 3RI usage scenario.

- If this is a Mail or Telephone (MOTO) payment authentication, the ACS may trigger a Decoupled Authentication between the issuer and cardholder (see Decoupled Authentication)
- If this is a follow-on payment from a previous 3DS authenticated transaction, you can include `PriorAuthenticationInfo` to link to the previous authentication and improve the likelihood of a successful result

Your server can utilize the fields `DeviceChannel`, `RiIndicator` and `MessageCategory` to inform Moneris if your merchant server is attempting to use the 3DS Requestor Initiated process.

### 6.2.3.1 Decoupled Authentication

For scenarios where a 3RI authentication requires challenge, instead of utilizing the standard challenge request and response the ACS authenticates the cardholder outside of the 3-D Secure protocol such as a banking app or mobile phone text to the cardholder. The Moneris 3DS Server waits for the ACS to authenticate the cardholder; this authentication can take up to 7 days. As this process relies on a cardholder action outside the 3DS flow, it occurs asynchronously to transaction processing.

Your server can utilize the fields `DecoupledRequestIndicator` and `DecoupledRequestAsyncUrl` to inform Moneris that you are opting in to accept a Decoupled Authentication attempt and where you want Moneris to POST the results asynchronously.

## 6.3 Implementing Card Lookup Request

The `CardLookup` request verifies the applicability of 3DS 2.2 on the card and returns the 3DS Method URL used for device fingerprinting if the card supports this feature. This request is optional, it may increase the chance of a frictionless flow.

The `threeDSMethodURL` & `threeDSMethodData` are returned to the merchant server on the `CardLookup` response, if supported.

- If you receive the `threeDSMethodURL`, you may send the `threeDSMethodData` to the `threeDSMethodURL` via a browser post in order to supplement the authentication request with device data pertaining to the cardholder's browser.
- If you do not receive the `threeDSMethodURL`, you may still proceed with 3DS Authentication.

The `threeDSMethodData` must be sent via HTTP POST to the `threeDSMethodURL` in a hidden `iFrame`.

In your implementation, use the following URLs as Host, depending on the development stage:

Testing:

`esqa.moneris.com`

Production:

`www3.moneris.com`

### 6.3.1 Card Lookup Request – `mpiCardLookup`

#### Card Lookup Request transaction object definition

```
$mpiCardLookup = new MpiCardLookup();
```

**HttpPostRequest object for Card Lookup Request transaction**

```
$mpgRequest = new mpgRequest($mpgTxn);
$mpgTxn = new mpgTransaction($mpiCardLookup);
```

**Core connection object fields (all API transactions)**

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

**Card Lookup Request transaction request fields – Required**

**NOTE:** Either a pan or a data\_key must be passed in the request

| Variable Name      | Type and Limits                                                         |                                                                 |
|--------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------|
| order ID           | String<br>50-character alpha-<br>numerica-Z A-Z 0-9 _ - : . @<br>spaces | 'order_id'=>\$order_id                                          |
| credit card number | String<br>max 20-character alpha-<br>numeric                            | 'pan'=>\$pan                                                    |
| data key           | String<br>25-character alphanumeric                                     | 'data_key'=>\$data_key                                          |
| notification URL   | String<br>256-character alpha-<br>numeric                               | \$mpiCardLookup->setNotificationURL<br>("HTTPS://YOURURL.COM"); |

**Sample Card Lookup Request**

```
<?php
```

```

require "../mpgClasses.php";
/***** Request Variables *****/
$store_id = "moneris";
$api_token = "hurgle";
$order_id = 'ord-' . date("dmy-G:i:s");
$span = "347668693641199";

$mpiCardLookup = new MpiCardLookup();
$mpiCardLookup->setOrderId($order_id);
$mpiCardLookup->setPan($span);
// $mpiCardLookup->setDataKey("800XGiwxgvfbZngigVFeld9d2"); //Optional - For Moneris Vault and
// Hosted Tokenization tokens in place of setPan
$mpiCardLookup->setNotificationUrl("https://yournotificationurl.com"); // (Website URL that
// will receive 3DS Method Completion response from ACS)
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($mpiCardLookup);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false or comment out this line for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
/***** Response *****/
$mpgResponse = $mpgHttpPost->getMpgResponse();
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nMessageType = " . $mpgResponse->getMpiMessageType());
print("\nThreeDSVersion = " . $mpgResponse->getThreeDSVersion());
print("\nThreeDSMethodURL = " . $mpgResponse->getMpiThreeDSMethodURL());
print("\nThreeDSMethodData = " . $mpgResponse->getMpiThreeDSMethodData());
print("\nThreeDSServerTransId = " . $mpgResponse->getMpiThreeDSServerTransId());
?>

```

## 6.4 Handling the 3DS Method for Device Fingerprinting

You can use the **threeDSMethodURL** & **threeDSMethodData** returned by a Card Lookup response to increase the probability of a frictionless 3DS flow for the cardholder. Transmitting the **threeDSMethodData** to the **threeDSMethodURL** via a browser HTTP POST allows the issuer use of a hidden iFrame on the merchant website to obtain details on the customer's device.

The results of the 3DS Method are returned to the merchant's **notificationURL** supplied in the preceding Card Lookup.

Below is a sample of a basic static form to help visualize the data and fields that need to be submitted.

### Device Fingerprinting request form (Merchant browser to ACS):

```

<form name="frm" method="POST" action="Rendering URL">
<input type="hidden" name="threeDSMethodData" value=
="eyJ0aHJlZURTU2VydmVyVHJh-
bnNJRCI6IjNhYzdjYWE3LWFiNDItMjY2My03OTFiLTJhYzA1YTU0MmM0YSIsInRocmVlRFNNZ-
XRob2R0b3RpZmljYXRpb25VUkwioiJ0aHJlZURTU2V0aG9kTm90aWZpY2F0aW9uVWJMin0">
</form>

```

### Decoded threeDSMethodData:

```

{"threeDSServerTransID":"3ac7caa7-aa42-2663-791b-2ac05a542c4a","-
threeDSMethodNotificationURL":"threeDSMethodNotificationURL"}

```

**Device Fingerprinting response form (ACS to Merchant notificationURL):**

```
<form name="frm" method="POST" action="threeDSMethodNotificationURL">
<input type="hidden" name="threeDSMethodData" value-
="eyJ0aHJlZURTU2VydmVyVHJh-
hbnNJRCI6IjNhYzdjYWE3LWFhNDItMjY2My03OTFiLTJhYzA1YTU0MmM0YSJ9">
</form>
```

**Decoded threeDSMethodData:**

```
{"threeDSServerTransID":"3ac7caa7-aa42-2663-791b-2ac05a542c4a"}
```

## 6.5 Implementing MPI 3DS Authentication Request

The MPI 3DS Authentication Request is used to start the validation process of the card. The result of this request determines whether 3DS 2.2 is supported by the card and what type of authentication is required.

In your implementation, use the following URLs as Host, depending on the development stage:

**Testing URLs:**

<https://mpg1t.moneris.io/mpi2/servlet/MpiServlet>

**Production URLs:**

<https://mpg1.moneris.io/mpi2/servlet/MpiServlet>

Below we detail three different scenarios for utilizing Moneris MPI 3DS Authentication. Each scenario has conditions for which fields are required or optional for the endpoint.

### 6.5.1 MPI 3DS Authentication Request - Browser Channel

The authentication request is used to start the validation process of the card.

The result of this request determines whether 3-D Secure 2.0 is supported by the card and what type of authentication is required.

In your implementation, use the following URLs as Host, depending on the development stage:

testing: [mpg1t.moneris.io](https://mpg1t.moneris.io)

production: [mpg1.moneris.io](https://mpg1.moneris.io)

**NOTE:** Billing-related request fields are required to be sent for this transaction, or else the authentication process may fail

**MPI 3DS Authentication Request transaction object definition**

```
$mpiThreeDSAuthentication = new MpiThreeDSAuthentication();
```

**HttpPostRequest object for MPI 3DS Authentication Request transaction**

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgTxn = new mpgTransaction($mpiThreeDSAuthentication);
```

**WARNING:** Do not send fields related to 3RI on browser-based authentications.

**Core connection object fields (all API transactions)**

| Variable Name | Type and Limits | Description                                                                                                                             |
|---------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| store ID      | <i>String</i>   | 'store_id'=>\$store_id                                                                                                                  |
| <store_id>    | N/A             | Unique identifier provided by Moneris upon merchant account setup                                                                       |
| API token     | <i>String</i>   | 'api_token'=>\$api_token                                                                                                                |
| <api_token>   | N/A             | Unique alphanumeric string assigned by Moneris upon merchant account activation                                                         |
|               |                 | To find your API token, refer to your test or production store's Admin settings in the Merchant Resource Center, at the following URLs: |
|               |                 | Testing: <a href="https://esqa.moneris.com/mpg/">https://esqa.moneris.com/mpg/</a>                                                      |
|               |                 | Production: <a href="https://www3.moneris.com/mpg/">https://www3.moneris.com/mpg/</a>                                                   |

**MPI 3DS Authentication Request transaction request fields – Required**

| Variable Name     | Type and Limits | Description                                           |
|-------------------|-----------------|-------------------------------------------------------|
| message category  | <i>String</i>   | \$mpiThreeDSAuthentication->setMessageCategory("02"); |
| <MessageCategory> | 2-character     |                                                       |

| Variable Name                     | Type and Limits                                                                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | numeric                                                                                  | Whether the authentication request is for a payment or non-payment use:<br><br>01 = payment authentication (PA)<br><br>02 = non-payment authentication (NPA)                                                                                                                                                                                                                                                                           |
| device channel<br><DeviceChannel> | <i>String</i><br><br>2-character<br>numeric                                              | <code>\$mpiThreeDSAuthentication-&gt;setDeviceChannel("02");</code><br><br>The interface used to initiate the authentication:<br><br>02 = Browser (BRW)<br><br>03 = 3DS Requestor Initiated (3RI)                                                                                                                                                                                                                                      |
| request type<br><RequestType>     | <i>String</i><br><br>2-character<br>alphanumeric                                         | <code>\$mpiThreeDSAuthentication-&gt;setRequestType("REQUEST_TYPE_VALUE");</code><br><br>Indicates the type of browser-based authentication request:<br><br>01 = cardholder initiated payment<br><br>02 = recurring transaction<br><br>03 = installment transaction<br><br>04 = add card<br><br>05 = maintain card<br><br>06 = cardholder verification as part of EMV token ID & V<br><br>Conditional. Required if device_channel = 02 |
| order ID<br><order_id>            | <i>String</i><br><br>50-character<br>alphanumeric<br><br>a-Z A-Z 0-9 _ - :<br>. @ spaces | <code>'order_id'=&gt;\$order_id;</code><br><br>Merchant-defined transaction identifier that must be unique for every Purchase, Pre-Authorization and Independent Refund transaction. No two transactions of these types may have the same order ID.<br><br>For Refund, Completion and Purchase Correction transactions, the order ID must be the same as that of the original transaction.                                             |
| data key                          | <i>String</i>                                                                            | <code>'data_key'=&gt;\$data_key;</code>                                                                                                                                                                                                                                                                                                                                                                                                |

| Variable Name                                               | Type and Limits                                                                                                                                                                                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <data_key><br><br>OR<br><br>credit card number<br><br><pan> | data key limits:<br><br>25-character alphanumeric<br><br>credit card number limits:<br><br>max 20-character alphanumeric                                                                                | <b>data key description:</b><br><br>Unique identifier for a Vault profile, and used in future Vault financial transactions to associate a transaction with that profile<br><br>Data key is generated by Moneris and returned to you in the Receipt object when the profile is first registered<br><br>'pan'=>\$pan<br><br><b>credit card number description:</b><br><br>Credit card number, usually 16 digits —field can be maximum 20 digits in support of future expansion of card number ranges.<br><br>Carries the token for network tokenization transactions. |
| expiry date<br><br><expdate>                                | String<br><br>4-character alphanumeric<br><br>YYYYMM                                                                                                                                                    | 'expiry_date'=>\$expiry_date<br><br>Expiry date of the credit card, in YYMM format.<br><br><div> <b>NOTE:</b> This is the reverse of the MMY date format that is presented on the card.         </div>                                                                                                                                                                                                                                                                                                                                                              |
| amount<br><br><amount>                                      | String<br><br>10-character decimal<br><br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><br><div> <b>EXAMPLE:</b><br/>           1234567.89         </div> | 'amount'=>\$amount<br><br>Transaction dollar amount<br><br>This must contain at least 3 digits, two of which are penny values<br><br>Minimum allowable value = \$0.01, maximum allowable value = \$9999999.99                                                                                                                                                                                                                                                                                                                                                       |
| cardholder name<br><br><CardholderName>                     | String<br><br>45-character                                                                                                                                                                              | \$mpiThreeDSAuthentication->setCardholderName("CARDHOLDER_NAME_VALUE");                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |



| Variable Name                                      | Type and Limits                                                 | Description                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | alphanumeric                                                    | Name of the cardholder                                                                                                                                                                                                                                                                                                                              |
|                                                    | <div> <b>NOTE:</b> Accented characters are not allowable </div> |                                                                                                                                                                                                                                                                                                                                                     |
| 3DS completion indicator<br><ThreeDSCompletionInd> | String<br>1-character alphabetic                                | <pre>\$mpiThreeDSAuthentication-&gt;setThreeDSCompletionInd("THREEDSCOMPLETION_VALUE");</pre> indicates whether 3ds method MpiCardLookup was successfully completed<br><br>Allowable values:<br><br>Y = Successfully completed<br><br>N = Did not successfully complete<br><br>U = Unavailable<br><br>Conditional. Required if card_lookup is used. |
| billing address<br><BillAddress1>                  | String<br>50-character alphanumeric                             | <pre>\$mpiThreeDSAuthentication-&gt;setBillAddress1("BILL_STREET_ADDRESS_VALUE");</pre> Cardholder billing address                                                                                                                                                                                                                                  |
| billing province<br><BillProvince>                 | String<br>3-character alphanumeric                              | <pre>\$mpiThreeDSAuthentication-&gt;setBillProvince("BILL_PROV_VALUE");</pre> Cardholder province or state<br><br>Defined in country subdivision ISO 3166-2                                                                                                                                                                                         |
| billing city<br><BillCity>                         | String<br>50-character alphanumeric                             | <pre>\$mpiThreeDSAuthentication-&gt;setBillCity("BILL_CITY_VALUE");</pre> Cardholder billing city                                                                                                                                                                                                                                                   |
| billing postal code<br><BillPostalCode>            | String<br>16-character alphanumeric                             | <pre>\$mpiThreeDSAuthentication-&gt;setBillPostalCode("BILL_POSTAL_CODE_VALUE");</pre> Cardholder billing postal code                                                                                                                                                                                                                               |
| billing country                                    | String                                                          | <pre>\$mpiThreeDSAuthentication-&gt;setBillCountry</pre>                                                                                                                                                                                                                                                                                            |

| Variable Name                                  | Type and Limits                             | Description                                                                                                                                                                                                                       |
|------------------------------------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <BillCountry>                                  | 3-character alphanumeric                    | <pre>("BILL_COUNTRY_VALUE");</pre> <p>Cardholder billing country</p> <p>Defined as 3 digit country code ISO 3166-1</p>                                                                                                            |
| shipping address<br><ShipAddress1>             | <i>String</i><br>50-character alphanumeric  | <pre>\$mpiThreeDSAuthentication-&gt;setShipAddress1 ("SHIP_STREET_ADDRESS_VALUE");</pre> <p>Shipping destination address</p>                                                                                                      |
| shipping province<br><ShipProvince>            | <i>String</i><br>3-character alphanumeric   | <pre>\$mpiThreeDSAuthentication-&gt;setShipProvince ("SHIP_PROV_VALUE");</pre> <p>Shipping destination province or state</p> <p>Defined in country subdivision ISO 3166-2</p>                                                     |
| shipping city<br><ShipCity>                    | <i>String</i><br>50-character alphanumeric  | <pre>\$mpiThreeDSAuthentication-&gt;setShipCity ("SHIP_CITY_VALUE");</pre> <p>Shipping destination city</p>                                                                                                                       |
| shipping postal code<br><ShipPostalCode>       | <i>String</i><br>16-character alphanumeric  | <pre>\$mpiThreeDSAuthentication-&gt;setShipPostalCode ("SHIP_POSTAL_CODE_VALUE");</pre> <p>Shipping destination postal or ZIP code</p>                                                                                            |
| shipping country<br><ShipCountry>              | <i>String</i><br>3-character alphanumeric   | <pre>\$mpiThreeDSAuthentication-&gt;setShipCountry ("SHIP_COUNTRY_VALUE");</pre> <p>Shipping destination country</p> <p>Defined as 3-digit country code in ISO 3166-1</p>                                                         |
| notification URL<br><NotificationURL>          | <i>String</i><br>256-character alphanumeric | <pre>\$mpiThreeDSAuthentication-&gt;setNotificationURL ("HTTPS://YOURURL.COM");</pre> <p>Notification URL for receiving the 3DS Method POST response from the issuer ACS.</p> <p>Conditional. Required if device_channel = 02</p> |
| challenge window size<br><ChallengeWindowSize> | <i>String</i><br>2-character                | <pre>\$mpiThreeDSAuthentication-&gt;setChallengeWindowSize ("CWS_VALUE");</pre>                                                                                                                                                   |

| Variable Name                                | Type and Limits                                                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| >                                            | alphanumeric                                                                    | <p>Relates to the rendering of the ACS challenge within the browser.</p> <p>Allowable values:</p> <p>01 = 250 x 400</p> <p>02 = 390 x 400</p> <p>03 = 500 x 600</p> <p>04 = 600 x 400</p> <p>05 = Full screen</p> <p>Conditional. Required if device_channel = 02</p>                                                                                                                                                  |
| browser IP Address<br><BrowserIP>            | <p><i>String</i></p> <p>Allows '.' and ':'</p> <p>45-character alphanumeric</p> | <pre>\$mpiThreeDSAuthentication-&gt;setBrowserIP("10.10.10.10") or ("011:0db8:85a3:0101:0101:8a2e:0370:7334"); // (IPv4 or IPv6)</pre> <p>IP address of the browser as returned by the HTTP headers to the 3DS Requestor.</p> <div> <p><b>NOTE:</b> This field is not mandatory, but it is required. It is highly recommended to provide. Lack of providing this field, might increase the risk of rejects.</p> </div> |
| browser user agent<br><BrowserUserAgent>     | <p><i>String</i></p> <p>2048-character alphanumeric</p>                         | <pre>\$mpiThreeDSAuthentication-&gt;setBrowserUserAgent("BROWSER_USER_AGENT_VALUE");</pre> <p>Browser User Agent</p> <p>Conditional. Required if device_channel = 02</p>                                                                                                                                                                                                                                               |
| browser java enabled<br><BrowserJavaEnabled> | <p><i>String</i></p> <p>1-character alphabetic</p>                              | <pre>\$mpiThreeDSAuthentication-&gt;setBrowserJavaEnabled(BROWSER_JAVA_VALUE);</pre> <p>Indicates whether Java is enabled in the browser</p> <p>Allowable values:</p> <p>T = True</p> <p>F = False</p> <p>Conditional. Required if device_channel = 02</p>                                                                                                                                                             |

| Variable Name                                  | Type and Limits                                    | Description                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| browser screen height<br><BrowserScreenHeight> | <i>String</i><br><br>6-character<br>numeric        | <pre>\$mpiThreeDSAuthentication-&gt;setBrowserScreenHeight (BROWSER_SCREEN_HEIGHT_VALUE) ;</pre> <p>Pixel height of cardholder screen</p> <p>Conditional. Required if device_channel = 02</p> <div> <b>NOTE:</b> This field is not mandatory, but it is required. It is highly recommended to provide. Lack of providing this field, might increase the risk of rejects. </div> |
| browser screen width<br><BrowserScreenWidth>   | <i>String</i><br><br>6-character<br>numeric        | <pre>\$mpiThreeDSAuthentication-&gt;setBrowserScreenWidth (BROWSER_SCREEN_WIDTH_VALUE) ;</pre> <p>Pixel width of cardholder screen</p> <p>Conditional. Required if device_channel = 02</p> <div> <b>NOTE:</b> This field is not mandatory, but it is required. It is highly recommended to provide. Lack of providing this field, might increase the risk of rejects. </div>    |
| browser language<br><BrowserLanguage>          | <i>String</i><br><br>8-character<br>alphanumeric   | <pre>\$mpiThreeDSAuthentication-&gt;setBrowserLanguage (BROWSER_LANGUAGE_VALUE) ;</pre> <p>As defined in IETF BCP47</p> <p>Conditional. Required if device_channel = 02</p>                                                                                                                                                                                                     |
| email<br><Email>                               | <i>String</i><br><br>254-character<br>alphanumeric | <pre>\$mpiThreeDSAuthentication-&gt;setEmail ("EMAIL_VALUE") ;</pre> <p>Cardholder email address</p> <div> <b>NOTE:</b> This field is not mandatory, but it is required. It is highly recommended to provide the cardholder's email address. Lack of providing the cardholder's address, might increase the risk of rejects. </div>                                             |
| cardholder work phone number<br><WorkPhone>    | <i>Object</i><br><br>N/A                           | <pre>\$mpiThreeDSAuthentication-&gt;setWorkPhone (\$workPhoneTemplate) ;</pre> <p>Cardholder work phone number</p>                                                                                                                                                                                                                                                              |

| Variable Name                                    | Type and Limits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                  |                 | <p><b>NOTE:</b> This field is not mandatory, but it is required. It is highly recommended to provide at least one of the Cardholder Phone Number. Lack of providing at least one of the Cardholder Phone Number, might increase the risk of rejects.</p> <p><b>NOTE:</b> This is a nested object within the transaction. For information about fields in the Cardholder Phone Number Info object, see Cardholder Phone Number Info Object and Variables.</p>                                                                                                                        |
| cardholder home phone number<br><home_phone>     | Object<br>N/A   | <pre>\$mpiThreeDSAuthentication-&gt;setHomePhone(\$homePhoneTemplate);</pre> <p>Cardholder home phone number</p> <p><b>NOTE:</b> This field is not mandatory, but it is required. It is highly recommended to provide at least one of the Cardholder Phone Number. Lack of providing at least one of the Cardholder Phone Number, might increase the risk of rejects.</p> <p><b>NOTE:</b> This is a nested object within the transaction. For information about fields in the Cardholder Phone Number Info object, see Cardholder Phone Number Info Object and Variables.</p>       |
| cardholder mobile phone number<br><mobile_phone> | Object<br>N/A   | <pre>\$mpiThreeDSAuthentication-&gt;setMobilePhone(\$mobilePhoneTemplate);</pre> <p>Cardholder mobile phone number</p> <p><b>NOTE:</b> This field is not mandatory, but it is required. It is highly recommended to provide at least one of the Cardholder Phone Number. Lack of providing at least one of the Cardholder Phone Number, might increase the risk of rejects.</p> <p><b>NOTE:</b> This is a nested object within the transaction. For information about fields in the Cardholder Phone Number Info object, see Cardholder Phone Number Info Object and Variables.</p> |

### MPI 3DS Cardholder Phone Number

| Variable Name | Type and Limits | Description                           |
|---------------|-----------------|---------------------------------------|
| country code  | String          | Country Code of phone number provided |

| Variable Name  | Type and Limits      | Description                                  |
|----------------|----------------------|----------------------------------------------|
| <country_code> | 3-character numeric  | by the Cardholder.                           |
| phone number   | <i>String</i>        | The phone number provided by the Cardholder. |
| <phone_number> | 15-character numeric |                                              |

### MPI 3DS Authentication Request transaction request fields – Optional

| Variable Name                           | Type and Limits                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| currency<br><currency>                  | <i>String</i><br>3-character numeric | <code>\$mpiThreeDSAuthentication-&gt;SetCurrency("CURRENCY_VALUE");</code><br><br>ISO 4217 3 digit currency code<br><br>CAD = 124<br><br>USD = 840<br><br><div style="border: 1px solid black; padding: 5px; background-color: #e0f2f1;"> <b>NOTE:</b> This field should not be sent unless Multi Currency Pricing is enabled on your merchant account         </div>                                                                                     |
| request challenge<br><RequestChallenge> | <i>String</i><br>2-character numeric | <code>\$mpiThreeDSAuthentication-&gt;setRequestChallenge ("CHALLENGE_VALUE");</code><br><br>Indicates whether a browser-based challenge is requested for this transaction. Standard is "01" <ul style="list-style-type: none"> <li>01 = No preference</li> <li>02 = No challenge requested</li> <li>03 = Challenge requested: 3DS Requestor Preference</li> <li>04 = Challenge requested: Mandate</li> </ul> Conditional. Required if device_channel = 02 |

### Sample MPI 3DS Authentication Request - Browser Channel

```
<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id = "moneris";
```

```
$api_token = "hurgle";
$mpiThreeDSAuthentication = new MpiThreeDSAuthentication();
$mpiThreeDSAuthentication->setOrderId("ord-110920-10:36:41"); //must be the same one that was
used in MpiCardLookup call
$mpiThreeDSAuthentication->setCardholderName("Moneris Test");
$mpiThreeDSAuthentication->setPan("347668693641199");
// $mpiThreeDSAuthentication->setDataKey("800XGiwxgvfbZngigVFeld9d2"); //Optional - For
Moneris Vault and Hosted Tokenization tokens in place of setPan
$mpiThreeDSAuthentication->setExpdate("2310");
$mpiThreeDSAuthentication->setAmount("1.00");
$mpiThreeDSAuthentication->setThreeDSCompletionInd("Y"); //(Y|N|U) indicates whether 3ds
method MpiCardLookup was successfully completed
$mpiThreeDSAuthentication->setRequestType("01"); //(01=payment|02=recur)
$mpiThreeDSAuthentication->setPurchaseDate("20200911035249"); //(YYYYMMDDHHMMSS)
$mpiThreeDSAuthentication->setNotificationURL("https://yournotificationurl.com"); //(Website
where response from RRes or CRes after challenge will go)
$mpiThreeDSAuthentication->setChallengeWindowSize("03"); //(01 = 250 x 400, 02 = 390 x 400,
03 = 500 x 600, 04 = 600 x 400, 05 = Full screen)
$mpiThreeDSAuthentication->setBrowserUserAgent("Mozilla/5.0 (Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/80.0.3987.132 Safari/537.36\\");

$mpiThreeDSAuthentication->setBrowserIP ("10.10.10.10") or
("011:0db8:85a3:0101:0101:8a2e:0370:7334"); //(IPv4 or IPv6)

$mpiThreeDSAuthentication->setBrowserJavaEnabled("true"); //(true|false)
$mpiThreeDSAuthentication->setBrowserScreenHeight("1000"); //(pixel height of cardholder
screen)
$mpiThreeDSAuthentication->setBrowserScreenWidth("1920"); //(pixel width of cardholder
screen)
$mpiThreeDSAuthentication->setBrowserLanguage("en-GB"); //(defined by IETF BCP47)
//Optional Methods
$mpiThreeDSAuthentication->setBillAddress1("3300 Bloor St W");
$mpiThreeDSAuthentication->setBillProvince("ON");
$mpiThreeDSAuthentication->setBillCity("Toronto");
$mpiThreeDSAuthentication->setBillPostalCode("M8X 2X2");
$mpiThreeDSAuthentication->setBillCountry("124");
$mpiThreeDSAuthentication->setShipAddress1("3300 Bloor St W");
$mpiThreeDSAuthentication->setShipProvince("ON");
$mpiThreeDSAuthentication->setShipCity("Toronto");
$mpiThreeDSAuthentication->setShipPostalCode("M8X 2X2");
$mpiThreeDSAuthentication->setShipCountry("124");
$mpiThreeDSAuthentication->setEmail("test@email.com");

$workPhoneTemplate = array('cc'=>'1','subscriber'=>'1111111111');
$mpiThreeDSAuthentication->setWorkPhone( $workPhoneTemplate );

$homePhoneTemplate = array('cc'=>'3','subscriber'=>'3333333333');
$mpiThreeDSAuthentication->setHomePhone( $homePhoneTemplate );

$mobilePhoneTemplate = array('cc'=>'2','subscriber'=>'2222222222');
$mpiThreeDSAuthentication->setMobilePhone( $mobilePhoneTemplate );

$mpiThreeDSAuthentication->setRequestChallenge("Y"); //(Y|N Requesting challenge regardless
of outcome)
$mpiThreeDSAuthentication->setMessageCategory("01");
$mpiThreeDSAuthentication->setDeviceChannel("03");
// $mpiThreeDSAuthentication->setDecoupledRequestIndicator("N");
// $mpiThreeDSAuthentication->setDecoupledRequestMaxTime("00010");
// $mpiThreeDSAuthentication->setDecoupledRequestAsyncUrl("localhost");
$mpiThreeDSAuthentication->setRiIndicator("08");
// $mpiThreeDSAuthentication->setRecurringFrequency("031");
// $mpiThreeDSAuthentication->setRecurringExpiry("20251230");
$paiTemplate = array(
'prior_request_auth_data'=>"d7c1ee99-9478-44a6-b1f2-391e29c6b340",
'prior_request_ref'=>"d7c1ee99-9478-44a6-b1f2-391e29c6b340",
```

```

'prior_request_auth_method'=>"01",
'prior_request_auth_timestamp'=>"201710282113"
);
//$mpiThreeDSAuthentication->setPriorAuthenticationInfo( $paiTemplate );
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($mpiThreeDSAuthentication);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false or comment out this line for production transactions
//print_r($mpgRequest);
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nMessageType = " . $mpgResponse->getMpiMessageType());
print("\nTransStatus = " . $mpgResponse->getMpiTransStatus());
print("\nTransStatusReason = " . $mpgResponse->getMpiTransStatusReason());
print("\nChallengeURL = " . $mpgResponse->getMpiChallengeURL());
print("\nChallengeData = " . $mpgResponse->getMpiChallengeData());
print("\nThreeDSServerTransId = " . $mpgResponse->getMpiThreeDSServerTransId());
print("\nDSTransId = " . $mpgResponse->getMpiDSTransId());
print("\nThreeDSVersion = " . $mpgResponse->getThreeDSVersion());
print("\nThreeDSAcTransID = " . $mpgResponse->getMpiThreeDSAcTransID());
print("\nThreeDSAuthTimeStamp = " . $mpgResponse->getMpiThreeDSAuthTimeStamp());
print("\nAuthenticationType = " . $mpgResponse->getMpiAuthenticationType());
print("\nCardHolderInfo = " . $mpgResponse->getMpiCardholderInfo());
//In Frictionless flow, you may receive TransStatus as "Y",
//in which case you can then proceed directly to Cavv Purchase/Preauth with values below'
if($mpgResponse->getMpiTransStatus() == "Y")
{
print("\nCavv = " . $mpgResponse->getMpiCavv());
print("\nECI = " . $mpgResponse->getMpiEci());
}
?>

```

## 6.5.2 MPI 3DS Authentication Request - 3RI with recurring

**NOTE:** Billing address request fields are recommended to be sent for this transaction, or else the authentication process may fail

### MPI 3DS Authentication Request transaction object definition

```
$mpiThreeDSAuthentication = new MpiThreeDSAuthentication();
```

### HttpPostRequest object for MPI 3DS Authentication Request transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgTxn = new mpgTransaction($mpiThreeDSAuthentication);
```



**Core connection object fields (all API transactions)**

| Variable Name | Type and Limits | Description                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| store ID      | <i>String</i>   | 'store_id'=>\$store_id                                                                                                                                                                                                                                                                                                                                                                                                       |
| <store_id>    | N/A             | Unique identifier provided by Moneris upon merchant account setup                                                                                                                                                                                                                                                                                                                                                            |
| API token     | <i>String</i>   | 'api_token'=>\$api_token                                                                                                                                                                                                                                                                                                                                                                                                     |
| <api_token>   | N/A             | <p>Unique alphanumeric string assigned by Moneris upon merchant account activation</p> <p>To find your API token, refer to your test or production store's Admin settings in the Merchant Resource Center, at the following URLs:</p> <p>Testing: <a href="https://esqa.moneris.com/mpg/">https://esqa.moneris.com/mpg/</a></p> <p>Production: <a href="https://www3.moneris.com/mpg/">https://www3.moneris.com/mpg/</a></p> |

**MPI 3DS Authentication Request transaction request fields – Required**

| Variable Name     | Type and Limits     | Description                                                                                                                                                         |
|-------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| message category  | <i>String</i>       | \$mpiThreeDSAuthentication->setMessageCategory("02");                                                                                                               |
| <MessageCategory> | 2-character numeric | <p>Whether the authentication request is for a payment or non-payment use:</p> <p>01 = payment authentication (PA)</p> <p>02 = non-payment authentication (NPA)</p> |
| device channel    | <i>String</i>       | \$mpiThreeDSAuthentication->setDeviceChannel("02");                                                                                                                 |
| <DeviceChannel>   | 2-character numeric | <p>The interface used to initiate the authentication:</p> <p>02 = Browser (BRW)</p>                                                                                 |

| Variable Name                               | Type and Limits                      | Description                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             |                                      | 03 = 3DS Requestor Initiated (3RI)                                                                                                                                                                                                                                                                                                                                           |
| recurring frequency<br><RecurringFrequency> | <i>String</i><br>4-character numeric | <pre>\$mpiThreeDSAuthentication-&gt;setRecurringFrequency("031");</pre> <p>The minimum number of days between recurring transactions. Numeric values between 1 and 9999, leading zeroes accepted.</p> <p>Conditional. Required if ri indicator = 01</p>                                                                                                                      |
| recurring expiry<br><RecurringExpiry>       | <i>String</i><br>8-character numeric | <pre>\$mpiThreeDSAuthentication-&gt;setRecurringExpiry("20221230");</pre> <p>End date after which no further recurring transactions shall be performed. Format is YYYYMMDD.</p> <p>Conditional. Required if ri indicator = 01</p>                                                                                                                                            |
| ri indicator<br><RiIndicator>               | <i>String</i><br>2-character numeric | <pre>\$mpiThreeDSAuthentication-&gt;setRiIndicator("03");</pre> <p>The type of 3DS Requestor Initiated (3RI) request:</p> <p>01 = Recurring</p> <p>02 = Installment</p> <p>03 = Add Card</p> <p>04 = Maintain Card Information</p> <p>05 = Account verification</p> <p>06 = Split/Delayed Shipment</p> <p>07 = Top-up</p> <p>08 = Mail Order</p> <p>09 = Telephone Order</p> |

**NOTE:** Visa Secure only support ri\_Indicator = 01, 02, 06, 08, 09, or 11 for Payment Transactions and ri Indicator = 03, 04, 05, 08, 09 or 10 for Non-Payment Transactions

| Variable Name      | Type and Limits                                           | Description                                                                                                                                                                                                                                                                                                                                        |
|--------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    |                                                           | <p>10 = Whitelist</p> <p>11 = Other Payment</p> <p>Conditional. Required if device_channel = 03</p>                                                                                                                                                                                                                                                |
| order ID           | <i>String</i>                                             | 'order_id'=>\$order_id;                                                                                                                                                                                                                                                                                                                            |
| <order_id>         | 50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces | <p>Merchant-defined transaction identifier that must be unique for every Purchase, Pre-Authorization and Independent Refund transaction. No two transactions of these types may have the same order ID.</p> <p>For Refund, Completion and Purchase Correction transactions, the order ID must be the same as that of the original transaction.</p> |
| data key           | <i>String</i>                                             | 'data_key'=>\$data_key;                                                                                                                                                                                                                                                                                                                            |
| <data_key>         | data key limits:                                          | <b>data key description:</b>                                                                                                                                                                                                                                                                                                                       |
| OR                 | 25-character alphanumeric                                 | Unique identifier for a Vault profile, and used in future Vault financial transactions to associate a transaction with that profile                                                                                                                                                                                                                |
| credit card number | credit card number limits:                                | Data key is generated by Moneris and returned to you in the Receipt object when the profile is first registered                                                                                                                                                                                                                                    |
| <pan>              | max 20-character alphanumeric                             | <p>'pan'=&gt;\$pan</p> <p><b>credit card number description:</b></p> <p>Credit card number, usually 16 digits —field can be maximum 20 digits in support of future expansion of card number ranges.</p> <p>Carries the token for network tokenization transactions.</p>                                                                            |

| Variable Name                       | Type and Limits                                                                                                                                                     | Description                                                                                                                                                                                                   |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| expiry date<br><expdate>            | String<br><br>4-character alphanumeric<br><br>YYMM                                                                                                                  | 'expiry_date'=>\$expiry_date<br><br>Expiry date of the credit card, in YYMM format.<br><br><b>NOTE:</b> This is the reverse of the MMYM date format that is presented on the card.                            |
| amount<br><amount>                  | String<br><br>10-character decimal<br><br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><br><b>EXAMPLE:</b> 1234567.89 | 'amount'=>\$amount<br><br>Transaction dollar amount<br><br>This must contain at least 3 digits, two of which are penny values<br><br>Minimum allowable value = \$0.01, maximum allowable value = \$9999999.99 |
| cardholder name<br><CardholderName> | String<br><br>45-character alphanumeric<br><br><b>NOTE:</b> Accented characters are not allowable                                                                   | \$mpiThreeDSAuthentication->setCardholderName ("CARDHOLDER_NAME_VALUE");<br><br>Name of the cardholder                                                                                                        |
| billing address<br><BillAddress1>   | String<br><br>50-character alphanumeric                                                                                                                             | \$mpiThreeDSAuthentication->setBillAddress1 ("BILL_STREET_ADDRESS_VALUE");<br><br>Cardholder billing address                                                                                                  |
| billing province<br><BillProvince>  | String<br><br>3-character alphanumeric                                                                                                                              | \$mpiThreeDSAuthentication->setBillProvince ("BILL_PROV_VALUE");<br><br>Cardholder province or state<br><br>Defined in country subdivision ISO 3166-2                                                         |
| billing city<br><BillCity>          | String<br><br>50-character alphanumeric                                                                                                                             | \$mpiThreeDSAuthentication->setBillCity ("BILL_CITY_VALUE");                                                                                                                                                  |

| Variable Name                            | Type and Limits                     | Description                                                                                                                                                     |
|------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          |                                     | Cardholder billing city                                                                                                                                         |
| billing postal code<br><BillPostalCode>  | String<br>16-character alphanumeric | \$mpiThreeDSAuthentication->setBillPostalCode ("BILL_POSTAL_CODE_VALUE");<br><br>Cardholder billing postal code                                                 |
| billing country<br><BillCountry>         | String<br>3-character alphanumeric  | \$mpiThreeDSAuthentication->setBillCountry ("BILL_COUNTRY_VALUE");<br><br>Cardholder billing country<br><br>Defined as 3 digit country code ISO 3166-1          |
| shipping address<br><ShipAddress1>       | String<br>50-character alphanumeric | \$mpiThreeDSAuthentication->setShipAddress1 ("SHIP_STREET_ADDRESS_VALUE");<br><br>Shipping destination address                                                  |
| shipping province<br><ShipProvince>      | String<br>3-character alphanumeric  | \$mpiThreeDSAuthentication->setShipProvince ("SHIP_PROV_VALUE");<br><br>Shipping destination province or state<br><br>Defined in country subdivision ISO 3166-2 |
| shipping city<br><ShipCity>              | String<br>50-character alphanumeric | \$mpiThreeDSAuthentication->setShipCity ("SHIP_CITY_VALUE");<br><br>Shipping destination city                                                                   |
| shipping postal code<br><ShipPostalCode> | String<br>16-character alphanumeric | \$mpiThreeDSAuthentication->setShipPostalCode ("SHIP_POSTAL_CODE_VALUE");<br><br>Shipping destination postal or ZIP code                                        |
| shipping country<br><ShipCountry>        | String<br>3-character alphanumeric  | \$mpiThreeDSAuthentication->setShipCountry ("SHIP_COUNTRY_VALUE");                                                                                              |

| Variable Name                                        | Type and Limits                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                      |                                           | Shipping destination country<br><br>Defined as 3-digit country code in ISO 3166-1                                                                                                                                                                                                                                                                                                                                                                                                   |
| email<br><Email>                                     | String<br><br>254-character alpha-numeric | <pre>\$mpiThreeDSAuthentication-&gt;setEmail("EMAIL_VALUE");</pre> <p>Cardholder email address</p> <div> <p><b>NOTE:</b> This field is not mandatory, but it is required. It is highly recommended to provide the cardholder's email address. Lack of providing the cardholder's address, might increase the risk of rejects.</p> </div>                                                                                                                                            |
| prior request auth info<br><PriorAuthenticationInfo> | Object<br><br>N/A                         | <pre>\$mpiThreeDSAuthentication-&gt;setPriorRequestAuthInfo(\$paiTemplate)</pre> <p>Object containing details for a prior 3DS authentication for this series of transactions. This is a nested object within the authentication transaction, and required when storing or using the information about the prior authentication for that card. For information about fields in the Prior Authentication Info object, see MPI 3DS Prior Authentication Info Object and Variables.</p> |

### MPI 3DS Prior Authentication Info

| Variable Name                                        | Type and Limits                         | Description                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| prior request auth data<br><prior_request_auth_data> | String<br><br>36-character alphanumeric | <pre>'prior_request_auth_data'=&gt;"d7c1ee99-9478-44a6-b1f2-391e29c6b340"</pre> <p>Refers to the DSTransID in the response of the previous 3DS authentication.</p> <div> <p><b>NOTE:</b> For MasterCard, the following strict formatting must be followed:<br/><b>dsTransID: UUID</b></p> </div> |

| Variable Name                                                  | Type and Limits                     | Description                                                                                                                                                                                                                                               |
|----------------------------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                |                                     | <div> dsTransID must be capitalized exactly as shown.<br/> A colon (:) must appear immediately after dsTransID with no space.<br/> A single space must follow the colon before the UUID value </div>                                                      |
| prior request ref<br><prior_request_auth_ref>                  | String<br>36-character alphanumeric | 'prior_request_ref'=>"d7c1ee99-9478-44a6-b1f2-391e29c6b340"<br><br>Refers to the 3DS ACS Transaction ID in the response of the previous 3DS authentication.                                                                                               |
| prior request auth method<br><prior_request_auth_method>       | String<br>2-character numeric       | 'prior_request_auth_method'=>"01",<br><br>Mechanism used by the cardholder to authenticate in the previous 3DS authentication:<br><br>01 = Frictionless authentication<br>02 = Challenge authentication<br>03 = AVS verified<br>04 = Other issuer methods |
| prior request auth timestamp<br><prior_request_auth_timestamp> | String<br>12-character numeric      | 'prior_request_auth_timestamp'=>"201710282113",<br><br>Date and time in UTC of the prior cardholder authentication. Found in the previous 3DS authentication response as 3DS Auth TimeStamp.<br><br>Format is YYYYMMDDHHMM.                               |

### MPI 3DS Authentication Request transaction request fields – Optional

| Variable Name          | Type and Limits               | Description                                                                                                                      |
|------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| currency<br><currency> | String<br>3-character numeric | \$mpiThreeDSAuthentication->SetCurrency ("CURRENCY_VALUE") ;<br><br>ISO 4217 3 digit currency code<br><br>CAD = 124<br>USD = 840 |

| Variable Name                                                | Type and Limits                             | Description                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                              |                                             | <div> <b>NOTE:</b> This field should not be sent unless Multi Currency Pricing is enabled on your merchant account </div>                                                                                                                                                                                                                                        |
| decoupled request indicator<br>< DecoupledRequestIndicator > | <i>String</i><br>1-character alphabetic     | <pre>\$mpiThreeDSAuthentication-&gt;setDecoupledRequestIndicator("Y");</pre> <p>Whether the request utilizes Decoupled Authentication or not, if the ACS confirms its use.</p> <p>Y = Decoupled Authentication is supported and preferred if challenge is necessary</p> <p>N = Do not use Decoupled Authentication (Default)</p> <p>Defaults to N if unused.</p> |
| decoupled request max time<br>< DecoupledRequestMaxTime >    | <i>String</i><br>5-character numeric        | <pre>\$mpiThreeDSAuthentication-&gt;setDecoupledRequestMaxTime("00010");</pre> <p>The maximum minutes that Moneris waits for an ACS to provide results.</p> <p>Numeric values between 1 and 10080. The max is equivalent to 7 days.</p> <p>Conditional. Required if device_channel = 03 and decoupled_request_indicator = Y</p>                                  |
| decoupled request async URL<br>< DecoupledRequestAsyncUrl >  | <i>String</i><br>256-character alphanumeric | <pre>\$mpiThreeDSAuthentication-&gt;setDecoupledRequestAsyncUrl("https://yourasyncnotificationurl.com");</pre> <p>Your URL where Moneris will POST the response back from ACS. Moneris reattempts 3 times to POST the response.</p> <p>Conditional. Only sent if decoupled_request_indicator = Y</p>                                                             |



**MPI 3DS Prior Authentication Info**

| Variable Name                                                      | Type and Limits                         | Description                                                                                                                                                                                                                                                           |
|--------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| prior request auth data<br><br><prior_request_auth_data>           | String<br><br>36-character alphanumeric | 'prior_request_auth_data'=>"d7c1ee99-9478-44a6-b1f2-391e29c6b340"<br><br>Refers to the DSTransID in the response of the previous 3DS authentication.                                                                                                                  |
| prior request ref<br><br><prior_request_auth_ref>                  | String<br><br>36-character alphanumeric | 'prior_request_ref'=>"d7c1ee99-9478-44a6-b1f2-391e29c6b340"<br><br>Refers to the 3DS ACS Transaction ID in the response of the previous 3DS authentication.                                                                                                           |
| prior request auth method<br><br><prior_request_auth_method>       | String<br><br>2-character numeric       | 'prior_request_auth_method'=>"01",<br><br>Mechanism used by the cardholder to authenticate in the previous 3DS authentication:<br><br>01 = Frictionless authentication<br><br>02 = Challenge authentication<br><br>03 = AVS verified<br><br>04 = Other issuer methods |
| prior request auth timestamp<br><br><prior_request_auth_timestamp> | String<br><br>12-character numeric      | 'prior_request_auth_timestamp'=>"201710282113",<br><br>Date and time in UTC of the prior cardholder authentication. Found in the previous 3DS authentication response as 3DS Auth TimeStamp.<br><br>Format is YYYYMMDDHHMM.                                           |

**Sample MPI 3DS Authentication Request - 3RI with recurring**

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id = "moneris";
$api_token = "hurgle";
$mpiThreeDSAuthentication = new MpiThreeDSAuthentication();
$mpiThreeDSAuthentication->setOrderId("ord-110920-10:36:41"); //must be the same one that was
used in MpiCardLookup call
$mpiThreeDSAuthentication->setCardholderName("Moneris Test");
$mpiThreeDSAuthentication->setPan("343427006265962");
// $mpiThreeDSAuthentication->setDataKey("800XGiwXgvfbZngigVFeld9d2"); //Optional - For
Moneris Vault and Hosted Tokenization tokens in place of setPan
$mpiThreeDSAuthentication->setExpdate("2310");
$mpiThreeDSAuthentication->setAmount("1.00");

```

```

$mpiThreeDSAuthentication->setThreeDSCompletionInd("Y"); //(Y|N|U) indicates whether 3ds
method MpiCardLookup was successfully completed
$mpiThreeDSAuthentication->setRequestType("02"); //(01=payment|02=recur)
$mpiThreeDSAuthentication->setPurchaseDate("20200911035249"); //(YYYYMMDDHHMMSS)
$mpiThreeDSAuthentication->setNotificationURL("https://yournotificationurl.com"); //(Website
where response from RRes or CRes after challenge will go)
$mpiThreeDSAuthentication->setChallengeWindowSize("03"); //(01 = 250 x 400, 02 = 390 x 400,
03 = 500 x 600, 04 = 600 x 400, 05 = Full screen)
$mpiThreeDSAuthentication->setBrowserUserAgent("Mozilla/5.0 (Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/80.0.3987.132 Safari/537.36\\");
$mpiThreeDSAuthentication->setBrowserJavaEnabled("true"); //(true|false)
$mpiThreeDSAuthentication->setBrowserScreenHeight("1000"); //(pixel height of cardholder
screen)
$mpiThreeDSAuthentication->setBrowserScreenWidth("1920"); //(pixel width of cardholder
screen)
$mpiThreeDSAuthentication->setBrowserLanguage("en-GB"); //(defined by IETF BCP47)
//Optional Methods
$mpiThreeDSAuthentication->setBillAddress1("3300 Bloor St W");
$mpiThreeDSAuthentication->setBillProvince("ON");
$mpiThreeDSAuthentication->setBillCity("Toronto");
$mpiThreeDSAuthentication->setBillPostalCode("M8X 2X2");
$mpiThreeDSAuthentication->setBillCountry("124");
$mpiThreeDSAuthentication->setShipAddress1("3300 Bloor St W");
$mpiThreeDSAuthentication->setShipProvince("ON");
$mpiThreeDSAuthentication->setShipCity("Toronto");
$mpiThreeDSAuthentication->setShipPostalCode("M8X 2X2");
$mpiThreeDSAuthentication->setShipCountry("124");
$mpiThreeDSAuthentication->setEmail("test@email.com");
$mpiThreeDSAuthentication->setRequestChallenge("Y"); //(Y|N Requesting challenge regardless
of outcome)
$mpiThreeDSAuthentication->setMessageCategory("01");
$mpiThreeDSAuthentication->setDeviceChannel("03");
//$mpiThreeDSAuthentication->setDecoupledRequestIndicator("N");
//$mpiThreeDSAuthentication->setDecoupledRequestMaxTime("00010");
//$mpiThreeDSAuthentication->setDecoupledRequestAsyncUrl("localhost");
$mpiThreeDSAuthentication->setRiIndicator("01");
$mpiThreeDSAuthentication->setRecurringFrequency("031");
$mpiThreeDSAuthentication->setRecurringExpiry("20251230");
$paiTemplate = array(
    'prior_request_auth_data'=>"d7clee99-9478-44a6-b1f2-391e29c6b340",
    'prior_request_ref'=>"d7clee99-9478-44a6-b1f2-391e29c6b340",
    'prior_request_auth_method'=>"01",
    'prior_request_auth_timestamp'=>"201710282113"
);
//$mpiThreeDSAuthentication->setPriorAuthenticationInfo( $paiTemplate );
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($mpiThreeDSAuthentication);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false or comment out this line for production transactions
//print_r($mpgRequest);
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nMessageType = " . $mpgResponse->getMpiMessageType());
print("\nTransStatus = " . $mpgResponse->getMpiTransStatus());
print("\nTransStatusReason = " . $mpgResponse->getMpiTransStatusReason());
print("\nChallengeURL = " . $mpgResponse->getMpiChallengeURL());
print("\nChallengeData = " . $mpgResponse->getMpiChallengeData());
print("\nThreeDSServerTransId = " . $mpgResponse->getMpiThreeDSServerTransId());

```

```

print("\nDSTransId = " . $mpgResponse->getMpiDSTransId());
print("\nThreeDSVersion = " . $mpgResponse->getThreeDSVersion());
print("\nThreeDSAcSTransID = " . $mpgResponse->getMpiThreeDSAcSTransID());
print("\nThreeDSAuthTimeStamp = " . $mpgResponse->getMpiThreeDSAuthTimeStamp());
print("\nAuthenticationType = " . $mpgResponse->getMpiAuthenticationType());
print("\nCardHolderInfo = " . $mpgResponse->getMpiCardholderInfo());
//In Frictionless flow, you may receive TransStatus as "Y",
//in which case you can then proceed directly to Cavv Purchase/Preauth with values below'
if($mpgResponse->getMpiTransStatus() == "Y")
{
print("\nCavv = " . $mpgResponse->getMpiCavv());
print("\nECI = " . $mpgResponse->getMpiEci());
}
?>

```

### 6.5.3 MPI 3DS Authentication Request - 3RI, non-recurring

**NOTE:** Billing address request fields are recommended to be sent for this transaction, or else the authentication process may fail

#### MPI 3DS Authentication Request transaction object definition

```
$mpiThreeDSAuthentication = new MpiThreeDSAuthentication();
```

#### HttpPostRequest object for MPI 3DS Authentication Request transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgTxn = new mpgTransaction($mpiThreeDSAuthentication);
```

**WARNING:** Do not send fields related to 3RI on browser-based authentications.

#### Core connection object fields (all API transactions)

| Variable Name | Type and Limits | Description                                                          |
|---------------|-----------------|----------------------------------------------------------------------|
| store ID      | <i>String</i>   | 'store_id'=>\$store_id                                               |
| <store_id>    | N/A             | Unique identifier provided by Moneris upon merchant account setup    |
| API token     | <i>String</i>   | 'api_token'=>\$api_token                                             |
| <api_token>   | N/A             | Unique alphanumeric string assigned by Moneris upon merchant account |

| Variable Name | Type and Limits | Description                                                                                                                                                                                                                                                                                                                                             |
|---------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |                 | <p>activation</p> <p>To find your API token, refer to your test or production store's Admin settings in the Merchant Resource Center, at the following URLs:</p> <p>Testing: <a href="https://esqa.moneris.com/mpg/">https://esqa.moneris.com/mpg/</a></p> <p>Production: <a href="https://www3.moneris.com/mpg/">https://www3.moneris.com/mpg/</a></p> |

### MPI 3DS Authentication Request transaction request fields – Required

| Variable Name                         | Type and Limits                      | Description                                                                                                                                                                                                                             |
|---------------------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| message category<br><MessageCategory> | <i>String</i><br>2-character numeric | <pre>\$mpiThreeDSAuthentication-&gt;setMessageCategory("02");</pre> <p>Whether the authentication request is for a payment or non-payment use:</p> <p>01 = payment authentication (PA)</p> <p>02 = non-payment authentication (NPA)</p> |
| device channel<br><DeviceChannel>     | <i>String</i><br>2-character numeric | <pre>\$mpiThreeDSAuthentication-&gt;setDeviceChannel("02");</pre> <p>The interface used to initiate the authentication:</p> <p>02 = Browser (BRW)</p> <p>03 = 3DS Requestor Initiated (3RI)</p>                                         |
| ri indicator<br><RiIndicator>         | <i>String</i><br>2-character numeric | <pre>\$mpiThreeDSAuthentication-&gt;setRiIndicator("03");</pre> <p>The type of 3DS Requestor Initiated (3RI) request:</p> <p>01 = Recurring</p>                                                                                         |

**NOTE:** Visa Secure only support ri\_Indicator = 01, 02, 06,

| Variable Name                                                                                                                     | Type and Limits                                                                                                        | Description                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div> 08, 09, or 11 for Payment Transactions and ri Indicator<br/> = 03, 04, 05, 08, 09 or 10 for Non-Payment Transactions </div> |                                                                                                                        | 02 = Installment<br>03 = Add Card<br>04 = Maintain Card Information<br>05 = Account verification<br>06 = Split/Delayed Shipment<br>07 = Top-up<br>08 = Mail Order<br>09 = Telephone Order<br>10 = Whitelist<br>11 = Other Payment                                                                                                                                          |
| order ID<br><order_id>                                                                                                            | String<br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                                                    | 'order_id'=>\$order_id;<br><br>Merchant-defined transaction identifier that must be unique for every Purchase, Pre-Authorization and Independent Refund transaction. No two transactions of these types may have the same order ID.<br><br>For Refund, Completion and Purchase Correction transactions, the order ID must be the same as that of the original transaction. |
| data key<br><data_key><br>OR<br>credit card number<br><pan>                                                                       | String<br>data key limits:<br>25-character alphanumeric<br>credit card number limits:<br>max 20-character alphanumeric | 'data_key'=>\$data_key;<br><br><b>data key description:</b><br><br>Unique identifier for a Vault profile, and used in future Vault financial transactions to associate a transaction with that profile<br><br>Data key is generated by Moneris and returned to you in the Receipt object when the profile is first registered                                              |

| Variable Name                       | Type and Limits                                                                                                                                                                                | Description                                                                                                                                                                                                                                                             |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     |                                                                                                                                                                                                | <p>'pan'=&gt;\$pan</p> <p><b>credit card number description:</b></p> <p>Credit card number, usually 16 digits —field can be maximum 20 digits in support of future expansion of card number ranges.</p> <p>Carries the token for network tokenization transactions.</p> |
| expiry date<br><expdate>            | <p><i>String</i></p> <p>4-character alphanumeric<br/>YYMM</p>                                                                                                                                  | <p>'expiry_date'=&gt;\$expiry_date</p> <p>Expiry date of the credit card, in YYMM format.</p> <div> <p><b>NOTE:</b> This is the reverse of the MMYM date format that is presented on the card.</p> </div>                                                               |
| amount<br><amount>                  | <p><i>String</i></p> <p>10-character decimal</p> <p>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point</p> <div> <p><b>EXAMPLE:</b> 1234567.89</p> </div> | <p>'amount'=&gt;\$amount</p> <p>Transaction dollar amount</p> <p>This must contain at least 3 digits, two of which are penny values</p> <p>Minimum allowable value = \$0.01, maximum allowable value = \$9999999.99</p>                                                 |
| cardholder name<br><CardholderName> | <p><i>String</i></p> <p>45-character alphanumeric</p> <div> <p><b>NOTE:</b> Accented characters are not allowable</p> </div>                                                                   | <p>\$mpiThreeDSAuthentication-&gt;setCardholderName ("CARDHOLDER_NAME_VALUE");</p> <p>Name of the cardholder</p>                                                                                                                                                        |
| billing address<br><BillAddress1>   | <p><i>String</i></p> <p>50-character alphanumeric</p>                                                                                                                                          | <p>\$mpiThreeDSAuthentication-&gt;setBillAddress1 ("BILL_STREET_ADDRESS_VALUE");</p> <p>Cardholder billing address</p>                                                                                                                                                  |

| Variable Name                           | Type and Limits                                | Description                                                                                                                                                                  |
|-----------------------------------------|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| billing province<br><BillProvince>      | <i>String</i><br><br>3-character alphanumeric  | <pre>\$mpiThreeDSAuthentication-&gt;setBillProvince("BILL_PROV_VALUE");</pre> <p>Cardholder province or state</p> <p>Defined in country subdivision ISO 3166-2</p>           |
| billing city<br><BillCity>              | <i>String</i><br><br>50-character alphanumeric | <pre>\$mpiThreeDSAuthentication-&gt;setBillCity("BILL_CITY_VALUE");</pre> <p>Cardholder billing city</p>                                                                     |
| billing postal code<br><BillPostalCode> | <i>String</i><br><br>16-character alphanumeric | <pre>\$mpiThreeDSAuthentication-&gt;setBillPostalCode("BILL_POSTAL_CODE_VALUE");</pre> <p>Cardholder billing postal code</p>                                                 |
| billing country<br><BillCountry>        | <i>String</i><br><br>3-character alphanumeric  | <pre>\$mpiThreeDSAuthentication-&gt;setBillCountry("BILL_COUNTRY_VALUE");</pre> <p>Cardholder billing country</p> <p>Defined as 3 digit country code ISO 3166-1</p>          |
| shipping address<br><ShipAddress1>      | <i>String</i><br><br>50-character alphanumeric | <pre>\$mpiThreeDSAuthentication-&gt;setShipAddress1("SHIP_STREET_ADDRESS_VALUE");</pre> <p>Shipping destination address</p>                                                  |
| shipping province<br><ShipProvince>     | <i>String</i><br><br>3-character alphanumeric  | <pre>\$mpiThreeDSAuthentication-&gt;setShipProvince("SHIP_PROV_VALUE");</pre> <p>Shipping destination province or state</p> <p>Defined in country subdivision ISO 3166-2</p> |
| shipping city<br><ShipCity>             | <i>String</i><br><br>50-character alphanumeric | <pre>\$mpiThreeDSAuthentication-&gt;setShipCity("SHIP_CITY_VALUE");</pre>                                                                                                    |

| Variable Name                            | Type and Limits                                 | Description                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shipping destination city                |                                                 |                                                                                                                                                                                                                                                                                                                                |
| shipping postal code<br><ShipPostalCode> | <i>String</i><br><br>16-character alphanumeric  | <code>\$mpiThreeDSAuthentication-&gt;setShipPostalCode ("SHIP_POSTAL_CODE_VALUE");</code><br><br>Shipping destination postal or ZIP code                                                                                                                                                                                       |
| shipping country<br><ShipCountry>        | <i>String</i><br><br>3-character alphanumeric   | <code>\$mpiThreeDSAuthentication-&gt;setShipCountry ("SHIP_COUNTRY_VALUE");</code><br><br>Shipping destination country<br><br>Defined as 3-digit country code in ISO 3166-1                                                                                                                                                    |
| email<br><Email>                         | <i>String</i><br><br>254-character alphanumeric | <code>\$mpiThreeDSAuthentication-&gt;setEmail ("EMAIL_VALUE");</code><br><br>Cardholder email address<br><br><b>NOTE:</b> This field is not mandatory, but it is required. It is highly recommended to provide the cardholder's email address. Lack of providing the cardholder's address, might increase the risk of rejects. |

### MPI 3DS Authentication Request transaction request fields – Optional

| Variable Name               | Type and Limits                          | Description                                                                                                                                                                                                                                                             |
|-----------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| currency<br><currency>      | <i>String</i><br><br>3-character numeric | <code>\$mpiThreeDSAuthentication-&gt;SetCurrency ("CURRENCY_VALUE");</code><br><br>ISO 4217 3 digit currency code<br><br>CAD = 124<br><br>USD = 840<br><br><b>NOTE:</b> This field should not be sent unless Multi Currency Pricing is enabled on your merchant account |
| decoupled request indicator | <i>String</i>                            | <code>\$mpiThreeDSAuthentication-&gt;setDecoupledRequestIndicator ("Y");</code>                                                                                                                                                                                         |



| Variable Name                                                            | Type and Limits                                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <<br>DecoupledRequestIndicator<br>>                                      | 1-character<br>alphabetic                          | Whether the request utilizes Decoupled Authentication or not, if the ACS confirms its use.<br><br>Y = Decoupled Authentication is supported and preferred if challenge is necessary<br><br>N = Do not use Decoupled Authentication (Default)<br><br>Defaults to N if unused.                                                                                                                                                                                                  |
| decoupled request max<br>time<br><br><<br>DecoupledRequestMaxTime<br>>   | <i>String</i><br><br>5-character<br>numeric        | <pre>\$mpiThreeDSAuthentication- &gt;setDecoupledRequestMaxTime("00010");</pre> The maximum minutes that Moneris waits for an ACS to provide results.<br><br>Numeric values between 1 and 10080. The max is equivalent to 7 days.<br><br>Conditional. Required if device_channel = 03 and decoupled_request_indicator = Y                                                                                                                                                     |
| decoupled request async<br>URL<br><br><<br>DecoupledRequestAsyncUrl<br>> | <i>String</i><br><br>256-character<br>alphanumeric | <pre>\$mpiThreeDSAuthentication- &gt;setDecoupledRequestAsyncUrl ("https://yourasyncnotificationurl.com");</pre> Your URL where Moneris will POST the response back from ACS. Moneris reattempts 3 times to POST the response.<br><br>Conditional. Only sent if decoupled-request-indicator = Y                                                                                                                                                                               |
| prior request auth info<br><PriorAuthenticationInfo>                     | <i>Object</i><br><br>N/A                           | <pre>\$mpiThreeDSAuthentication- &gt;setPriorRequestAuthInfo(\$paiTemplate)</pre> Object containing details for a prior 3DS authentication for this series of transactions. This is a nested object within the authentication transaction, and required when storing or using the information about the prior authentication for that card. For information about fields in the Prior Authentication Info object, see MPI 3DS Prior Authentication Info Object and Variables. |

**MPI 3DS Prior Authentication Info**

| Variable Name                                                      | Type and Limits                         | Description                                                                                                                                                                                                                                                           |
|--------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| prior request auth data<br><br><prior_request_auth_data>           | String<br><br>36-character alphanumeric | 'prior_request_auth_data'=>"d7c1ee99-9478-44a6-b1f2-391e29c6b340"<br><br>Refers to the DSTransID in the response of the previous 3DS authentication.                                                                                                                  |
| prior request ref<br><br><prior_request_auth_ref>                  | String<br><br>36-character alphanumeric | 'prior_request_ref'=>"d7c1ee99-9478-44a6-b1f2-391e29c6b340"<br><br>Refers to the 3DS ACS Transaction ID in the response of the previous 3DS authentication.                                                                                                           |
| prior request auth method<br><br><prior_request_auth_method>       | String<br><br>2-character numeric       | 'prior_request_auth_method'=>"01",<br><br>Mechanism used by the cardholder to authenticate in the previous 3DS authentication:<br><br>01 = Frictionless authentication<br><br>02 = Challenge authentication<br><br>03 = AVS verified<br><br>04 = Other issuer methods |
| prior request auth timestamp<br><br><prior_request_auth_timestamp> | String<br><br>12-character numeric      | 'prior_request_auth_timestamp'=>"201710282113",<br><br>Date and time in UTC of the prior cardholder authentication. Found in the previous 3DS authentication response as 3DS Auth TimeStamp.<br><br>Format is YYYYMMDDHHMM.                                           |

**Sample MPI 3DS Authentication Request - 3RI without recurring**

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id = "moneris";
$api_token = "hurgle";
$mpiThreeDSAuthentication = new MpiThreeDSAuthentication();
$mpiThreeDSAuthentication->setOrderId("ord-110920-10:36:41"); //must be the same one that was
used in MpiCardLookup call
$mpiThreeDSAuthentication->setCardholderName("Moneris Test");
$mpiThreeDSAuthentication->setPan("347668693641199");
// $mpiThreeDSAuthentication->setDataKey("800XGiwXgvfbZngigVFeld9d2"); //Optional - For
Moneris Vault and Hosted Tokenization tokens in place of setPan
$mpiThreeDSAuthentication->setExpdate("2310");
$mpiThreeDSAuthentication->setAmount("1.00");

```

```

$mpiThreeDSAuthentication->setThreeDSCompletionInd("Y"); //(Y|N|U) indicates whether 3ds
method MpiCardLookup was successfully completed
$mpiThreeDSAuthentication->setRequestType("01"); //(01=payment|02=recur)
$mpiThreeDSAuthentication->setPurchaseDate("20200911035249"); //(YYYYMMDDHHMMSS)
$mpiThreeDSAuthentication->setNotificationURL("https://yournotificationurl.com"); //(Website
where response from RRes or CRes after challenge will go)
$mpiThreeDSAuthentication->setChallengeWindowSize("03"); //(01 = 250 x 400, 02 = 390 x 400,
03 = 500 x 600, 04 = 600 x 400, 05 = Full screen)
$mpiThreeDSAuthentication->setBrowserUserAgent("Mozilla/5.0 (Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/80.0.3987.132 Safari/537.36\\");
$mpiThreeDSAuthentication->setBrowserJavaEnabled("true"); //(true|false)
$mpiThreeDSAuthentication->setBrowserScreenHeight("1000"); //(pixel height of cardholder
screen)
$mpiThreeDSAuthentication->setBrowserScreenWidth("1920"); //(pixel width of cardholder
screen)
$mpiThreeDSAuthentication->setBrowserLanguage("en-GB"); //(defined by IETF BCP47)
//Optional Methods
$mpiThreeDSAuthentication->setBillAddress1("3300 Bloor St W");
$mpiThreeDSAuthentication->setBillProvince("ON");
$mpiThreeDSAuthentication->setBillCity("Toronto");
$mpiThreeDSAuthentication->setBillPostalCode("M8X 2X2");
$mpiThreeDSAuthentication->setBillCountry("124");
$mpiThreeDSAuthentication->setShipAddress1("3300 Bloor St W");
$mpiThreeDSAuthentication->setShipProvince("ON");
$mpiThreeDSAuthentication->setShipCity("Toronto");
$mpiThreeDSAuthentication->setShipPostalCode("M8X 2X2");
$mpiThreeDSAuthentication->setShipCountry("124");
$mpiThreeDSAuthentication->setEmail("test@email.com");
$mpiThreeDSAuthentication->setRequestChallenge("Y"); //(Y|N Requesting challenge regardless
of outcome)
$mpiThreeDSAuthentication->setMessageCategory("01");
$mpiThreeDSAuthentication->setDeviceChannel("03");
//$mpiThreeDSAuthentication->setDecoupledRequestIndicator("N");
//$mpiThreeDSAuthentication->setDecoupledRequestMaxTime("00010");
//$mpiThreeDSAuthentication->setDecoupledRequestAsyncUrl("localhost");
$mpiThreeDSAuthentication->setRiIndicator("01");
// $mpiThreeDSAuthentication->setRecurringFrequency("031");
// $mpiThreeDSAuthentication->setRecurringExpiry("20251230");
$paiTemplate = array(
    'prior_request_auth_data'=>"d7clee99-9478-44a6-b1f2-391e29c6b340",
    'prior_request_ref'=>"d7clee99-9478-44a6-b1f2-391e29c6b340",
    'prior_request_auth_method'=>"01",
    'prior_request_auth_timestamp'=>"201710282113"
);
$mpiThreeDSAuthentication->setPriorAuthenticationInfo( $paiTemplate );
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($mpiThreeDSAuthentication);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false or comment out this line for production transactions
//print_r($mpgRequest);
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nMessageType = " . $mpgResponse->getMpiMessageType());
print("\nTransStatus = " . $mpgResponse->getMpiTransStatus());
print("\nTransStatusReason = " . $mpgResponse->getMpiTransStatusReason());
print("\nChallengeURL = " . $mpgResponse->getMpiChallengeURL());
print("\nChallengeData = " . $mpgResponse->getMpiChallengeData());
print("\nThreeDSServerTransId = " . $mpgResponse->getMpiThreeDSServerTransId());

```

```

print("\nDSTransID = " . $mpgResponse->getMpiDSTransID());
print("\nThreeDSVersion = " . $mpgResponse->getThreeDSVersion());
print("\nThreeDSAcSTransID = " . $mpgResponse->getMpiThreeDSAcSTransID());
print("\nThreeDSAuthTimeStamp = " . $mpgResponse->getMpiThreeDSAuthTimeStamp());
print("\nAuthenticationType = " . $mpgResponse->getMpiAuthenticationType());
print("\nCardHolderInfo = " . $mpgResponse->getMpiCardholderInfo());
//In Frictionless flow, you may receive TransStatus as "Y",
//in which case you can then proceed directly to Cavv Purchase/Preauth with values below'
if($mpgResponse->getMpiTransStatus() == "Y")
{
print("\nCavv = " . $mpgResponse->getMpiCavv());
print("\nECI = " . $mpgResponse->getMpiEci());
}
?>

```

## 6.6 Handling the Challenge Flow

If you get a TransStatus = “C” in your threeDSAuthentication Response, then a form must be built and POSTed to the URL provided.

The form can be dynamically generated and added to the DOM and submitted or created and submitted in a manner that suits your environment. This can be built as a full page redirect or presented as an inline iframe or as a lightbox.

If you wish for this to be loaded inside a defined space it must conform to the size specified in the ChallengeWindowSize from the request. The “action” is retrieved from the ChallengeURL and the “creq” field is retrieved from the ChallengeData.

Below is a sample of a basic static form to help visualize the data and fields that need to be submitted.

```

<form method="POST" action="https://3dsurl.example.com/do3DS">
<input name="creq" value="thisissamplechallengedata1234567890">
</form>

```

### 6.6.1 Cavv Lookup Request – mpiCavvLookup

(Challenge Flow Only)

In the challenge flow, the 3DS server will POST a **creq** value back to the notificationURL provided in the threeDSAuthentication request once the cardholder has completed the challenge. The “creq” is then posted to the Moneris 3DS server in the CavvLookup request, the response to this request will include the result of the challenge, which will include the eci and the cavv if the challenge was successful.

#### Cavv Lookup Request transaction object definition

```
$mpiCavvLookup = new MpiCavvLookup();
```

#### HttpPostRequest object for Cavv Lookup Request transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgTxn = new mpgTransaction($mpiCavvLookup);
```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String          | 'store_id'=>\$store_id   |
|               | N/A             |                          |
| API token     | String          | 'api_token'=>\$api_token |
|               | N/A             |                          |

### Cavv Lookup Request transaction request fields – Required

| Variable Name | Type and Limits                              |                                                   |
|---------------|----------------------------------------------|---------------------------------------------------|
| cres          | <i>String</i><br>200-character alpha-numeric | <code>\$mpiCavvLookup-&gt;setCRes(\$cres);</code> |

## Sample Cavv Lookup Request

```
<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id = "moneris";
$sapi_token = "hurgle";
//BASE64 Encoded CRes value returned from response at completion of challenge flow.
$cres =
"eyJhY3NUcmFuc0EiIjoiazQ0ZDI2NjUtnjU2Yy00ZGNlTg3MWUtYTBkYmMwODA0OTYzIiwibWVzc2FnZVR5cGUOiJD
UmVzIiwiaY2hhbGxlbmdldQ29tcGxldGlvbkluZCI6IlkiLCJtZXNzYWdlVmVyc2l2b2I6IjIuMS4wIiwidHJhbnNtDGfOd
XmiOiJZiIiwidGhyZWVUeUlnclnZlclRyYw5zSUQiOiJlMTFkNDk4NS04ZDI1LTQwZWZwOT1kNiJmZgM2ZlNWU2OGYifQ
==";

$mpiCavvLookup = new MpiCavvLookup();
$mpiCavvLookup->setCRes($cres);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($mpiCavvLookup);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false or comment out this line for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id, $sapi_token, $mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nMessageType = " . $mpgResponse->getMpiMessageType());
print("\nThreeDSServerTransId = " . $mpgResponse->getMpiThreeDSServerTransId());
print("\nTransStatus = " . $mpgResponse->getMpiTransStatus());
print("\nChallengeCompletionIndicator = " . $mpgResponse->getMpiChallengeCompletionIndicator
());
print("\nCavv = " . $mpgResponse->getMpiCavv());
print("\nECI = " . $mpgResponse->getMpiEci());
?>
```

## 6.7 Handling the Decoupled Authentication Flow

If you get a TransStatus = “D” in your threeDSAuthentication Response, then your server must be prepared to accept a second asynchronous response from Moneris.

The cardholder will be engaged by their issuer for cardholder authentication outside the 3DS protocol. This may involve alternate authentication applications or SMS prompts to the cardholder to confirm.

The cardholder is given up to 7 days to complete this decoupled challenge. Once completed, the issuer will communicate to Moneris and our MPI system sends a second 3DS Authentication Response to the address you define in <decoupled\_request\_async\_url>

### Sample Authentication Request – 3RI With Decoupled Authentication for MOTO

```
<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id = "moneris";
$sapi_token = "hurgle";
$mpiThreeDSAuthentication = new MpiThreeDSAuthentication();
$mpiThreeDSAuthentication->setOrderId("ord-110920-10:36:41"); //must be the same one that was
used in MpiCardLookup call
$mpiThreeDSAuthentication->setCardholderName("Moneris Test");
$mpiThreeDSAuthentication->setPan("347668693641199");
// $mpiThreeDSAuthentication->setDataKey("800XGiwxgvfbZngigVFeld9d2"); //Optional - For
Moneris Vault and Hosted Tokenization tokens in place of setPan
$mpiThreeDSAuthentication->setExpdate("2310");
$mpiThreeDSAuthentication->setAmount("1.00");
$mpiThreeDSAuthentication->setThreeDSCompletionInd("Y"); // (Y|N|U) indicates whether 3ds
method MpiCardLookup was successfully completed
$mpiThreeDSAuthentication->setRequestType("01"); // (01=payment|02=recur)
$mpiThreeDSAuthentication->setPurchaseDate("20200911035249"); // (YYYYMMDDHHMMSS)
$mpiThreeDSAuthentication->setNotificationURL("https://yournotificationurl.com"); // (Website
where response from RRes or CRes after challenge will go)
$mpiThreeDSAuthentication->setChallengeWindowSize("03"); // (01 = 250 x 400, 02 = 390 x 400,
03 = 500 x 600, 04 = 600 x 400, 05 = Full screen)
$mpiThreeDSAuthentication->setBrowserUserAgent("Mozilla/5.0 (Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/80.0.3987.132 Safari/537.36");
$mpiThreeDSAuthentication->setBrowserJavaEnabled("true"); // (true|false)
$mpiThreeDSAuthentication->setBrowserScreenHeight("1000"); // (pixel height of cardholder
screen)
$mpiThreeDSAuthentication->setBrowserScreenWidth("1920"); // (pixel width of cardholder
screen)
$mpiThreeDSAuthentication->setBrowserLanguage("en-GB"); // (defined by IETF BCP47)
// Optional Methods
$mpiThreeDSAuthentication->setBillAddress1("3300 Bloor St W");
$mpiThreeDSAuthentication->setBillProvince("ON");
$mpiThreeDSAuthentication->setBillCity("Toronto");
$mpiThreeDSAuthentication->setBillPostalCode("M8X 2X2");
$mpiThreeDSAuthentication->setBillCountry("124");
$mpiThreeDSAuthentication->setShipAddress1("3300 Bloor St W");
$mpiThreeDSAuthentication->setShipProvince("ON");
$mpiThreeDSAuthentication->setShipCity("Toronto");
$mpiThreeDSAuthentication->setShipPostalCode("M8X 2X2");
$mpiThreeDSAuthentication->setShipCountry("124");
$mpiThreeDSAuthentication->setEmail("test@email.com");
$mpiThreeDSAuthentication->setRequestChallenge("Y"); // (Y|N Requesting challenge regardless
of outcome)
$mpiThreeDSAuthentication->setMessageCategory("01");
$mpiThreeDSAuthentication->setDeviceChannel("03");
$mpiThreeDSAuthentication->setDecoupledRequestIndicator("N");
$mpiThreeDSAuthentication->setDecoupledRequestMaxTime("00010");
```

```

$mpiThreeDSAuthentication->setDecoupledRequestAsyncUrl("localhost");
$mpiThreeDSAuthentication->setRiIndicator("08");
// $mpiThreeDSAuthentication->setRecurringFrequency("031");
// $mpiThreeDSAuthentication->setRecurringExpiry("20251230");
$paiTemplate = array(
    'prior_request_auth_data'=>"d7clee99-9478-44a6-blf2-391e29c6b340",
    'prior_request_ref'=>"d7clee99-9478-44a6-blf2-391e29c6b340",
    'prior_request_auth_method'=>"01",
    'prior_request_auth_timestamp'=>"201710282113"
);
// $mpiThreeDSAuthentication->setPriorAuthenticationInfo( $paiTemplate );
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($mpiThreeDSAuthentication);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false or comment out this line for production transactions
// print_r($mpgRequest);
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nMessageType = " . $mpgResponse->getMpiMessageType());
print("\nTransStatus = " . $mpgResponse->getMpiTransStatus());
print("\nTransStatusReason = " . $mpgResponse->getMpiTransStatusReason());
print("\nChallengeURL = " . $mpgResponse->getMpiChallengeURL());
print("\nChallengeData = " . $mpgResponse->getMpiChallengeData());
print("\nThreeDSServerTransId = " . $mpgResponse->getMpiThreeDSServerTransId());
print("\nDSTransId = " . $mpgResponse->getMpiDSTransId());
print("\nThreeDSVersion = " . $mpgResponse->getThreeDSVersion());
print("\nThreeDSAcTransID = " . $mpgResponse->getMpiThreeDSAcTransID());
print("\nThreeDSAuthTimeStamp = " . $mpgResponse->getMpiThreeDSAuthTimeStamp());
print("\nAuthenticationType = " . $mpgResponse->getMpiAuthenticationType());
print("\nCardHolderInfo = " . $mpgResponse->getMpiCardholderInfo());
// In Frictionless flow, you may receive TransStatus as "Y",
// in which case you can then proceed directly to Cavv Purchase/Preauth with values below'
if($mpgResponse->getMpiTransStatus() == "Y")
{
    print("\nCavv = " . $mpgResponse->getMpiCavv());
    print("\nECI = " . $mpgResponse->getMpiEci());
}
?>

```

## Sample Authentication Request – 3RI With Prior Authentication and Decoupled Authentication

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id = "moneris";
$api_token = "hurgle";
$mpiThreeDSAuthentication = new MpiThreeDSAuthentication();
$mpiThreeDSAuthentication->setOrderId("ord-110920-10:36:41"); // must be the same one that was
used in MpiCardLookup call
$mpiThreeDSAuthentication->setCardholderName("Moneris Test");
$mpiThreeDSAuthentication->setPan("347668693641199");
// $mpiThreeDSAuthentication->setDataKey("800XGiwXgvfbZngigVFeld9d2"); // Optional - For
Moneris Vault and Hosted Tokenization tokens in place of setPan
$mpiThreeDSAuthentication->setExpdate("2310");
$mpiThreeDSAuthentication->setAmount("1.00");
$mpiThreeDSAuthentication->setThreeDSCompletionInd("Y"); // (Y|N|U) indicates whether 3ds
method MpiCardLookup was successfully completed

```

```

$mpiThreeDSAuthentication->setRequestType("01"); //(01=payment|02=recur)
$mpiThreeDSAuthentication->setPurchaseDate("20200911035249"); //(YYYYMMDDHHMMSS)
$mpiThreeDSAuthentication->setNotificationURL("https://yournotificationurl.com"); //(Website
where response from RRes or CRes after challenge will go)
$mpiThreeDSAuthentication->setChallengeWindowSize("03"); //(01 = 250 x 400, 02 = 390 x 400,
03 = 500 x 600, 04 = 600 x 400, 05 = Full screen)
$mpiThreeDSAuthentication->setBrowserUserAgent("Mozilla/5.0 (Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/80.0.3987.132 Safari/537.36\\");
$mpiThreeDSAuthentication->setBrowserJavaEnabled("true"); //(true|false)
$mpiThreeDSAuthentication->setBrowserScreenHeight("1000"); //(pixel height of cardholder
screen)
$mpiThreeDSAuthentication->setBrowserScreenWidth("1920"); //(pixel width of cardholder
screen)
$mpiThreeDSAuthentication->setBrowserLanguage("en-GB"); //(defined by IETF BCP47)
//Optional Methods
$mpiThreeDSAuthentication->setBillAddress1("3300 Bloor St W");
$mpiThreeDSAuthentication->setBillProvince("ON");
$mpiThreeDSAuthentication->setBillCity("Toronto");
$mpiThreeDSAuthentication->setBillPostalCode("M8X 2X2");
$mpiThreeDSAuthentication->setBillCountry("124");
$mpiThreeDSAuthentication->setShipAddress1("3300 Bloor St W");
$mpiThreeDSAuthentication->setShipProvince("ON");
$mpiThreeDSAuthentication->setShipCity("Toronto");
$mpiThreeDSAuthentication->setShipPostalCode("M8X 2X2");
$mpiThreeDSAuthentication->setShipCountry("124");
$mpiThreeDSAuthentication->setEmail("test@email.com");
$mpiThreeDSAuthentication->setRequestChallenge("Y"); //(Y|N Requesting challenge regardless
of outcome)
$mpiThreeDSAuthentication->setMessageCategory("01");
$mpiThreeDSAuthentication->setDeviceChannel("03");
$mpiThreeDSAuthentication->setDecoupledRequestIndicator("N");
$mpiThreeDSAuthentication->setDecoupledRequestMaxTime("00010");
$mpiThreeDSAuthentication->setDecoupledRequestAsyncUrl("localhost");
$mpiThreeDSAuthentication->setRiIndicator("08");
// $mpiThreeDSAuthentication->setRecurringFrequency("031");
// $mpiThreeDSAuthentication->setRecurringExpiry("20251230");
$paiTemplate = array(
'prior_request_auth_data'=>"d7clee99-9478-44a6-b1f2-391e29c6b340",
'prior_request_ref'=>"d7clee99-9478-44a6-b1f2-391e29c6b340",
'prior_request_auth_method'=>"01",
'prior_request_auth_timestamp'=>"201710282113"
);
$mpiThreeDSAuthentication->setPriorAuthenticationInfo( $paiTemplate );
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($mpiThreeDSAuthentication);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false or comment out this line for production transactions
//print_r($mpgRequest);
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nMessageType = " . $mpgResponse->getMpiMessageType());
print("\nTransStatus = " . $mpgResponse->getMpiTransStatus());
print("\nTransStatusReason = " . $mpgResponse->getMpiTransStatusReason());
print("\nChallengeURL = " . $mpgResponse->getMpiChallengeURL());
print("\nChallengeData = " . $mpgResponse->getMpiChallengeData());
print("\nThreeDSServerTransId = " . $mpgResponse->getMpiThreeDSServerTransId());
print("\nDSTransId = " . $mpgResponse->getMpiDSTransId());
print("\nThreeDSVersion = " . $mpgResponse->getThreeDSVersion());
print("\nThreeDSAcTransID = " . $mpgResponse->getMpiThreeDSAcTransID());

```



```
print("\nThreeDSAuthTimeStamp = " . $mpgResponse->getMpiThreeDSAuthTimeStamp());
print("\nAuthenticationType = " . $mpgResponse->getMpiAuthenticationType());
print("\nCardHolderInfo = " . $mpgResponse->getMpiCardholderInfo());
//In Frictionless flow, you may receive TransStatus as "Y",
//in which case you can then proceed directly to Cavv Purchase/Preauth with values below'
if($mpgResponse->getMpiTransStatus() == "Y")
{
    print("\nCavv = " . $mpgResponse->getMpiCavv());
    print("\nECI = " . $mpgResponse->getMpiEci());
}
?>
```

## 6.8 Performing the Authorization

Once the authentication is complete and a CAVV and ECI value are retrieved, these values can be sent to Moneris using the transactions Purchase with 3-D Secure – cavv\_purchase or Pre-Authorization with 3-D Secure – cavv\_preauth.

### 6.8.1 Purchase with 3-D Secure – cavv\_purchase

The Purchase with 3-D Secure transaction follows a 3-D Secure MPI authentication. After receiving confirmation from the MPI ACS transaction, this Purchase verifies funds on the customer's card, removes the funds and prepares them for deposit into the merchant's account.

To perform the 3-D Secure authentication, the Moneris MPI or any third-party MPI may be used.

In addition to 3-D Secure transactions, this transaction can also be used to process Apple Pay and Google Pay™ transactions. This transaction is applicable only if choosing to integrate directly to Apple Wallet or Google Wallet (if not using the Moneris Apple Pay or Google Pay™ SDKs).

Refer to Apple or Google developer portals for details on integrating directly to their wallets to retrieve the payload data.

**WARNING:** Moneris strongly discourages the use of frames as part of a 3-D Secure implementation, and cannot guarantee their reliability when processing transactions in the production environment.

```
$txnArray = array('type'=>'cavv_purchase', ...);

$mpgTxn = new mpgTransaction($txnArray);

$mpgRequest = new mpgRequest($mpgTxn);

$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

**Core connection object fields (all API transactions)**

| Variable Name | Type and Limits      |                          |
|---------------|----------------------|--------------------------|
| store ID      | <i>String</i><br>N/A | 'store_id'=>\$store_id   |
| API token     | <i>String</i><br>N/A | 'api_token'=>\$api_token |

**Purchase with 3-D Secure transaction request fields – Required**

| Variable Name                                       | Type and Limits                                                                                                                                                    |                              |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| order ID                                            | <i>String</i><br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                                                                                         | 'order_id'=>\$order_id       |
| amount                                              | <i>String</i><br>10-character decimal<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><div>EXAMPLE: 1234567.89</div> | 'amount'=>\$amount           |
| credit card number                                  | <i>String</i><br>max 20-character alphanumeric                                                                                                                     | 'pan'=>\$pan                 |
| expiry date                                         | <i>String</i><br>4-character alphanumeric<br>YYMM                                                                                                                  | 'expiry_date'=>\$expiry_date |
| Cardholder Authentication Verification Value (CAVV) | <i>String</i><br>50-character alphanumeric                                                                                                                         | cavv=>\$cavv                 |

| Variable Name                                                                                                                                                                                                                                                                                                                                                                                                       | Type and Limits                           |                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-----------------------|
| <b>NOTE:</b> For Apple Pay and Google Pay™ Cavv Purchase and Cavv Pre-Authorization transactions, CAVV field contains the decrypted cryptogram. For more, see Appendix A Definition of Request Fields.                                                                                                                                                                                                              |                                           |                       |
|                                                                                                                                                                                                                                                                                                                                                                                                                     |                                           |                       |
| electronic commerce indicator                                                                                                                                                                                                                                                                                                                                                                                       | <i>String</i><br>1-character alphanumeric | 'crypt_type'=>\$crypt |
| <b>NOTE:</b> For Apple Pay and Google Pay™ Cavv Purchase and Cavv Pre-Authorization transactions, the e-commerce indicator is a mandatory field containing the value received from the decrypted payload or a default value of 5. If you get a 2-character value (e.g., 05 or 07) from the payload, remove the initial 0 and just send us the 2nd character. For more, see Appendix A Definition of Request Fields. |                                           |                       |
|                                                                                                                                                                                                                                                                                                                                                                                                                     |                                           |                       |

### 3-D Secure 2.2 -specific fields – Required

| Variable Name                                                                          | Type and Limits                       |                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3DS version                                                                            | <i>String</i><br>10-character numeric | 'threeds_version'=>\$threeds_version<br><br>Acceptable values:<br><br>2.0.0 = 3DS protocol 2.0.0<br><br>2.1.0 = 3DS protocol 2.1.0<br><br>2.2.0 = 3DS protocol 2.2.0<br><br>2.3.0 = 3DS protocol 2.3.0 |
| <b>NOTE:</b> Mandatory for financial transactions using 3rd Party 3-D Secure services. |                                       |                                                                                                                                                                                                        |
|                                                                                        |                                       |                                                                                                                                                                                                        |
| 3DS server transaction ID                                                              | <i>String</i>                         | 'threeds_server_trans_                                                                                                                                                                                 |

| Variable Name                                                                                                                                                   | Type and Limits      |                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------------------------------------|
| <b>NOTE:</b> Mandatory for financial transactions using 3rd Party 3-D Secure services - obtained from the Cavv Lookup request or MPI 3DS Authentication request | 36-character numeric | <code>id'=&gt;\$threeds_server_trans_id</code> |

Following fields are required for Apple Pay and Google Pay only:

| Variable Name | Type and Limits                    |  |
|---------------|------------------------------------|--|
| network       | String<br>alphanumeric             |  |
| data type     | String<br>3-character alphanumeric |  |

#### Purchase with 3-D Secure transaction request fields – Optional

| Variable Name      | Type and Limits                                                                                                                                                                                          |                                                            |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| customer ID        | String<br>50-character alphanumeric<br><b>NOTE:</b><br>Some special characters are not allowed:<br><code>&lt;&gt;\$%=?^{}[]\</code>                                                                      | <code>'cust_id'=&gt;\$cust_id</code>                       |
| dynamic descriptor | String<br>20-character alphanumeric<br>total of 22 characters including your merchant name and separator<br><b>NOTE:</b><br>Some special characters are not allowed:<br><code>&lt;&gt;\$%=?^{}[]\</code> | <code>'dynamic_descriptor'=&gt;\$dynamic_descriptor</code> |

| Variable Name                                                                                                                                                       | Type and Limits                     |                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Surcharge Information</b><br><br><div> <b>NOTE:</b> This object requires use of the Vault Surcharge Lookup transaction prior to confirm card eligibility. </div> | <i>Object</i><br><br>N/A            | \$mpgTxn->setSurchargeInfo (\$surchargeInfo) ;<br><br>Contains fields related to Surcharge feature. |
| foreign indicator                                                                                                                                                   | <i>Boolean</i><br><br>true or false | 'foreign_indicator'=>\$foreign_indicator                                                            |

### 3-D Secure 2.2 -specific fields – Optional

| Variable Name                                                                                                                      | Type and Limits                                |                                |
|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|--------------------------------|
| <b>DS transaction ID</b><br><br><div> <b>NOTE:</b> Only used in financial transactions using 3rd Party 3-D Secure services. </div> | <i>String</i><br><br>36-character alphanumeric | 'ds_trans_id' => \$ds_trans_id |

### Sample Purchase with 3-D Secure – cavv\_purchase

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='monca03650';
$api_token='7Yw0MPTlhjBRcZiE6837';
/***** Transactional Variables *****/
$type='cavv_purchase';
$order_id='ord-' .date("dmy-G:i:s");
$cust_id='CUST887763';
$amount='6000.00';
$pan='4622943127023886';
$expiry_date='2212';
$cavv='AAABBJg0VhIOVniQEjRWAAAAAA=';
$crypt_type = '7';
$wallet_indicator = "APP";
$dynamic_descriptor='123456';
$foreign_indicator='true';

// TrId and TokenCryptogram are optional, refer documentation for more details.
$tr_id = '50189815682';
$token_cryptogram = 'APmbM/41le0uAAH+s6xMAAADFA==';
'foreign_indicator'=>$foreign_indicator
/***** Transaction Associative Array *****/
$txnArray=array(
    'type'=>$type,

```

```

'order_id'=>$order_id,
'cust_id'=>$cust_id,
'amount'=>$amount,
'pan'=>$pan,
'expdate'=>$expiry_date,
'cavv'=>$cavv,
'crypt_type'=>$crypt_type, //mandatory for AMEX only
// 'wallet_indicator'=>$wallet_indicator, //set only for wallet transactions. e.g. APPLE PAY
// 'network'=> "Interac", //set only for Interac e-commerce
// 'data_type'=> "3DSecure", //set only for Interac e-commerce
'dynamic_descriptor'=>$dynamic_descriptor,
'threads_version' => '2.2.0', //Mandatory for financial transactions using 3rd Party 3-D
Secure services
'threads_server_trans_id' => 'e11d4985-8d25-40ed-99d6-c3803fe5e68f' Mandatory for financial
transactions using 3rd Party 3-D Secure services - obtained from MpiCavvLookup or
MpiThreeDSAuthentication
//, 'cm_id' => '8nAK8712sGaAkls56' //set only for usage with Offlinx - Unique max 50
alphanumeric characters transaction id generated by merchant
//, 'ds_trans_id' => '12345' //Optional - to be used only if you are using 3rd party 3ds
service
//, 'tr_id' => $tr_id
//, 'token_cryptogram' => $token_cryptogram
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("U");
$cof->setPaymentInformation("2");
$cof->setIssuerId("139X3130ASCXAS9");
$mpgTxn->setCofInfo($cof);
/***** Installment Info *OPTIONAL* *****/
$installmentInfo = new InstallmentInfo();
$installmentInfo->setPlanId("ae859ef1-eb91-b708-8b80-1dd481746401");
$installmentInfo->setPlanIdRef("0000000065");
$installmentInfo->setTacVersion("2");
// $mpgTxn->setInstallmentInfo($installmentInfo);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
/***** Response *****/
$mpgResponse = $mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nCavvResultCode = " . $mpgResponse->getCavvResultCode());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\nThreeDSVersion = " . $mpgResponse->getThreeDSVersion());
print("\nSourcePanLast4 = " . $mpgResponse->getSourcePanLast4());
print("\nAdviceCode = " . $mpgResponse->getAdviceCode());

```

```
// $installmentResults = $mpgResponse->getInstallmentResults();
// print("\nPlanId = " . $installmentResults->getPlanId());
// print("\nPlanIDRef = " . $installmentResults->getPlanIDRef());
// print("\nTacVersion = " . $installmentResults->getTacVersion());
// print("\nPlanAcceptanceId = " . $installmentResults->getPlanAcceptanceId());
// print("\nPlanStatus = " . $installmentResults->getPlanStatus());
// print("\nPlanResponse = " . $installmentResults->getPlanResponse());
?>
```

## 6.8.2 Pre-Authorization with 3-D Secure – cavv\_preauth

The Pre-Authorization with 3-D Secure transaction follows a 3-D Secure MPI authentication. After receiving confirmation from the MPI ACS Request transaction, this Pre-Authorization verifies funds on the customer's card, removes the funds and prepares them for deposit into the merchant's account.

To perform the 3-D Secure authentication, the Moneris MPI or any third-party MPI may be used.

In addition to 3-D Secure transactions, this transaction can also be used to process Apple Pay and Google Pay™ transactions. This transaction is applicable only if choosing to integrate directly to Apple Wallet or Google Wallet (if not using the Moneris Apple Pay or Google Pay™ SDKs).

Refer to Apple or Google developer portals for details on integrating directly to their wallets to retrieve the payload data.

**WARNING:** Moneris strongly discourages the use of frames as part of a 3-D Secure implementation, and cannot guarantee their reliability when processing transactions in the production environment.

```
$txnArray = array('type'=>'cavv_preauth', ...);
$mpgTxn = new mpgTransaction($txnArray);
$mpgRequest = new mpgRequest($mpgTxn);
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

## Pre-Authorization with 3-D Secure transaction request fields – Required

| Variable Name                                       | Type and Limits                                                                                                                                                                                                                                          |                              |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| order ID                                            | <i>String</i><br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                                                                                                                                                                               | 'order_id'=>\$order_id       |
| amount                                              | <i>String</i><br>10-character decimal<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><div>EXAMPLE: 1234567.89</div>                                                                                       | 'amount'=>\$amount           |
| credit card number                                  | <i>String</i><br>max 20-character alphanumeric                                                                                                                                                                                                           | 'pan'=>\$pan                 |
| Cardholder Authentication Verification Value (CAVV) | <i>String</i><br>50-character alphanumeric<br><div>NOTE: For Apple Pay and Google Pay™ Cavv Purchase and Cavv Pre-Authorization transactions, CAVV field contains the decrypted cryptogram. For more, see Appendix A Definition of Request Fields.</div> | cavv=>\$cavv                 |
| expiry date                                         | <i>String</i><br>4-character alphanumeric<br>YYMM                                                                                                                                                                                                        | 'expiry_date'=>\$expiry_date |
| electronic commerce indicator                       | <i>String</i><br>1-character alphanumeric                                                                                                                                                                                                                | 'crypt_type'=>\$crypt        |



| Variable Name                                                                                                                                                                                                                                                                                                                                                                                                              | Type and Limits |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--|
| <p><b>NOTE:</b> For Apple Pay and Google Pay™ Cavv Purchase and Cavv Pre-Authorization transactions, the e-commerce indicator is a mandatory field containing the value received from the decrypted payload or a default value of 5. If you get a 2-character value (e.g., 05 or 07) from the payload, remove the initial 0 and just send us the 2nd character. For more, see Appendix A Definition of Request Fields.</p> |                 |  |

### 3-D Secure 2.2 -specific fields – Required

| Variable Name                                                                                                                                                                                           | Type and Limits                           |                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3DS version<br><br><p><b>NOTE:</b> Mandatory for financial transactions using 3rd Party 3-D Secure services.</p>                                                                                        | <i>String</i><br><br>10-character numeric | 'threads_version'=>\$threads_version<br><br>Acceptable values:<br><br>2.0.0 = 3DS protocol 2.0.0<br><br>2.1.0 = 3DS protocol 2.1.0<br><br>2.2.0 = 3DS protocol 2.2.0<br><br>2.3.0 = 3DS protocol 2.3.0 |
| 3DS server transaction ID<br><br><p><b>NOTE:</b> Mandatory for financial transactions using 3rd Party 3-D Secure services - obtained from the Cavv Lookup request or MPI 3DS Authentication request</p> | <i>String</i><br><br>36-character numeric | 'threads_server_trans_id'=>\$threads_server_trans_id                                                                                                                                                   |

## Pre-Authorization with 3-D Secure transaction request fields – Optional

| Variable Name      | Type and Limits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| status check       | <i>Boolean</i><br><br>true/false                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <pre>\$mpgHttpPost =new<br/>mpgHttpsPostStatus(\$store_<br/>id,\$api_<br/>token,\$status,\$mpgRequest);</pre>                                                                                                                                                                                                                                                                                                                                                                                   |
| customer ID        | <i>String</i><br><br>50-character alphanumeric<br><br><b>NOTE:</b><br>Some special characters are not allowed:<br><>\$%=?^{}[]\                                                                                                                                                                                                                                                                                                                                                                                                   | <pre>'cust_id'=&gt;\$cust_id</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| dynamic descriptor | <i>String</i><br><br>20-character alphanumeric<br><br>total of 22 characters including your merchant name and separator<br><br><b>NOTE:</b><br>Some special characters are not allowed:<br><>\$%=?^{}[]\<br><b>NOTE: For Pre-Authorization transactions:</b> the value in the dynamic descriptor field will only be carried over to a Pre-Authorization Completion when executing the latter via the Merchant Resource Center; otherwise, the value for dynamic descriptor must be sent again in the Pre-Authorization Completion | <pre>'dynamic_descriptor'=&gt;\$dynamic_<br/>descriptor</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| is incremental     | <i>Boolean</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <pre>'is_incremental'=&gt;\$is_incremental</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| is_incremental     | true/false                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Indicates if this preauthorization is using an estimated amount. Estimations allow for incrementing the amount held via subsequent incrementalAuth requests. Defaults to false.<br><br><b>NOTE:</b> Please note that if this field is true, the preauthorization is only eligible for a single Preauthorization Completion. Any completion sent for partial completion is treated as a full completion (ship_indicator= P is treated as = F when is_incremental= true on the original preauth). |

| Variable Name                                                                                                                                                                                                                                                                             | Type and Limits                     |                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                           |                                     | <b>NOTE:</b> For preauthorizations using Account-Based Payments, as part of Pay By Bank, <code>is_incremental = true</code> indicates the transaction may be completed for up to 15% more than the original preauthorization amount. |
| Surcharge Information<br><br><b>NOTE:</b> This object requires use of the Vault Surcharge Lookup transaction prior to confirm card eligibility.                                                                                                                                           | <i>Object</i><br><br>N/A            | <pre>\$mpgTxn-&gt;setSurchargeInfo(\$surchargeInfo);</pre><br>Contains fields related to Surcharge feature.                                                                                                                          |
| foreign indicator                                                                                                                                                                                                                                                                         | <i>Boolean</i><br><br>true or false | <pre>'foreign_indicator'=&gt;\$foreign_indicator</pre>                                                                                                                                                                               |
| AVS Information                                                                                                                                                                                                                                                                           | <i>Object</i><br><br>N/A            | <pre>\$mpgTxn-&gt;setAvsInfo(\$mpgAvsInfo);</pre>                                                                                                                                                                                    |
| CVD Information<br><br><b>NOTE:</b> When storing credentials on the initial transaction, the CVD object must be sent; for subsequent transactions using stored credentials, CVD can be sent with cardholder-initiated transactions only— <b>merchants must not store CVD information.</b> | <i>Object</i><br><br>N/A            | <pre>\$mpgTxn-&gt;setCvdInfo(\$mpgCvdInfo);</pre>                                                                                                                                                                                    |
| final authorization<br><br><b>NOTE:</b> Applies to Mastercard transactions only                                                                                                                                                                                                           | <i>String</i><br><br>true/false     | <pre>'final_auth' =&gt; 'true'</pre>                                                                                                                                                                                                 |

### 3-D Secure 2.2 -specific fields – Optional

| Variable Name                                                                         | Type and Limits                                |                                |
|---------------------------------------------------------------------------------------|------------------------------------------------|--------------------------------|
| DS transaction ID                                                                     | <i>String</i><br><br>36-character alphanumeric | 'ds_trans_id' => \$ds_trans_id |
| <b>NOTE:</b> Only used in financial transactions using 3rd Party 3-D Secure services. |                                                |                                |

### Sample Pre-Authorization with 3-D Secure – cavv\_preauth

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='monca03650';
$api_token='7Yw0MPTlhjBRcZiE6837';
/***** Transactional Variables *****/
$type='cavv_preauth';
$order_id='ord-' . date("dmy-G:i:s");
$cust_id='CUST887763';
$amount='4840.00';
$pan='5454545454545454';
$expiry_date='2212';
$cavv='AAABBJg0VhI0VniQEjRWAAAAA=';
$crypt_type = '7';
$wallet_indicator = "APP";
$dynamic_descriptor='123456';
// TrId and TokenCryptogram are optional, refer documentation for more details.
$str_id = '50189815682';
$token_cryptogram = 'APmbM/41le0uAAH+s6xMAAADFA==';
/***** Transaction Associative Array *****/
$txnArray=array(
    'type'=>$type,
    'order_id'=>$order_id,
    'cust_id'=>$cust_id,
    'amount'=>$amount,
    'pan'=>$pan,
    'expdate'=>$expiry_date,
    'cavv'=>$cavv,
    'crypt_type'=>$crypt_type, //mandatory for AMEX only
    //wallet_indicator'=>$wallet_indicator, //set only for wallet transactions. e.g. APPLE PAY
    //'network'=> "Interac", //set only for Interac e-commerce
    //'data_type'=> "3DSecure", //set only for Interac e-commerce
    'dynamic_descriptor'=>$dynamic_descriptor,
    'threads_version' => '2', //Mandatory for 3DS Version 2.0+
    'threads_server_trans_id' => 'e11d4985-8d25-40ed-99d6-c3803fe5e68f' //Mandatory for 3DS
    Version 2.0+ - obtained from MpiCavvLookup or MpiThreeDSAuthentication
    //,'cm_id' => '8nAK8712sGaAkls56' //set only for usage with Offlinx - Unique max 50
    alphanumeric characters transaction id generated by merchant
    //,'ds_trans_id' => '12345' //Optional - to be used only if you are using 3rd party 3ds 2.0
    service
    //,'tr_id' => $str_id
    //,'token_cryptogram' => $token_cryptogram
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("U");
$cof->setPaymentInformation("2");

```

```

$cof->setIssuerId("139X3130ASCXAS9");
$mpgTxn->setCofInfo($cof);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false or comment out this line for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
/***** Response *****/
$mpgResponse = $mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nCavvResultCode = " . $mpgResponse->getCavvResultCode());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\nThreeDSVersion = " . $mpgResponse->getThreeDSVersion());
print("\nSourcePanLast4 = " . $mpgResponse->getSourcePanLast4());
print("\nAdviceCode = " . $mpgResponse->getAdviceCode());
?>

```

### 6.8.3 Purchase with Vault and 3-D Secure

**NOTE:** This transaction supports both temporary and permanent tokens.

**WARNING:** Moneris strongly discourages the use of frames as part of a 3-D Secure implementation, and cannot guarantee their reliability when processing transactions in the production environment.

```

$txnArray = array('type'=>'res_cavv_purchase_cc', ...);
$mpgTxn = new mpgTransaction($txnArray);
$mpgRequest = new mpgRequest($mpgTxn);
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);

```

**Core connection object fields (all API transactions)**

| Variable Name | Type and Limits      |                          |
|---------------|----------------------|--------------------------|
| store ID      | <i>String</i><br>N/A | 'store_id'=>\$store_id   |
| API token     | <i>String</i><br>N/A | 'api_token'=>\$api_token |

**Purchase with Vault and 3-D Secure transaction request fields – Required**

| Variable Name                                       | Type and Limits                                                                                                                                                    |                        |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| data key                                            | <i>String</i><br>25-character alphanumeric                                                                                                                         | 'data_key'=>\$data_key |
| order ID                                            | <i>String</i><br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                                                                                         | 'order_id'=>\$order_id |
| amount                                              | <i>String</i><br>10-character decimal<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><div>EXAMPLE: 1234567.89</div> | 'amount'=>\$amount     |
| Cardholder Authentication Verification Value (CAVV) | <i>String</i><br>50-character alphanumeric                                                                                                                         | cavv=>\$cavv           |
| electronic commerce indicator                       | <i>String</i><br>1-character alphanumeric                                                                                                                          | 'crypt_type'=>\$crypt  |

**3-D Secure 2.2 -specific fields – Required**

| Variable Name                                                                                                                                                                                                                                                                 | Type and Limits                       |                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3DS version                                                                                                                                                                                                                                                                   | <i>String</i><br>10-character numeric | 'threads_version'=>\$threads_version<br><br>Acceptable values:<br><br>2.0.0 = 3DS protocol 2.0.0<br><br>2.1.0 = 3DS protocol 2.1.0<br><br>2.2.0 = 3DS protocol 2.2.0<br><br>2.3.0 = 3DS protocol 2.3.0 |
| 3DS server transaction ID                                                                                                                                                                                                                                                     | <i>String</i><br>36-character numeric | 'threads_server_trans_id'=>\$threads_server_trans_id                                                                                                                                                   |
| <div> <b>NOTE:</b> Mandatory for financial transactions using 3rd Party 3-D Secure services.<br/><br/> <b>NOTE:</b> Mandatory for financial transactions using 3rd Party 3-D Secure services - obtained from the Cavv Lookup request or MPI 3DS Authentication request </div> |                                       |                                                                                                                                                                                                        |

**Purchase with Vault and 3-D Secure transaction request fields – Optional**

| Variable Name                                                                                             | Type and Limits                                   |                                                                                          |
|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------|
| customer ID                                                                                               | <i>String</i><br>50-character alphanumeric        | 'cust_id'=>\$cust_id                                                                     |
| <div> <b>NOTE:</b><br/>Some special characters are not allowed:<br/>&lt; &gt; \$ % = ? ^ { } [ ] \ </div> |                                                   |                                                                                          |
| expiry date                                                                                               | <i>String</i><br>4-character alphanumeric<br>YYMM | 'expiry_date'=>\$expiry_date                                                             |
| Surcharge Information                                                                                     | <i>Object</i><br>N/A                              | \$mpgTxn->setSurchargeInfo(\$surchargeInfo);<br><br>Contains fields related to Surcharge |
| <div> <b>NOTE:</b> This object requires </div>                                                            |                                                   |                                                                                          |

| Variable Name                                                                                           | Type and Limits |                        |
|---------------------------------------------------------------------------------------------------------|-----------------|------------------------|
| <div> use of the Vault Surcharge<br/> Lookup transaction prior to<br/> confirm card eligibility. </div> |                 | feature.               |
| final authorization                                                                                     | String          | 'final_auth' => 'true' |
| <div> <b>NOTE:</b> Applies to Mastercard<br/> transactions only </div>                                  | true/false      |                        |

### 3-D Secure 2.2 -specific fields – Optional

| Variable Name                                                                                                | Type and Limits           |                                |
|--------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------------|
| DS transaction ID                                                                                            | String                    | 'ds_trans_id' => \$ds_trans_id |
| <div> <b>NOTE:</b> Only used in financial<br/> transactions using 3rd Party<br/> 3-D Secure services. </div> | 36-character alphanumeric |                                |

### Sample Purchase with Vault and 3-D Secure

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='monca00597';
$api_token='O27AbCbXQorPggMQe6hU';
/***** Transaction Variables *****/
$data_key='4HIme0ZGURXE3NRBXHUj6nSc4';
$orderid='res-preauth-' . date("dmy-G:i:s");
$amount='1.00';
$cavv='AAABBJg0VhI0VniQEjRWAAAAA';
$custid='customer1'; //if sent will be submitted, otherwise cust_id from profile will be used
$expdate = '2301'; //YYMM - used only for temp token
//$crypt_type = '6'; //value obtained from MpiACS transaction

//NT Response Option
$get_nt_response = 'false'; //Optional - set it true only if you want to get network
tokenization response.
/***** Transaction Array *****/
$txnArray = array('type'=>'res_cavv_purchase_cc',
'data_key'=>$data_key,
'order_id'=>$orderid,
'cust_id'=>$custid,
'amount'=>$amount,
'cavv'=>$cavv,
'expdate'=>$expdate, //mandatory for temp tokens only
'crypt_type'=>$crypt_type, //set for AMEX SafeKey only
'dynamic_descriptor'=>'12346',
'threads_version' => '2.2.0', //Mandatory for financial transactions using 3rd Party 3-D
Secure services

```



```

'threads_server_trans_id' => 'e11d4985-8d25-40ed-99d6-c3803fe5e68f', //Mandatory for
financial transactions using 3rd Party 3-D Secure services - obtained from MpiCavvLookup or
MpiThreeDSAuthentication
'ds_trans_id' => '12345', //Optional - to be used only if you are using 3rd party 3ds service
'get_nt_response'=>$get_nt_response
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("U");
$cof->setPaymentInformation("2");
$cof->setIssuerId("139X3130ASCXAS9");
$mpgTxn->setCofInfo($cof);
/***** Installment Info *OPTIONAL* *****/
$installmentInfo = new InstallmentInfo();
$installmentInfo->setPlanId("ae859ef1-eb91-b708-8b80-1dd481746401");
$installmentInfo->setPlanIdRef("0000000065");
$installmentInfo->setTacVersion("2");
// $mpgTxn->setInstallmentInfo($installmentInfo);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
/***** mpgHttpPost Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response Object *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nAVSResponse = " . $mpgResponse->getAvsResultCode());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
print("\nCavvResultCode = " . $mpgResponse->getCavvResultCode());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\nThreeDSVersion = " . $mpgResponse->getThreeDSVersion());
print("\n\nSourcePanLast4 = " . $mpgResponse->getSourcePanLast4());

$installmentResults = $mpgResponse->getInstallmentResults();
print("\nPlanId = " . $installmentResults->getPlanId());
print("\nPlanIDRef = " . $installmentResults->getPlanIDRef());
print("\nTacVersion = " . $installmentResults->getTacVersion());
print("\nPlanAcceptanceId = " . $installmentResults->getPlanAcceptanceId());
print("\nPlanStatus = " . $installmentResults->getPlanStatus());
print("\nPlanResponse = " . $installmentResults->getPlanResponse());

if($get_nt_response == 'true')
{
print("\n\nNTResponseCode = " . $mpgResponse->getNTResponseCode());
print("\nNTMessage = " . $mpgResponse->getNTMessage());
print("\nNTUsed = " . $mpgResponse->getNTUsed());
print("\nNTTokenBin = " . $mpgResponse->getNTTokenBin());
print("\nNTTokenLast4 = " . $mpgResponse->getNTTokenLast4());
}

```

```

print("\nNTTokenExpDate = " . $mpgResponse->getNTTokenExpDate());
}
//----- ResolveData -----
print("\nCust ID = " . $mpgResponse->getResDataCustId());
print("\nPhone = " . $mpgResponse->getResDataPhone());
print("\nEmail = " . $mpgResponse->getResDataEmail());
print("\nNote = " . $mpgResponse->getResDataNote());
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\nCrypt Type = " . $mpgResponse->getResDataCryptType());
print("\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
print("\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
?>

```

### 6.8.4 Pre-Authorization with Vault & 3-D Secure

**NOTE:** This transaction supports both temporary and permanent tokens.

**WARNING:** Moneris strongly discourages the use of frames as part of a 3-D Secure implementation, and cannot guarantee their reliability when processing transactions in the production environment.

#### Pre-Authorization with Vault & 3-D Secure transaction object definition

```

$txnArray = array('type'=>'res_cavv_preauth_cc', ...);

$mpgTxn = new mpgTransaction($txnArray);

```

#### HttpPostRequest object for Pre-Authorization with Vault & 3-D Secure

```

$mpgRequest = new mpgRequest($mpgTxn);

$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);

```

#### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

**Pre-Authorization with Vault & 3-D Secure transaction request fields – Required**

| Variable Name                                       | Type and Limits                                                                                                                                                    | Set Method             |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| data key                                            | <i>String</i><br>25-character alphanumeric                                                                                                                         | 'data_key'=>\$data_key |
| order ID                                            | <i>String</i><br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                                                                                         | 'order_id'=>\$order_id |
| amount                                              | <i>String</i><br>10-character decimal<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><div>EXAMPLE: 1234567.89</div> | 'amount'=>\$amount     |
| Cardholder Authentication Verification Value (CAVV) | <i>String</i><br>50-character alphanumeric                                                                                                                         | cavv=>\$cavv           |
| electronic commerce indicator                       | <i>String</i><br>1-character alphanumeric                                                                                                                          | 'crypt_type'=>\$crypt  |

**Pre-Authorization with Vault & 3-D Secure transaction request fields – Optional**

| Variable Name      | Type and Limits                                                                                                                                | Set Method                                 |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| customer ID        | <i>String</i><br>50-character alphanumeric<br><div>NOTE:<br/>Some special characters are not allowed:<br/>&lt; &gt; \$ % = ? ^ { } [ ] \</div> | 'cust_id'=>\$cust_id                       |
| dynamic descriptor | <i>String</i>                                                                                                                                  | 'dynamic_descriptor'=>\$dynamic_descriptor |

| Variable Name                                                                                                                                                                                                                                                                                                                | Type and Limits                                                                                                                                                                                               | Set Method                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>NOTE: For Pre-Authorization transactions:</b> the value in the dynamic descriptor field will only be carried over to a Pre-Authorization Completion when executing the latter via the Merchant Resource Center; otherwise, the value for dynamic descriptor must be sent again in the Pre-Authorization Completion</p> | <p>20-character alphanumeric</p> <p>total of 22 characters including your merchant name and separator</p> <p><b>NOTE:</b><br/>Some special characters are not allowed:<br/>&lt; &gt; \$ % = ? ^ { } [ ] \</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <p>is_incremental</p> <p>is_incremental</p>                                                                                                                                                                                                                                                                                  | <p><i>Boolean</i></p> <p>true/false</p>                                                                                                                                                                       | <p>'is_incremental'=&gt;\$is_incremental</p> <p>Indicates if this preauthorization is using an estimated amount. Estimations allow for incrementing the amount held via subsequent incrementalAuth requests. Defaults to false.</p> <p><b>NOTE:</b> Please note that if this field is true, the preauthorization is only eligible for a single Preauthorization Completion. Any completion sent for partial completion is treated as a full completion (ship_indicator= P is treated as = F when is_incremental= true on the original preauth).</p> <p><b>NOTE:</b> For preauthorizations using Account-Based Payments, as part of Pay By Bank, is_incremental = true indicates the transaction may be completed for up to 15% more than the original preauthorization amount.</p> |
| <p>Surcharge Information</p> <p><b>NOTE:</b> This object requires use of the Vault Surcharge Lookup transaction prior to confirm card eligibility.</p>                                                                                                                                                                       | <p><i>Object</i></p> <p>N/A</p>                                                                                                                                                                               | <p>\$mpgTxn-&gt;setSurchargeInfo(\$surchargeInfo);</p> <p>Contains fields related to Surcharge feature.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Variable Name                                                                         | Type and Limits                            | Set Method                     |
|---------------------------------------------------------------------------------------|--------------------------------------------|--------------------------------|
| expiry date                                                                           | String<br>4-character alphanumeric<br>YYMM | 'expiry_date'=>\$expiry_date   |
| DS transaction ID                                                                     | String<br>36-character alphanumeric        | 'ds_trans_id' => \$ds_trans_id |
| <b>NOTE:</b> Only used in financial transactions using 3rd Party 3-D Secure services. |                                            |                                |
| final authorization                                                                   | String<br>true/false                       | 'final_auth' => 'true'         |
| <b>NOTE:</b> Applies to Mastercard transactions only                                  |                                            |                                |

### Sample Pre-Authorization with Vault & 3-D Secure

```
<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='monca00597';
$api_token='O27AbCbXQorPggMQe6hU';
/***** Transaction Variables *****/
$data_key='4HIme0ZGURXE3NRBXHUj6nSc4';
$orderid='res-preauth-' . date("dmy-G:i:s");
$amount='1.00';
$cavv='AAABBJg0VhI0VniQEjRWAAAAA';
$custid='customer1'; //if sent will be submitted, otherwise cust_id from profile will be used
$expdate = '2301'; //YYMM - used only for temp token
//$crypt_type = '6'; //value obtained from MpiACS transaction

//NT Response Option
$get_nt_response = 'false'; //Optional - set it true only if you want to get network
tokenization response.
/***** Transaction Array *****/
$txnArray = array('type'=>'res_cavv_preauth_cc',
'data_key'=>$data_key,
'order_id'=>$orderid,
'cust_id'=>$custid,
'amount'=>$amount,
'cavv'=>$cavv,
'expdate'=>$expdate, //mandatory for temp tokens only
'crypt_type'=>$crypt_type, //set for AMEX SafeKey only
'dynamic_descriptor'=>'12346',
'threads_version' => '2.2.0', //Mandatory for financial transactions using 3rd Party 3-D
Secure services
'threads_server_trans_id' => 'e11d4985-8d25-40ed-99d6-c3803fe5e68f', //Mandatory for
financial transactions using 3rd Party 3-D Secure services - obtained from MpiCavvLookup or
MpiThreeDSAuthentication
'ds_trans_id' => '12345', //Optional - to be used only if you are using 3rd party 3ds service
'get_nt_response'=>$get_nt_response
);
```

```

/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("U");
$cof->setPaymentInformation("2");
$cof->setIssuerId("139X3130ASCXAS9");
$mpgTxn->setCofInfo($cof);
/***** Installment Info *OPTIONAL* *****/
$installmentInfo = new InstallmentInfo();
$installmentInfo->setPlanId("ae859ef1-eb91-b708-8b80-1dd481746401");
$installmentInfo->setPlanIdRef("0000000065");
$installmentInfo->setTacVersion("2");
// $mpgTxn->setInstallmentInfo($installmentInfo);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
/***** mpgHttpPost Object *****/
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
/***** Response Object *****/
$mpgResponse = $mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nAVSResponse = " . $mpgResponse->getAvsResultCode());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
print("\nCavvResultCode = " . $mpgResponse->getCavvResultCode());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\nThreeDSVersion = " . $mpgResponse->getThreeDSVersion());
print("\n\nSourcePanLast4 = " . $mpgResponse->getSourcePanLast4());

// $installmentResults = $mpgResponse->getInstallmentResults();
// print("\nPlanId = " . $installmentResults->getPlanId());
// print("\nPlanIDRef = " . $installmentResults->getPlanIDRef());
// print("\nTacVersion = " . $installmentResults->getTacVersion());
// print("\nPlanAcceptanceId = " . $installmentResults->getPlanAcceptanceId());
// print("\nPlanStatus = " . $installmentResults->getPlanStatus());
// print("\nPlanResponse = " . $installmentResults->getPlanResponse());

if($get_nt_response == 'true')
{
print("\n\nNTResponseCode = " . $mpgResponse->getNTResponseCode());
print("\nNTMessage = " . $mpgResponse->getNTMessage());
print("\nNTUsed = " . $mpgResponse->getNTUsed());
print("\nNTTokenBin = " . $mpgResponse->getNTTokenBin());
print("\nNTTokenLast4 = " . $mpgResponse->getNTTokenLast4());
print("\nNTTokenExpDate = " . $mpgResponse->getNTTokenExpDate());
}
//----- ResolveData -----
print("\n\nCust ID = " . $mpgResponse->getResDataCustId());
print("\nPhone = " . $mpgResponse->getResDataPhone());
print("\nEmail = " . $mpgResponse->getResDataEmail());

```

```
print("\nNote = " . $mpgResponse->getResDataNote());
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\nCryp Type = " . $mpgResponse->getResDataCryptType());
print("\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
print("\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
?>
```

### 6.8.5 Purchase with 3-D Secure and Recurring Billing

**WARNING:** Moneris strongly discourages the use of frames as part of a 3-D Secure implementation, and cannot guarantee their reliability when processing transactions in the production environment.

The example below illustrates the Purchase with 3-D Secure when also sending the Recurring Billing Info object in the transaction.

#### Purchase with 3-D Secure and Recurring Billing

```
<?php
## Example php -q TestPurchase-VBV.php "moneris" store
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
/***** Transactional Variables *****/
$type='cavv_purchase';
$order_id='ord-' . date("dmy-G:i:s");
$cust_id='CUST887763';
$amount='10.00';
$span="4242424242424242";
$expiry_date="1511";
$cavv='AAABBJg0VhIOVniQEjRWAAAAA=';
$crypt_type = '7';
$wallet_indicator = "APP";
$dynamic_descriptor='123456';
/***** Recur Variables *****/
$recurUnit = 'month'; //eom - end of month
$startDate = '2018/02/06';
$numRecurs = '4';
$recurInterval = '10';
$recurAmount = '31.00';
$startNow = 'true';
/***** Transaction Associative Array *****/
$txnArray=array(
    'type'=>$type,
    'order_id'=>$order_id,
    'cust_id'=>$cust_id,
    'amount'=>$amount,
    'pan'=>$span,
    'expdate'=>$expiry_date,
    'cavv'=>$cavv,
    'crypt_type'=>$crypt_type, //mandatory for AMEX only
    //'wallet_indicator'=>$wallet_indicator, //set only for wallet transactions. e.g. APPLE PAY
    //'network'=> "Interac", //set only for Interac e-commerce
    //'data_type'=> "3DSecure", //set only for Interac e-commerce
```

```

'dynamic_descriptor'=>$dynamic_descriptor,
'threads_version' => '2.2.0', //Mandatory for financial transactions using 3rd Party 3-D
Secure services
'threads_server_trans_id' => 'e11d4985-8d25-40ed-99d6-c3803fe5e68f', //Mandatory for
financial transactions using 3rd Party 3-D Secure services - obtained from MpiCavvLookup or
MpiThreeDSAuthentication
//ds_trans_id' => '12345' //Optional - to be used only if you are using 3rd party 3ds
service
);
/***** Transaction Object *****/
$recurArray = array('recur_unit'=>$recurUnit, // (day | week | month)
'start_date'=>$startDate, //yyyy/mm/dd
'num_recur'=>$numRecur,
'start_now'=>$startNow,
'period' => $recurInterval,
'recur_amount'=> $recurAmount
);
$mpgRecur = new mpgRecur($recurArray);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Transaction Object *****/
$mpgTxn->setRecur($mpgRecur);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("R");
$cof->setPaymentInformation("2");
$cof->setIssuerId("168451306048014");
$mpgTxn->setCofInfo($cof);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nCavvResultCode = " . $mpgResponse->getCavvResultCode());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\nThreeDSVersion = " . $mpgResponse->getThreeDSVersion());
?>

```

## 6.9 Testing Your 3-D Secure 2.2 Integration

In the testing stage of development:



1. Use the testing URL as Host for your requests:  
esqa.moneris.com
2. In all Card Lookup Request transactions, make sure that you are using the testing version of your credentials for store ID and API token
3. In all MPI 3DS Authentication Request transactions, make sure that you are using the testing version of your credentials for store ID and API token
4. In all Cavv Lookup Request transactions, make sure that you are using the testing version of your credentials for store ID and API token

## 6.10 Moving to Production With 3-D Secure 2.2

Once you have finished testing your 3D Secure 2.2 integration, do the following to move the integration into production:

1. Use the production URL as Host for your requests:  
www3.moneris.com
2. In all Card Lookup Request transactions, make sure that you are using the production version of your credentials for store ID and API token
3. In all MPI 3DS Authentication Request transactions, make sure that you are using the production version of your credentials for store ID and API token
4. In all Cavv Lookup Request transactions, make sure that you are using the production version of your credentials for store ID and API token

## 6.11 3-D Secure 2.2 TransStatus Codes

| Value | Description                    | Comments                                                       |
|-------|--------------------------------|----------------------------------------------------------------|
| Y     | Authenticated                  | Cardholder has been fully authenticated                        |
| D     | Challenge Required (Decoupled) | Cardholder requires a challenge using Decoupled Authentication |
| A     | Authentication Attempt         | A proof of authentication attempt was generated                |

| Value | Description        | Comments                                                                                                            |
|-------|--------------------|---------------------------------------------------------------------------------------------------------------------|
| C     | Challenge Required | Cardholder requires a challenge to complete authentication                                                          |
| U     | Not Authenticated  | Authentication could not be performed due to technical or other issue                                               |
| N     | Not Authenticated  | Not authenticated                                                                                                   |
| R     | Not Authenticated  | Not authenticated because the Issuer is rejecting authentication and requesting that authorisation not be attempted |

## 6.12 3-D Secure 2.2 Commons TransStatusReason Decline Codes

The following codes are returned by the 3-D Secure service in order to provide additional information about the 3-D Secure transaction status.

| TransStatusReason Code | Description                            |
|------------------------|----------------------------------------|
| 01                     | Card authentication failed             |
| 02                     | Unknown Device                         |
| 03                     | Unsupported Device                     |
| 04                     | Exceeds authentication frequency limit |
| 05                     | Expired card                           |
| 06                     | Invalid card number                    |
| 07                     | Invalid transaction                    |
| 08                     | No Card record                         |
| 09                     | Security failure                       |
| 10                     | Stolen card                            |
| 11                     | Suspected fraud                        |

| TransStatusReason Code | Description                                                                                                   |
|------------------------|---------------------------------------------------------------------------------------------------------------|
| 12                     | Transaction not permitted to cardholder                                                                       |
| 13                     | Cardholder not enrolled in service                                                                            |
| 14                     | Transaction timed out at the ACS                                                                              |
| 15                     | Low confidence                                                                                                |
| 16                     | Medium confidence                                                                                             |
| 17                     | High confidence                                                                                               |
| 18                     | Very High confidence                                                                                          |
| 19                     | Exceeds ACS maximum challenges                                                                                |
| 20                     | Non-Payment transaction not supported                                                                         |
| 21                     | 3RI transaction not supported                                                                                 |
| 22                     | ACS technical issue                                                                                           |
| 23                     | Decoupled Authentication required by ACS but not requested by 3DS Requestor                                   |
| 24                     | 3DS Requestor Decoupled Max Expiry Time exceeded                                                              |
| 25                     | Decoupled Authentication was provided insufficient time to authenticate cardholder. ACS will not make attempt |
| 26                     | Authentication attempted but not performed by the cardholder                                                  |

**NOTE:** For a list of all TransStatus Decline Codes, please see Reference section of 3D Secure 2.2 at <https://developer.moneris.com>.

## 6.13 CAVV Result Codes

The Cardholder Authentication Verification Value (CAVV), the Accountholder Authentication Value (AAV), and the American Express Verification Value (AEVV), are the values that allows Visa, Mastercard and American Express to validate the integrity of the Visa Secure, Mastercard Identity Check and American Express SafeKey transaction data. These values are passed back from the issuer to the merchant after the authentication has taken place. The merchant then integrates the CAVV/AAV/AEVV value into the authorization request using the Purchase or Pre-Authorization with 3-D Secure transaction type.

To summarize this process:

1. Merchant conducts 3-D Secure authentication request and receives CAVV/AAV/AEVV value in response
2. Merchant sends the CAVV/AAV/AEVV value to Moneris using the Purchase or Pre-Authorization with 3-D Secure transaction type and receives the CAVV result code in the response

The following tables describe the contents of the CAVV data response and what it means to the merchant.

### 6.13.1 Visa CAVV Result Codes

#### Visa CAVV result codes

| Result Code | Message                                 | Significance to Merchants                                                                                                                                                                  |
|-------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Blank       | CAVV not present or not verified        | Not a Visa Secure transaction. No liability shift and merchant is not protected from chargebacks                                                                                           |
| 0           | CAVV authentication results invalid     | Not a Visa Secure transaction. No liability shift and merchant is not protected from chargebacks                                                                                           |
| 1           | CAVV failed validation (authentication) | Provided that you have implemented the Visa Secure process correctly, the liability for this transaction should remain with the Issuer for chargeback reason codes covered by Visa Secure. |
| 2           | CAVV passed validation (authentication) | Fully authenticated transaction. There is a liability shift and the merchant is protected from chargebacks.                                                                                |
| 3, 8, A     | CAVV passed validation (attempt)        | Visa Secure has been attempted. There is a liability shift and the merchant is protected from certain card fraud-related chargebacks.                                                      |

| Result Code | Message                                       | Significance to Merchants                                                                                                             |
|-------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 4, 7, 9     | CAVV failed validation (attempt)              | Visa Secure has been attempted. There is a liability shift and the merchant is protected from certain card fraud-related chargebacks. |
| 6           | CAVV not validated - Issuer not participating | Visa Secure has been attempted. There is a liability shift and the merchant is protected from certain card fraud-related chargebacks. |
| B           | CAVV passed validation; information only      | Not a Visa Secure transaction. No liability shift and merchant is not protected from chargebacks                                      |
| C           | CAVV was not validated (attempt)              | Visa Secure has been attempted. There is a liability shift and the merchant is protected from certain card fraud-related chargebacks. |
| D           | CAVV was not validated (authentication)       | Visa Secure has been attempted. There is a liability shift and the merchant is protected from certain card fraud-related chargebacks. |

### 6.13.2 Mastercard CAVV Result Codes

#### Mastercard CAVV result codes

| Result Code | Message                   | Significance to Merchants                                                                                                                                                                     |
|-------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0           | Authentication failed     | Not a Mastercard Identity Check transaction. No liability shift and merchant is not protected from chargebacks                                                                                |
| 1           | Authentication attempted  | Mastercard Identity Check has been attempted. There is a liability shift and the merchant is protected from certain card fraud-related chargebacks (international commercial cards excluded). |
| 2           | Authentication successful | Fully authenticated transaction. There is a liability shift and the merchant is protected from chargebacks.                                                                                   |

### 6.13.3 American Express CAVV Result Codes

#### American Express CAVV result codes

**NOTE:** American Express SafeKey is only available to American Express direct acquired merchants (i.e., not OptBlue merchants). Any questions pertaining to chargebacks, liability and disputes should be addressed to your American Express representative given that American Express is the acquirer of record for these merchants.

| Result Code | Description                                                                                  |
|-------------|----------------------------------------------------------------------------------------------|
| 1           | AEVV Failed - Authentication, Issuer Key                                                     |
| 2           | AEVV Passed - Authentication, Issuer Key                                                     |
| 3           | AEVV Passed - Attempt, Issuer Key                                                            |
| 4           | AEVV Failed - Attempt, Issuer Key                                                            |
| 7           | AEVV Failed - Attempt, Issuer not participating, Network Key                                 |
| 8           | AEVV Passed - Attempt, Issuer not participating, Network Key                                 |
| 9           | AEVV Failed - Attempt, Participating, Access Control Server (ACS) not available, Network Key |
| A           | AEVV Passed - Attempt, Participating, Access Control Server (ACS) not available, Network Key |
| U           | AEVV Unchecked                                                                               |

## 7 Pay By Bank Transactions

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|                                                  |            |
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### 7.1 About Pay By Bank (Interac Konek)

The Moneris Gateway API supports Pay By Bank wallets in Canada via Interac Konek. This service provides a single-point of integration for retail websites and applications allowing customers to quickly move through the checkout experience. The customer log-ins to Konek with the credentials for their financial institution and selects from any payment options attached.

Although Konek is supported via Interac, the payment options accessible to the customer can include other card products from their bank through Visa, Mastercard, or American Express. Konek also allows for payment directly from bank accounts instead. Regardless of the actual payment method chosen by the customer, the data is tokenized in the response to the merchant to protect against fraud.

### 7.2 Building Your Pay By Bank Integration

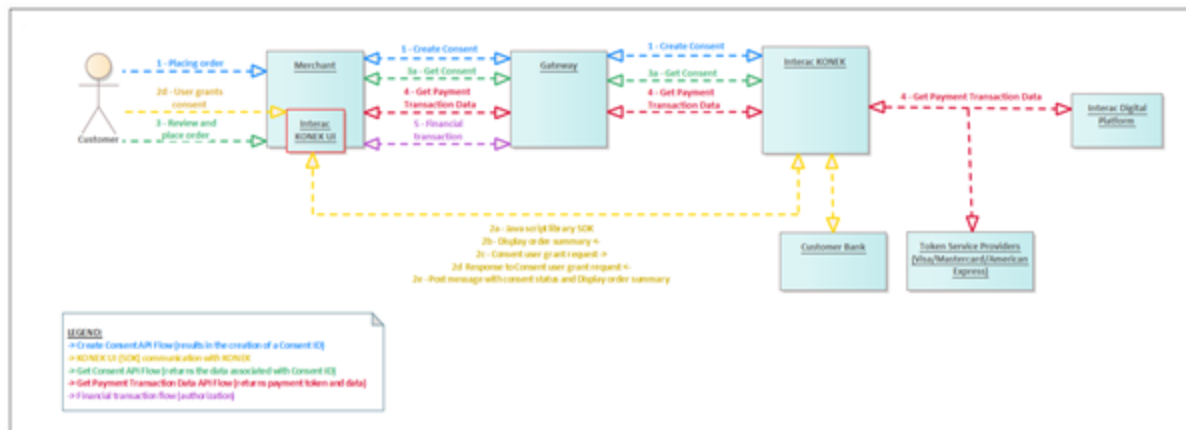
Implementing Pay By Bank with your merchant website requires two major steps.

- Install the Interac Konek SDK
  - Reach out to your Client Consultant from Moneris to receive assistance with access to Interac Konek documentation.
  - If you do not have a Consultant, please register on our Moneris Developer Portal.
- Implement the three Moneris Gateway API transactions outlined within this document
  - Create Consent
  - Get Consent ID
  - Get Payment Transaction Data

### 7.2.1 Transaction Flow for Pay By Bank

The Moneris Gateway API supports three transactions for the Pay By Bank feature. In addition, you must install an Interac Konek SDK on your merchant server for handling part of the Create Consent process. The diagram below displays the proper sequence to follow for a Pay By Bank transaction.

**Figure 2: Flow Diagram for Pay By Bank**



## 1. Creating Cardholder Consent Form

In the first phase, the cardholder places their order on the merchant's website and selects Konek as the payment method. The merchant server uses the Moneris Gateway API call Create Consent to obtain an Interac Konek Consent ID (consent\_id). Moneris Gateway communicates to Interac Konek on your behalf and relays details about the order to Interac.



This Consent ID is used within the Interac Konek SDK to ensure only authenticated merchants can display the log-in prompt for the Konek frame.

## 2. Cardholder Granting Consent

In the second phase, the merchant website initiates the Javascript library SDK for Interac Konek. This calls the Interac Konek systems to display a login page for the user. The cardholder will input the credentials for the financial institution (FI) to log-in to Konek. The UI will display their available payment methods attached to their profile for selection, including both credit card products and bank accounts.

The cardholder grants consent to the payment method within the user interface, which is communicated back to the Interac Konek system. The merchant website receives a POST message with the status of the consent form and order.

**NOTE:** Integration of the Interac Konek SDK is outside the scope of this guide. See "Building Your Pay By Bank Integration" on page 315 for details on obtaining documentation on the Konek SDK.

## 3. Get Consent Data

The third phase begins as the cardholder reviews their order in the merchant website. The merchant server sends a Get Consent API call with the Consent ID from the first step to the Moneris Gateway to validate the consent form; Gateway engages Interac Konek to return data related to the consent obtained within the Konek frame.

The merchant website can redisplay order details obtained from this call as part of the cardholder confirming payment.

**Things to Consider:** Moneris recommends developers retrieve the last four digits of either the bank account number or credit card number (PAN) used as the tokenized payment method. You can display these digits to the customer as part of the order confirmation process to assure cardholders their payment method matches their selection within the Konek interface.

## 4. Get Payment Transaction Data

Once the customer has consented to using their payment method, the merchant can use the Get Payment Transaction Data API call to the Moneris Gateway to retrieve the payment token (PaymentToken) representing that account or card within the cardholder's Konek profile. This payment token can be used in place of a card number in a financial transaction. The merchant will also receive a cryptogram (Cryptogram) for use in the financial transaction.

## 5. Financial Transaction (Authorization)

With the payment token and cryptogram, you can proceed with processing a financial transaction such as a Purchase or Preauthorization supplemented with the info from the PBB authentication. Refer to "Performing the Authorization" on page 327

## 7.3 Implementing Pay By Bank Transactions

In your implementation of the following transactions, use the following URLs as Host, depending on the development stage:

Testing:

esqa.moneris.com

Production:

www3.moneris.com

### 7.3.1 Create Consent

Creates a Consent ID within Interac Konek systems for generating a cardholder consent frame via the Interac Konek SDK.

This call supplies Moneris Gateway and Interac Konek with the payment details of the order, the supported payment methods of the merchant, and the required contact information for the cardholder

#### Create Consent Request transaction object definition

```
$pbbCreateConsent = new PbbCreateConsent();
```

#### HttpPostRequest object for Get Consent transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

#### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

**Create Consent Request transaction request fields – Required**

| Variable Name             | Type and Limits                     |                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pay By Bank Merchant ID   | <i>String</i><br>10-50 alphanumeric | <pre>\$pbbCreateConsent-&gt;setPbbMerchantId(\$pbb_merchant_id);</pre> <p>Unique Merchant identifier assigned by Interac Konek during onboarding for Pay By Bank</p>                                                                                                                                                                                                                                              |
| Required Contact Fields   | <i>Array of strings</i><br>N/A      | <pre>\$pbbCreateConsent-&gt;setRequiredContactFields(\$required_contact_fields);</pre> <p>List of cardholder contact fields flagged as required by the merchant. Array contains contact_field strings for each field flagged. Possible values are:</p> <p>NAME, EMAIL, PHONE, SHIPPING_ADDRESS</p>                                                                                                                |
| Supported Payment Methods | <i>Array of strings</i><br>N/A      | <pre>\$pbbCreateConsent-&gt;setSupportedPaymentMethods(\$supported_payment_methods);</pre> <p>List of payment methods (payment_method) the merchant supports for the transaction. Values are of the type AccountType, with the possible values as:</p> <p>BANK_ACCOUNT_SAVINGS, BANK_ACCOUNT_CHEQUING, BANK_ACCOUNT_LINE_OF_CREDIT, VISA_CREDIT, VISA_DEBIT, MASTERCARD_CREDIT, MASTERCARD_DEBIT, AMEX_CREDIT</p> |
| Payment Details           |                                     | <pre>\$pbbCreateConsent-&gt;setPbbPayment(\$currency, \$amount, \$sub_total, \$tax, \$shipping_cost, \$additional_fees);</pre> <p>Payment details of the order. For recurring transactions, this field represents the first payment of the order.</p> <p>See the " Pay By Bank - Payment Object" on page 584 for details on nested fields.</p>                                                                    |

| Variable Name   | Type and Limits  |                                  |
|-----------------|------------------|----------------------------------|
| Additional Fees | Array            |                                  |
|                 | Array of strings | List of additional fee elements. |

**Order Details**

```
$pbbCreateConsent->setPbbOrder
($shipment_type, $merchant_order_
ref, $placement_mode, $items);
```

Order details from the merchant for the consent.

See the "Pay By Bank - Order Object" on page 585 for information on its nested fields.

**Sample Create Consent Request**

```
<?php
require "../mpgClasses.php";
$store_id = "monca03650";
$api_token = "7Yw0MPTlhjBRcZiE6837";
$processing_country_code = "CA";
$pbb_merchant_id = "CAM0000001";
$required_contact_fields = array("NAME", "PHONE", "EMAIL", "SHIPPING_ADDRESS");
$supported_payment_methods = array("MASTERCARD_DEBIT", "VISA_DEBIT", "AMEX_CREDIT", "VISA_
CREDIT", "MASTERCARD_CREDIT");
//payment
$currency = "CAD";
$amount = "72.00";
$sub_total = "2.00";
$tax = "0.00";
$shipping_cost = "0.00";
$additional_fee1 = array("name" => "java", "amount" => 50.00);
$additional_fee2 = array("name" => "Dotnet", "amount" => 20.00);
$additional_fees = array();
array_push($additional_fees, $additional_fee1, $additional_fee2);
//order
$shipment_type = "NONE";
$merchant_order_ref = "1695401825L1CLwAJ98p7JwAc";
$placement_mode = "STANDARD";
$item1 = array(
    "item_payment_type" => "SINGLE",
    "amount" => 1.00,
    "qty" => 1,
    "description" => "Item1",
    "item_ref" => "Testing"
);
$item2 = array(
    "item_payment_type" => "SINGLE",
    "amount" => 1.00,
    "qty" => 1,
    "description" => "Item2",
    "item_ref" => "Testing2"
);
$items = array();
array_push($items, $item1, $item2);
$pbbCreateConsent = new PbbCreateConsent();
$pbbCreateConsent->setPbbMerchantId($pbb_merchant_id);
$pbbCreateConsent->setRequiredContactFields($required_contact_fields);
```

```

$pbbsCreateConsent->setSupportedPaymentMethods($supported_payment_methods);
$pbbsCreateConsent->setPbbPayment($currency, $amount, $sub_total, $tax, $shipping_cost,
$additional_fees);
$pbbsCreateConsent->setPbbOrder($shipment_type, $merchant_order_ref, $placement_mode, $items);
$mpgTxn = new mpgTransaction($pbbsCreateConsent);
// $mpgTxn->setPbbPayment($payment);
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode($processing_country_code);
$mpgRequest->setTestMode(true);
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nConsentId = " . $mpgResponse->getMpgResponseData()["ConsentId"] );
print("\nRequestId = " . $mpgResponse->getMpgResponseData()["RequestId"] );
print("\nCorrelationId = " . $mpgResponse->getMpgResponseData()["CorrelationId"] );
print("\nDeviceProfileSessionId = " . $mpgResponse->getMpgResponseData(
["DeviceProfileSessionId"]);
?>

```

### 7.3.2 Get Consent ID

With the Consent ID from your previous Create Consent call, the merchant can use this Moneris Gateway API call to return the details about the consent process and populate your order confirmation page. This call is used after the cardholder has logged into their Konek profile and selected their payment method, but just before the cardholder has finally confirmed the order for payment processing.

#### Get Consent Request transaction object definition

```
$pbbs_consent_id = new PbbGetConsentId();
```

#### HttpPostRequest object for Get Consent transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

#### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

#### Get Consent Request transaction request fields – Required

| Variable Name | Type and Limits |                    |
|---------------|-----------------|--------------------|
| Consent ID    | String          | \$pbbs_consent_id- |

| Variable Name | Type and Limits                     |                                                                                                                              |
|---------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| consent_id    | 36-character alphanumeric, variable | >setConsentId(\$consent_id);<br><br>Consent token obtained from the Interac Konek SDK. Used to authenticate PBB connections. |

### Sample Get Consent Data Request

```

<?php
require "../mpgClasses.php";
$type = "pbb_get_consent_id";
$store_id = "monca03650";
$api_token = "7Yw0MPTlhjBRcZiE6837";
$processing_country_code = "CA";
$consent_id = 'eff5969e-6da9-406e-b5bf-603707f11195';
$correlation_id = '37b77ce9-2ef6-4d08-a8d0-6748b87c4db4';
$ext_id = -1;
$is_phoenix = true;
$pbb_consent_id = new PbbGetConsentId();
$pbb_consent_id->setConsentId($consent_id);
$pbb_consent_id->setExtId($ext_id);
$pbb_consent_id->setIsPhoenix($is_phoenix);
$mpgTxn = new mpgTransaction($pbb_consent_id);
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode($processing_country_code);
$mpgRequest->setTestMode(true);
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nMessage = " . $mpgResponse->getMessage());
print("\nCreatedOn = " . $mpgResponse->getCreatedOn());
print("\nConsentId = " . $mpgResponse->getConsentId());
//CustomerInfo
print("\nName = " . $mpgResponse->getName());
print("\nEmail = " . $mpgResponse->getEmail());
print("\nPhoneNumber = " . $mpgResponse->getPhoneNumber());
print("\nAuthMethod = " . $mpgResponse->getAuthMethod());
print("\nUserType = " . $mpgResponse->getUserType());
//Status
print("\nStatus = " . $mpgResponse->getStatus());
// Order
print("\nMerchantOrderRef = " . $mpgResponse->getMerchantOrderRef());
print("\nMerchantCustomerRef = " . $mpgResponse->getMerchantCustomerRef());
print("\nPlacementMode = " . $mpgResponse->getPlacementMode());
print("\nShipmentType = " . $mpgResponse->getShipmentType());
// Items
foreach ($mpgResponse->items as $item) {
print("\n\nItem ItemRef = " . ifPresentReturnValue($item, "ItemRef"));
print("\nItem ItemPaymentType = " . ifPresentReturnValue($item, "ItemPaymentType"));
print("\nItem Amount = " . ifPresentReturnValue($item, "Amount"));
print("\nItem Quantity = " . ifPresentReturnValue($item, "Quantity"));
print("\nItem Description = " . ifPresentReturnValue($item, "Description"));
print("\nItem Type = " . ifPresentReturnValue($item, "ItemType"));
}
// Payment
print("\n\nSubTotal = " . $mpgResponse->getSubTotal());
print("\nAmount = " . $mpgResponse->getAmount());
print("\nTax = " . $mpgResponse->getTax());
print("\nShippingCost = " . $mpgResponse->getShippingCost());

```

```

print("\nCurrency = " . $mpgResponse->getCurrency() );
// Additional Fees
foreach($mpgResponse->additionalFees as $fee) {
print("\n\nFee Name = " . ifPresentReturnValue($fee, "Name") );
print("\nFee Amount = " . ifPresentReturnValue($fee, "Amount") );
}
// MerchantRequiredFields
foreach($mpgResponse->merchantRequiredContactFields as $field) {
print("\nMerchant Required Contact Fields = " . $field );
}
// MerchantSupportedPaymentMethods
foreach($mpgResponse->merchantSupportedPaymentMethods as $method) {
print("\nMerchant Supported Payment Methods = " . $method );
}
function ifPresentReturnValue($array, $key) {
if (array_key_exists($key, $array)) {
return $array[$key];
}
return "";
}
?>
<?php

?>

```

### 7.3.3 Get Payment Transaction Data Request

After customer review of their order, the merchant's checkout webpage must use the Consent ID to initiate a request for payment transaction data. The data includes the Payment Token for use in the subsequent financial transaction (Purchase or Preauthorization) and any cryptogram provided by the cardholder's bank if the payment method is not a credit card product.

#### Get Payment Transaction Data transaction object definition

```
$pbb_get_transaction_data = new PbbGetTransactionData();
```

#### HttpPostRequest object for Get Payment Transaction Data transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

#### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

**Get Payment Transaction Data Request transaction request fields – Required**

| Variable Name    | Type and Limits                                                                         |                                                                                                                                                                                                       |
|------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Consent ID       | <i>String</i><br>36-character alphanumeric, variable                                    | <pre>\$pbb_get_transaction_data-&gt;setConsentId(\$consent_id);</pre> <p>Consent token obtained from the Interac Konek SDK. Used to authenticate PBB connections.</p>                                 |
| Amount           | <i>Number</i><br>10-character numeric, variable<br>minimum: 0.01<br>maximum: 9999999.99 | <pre>\$pbb_get_transaction_data-&gt;setAmount(\$amount);</pre> <p>Amount for use in subsequent financial transaction.</p>                                                                             |
| Currency         | <i>String</i><br>3-character alphanumeric                                               | <pre>\$pbb_get_transaction_data-&gt;setCurrency(\$currency);</pre> <p>ISO 4217 currency code used for the payment. Possible values</p> <p>CAD</p>                                                     |
| Payment Type     | <i>String</i><br>9-character alphanumeric, variable                                     | <pre>\$pbb_get_transaction_data-&gt;setTransactionType(\$transaction_type);</pre> <p>Payment type.</p> <p>Enums: SINGLE, RECURRING</p>                                                                |
| Transaction Type | <i>String</i><br>9-character alphanumeric, variable                                     | <pre>\$pbb_get_transaction_data-&gt;setPaymentType(\$payment_type);</pre> <p>Intended transaction type for the following financial transaction.</p> <p>Enums: PURCHASE, PRE_AUTH, REFUND, CAPTURE</p> |

**Sample Get Transaction Data Request**



```
<?php
require "../mpgClasses.php";
$store_id = "monca03650";
$api_token = "7YwOMPTlhjBRcZiE6837";
$processing_country_code = "CA";
$consent_id = 'f4aa6f89-6438-4bd2-89b7-d805e280b4a8';
$amount = "21.00";
$currency = "CAD";
$transaction_type = "PRE_AUTH";
$payment_type = "SINGLE";
$pbbs_get_transaction_data = new PbbGetTransactionData();
$pbbs_get_transaction_data->setConsentId($consent_id);
$pbbs_get_transaction_data->setAmount($amount);
$pbbs_get_transaction_data->setCurrency($currency);
$pbbs_get_transaction_data->setTransactionType($transaction_type);
$pbbs_get_transaction_data->setPaymentType($payment_type);
$mpgTxn = new mpgTransaction($pbbs_get_transaction_data);
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode($processing_country_code);
$mpgRequest->setTestMode(true);
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransactionRef = " . ifPresentReturnValue($mpgResponse->getMpgResponseData(),
"TransactionRef"));
print("\nTransaction Data - PaymentMethod = " . ifPresentReturnValue($mpgResponse-
>getMpgResponseData(), "PaymentMethod"));
print("\nTransaction Data - TransactionType = " . ifPresentReturnValue($mpgResponse-
>getMpgResponseData(), "TransactionType"));
print("\nTransaction Data - Currency = " . ifPresentReturnValue($mpgResponse-
>getMpgResponseData(), "Currency"));
print("\nTransaction Data - Amount = " . ifPresentReturnValue($mpgResponse-
>getMpgResponseData(), "Amount"));
print("\nTransaction Data - TokenExpiry = " . ifPresentReturnValue($mpgResponse-
>getMpgResponseData(), "TokenExpiry"));
print("\nTransaction Data - PaymentToken = " . ifPresentReturnValue($mpgResponse-
>getMpgResponseData(), "PaymentToken"));
print("\nTransaction Data - Cryptogram = " . ifPresentReturnValue($mpgResponse-
>getMpgResponseData(), "Cryptogram"));
print("\nTransaction Data - CryptogramExpiry = " . ifPresentReturnValue($mpgResponse-
>getMpgResponseData(), "CryptogramExpiry"));
print("\nTokenPanLastDigits = " . ifPresentReturnValue($mpgResponse->getMpgResponseData(),
"TokenPanLastDigits"));
function ifPresentReturnValue($array, $key) {
if (array_key_exists($key, $array)) {
return $array[$key];
}
return "";
}
?>
```

### 7.3.4 Cancel Consent

Cancels a Consent ID within Interac Konek systems previously created using Create Consent.

This call supplies Moneris Gateway and Interac Konek with the consent ID for cancellation, as well as a reason for the cancellation.

**Cancel Consent Request transaction object definition**

```
$pbbCancelConsent = new PbbCancelConsent();
```

**HttpPostRequest object for Get Consent transaction**

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

**Core connection object fields (all API transactions)**

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

**Cancel Consent Request transaction request fields – Required**

| Variable Name    | Type and Limits                                 |                                                                                                                                                         |
|------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Consent ID       | String<br>36-character alpha-numeric, variable  | \$pbb_cancel_consent->setConsentId(\$consent_id);<br><br>Consent token obtained from the Interac Konek SDK. Used to authenticate PBB connections.       |
| Cancel Reference | String<br>36-character alpha-numeric, variable  | \$pbb_cancel_consent->setCancelRef(\$cancel_ref);<br><br>Uniquely identifying reference for the consent cancellation request. Assigned by the merchant. |
| Cancel Reason    | String<br>200-character alpha-numeric, variable | \$pbb_cancel_consent->setCancelReason(\$cancel_reason);<br><br>Specifies reason for cancellation                                                        |

**Sample Cancel Consent Request**

```
<?php
require "../mpgClasses.php";
$store_id = "monca03650";
$api_token = "7Yw0MPTlhjBRcZiE6837";
```

```

$processing_country_code = "CA";
$consent_id = 'ea534ba8-8d92-4510-a728-2b40e1bde735';
$cancel_ref = 'e32f9a6b-843c-4fbb-a8ac-2b11bbca7fd3';
$cancel_reason = 'this is the optional cancel';
$pbbs_cancel_consent = new PbbCancelConsent();
$pbbs_cancel_consent->setConsentId($consent_id);
$pbbs_cancel_consent->setCancelReference($cancel_ref);
$pbbs_cancel_consent->setCancelReason($cancel_reason);
$mpgTxn = new mpgTransaction($pbbs_cancel_consent);
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode($processing_country_code);
$mpgRequest->setTestMode(true);
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
$correlation_id = ifPresentReturnValue($mpgResponse->getMpgResponseData(), "CorrelationId");
if ($correlation_id != null && $correlation_id != "") {
    print("\nRequestId = " . ifPresentReturnValue($mpgResponse->getMpgResponseData(),
    "RequestId"));
    print("\nCorrelationId = " . $correlation_id);
} else {
    print("\nCode = " . ifPresentReturnValue($mpgResponse->getMpgResponseData(), "Code"));
    print("\nType = " . ifPresentReturnValue($mpgResponse->getMpgResponseData(), "Type"));
    print("\nMessage = " . $mpgResponse->getMessage());
}
function ifPresentReturnValue($array, $key) {
    if (array_key_exists($key, $array)) {
        return $array[$key];
    }
    return "";
}
?>

```

## 7.4 Performing the Authorization

Financial transactions will need some level of Pay By Bank data included. This section covers how to map PBB information into Purchases, Preauthorizations, Completions, Refunds, and Voids. We breakdown the differences between Card Based Payments versus Account Based Payments, as well as including specifics for how Visa, Mastercard, and Amex handle their PBB information.

### 7.4.1 Card Based Payments

| Transaction Type | Fields                                                                                                                                 | Notes                                                                                                                                                                                  |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purchase</b>  | <p>pbb_info object with the following fields:</p> <ul style="list-style-type: none"> <li>consent_id</li> <li>payment_method</li> </ul> | <p>Execute a <code>getTransactionData</code> call prior to the Purchase to obtain the necessary payment information.</p> <p>See "Get Payment Transaction Data Request" on page 323</p> |
| <b>Preauth</b>   | <p>pbb_info object with the following fields:</p>                                                                                      | <p>Execute a <code>getTransactionData</code> call</p>                                                                                                                                  |

| Transaction Type  | Fields                                                                                                                                                                | Notes                                                                                                                                                                                                                                                                                                                                 |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | <ul style="list-style-type: none"> <li>consent_id</li> <li>payment_method</li> </ul> <p>and is_incremental field = true when processing an Account-Based Payment.</p> | <p>prior to the Preauthorization to obtain the necessary payment information.</p> <p>See "Get Payment Transaction Data Request" on page 323</p> <p>Account-Based Payments follow Interac rules for pre-authorization and completion; merchants may complete the transaction for up to 15% more than the original pre-auth amount.</p> |
| <b>Completion</b> | N/A                                                                                                                                                                   | Do not pass pbb_info object. Process as any other credit card completion.                                                                                                                                                                                                                                                             |
| <b>Refund</b>     | N/A                                                                                                                                                                   | Do not pass pbb_info object. Process as any other credit card refund.                                                                                                                                                                                                                                                                 |
| <b>Void</b>       | N/A                                                                                                                                                                   | Do not pass pbb_info object. Process as any other credit card void.                                                                                                                                                                                                                                                                   |

#### 7.4.1.1 Visa and Mastercard Card Based Payments

In addition to the the transaction specific rules above, Visa and Mastercard also have general rules for all financial transactions. Populate fields using the following:

| Fields                       | Populate With...                                                                                                                |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| expdate                      | TokenExpiry from the Transaction Data object in your <b>Get Payment Transaction Data</b> response. Expiry format is <b>YYMM</b> |
| pan                          | PaymentToken from the Transaction Data object in your <b>Get Payment Transaction Data</b> response.                             |
| wallet_indicator             | PBB for "Pay By Bank"                                                                                                           |
| token_cryptogram             | Cryptogram from the Transaction Data object in your Get Payment Transaction Data response.                                      |
| <b>NOTE: MasterCard Only</b> | <b>NOTE: token_cryptogram is found outside the pbb_info object.</b>                                                             |

### 7.4.1.2 Amex Card Based Payments

In addition to the the transaction specific rules above, Amex also has general rules for all financial transactions. Populate fields using the following:

| Fields           | Populate With...                                                                                                                |
|------------------|---------------------------------------------------------------------------------------------------------------------------------|
| expdate          | TokenExpiry from the Transaction Data object in your <b>Get Payment Transaction Data</b> response. Expiry format is <b>YYMM</b> |
| pan              | PaymentToken from the Transaction Data object in your <b>Get Payment Transaction Data</b> response.                             |
| wallet_indicator | PBB for "Pay By Bank"                                                                                                           |

### 7.4.2 Account Based Payments

| Transaction Type  | Fields                                                                                                                                                                                                           | Notes                                                                                                                                                                                          |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purchase</b>   | <p>pbb_info object with the following fields:</p> <ul style="list-style-type: none"> <li>• consent_id</li> <li>• payment_method</li> <li>• cryptogram</li> <li>• cryptogram_expiry</li> <li>• channel</li> </ul> | <p>Execute a <code>getTransactionData</code> call prior to the Purchase to obtain the necessary payment information.</p> <p>See "Get Payment Transaction Data Request" on page 323</p>         |
| <b>Preauth</b>    | <p>pbb_info object with the following fields:</p> <ul style="list-style-type: none"> <li>• consent_id</li> <li>• payment_method</li> <li>• cryptogram</li> <li>• cryptogram_expiry</li> <li>• channel</li> </ul> | <p>Execute a <code>getTransactionData</code> call prior to the Preauthorization to obtain the necessary payment information.</p> <p>See "Get Payment Transaction Data Request" on page 323</p> |
| <b>Completion</b> | <p>pbb_info object with the following fields:</p>                                                                                                                                                                | <p>Execute a <code>getTransactionData</code> call prior to the Preauthorization to obtain the</p>                                                                                              |

| Transaction Type              | Fields                                                                                                                                                                                                                                              | Notes                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <ul style="list-style-type: none"> <li>• consent_id</li> <li>• payment_method</li> <li>• cryptogram</li> <li>• cryptogram_expiry</li> <li>• channel</li> </ul>                                                                                      | necessary payment information, with transaction_type = CAPTURE                                                                                                                                                                                                                                                                                                  |
| <b>Multiple Completion</b>    | <p>pbb_info object with the following fields:</p> <ul style="list-style-type: none"> <li>• consent_id</li> <li>• payment_method</li> <li>• cryptogram</li> <li>• cryptogram_expiry</li> <li>• channel</li> </ul> <p>shipping_indicator = P or F</p> | <p>Execute a getTransactionData call prior to the Preauthorization to obtain the necessary payment information, with transaction_type = CAPTURE</p> <div> <p><b>NOTE:</b> The amount passed in getTransactionData call initially should match that of the original Preauth transaction.</p> </div>                                                              |
| <b>Zero-Dollar Completion</b> | <p>pbb_info object with the following fields:</p> <ul style="list-style-type: none"> <li>• consent_id</li> <li>• payment_method</li> <li>• cryptogram</li> <li>• cryptogram_expiry</li> <li>• channel</li> </ul>                                    | <p>Execute a getTransactionData call prior to the Preauthorization to obtain the necessary payment information, with transaction_type = CAPTURE</p> <div> <p><b>NOTE:</b> The amount passed in getTransactionData call initially should match that of the original Preauth transaction. However, when performing the actual completion amount = 0.00</p> </div> |
| <b>Refund</b>                 | <p>pbb_info object with the following fields:</p> <ul style="list-style-type: none"> <li>• consent_id</li> <li>• life_cycle_trace</li> <li>• channel</li> </ul>                                                                                     | Obtain Life Cycle Trace ID from the response to the initial Purchase or Preauthorization transaction.                                                                                                                                                                                                                                                           |

## 7.5 Testing Your Pay By Bank Integration

In the testing stage of development:

1. Use the testing URL as Host for your requests:  
esqa.moneris.com
2. In all Get Consent Data transactions, make sure that you are using the testing version of your credentials for store ID and API token
3. In all Get Payment Transaction Data transactions, make sure that you are using the testing version of your credentials for store ID and API token

## 7.6 Moving to Production With Pay By Bank

Once you have finished testing your Pay By Bank integration, do the following to move the integration into production:

1. Use the production URL as Host for your requests:  
www3.moneris.com
2. In all Get Consent Data transactions, make sure that you are using the production version of your credentials for store ID and API token
3. In all Get Payment Transaction Data transactions, make sure that you are using the production version of your credentials for store ID and API token

## 8 Multi-Currency Pricing (MCP)

- 8.1 About Multi-Currency Pricing (MCP)
- 8.2 Methods of Processing MCP Transactions
- 8.3 MCP Get Rate
- 8.4 MCP Purchase
- 8.5 MCP Purchase with 3-D Secure
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- 8.10 MCP Pre-Authorization Completion
- 8.11 MCP Purchase Correction
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- 8.13 MCP Independent Refund
- 8.14 MCP Purchase with Vault
- 8.15 MCP Pre-Authorization with Vault
- 8.16 MCP Independent Refund with Vault
- 8.17 MCP Currency Codes
- 8.18 MCP Error Codes

### 8.1 About Multi-Currency Pricing (MCP)

Multi-currency pricing (MCP) is a financial service which allows businesses to price goods and services in a variety of foreign currencies, while continuing to receive settlement and reporting in Canadian dollars. MCP allows cardholders to shop, view prices and pay in the currency of their choice.

MCP is only available when processing Visa and Mastercard transactions.

**NOTE:** Use MCP only when processing transactions that involve foreign currency exchange; for transactions strictly in Canadian dollars, use the basic financial transaction requests



## 8.2 Methods of Processing MCP Transactions

There are two methods of processing multi-currency pricing transactions via the Moneris Gateway:

1. **Using the MCP Get Rate transaction** – this method is used to obtain a foreign exchange rate and locks that specific rate in for a limited time, and is applied in a subsequent transaction
2. **Without using MCP Get Rate** – this method sends a MCP transaction without performing the Get Rate request, and the foreign exchange rate is obtained at processing time

## 8.3 MCP Get Rate

Performs a foreign currency exchange rate look-up, and secures that exchange rate for use in a subsequent MCP financial transaction.

The exchange rate retrieved by this transaction request is represented in the response as the **RateToken**, and the underlying exchange rate is locked in for a limited time period.

### MCP Get Rate transaction object definition

```
$txnArray = array('type'=>'mcp_get_rate', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for MCP Get Rate transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### MCP Get Rate transaction request fields – Required

| Variable Name         | Type and Limits                             | Set Method                           |
|-----------------------|---------------------------------------------|--------------------------------------|
| MCP version number    | String<br>numeric<br>current version is 1.0 | 'mcp_version'=> \$mcp_version        |
| rate transaction type | String<br>1-character alphabetic            | 'rate_txn_type'=>\$rate_txn_type     |
| MCP Rate Info         | Object<br>N/A                               | \$mpgTxn->setMCPRateInfo(\$mcpRate); |

## MCP Rate Info object request fields

At least one of the following variables must be sent:

| Variable Name                  | Type and Limits                                                                                                                         | Set Method                                                                                     |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| add cardholder amount          | <i>String array</i><br><br>12-character numeric, 3-character numeric<br><br>(smallest discrete unit of foreign currency, currency code) | <code>\$mcpRate-&gt;setCardholderAmount('FOREIGN_AMT', 'FOREIGN_CURRENCY_CODE');</code>        |
| add merchant settlement amount | <i>String array</i><br><br>12-character numeric, 3-character numeric<br><br>(amount in CAD pennies, currency code)                      | <code>\$mcpRate-&gt;setMerchantSettlementAmount('CAD_AMOUNT', 'FOREIGN_CURRENCY_CODE');</code> |

## Sample MCP Get Rate

```
<?php
##
## Example php -q TestPurchase.php store1
##
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
$type = 'mcp_get_rate';
$mcp_version = '1.0';
$rate_txn_type = 'P';
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'mcp_version'=>$mcp_version,
'rate_txn_type'=>$rate_txn_type
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Credential on File *****/
$mcpRate = new MCPRate();
$mcpRate->setCardholderAmount('100', '840');
$mcpRate->setMerchantSettlementAmount('200', '826');
$mpgTxn->setMCPRateInfo($mcpRate);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
/* Status Check Example
$mpgHttpPost =new mpgHttpPostStatus($store_id,$api_token,$status_check,$mpgRequest);
*/
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nRateTxnType = " . $mpgResponse->getRateTxnType());
print("\nMCPRateToken = " . $mpgResponse->getMCPRateToken());
print("\nRateInqStartTime = " . $mpgResponse->getRateInqStartTime()); //The time (unix UTC)
of when the rate is requested
print("\nRateInqEndTime = " . $mpgResponse->getRateInqEndTime()); //The time (unix UTC) of
when the rate is returned
print("\nRateValidityStartTime = " . $mpgResponse->getRateValidityStartTime()); //The time
(unix UTC) of when the rate is valid from
print("\nRateValidityEndTime = " . $mpgResponse->getRateValidityEndTime()); //The time (unix
UTC) of when the rate is valid until
print("\nRateValidityPeriod = " . $mpgResponse->getRateValidityPeriod()); //The time in
minutes this rate is valid for
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
//RateData
for ($index = 0; $index < $mpgResponse->getRatesCount(); $index++)
{
    print("\nMCPRate = " . $mpgResponse->getMCPRate($index));
    print("\nMerchantSettlementCurrency = " . $mpgResponse->getMerchantSettlementCurrency
($index));
    print("\nMerchantSettlementAmount = " . $mpgResponse->getMerchantSettlementAmount($index));
    //Domestic (CAD) amount
    print("\nCardholderCurrencyCode = " . $mpgResponse->getCardholderCurrencyCode($index));
    print("\nCardholderAmount = " . $mpgResponse->getCardholderAmount($index)); //Foreign amount

    print("\nMCPErrorStatusCode = " . $mpgResponse->getMCPErrorStatusCode($index));
    print("\nMCPErrorMessage = " . $mpgResponse->getMCPErrorMessage($index));
}
?>
```

## 8.4 MCP Purchase

Verifies funds on the customer's card, removes the funds and prepares them for deposit into the merchant's account.

This transaction request is the multi-currency pricing (MCP) enabled version of the equivalent financial transaction.

### MCP Purchase transaction object definition

```
$txnArray = array('type'=>'mcp_purchase', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for MCP Purchase transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

**Core connection object fields (all API transactions)**

| Variable Name | Type and Limits      | Set Method               |
|---------------|----------------------|--------------------------|
| store ID      | <i>String</i><br>N/A | 'store_id'=>\$store_id   |
| API token     | <i>String</i><br>N/A | 'api_token'=>\$api_token |

**MCP Purchase transaction request fields – Required**

| Variable Name                 | Type and Limits                                                            | Set Method                   |
|-------------------------------|----------------------------------------------------------------------------|------------------------------|
| order ID                      | <i>String</i><br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces | 'order_id'=>\$order_id       |
| credit card number            | <i>String</i><br>max 20-character alphanumeric                             | 'pan'=>\$pan                 |
| expiry date                   | <i>String</i><br>4-character alphanumeric<br>YYMM                          | 'expiry_date'=>\$expiry_date |
| electronic commerce indicator | <i>String</i><br>1-character alphanumeric                                  | 'crypt_type'=>\$crypt        |

**MCP-specific request fields – Required**

| Variable Name      | Type and Limits                                    | Set Method                                    |
|--------------------|----------------------------------------------------|-----------------------------------------------|
| MCP version number | <i>String</i><br>numeric<br>current version is 1.0 | 'mcp_version'=> \$mcp_version                 |
| cardholder amount  | <i>String</i>                                      | 'cardholder_amount' =><br>\$cardholder_amount |

| Variable Name            | Type and Limits                            | Set Method                                               |
|--------------------------|--------------------------------------------|----------------------------------------------------------|
|                          | 12-character numeric                       |                                                          |
|                          | smallest discrete unit of foreign currency |                                                          |
| cardholder currency code | <i>String</i><br>3-character numeric       | 'cardholder_currency_code' => \$cardholder_currency_code |

**MCP Purchase transaction request fields – Optional**

| Variable Name                                                    | Type and Limits                                                                                                                                                                                      | Set Method                                 |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| customer ID                                                      | <i>String</i><br>50-character alphanumeric<br><br><b>NOTE:</b><br>Some special characters are not allowed:<br><>\$%=?^{}[]\                                                                          | 'cust_id'=>\$cust_id                       |
| dynamic descriptor                                               | <i>String</i><br>20-character alphanumeric<br><br>total of 22 characters including your merchant name and separator<br><br><b>NOTE:</b><br>Some special characters are not allowed:<br><>\$%=?^{}[]\ | 'dynamic_descriptor'=>\$dynamic_descriptor |
| wallet indicator                                                 | <i>String</i><br>3-character alphanumeric                                                                                                                                                            | 'wallet_indicator'=>\$wallet_indicator     |
| Credential on File Info                                          | <i>Object</i>                                                                                                                                                                                        | \$mpgTxn->setCofInfo(\$cof);               |
| cof                                                              | N/A                                                                                                                                                                                                  |                                            |
| <b>NOTE:</b> This is a nested object within the transaction, and |                                                                                                                                                                                                      |                                            |

| Variable Name                                                                                                                                                                                               | Type and Limits | Set Method                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------------------------|
| <div> <p>required when storing or using the customer's stored credentials. For information about fields in the Credential on File Info object, see Credential on File Info Object and Variables.</p> </div> |                 |                                     |
| AVS Information                                                                                                                                                                                             | Object<br>N/A   | \$mpgTxn->setAvsInfo(\$mpgAvsInfo); |
| CVD Information                                                                                                                                                                                             | Object<br>N/A   | \$mpgTxn->setCvdInfo(\$mpgCvdInfo); |

### MCP-specific request fields – Optional

| Variable Name  | Type and Limits | Set Method                           |
|----------------|-----------------|--------------------------------------|
| MCP rate token | String<br>N/A   | 'mcp_rate_token' => \$mcp_rate_token |

### Sample MCP Purchase

```

<?php
##
## Example php -q TestPurchase.php store1
##
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
/***** Transactional Variables *****/
$type='mcp_purchase';
$cust_id='cust id';
$order_id='ord-'.date("dmy-G:i:s");
$pan='4242424242424242';
$expiry_date='2011';
$crypt='7';
$dynamic_descriptor='123';
$status_check = 'false';
$mcp_version = '1.0';
$cardholder_amount = '100';
$cardholder_currency_code = '840';
$mcp_rate_token = 'P1536163325404090';
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'cust_id'=>$cust_id,
'pan'=>$pan,
'expdate'=>$expiry_date,
```

```

'crypt_type'=>$crypt,
'dynamic_descriptor'=>$dynamic_descriptor,
'mcp_version'=> $mcp_version,
'cardholder_amount' => $cardholder_amount,
'cardholder_currency_code' => $cardholder_currency_code,
'mcp_rate_token' => $mcp_rate_token
//,'wallet_indicator' => '' //Refer to documentation for details
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("U");
$cof->setPaymentInformation("2");
$cof->setIssuerId("168451306048014");
$mpgTxn->setCofInfo($cof);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
/* Status Check Example
$mpgHttpPost =new mpgHttpPostStatus($store_id,$api_token,$status_check,$mpgRequest);
*/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nStatusCode = " . $mpgResponse->getStatusCode());
print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
print("\nHostId = " . $mpgResponse->getHostId());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\nMerchantSettlementAmount = " . $mpgResponse->getMerchantSettlementAmount());
print("\nCardholderAmount = " . $mpgResponse->getCardholderAmount());
print("\nCardholderCurrencyCode = " . $mpgResponse->getCardholderCurrencyCode());
print("\nMCPRate = " . $mpgResponse->getMCPRate());
print("\nMCPErrrorStatusCode = " . $mpgResponse->getMCPErrrorStatusCode());
print("\nMCPErrrorMessage = " . $mpgResponse->getMCPErrrorMessage());
print("\nHostId = " . $mpgResponse->getHostId());
?>

```

## 8.5 MCP Purchase with 3-D Secure

### MCP Purchase with 3-D Secure transaction object definition

```

$txnArray = array('type'=>'mcp_cavv_purchase', ...);

$mpgTxn = new mpgTransaction($txnArray);

```

**HttpPostRequest object for MCP Purchase with 3-D Secure transaction**

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

**Core connection object fields (all API transactions)**

| Variable Name | Type and Limits      |                          |
|---------------|----------------------|--------------------------|
| store ID      | <i>String</i><br>N/A | 'store_id'=>\$store_id   |
| API token     | <i>String</i><br>N/A | 'api_token'=>\$api_token |

**MCP Purchase with 3-D Secure transaction request fields – Required**

| Variable Name                                       | Type and Limits                                                            | Set Method                   |
|-----------------------------------------------------|----------------------------------------------------------------------------|------------------------------|
| order ID                                            | <i>String</i><br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces | 'order_id'=>\$order_id       |
| credit card number                                  | <i>String</i><br>max 20-character alphanumeric                             | 'pan'=>\$pan                 |
| expiry date                                         | <i>String</i><br>4-character alphanumeric<br>YYMM                          | 'expiry_date'=>\$expiry_date |
| Cardholder Authentication Verification Value (CAVV) | <i>String</i><br>50-character alphanumeric                                 | cavv=>\$cavv                 |
| electronic commerce indicator                       | <i>String</i><br>1-character alphanumeric                                  | 'crypt_type'=>\$crypt        |



**MCP-specific request fields – Required**

| Variable Name            | Type and Limits                                                              | Set Method                                               |
|--------------------------|------------------------------------------------------------------------------|----------------------------------------------------------|
| MCP version number       | String<br>numeric<br>current version is 1.0                                  | 'mcp_version'=> \$mcp_version                            |
| cardholder amount        | String<br>12-character numeric<br>smallest discrete unit of foreign currency | 'cardholder_amount' => \$cardholder_amount               |
| cardholder currency code | String<br>3-character numeric                                                | 'cardholder_currency_code' => \$cardholder_currency_code |

**3-D Secure 2.2 -specific fields – Required**

| Variable Name                                                                                                                                                                                    | Type and Limits                | Set Method                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3DS version<br><br><b>NOTE:</b> Mandatory for financial transactions using 3rd Party 3-D Secure services.                                                                                        | String<br>10-character numeric | 'threeds_version'=>\$threeds_version<br><br>Acceptable values:<br><br>2.0.0 = 3DS protocol 2.0.0<br><br>2.1.0 = 3DS protocol 2.1.0<br><br>2.2.0 = 3DS protocol 2.2.0<br><br>2.3.0 = 3DS protocol 2.3.0 |
| 3DS server transaction ID<br><br><b>NOTE:</b> Mandatory for financial transactions using 3rd Party 3-D Secure services - obtained from the Cavv Lookup request or MPI 3DS Authentication request | String<br>36-character numeric | 'threeds_server_trans_id'=>\$threeds_server_trans_id                                                                                                                                                   |

## MCP Purchase with 3-D Secure transaction request fields – Optional

| Variable Name                                                                                                                                                                                                                                                                 | Type and Limits                                                                                                                                                                                                                                            | Set Method                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| customer ID                                                                                                                                                                                                                                                                   | <i>String</i><br>50-character alphanumeric<br><div> <b>NOTE:</b><br/>           Some special characters are not allowed:<br/>           &lt; &gt; \$ % = ? ^ { } [ ] \         </div>                                                                      | 'cust_id'=>\$cust_id                       |
| dynamic descriptor                                                                                                                                                                                                                                                            | <i>String</i><br>20-character alphanumeric<br>total of 22 characters including your merchant name and separator<br><div> <b>NOTE:</b><br/>           Some special characters are not allowed:<br/>           &lt; &gt; \$ % = ? ^ { } [ ] \         </div> | 'dynamic_descriptor'=>\$dynamic_descriptor |
| wallet indicator                                                                                                                                                                                                                                                              | <i>String</i><br>3-character alphanumeric                                                                                                                                                                                                                  | 'wallet_indicator'=>\$wallet_indicator     |
| Credential on File Info                                                                                                                                                                                                                                                       | <i>Object</i>                                                                                                                                                                                                                                              | \$mpgTxn->setCofInfo(\$cof);               |
| cof                                                                                                                                                                                                                                                                           | N/A                                                                                                                                                                                                                                                        |                                            |
| <div> <b>NOTE:</b> This is a nested object within the transaction, and required when storing or using the customer's stored credentials. For information about fields in the Credential on File Info object, see Credential on File Info Object and Variables.         </div> |                                                                                                                                                                                                                                                            |                                            |
| AVS Information                                                                                                                                                                                                                                                               | <i>Object</i><br>N/A                                                                                                                                                                                                                                       | \$mpgTxn->setAvsInfo(\$mpgAvsInfo);        |
| CVD Information                                                                                                                                                                                                                                                               | <i>Object</i>                                                                                                                                                                                                                                              | \$mpgTxn->setCvdInfo                       |

| Variable Name | Type and Limits | Set Method      |
|---------------|-----------------|-----------------|
|               | N/A             | (\$mpgCvdInfo); |

### MCP-specific request fields – Optional

| Variable Name  | Type and Limits | Set Method                           |
|----------------|-----------------|--------------------------------------|
| MCP rate token | String<br>N/A   | 'mcp_rate_token' => \$mcp_rate_token |

### 3-D Secure 2.2 -specific fields – Optional

| Variable Name                                                                         | Type and Limits                     | Set Method                     |
|---------------------------------------------------------------------------------------|-------------------------------------|--------------------------------|
| DS transaction ID                                                                     | String<br>36-character alphanumeric | 'ds_trans_id' => \$ds_trans_id |
| <b>NOTE:</b> Only used in financial transactions using 3rd Party 3-D Secure services. |                                     |                                |

### Sample MCP Purchase with 3-D Secure

```
<?php
## Example php -q TestPurchase-VBV.php "moneris" store
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='monca02932';
$api_token='CG8kYzGgzVU5z23irgMx';
/***** Transactional Variables *****/
$type='mcp_cavv_purchase';
$order_id='ord-' . date("dmy-G:i:s");
$cust_id='CUST887763';
$amount='10.00';
$pan="4740611374762707";
$expiry_date="2211";
$cavv='BwABApFSYyd4l2eQQFJjAAAAAAA=';
$crypt_type = '7';
$wallet_indicator = "APP";
$dynamic_descriptor='123456';
$mcp_version = '1.0';
$cardholder_amount = '100';
$cardholder_currency_code = '840';
$mcp_rate_token = 'P1623680755788776';
/***** Transaction Associative Array *****/
$txnArray=array(
    'type'=>$type,
    'order_id'=>$order_id,
    'cust_id'=>$cust_id,
    'amount'=>$amount,
    'pan'=>$pan,
    'expdate'=>$expiry_date,
    'cavv'=>$cavv,
    'crypt_type'=>$crypt_type, //mandatory for AMEX only
```

```
//wallet_indicator'=>$wallet_indicator, //set only for wallet transactions. e.g. APPLE PAY
//network'=> "Interac", //set only for Interac e-commerce
'data_type'=> "3DSecure", //set only for Interac e-commerce
'dynamic_descriptor'=>$dynamic_descriptor,
'threads_version' => '2.2.0', //Mandatory for financial transactions using 3rd Party 3-D
Secure services
'threads_server_trans_id' => 'e11d4985-8d25-40ed-99d6-c3803fe5e68f', //Mandatory for
financial transactions using 3rd Party 3-D Secure services - obtained from MpiCavvLookup or
MpiThreeDSAuthentication
//cm_id' => '8nAK8712sGaAkl56', //set only for usage with Offlinx - Unique max 50
alphanumeric characters transaction id generated by merchant
//ds_trans_id' => '12345', //Optional - to be used only if you are using 3rd party 3ds
service
'mcp_version'=> $mcp_version,
'cardholder_amount' => $cardholder_amount,
'cardholder_currency_code' => $cardholder_currency_code,
'mcp_rate_token' => $mcp_rate_token
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("U");
$cof->setPaymentInformation("2");
$cof->setIssuerId("139X3130ASCXAS9");
$mpgTxn->setCofInfo($cof);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //US for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nCavvResultCode = " . $mpgResponse->getCavvResultCode());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\nThreeDSVersion = " . $mpgResponse->getThreeDSVersion());
print("\nMerchantSettlementAmount = " . $mpgResponse->getMerchantSettlementAmount());
print("\nCardholderAmount = " . $mpgResponse->getCardholderAmount());
print("\nCardholderCurrencyCode = " . $mpgResponse->getCardholderCurrencyCode());
print("\nMCPRate = " . $mpgResponse->getMCPRate());
print("\nMCPErrrorStatusCode = " . $mpgResponse->getMCPErrrorStatusCode());
print("\nMCPErrrorMessage = " . $mpgResponse->getMCPErrrorMessage());
print("\nHostId = " . $mpgResponse->getHostId());
?>
```

## 8.6 MCP Purchase with 3-D Secure and Vault

### MCP Purchase with 3-D Secure and Vault transaction object definition

```
$txnArray = array('type'=>'mcp_cavv_res_purchase_cc', ...);  
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for MCP Purchase with 3-D Secure and Vault transaction

```
$mpgRequest = new mpgRequest($mpgTxn);  
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

### MCP Purchase with 3-D Secure and Vault transaction request fields – Required

| Variable Name      | Type and Limits                                                     | Set Method                   |
|--------------------|---------------------------------------------------------------------|------------------------------|
| data key           | String<br>25-character alphanumeric                                 | 'data_key'=>\$data_key       |
| order ID           | String<br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces | 'order_id'=>\$order_id       |
| credit card number | String<br>max 20-character alpha-numeric                            | 'pan'=>\$pan                 |
| expiry date        | String<br>4-character alphanumeric<br>YYMM                          | 'expiry_date'=>\$expiry_date |

| Variable Name                                       | Type and Limits                            | Set Method            |
|-----------------------------------------------------|--------------------------------------------|-----------------------|
| Cardholder Authentication Verification Value (CAVV) | <i>String</i><br>50-character alphanumeric | cavv=>\$cavv          |
| electronic commerce indicator                       | <i>String</i><br>1-character alphanumeric  | 'crypt_type'=>\$crypt |

**MCP-specific request fields – Required**

| Variable Name            | Type and Limits                                                                     | Set Method                                                  |
|--------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------|
| MCP version number       | <i>String</i><br>numeric<br>current version is 1.0                                  | 'mcp_version'=> \$mcp_version                               |
| cardholder amount        | <i>String</i><br>12-character numeric<br>smallest discrete unit of foreign currency | 'cardholder_amount' =><br>\$cardholder_amount               |
| cardholder currency code | <i>String</i><br>3-character numeric                                                | 'cardholder_currency_code'<br>=> \$cardholder_currency_code |

**3-D Secure 2.2 -specific fields – Required**

| Variable Name             | Type and Limits                       | Set Method                                                                                                                                                                                             |
|---------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3DS version               | <i>String</i><br>10-character numeric | 'threeds_version'=>\$threeds_version<br><br>Acceptable values:<br><br>2.0.0 = 3DS protocol 2.0.0<br><br>2.1.0 = 3DS protocol 2.1.0<br><br>2.2.0 = 3DS protocol 2.2.0<br><br>2.3.0 = 3DS protocol 2.3.0 |
| 3DS server transaction ID | <i>String</i>                         | 'threeds_server_trans_                                                                                                                                                                                 |

**NOTE:** Mandatory for financial transactions using 3rd Party 3-D Secure services.

| Variable Name                                                                                                                                                   | Type and Limits      | Set Method                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------------------------------------|
| <b>NOTE:</b> Mandatory for financial transactions using 3rd Party 3-D Secure services - obtained from the Cavv Lookup request or MPI 3DS Authentication request | 36-character numeric | <code>id'=&gt;\$threeds_server_trans_id</code> |

### MCP Purchase with 3-D Secure and Vault transaction request fields – Optional

| Variable Name                                                                                                         | Type and Limits                                                                                                                                                                                                            | Set Method                                                 |
|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| customer ID                                                                                                           | <i>String</i><br>50-character alphanumeric<br><b>NOTE:</b><br>Some special characters are not allowed:<br><code>&lt; &gt; \$ % = ? ^ { } [ ] \</code>                                                                      | <code>'cust_id'=&gt;\$cust_id</code>                       |
| dynamic descriptor                                                                                                    | <i>String</i><br>20-character alphanumeric<br>total of 22 characters including your merchant name and separator<br><b>NOTE:</b><br>Some special characters are not allowed:<br><code>&lt; &gt; \$ % = ? ^ { } [ ] \</code> | <code>'dynamic_descriptor'=&gt;\$dynamic_descriptor</code> |
| wallet indicator                                                                                                      | <i>String</i><br>3-character alphanumeric                                                                                                                                                                                  | <code>'wallet_indicator'=&gt;\$wallet_indicator</code>     |
| Credential on File Info                                                                                               | <i>Object</i>                                                                                                                                                                                                              | <code>\$mpgTxn-&gt;setCofInfo(\$cof);</code>               |
| cof                                                                                                                   | N/A                                                                                                                                                                                                                        |                                                            |
| <b>NOTE:</b> This is a nested object within the transaction, and required when storing or using the customer's stored |                                                                                                                                                                                                                            |                                                            |

| Variable Name                                                                                                                                          | Type and Limits | Set Method                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------------------------|
| <div> <p>credentials. For information about fields in the Credential on File Info object, see Credential on File Info Object and Variables.</p> </div> |                 |                                     |
| AVS Information                                                                                                                                        | Object<br>N/A   | \$mpgTxn->setAvsInfo(\$mpgAvsInfo); |
| CVD Information                                                                                                                                        | Object<br>N/A   | \$mpgTxn->setCvdInfo(\$mpgCvdInfo); |

### 3-D Secure 2.2 -specific fields – Optional

| Variable Name                                                                                             | Type and Limits                     | Set Method                     |
|-----------------------------------------------------------------------------------------------------------|-------------------------------------|--------------------------------|
| DS transaction ID                                                                                         | String<br>36-character alphanumeric | 'ds_trans_id' => \$ds_trans_id |
| <div> <p><b>NOTE:</b> Only used in financial transactions using 3rd Party 3-D Secure services.</p> </div> |                                     |                                |

### Sample MCP Purchase with 3-D Secure and Vault

```
<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store1';
$api_token='yesguyl';
/***** Transaction Variables *****/
$orderid='res-purchase-' . date("dmy-G:i:s");
$data_key='4INQR1A8ocxD0oafSz50LADxy';
$amount='1.00';
$cavv='AAABBJg0VhI0VniQEjRWAAAAA';
$custid='customer1'; //if sent will be submitted, otherwise cust_id from profile will be used
$expdate = '1901'; //YYMM - used only for temp token
$crypt_type = '7'; //value obtained from MpiACS transaction
$mcp_version = '1.0';
$cardholder_amount = '100';
$cardholder_currency_code = '840';
$mcp_rate_token = 'P1623436475143166';
/***** Transaction Array *****/
$txnArray =array('type'=>'mcp_res_cavv_purchase_cc',
'data_key'=>$data_key,
'order_id'=>$orderid,
'cust_id'=>$custid,
'amount'=>$amount,
'cavv'=>$cavv,
'expdate'=>$expdate, //mandatory for temp tokens only
```



```

'crypt_type'=>$crypt_type, //set for AMEX SafeKey only
//'dynamic_descriptor'=>'12346',
'threads_version' => '2.2.0', //Mandatory for financial transactions using 3rd Party 3-D
Secure services.
'threads_server_trans_id' => 'e11d4985-8d25-40ed-99d6-c3803fe5e68f', //Mandatory for
financial transactions using 3rd Party 3-D Secure services - obtained from MpiCavvLookup or
MpiThreeDSAuthentication
//'ds_trans_id' => '12345', //Optional - to be used only if you are using 3rd party 3ds
service
'mcp_version'=> $mcp_version,
'cardholder_amount' => $cardholder_amount,
'cardholder_currency_code' => $cardholder_currency_code,
'mcp_rate_token' => $mcp_rate_token
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("U");
$cof->setPaymentInformation("2");
$cof->setIssuerId("168451306048014");
$mpgTxn->setCofInfo($cof);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** mpgHttpPost Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response Object *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nAVSResponse = " . $mpgResponse->getAvsResultCode());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
print("\nCavvResultCode = " . $mpgResponse->getCavvResultCode());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\nThreeDSVersion = " . $mpgResponse->getThreeDSVersion());
//----- ResolveData -----
print("\n\nCust ID = " . $mpgResponse->getResDataCustId());
print("\n\nPhone = " . $mpgResponse->getResDataPhone());
print("\n\nEmail = " . $mpgResponse->getResDataEmail());
print("\n\nNote = " . $mpgResponse->getResDataNote());
print("\n\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\n\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\n\nCrypt Type = " . $mpgResponse->getResDataCryptType());
print("\n\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
print("\n\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\n\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
?>

```

## 8.7 MCP Pre-Authorization

Verifies and locks funds on the customer's credit card. The funds are locked for a specified amount of time based on the card issuer. For situations where the merchant does not know the precise amount when the transaction begins, a Pre-Authorization transaction may use an estimated amount instead. If the cardholder incurs additional charges during the transaction period, the merchant is able to increase the total amount authorized via Incremental Pre-Authorization; these increments are added to the previously authorized amount. To retrieve the funds that have been locked by a Pre-Authorization transaction so that they may be settled in the merchant's account, a Pre-Authorization Completion transaction must be performed. A Pre-Authorization transaction may only be "completed" once. If the Pre-Authorization is not completed within the allowable chargeback protection period then a new authorization should be obtained. For Mastercard, the merchant can instead submit an Incremental Pre-Authorization with a \$0 amount field associated to the original Pre-Authorization transaction; if approved, this extends the protection period.

This transaction request is the multi-currency pricing (MCP) enabled version of the equivalent financial transaction.

### MCP Pre-Authorization transaction object definition

```
$txnArray = array('type'=>'mcp_preauth', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for MCP Pre-Authorization transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

### MCP Pre-Authorization transaction request fields – Required

| Variable Name | Type and Limits                     | Set Method             |
|---------------|-------------------------------------|------------------------|
| order ID      | String<br>50-character alphanumeric | 'order_id'=>\$order_id |

| Variable Name                 | Type and Limits                                           | Set Method                   |
|-------------------------------|-----------------------------------------------------------|------------------------------|
|                               | a-Z A-Z 0-9 _ - : . @ spaces                              |                              |
| credit card number            | <i>String</i><br><br>max 20-character alpha-numeric       | 'pan'=>\$pan                 |
| expiry date                   | <i>String</i><br><br>4-character alphanumeric<br><br>YYMM | 'expiry_date'=>\$expiry_date |
| electronic commerce indicator | <i>String</i><br><br>1-character alphanumeric             | 'crypt_type'=>\$crypt        |

**MCP-specific request fields – Required**

| Variable Name            | Type and Limits                                                                             | Set Method                                                  |
|--------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| MCP version number       | <i>String</i><br><br>numeric<br><br>current version is 1.0                                  | 'mcp_version'=> \$mcp_version                               |
| cardholder amount        | <i>String</i><br><br>12-character numeric<br><br>smallest discrete unit of foreign currency | 'cardholder_amount' =><br>\$cardholder_amount               |
| cardholder currency code | <i>String</i><br><br>3-character numeric                                                    | 'cardholder_currency_code'<br>=> \$cardholder_currency_code |

**MCP Pre-Authorization transaction request fields – Optional**

| Variable Name | Type and Limits                                | Set Method           |
|---------------|------------------------------------------------|----------------------|
| customer ID   | <i>String</i><br><br>50-character alphanumeric | 'cust_id'=>\$cust_id |

| Variable Name                  | Type and Limits                                                                                                                                                                                                                                                                                                       | Set Method                                 |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
|                                | <b>NOTE:</b><br>Some special characters are not allowed:<br><code>&lt;&gt;\$%=?^{}[]\</code>                                                                                                                                                                                                                          |                                            |
| dynamic descriptor             | <i>String</i><br><br>20-character alphanumeric<br><br>total of 22 characters including your merchant name and separator<br><br><b>NOTE:</b><br>Some special characters are not allowed:<br><code>&lt;&gt;\$%=?^{}[]\</code>                                                                                           | 'dynamic_descriptor'=>\$dynamic_descriptor |
|                                | <b>NOTE: For Pre-Authorization transactions:</b> the value in the dynamic descriptor field will only be carried over to a Pre-Authorization Completion when executing the latter via the Merchant Resource Center; otherwise, the value for dynamic descriptor must be sent again in the Pre-Authorization Completion |                                            |
| wallet indicator               | <i>String</i><br><br>3-character alphanumeric                                                                                                                                                                                                                                                                         | 'wallet_indicator'=>\$wallet_indicator     |
| final authorization            | <i>String</i><br><br>true/false                                                                                                                                                                                                                                                                                       | 'final_auth' => 'true'                     |
|                                | <b>NOTE:</b> Applies to Mastercard transactions only                                                                                                                                                                                                                                                                  |                                            |
| Credential on File Info<br>cof | <i>Object</i><br><br>N/A                                                                                                                                                                                                                                                                                              | \$mpgTxn->setCofInfo(\$cof);               |
|                                | <b>NOTE:</b> This is a nested object within the transaction, and required when storing or using the customer's stored credentials. For information about fields in the Credential on File Info object, see Credential on File Info Object and Variables.                                                              |                                            |
| AVS Information                | <i>Object</i><br><br>N/A                                                                                                                                                                                                                                                                                              | \$mpgTxn->setAvsInfo(\$mpgAvsInfo);        |

| Variable Name   | Type and Limits      | Set Method                              |
|-----------------|----------------------|-----------------------------------------|
| CVD Information | <i>Object</i><br>N/A | \$mpgTxn->setCvdInfo<br>(\$mpgCvdInfo); |

**MCP-specific request fields – Optional**

| Variable Name  | Type and Limits      | Set Method                           |
|----------------|----------------------|--------------------------------------|
| MCP rate token | <i>String</i><br>N/A | 'mcp_rate_token' => \$mcp_rate_token |

**Sample MCP Pre-Authorization**

```

<?php
##
## This program takes 3 arguments from the command line:
## 1. Store id
## 2. api token
## 3. order id
##
## Example php -q TestPreAuth.php store1 yesguy
##
require "../mpgClasses.php";
$store_id='store5';
$api_token="yesguy";
/***** Transactional Variables *****/
$type='mcp_preauth';
$cust_id='cust id';
$order_id='ord-'.date("dmy-G:i:s");
$pan='4242424242424242';
$expiry_date='2011';
$crypt='7';
$dynamic_descriptor='123';
$status_check = 'false';
$mcp_version = '1.0';
$cardholder_amount = '100';
$cardholder_currency_code = '840';
$mcp_rate_token = 'P1536170825312107';
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'cust_id'=>$cust_id,
'pan'=>$pan,
'expdate'=>$expiry_date,
'crypt_type'=>$crypt,
'dynamic_descriptor'=>$dynamic_descriptor,
'mcp_version'=> $mcp_version,
'cardholder_amount' => $cardholder_amount,
'cardholder_currency_code' => $cardholder_currency_code,
'mcp_rate_token' => $mcp_rate_token
//, 'wallet_indicator' => '' //Refer to documentation for details
// 'final_auth'=>'true'
);
$mpgTxn = new mpgTransaction($txnArray);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("U");
$cof->setPaymentInformation("2");

```

```

$cof->setIssuerId("168451306048014");
$mpgTxn->setCofInfo($cof);
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\nMerchantSettlementAmount = " . $mpgResponse->getMerchantSettlementAmount());
print("\nCardholderAmount = " . $mpgResponse->getCardholderAmount());
print("\nCardholderCurrencyCode = " . $mpgResponse->getCardholderCurrencyCode());
print("\nMCPRate = " . $mpgResponse->getMCPRate());
print("\nMCPErrStatusCode = " . $mpgResponse->getMCPErrStatusCode());
print("\nMCPErrMessage = " . $mpgResponse->getMCPErrMessage());
print("\nHostId = " . $mpgResponse->getHostId());
?>

```

## 8.8 MCP Pre-Authorization with 3-D Secure

### MCP Pre-Authorization with 3-D Secure transaction object definition

```
$txnArray = array('type'=>'mcp_cavv_preauth', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for MCP Pre-Authorization with 3-D Secure transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

**MCP Pre-Authorization with 3-D Secure transaction request fields – Required**

| Variable Name                                       | Type and Limits                                                            | Set Method                   |
|-----------------------------------------------------|----------------------------------------------------------------------------|------------------------------|
| order ID                                            | <i>String</i><br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces | 'order_id'=>\$order_id       |
| credit card number                                  | <i>String</i><br>max 20-character alpha-numeric                            | 'pan'=>\$pan                 |
| expiry date                                         | <i>String</i><br>4-character alphanumeric<br>YYMM                          | 'expiry_date'=>\$expiry_date |
| Cardholder Authentication Verification Value (CAVV) | <i>String</i><br>50-character alphanumeric                                 | cavv=>\$cavv                 |
| electronic commerce indicator                       | <i>String</i><br>1-character alphanumeric                                  | 'crypt_type'=>\$crypt        |

**MCP-specific request fields – Required**

| Variable Name            | Type and Limits                                                                     | Set Method                                               |
|--------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------|
| MCP version number       | <i>String</i><br>numeric<br>current version is 1.0                                  | 'mcp_version'=> \$mcp_version                            |
| cardholder amount        | <i>String</i><br>12-character numeric<br>smallest discrete unit of foreign currency | 'cardholder_amount' => \$cardholder_amount               |
| cardholder currency code | <i>String</i><br>3-character numeric                                                | 'cardholder_currency_code' => \$cardholder_currency_code |

**3-D Secure 2.2 -specific fields – Required**

| Variable Name                                                                                                                                                                                                         | Type and Limits                           | Set Method                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3DS version<br><br><div> <b>NOTE:</b> Mandatory for financial transactions using 3rd Party 3-D Secure services.         </div>                                                                                        | <i>String</i><br><br>10-character numeric | <pre>'threads_version'=&gt;\$threads_version</pre><br>Acceptable values:<br><br>2.0.0 = 3DS protocol 2.0.0<br><br>2.1.0 = 3DS protocol 2.1.0<br><br>2.2.0 = 3DS protocol 2.2.0<br><br>2.3.0 = 3DS protocol 2.3.0 |
| 3DS server transaction ID<br><br><div> <b>NOTE:</b> Mandatory for financial transactions using 3rd Party 3-D Secure services - obtained from the Cavv Lookup request or MPI 3DS Authentication request         </div> | <i>String</i><br><br>36-character numeric | <pre>'threads_server_trans_id'=&gt;\$threads_server_trans_id</pre>                                                                                                                                               |

**MCP Pre-Authorization with 3-D Secure transaction request fields – Optional**

| Variable Name      | Type and Limits                                                                                                                                                                                                 | Set Method                                               |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| customer ID        | <i>String</i><br><br>50-character alphanumeric<br><br><div> <b>NOTE:</b><br/>           Some special characters are not allowed:<br/>           &lt;&gt;\$%=?^{}[]\         </div>                              | <pre>'cust_id'=&gt;\$cust_id</pre>                       |
| dynamic descriptor | <i>String</i><br><br>20-character alphanumeric<br><br>total of 22 characters including your merchant name and separator<br><br><div> <b>NOTE:</b><br/>           Some special characters are not         </div> | <pre>'dynamic_descriptor'=&gt;\$dynamic_descriptor</pre> |



| Variable Name                                                                                                                                                                                                                                                       | Type and Limits                                    | Set Method                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------|
|                                                                                                                                                                                                                                                                     | <div>allowed:<br/>&lt;&gt;\$%= ? ^ { } [ ] \</div> |                                        |
| wallet indicator                                                                                                                                                                                                                                                    | <i>String</i><br>3-character alphanumeric          | 'wallet_indicator'=>\$wallet_indicator |
| Credential on File Info<br>cof                                                                                                                                                                                                                                      | <i>Object</i><br>N/A                               | \$mpgTxn->setCofInfo(\$cof);           |
| <div><b>NOTE:</b> This is a nested object within the transaction, and required when storing or using the customer's stored credentials. For information about fields in the Credential on File Info object, see Credential on File Info Object and Variables.</div> |                                                    |                                        |
| AVS Information                                                                                                                                                                                                                                                     | <i>Object</i><br>N/A                               | \$mpgTxn->setAvsInfo(\$mpgAvsInfo);    |
| CVD Information                                                                                                                                                                                                                                                     | <i>Object</i><br>N/A                               | \$mpgTxn->setCvdInfo(\$mpgCvdInfo);    |

### MCP-specific request fields – Optional

| Variable Name  | Type and Limits      | Set Method                           |
|----------------|----------------------|--------------------------------------|
| MCP rate token | <i>String</i><br>N/A | 'mcp_rate_token' => \$mcp_rate_token |

### 3-D Secure 2.2 -specific fields – Optional

| Variable Name                                  | Type and Limits                            | Set Method                     |
|------------------------------------------------|--------------------------------------------|--------------------------------|
| DS transaction ID                              | <i>String</i><br>36-character alphanumeric | 'ds_trans_id' => \$ds_trans_id |
| <div><b>NOTE:</b> Only used in financial</div> |                                            |                                |

| Variable Name                                     | Type and Limits | Set Method |
|---------------------------------------------------|-----------------|------------|
| transactions using 3rd Party 3-D Secure services. |                 |            |

### Sample MCP Pre-Authorization with 3-D Secure

```
<?php
## Example php -q TestPurchase-VBV.php "moneris" store
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='monca02932';
$api_token='CG8kYzGgzVU5z23irgMx';
/***** Transactional Variables *****/
$type='mcp_cavv_preauth';
$order_id='ord-' . date("dmy-G:i:s");
$cust_id='CUST887763';
$amount='10.00';
$pan="4242424242424242";
$expiry_date="0812";
$cavv='AAABBJg0VhI0VniQEjRWAAAAA=';
$crypt_type = '7';
$wallet_indicator = "APP";
$dynamic_descriptor='123456';
$mcp_version = '1.0';
$cardholder_amount = '100';
$cardholder_currency_code = '840';
$mcp_rate_token = 'P1623437375175657';
/***** Transaction Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'cust_id'=>$cust_id,
'amount'=>$amount,
'pan'=>$pan,
'expdate'=>$expiry_date,
'cavv'=>$cavv,
'crypt_type'=>$crypt_type, //mandatory for AMEX only
// 'wallet_indicator'=>$wallet_indicator, //set only for wallet transactions. e.g. APPLE PAY
'dynamic_descriptor'=>$dynamic_descriptor,
'threads_version' => '2.2.0', //Mandatory for financial transactions using 3rd Party 3-D
Secure services
'threads_server_trans_id' => 'e11d4985-8d25-40ed-99d6-c3803fe5e68f', //Mandatory for
financial transactions using 3rd Party 3-D Secure services - obtained from MpiCavvLookup or
MpiThreeDSAuthentication
// 'cm_id' => '8nAK8712sGaAkls56', //set only for usage with Offlinx - Unique max 50
alphanumeric characters transaction id generated by merchant
// 'ds_trans_id' => '12345', //Optional - to be used only if you are using 3rd party 3ds
service
'mcp_version'=> $mcp_version,
'cardholder_amount' => $cardholder_amount,
'cardholder_currency_code' => $cardholder_currency_code,
'mcp_rate_token' => $mcp_rate_token
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("U");
$cof->setPaymentInformation("2");
$cof->setIssuerId("139X3130ASCXAS9");
$mpgTxn->setCofInfo($cof);
```

```

/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
/***** Response *****/
$mpgResponse = $mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nCavvResultCode = " . $mpgResponse->getCavvResultCode());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\nThreeDSVersion = " . $mpgResponse->getThreeDSVersion());
print("\nMerchantSettlementAmount = " . $mpgResponse->getMerchantSettlementAmount());
print("\nCardholderAmount = " . $mpgResponse->getCardholderAmount());
print("\nCardholderCurrencyCode = " . $mpgResponse->getCardholderCurrencyCode());
print("\nMCPRate = " . $mpgResponse->getMCPRate());
print("\nMCPErrStatusCode = " . $mpgResponse->getMCPErrStatusCode());
print("\nMCPErrErrorMessage = " . $mpgResponse->getMCPErrErrorMessage());
print("\nHostId = " . $mpgResponse->getHostId());
?>

```

## 8.9 MCP Pre-Authorization with 3-D Secure and Vault

### MCP Pre-Authorization with 3-D Secure and Vault transaction object definition

```
$txnArray = array('type'=>'mcp_res_cavv_preauth_cc', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for MCP Pre-Authorization with 3-D Secure and Vault transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String          | 'api_token'=>\$api_token |

| Variable Name | Type and Limits |  |
|---------------|-----------------|--|
|               | N/A             |  |

**MCP Pre-Authorization with 3-D Secure and Vault transaction request fields – Required**

| Variable Name                                       | Type and Limits                                                            | Set Method                   |
|-----------------------------------------------------|----------------------------------------------------------------------------|------------------------------|
| data key                                            | <i>String</i><br>25-character alphanumeric                                 | 'data_key'=>\$data_key       |
| order ID                                            | <i>String</i><br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces | 'order_id'=>\$order_id       |
| credit card number                                  | <i>String</i><br>max 20-character alphanumeric                             | 'pan'=>\$pan                 |
| expiry date                                         | <i>String</i><br>4-character alphanumeric<br>YYMM                          | 'expiry_date'=>\$expiry_date |
| Cardholder Authentication Verification Value (CAVV) | <i>String</i><br>50-character alphanumeric                                 | cavv=>\$cavv                 |
| electronic commerce indicator                       | <i>String</i><br>1-character alphanumeric                                  | 'crypt_type'=>\$crypt        |

**MCP-specific request fields – Required**

| Variable Name      | Type and Limits                                    | Set Method                                 |
|--------------------|----------------------------------------------------|--------------------------------------------|
| MCP version number | <i>String</i><br>numeric<br>current version is 1.0 | 'mcp_version'=> \$mcp_version              |
| cardholder amount  | <i>String</i>                                      | 'cardholder_amount' => \$cardholder_amount |

| Variable Name            | Type and Limits                            | Set Method                                               |
|--------------------------|--------------------------------------------|----------------------------------------------------------|
|                          | 12-character numeric                       |                                                          |
|                          | smallest discrete unit of foreign currency |                                                          |
| cardholder currency code | <i>String</i><br>3-character numeric       | 'cardholder_currency_code' => \$cardholder_currency_code |

**3-D Secure 2.2 -specific fields – Required**

| Variable Name             | Type and Limits                       | Set Method                                                                                                                                                                                             |
|---------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3DS version               | <i>String</i><br>10-character numeric | 'threeds_version'=>\$threeds_version<br><br>Acceptable values:<br><br>2.0.0 = 3DS protocol 2.0.0<br><br>2.1.0 = 3DS protocol 2.1.0<br><br>2.2.0 = 3DS protocol 2.2.0<br><br>2.3.0 = 3DS protocol 2.3.0 |
| 3DS server transaction ID | <i>String</i><br>36-character numeric | 'threeds_server_trans_id'=>\$threeds_server_trans_id                                                                                                                                                   |

**NOTE:** Mandatory for financial transactions using 3rd Party 3-D Secure services.

**NOTE:** Mandatory for financial transactions using 3rd Party 3-D Secure services - obtained from the Cavv Lookup request or MPI 3DS Authentication request

**MCP Pre-Authorization with 3-D Secure and Vault transaction request fields – Optional**

| Variable Name | Type and Limits                                          | Set Method           |
|---------------|----------------------------------------------------------|----------------------|
| customer ID   | <i>String</i><br>50-character alphanumeric               | 'cust_id'=>\$cust_id |
|               | <b>NOTE:</b><br>Some special characters are not allowed: |                      |

| Variable Name           | Type and Limits                                                                                                                                                                                                                                                 | Set Method                                 |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
|                         | <>\$%=?^{}[]\                                                                                                                                                                                                                                                   |                                            |
| dynamic descriptor      | <p><i>String</i></p> <p>20-character alphanumeric</p> <p>total of 22 characters including your merchant name and separator</p> <p><b>NOTE:</b><br/>Some special characters are not allowed:<br/>&lt;&gt;\$%=?^{}[]\</p>                                         | 'dynamic_descriptor'=>\$dynamic_descriptor |
| wallet indicator        | <p><i>String</i></p> <p>3-character alphanumeric</p>                                                                                                                                                                                                            | 'wallet_indicator'=>\$wallet_indicator     |
| Credential on File Info | <i>Object</i>                                                                                                                                                                                                                                                   | \$mpgTxn->setCofInfo(\$cof);               |
| cof                     | N/A                                                                                                                                                                                                                                                             |                                            |
|                         | <p><b>NOTE:</b> This is a nested object within the transaction, and required when storing or using the customer's stored credentials. For information about fields in the Credential on File Info object, see Credential on File Info Object and Variables.</p> |                                            |
| AVS Information         | <p><i>Object</i></p> <p>N/A</p>                                                                                                                                                                                                                                 | \$mpgTxn->setAvsInfo(\$mpgAvsInfo);        |
| CVD Information         | <p><i>Object</i></p> <p>N/A</p>                                                                                                                                                                                                                                 | \$mpgTxn->setCvdInfo(\$mpgCvdInfo);        |

**MCP-specific request fields – Optional**

| Variable Name  | Type and Limits   | Set Method                           |
|----------------|-------------------|--------------------------------------|
| MCP rate token | String<br><br>N/A | 'mcp_rate_token' => \$mcp_rate_token |

**3-D Secure 2.2 -specific fields – Optional**

| Variable Name                                                                                                                                                                 | Type and Limits                         | Set Method                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|--------------------------------|
| DS transaction ID                                                                                                                                                             | String<br><br>36-character alphanumeric | 'ds_trans_id' => \$ds_trans_id |
| <div style="border: 1px solid black; padding: 5px; width: fit-content;"> <b>NOTE:</b> Only used in financial transactions using 3rd Party 3-D Secure services.         </div> |                                         |                                |

**Sample MCP Pre-Authorization with 3-D Secure and Vault**

```
<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store1';
$api_token='yesguy1';
/***** Transaction Variables *****/
$data_key='4INQR1A8ocxD0oafSz50LADxy';
$orderid='res-preauth-' . date("dmy-G:i:s");
$amount='1.00';
$cavv='AAABBJg0VhI0VniQEjRWAAAAA';
$custid='customer1'; //if sent will be submitted, otherwise cust_id from profile will be used
$expdate = '1901'; //YYMM - used only for temp token
$crypt_type = '7'; //value obtained from MpiACS transaction
$mcp_version = '1.0';
$cardholder_amount = '100';
$cardholder_currency_code = '840';
$mcp_rate_token = 'P1623438275728112';
/***** Transaction Array *****/
$txnArray =array('type'=>'mcp_res_cavv_preauth_cc',
'data_key'=>$data_key,
'order_id'=>$orderid,
'cust_id'=>$custid,
'amount'=>$amount,
'cavv'=>$cavv,
'expdate'=>$expdate, //mandatory for temp tokens only
'crypt_type'=>$crypt_type, //set for AMEX SafeKey only
'dynamic_descriptor'=>'12346',
'threads_version' => '2.2.0', //Mandatory for financial transactions using 3rd Party 3-D
Secure services
'threads_server_trans_id' => 'e11d4985-8d25-40ed-99d6-c3803fe5e68f', //Mandatory for
financial transactions using 3rd Party 3-D Secure services - obtained from MpiCavvLookup or
MpiThreeDSAuthentication
'ds_trans_id' => '12345', //Optional - to be used only if you are using 3rd party 3ds
service
'mcp_version'=> $mcp_version,
'cardholder_amount' => $cardholder_amount,
'cardholder_currency_code' => $cardholder_currency_code,
```

```

'mcp_rate_token' => $mcp_rate_token
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("U");
$cof->setPaymentInformation("2");
$cof->setIssuerId("139X3130ASCXAS9");
$mpgTxn->setCofInfo($cof);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
/***** mpgHttpPost Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response Object *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nAVSResponse = " . $mpgResponse->getAvsResultCode());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
print("\nCavvResultCode = " . $mpgResponse->getCavvResultCode());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\nThreeDSVersion = " . $mpgResponse->getThreeDSVersion());
//----- ResolveData -----
print("\nCust ID = " . $mpgResponse->getResDataCustId());
print("\nPhone = " . $mpgResponse->getResDataPhone());
print("\nEmail = " . $mpgResponse->getResDataEmail());
print("\nNote = " . $mpgResponse->getResDataNote());
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\nCryp Type = " . $mpgResponse->getResDataCryptType());
print("\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
print("\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
?>

```

## 8.10 MCP Pre-Authorization Completion

Retrieves funds that have been locked by an MCP Pre-Authorization transaction, and prepares them for settlement into the merchant's account.

This transaction request is the multi-currency pricing (MCP) enabled version of the equivalent financial transaction.



**MCP Pre-Authorization Completion transaction object definition**

```
$txnArray = array('type'=>'mcp_completion', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

**HttpPostRequest object for MCP Pre-Authorization Completion transaction**

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

**Core connection object fields (all API transactions)**

| Variable Name | Type and Limits      |                          |
|---------------|----------------------|--------------------------|
| store ID      | <i>String</i><br>N/A | 'store_id'=>\$store_id   |
| API token     | <i>String</i><br>N/A | 'api_token'=>\$api_token |

**MCP Pre-Authorization Completion transaction request fields – Required**

| Variable Name                 | Type and Limits                                                                           | Set Method                 |
|-------------------------------|-------------------------------------------------------------------------------------------|----------------------------|
| order ID                      | <i>String</i><br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                | 'order_id'=>\$order_id     |
| transaction number            | <i>String</i><br>255-character, alpha-numeric, hyphens or under-scores<br>variable length | 'txn_number'=>\$txn_number |
| electronic commerce indicator | <i>String</i><br>1-character alphanumeric                                                 | 'crypt_type'=>\$crypt      |

**MCP-specific request fields – Required**

| Variable Name            | Type and Limits                                                                      | Set Method                                               |
|--------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------|
| MCP version number       | String<br><br>numeric<br><br>current version is 1.0                                  | 'mcp_version'=> \$mcp_version                            |
| cardholder amount        | String<br><br>12-character numeric<br><br>smallest discrete unit of foreign currency | 'cardholder_amount' => \$cardholder_amount               |
| cardholder currency code | String<br><br>3-character numeric                                                    | 'cardholder_currency_code' => \$cardholder_currency_code |

**MCP Pre-Authorization Completion transaction request fields – Optional**

| Variable Name      | Type and Limits                                                                                                                                                                                              | Set Method                                 |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| customer ID        | String<br><br>50-character alphanumeric<br><br><b>NOTE:</b><br>Some special characters are not allowed:<br>< > \$ % = ? ^ { } [ ] \                                                                          | 'cust_id'=>\$cust_id                       |
| dynamic descriptor | String<br><br>20-character alphanumeric<br><br>total of 22 characters including your merchant name and separator<br><br><b>NOTE:</b><br>Some special characters are not allowed:<br>< > \$ % = ? ^ { } [ ] \ | 'dynamic_descriptor'=>\$dynamic_descriptor |

**NOTE: For Pre-Authorization transactions:** the value in the dynamic descriptor field will only be carried over to a Pre-Authorization Completion when executing the latter via the Merchant Resource Center; otherwise, the value for dynamic descriptor must be sent again in the Pre-Authorization Completion

**MCP-specific request fields – Optional**

| Variable Name  | Type and Limits          | Set Method                           |
|----------------|--------------------------|--------------------------------------|
| MCP rate token | <i>String</i><br><br>N/A | 'mcp_rate_token' => \$mcp_rate_token |

**Sample MCP Pre-Authorization Completion**

```

<?php
require "../mpgClasses.php";
$store_id='store5';
$api_token='yesguy';
$orderid='ord-050918-14:11:30';
$txnnumber='407995-0_11';
$dynamic_descriptor='123';
$ship_indicator = "F"; //optional
$mcp_version = '1.0';
$cardholder_amount = '100';
$cardholder_currency_code = '840';
$mcp_rate_token = 'P1536170825312107';
## step 1) create transaction array ###
$txnArray= array('type'=>'mcp_completion',
'txn_number'=>$txnnumber,
'order_id'=>$orderid,
'crypt_type'=>'7',
'cust_id'=>'customer ID',
// 'ship_indicator'=>$ship_indicator, //optional
'dynamic_descriptor'=>$dynamic_descriptor,
'mcp_version'=> $mcp_version,
'cardholder_amount' => $cardholder_amount,
'cardholder_currency_code' => $cardholder_currency_code,
'mcp_rate_token' => $mcp_rate_token
);
## step 2) create a transaction object passing the hash created in
## step 1.
$mpgTxn = new mpgTransaction($txnArray);
## step 3) create a mpgRequest object passing the transaction object created
## in step 2
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
## step 4) create mpgHttpPost object which does an https post ##
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
## step 5) get an mpgResponse object ##
$mpgResponse=$mpgHttpPost->getMpgResponse();
## step 6) retrieve data using get methods
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());

```

```

print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nMerchantSettlementAmount = " . $mpgResponse->getMerchantSettlementAmount());
print("\nCardholderAmount = " . $mpgResponse->getCardholderAmount());
print("\nCardholderCurrencyCode = " . $mpgResponse->getCardholderCurrencyCode());
print("\nMCPRate = " . $mpgResponse->getMCPRate());
print("\nMCPErrorsStatusCode = " . $mpgResponse->getMCPErrorsStatusCode());
print("\nMCPErrorsMessage = " . $mpgResponse->getMCPErrorsMessage());
print("\nHostId = " . $mpgResponse->getHostId());
?>

```

## 8.11 MCP Purchase Correction

Restores the full amount of a previous MCP Purchase or MCP Pre-Authorization Completion transaction to the cardholder's card, and removes any record of it from the cardholder's statement.

This transaction can be used against a Purchase or Pre-Authorization Completion transaction that occurred same day provided that the batch containing the original transaction remains open.

MCP processing uses the automated closing feature, and Batch Close occurs daily between 10 and 11 pm Eastern Time.

This transaction request is the multi-currency pricing (MCP) enabled version of the equivalent financial transaction.

### MCP Purchase Correction transaction object definition

```
$txnArray = array('type'=>'mcp_purchaseCorrection', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for MCP Purchase Correction transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

**MCP Purchase Correction transaction request fields – Required**

| Variable Name                 | Type and Limits                                                                                   | Set Method                 |
|-------------------------------|---------------------------------------------------------------------------------------------------|----------------------------|
| order ID                      | <i>String</i><br><br>50-character alpha-numeric-A-Z 0-9 _ - : . @ spaces                          | 'order_id'=>\$order_id     |
| transaction number            | <i>String</i><br><br>255-character, alpha-numeric, hyphens or under-scores<br><br>variable length | 'txn_number'=>\$txn_number |
| electronic commerce indicator | <i>String</i><br><br>1-character alphanumeric                                                     | 'crypt_type'=>\$crypt      |

**MCP Purchase Correction transaction request fields – Optional**

| Variable Name      | Type and Limits                                                                                                                                                                                                                        | Description                                |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| dynamic descriptor | <i>String</i><br><br>20-character alphanumeric<br><br>total of 22 characters including your merchant name and separator<br><br><div><b>NOTE:</b><br/>Some special characters are not allowed:<br/>&lt; &gt; \$ % = ? ^ { } [ ] \</div> | 'dynamic_descriptor'=>\$dynamic_descriptor |
| customer ID        | <i>String</i><br><br>50-character alphanumeric<br><br><div><b>NOTE:</b><br/>Some special characters are not allowed:<br/>&lt; &gt; \$ % = ? ^ { } [ ] \</div>                                                                          | 'cust_id'=>\$cust_id                       |

## Sample MCP Purchase Correction

```
<?php
require "../mpgClasses.php";
$store_id='store5';
$api_token='yesguy';
$orderid='ord-150816-12:36:20';
$txnnumber='117816-0_10';
$dynamic_descriptor='1234';
## step 1) create transaction hash ###
$txnArray=array('type'=>'mcp_purchaseCorrection',
'txn_number'=>$txnnumber,
'order_id'=>$orderid,
'crypt_type'=>'7',
'cust_id'=>'customer ID',
'dynamic_descriptor'=>$dynamic_descriptor
);
## step 2) create a transaction object passing the array created in
## step 1.
$mpgTxn = new mpgTransaction($txnArray);
## step 3) create a mpgRequest object passing the transaction object created
## in step 2
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
## step 4) create mpgHttpPost object which does an https post ##
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
## step 5) get an mpgResponse object ##
$mpgResponse=$mpgHttpPost->getMpgResponse();
## step 6) retrieve data using get methods
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nMerchantSettlementAmount = " . $mpgResponse->getMerchantSettlementAmount());
print("\nCardholderAmount = " . $mpgResponse->getCardholderAmount());
print("\nCardholderCurrencyCode = " . $mpgResponse->getCardholderCurrencyCode());
print("\nMCPRate = " . $mpgResponse->getMCPRate());
print("\nMCPErrrorStatusCode = " . $mpgResponse->getMCPErrrorStatusCode());
print("\nMCPErrrorMessage = " . $mpgResponse->getMCPErrrorMessage());
print("\nHostId = " . $mpgResponse->getHostId());
?>
```

## 8.12 MCP Refund

Restores all or part of the funds from a MCP Purchase or MCP Pre-Authorization Completion transaction to the cardholder's card.

Unlike a MCP Purchase Correction, there is a record of both the initial charge and the refund on the card-holder's statement.

For processing refunds on a different card than the one used in the original transaction, the MCP Independent Refund transaction should be used instead.

This transaction request is the multi-currency pricing (MCP) enabled version of the equivalent financial transaction.

### MCP Refund transaction object definition

```
$txnArray = array('type'=>'mcp_refund', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for MCP Refund transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

### MCP Refund transaction request fields – Required

| Variable Name      | Type and Limits                                                                            | Set Method                 |
|--------------------|--------------------------------------------------------------------------------------------|----------------------------|
| order ID           | String<br>50-character alpha-<br>numerica-Z A-Z 0-9 _ - : . @<br>spaces                    | 'order_id'=>\$order_id     |
| transaction number | String<br>255-character, alpha-<br>numeric, hyphens or under-<br>scores<br>variable length | 'txn_number'=>\$txn_number |

| Variable Name                 | Type and Limits                           | Set Method            |
|-------------------------------|-------------------------------------------|-----------------------|
| electronic commerce indicator | <i>String</i><br>1-character alphanumeric | 'crypt_type'=>\$crypt |

**MCP-specific request fields – Required**

| Variable Name            | Type and Limits                                                                     | Set Method                                               |
|--------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------|
| MCP version number       | <i>String</i><br>numeric<br>current version is 1.0                                  | 'mcp_version'=> \$mcp_version                            |
| cardholder amount        | <i>String</i><br>12-character numeric<br>smallest discrete unit of foreign currency | 'cardholder_amount' => \$cardholder_amount               |
| cardholder currency code | <i>String</i><br>3-character numeric                                                | 'cardholder_currency_code' => \$cardholder_currency_code |

**MCP Refund transaction request fields – Optional**

| Variable Name      | Type and Limits                                                                                                                                                                                                            | Description                                |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| dynamic descriptor | <i>String</i><br>20-character alphanumeric<br>total of 22 characters including your merchant name and separator<br><div><b>NOTE:</b><br/>Some special characters are not allowed:<br/>&lt; &gt; \$ % = ? ^ { } [ ] \</div> | 'dynamic_descriptor'=>\$dynamic_descriptor |
| customer ID        | <i>String</i><br>50-character alphanumeric                                                                                                                                                                                 | 'cust_id'=>\$cust_id                       |



| Variable Name | Type and Limits | Description |
|---------------|-----------------|-------------|
|---------------|-----------------|-------------|

**NOTE:**  
Some special characters are not allowed:  
<>\$%=?^{}[]\

### MCP-specific request fields – Optional

| Variable Name  | Type and Limits   | Set Method                           |
|----------------|-------------------|--------------------------------------|
| MCP rate token | String<br><br>N/A | 'mcp_rate_token' => \$mcp_rate_token |

### Sample MCP Refund

```
<?php
##
## This program takes 4 arguments from the command line:
## 1. Store id
## 2. api token
## 3. order id
## 4. trans number
##
## Example php -q TestRefund.php store1 yesguy my_order_id 45109-89-0
##
require "../mpgClasses.php";
$store_id='store5';
$api_token='yesguy';
$orderid='ord-050918-12:03:26';
$txnnumber='407512-0_11';
$script_type = '7';
$dynamic_descriptor='123';
$mcp_version = '1.0';
$cardholder_amount = '100';
$cardholder_currency_code = '840';
$mcp_rate_token = 'R1536163085399771';
## step 1) create transaction array ###
$txnArray=array('type'=>'mcp_refund',
'txn_number'=>$txnnumber,
'order_id'=>$orderid,
'crypt_type'=>$script_type,
'cust_id'=> 'Customer ID',
'dynamic_descriptor'=>$dynamic_descriptor,
'mcp_version'=> $mcp_version,
'cardholder_amount' => $cardholder_amount,
'cardholder_currency_code' => $cardholder_currency_code,
'mcp_rate_token' => $mcp_rate_token //optional
);
## step 2) create a transaction object passing the array created in
## step 1.
$mpgTxn = new mpgTransaction($txnArray);
## step 3) create a mpgRequest object passing the transaction object created
## in step 2
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
## step 4) create mpgHttpPost object which does an https post ##
```

```

$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
## step 5) get an mpgResponse object ##
$mpgResponse=$mpgHttpPost->getMpgResponse();
## step 6) retrieve data using get methods
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nMerchantSettlementAmount = " . $mpgResponse->getMerchantSettlementAmount());
print("\nCardholderAmount = " . $mpgResponse->getCardholderAmount());
print("\nCardholderCurrencyCode = " . $mpgResponse->getCardholderCurrencyCode());
print("\nMCPRate = " . $mpgResponse->getMCPRate());
print("\nMCPErrrorStatusCode = " . $mpgResponse->getMCPErrrorStatusCode());
print("\nMCPErrrorMessage = " . $mpgResponse->getMCPErrrorMessage());
print("\nHostId = " . $mpgResponse->getHostId());
?>

```

## 8.13 MCP Independent Refund

Credits a specified amount to the cardholder's credit card. The credit card number and expiry date are mandatory.

It is not necessary for the transaction that you are refunding to have been processed via the Moneris Gateway.

This transaction request is the multi-currency pricing (MCP) enabled version of the equivalent financial transaction.

### Things to Consider:

- Because of the potential for fraud, permission for this transaction is not granted to all accounts by default. If it is required for your business, it must be requested via your account manager.

### MCP Independent Refund transaction object definition

```
$txnArray = array('type'=>'mcp_ind_refund', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

**HttpPostRequest object for MCP Independent Refund transaction**

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

**Core connection object fields (all API transactions)**

| Variable Name | Type and Limits      |                          |
|---------------|----------------------|--------------------------|
| store ID      | <i>String</i><br>N/A | 'store_id'=>\$store_id   |
| API token     | <i>String</i><br>N/A | 'api_token'=>\$api_token |

**MCP Independent Refund transaction request fields – Required**

| Variable Name                 | Type and Limits                                                                | Set Method                   |
|-------------------------------|--------------------------------------------------------------------------------|------------------------------|
| order ID                      | <i>String</i><br>50-character alpha-<br>numerica-Z A-Z 0-9 _ - : . @<br>spaces | 'order_id'=>\$order_id       |
| credit card number            | <i>String</i><br>max 20-character alpha-<br>numeric                            | 'pan'=>\$pan                 |
| expiry date                   | <i>String</i><br>4-character alphanumeric<br>YYMM                              | 'expiry_date'=>\$expiry_date |
| electronic commerce indicator | <i>String</i><br>1-character alphanumeric                                      | 'crypt_type'=>\$crypt        |

**MCP-specific request fields – Required**

| Variable Name      | Type and Limits | Set Method                    |
|--------------------|-----------------|-------------------------------|
| MCP version number | <i>String</i>   | 'mcp_version'=> \$mcp_version |

| Variable Name            | Type and Limits                                                                     | Set Method                                                  |
|--------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------|
|                          | numeric<br>current version is 1.0                                                   |                                                             |
| cardholder amount        | <i>String</i><br>12-character numeric<br>smallest discrete unit of foreign currency | 'cardholder_amount' =><br>\$cardholder_amount               |
| cardholder currency code | <i>String</i><br>3-character numeric                                                | 'cardholder_currency_code' =><br>\$cardholder_currency_code |

#### MCP Independent Refund transaction request fields – Optional

| Variable Name      | Type and Limits                                                                                                                                                                                                            | Set Method                                 |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| customer ID        | <i>String</i><br>50-character alphanumeric<br><div><b>NOTE:</b><br/>Some special characters are not allowed:<br/>&lt; &gt; \$ % = ? ^ { } [ ] \</div>                                                                      | 'cust_id'=>\$cust_id                       |
| dynamic descriptor | <i>String</i><br>20-character alphanumeric<br>total of 22 characters including your merchant name and separator<br><div><b>NOTE:</b><br/>Some special characters are not allowed:<br/>&lt; &gt; \$ % = ? ^ { } [ ] \</div> | 'dynamic_descriptor'=>\$dynamic_descriptor |
|                    |                                                                                                                                                                                                                            |                                            |

**MCP-specific request fields – Optional**

| Variable Name  | Type and Limits          | Set Method                           |
|----------------|--------------------------|--------------------------------------|
| MCP rate token | <i>String</i><br><br>N/A | 'mcp_rate_token' => \$mcp_rate_token |

**Sample MCP Independent Refund**

```

<?php
##
## This program takes 3 arguments from the command line:
## 1. Store id
## 2. api token
## 3. order id
##
## Example php -q TestIndependentRefund.php store1 yesguy unique_order_id
##
require "../mpgClasses.php";
$store_id='store5';
$api_token='yesguy';
$orderid='ord-'.date("dmy-G:i:s");
$pan='4242424242424242';
$expiry_date='2011';
$script='7';
$dynamic_descriptor='123456';
$mcp_version = '1.0';
$cardholder_amount = '100';
$cardholder_currency_code = '840';
$mcp_rate_token = 'R1536163085399771';
## step 1) create transaction array ###
$txnArray=array('type'=>'mcp_ind_refund',
'order_id'=>$orderid,
'cust_id'=>'my cust id',
'pan'=>$pan,
'expdate'=>$expiry_date,
'crypt_type'=>$script,
'dynamic_descriptor'=>$dynamic_descriptor,
'mcp_version'=> $mcp_version,
'cardholder_amount' => $cardholder_amount,
'cardholder_currency_code' => $cardholder_currency_code,
'mcp_rate_token' => $mcp_rate_token
);
## step 2) create a transaction object passing the array created in
## step 1.
$mpgTxn = new mpgTransaction($txnArray);
## step 3) create a mpgRequest object passing the transaction object created
## in step 2
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
## step 4) create mpgHttpPost object which does an https post ##
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
## step 5) get an mpgResponse object ##
$mpgResponse=$mpgHttpPost->getMpgResponse();
## step 6) retrieve data using get methods
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());

```

```

print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nMerchantSettlementAmount = " . $mpgResponse->getMerchantSettlementAmount());
print("\nCardholderAmount = " . $mpgResponse->getCardholderAmount());
print("\nCardholderCurrencyCode = " . $mpgResponse->getCardholderCurrencyCode());
print("\nMCPRate = " . $mpgResponse->getMCPRate());
print("\nMCPErrorStatusCode = " . $mpgResponse->getMCPErrorStatusCode());
print("\nMCPErrorMessage = " . $mpgResponse->getMCPErrorMessage());
print("\nHostId = " . $mpgResponse->getHostId());
?>

```

## 8.14 MCP Purchase with Vault

This transaction uses the data key to identify a previously registered credit card profile in Vault. The details saved within the profile are then submitted to perform a Purchase transaction.

The data key may be a temporary one generated used Hosted Tokenization, or may be a permanent one from the Vault.

This transaction request is the multi-currency pricing (MCP) enabled version of the equivalent financial transaction.

### MCP Purchase with Vault transaction object definition

```

$txnArray = array('type'=>'mcp_res_purchase_cc', ...);

$mpgTxn = new mpgTransaction($txnArray);

```

### HttpPostRequest object for MCP Purchase with Vault transaction

```

$mpgRequest = new mpgRequest($mpgTxn);

$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);

```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

**MCP Purchase with Vault transaction request fields – Required**

| Variable Name                 | Type and Limits                                                                | Set Method             |
|-------------------------------|--------------------------------------------------------------------------------|------------------------|
| data key                      | <i>String</i><br>25-character alphanumeric                                     | 'data_key'=>\$data_key |
| order ID                      | <i>String</i><br>50-character alpha-<br>numerica-Z A-Z 0-9 _ - : . @<br>spaces | 'order_id'=>\$order_id |
| electronic commerce indicator | <i>String</i><br>1-character alphanumeric                                      | 'crypt_type'=>\$crypt  |

**MCP-specific request fields – Required**

| Variable Name            | Type and Limits                                                                     | Set Method                                                  |
|--------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------|
| MCP version number       | <i>String</i><br>numeric<br>current version is 1.0                                  | 'mcp_version'=> \$mcp_version                               |
| cardholder amount        | <i>String</i><br>12-character numeric<br>smallest discrete unit of foreign currency | 'cardholder_amount' =><br>\$cardholder_amount               |
| cardholder currency code | <i>String</i><br>3-character numeric                                                | 'cardholder_currency_code'<br>=> \$cardholder_currency_code |

**MCP Purchase with Vault transaction request fields – Optional**

| Variable Name | Type and Limits                                                | Set Method           |
|---------------|----------------------------------------------------------------|----------------------|
| customer ID   | <i>String</i><br>50-character alphanumeric<br><div>NOTE:</div> | 'cust_id'=>\$cust_id |

| Variable Name                                                                                                                                                                                                                                                       | Type and Limits                                                                        | Set Method                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------|
|                                                                                                                                                                                                                                                                     | <div>Some special characters are not allowed:<br/>&lt; &gt; \$ % = ? ^ { } [ ] \</div> |                                       |
| Credential on File Info                                                                                                                                                                                                                                             | <i>Object</i>                                                                          | \$mpgTxn->setCofInfo (\$cof) ;        |
| cof                                                                                                                                                                                                                                                                 | N/A                                                                                    |                                       |
| <div><b>NOTE:</b> This is a nested object within the transaction, and required when storing or using the customer's stored credentials. For information about fields in the Credential on File Info object, see Credential on File Info Object and Variables.</div> |                                                                                        |                                       |
| AVS Information                                                                                                                                                                                                                                                     | <i>Object</i>                                                                          | \$mpgTxn->setAvsInfo (\$mpgAvsInfo) ; |
|                                                                                                                                                                                                                                                                     | N/A                                                                                    |                                       |
| CVD Information                                                                                                                                                                                                                                                     | <i>Object</i>                                                                          | \$mpgTxn->setCvdInfo (\$mpgCvdInfo) ; |
|                                                                                                                                                                                                                                                                     | N/A                                                                                    |                                       |

### MCP-specific request fields – Optional

| Variable Name  | Type and Limits | Set Method                           |
|----------------|-----------------|--------------------------------------|
| MCP rate token | <i>String</i>   | 'mcp_rate_token' => \$mcp_rate_token |
|                | N/A             |                                      |

### Sample MCP Purchase with Vault

```
<?php
##
## This program takes 3 arguments from the command line:
## 1. Store id
## 2. api token
## 3. order id
##
## Example php -q TestResPurchaseCC.php store3 yesguy unique_order_id 1.00
##
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
/***** Transaction Variables *****/
```



```

$data_key='uX6jFwbvCytmG7oT70YVNm0e2';
$orderid='res-purch-' . date("dmy-G:i:s");
$custid='cust';
$script_type='1';
$expdate='1911'; //For Temp Tokens only
$mcp_version = '1.0';
$cardholder_amount = '100';
$cardholder_currency_code = '840';
$mcp_rate_token = 'P1536163745116323';
/***** Transaction Array *****/
$txnArray=array('type'=>'mcp_res_purchase_cc',
'data_key'=>$data_key,
'order_id'=>$orderid,
'cust_id'=>$custid,
'crypt_type'=>$script_type,
// 'expdate'=>$expdate,
'dynamic_descriptor'=>'12484',
'mcp_version'=> $mcp_version,
'cardholder_amount' => $cardholder_amount,
'cardholder_currency_code' => $cardholder_currency_code,
'mcp_rate_token' => $mcp_rate_token
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("U");
$cof->setPaymentInformation("2");
$cof->setIssuerId("168451306048014");
$mpgTxn->setCofInfo($cof);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
/***** mpgHttpPost Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response Object *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nAVSResponse = " . $mpgResponse->getAvsResultCode());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
//----- ResolveData -----
print("\n\nCust ID = " . $mpgResponse->getResDataCustId());
print("\nPhone = " . $mpgResponse->getResDataPhone());
print("\nEmail = " . $mpgResponse->getResDataEmail());
print("\nNote = " . $mpgResponse->getResDataNote());
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\nCrypt Type = " . $mpgResponse->getResDataCryptType());
print("\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());

```

```

print("\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
print("\nMerchantSettlementAmount = " . $mpgResponse->getMerchantSettlementAmount());
print("\nCardholderAmount = " . $mpgResponse->getCardholderAmount());
print("\nCardholderCurrencyCode = " . $mpgResponse->getCardholderCurrencyCode());
print("\nMCPRate = " . $mpgResponse->getMCPRate());
print("\nMCPErrorsStatuscode = " . $mpgResponse->getMCPErrorsStatuscode());
print("\nMCPErrorsMessage = " . $mpgResponse->getMCPErrorsMessage());
print("\nHostId = " . $mpgResponse->getHostId());
?>

```

## 8.15 MCP Pre-Authorization with Vault

This transaction uses the data key to identify a previously registered credit card profile in Vault. The details saved within the profile are then submitted to perform a Pre-Authorization transaction.

The data key may be a temporary one generated used Hosted Tokenization, or may be a permanent one from the Vault.

This transaction request is the multi-currency pricing (MCP) enabled version of the equivalent financial transaction.

### MCP Pre-Authorization with Vault transaction object definition

```

$txnArray = array('type'=>'mcp_res_preauth_cc', ...);

$mpgTxn = new mpgTransaction($txnArray);

```

### HttpPostRequest object for MCP Pre-Authorization with Vault transaction

```

$mpgRequest = new mpgRequest($mpgTxn);

$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);

```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

### MCP Pre-Authorization with Vault transaction request fields – Required

| Variable Name | Type and Limits | Set Method             |
|---------------|-----------------|------------------------|
| data key      | String          | 'data_key'=>\$data_key |

| Variable Name                 | Type and Limits                                                             | Set Method             |
|-------------------------------|-----------------------------------------------------------------------------|------------------------|
|                               | 25-character alphanumeric                                                   |                        |
| order ID                      | String<br><br>50-character alpha-<br>numerica-Z A-Z 0-9 _ - : . @<br>spaces | 'order_id'=>\$order_id |
| electronic commerce indicator | String<br><br>1-character alphanumeric                                      | 'crypt_type'=>\$crypt  |

**MCP-specific request fields – Required**

| Variable Name            | Type and Limits                                                                      | Set Method                                                  |
|--------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------|
| MCP version number       | String<br><br>numeric<br><br>current version is 1.0                                  | 'mcp_version'=> \$mcp_version                               |
| cardholder amount        | String<br><br>12-character numeric<br><br>smallest discrete unit of foreign currency | 'cardholder_amount' =><br>\$cardholder_amount               |
| cardholder currency code | String<br><br>3-character numeric                                                    | 'cardholder_currency_code'<br>=> \$cardholder_currency_code |

**MCP Pre-Authorization with Vault transaction request fields – Optional**

| Variable Name | Type and Limits                                                                                                                                        | Set Method           |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| customer ID   | String<br><br>50-character alphanumeric<br><br><div><b>NOTE:</b><br/>Some special characters are not allowed:<br/>&lt; &gt; \$ % = ? ^ { } [ ] \</div> | 'cust_id'=>\$cust_id |

| Variable Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Type and Limits                                                                                                 | Set Method                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| dynamic descriptor                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>String</i><br>20-character alphanumeric<br>total of 22 characters including your merchant name and separator | 'dynamic_descriptor'=>\$dynamic_descriptor |
| <div> <b>NOTE: For Pre-Authorization transactions:</b> the value in the dynamic descriptor field will only be carried over to a Pre-Authorization Completion when executing the latter via the Merchant Resource Center; otherwise, the value for dynamic descriptor must be sent again in the Pre-Authorization Completion         </div> <div> <b>NOTE:</b><br/>           Some special characters are not allowed:<br/>           &lt; &gt; \$ % = ? ^ { } [ ] \         </div> |                                                                                                                 |                                            |
| final authorization                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>String</i><br>true/false                                                                                     | 'final_auth' => 'true'                     |
| <div> <b>NOTE:</b> Applies to Mastercard transactions only         </div>                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                 |                                            |
| Credential on File Info                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <i>Object</i>                                                                                                   | \$mpgTxn->setCofInfo(\$cof);               |
| cof                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | N/A                                                                                                             |                                            |
| <div> <b>NOTE:</b> This is a nested object within the transaction, and required when storing or using the customer's stored credentials. For information about fields in the Credential on File Info object, see Credential on File Info Object and Variables.         </div>                                                                                                                                                                                                      |                                                                                                                 |                                            |
| AVS Information                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <i>Object</i><br>N/A                                                                                            | \$mpgTxn->setAvsInfo(\$mpgAvsInfo);        |
| CVD Information                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <i>Object</i><br>N/A                                                                                            | \$mpgTxn->setCvdInfo(\$mpgCvdInfo);        |

**MCP-specific request fields – Optional**

| Variable Name  | Type and Limits          | Set Method                           |
|----------------|--------------------------|--------------------------------------|
| MCP rate token | <i>String</i><br><br>N/A | 'mcp_rate_token' => \$mcp_rate_token |

**Sample MCP Pre-Authorization with Vault**

```

<?php
##
## This program takes 3 arguments from the command line:
## 1. Store id
## 2. api token
## 3. order id
##
## Example php -q TestResPreauthCC.php store3 yesguy unique_order_id cust_id 15.00 1
##
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
/***** Transaction Variables *****/
$data_key='uX6jFwbvCytmG7oT70YVNm0e2';
$orderid='res-preauth-'.date("dmy-G:i:s");
$amount='1.00';
$custid='cust'; //if sent will be submitted, otherwise cust_id from profile will be used
$crypt_type='1';
//$expdate='1512';
$mcp_version = '1.0';
$cardholder_amount = '100';
$cardholder_currency_code = '840';
$mcp_rate_token = 'P1536170825312107';
/***** Transaction Array *****/
$txnArray =array('type'=>'mcp_res_preauth_cc',
'data_key'=>$data_key,
'order_id'=>$orderid,
'cust_id'=>$custid,
'amount'=>$amount,
'crypt_type'=>$crypt_type,
//'expdate'=>$expdate,
'dynamic_descriptor'=>'12424',
'mcp_version'=> $mcp_version,
'cardholder_amount' => $cardholder_amount,
'cardholder_currency_code' => $cardholder_currency_code,
'mcp_rate_token' => $mcp_rate_token
//'final_auth'=>'true'
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("U");
$cof->setPaymentInformation("2");
$cof->setIssuerId("168451306048014");
$mpgTxn->setCofInfo($cof);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** mpgHttpPost Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);

```

```

/***** Response Object *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nAVSResponse = " . $mpgResponse->getAvsResultCode());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
//----- ResolveData -----
print("\n\nCust ID = " . $mpgResponse->getResDataCustId());
print("\nPhone = " . $mpgResponse->getResDataPhone());
print("\nEmail = " . $mpgResponse->getResDataEmail());
print("\nNote = " . $mpgResponse->getResDataNote());
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\nCrypt Type = " . $mpgResponse->getResDataCryptType());
print("\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
print("\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
print("\nMerchantSettlementAmount = " . $mpgResponse->getMerchantSettlementAmount());
print("\nCardholderAmount = " . $mpgResponse->getCardholderAmount());
print("\nCardholderCurrencyCode = " . $mpgResponse->getCardholderCurrencyCode());
print("\nMCPRate = " . $mpgResponse->getMCPRate());
print("\nMCPErrorStatusCode = " . $mpgResponse->getMCPErrorStatusCode());
print("\nMCPErrorMessage = " . $mpgResponse->getMCPErrorMessage());
print("\nHostId = " . $mpgResponse->getHostId());
?>

```

## 8.16 MCP Independent Refund with Vault

This transaction uses the data key to identify a previously registered credit card profile in Vault. The details saved within the profile are then submitted to perform an Independent Refund transaction.

This transaction request is the multi-currency pricing (MCP) enabled version of the equivalent financial transaction.

### Things to Consider:

- Because of the potential for fraud, permission for this transaction is not granted to all accounts by default. If it is required for your business, it must be requested via your account manager.

**MCP Independent Refund with Vault transaction object definition**

```
$txnArray = array('type'=>'mcp_res_ind_refund_cc', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

**HttpPostRequest object for MCP Independent Refund with Vault transaction**

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

**Core connection object fields (all API transactions)**

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

**MCP Independent Refund with Vault transaction request fields – Required**

| Variable Name                 | Type and Limits                                                         | Set Method             |
|-------------------------------|-------------------------------------------------------------------------|------------------------|
| data key                      | String<br>25-character alphanumeric                                     | 'data_key'=>\$data_key |
| order ID                      | String<br>50-character alpha-<br>numerica-Z A-Z 0-9 _ - : . @<br>spaces | 'order_id'=>\$order_id |
| electronic commerce indicator | String<br>1-character alphanumeric                                      | 'crypt_type'=>\$crypt  |

**MCP-specific request fields – Required**

| Variable Name      | Type and Limits   | Set Method                    |
|--------------------|-------------------|-------------------------------|
| MCP version number | String<br>numeric | 'mcp_version'=> \$mcp_version |

| Variable Name            | Type and Limits                                                                      | Set Method                                               |
|--------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------|
|                          | current version is 1.0                                                               |                                                          |
| cardholder amount        | String<br><br>12-character numeric<br><br>smallest discrete unit of foreign currency | 'cardholder_amount' => \$cardholder_amount               |
| cardholder currency code | String<br><br>3-character numeric                                                    | 'cardholder_currency_code' => \$cardholder_currency_code |

### MCP Independent Refund with Vault transaction request fields – Optional

| Variable Name | Type and Limits                                                                                                                                        | Set Method           |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| customer ID   | String<br><br>50-character alphanumeric<br><br><div><b>NOTE:</b><br/>Some special characters are not allowed:<br/>&lt; &gt; \$ % = ? ^ { } [ ] \</div> | 'cust_id'=>\$cust_id |

### MCP-specific request fields – Optional

| Variable Name  | Type and Limits   | Set Method                           |
|----------------|-------------------|--------------------------------------|
| MCP rate token | String<br><br>N/A | 'mcp_rate_token' => \$mcp_rate_token |

### Sample MCP Independent Refund with Vault

```
<?php
##
## This program takes 3 arguments from the command line:
## 1. Store id
## 2. api token
## 3. order id
##
## Example php -q TestResIndRefundCC.php store3 yesguy unique_order_id cust_id 15.00 1
##
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
/***** Transaction Variables *****/
```



```

$data_key='uX6jFwbvCytmG7oT70YVNm0e2';
$orderid='res-ind-refund-' . date("dmy-G:i:s");
$custid='';
$script_type='1';
$mcp_version = '1.0';
$cardholder_amount = '100';
$cardholder_currency_code = '840';
$mcp_rate_token = 'R1536165545730980';
/***** Transaction Array *****/
$txnArray = array('type'=>'mcp_res_ind_refund_cc',
'data_key'=>$data_key,
'order_id'=>$orderid,
'cust_id'=>$custid,
'crypt_type'=>$script_type,
'dynamic_descriptor'=>'12346',
'mcp_version'=> $mcp_version,
'cardholder_amount' => $cardholder_amount,
'cardholder_currency_code' => $cardholder_currency_code,
'mcp_rate_token' => $mcp_rate_token
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
/***** mpgHttpPost Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response Object *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nResSuccess = " . $mpgResponse->getResSuccess());
print("\nPaymentType = " . $mpgResponse->getPaymentType());
//----- ResolveData -----
print("\n\nCust ID = " . $mpgResponse->getResDataCustId());
print("\nPhone = " . $mpgResponse->getResDataPhone());
print("\nEmail = " . $mpgResponse->getResDataEmail());
print("\nNote = " . $mpgResponse->getResDataNote());
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\nCrypt Type = " . $mpgResponse->getResDataCryptType());
print("\nAvs Street Number = " . $mpgResponse->getResDataAvsStreetNumber());
print("\nAvs Street Name = " . $mpgResponse->getResDataAvsStreetName());
print("\nAvs Zipcode = " . $mpgResponse->getResDataAvsZipcode());
print("\nMerchantSettlementAmount = " . $mpgResponse->getMerchantSettlementAmount());
print("\nCardholderAmount = " . $mpgResponse->getCardholderAmount());
print("\nCardholderCurrencyCode = " . $mpgResponse->getCardholderCurrencyCode());
print("\nMCPRate = " . $mpgResponse->getMCPRate());
print("\nMCPErrrorStatusCode = " . $mpgResponse->getMCPErrrorStatusCode());
print("\nMCPErrrorMessage = " . $mpgResponse->getMCPErrrorMessage());
print("\nHostId = " . $mpgResponse->getHostId());
?>

```

## 8.17 MCP Currency Codes

For currency symbols, see <https://justforex.com/education/currencies>

**NOTE:** This documentation contains links to websites owned and operated by third parties. If you use these links, you leave our website. These links are provided for your information and convenience only and are not an endorsement by Moneris Solutions of the content of such linked websites or third parties. Moneris Solutions has no control over the contents of any linked website and is not responsible for these websites or their content or availability. If you decide to access any third party websites and make use of the information contained on them, you do so entirely at your own risk.

| Numeric Currency Code (ISO) | Currency Name/Acronym        |
|-----------------------------|------------------------------|
| 008                         | Albanian Lek (ALL)           |
| 012                         | Algerian Dinar (DZD)         |
| 032                         | Argentine Peso (ARS)         |
| 036                         | Australian Dollar (AUD)      |
| 048                         | Bahraini Dinar (BHD)         |
| 050                         | Bangladeshi Taka (BDT)       |
| 052                         | Barbados Dollar (BBD)        |
| 060                         | Bermudian Dollar (BMD)       |
| 064                         | Bhutan Ngultrum (BTN)        |
| 068                         | Bolivia Boliviano (BOB)      |
| 084                         | Belize Dollar (BZD)          |
| 090                         | Solomon Islands Dollar (SBD) |
| 096                         | Brunei Dollar (BND)          |
| 108                         | Burundi Franc (BIF)          |
| 132                         | Cabo Verde Escudo (CVE)      |

| Numeric Currency Code (ISO) | Currency Name/Acronym       |
|-----------------------------|-----------------------------|
| 136                         | Cayman Islands Dollar (KYD) |
| 144                         | Sri Lanka Rupee (LKR)       |
| 152                         | Chilean Peso (CLP)          |
| 156                         | Chinese Yuan (CNY)          |
| 170                         | Colombian Peso (COP)        |
| 174                         | Comorian Franc (KMF)        |
| 188                         | Costa Rican Colon (CRC)     |
| 191                         | Croatian Kuna (HRK)         |
| 192                         | Cuban Peso (CUP)            |
| 203                         | Czech Koruna (CZK)          |
| 208                         | Danish Krone (DKK)          |
| 214                         | Dominican Republic Peso     |
| 222                         | Salvadoran Colon (SVC)      |
| 242                         | Fijian Dollar (FJD)         |
| 262                         | Djiboutian Franc (DJF)      |
| 270                         | Gambian Dalasi (GMD)        |
| 292                         | Gibraltar Pound (GIP)       |
| 320                         | Guatemala Quetzal (GTQ)     |
| 324                         | Guinean Franc (GNF)         |
| 328                         | Guyanese Dollar (GYD)       |
| 332                         | Haitian Gourde (HTG)        |
| 340                         | Honduran Lempira (HNL)      |

| Numeric Currency Code (ISO) | Currency Name/Acronym   |
|-----------------------------|-------------------------|
| 344                         | Hong Kong Dollar (HKD)  |
| 348                         | Hungarian Forint (HUF)  |
| 352                         | Iceland Krona (ISK)     |
| 356                         | Indian Rupee (INR)      |
| 360                         | Indonesian Rupiah (IDR) |
| 376                         | Israeli Shekel (ILS)    |
| 388                         | Jamaican Dollar (JMD)   |
| 392                         | Japanese Yen (JPY)      |
| 398                         | Kazakh Tenge (KZT)      |
| 400                         | Jordanian Dinar (JOD)   |
| 404                         | Kenyan Shilling (KES)   |
| 410                         | South Korean Won (KRW)  |
| 414                         | Kuwaiti Dinar (KWD)     |
| 418                         | Laotian Kip (LAK)       |
| 426                         | Lesotho Loti (LSL)      |
| 430                         | Liberian Dollar (LRD)   |
| 446                         | Macanese Pataca (MOP)   |
| 454                         | Malawian Kwacha (MWK)   |
| 458                         | Malaysian Ringgit (MYR) |
| 462                         | Maldivian Rufiyaa (MVR) |
| 480                         | Mauritius Rupee (MUR)   |
| 484                         | Mexican Peso (MXN)      |

| Numeric Currency Code (ISO) | Currency Name/Acronym               |
|-----------------------------|-------------------------------------|
| 498                         | Moldovan Leu (MDL)                  |
| 504                         | Moroccan Dirham (MAD)               |
| 512                         | Omani Rial (OMR)                    |
| 516                         | Namibian Dollar (NAD)               |
| 524                         | Nepalese Rupee (NPR)                |
| 532                         | Netherlands Antillean Guilder (ANG) |
| 533                         | Aruban Guilder (AWG)                |
| 548                         | Vanuatu Vatu (VUV)                  |
| 554                         | New Zealand Dollar (NZD)            |
| 558                         | Nicaraguan Cordoba (NIO)            |
| 566                         | Nigerian Naira (NGN)                |
| 578                         | Norwegian Krone (NOK)               |
| 586                         | Pakistan Rupee (PKR)                |
| 598                         | Papua New Guinean Kina (PGK)        |
| 600                         | Paraguayan Guarani (PYG)            |
| 604                         | Peruvian Nuevo Sol (PEN)            |
| 608                         | Philippine Peso (PHP)               |
| 634                         | Qatari Rial (QAR)                   |
| 643                         | Russian Ruble (RUB)                 |
| 646                         | Rwandan Franc (RWF)                 |
| 654                         | Saint Helena Pound (SHP)            |
| 682                         | Saudi Riyal (SAR)                   |

| Numeric Currency Code (ISO) | Currency Name/Acronym          |
|-----------------------------|--------------------------------|
| 690                         | Seychelles Rupee (SCR)         |
| 694                         | Sierra Leonean Leone (SLL)     |
| 702                         | Singapore Dollar (SGD)         |
| 704                         | Vietnamese Dong (VND)          |
| 710                         | South African Rand (ZAR)       |
| 748                         | Swaziland Lilangeni (SZL)      |
| 752                         | Swedish Krona (SEK)            |
| 756                         | Swiss Franc (CHF)              |
| 764                         | Thai Baht (THB)                |
| 780                         | Trinidad & Tobago Dollar (TTD) |
| 784                         | UAE Dirham (AED)               |
| 788                         | Tunisian Dinar (TND)           |
| 800                         | Ugandan Shilling (UGX)         |
| 807                         | Macedonian Denar (MKD)         |
| 818                         | Egyptian Pound (EGP)           |
| 826                         | UK Pound Sterling (GBP)        |
| 834                         | Tanzanian Shilling (TZS)       |
| 840                         | US Dollar (USD)                |
| 858                         | Uruguayan Peso (UYU)           |
| 860                         | Uzbekistani Sum (UZS)          |
| 882                         | Samoaan Tala (WST)             |
| 901                         | New Taiwan Dollar (TWD)        |

| Numeric Currency Code (ISO) | Currency Name/Acronym                         |
|-----------------------------|-----------------------------------------------|
| 929                         | Mauritanian Ouguiya (MRU)                     |
| 933                         | Belarusian Ruble (BYN)                        |
| 934                         | Turkmenistan Manat (TMT)                      |
| 941                         | Serbian Dinar (RSD)                           |
| 943                         | Mozambique Metical (MZN)                      |
| 944                         | Azerbaijani Manat (AZN)                       |
| 946                         | Romanian New Leu (RON)                        |
| 949                         | New Turkish Lira (TRY)                        |
| 951                         | East Caribbean Dollar (XCD)                   |
| 952                         | West African CFA Franc BCEAO (XOF)            |
| 953                         | CFP Franc (XPF)                               |
| 967                         | Zambian Kwacha (ZMW)                          |
| 968                         | Surinamese Dollar (SRD)                       |
| 969                         | Malagasy Ariary (MGA)                         |
| 971                         | Afghan Afghani (AFN)                          |
| 972                         | Tajikistan Somoni (TJS)                       |
| 973                         | Angola Kwanza (AOA)                           |
| 975                         | Bulgarian Lev (BGN)                           |
| 977                         | Bosnia and Herzegovina Convertible Mark (BAM) |
| 978                         | Euro (EUR)                                    |
| 981                         | Georgian Lari (GEL)                           |
| 985                         | Polish New Zloty (PLN)                        |

| Numeric Currency Code (ISO) | Currency Name/Acronym |
|-----------------------------|-----------------------|
| 986                         | Brazilian Real (BRL)  |

## 8.18 MCP Error Codes

| Error Code | Description                                                                                 |
|------------|---------------------------------------------------------------------------------------------|
| 200        | OK (there will be no value returned in the MCP error message)                               |
| 500        | Upstream error                                                                              |
| 1000       | Invalid JSON format                                                                         |
| 1003       | Invalid txnType detected: <invalid txnType> please enter PURCHASE or REFUND                 |
| 1005       | Invalid rateInquiryId-txnType combination.                                                  |
| 1007       | Warning: at least one of cardHolderCurrency or merchantSettlementCurrency must be non-zero. |
| 1008       | Card-holder amount must be non-zero.                                                        |
| 1009       | Negative amounts detected                                                                   |
| 1010       | Unsupported cardholder currency detected: <unsupported currency>                            |
| 1015       | invalid rateInquiryId                                                                       |
| 1016       | Unsupported merchant id                                                                     |



## 9 Installments by Visa

- 9.1 About Installments by Visa
- 9.2 Installments by Visa Transaction Types
- 9.3 Sending Transactions with Installments by Visa
- 9.4 Installment Plan Lookup
- 9.5 Vault Installment Plan Lookup
- 9.6 Installment Info Object

### 9.1 About Installments by Visa

Installments by Visa enables issuers the ability to offer cardholders installment payment plans at the time of purchase. When a cardholder accepts an installment plan option, the merchant receives the payment in full, and the cardholder pays the issuer according to the plan.

For a full list of definitions of the request and response fields see B.3 Definition of Response Fields – Installments by Visa

### 9.2 Installments by Visa Transaction Types

Financial transactions that support Installments by Visa include the following:

- Purchase
- Pre-Authorization
- Pre-Authorization Completion
- Purchase Correction
- Refund

- Purchase with Vault – ResPurchaseCC
- Pre-Authorization with Vault – ResPreauthCC

**NOTE:** Independent Refund transactions do not support Installments by Visa

**WARNING:** Do not send the Installment Info object on any transaction that is not intended to offer Installments by Visa functionality; doing so may cause the transaction to fail.

### 9.3 Sending Transactions with Installments by Visa

Sending transactions with Installments by Visa functionality involves the following steps:

1. Send the Installment Plan Lookup or Vault Installment Plan Lookup (for Vault transactions) transaction request to obtain the **installment plan ID**, **installment plan reference** and **terms and conditions version** data in the response
2. Present the offered installment plan(s) to the cardholder and obtain their agreement to a particular plan.
3. Using the data obtained in the response above, send the Installment Info object in the Purchase or Pre-Authorization; for Vault transactions, use Purchase with Vault or Pre-Authorization with Vault

When completing the transaction with a Pre-Authorization Completion, or when doing a Purchase Correction or Refund, as in the rest of the Unified API, the previous transactions are referenced using the **order ID** and **transaction number**, or for Vault transactions, using the **data key**.

**NOTE:** Independent Refund transactions do not support Installments by Visa

### 9.4 Installment Plan Lookup

Used to obtain information required to do financial transactions with Installments by Visa.

**Installment Plan Lookup transaction object definition**

```
$txnArray = array('type'=>'installmentLookup', ...);

$mpgTxn = new mpgTransaction($txnArray);

$mpgRequest = new mpgRequest($mpgTxn);

$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

**Core connection object fields (all API transactions)**

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

**Installment Plan Lookup transaction request fields – Required**

| Variable Name      | Type and Limits                                                                                                                                             |                              |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| order ID           | String<br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                                                                                         | 'order_id'=>\$order_id       |
| amount             | String<br>10-character decimal<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><div>EXAMPLE: 1234567.89</div> | 'amount'=>\$amount           |
| credit card number | String<br>max 20-character alpha-numeric                                                                                                                    | 'pan'=>\$pan                 |
| expiry date        | String                                                                                                                                                      | 'expiry_date'=>\$expiry_date |

| Variable Name | Type and Limits |
|---------------|-----------------|
|---------------|-----------------|

4-character alphanumeric

YYMM

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='monca03650';
$api_token='7Yw0MPTlhjBRcZiE6837';
/***** Transaction Variables *****/
$type='installment_lookup';
$order_id='Test'.date("dmy-G:i:s");
$amount='600.00';
$pan='4761270070000310';
$expdate='2212';
/***** Transaction Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'amount'=>$amount,
'pan'=>$pan,
'expdate'=>$expdate
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("U");
$cof->setPaymentInformation("2");
$cof->setIssuerId("168451306048014");
$mpgTxn->setCofInfo($cof);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
/* Status Check Example
$mpgHttpPost =new mpgHttpPostStatus($store_id,$api_token,$status_check,$mpgRequest);
*/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nBankTotals = " . $mpgResponse->getBankTotals());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print("\nSourcePanLast4 = " . $mpgResponse->getSourcePanLast4());
$eligibleInstallmentPlans = $mpgResponse->getEligibleInstallmentPlans();

$planCount = $eligibleInstallmentPlans->getPlanCount();

```

```

$installmentPlans = $eligibleInstallmentPlans->getInstallmentPlans();

for ($i = 0; $i < $planCount; $i++)
{
    print("\nPlanId = " . $installmentPlans[$i]->getPlanId());
    print("\nPlanIdRef = " . $installmentPlans[$i]->getPlanIdRef());
    print("\nName = " . $installmentPlans[$i]->getName());
    print("\nType = " . $installmentPlans[$i]->getType());
    print("\nNumInstallments = " . $installmentPlans[$i]->getNumInstallments());
    print("\nInstallmentFrequency = " . $installmentPlans[$i]->getInstallmentFrequency());
    $tac = $installmentPlans[$i]->getTac();
    $tacDetailsList = $tac->getTacDetailsList();
    $tacCount = $tac->getTacCount();
    print("\ntacCount = " . $tacCount);
    for ($j = 0; $j < $tacCount; $j++)
    {
        $tacDetails = $tacDetailsList[$j];

        print("\nText = " . $tacDetails->getText());
        print("\nUrl = " . $tacDetails->getUrl());
        print("\nVersion = " . $tacDetails->getVersion());
        print("\nLanguageCode = " . $tacDetails->getLanguageCode());
    }
    $promotionInfo = $installmentPlans[$i]->getPromotionInfo();
    print("\nPromotionCode = " . $promotionInfo->getPromotionCode());
    print("\nPromotionId = " . $promotionInfo->getPromotionId());
    $firstInstallment = $installmentPlans[$i]->getFirstInstallment();
    print("\nUpfrontFee = " . $firstInstallment->getUpfrontFee());
    print("\nInstallmentFee = " . $firstInstallment->getInstallmentFee());
    print("\nAmount = " . $firstInstallment->getAmount());
    $lastInstallment = $installmentPlans[$i]->getLastInstallment();
    print("\nInstallmentFee = " . $lastInstallment->getInstallmentFee());
    print("\nAmount = " . $lastInstallment->getAmount());
    print("\nAPR = " . $installmentPlans[$i]->getAPR());
    print("\nTotalFees = " . $installmentPlans[$i]->getTotalFees());
    print("\nTotalPlanCost = " . $installmentPlans[$i]->getTotalPlanCost());
}
?>

```

## 9.5 Vault Installment Plan Lookup

Used to obtain information required to do financial transactions with installments when using a token stored in the Moneris Vault.

### Vault Installment Plan Lookup transaction object definition

```

$txnArray = array('type'=>'resInstallmentLookup', ...);

$mpgTxn = new mpgTransaction($txnArray);

$mpgRequest = new mpgRequest($mpgTxn);

$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);

```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                        |
|---------------|-----------------|------------------------|
| store ID      | <i>String</i>   | 'store_id'=>\$store_id |

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
|               | N/A             |                          |
| API token     | String          | 'api_token'=>\$api_token |
|               | N/A             |                          |

### Vault Installment Plan Lookup transaction request fields – Required

| Variable Name                                                                             | Type and Limits                                                                                                                                                      |                              |
|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| order ID                                                                                  | String<br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                                                                                                  | 'order_id'=>\$order_id       |
| amount                                                                                    | String<br>10-character decimal<br>Up to 7 digits (dollars) +<br>decimal point (.) + 2 digits<br>(cents) after the decimal<br>point<br><div>EXAMPLE: 1234567.89</div> | 'amount'=>\$amount           |
| data key                                                                                  | String<br>25-character alphanumeric                                                                                                                                  | 'data_key'=>\$data_key       |
| expiry date                                                                               | String<br>4-character alphanumeric<br>YYMM                                                                                                                           | 'expiry_date'=>\$expiry_date |
| <div>NOTE: Only send this field if using a temporary token; if not, omit this field</div> |                                                                                                                                                                      |                              |

```
<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='monca03650';
$api_token='7Yw0MPTlhjBRcZiE6837';
/***** Transaction Variables *****/
$type='res_installment_lookup';
$order_id='Test'.date("dmy-G:i:s");
$amount='600.00';
$data_key='8cwY6hotzkM362ygNiytlBtY0';
$expdate='2212'; //For Temp Token
```

```

/***** Transaction Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'amount'=>$amount,
'data_key'=>$data_key,
'expdate'=>$expdate
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
/* Status Check Example
$mpgHttpPost =new mpgHttpPostStatus($store_id,$api_token,$status_check,$mpgRequest);
*/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nBankTotals = " . $mpgResponse->getBankTotals());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
$eligibleInstallmentPlans = $mpgResponse->getEligibleInstallmentPlans();

$planCount = $eligibleInstallmentPlans->getPlanCount();
$installmentPlans = $eligibleInstallmentPlans->getInstallmentPlans();

for ($i = 0; $i < $planCount; $i++)
{
print("\nPlanId = " . $installmentPlans[$i]->getPlanId());
print("\nPlanIdRef = " . $installmentPlans[$i]->getPlanIdRef());
print("\nName = " . $installmentPlans[$i]->getName());
print("\nType = " . $installmentPlans[$i]->getType());
print("\nNumInstallments = " . $installmentPlans[$i]->getNumInstallments());
print("\nInstallmentFrequency = " . $installmentPlans[$i]->getInstallmentFrequency());
$tac = $installmentPlans[$i]->getTac();
$tacDetailsList = $tac->getTacDetailsList();
$tacCount = $tac->getTacCount();
print("\ntacCount = " . $tacCount);
for ($j = 0; $j < $tacCount; $j++)
{
$tacDetails = $tacDetailsList[$j];

print("\nText = " . $tacDetails->getText());
print("\nUrl = " . $tacDetails->getUrl());
print("\nVersion = " . $tacDetails->getVersion());
print("\nLanguageCode = " . $tacDetails->getLanguageCode());
}
$promotionInfo = $installmentPlans[$i]->getPromotionInfo();
print("\nPromotionCode = " . $promotionInfo->getPromotionCode());
print("\nPromotionId = " . $promotionInfo->getPromotionId());

```

```

$firstInstallment = $installmentPlans[$i]->getFirstInstallment();
print("\nUpfrontFee = " . $firstInstallment->getUpfrontFee());
print("\nInstallmentFee = " . $firstInstallment->getInstallmentFee());
print("\nAmount = " . $firstInstallment->getAmount());
$lastInstallment = $installmentPlans[$i]->getLastInstallment();
print("\nInstallmentFee = " . $lastInstallment->getInstallmentFee());
print("\nAmount = " . $lastInstallment->getAmount());
print("\nAPR = " . $installmentPlans[$i]->getAPR());
print("\nTotalFees = " . $installmentPlans[$i]->getTotalFees());
print("\nTotalPlanCost = " . $installmentPlans[$i]->getTotalPlanCost());
}
//Resolve Data
print("\ncust_id = " . $mpgResponse->getResDataCustId());
print("\nphone = " . $mpgResponse->getResDataPhone());
print("\nemail = " . $mpgResponse->getResDataEmail());
print("\nnote = " . $mpgResponse->getResDataNote());
print("\nexdate = " . $mpgResponse->getResDataExpdate());
print("\nmasked_pan = " . $mpgResponse->getResDataMaskedPan());
?>

```

## 9.6 Installment Info Object

When sending Purchase or Pre-Authorization transactions with Installments by Visa, the Installment Info object is included in the request. The Installment Info object uses information received in the response to the Installment Plan Lookup transaction.

For a full list of definitions of the request and response fields see B.3 Definition of Response Fields – Installments by Visa

### Installment Info object request fields

| Variable Name                | Type and Limits                                                                 |                                                            |
|------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------|
| installment plan ID          | <i>String</i><br>36-character alphanumeric<br>fixed length                      | <code>\$installmentInfo-&gt;setPlanId("VALUE");</code>     |
| installment plan reference   | <i>String</i><br>10-character alphanumeric<br>fixed length                      | <code>\$installmentInfo-&gt;setPlanIdRef("VALUE");</code>  |
| terms and conditions version | <i>String</i><br>10-character alphanumeric<br>variable length (1-10 characters) | <code>\$installmentInfo-&gt;setTacVersion("VALUE");</code> |



**WARNING:** Do not send the Installment Info object on any transaction that is not intended to offer Installments by Visa functionality; doing so may cause the transaction to fail.

## 10 e-Fraud Tools

- 10.1 Address Verification Service
- 10.2 Card Validation Digits (CVD)
- 10.3 Transaction Risk Management Tool



## 10.1 Address Verification Service

- 10.1.1 About Address Verification Service (AVS)
- 10.1.2 AVS Info Object
- 10.1.3 AVS Response Codes
- 10.1.4 AVS Sample Code

### 10.1.1 About Address Verification Service (AVS)

Address Verification Service (AVS) is an optional fraud-prevention tool offered by issuing banks whereby a cardholder's address is submitted as part of the transaction authorization. The AVS address is then compared to the address kept on file at the issuing bank. AVS checks whether the street number, street name and zip/postal code match. The issuing bank returns an AVS result code indicating whether the data was matched successfully. Regardless of the AVS result code returned, the credit card is authorized by the issuing bank.

The response that is received from AVS verification is intended to provide added security and fraud prevention, but the response itself does not affect the completion of a transaction. Upon receiving a response, the choice to proceed with a transaction is left entirely to the merchant. The responses is **not** a strict guideline of whether a transaction will be approved or declined.

The following transactions support AVS:

- Purchase (Basic and Mag Swipe)
- Pre-Authorization (Basic)
- Re-Authorization (Basic)
- ResAddCC (Vault)
- ResUpdateCC (Vault)

#### Things to Consider:

- AVS is supported by Visa, MasterCard, American Express, Discover and JCB.
- When testing AVS, you must **only** use the Visa test card numbers 4242424242424242 or 4005554444444403, and the amounts described in the Simulator eFraud Response Codes document available at the Moneris developer portal (<https://developer-moneris.com>).
- Store ID "store5" is set up to support AVS testing.

10.1.2 AVS Info Object

AVSInfo object definition

```
$avsTemplate = array(
    'avs_street_number'=>$avs_street_number,
    'avs_street_name' =>$avs_street_name,
    'avs_zipcode' => $avs_zipcode,
    'avs_hostname'=>$avs_hostname,
    'avs_browser' =>$avs_browser,
    'avs_shiptocountry' => $avs_shiptocountry,
    'avs_merchprodsku' => $avs_merchprodsku,
    'avs_custip'=>$avs_custip,
    'avs_custphone' => $avs_custphone
);

$mpgAvsInfo = new mpgAvsInfo ($avsTemplate);
```

Transaction object set method

```
$mpgTxn->setAvsInfo ($mpgAvsInfo);
```

| Variable Name     | Type and Limits                                                                                                                                                            | Set Method                         | Description              |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|--------------------------|
| AVS street number | <div>String</div> <div>19-character alpha-numeric</div> <div><b>NOTE:</b> this character limit is a combined total allowed for AVS street number and AVS street name</div> | 'avs_street_number'=>'212'         | Cardholder street number |
| AVS street name   | <div>String</div> <div>19-character alpha-numeric</div> <div><b>NOTE:</b> this character</div>                                                                             | 'avs_street_name'=>'Payton Street' | Cardholder street name   |

| Variable Name       | Type and Limits                                                                          | Set Method                    | Description                |
|---------------------|------------------------------------------------------------------------------------------|-------------------------------|----------------------------|
|                     | <div>limit is the combined total allowed for AVS street number and AVS street name</div> |                               |                            |
| AVS zip/postal code | <i>String</i><br>9-character alpha-numeric                                               | 'avs_zipcode'<br>=> 'M1M1M1 ' | Cardholder zip/postal code |

### 10.1.3 AVS Response Codes

Below is a full list of possible AVS response codes. These can be returned when you call the `$mp-gResponse->getAvsResultCode()` method.

| Code | Visa                                    | Mastercard/Discover                                            | American Express/ JCB                                              |
|------|-----------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------------|
| A    | AVS street address only (partial match) | Address matches, zip/ postal code does not                     | Billing address matches, zip/postal code does not                  |
| D    | No longer applicable to Visa            | N/A                                                            | Customer name incorrect; zip/postal code matches                   |
| E    | N/A                                     | N/A                                                            | Customer name incorrect, billing address and zip/postal code match |
| F    | No longer applicable to Visa            | N/A                                                            | Customer name incorrect; billing address matches                   |
| G    | No longer applicable to Visa            | Address information not verified for international transaction | N/A                                                                |
| K    | N/A                                     | N/A                                                            | Customer name matches                                              |

| Code | Visa                              | Mastercard/Discover                                                                                                                               | American Express/ JCB                                                       |
|------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| L    | N/A                               | N/A                                                                                                                                               | Customer name and zip/postal code match                                     |
| M    | No longer applicable to Visa      | N/A                                                                                                                                               | Customer name, billing address, and zip/postal code match                   |
| N    | AVS non-match                     | Neither address nor zip/postal code matches                                                                                                       | Billing address and zip/postal code do not match                            |
| O    | N/A                               | N/A                                                                                                                                               | Customer name and billing address match                                     |
| R    | AVS indeterminate outcome (retry) | Retry; system unable to process                                                                                                                   | System unavailable; retry                                                   |
| S    | No longer applicable to Visa      | AVS currently not supported                                                                                                                       | AVS currently not supported                                                 |
| T    | N/A                               | Nine-digit zip code matches; address does not match                                                                                               | N/A                                                                         |
| U    | AVS unable to verify              | No data from issuer-authorization system                                                                                                          | Information is unavailable                                                  |
| W    | No longer applicable to Visa      | For U.S. addresses, nine-digit postal code matches, address does not<br><br>For addresses outside the U.S., postal code matches, address does not | Customer name, billing address, and zip/postal code are all correct matches |
| X    | No longer applicable to Visa      | For U.S. addresses, nine-digit postal code and address match<br><br>For addresses outside the U.S., postal code and address match                 | N/A                                                                         |
| Y    | AVS full match                    | Billing address and zip/postal code both match                                                                                                    | Billing address and zip/postal code both match                              |

| Code | Visa                                     | Mastercard/Discover                                                     | American Express/ JCB                             |
|------|------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------|
| Z    | AVS zip/postal code only (partial match) | For U.S. addresses, five-digit zip code matches, address does not match | Zip/postal code matches, billing address does not |

### 10.1.4 AVS Sample Code

This is a sample of Java code illustrating how AVS is implemented with a Purchase transaction. Purchase object information that is not relevant to AVS has been removed.

For more about Purchase transactions, see 2.1 Purchase.

| Sample Purchase with AVS information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> \$avs_street_number = '201'; \$avs_street_name = 'Michigan Ave'; \$avs_zipcode = 'M1M1M1'; \$avs_email = 'test@host.com'; \$avs_hostname = "www.testhost.com"; \$avs_browser = 'Mozilla'; \$avs_shiptocountry = 'Canada'; \$avs_merchprodsku = '123456'; \$avs_custip = '192.168.0.1'; \$avs_custphone = '5556667777'; \$avsTemplate = array(     'avs_street_number'=&gt;\$avs_street_number,     'avs_street_name' =&gt;\$avs_street_name,     'avs_zipcode' =&gt; \$avs_zipcode,     'avs_hostname'=&gt;\$avs_hostname,     'avs_browser' =&gt;\$avs_browser,     'avs_shiptocountry' =&gt; \$avs_shiptocountry,     'avs_merchprodsku' =&gt; \$avs_merchprodsku,     'avs_custip'=&gt;\$avs_custip,     'avs_custphone' =&gt; \$avs_custphone ); \$mpgAvsInfo = new mpgAvsInfo (\$avsTemplate); \$txnArray=array(     'type'=&gt;'purchase',     'order_id'=&gt;\$order_id,     'cust_id'=&gt;\$cust_id,     'amount'=&gt;\$amount,     'pan'=&gt;\$pan,     'expdate'=&gt;\$expiry_date,     'crypt_type'=&gt;\$crypt ); \$mpgTxn = new mpgTransaction(\$txnArray); \$mpgTxn-&gt;setAvsInfo(\$mpgAvsInfo); </pre> |



## 10.2 Card Validation Digits (CVD)

- 10.2.1 About Card Validation Digits (CVD)
- 10.2.3 CVD Information Object
- 10.2.4 CVD Result Codes
- 10.2.5 Sample Purchase with CVD Info Object

### 10.2.1 About Card Validation Digits (CVD)

The Card Validation Digits (CVD) value is an additional number printed on credit cards that is used as an additional check when verifying cardholder credentials during a transaction.

The response that is received from CVD verification is intended to provide added security and fraud prevention, but the response itself does not affect the completion of a transaction. Upon receiving a response, the choice whether to proceed with a transaction is left entirely to the merchant. The responses is **not** a strict guideline of which transaction will approve or decline.

The following transactions support CVD:

- Purchase (Basic, Vault and Mag Swipe)
- Pre-Authorization (Basic and Vault)
- Re-Authorization

#### Things to Consider:

- CVD is only supported by Visa, MasterCard, American Express, Discover, JCB and UnionPay.
- For UnionPay cards, the CVD response will not be returned; the issuer will approve or decline based on the CVD result.
- When testing CVD, you must **only** use the Visa test card numbers 4242424242424242 or 4005554444444403, and the amounts described in the Simulator eFraud Response Codes document available at the Moneris developer portal (<https://developer-moneris.com>).
- Test store\_id “store5” is set up to support CVD testing.

### 10.2.2 Transactions Where CVD Is Required

The Card Validation Digits (CVD) object is required in transaction requests in the following scenarios:

- Initial transactions when storing cardholder credentials in Credential on File scenarios; subsequent follow-on transactions do not use CVD
- Any Purchase, Pre-Authorization or Card Verification where you are not storing cardholder credentials

### 10.2.3 CVD Information Object

**NOTE:** The CVD value must only be passed to the Moneris Gateway. Under **no** circumstances may it be stored for subsequent uses or displayed as part of the receipt information.

#### CvdInfo object definition

```
CvdInfo cvdCheck = new CvdInfo();  
  
$cvdTemplate = array(  
    'cvd_indicator' => $cvd_indicator,  
    'cvd_value' => $cvd_value  
);  
  
$mpgCvdInfo = new mpgCvdInfo ($cvdTemplate);
```

#### Transaction object set method

```
transaction.setCvdInfo(cvdCheck);  
  
$mpgTxn->setCvdInfo($mpgCvdInfo);
```

**Table 1 CVD Info Object – Required Fields**

| Variable Name | Type and Limits                          | Set Method               | Description                                                                                                                                                        |
|---------------|------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CVD indicator | <i>String</i><br><br>1-character numeric | 'cvd_indicator'<br>=>'1' | Indicates presence of CVD<br><br>Possible values:<br><br>0: CVD value is deliberately bypassed or is not provided by the merchant.<br><br>1: CVD value is present. |

| Variable Name | Type and Limits | Set Method | Description                                                                                                  |
|---------------|-----------------|------------|--------------------------------------------------------------------------------------------------------------|
|               |                 |            | 2: CVD value is on the card, but is illegible.<br><br>9: Cardholder states that the card has no CVD imprint. |

CVD value

*String*

4-character numeric

'cvd\_value'  
=>'123'

CVD value located on credit card

**NOTE:** The CVD value must only be passed to the Moneris Gateway. Under **no** circumstances may it be stored for subsequent uses or displayed as part of the receipt information.

### 10.2.4 CVD Result Codes

CVD verification is available for Visa, Mastercard, Discover, American Express, JCB and UnionPay transactions.

| Code  | Description                                                                   |
|-------|-------------------------------------------------------------------------------|
| M     | Match                                                                         |
| N     | No match                                                                      |
| P     | Not processed                                                                 |
| S     | CVD should be on the card, but Merchant has indicated that CVD is not present |
| U     | Issuer is not a CVD participant                                               |
| Y     | Match for American Express/JCB only                                           |
| D     | Invalid security code for American Express or JCB only                        |
| Other | Invalid response code                                                         |

### 10.2.5 Sample Purchase with CVD Info Object

This is a sample of Java code illustrating how CVD is implemented with a Purchase transaction. Purchase object information that is not relevant to CVD has been removed.

#### Sample Purchase with CVD Information

```
$cvdTemplate = array(
    'cvd_indicator' => '1',
    'cvd_value' => '123'
);
$mpgCvdInfo = new mpgCvdInfo ($cvdTemplate);
$txnArray=array(
    'type'=>'purchase',
    'order_id'=>$order_id,
    'cust_id'=>$cust_id,
    'amount'=>$amount,
    'pan'=>$pan,
    'expdate'=>$expiry_date,
    'crypt_type'=>$crypt
);
$mpgTxn = new mpgTransaction($txnArray);
$mpgTxn->setCvdInfo ($mpgCvdInfo);
```

## 10.3 Transaction Risk Management Tool

- 10.3.1 About the Transaction Risk Management Tool
- 10.3.2 Introduction to Queries
- 10.3.3 Session Query
- 10.3.4 Attribute Query
- 10.3.6 Inserting the Profiling Tags Into Your Website
- 10.3.6 Inserting the Profiling Tags Into Your Website

The Transaction Risk Management Tool (TRMT) is available to **Canadian integrations** only.

### 10.3.1 About the Transaction Risk Management Tool

The Transaction Risk Management Tool provides additional information to assist in identifying fraudulent transactions. To maximize the benefits from the Transaction Risk Management Tool, it is highly recommended that you:

- Carefully consider the business logic and processes that you need to implement surrounding the handling of response information the Transaction Risk Management Tool provides.
- Implement the other fraud tools available through Moneris Gateway (such as AVS, CVD, Verified by Visa, MasterCard SecureCode and American Express SafeKey).

### 10.3.2 Introduction to Queries

There are two types of transactions associated with the Transaction Risk Management Tool (TRMT):

- Session Query (page 418)
- Attribute Query (page 425)

The Session Query and Attribute Query are used at the time of the transaction to obtain the risk assessment.

Moneris recommends that you use the Session Query as much as possible for obtaining your risk assessment because it uses the device fingerprint as well as other transaction information when providing the risk scores.

To use the Session Query, you must implement two components:

- Tags on your website to collect the device fingerprinting information
- Session Query transaction.

If you are not able to collect the necessary information for the Session Query (such as the device fingerprint), then use the Attribute Query.

### 10.3.3 Session Query

Once a device profiling session has been initiated upon a client device, the Session Query API is used at the time of the transaction or even to obtain a device identifier or 'fingerprint', attribute list and risk assessment for the client device.

#### Session Query transaction object definition

```
$riskTxn = new riskTransaction($txnArray);
```

#### HttpPostRequest object for Session Query transaction

```
$riskHttpPost = new riskHttpPost($store_id, $api_token, $riskRequest);
```

#### Session Query transaction values

Table 3: Session Query transaction object mandatory values

| Value        | Type                                                                                                        | Limits                                                               | Set method                     |
|--------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------|
|              | Description                                                                                                 |                                                                      |                                |
| Session ID   | String                                                                                                      | 9-character decimal<br>Permitted characters: [a-z], [A-Z], 0-9, _, - | 'session_id'=>\$session_id     |
|              | Web server session identifier generated when device profiling was initiated.                                |                                                                      |                                |
| Service type | String                                                                                                      | 9-character decimal                                                  | 'service_type'=>\$service_type |
|              | Which output fields are returned.<br>session -- returns IP and device related attributes.                   |                                                                      |                                |
| Event type   | String                                                                                                      | payment                                                              | 'event_type'=>\$event_type     |
|              | Defines the type of transaction or event for reporting purposes.<br>payment - Purchasing of goods/services. |                                                                      |                                |

**Table 3: Session Query transaction object mandatory values (continued)**

| Value                             | Type                                                                                                                                                                                                                                                          | Limits                                      | Set method                                             |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|--------------------------------------------------------|
|                                   | Description                                                                                                                                                                                                                                                   |                                             |                                                        |
| Credit card number (PAN)          | String                                                                                                                                                                                                                                                        | 20-character numeric<br>No spaces or dashes | 'pan'=>\$pan                                           |
|                                   | Most credit card numbers today are 16 digits, but some 13-digit numbers are still accepted by some issuers. This field has been intentionally expanded to 20 digits in consideration for future expansion and potential support of private label card ranges. |                                             |                                                        |
| Account address street 1          | String                                                                                                                                                                                                                                                        | 32-character alphanumeric                   | 'account_address_street1'=>\$account_address_street1   |
|                                   | First portion of the street address component of the billing address.                                                                                                                                                                                         |                                             |                                                        |
| Account Address street 2          | String                                                                                                                                                                                                                                                        | 32-character alphanumeric                   | 'account_address_street2'=>\$account_address_street2   |
|                                   | Second portion of the street address component of the billing address.                                                                                                                                                                                        |                                             |                                                        |
| Account address city              | String                                                                                                                                                                                                                                                        | 50-character alphanumeric                   | 'account_address_city'=>\$account_address_city         |
|                                   | The city component of the billing address.                                                                                                                                                                                                                    |                                             |                                                        |
| Account address state/- province  | String                                                                                                                                                                                                                                                        | 64-character alphanumeric                   | 'account_address_state'=>\$account_address_state       |
|                                   | The state/province component of the billing address.                                                                                                                                                                                                          |                                             |                                                        |
| Account address country           | String                                                                                                                                                                                                                                                        | 2-character alphanumeric                    | 'account_address_country'=>\$account_address_country   |
|                                   | ISO2 country code of the billing addresses.                                                                                                                                                                                                                   |                                             |                                                        |
| Account address ZIP/- postal code | String                                                                                                                                                                                                                                                        | 8-character alphanumeric                    | 'account_address_zip'=>\$account_address_zip           |
|                                   | ZIP/postal code of the billing address.                                                                                                                                                                                                                       |                                             |                                                        |
| Shipping address street 1         | String                                                                                                                                                                                                                                                        | 32-character alphanumeric                   | 'shipping_address_street1'=>\$shipping_address_street1 |
|                                   | First portion of the street address component of the shipping address.                                                                                                                                                                                        |                                             |                                                        |

**Table 3: Session Query transaction object mandatory values (continued)**

| Value                             | Type                                                                                                                                                                                                                                                    | Limits                                                      | Set method                                             |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------|
|                                   | Description                                                                                                                                                                                                                                             |                                                             |                                                        |
| Shipping address street 2         | String                                                                                                                                                                                                                                                  | 32-character alphanumeric                                   | 'shipping_address_street2'=>\$shipping_address_street2 |
|                                   | Second portion of the street address component of the shipping address.                                                                                                                                                                                 |                                                             |                                                        |
| Shipping address city             | String                                                                                                                                                                                                                                                  | 50-character alphanumeric                                   | 'shipping_address_city'=>\$shipping_address_city       |
|                                   | City component of the shipping address.                                                                                                                                                                                                                 |                                                             |                                                        |
| Shipping address state/- province | String                                                                                                                                                                                                                                                  | 64-character alphanumeric                                   | 'shipping_address_state'=>\$shipping_address_state     |
|                                   | The state/province component of the shipping address.                                                                                                                                                                                                   |                                                             |                                                        |
| Shipping address country          | String                                                                                                                                                                                                                                                  | 2-character alphanumeric                                    | 'shipping_address_country'=>\$shipping_address_country |
|                                   | ISO2 country code of the account address country.                                                                                                                                                                                                       |                                                             |                                                        |
| Shipping address ZIP              | String                                                                                                                                                                                                                                                  | 8-character alphanumeric                                    | 'shipping_address_zip'=>\$shipping_address_zip         |
|                                   | The ZIP/postal code component of the shipping address.                                                                                                                                                                                                  |                                                             |                                                        |
| Local attribute 1-5               | String                                                                                                                                                                                                                                                  | 255-character alphanumeric                                  |                                                        |
|                                   | These five attributes can be used to pass custom attribute data. These are used if you wish to correlate some data with the returned device information.                                                                                                |                                                             |                                                        |
| Transaction amount                | String                                                                                                                                                                                                                                                  | 255-character alphanumeric<br>Must contain 2 decimal places |                                                        |
|                                   | The numeric currency amount.                                                                                                                                                                                                                            |                                                             |                                                        |
| Transaction currency              | String                                                                                                                                                                                                                                                  | 10-character numeric                                        |                                                        |
|                                   | <p>The currency type that the transaction was denominated in. If TransactionAmount is passed, the TransactionCurrency is required.</p> <p>Values to be used are:</p> <ul style="list-style-type: none"> <li>• CAD – 124</li> <li>• USD – 840</li> </ul> |                                                             |                                                        |



**Table 4: Session Query transaction object optional values**

| Value                    | Type Limits                                                                                                                                                  |                            | Set method                           |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------------------------------------|
|                          | Description                                                                                                                                                  |                            |                                      |
| Account login            | String                                                                                                                                                       | 255-character alphanumeric | 'account_login'=>\$account_login     |
|                          | The Account Login name.                                                                                                                                      |                            |                                      |
| Password hash            | String                                                                                                                                                       | 40-character alphanumeric  | 'password_hash' =>\$password_hash    |
|                          | The input must be a SHA-2 hash of the password in hexadecimal format. Used to check if it is on a watch list.                                                |                            |                                      |
| Account number           | String                                                                                                                                                       | 255-character alphanumeric | 'account_number' => \$account_number |
|                          | The account number for the account.                                                                                                                          |                            |                                      |
| Account name             | String                                                                                                                                                       | 255-character alphanumeric | 'account_name' => \$account_name     |
|                          | Account name (or concatenation of first and last name of account holder).                                                                                    |                            |                                      |
| Account email            | String                                                                                                                                                       | 100-character alphanumeric | 'account_email'=>\$account_email     |
|                          | The email address entered into the form for this contact. Used to check if this is a high risk account email id.                                             |                            |                                      |
| Account telephone        | String                                                                                                                                                       | 32-character alphanumeric  |                                      |
|                          | Contact telephone number including country and city codes. All whitespace is removed.<br><br>Must be in format: 0..9,<space>,(,),[,] braces must be matched. |                            |                                      |
| Address street 1         | String                                                                                                                                                       | 32-character alphanumeric  |                                      |
|                          | The first portion of the street address component of the account address.                                                                                    |                            |                                      |
| Address street 2         | String                                                                                                                                                       | 32-character alphanumeric  |                                      |
|                          | The second portion of the street address component of the account address.                                                                                   |                            |                                      |
| Address city             | String                                                                                                                                                       | 50-character alphanumeric  |                                      |
|                          | The city component of the account address.                                                                                                                   |                            |                                      |
| Address state/- province | String                                                                                                                                                       | 64-character alphanumeric  |                                      |
|                          | The state/province component of the account address                                                                                                          |                            |                                      |

**Table 4: Session Query transaction object optional values (continued)**

| Value                       | Type                                                                                             | Limits                     | Set method |
|-----------------------------|--------------------------------------------------------------------------------------------------|----------------------------|------------|
|                             | Description                                                                                      |                            |            |
| Address country             | String                                                                                           | 2-character alphanumeric   |            |
|                             | The 2 character ISO2 country code of the account address country                                 |                            |            |
| Address ZIP                 | String                                                                                           | 8-character alphanumeric   |            |
|                             | The ZIP/postal code of the account address.                                                      |                            |            |
| Ship Address Street 1       | String                                                                                           | 32-character alphanumeric  |            |
|                             | The first portion of the street address component of the shipping address                        |                            |            |
| Ship Address Street 2       | String                                                                                           | 32-character alphanumeric  |            |
|                             | The second portion of the street address component of the shipping address                       |                            |            |
| Ship Address City           | String                                                                                           | 50-character alphanumeric  |            |
|                             | The city component of the shipping address                                                       |                            |            |
| Ship Address State/Province | String                                                                                           | 64-character alphanumeric  |            |
|                             | The state/province component of the shipping address                                             |                            |            |
| Ship Address Country        | String                                                                                           | 2-character alphanumeric   |            |
|                             | The 2 character ISO2 country code of the shipping address country                                |                            |            |
| Ship Address ZIP            | String                                                                                           | 8-character alphanumeric   |            |
|                             | The ZIP/postal code of the shipping address                                                      |                            |            |
| CC Number Hash              | String                                                                                           | 255-character alphanumeric |            |
|                             | This is a SHA-2 hash (in hexadecimal format) of the credit card number.                          |                            |            |
| Custom Attribute 1-8        | String                                                                                           | 255-character alphanumeric |            |
|                             | These 8 attributes can be used to pass custom attribute data which can be used within the rules. |                            |            |

## Sample Session Query - CA

```
<?php
require "../..//mpgClasses.php";
/***** Request Variables *****/
$store_id='moneris';
$api_token='hurgle';
/***** Transactional Variables *****/
$type='session_query';
$order_id='risktest-' . date("dmy-G:i:s");
$session_id='abc123';
$service_type='session';
//$event_type='login';
/***** SessionAccountInfo Variables *****/
$policy = '';
$device_id = '4EC40DE5-0770-4fa0-BE53-981C067C598D';
$account_login = '13195417-8CA0-46cd-960D-14C158E4DBB2';
$password_hash = '489c830f10f7c601d30599a0deaf66e64d2aa50a';
$account_number = '3E17A905-AC8A-4c8d-A417-3DADA2A55220';
$account_name = '4590FCC0-DF4A-44d9-A57B-AF9DE98B84DD';
$account_email = '3CAE72EF-6B69-4a25-93FE-2674735E78E8@test.threatmetrix.com';
$account_telephone = '5556667777';
$pan = '4242424242424242';
$account_address_street1 = '3300 Bloor St W';
$account_address_street2 = '4th Flr West Tower';
$account_address_city = 'Toronto';
$account_address_state = 'Ontario';
$account_address_country = 'CA';
$account_address_zip = 'M8X2X2';
$shipping_address_street1 = '3300 Bloor St W';
$shipping_address_street2 = '4th Flr West Tower';
$shipping_address_city = 'Toronto';
$shipping_address_state = 'Ontario';
$shipping_address_country = 'CA';
$shipping_address_zip = 'M8X2X2';
$local_attrib_1 = 'a';
$local_attrib_2 = 'b';
$local_attrib_3 = 'c';
$local_attrib_4 = 'd';
$local_attrib_5 = 'e';
$online_tld = 'Facebook';
$online_id_handle = 'Moneris';
$transaction_amount = '1.00';
$transaction_currency = '124';
/***** SessionAccountInfo Associative Array *****/
$sessionAccountInfoTemplate = array
(
    'account_login'=>$account_login,
    'password_hash' =>$password_hash,
    'account_number' => $account_number,
    'account_name' => $account_name,
    'account_email'=>$account_email,
    'pan' =>$pan
);
/***** SessionAccountInfo Object *****/
$mpgSessionAccountInfo = new mpgSessionAccountInfo ($sessionAccountInfoTemplate);
/***** Transactional Associative Array *****/
$txnArray=array(
    'type'=>$type,
    'order_id'=>$order_id,
    'session_id'=>$session_id,
```

## Sample Session Query - CA

```

'service_type'=>$service_type
);
/***** Transaction Object *****/
$riskTxn = new riskTransaction($txnArray);
/***** Set SessionAccountInfo *****/
$riskTxn->setSessionAccountInfo($mpgSessionAccountInfo);
/***** Request Object *****/
$riskRequest = new riskRequest($riskTxn);
$riskRequest->setTestMode(true);
/***** HTTPS Post Object *****/
$riskHttpPost = new riskHttpPost($store_id,$api_token,$riskRequest);
/***** Response *****/
$riskResponse=$riskHttpPost->getRiskResponse();
//print("\nResponse = " . $riskResponse);
print("\nResponseCode = " . $riskResponse->getResponseCode());
print("\nMessage = " . $riskResponse->getMessage());
$results = $riskResponse->getResults();
foreach($results as $key => $value)
{
print("\n".$key ." = ". $value);
}
$rules = $riskResponse->getRules();
//print_r($rules);
foreach ($rules as $i)
{
foreach ($i as $key => $value)
{
echo "\n$key = $value";
}
}
}
?>

```

## 10.3.3.1 Session Query Transaction Flow

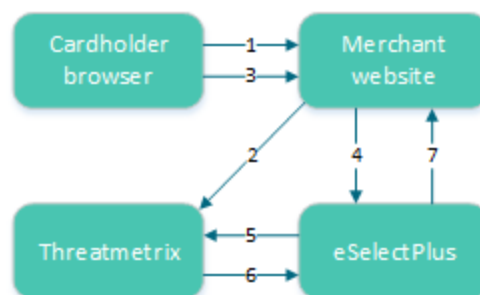


Figure 3: Session Query transaction flow

1. Cardholder logs onto the merchant website.
2. When the page has loaded in the cardholder's browser, special tags within the site allow information from the device to be gathered and sent to ThreatMetrix as the device fingerprint.  
The HTML tags should be placed where the cardholder is resident on the page for a couple of seconds to get the broadest data possible.
3. Customer submits a transaction.

4. Merchant's web application makes a Session Query transaction to the Moneris Gateway using the same session id that was included in the device fingerprint. This call must be made within 30 minutes of profiling (2).
5. Moneris Gateway submits the Session Query data to ThreatMetrix.
6. ThreatMetrix uses the Session Query data and the device fingerprint information to assess the transaction against the rules. A score is generated based on the rules.
7. The merchant uses the returned device information in its risk analysis to make a business decision. The merchant may wish to continue or cancel with the cardholder's payment transaction.

### 10.3.4 Attribute Query

The Attribute Query is used to obtain a risk assessment of transaction-related identifiers such as the email address and the card number. Unlike the Session Query, the Attribute Query does not require the device fingerprinting information to be provided.

#### AttributeQuery transaction object definition

```
$riskTxn = new riskTransaction($txnArray);
```

#### HttpPostRequest object for AttributeQuery transaction

```
$riskHttpPost = new riskHttpPost($store_id, $api_token, $riskRequest);
```

#### Attribute Query transaction values

**Table 5: Attribute Query transaction object mandatory values**

| Value              | Type                                                                                                                                                                                                                                                          |                                             | Limits | Set method                     |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|--------|--------------------------------|
|                    | Description                                                                                                                                                                                                                                                   |                                             |        |                                |
| Service type       | String                                                                                                                                                                                                                                                        | N/A                                         |        | 'service_type'=>\$service_type |
|                    | Which output fields are returned.<br>session -- returns IP and device related attributes.                                                                                                                                                                     |                                             |        |                                |
| Device ID          | String                                                                                                                                                                                                                                                        | 36-character alphanumeric                   |        | 'device_id'=>\$device_id       |
|                    | Unique device identifier generated by a previous call to the ThreatMetrix session-query API.                                                                                                                                                                  |                                             |        |                                |
| Credit card number | String                                                                                                                                                                                                                                                        | 20-character numeric<br>No spaces or dashes |        | 'pan'=>\$pan                   |
|                    | Most credit card numbers today are 16 digits, but some 13-digit numbers are still accepted by some issuers. This field has been intentionally expanded to 20 digits in consideration for future expansion and potential support of private label card ranges. |                                             |        |                                |

**Table 5: Attribute Query transaction object mandatory values (continued)**

| Value                             | Type                                                                                                                                                                                                                                                                                                 | Limits                    | Set method                                             |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------------------------------------|
|                                   | Description                                                                                                                                                                                                                                                                                          |                           |                                                        |
| IP address                        | String                                                                                                                                                                                                                                                                                               | 64-character alphanumeric | 'ip_address'=>\$ip_address                             |
|                                   | True IP address. Results will be returned as true_ip_geo, true_ip_score and so on.                                                                                                                                                                                                                   |                           |                                                        |
| IP forwarded                      | String                                                                                                                                                                                                                                                                                               | 64-character alphanumeric | 'ip_forwarded'=>\$ip_forwarded                         |
|                                   | <p>The IP address of the proxy. If the IPAddress is supplied, results will be returned as proxy_ip_geo and proxy_ip_score.</p> <p>If the IP Address is not supplied, this IP address will be treated as the true IP address and results will be returned as true_ip_geo, true_ip_score and so on</p> |                           |                                                        |
| Account address street 1          | String                                                                                                                                                                                                                                                                                               | 32-character alphanumeric | 'account_address_street1'=>\$account_address_street1   |
|                                   | First portion of the street address component of the billing address.                                                                                                                                                                                                                                |                           |                                                        |
| Account Address Street 2          | String                                                                                                                                                                                                                                                                                               | 32-character alphanumeric | 'account_address_street2'=>\$account_address_street2   |
|                                   | Second portion of the street address component of the billing address.                                                                                                                                                                                                                               |                           |                                                        |
| Account address city              | String                                                                                                                                                                                                                                                                                               | 50-character alphanumeric | 'account_address_city'=>\$account_address_city         |
|                                   | The city component of the billing address.                                                                                                                                                                                                                                                           |                           |                                                        |
| Account address state/- province  | String                                                                                                                                                                                                                                                                                               | 64-character alphanumeric | 'account_address_state'=>\$account_address_state       |
|                                   | The state component of the billing address.                                                                                                                                                                                                                                                          |                           |                                                        |
| Account address country           | String                                                                                                                                                                                                                                                                                               | 2-character alphanumeric  | 'account_address_country'=>\$account_address_country   |
|                                   | ISO2 country code of the billing addresses.                                                                                                                                                                                                                                                          |                           |                                                        |
| Account address zip/- postal code | String                                                                                                                                                                                                                                                                                               | 8-character alphanumeric  | 'account_address_zip'=>\$account_address_zip           |
|                                   | Zip/postal code of the billing address.                                                                                                                                                                                                                                                              |                           |                                                        |
| Shipping address street 1         | String                                                                                                                                                                                                                                                                                               | 32-character alphanumeric | 'shipping_address_street1'=>\$shipping_address_street1 |
|                                   | Account address country                                                                                                                                                                                                                                                                              |                           |                                                        |

**Table 5: Attribute Query transaction object mandatory values (continued)**

| Value                             | Type                                                                    | Limits                    | Set method                                             |
|-----------------------------------|-------------------------------------------------------------------------|---------------------------|--------------------------------------------------------|
|                                   | Description                                                             |                           |                                                        |
| Shipping Address Street 2         | String                                                                  | 32-character alphanumeric | 'shipping_address_street2'=>\$shipping_address_street2 |
|                                   | Second portion of the street address component of the shipping address. |                           |                                                        |
| Shipping Address City             | String                                                                  | 50-character alphanumeric | 'shipping_address_city'=>\$shipping_address_city       |
|                                   | City component of the shipping address.                                 |                           |                                                        |
| Shipping Address State/Province   | String                                                                  | 64-character alphanumeric | 'shipping_address_state'=>\$shipping_address_state     |
|                                   | State/Province component of the shipping address.                       |                           |                                                        |
| Shipping Address Country          | String                                                                  | 2-character alphanumeric  | 'shipping_address_country'=>\$shipping_address_country |
|                                   | ISO2 country code of the account address country.                       |                           |                                                        |
| Shipping Address zip/-postal code | String                                                                  | 8-character alphanumeric  | 'shipping_address_zip'=>\$shipping_address_zip         |
|                                   | The zip/postal code component of the shipping address.                  |                           |                                                        |

### Sample Attribute Query

```
<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='moneris';
$sapi_token='hurgle';
/***** Transactional Variables *****/
$type='session_query';
$order_id='risktest-'.date("dmy-G:i:s");
$session_id='abc123';
$service_type='session';
//$event_type='login';
/***** SessionAccountInfo Variables *****/
$policy = '';
$device_id = '4EC40DE5-0770-4fa0-BE53-981C067C598D';
$account_login = '13195417-8CA0-46cd-960D-14C158E4DBB2';
$password_hash = '489c830f10f7c601d30599a0deaf66e64d2aa50a';
$account_number = '3E17A905-AC8A-4c8d-A417-3DADA2A55220';
$account_name = '4590FCC0-DF4A-44d9-A57B-AF9DE98B84DD';
$account_email = '3CAE72EF-6B69-4a25-93FE-2674735E78E8@test.threatmetrix.com';
$account_telephone = '5556667777';
$pan = '4242424242424242';
$account_address_street1 = '3300 Bloor St W';
$account_address_street2 = '4th Flr West Tower';
```

## Sample Attribute Query

```

$account_address_city = 'Toronto';
$account_address_state = 'Ontario';
$account_address_country = 'CA';
$account_address_zip = 'M8X2X2';
$shipping_address_street1 = '3300 Bloor St W';
$shipping_address_street2 = '4th Flr West Tower';
$shipping_address_city = 'Toronto';
$shipping_address_state = 'Ontario';
$shipping_address_country = 'CA';
$shipping_address_zip = 'M8X2X2';
$local_attr1 = 'a';
$local_attr2 = 'b';
$local_attr3 = 'c';
$local_attr4 = 'd';
$local_attr5 = 'e';
$online_tld = 'Facebook';
$online_id_handle = 'Moneris';
$transaction_amount = '1.00';
$transaction_currency = '124';
/***** SessionAccountInfo Associative Array *****/
$sessionAccountInfoTemplate = array
(
    'account_login'=>$account_login,
    'password_hash' =>$password_hash,
    'account_number' => $account_number,
    'account_name' => $account_name,
    'account_email'=>$account_email,
    'pan' =>$pan
);
/***** SessionAccountInfo Object *****/
$mpgSessionAccountInfo = new mpgSessionAccountInfo ($sessionAccountInfoTemplate);
/***** Transactional Associative Array *****/
$txnArray=array(
    'type'=>$type,
    'order_id'=>$order_id,
    'session_id'=>$session_id,
    'service_type'=>$service_type
);
/***** Transaction Object *****/
$riskTxn = new riskTransaction($txnArray);
/***** Set SessionAccountInfo *****/
$riskTxn->setSessionAccountInfo($mpgSessionAccountInfo);
/***** Request Object *****/
$riskRequest = new riskRequest($riskTxn);
$riskRequest->setTestMode(true);
/***** HTTPS Post Object *****/
$riskHttpPost =new riskHttpPost($store_id,$api_token,$riskRequest);
/***** Response *****/
$riskResponse=$riskHttpPost->getRiskResponse();
//print("\nResponse = " . $riskResponse);
print("\nResponseCode = " . $riskResponse->getResponseCode());
print("\nMessage = " . $riskResponse->getMessage());
$results = $riskResponse->getResults();
foreach($results as $key => $value)
{
    print("\n".$key ." = ". $value);
}
$rules = $riskResponse->getRules();
//print_r($rules);

```



| Sample Attribute Query                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------|
| <pre>foreach (\$rules as \$i) {   foreach (\$i as \$key =&gt; \$value)   {     echo "\n\$key = \$value";   } } ?&gt;</pre> |

10.3.4.1 Attribute Query Transaction Flow

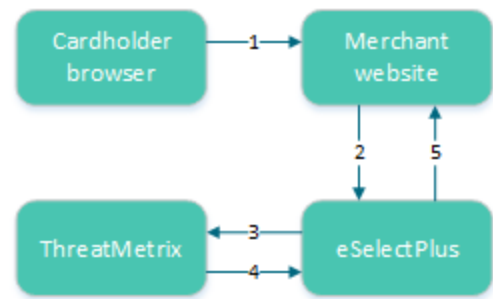


Figure 4: Attribute query transaction flow

1. Cardholder logs onto merchant website and submits a transaction.
2. The merchant’s web application makes an Attribute Query transaction that includes the session ID to the Moneris Gateway.
3. Moneris Gateway submits Attribute Query data to ThreatMetrix.
4. ThreatMetrix uses the Attribute Query data to assess the transaction against the rules. A score is generated based on the rules.
5. The merchant uses the returned device information in its risk analysis to make a business decision. The merchant may wish to continue or cancel with the cardholder's payment transaction.

10.3.5 Handling Response Information

When reviewing the response information and determining how to handle the transaction, it is recommended that you (either manually or through automated logic on your site) use the following pieces of information:

- Risk score
- Rules triggered (such as Rule Codes, Rule Names, Rule Messages)
- Results obtained from Verified by Visa, MasterCard Secure Code, AVS, CVD and the financial transaction authorization

- Response codes for the Transaction Risk Management Transaction that are included by automated processes.

### 10.3.5.1 TRMT Response Fields

**Table 6: Receipt object response values for TRMT**

| Value         | Type                                                                                                                                                                                                                                                                   | Limits                   | Get method                                          |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-----------------------------------------------------|
|               | Definition                                                                                                                                                                                                                                                             |                          |                                                     |
| Response Code | String                                                                                                                                                                                                                                                                 | 3-character alphanumeric | <code>\$mpgResponse-&gt;getResponseCode () ;</code> |
|               | 001 – Success<br>981 – Data error<br>982 – Duplicate Order ID<br>983 – Invalid Transaction<br>984 – Previously asserted<br>985 – Invalid activity description<br>986- Invalid impact description<br>987 – Invalid Confidence description<br>988 - Cannot find Previous |                          |                                                     |
| Message       | String                                                                                                                                                                                                                                                                 | N/A                      | <code>\$mpgResponse-&gt;getMessage () ;</code>      |
|               | Response message                                                                                                                                                                                                                                                       |                          |                                                     |
| Event type    | String                                                                                                                                                                                                                                                                 | N/A                      |                                                     |
|               | Type of transaction or event returned in the response.                                                                                                                                                                                                                 |                          |                                                     |
| Org ID        | String                                                                                                                                                                                                                                                                 | N/A                      |                                                     |
|               | ThreatMetrix-defined unique transaction identifier                                                                                                                                                                                                                     |                          |                                                     |
| Policy        | String                                                                                                                                                                                                                                                                 | N/A                      |                                                     |
|               | Policy used for the Session Query will be returned with the return request. If the Policy was not included, then the Policy name default is returned.                                                                                                                  |                          |                                                     |

**Table 6: Receipt object response values for TRMT (continued)**

| Value              | Type                                                                                                       | Limits | Get method |
|--------------------|------------------------------------------------------------------------------------------------------------|--------|------------|
|                    | Definition                                                                                                 |        |            |
| Policy score       | String                                                                                                     | N/A    |            |
|                    | The sum of all the risks weights from triggered rules within the selected policy in the range [-100...100] |        |            |
| Request duration   | String                                                                                                     | N/A    |            |
|                    | Length of time it takes for the transaction to be processed.                                               |        |            |
| Request ID         | String                                                                                                     | N/A    |            |
|                    | Unique number and will always be returned with the return request.                                         |        |            |
| Request result     | String                                                                                                     | N/A    |            |
|                    | See 10.3.5.1 (page 430).                                                                                   |        |            |
| Review status      | String                                                                                                     | N/A    |            |
|                    | The transaction status based on the assessments and risk scores.                                           |        |            |
| Risk rating        | String                                                                                                     | N/A    |            |
|                    | The rating based on the assessments and risk scores.                                                       |        |            |
| Service type       | String                                                                                                     | N/A    |            |
|                    | The service type will be returned in the attribute query response.                                         |        |            |
| Session ID         | String                                                                                                     | N/A    |            |
|                    | Temporary identifier unique to the visitor will be returned in the return request.                         |        |            |
| Summary risk score | String                                                                                                     | N/A    |            |
|                    | Based on all of the returned values in the range [-100 ... 100]                                            |        |            |
| Transaction ID     | String                                                                                                     | N/A    |            |
|                    | This is the transaction identifier and will always be returned in the response when supplied as input.     |        |            |
| Unknown session    | String                                                                                                     | N/A    |            |
|                    | If present, the value is "yes". It indicates the session ID that was passed was not found.                 |        |            |

**Table 7: Response code descriptions**

| Value | Definition |
|-------|------------|
| 001   | Success    |

| Value | Definition                     |
|-------|--------------------------------|
| 981   | Data error                     |
| 982   | Duplicate order ID             |
| 983   | Invalid transaction            |
| 984   | Previously asserted            |
| 985   | Invalid activity description   |
| 986   | Invalid impact description     |
| 987   | Invalid confidence description |
| 988   | Cannot find previous           |

**Table 8: Request result values and descriptions**

| Value                                | Definition                                                                               |
|--------------------------------------|------------------------------------------------------------------------------------------|
| fail_duplicate_entities_of_same_type | More than one entity of the same was specified, e.g. password_hash was specified twice.  |
| fail_incomplete                      | ThreatMetrix was unable to process the request due to incomplete or incorrect input data |
| fail_invalid_account_number          | The format of the supplied account number was invalid                                    |
| fail_invalid_characters              | Invalid characters submitted                                                             |
| fail_invalid_charset                 | The value of character set was invalid                                                   |
| fail_invalid_currency_code           | The format of the currency_code was invalid                                              |
| fail_invalid_currency_format         | The format of the currency_format was invalid                                            |
| fail_invalid_telephone_number        | Format of the supplied telephone number was invalid                                      |
| fail_access                          | ThreatMetrix was unable to process the request because of API verification failing       |
| fail_internal_error                  | ThreatMetrix encountered an error while processing the request                           |
| fail_invalid_device_id               | Format of the supplied device_id was invalid                                             |

| Value                                     | Definition                                                                                     |
|-------------------------------------------|------------------------------------------------------------------------------------------------|
| fail_invalid_email_address                | Format of the supplied email address was invalid                                               |
| fail_invalid_fuzzy_device_id              | The format of fuzzy_device_id was invalid                                                      |
| fail_invalid_ip_address_parameter         | Format of a supplied ip_address parameter was invalid                                          |
| fail_invalid_parameter                    | The format of the parameter was invalid, or the value is out of boundary                       |
| fail_invalid_sha_hash                     | The format of a parameter specified as a sha hash was invalid, sha hash included sha1/2/3 hash |
| fail_invalid_submitter_id                 | The format of the submitter id was invalid or the value is out of boundary                     |
| fail_no_policy_configured                 | No policy was configured against the org_id                                                    |
| fail_not_enough_params                    | Not enough device attributes were collected during profiling to perform a fingerprint match    |
| fail_parameter_overlength                 | The value of the parameter was overlength                                                      |
| fail_temporarily_unavailable              | Request failed because the service is temporarily unavailable                                  |
| fail_too_many_instances_of_same_parameter | Multiple values for some parameters which only allow one instance                              |
| fail_verification                         | API query limit reached                                                                        |
| success                                   | ThreatMetrix was able to process the request successfully                                      |

### 10.3.5.2 Understanding the Risk Score

For each Session Query or Attribute Query, a score with a value between -100 and +100 is returned based on the rules that were triggered for the transaction.

Table 9 defines the risk scores ranges.

**Table 9: Session Query and Attribute Query risk score definitions**

| Risk score | Visa definition                                                                                                                                                                                                                                            |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -100 to -1 | A lower score indicates a higher probability that the transaction is fraudulent.                                                                                                                                                                           |
| 0          | Neutral transaction                                                                                                                                                                                                                                        |
| 1 to 100   | <p>A higher score indicates a lower probability that the transaction is fraudulent.</p> <p><b>Note:</b> All e-commerce transactions have some level of risk associated with them. Therefore, it is rare to see risk score in the high positive values.</p> |

When evaluating the risk of a transaction, the risk score gives an initial indicator of the potential risk that the transaction is fraudulent. Because some of the rules that are evaluated on each transaction may not be relevant to your business scenario, review the rules that were triggered for the transaction before determining how to handle the transaction.

### 10.3.5.3 Understanding the Rule Codes, Rule Names and Rule Messages

The rule codes, rule names and rule messages provide details about what rules were triggered during the assessment of the information provided in the Session or Attribute Query. Each rule code has a rule name and rule message. The rule name and rule message are typically similar. Table 10 provides additional information on each rule.

When evaluating the risk of a transaction, it is recommended that you review the rules that were triggered for the transaction and assess the relevance to your business. (That is, how does it relate to the typical buying habits of your customer base?)

If you are automating some or all of the decision-making processes related to handling the responses, you may want to use the rule codes. If you are documenting manual processes, you may want to refer to the more user-friendly rule name or rule message.

**Table 10: Rule names, numbers and messages**

| Rule name         | Rule number                                                                                                                                     | Rule message        |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
|                   | Rule explanation                                                                                                                                |                     |
| White lists       |                                                                                                                                                 |                     |
| DeviceWhitelisted | WL001                                                                                                                                           | Device White Listed |
|                   | Device is on the white list. This indicates that the device has been flagged as always "ok".<br><b>Note:</b> This rule is currently not in use. |                     |

Table 10: Rule names, numbers and messages (continued)

| Rule name             | Rule number                                                                                                                                              |                                         | Rule message |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|--------------|
|                       | Rule explanation                                                                                                                                         |                                         |              |
| IPWhitelisted         | WL002                                                                                                                                                    | IP White Listed                         |              |
|                       | IP address is on the white list. This indicates the device has been flagged as always "ok".                                                              |                                         |              |
|                       | <b>Note:</b> This rule is currently not in use.                                                                                                          |                                         |              |
| EmailWhitelisted      | WL003                                                                                                                                                    | Email White Listed                      |              |
|                       | Email address is on the white list. This indicates that the device has been flagged as always "ok".                                                      |                                         |              |
|                       | <b>Note:</b> This rule is currently not in use.                                                                                                          |                                         |              |
| Event velocity        |                                                                                                                                                          |                                         |              |
| 2DevicePayment        | EV003                                                                                                                                                    | 2 Device Payment Velocity               |              |
|                       | Multiple payments were detected from this device in the past 24 hours.                                                                                   |                                         |              |
| 2IPPaymentVelocity    | EV006                                                                                                                                                    | 2 IP Payment Velocity                   |              |
|                       | Multiple payments were detected from this IP within the past 24 hours.                                                                                   |                                         |              |
| 2ProxyPaymentVelocity | EV008                                                                                                                                                    | 2 Proxy Payment Velocity                |              |
|                       | The device has used 3 or more different proxies during a 24 hour period. This could be a risk or it could be someone using a legitimate corporate proxy. |                                         |              |
| Email                 |                                                                                                                                                          |                                         |              |
| 3EmailPerDeviceDay    | EM001                                                                                                                                                    | 3 Emails for the Device ID in 1 Day     |              |
|                       | This device has presented 3 different email IDs within the past 24 hours.                                                                                |                                         |              |
| 3EmailPerDeviceWeek   | EM002                                                                                                                                                    | 3 emails for the Device ID in 1 week    |              |
|                       | This device has presented 3 different email IDs within the past week.                                                                                    |                                         |              |
| 3DevciePerEmailDay    | EM003                                                                                                                                                    | 3 Device Ids for email address in 1 day |              |
|                       | This email has been presented from three different devices in the past 24 hours.                                                                         |                                         |              |

Table 10: Rule names, numbers and messages (continued)

| Rule name                     | Rule number                      Rule message                                                                                                                                                                                  |                                          |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
|                               | Rule explanation                                                                                                                                                                                                               |                                          |
| 3DevciePerEmailWeek           | EM004                                                                                                                                                                                                                          | 3 Device Ids for email address in 1 week |
|                               | This email has been presented from three different devices in the past week.                                                                                                                                                   |                                          |
| EmailDistanceTravelled        | EM005                                                                                                                                                                                                                          | Email Distance Travelled                 |
|                               | This email address has been associated with different physical locations in a short period of time.                                                                                                                            |                                          |
| 3EmailPerSmartIDHour          | EM006                                                                                                                                                                                                                          | 3 Emails for SmartID in 1 Hour           |
|                               | The SmartID for this device has been associated with 3 different email addresses in 1 hour.                                                                                                                                    |                                          |
| GlobalEMailOverOneMonth       | EM007                                                                                                                                                                                                                          | Global Email over 1 month                |
|                               | The e-mail address involved in the transaction over 30 days ago. This generally indicates that the transaction is less risky.<br><br><b>Note:</b> This rule is set so that it does not impact the policy score or risk rating. |                                          |
| ComputerGeneratedEmailAddress | EM008                                                                                                                                                                                                                          | Computer Generated Email Address         |
|                               | This transaction used a computer-generated email address.                                                                                                                                                                      |                                          |
| Account Number                |                                                                                                                                                                                                                                |                                          |
| 3AccountNumberPerDeviceDay    | AN001                                                                                                                                                                                                                          | 3 Account Numbers for device in 1 day    |
|                               | This device has presented 3 different user accounts within the past 24 hours.                                                                                                                                                  |                                          |
| 3AccountNumberPerDeviceWeek   | AN002                                                                                                                                                                                                                          | 3 Account Numbers for device in 1 week   |
|                               | This device has presented 3 different user accounts within the past week.                                                                                                                                                      |                                          |
| 3DevciePerAccountNumberDay    | AN003                                                                                                                                                                                                                          | 3 Device IDs for account number in 1 day |
|                               | This user account been used from three different devices in the past 24 hours.                                                                                                                                                 |                                          |



**Table 10: Rule names, numbers and messages (continued)**

| Rule name                        | Rule number                                                                                               |                                              | Rule message |  |
|----------------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------|--------------|--|
|                                  | Rule explanation                                                                                          |                                              |              |  |
| 3DevciePerAccountNumberWeek      | AN004                                                                                                     | 3 Device IDs for account number in 1 week    |              |  |
|                                  | This card number has been used from three different devices in the past week.                             |                                              |              |  |
| AccountNumberDistanceTravelled   | AN005                                                                                                     | Account Number distance travelled            |              |  |
|                                  | This card number has been used from a number of physically different locations in a short period of time. |                                              |              |  |
| Credit card/payments             |                                                                                                           |                                              |              |  |
| 3CreditCardPerDeviceDay          | CP001                                                                                                     | 3 credit cards for device in 1 day           |              |  |
|                                  | This device has used three credit cards within 24 hours.                                                  |                                              |              |  |
| 3CreditCardPerDeviceWeek         | CP002                                                                                                     | 3 credit cards for device in 1 week          |              |  |
|                                  | This device has used three credit cards within 1 week.                                                    |                                              |              |  |
| 3DevicePerCreditCardDay          | CP003                                                                                                     | 3 device ids for credit card in 1 day        |              |  |
|                                  | This credit card has been used on three different devices in 24 hours.                                    |                                              |              |  |
| 3DevciePerCreditCardWeek         | CP004                                                                                                     | 3 device ids for credit card in 1 week       |              |  |
|                                  | This credit card has been used on three different devices in 1 week.                                      |                                              |              |  |
| CredtCardDistanceTravelled       | CP005                                                                                                     | Credit Card has travelled                    |              |  |
|                                  | The credit card has been used at a number of physically different locations in a short period of time.    |                                              |              |  |
| CreditCardShipAddressGeoMismatch | CP006                                                                                                     | Credit Card and Ship Address do not match    |              |  |
|                                  | The credit card was issued in a region different from the Ship To Address information provided.           |                                              |              |  |
| CreditCardBillAddressGeoMismatch | CP007                                                                                                     | Credit Card and Billing Address do not match |              |  |
|                                  | The credit card was issued in a region different from the Billing Address information provided.           |                                              |              |  |
| CreditCardDeviceGeoMismatch      | CP008                                                                                                     | Credit Card and device location do not match |              |  |
|                                  | The device is located in a region different from where the card was issued.                               |                                              |              |  |

**Table 10: Rule names, numbers and messages (continued)**

| Rule name                           | Rule number                                                                                     |                                                                      | Rule message |  |
|-------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|--------------|--|
|                                     | Rule explanation                                                                                |                                                                      |              |  |
| CreditCardBINShipAddressGeoMismatch | CP009                                                                                           | Credit Card issuing location and Shipping address do not match       |              |  |
|                                     | The credit card was issued in a region different from the Ship To Address information provided. |                                                                      |              |  |
| CreditCardBINBillAddressGeoMismatch | CP010                                                                                           | Credit Card issuing location and Billing address do not match        |              |  |
|                                     | The credit card was issued in a region different from the Billing Address information provided. |                                                                      |              |  |
| CreditCardBINDeviceGeoMismatch      | CP011                                                                                           | Credit Card issuing location and location of the device do not match |              |  |
|                                     | The device is located in a region different from where the card was issued.                     |                                                                      |              |  |
| TransactionValueDay                 | CP012                                                                                           | Daily Transaction Value Threshold                                    |              |  |
|                                     | The transaction value exceeds the daily threshold.                                              |                                                                      |              |  |
| TransactionValueWeek                | CP013                                                                                           | Weekly Transaction Value Threshold                                   |              |  |
|                                     | The transaction value exceeds the weekly threshold.                                             |                                                                      |              |  |
| Proxy rules                         |                                                                                                 |                                                                      |              |  |
| 3ProxyPerDeviceDay                  | PX001                                                                                           | 3 Proxy Ips in 1 day                                                 |              |  |
|                                     | This device has used three different proxy servers in the past 24 hours.                        |                                                                      |              |  |
| AnonymousProxy                      | PX002                                                                                           | Anonymous Proxy IP                                                   |              |  |
|                                     | This device is using an anonymous proxy                                                         |                                                                      |              |  |
| UnusualProxyAttributes              | PX003                                                                                           | Unusual Proxy Attributes                                             |              |  |
|                                     | This transaction is coming from a source with unusual proxy attributes.                         |                                                                      |              |  |
| AnonymousProxy                      | PX004                                                                                           | Anonymous Proxy                                                      |              |  |
|                                     | This device is connecting through an anonymous proxy connection.                                |                                                                      |              |  |
| HiddenProxy                         | PX005                                                                                           | Hidden Proxy                                                         |              |  |
|                                     | This device is connecting via a hidden proxy server.                                            |                                                                      |              |  |

**Table 10: Rule names, numbers and messages (continued)**

| Rule name                     | Rule number                                                                                            |                                              | Rule message |
|-------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------|--------------|
|                               | Rule explanation                                                                                       |                                              |              |
| OpenProxy                     | PX006                                                                                                  | Open Proxy                                   |              |
|                               | This transaction is coming from a source that is using an open proxy.                                  |                                              |              |
| TransparentProxy              | PX007                                                                                                  | Transparent Proxy                            |              |
|                               | This transaction is coming from a source that is using a transparent proxy.                            |                                              |              |
| DeviceProxyGeoMismatch        | PX008                                                                                                  | Proxy and True GEO Match                     |              |
|                               | This device is connecting through a proxy server that didn't match the devices geo-location.           |                                              |              |
| ProxyTrueISPMismatch          | PX009                                                                                                  | Proxy and True ISP Match                     |              |
|                               | This device is connecting through a proxy server that doesn't match the true IP address of the device. |                                              |              |
| ProxyTrueOrganizationMismatch | PX010                                                                                                  | Proxy and True Org Match                     |              |
|                               | The Proxy information and True ISP information for this source do not match.                           |                                              |              |
| DeviceProxyRegionMismatch     | PX011                                                                                                  | Proxy and True Region Match                  |              |
|                               | The proxy and device region location information do not match.                                         |                                              |              |
| ProxyNegativeReputation       | PX012                                                                                                  | Proxy IP Flagged Risky in Reputation Network |              |
|                               | This device is connecting from a proxy server with a known negative reputation.                        |                                              |              |
| SatelliteProxyISP             | PX013                                                                                                  | Satellite Proxy                              |              |
|                               | This transaction is coming from a source that is using a satellite proxy.                              |                                              |              |
| GEO                           |                                                                                                        |                                              |              |
| DeviceCountriesNotAllowed     | GE001                                                                                                  | True GEO in Countries Not Allowed blacklist  |              |
|                               | This device is connecting from a high-risk geographic location.                                        |                                              |              |

Table 10: Rule names, numbers and messages (continued)

| Rule name                 | Rule number                                                                                                          |                                                        | Rule message |
|---------------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|--------------|
|                           | Rule explanation                                                                                                     |                                                        |              |
| DeviceCountriesNotAllowed | GE002                                                                                                                | True GEO in Countries Not Allowed (negative whitelist) |              |
|                           | The device is from a region that is not on the whitelist of regions that are accepted.                               |                                                        |              |
| DeviceProxyGeoMismatch    | GE003                                                                                                                | True GEO different from Proxy GEO                      |              |
|                           | The true geographical location of this device is different from the proxy geographical location.                     |                                                        |              |
| DeviceAccountGeoMismatch  | GE004                                                                                                                | Account Address different from True GEO                |              |
|                           | This device has presented an account billing address that doesn't match the devices geolocation.                     |                                                        |              |
| DeviceShipGeoMismatch     | GE005                                                                                                                | Device and Ship Geo mismatch                           |              |
|                           | The location of the device and the shipping address do not match.                                                    |                                                        |              |
| DeviceShipGeoMismatch     | GE006                                                                                                                | Device and Ship Geo mismatch                           |              |
|                           | The location of the device and the shipping address do not match.                                                    |                                                        |              |
| Device                    |                                                                                                                      |                                                        |              |
| SatelliteISP              | DV001                                                                                                                | Satellite ISP                                          |              |
|                           | This transaction is from a source that is using a satellite ISP.                                                     |                                                        |              |
| MidsessionChange          | DV002                                                                                                                | Session Changed Mid-session                            |              |
|                           | This device changed session details and identifiers in the middle of a session.                                      |                                                        |              |
| LanguageMismatch          | DV003                                                                                                                | Language Mismatch                                      |              |
|                           | The language of the user does not match the primary language spoken in the location where the True IP is registered. |                                                        |              |
| NoDeviceID                | DV004                                                                                                                | No Device ID                                           |              |
|                           | No device ID was available for this transaction.                                                                     |                                                        |              |
| Dial-upConnection         | DV005                                                                                                                | Dial-up connection                                     |              |
|                           | This device uses a less identifiable dial-up connection.                                                             |                                                        |              |

**Table 10: Rule names, numbers and messages (continued)**

| Rule name                  | Rule number                                                                          | Rule message                               |
|----------------------------|--------------------------------------------------------------------------------------|--------------------------------------------|
|                            | Rule explanation                                                                     |                                            |
| DeviceNegativeReputation   | DV006                                                                                | Device Blacklisted in Reputational Network |
|                            | This device has a known negative reputation as reported to the fraud network.        |                                            |
| DeviceGlobalBlacklist      | DV007                                                                                | Device on the Global Black List            |
|                            | This device has been flagged on the global blacklist of known problem devices.       |                                            |
| DeviceCompromisedDay       | DV008                                                                                | Device compromised in last day             |
|                            | This device has been reported as compromised in the last 24 hours.                   |                                            |
| DeviceCompromisedHour      | DV009                                                                                | Device compromised in last hour            |
|                            | This device has been reported as compromised in the last hour.                       |                                            |
| FlashImagesCookiesDisabled | DV010                                                                                | Flash Images Cookies Disabled              |
|                            | Key browser functions/identifiers have been disabled on this device.                 |                                            |
| FlashCookiesDisabled       | DV011                                                                                | Flash Cookies Disabled                     |
|                            | Key browser functions/identifiers have been disabled on this device.                 |                                            |
| FlashDisabled              | DV012                                                                                | Flash Disabled                             |
|                            | Key browser functions/identifiers have been disabled on this device.                 |                                            |
| ImagesDisabled             | DV013                                                                                | Images Disabled                            |
|                            | Key browser functions/identifiers have been disabled on this device.                 |                                            |
| CookiesDisabled            | DV014                                                                                | Cookies Disabled                           |
|                            | Key browser functions/identifiers have been disabled on this device.                 |                                            |
| DeviceDistanceTravelled    | DV015                                                                                | Device Distance Travelled                  |
|                            | The device has been used from multiple physical locations in a short period of time. |                                            |

**Table 10: Rule names, numbers and messages (continued)**

| Rule name             | Rule number                                                    | Rule message                    |
|-----------------------|----------------------------------------------------------------|---------------------------------|
|                       | Rule explanation                                               |                                 |
| PossibleCookieWiping  | DV016                                                          | Cookie Wiping                   |
|                       | This device appears to be deleting cookies after each session. |                                 |
| PossibleCookieCopying | DV017                                                          | Possible Cookie Copying         |
|                       | This device appears to be copying cookies.                     |                                 |
| PossibleVPNConnection | DV018                                                          | Possibly using a VPN Connection |
|                       | This device may be using a VPN connection                      |                                 |

### 10.3.5.4 Examples of Risk Response

#### Session Query

| Sample Risk Response - Session Query                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>&lt;?xml version="1.0"?&gt; &lt;response&gt; &lt;receipt&gt; &lt;ResponseCode&gt;001&lt;/ResponseCode&gt; &lt;Message&gt;Success&lt;/Message&gt; &lt;Result&gt; &lt;session_id&gt;abcl23&lt;/session_id&gt; &lt;unknown_session&gt;yes&lt;/unknown_session&gt; &lt;event_type&gt;payment&lt;/event_type&gt; &lt;service_type&gt;session&lt;/service_type&gt; &lt;policy_score&gt;-25&lt;/policy_score&gt; &lt;transaction_id&gt;riskcheck42&lt;/transaction_id&gt; &lt;org_id&gt;11kue096&lt;/org_id&gt; &lt;request_id&gt;91C1879B-33D4-4D72-8FCB-B60A172B3CAC&lt;/request_id&gt; &lt;risk_rating&gt;medium&lt;/risk_rating&gt; &lt;request_result&gt;success&lt;/request_result&gt; &lt;summary_risk_score&gt;-25&lt;/summary_risk_score&gt; &lt;Policy&gt;default&lt;/policy&gt; &lt;review_status&gt;review&lt;/review_status&gt; &lt;/Result&gt; &lt;Rule&gt; &lt;RuleName&gt;ComputerGeneratedEMail&lt;/RuleName&gt; &lt;RuleCode&gt;UN001&lt;/RuleCode&gt; &lt;RuleMessageEn&gt;Unknown Rule&lt;/RuleMessageEn&gt; &lt;RuleMessageFr&gt;Regle Inconnus&lt;/RuleMessageFr&gt; &lt;/Rule&gt; &lt;Rule&gt; &lt;RuleName&gt;NoDeviceID&lt;/RuleName&gt; &lt;RuleCode&gt;DV004&lt;/RuleCode&gt; &lt;RuleMessageEn&gt;No Device ID&lt;/RuleMessageEn&gt; &lt;RuleMessageFr&gt;null&lt;/RuleMessageFr&gt; &lt;/Rule&gt; &lt;/receipt&gt; &lt;/response&gt;</pre> |

#### Attribute Query

| Sample Risk Response - Attribute Query                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>&lt;?xml version="1.0"?&gt; &lt;response&gt; &lt;receipt&gt; &lt;ResponseCode&gt;001&lt;/ReponseCode&gt; &lt;Message = Success&lt;/Message&gt; &lt;Result&gt; &lt;org_id&gt;11kue096&lt;/org_id&gt; &lt;request_id&gt;443D7FB5-CC5C-4917-A57E-27EAC824069C&lt;/request_id&gt; &lt;service_type&gt;session&lt;/service_type&gt; &lt;risk_rating&gt;medium&lt;/risk_rating&gt; &lt;summary_risk_score&gt;-25&lt;/summary_risk_score&gt; &lt;request_result&gt;success&lt;/request_result&gt;</pre> |

**Sample Risk Response - Attribute Query**

```

<policy>default</policy>
<policy_score>-25</policy_score>
<transaction_id>riskcheck19</transaction_id>
<review_status>review</review_status>
</Result>
<Rule>
<RuleName>ComputerGeneratedEmail</RuleName>
<RuleCode>UN001</RuleCode>
<RuleMessageEn>Unknown Rule</RuleMessageEn>
<RuleMessageFr>Regle Inconnus</RuleMessageFr>
</Rule>
<Rule>
<RuleName>NoDeviceID</RuleName>
<RuleCode>DV004</RuleCode>
<RuleMessageEn>No Device ID</RuleMessageEn>
<RuleMessageFr>null</RuleMessageFr>
</Rule>
</receipt>
</response>

```

### 10.3.6 Inserting the Profiling Tags Into Your Website

Place the profiling tags on an HTML page served by your web application such that ThreatMetrix can collect device information from the customer's web browser. The tags must be placed on a page that a visitor would display in a browser window for 3-5 seconds (such as a page that requires a user to input data). After the device is profiled, a Session Query may be used to obtain the detail device information for risk assessment before submitting a financial payment transaction.

There are two profiling tags that require two variables. Those tags are `org_id` and `session_id`. `session_id` must match the session ID value that is to be passed in the Session Query transaction. The valid `org_id` values are:

**11kue096**

QA testing environment.

**lbhqgx47**

Production environment.

Below is an HTML sample of the profiling tags.

**NOTE:** Your site must replace `<my_session_id>` in the sample code with a unique alpha-numeric value each time you fingerprint a new customer.

```

<p style="background:url(https://h.online-metrix.net/fp/clear.png?org_id=11kue096&session_id=<my_session_id>&m=1)">
</p>

```

```



```



---

```
<script src="https://h.onlinemetrix.net/fp/check.js?org_id=11kue096&session_id=<my_session_id>"
type="text/javascript">
</script>

<object type="application/x-shockwave-flash"

data="https://h.onlinemetrix.net/fp/fp.swf?org_id=11kue096&session_id=<my_session_id>"
width="1" height="1" id="obj_id">
<param name="movie"
value="https://h.onlinemetrix.net/fp/fp.swf?org_id=11kue096&session_id=<my_session_id>" />
<div></div>
</object>
```

## 10.4 Incorporating All Available Fraud Tools

- 10.4.1 Implementation Options for TRMT
- 10.4.2 Implementation Checklist
- 10.4.3 Making a Decision

To minimize fraudulent activity in online transactions, Moneris recommends that you implement all of the fraud tools available through the Moneris Gateway. These are explained below:

### **Address Verification Service (AVS)**

Verifies the cardholder's billing address information.

### **Verified by Visa, MasterCard Secure Code and Amex SafeKey (VbV/MCSC/SafeKey)**

Authenticates the cardholder at the time of an online transaction.

### **Card Validation Digit (CVD)**

Validates that cardholder is in possession of a genuine credit card during the transaction.

Note that all responses that are returned from these verification methods are intended to provide added security and fraud prevention. The response itself does not affect the completion of a transaction. Upon receiving a response, the choice to proceed with a transaction is left entirely to the merchant.

### 10.4.1 Implementation Options for TRMT

#### **Option A**

Process a Transaction Risk Management Tool query and obtain the response. You can then decide whether to continue with the transaction, abort the transaction, or use additional eFraud features.

If you want to use additional eFraud features, perform one or both of the following to help make your decision about whether to continue with the transaction or abort it:

- Process a VbV/MCSC/SafeKey transaction and obtain the response. The merchant then makes the decision whether to continue with the transaction or to abort it.
- Process a financial transaction including AVS/CVD details and obtain the response. The merchant then makes a decision whether to continue with the transaction or to abort it.

#### **Option B**

1. Process a Transaction Risk Management Tool query and obtain the response.
2. Process a VbV/MCSC/SafeKey transaction and obtain the response.
3. Process a financial transaction including AVS/CVD details and obtain the response.
4. Merchant then makes a one-time decision based on the responses received from the eFraud tools.

## 10.4.2 Implementation Checklist

The following checklists provide high-level tasks that are required as part of your implementation of the Transaction Risk Management Tool. Because each organization has certain project requirements for implementing system and process changes, this list is only a guideline, and does not cover all aspects of your project.

### Download and review all of the applicable APIs and Integration Guides

Please review the sections outlined within this document that refers to the following feature

**Table 11: API documentation**

| Document/API                                                                                               | Use the document if you are....                                                              |
|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Transaction Risk Management Tool Integration Guide (Section #)                                             | Implementing or updating your integration for the Transaction Risk Management Tool           |
| Moneris MPI – Verified by Visa/MasterCard SecureCode/American Express SafeKey – Java API Integration Guide | Implementing or updating Verified by Visa, MasterCard SecureCode or American Express SafeKey |
| Basic transaction with VS and CVD (Section#)                                                               | Implementing or updating transaction processing, AVS or CVD                                  |

### Design your transaction flow and business processes

When designing your transaction flow, think about which scenarios you would like to have automated, and which scenarios you would like to have handled manually by your employees.

The “Understand Transaction Risk Management Transaction Flow” and Handling Response Information (page 429) sections can help you work through the design of your transaction and process flows.

Things to consider when designing your process flows:

- Processes for notifying people within your organization when there is scheduled maintenance for Moneris Gateway.
- Handling refunds, canceled orders and so on.
- Communicating with customers when you will not be shipping the goods because of suspected fraud, back-ordered goods and so on.

---

## Complete your development and testing

- The Moneris Gateway API - Integration Guide - PHP provides the technical details required for the development and testing. Ensure that you follow the testing instructions and data provided.

### If you are an integrator

- Ensure that your solution meets the requirements for PCI-DSS/PA-DSS as applicable.
- Send an email to [eproducts@moneris.com](mailto:eproducts@moneris.com) with the subject line "Certification Request".
- Develop material to set up your customers as quickly as possible with your solution and a Moneris account. Include information such as:
  - Steps they must take to enter their store ID or API token information into your solution.
  - Any optional services that you support via Moneris Gateway (such as TRMT, AVS, CVD, VBV/MCSC/SafeKey and so on) so that customers can request these features.

### 10.4.3 Making a Decision

Depending on your business policies and processes, the information obtained from the fraud tools (such as AVS, CVD, VbV/MCSC/SafeKey and TRMT) can help you make an informed decision about whether to accept a transaction or deny it because it is potentially fraudulent.

If you do not want to continue with a likely fraudulent transaction, you must inform the customer that you are not proceeding with their transaction.

If you are attempting to do further authentication by using the available fraud tools, but you have received an approval response instead, cancel the financial transaction by doing one of the following:

- If the original transaction is a Purchase, use a Purchase Correction or Refund transaction. You will need the original order ID and transaction number.
- If the original transaction is a Pre-Authorization, use a Completion transaction for \$0.00.

# 11 Apple Pay and Google Pay™ Integrations

- 11.1 About Apple Pay and Google Pay™ Integration
- 11.2 Apple Pay Transaction Process Overview
- 1 Google Pay™ Transaction Process Overview
- 11.4 About API Integration of Apple Pay and Google Pay™
- 11.5 Cavv Purchase – Apple Pay and Google Pay™
- 11.6 Cavv Pre-Authorization – Apple Pay & Google Pay™

## 11.1 About Apple Pay and Google Pay™ Integration

The Moneris Gateway enables merchants to process in-app transactions within applications running on iOS (Apple Pay) or Android (Google Pay™) mobile devices, and in-browser when using the Safari web browser (Apple Pay, using Apple devices only) or the Chrome web browser on (Google Pay™).

Moneris Solutions offers two integration methods for processing Apple Pay and Google Pay™ transactions. Merchants can choose to use one of two methods:

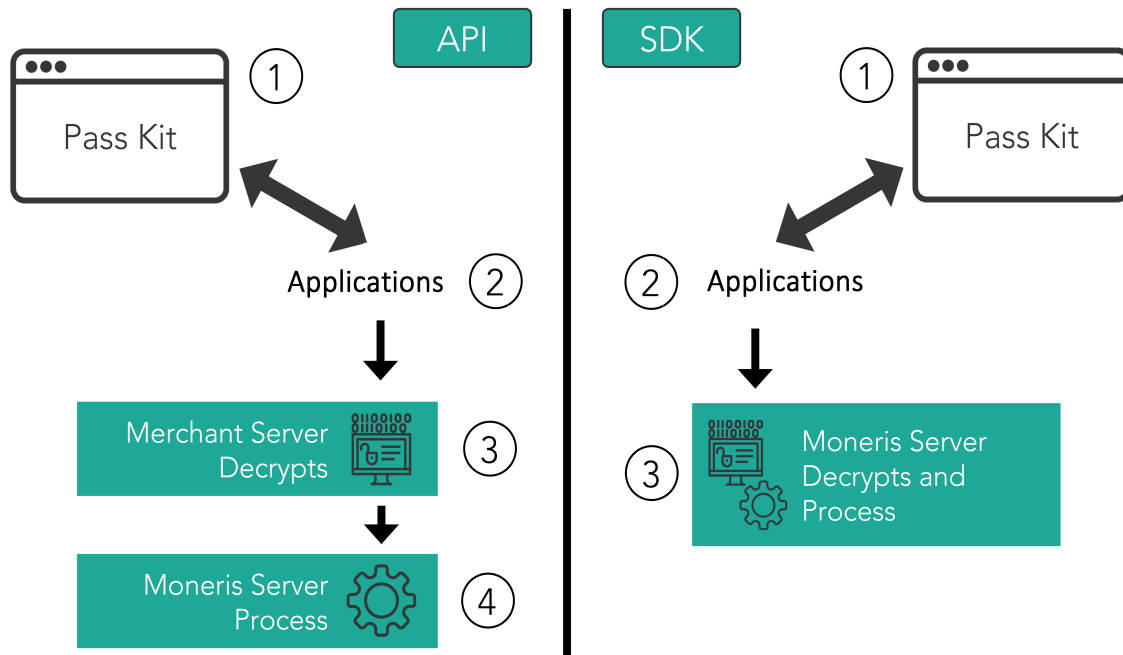
- Software Development Kit (SDK), or
- API method (where decryption of the transaction payload is handled by merchants)

While both methods provide the same basic payment features, there are differences in their implementations.

This guide mainly focuses on the API method, but some of its API requests are utilized in SDK implementation as well; for detailed information about the SDK method of integration, see the Moneris Developer Portal at <https://developer.moneris.com>.

## 11.2 Apple Pay Transaction Process Overview

For both API and SDK methods of mobile in-app integration, the merchant's iOS app uses Apple's PassKit Framework to request and receive encrypted payment details from Apple. When payment details are returned in their encrypted form, they can be decrypted and processed by the Moneris Gateway in one of two ways: SDK or API.



## Steps in the Apple Pay payment process

### API

1. Merchant's mobile application or web page requests and receives the encrypted payload
2. Encrypted payload is sent to the merchant's server, where it is decrypted
3. Moneris Gateway receives the decrypted payload from the merchant's server, and processes the Cavv Purchase – Apple Pay and Google Pay™ or "Cavv Pre-Authorization – Apple Pay & Google Pay™" on page 461transaction
  - a. Please ensure the wallet indicator is properly populated with the correct value (APP for Apple Pay In-App or APW for Apple Pay on the Web)

### SDK

1. Merchant's mobile application or web page requests and receives the encrypted payload
2. Encrypted payload is sent from the merchant's server to the Moneris Gateway, and the payload is decrypted and processed

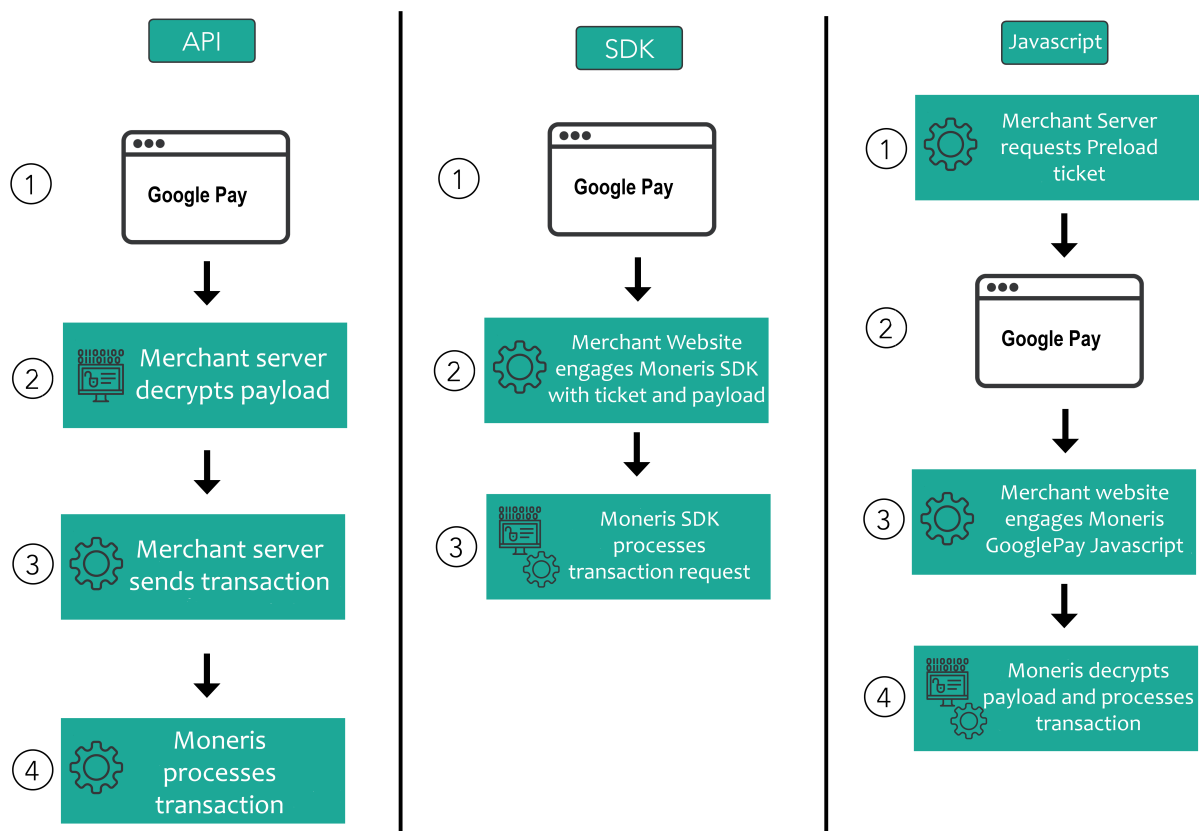
This guide mainly focuses on the API method, but some of its API requests are utilized in SDK implementation as well; for detailed information about the SDK method of integration, see the Moneris Developer Portal at <https://developer.moneris.com>.

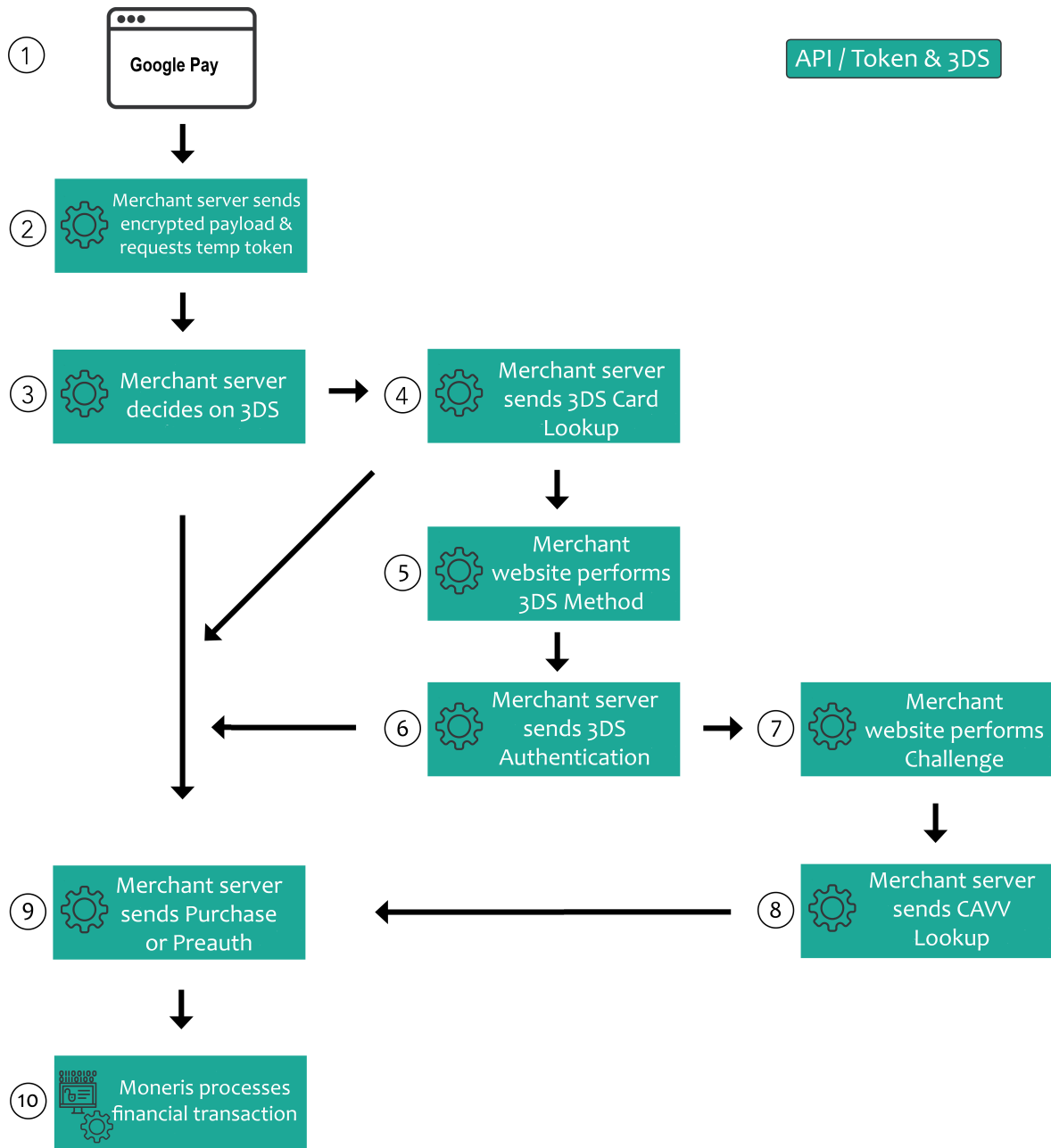
### 11.3 Google Pay™ Transaction Process Overview

Moneris offers four integration methods for processing transactions with Google Pay wallets. All integrations use the Google Pay™ Framework to request and receive encrypted payment details from Google. When payment details are returned in their encrypted form, the merchant can decrypt the payload on their server or transmit the encrypted payload to Moneris for decryption.

Moneris recommends merchants utilize the 3DS process on Google Pay temporary tokens with an underlying card type of FPAN to reduce risk of fraud and chargebacks. Merchants may attempt or skip the 3DS process if the underlying card type is DPAN; the card issuer may not support 3DS for the device PAN, however, so Moneris recommends using the 3DS Card Lookup to ensure support for 3DS.

#### Google Pay™ Integration Methods & Process







### Decrypted Payload (Merchant server to Moneris Gateway API)

Merchant decrypts the Google Pay encrypted payload locally then processes a standard financial transaction via Moneris Gateway API call with card data. Used by both in-app and web solutions.

**NOTE:** In this API scenario where merchant's server is responsible for decrypting the payload, merchants must sign agreement with Google directly. Google can then provide you with the keys to decrypt the payload.

1. Merchant's mobile application or web page requests and receives the encrypted payload from Google
2. Encrypted payload is sent from the merchant's website to Moneris Gateway via the SDK, and the payload is decrypted and processed

### Encrypted Payload (Merchant website to Moneris SDK)

Merchant passes the encrypted payload to the Moneris Google Pay SDK. Used for in-app solutions only. The SDK files are located on the [Moneris Github](#) and instructions on integration found on our [Moneris Developer Portal](#).

1. Merchant's mobile application or web page requests and receives the encrypted payload from Google
2. Encrypted payload is sent from the merchant's website to Moneris Gateway via the SDK, and the payload is decrypted and processed

### Encrypted Payload (Merchant website to Moneris JavaScript)

Merchant edits the Google Pay Javascript to utilize Moneris Google Pay for processing the payment. Includes an optional Preload via Moneris Gateway API specific to this integration method. See the full guide located on our [Moneris Github](#) or instructions on our [Moneris Developer Portal](#).

1. Merchant's server populates a Preload request and receives a ticket (optional step)
2. Merchant's mobile application or web page requests and receives the encrypted payload from Google
3. Encrypted payload is sent from the merchant's application or website to the via embedded Moneris Javascript, and the payload is decrypted and processed.

### Encrypted Payload With 3DS (Merchant server to Moneris Gateway API)

Merchant transmits the encrypted payload via Moneris Gateway API to decrypt and tokenize the card data temporarily. This temporary token is usable for performing 3D-Secure authentication and the subsequent financial transaction.

**NOTE:** Moneris recommends merchants utilize the 3DS process on Google Pay temporary tokens with an underlying card type of FPAN to reduce risk of fraud and chargebacks. Merchants may attempt or skip the 3DS process if the underlying card type is DPAN; the card issuer may not support 3DS for the device PAN, however, so Moneris recommends using the 3DS Card Lookup to ensure support for 3DS.

1. Merchant's app or web page requests and receives the encrypted payload from Google
2. Encrypted payload is sent from the merchant's server to the via a GooglePay Temporary Token Add. Moneris returns a temporary payment token in the response and a GooglePayPaymentMethod indicating the type of underlying card data (FPAN or DPAN)
3. Merchant server elects whether to perform 3DS Authentication or not. If electing to skip 3DS Authentication, the merchant server can skip to Step 9 and immediately perform a financial transaction.

4. Merchant server sends the temporary token in a 3DS Card Lookup request. Moneris responds with whether the underlying card supports 3DS authentication or not and details for the 3DS Method, if available.

For cards that do not support 3DS, the merchant server should skip the rest of the 3DS Authentication flow and move to Step 9.

5. If available, merchant server and app performs the 3DS Method using the 3DSMethodData and 3DSMethodURL. See "Handling the 3DS Method for Device Fingerprinting" on page 248
6. Merchant server performs a 3DS Authentication request (Browser Channel) to the Moneris Gateway. See "Implementing MPI 3DS Authentication Request" on page 249

If Moneris responds with a successful result (frictionless), the merchant server receives the CAVV and ECI values from the 3DS authentication response itself. Skip the challenge flow and move to Step 9.

If Moneris responds that a challenge prompt (friction) is required, continue with the next step.

7. Merchant server and application proceed with the 3DS Challenge. See "Handling the Challenge Flow" on page 280
8. Merchant server sends CAVV Lookup request to retrieve the authentication value (CAVV) and e-commerce indicator (crypt\_type) after the challenge is completed.
9. Merchant server performs either a GooglePayTokenPreauth or GooglePayTokenPurchase as the financial transaction. The CAVV and electronic commerce indicator (crypt\_type) are included as follows:

If 3DS was skipped earlier, omit the CAVV field and use the appropriate ECI for your transaction type.

If 3DS authentication was performed successfully, supply the CAVV from your 3DS Authentication or CAVV Lookup response.

## 11.4 About API Integration of Apple Pay and Google Pay™

An API integration works to provide a communication link between your merchant server and Moneris' server. APIs are required to complete any transaction, and therefore the APIs for Apple Pay and Google Pay™ are also included in SDK integration.

If the merchant chooses to use the API-only integration method, the merchant must decrypt payload information themselves before sending the decrypted information to the Moneris Gateway to be processed. Because this process is complicated, Moneris recommends only businesses with expertise and a previously integrated payment processing system use the API integration method; all other merchants should use the Moneris Apple Pay or Google Pay™ SDK as the integration method.

### 11.4.1 Transaction Types Used for Apple Pay and Google Pay™

In the Moneris Gateway API, there are two transaction types that allow you to process decrypted transaction payload information from Apple Pay and Google Pay™:

- 11.5 Cavv Purchase – Apple Pay and Google Pay™
- 11.6 Cavv Pre-Authorization – Apple Pay & Google Pay™

Note: In the event that a decrypted payment token contains a Device Primary Account Number (DPAN), a 3DS authentication attempt is not recommended as it may not be supported. A DPAN is a device-specific token from the wallet provider to identify the underlying card that is associated with a cardholder's Funding Primary Account Number (FPAN).

Once you have processed the initial transaction using Cavv Purchase or Cavv Pre-Authorization if required you can then process any of the following transactions:

- Refund
- Pre-Authorization Completion
- Purchase Correction

## 11.5 Cavv Purchase – Apple Pay and Google Pay™

The Cavv Purchase for Apple Pay and Google Pay™ transaction follows a 3-D Secure model but it does not require an MPI. Once the transaction payload has been decrypted, this transaction verifies funds on the customer's card, removes the funds and prepares them for deposit into the merchant's account.

To perform the 3-D Secure authentication, the Moneris MPI or any third-party MPI may be used.

In addition to 3-D Secure transactions, this transaction can also be used to process Apple Pay and Google Pay™ transactions. This transaction is applicable only if choosing to integrate directly to Apple Wallet or Google Wallet (if not using the Moneris Apple Pay or Google Pay™ SDKs).

Refer to Apple or Google developer portals for details on integrating directly to their wallets to retrieve the payload data.

**WARNING:** Moneris strongly discourages the use of frames as part of a 3-D Secure implementation, and cannot guarantee their reliability when processing transactions in the production environment.

### Cavv Purchase for Apple Pay & Google Pay™ transaction object definition

```
$txnArray = array('type'=>'cavv_purchase', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for Cavv Purchase for Apple Pay & Google Pay™ transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Cavv Purchase for Apple Pay & Google Pay™ transaction request fields – Required

| Variable Name      | Type and Limits                                                                                                                                             | Set Method             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| order ID           | String<br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                                                                                         | 'order_id'=>\$order_id |
| amount             | String<br>10-character decimal<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><div>EXAMPLE: 1234567.89</div> | 'amount'=>\$amount     |
| credit card number | String<br>max 20-character alpha-                                                                                                                           | 'pan'=>\$pan           |

| Variable Name                                                                                                                                                                                                                                                                                                                                                                                                       | Type and Limits                                   | Set Method                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                     | numeric                                           |                              |
| expiry date                                                                                                                                                                                                                                                                                                                                                                                                         | <i>String</i><br>4-character alphanumeric<br>YYMM | 'expiry_date'=>\$expiry_date |
| Cardholder Authentication Verification Value (CAVV)                                                                                                                                                                                                                                                                                                                                                                 | <i>String</i><br>50-character alphanumeric        | cavv=>\$cavv                 |
| <b>NOTE:</b> For Apple Pay and Google Pay™ Cavv Purchase and Cavv Pre-Authorization transactions, CAVV field contains the decrypted cryptogram. For more, see Appendix A Definition of Request Fields.                                                                                                                                                                                                              |                                                   |                              |
| electronic commerce indicator                                                                                                                                                                                                                                                                                                                                                                                       | <i>String</i><br>1-character alphanumeric         | 'crypt_type'=>\$crypt        |
| <b>NOTE:</b> For Apple Pay and Google Pay™ Cavv Purchase and Cavv Pre-Authorization transactions, the e-commerce indicator is a mandatory field containing the value received from the decrypted payload or a default value of 5. If you get a 2-character value (e.g., 05 or 07) from the payload, remove the initial 0 and just send us the 2nd character. For more, see Appendix A Definition of Request Fields. |                                                   |                              |

Following fields are required for Apple Pay and Google Pay only:

| Variable Name | Type and Limits             |  |
|---------------|-----------------------------|--|
| network       | <i>String</i><br>alphabetic |  |

| Variable Name | Type and Limits          |
|---------------|--------------------------|
| data type     | <i>String</i>            |
|               | 3-character alphanumeric |

### Cavv Purchase for Apple Pay & Google Pay™ transaction request fields – Optional

| Variable Name        | Type and Limits                                                                                                                                                                                                            | Set Method                                                                                                     |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| status check         | <i>Boolean</i><br>true/false                                                                                                                                                                                               | <code>\$mpgHttpPost =new<br/>mpgHttpPostStatus(\$store_<br/>id,\$api_<br/>token,\$status,\$mpgRequest);</code> |
| customer ID          | <i>String</i><br>50-character alphanumeric<br><div><b>NOTE:</b><br/>Some special characters are not allowed:<br/>&lt; &gt; \$ % = ? ^ { } [ ] \</div>                                                                      | <code>'cust_id'=&gt;\$cust_id</code>                                                                           |
| dynamic descriptor   | <i>String</i><br>20-character alphanumeric<br>total of 22 characters including your merchant name and separator<br><div><b>NOTE:</b><br/>Some special characters are not allowed:<br/>&lt; &gt; \$ % = ? ^ { } [ ] \</div> | <code>'dynamic_descriptor'=&gt;\$dynamic_<br/>descriptor</code>                                                |
| card match ID        | <i>String</i><br>50-character alphanumeric<br><div><b>NOTE:</b> Applies to Offlinx™ only; must be unique value for each transaction</div>                                                                                  | <code>'cm_id' =&gt; \$transaction_id</code>                                                                    |
| Customer Information | <i>Object</i><br>N/A                                                                                                                                                                                                       | <code>\$mpgTxn-&gt;setCustInfo<br/>(\$mpgCustInfo);</code>                                                     |

## Sample Cavv Purchase for Apple Pay &amp; Google Pay™

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='monca03650';
$api_token='7Yw0MPTlhjBRcZiE6837';
/***** Transactional Variables *****/
$type='cavv_purchase';
$order_id='ord-'.date("dmy-G:i:s");
$cust_id='CUST887763';
$amount='6000.00';
$pan='4622943127023886';
$expiry_date='2212';
$cavv='AAABBJg0VhI0VniQEjRWAAAAA=';
$crypt_type = '7';
$wallet_indicator = "APP";
$dynamic_descriptor='123456';
$foreign_indicator='true';

// TrId and TokenCryptogram are optional, refer documentation for more details.
$str_id = '50189815682';
$token_cryptogram = 'APbmB/411e0uAAH+s6xMAAADFA==';
'foreign_indicator'=>$foreign_indicator
/***** Transaction Associative Array *****/
$txnArray=array(
    'type'=>$type,
    'order_id'=>$order_id,
    'cust_id'=>$cust_id,
    'amount'=>$amount,
    'pan'=>$pan,
    'expdate'=>$expiry_date,
    'cavv'=>$cavv,
    'crypt_type'=>$crypt_type, //mandatory for AMEX only
    //'wallet_indicator'=>$wallet_indicator, //set only for wallet transactions. e.g. APPLE PAY
    //'network'=> "Interac", //set only for Interac e-commerce
    //'data type'=> "3DSecure", //set only for Interac e-commerce
    'dynamic_descriptor'=>$dynamic_descriptor,
    'threeds_version' => '2.2.0', //Mandatory for financial transactions using 3rd Party 3-D
    Secure services
    'threeds_server_trans_id' => 'e11d4985-8d25-40ed-99d6-c3803fe5e68f' Mandatory for financial
    transactions using 3rd Party 3-D Secure services - obtained from MpiCavvLookup or
    MpiThreeDSAuthentication
    //,'cm_id' => '8nAK8712sGaAkls56' //set only for usage with Offlinx - Unique max 50
    alphanumeric characters transaction id generated by merchant
    //,'ds_trans_id' => '12345' //Optional - to be used only if you are using 3rd party 3ds
    service
    //,'tr_id' => $str_id
    //,'token_cryptogram' => $token_cryptogram
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("U");
$cof->setPaymentInformation("2");
$cof->setIssuerId("139X3130ASCXAS9");
$mpgTxn->setCofInfo($cof);
/***** Installment Info *OPTIONAL* *****/
$installmentInfo = new InstallmentInfo();
$installmentInfo->setPlanId("ae859ef1-eb91-b708-8b80-1dd481746401");
$installmentInfo->setPlanIdRef("0000000065");
$installmentInfo->setTacVersion("2");
//$mpgTxn->setInstallmentInfo($installmentInfo);
/***** Request Object *****/

```



```

$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTP Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nCavvResultCode = " . $mpgResponse->getCavvResultCode());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\nThreeDSVersion = " . $mpgResponse->getThreeDSVersion());
print("\nSourcePanLast4 = " . $mpgResponse->getSourcePanLast4());
print("\nAdviceCode = " . $mpgResponse->getAdviceCode());

// $installmentResults = $mpgResponse->getInstallmentResults();
// print("\nPlanId = " . $installmentResults->getPlanId());
// print("\nPlanIDRef = " . $installmentResults->getPlanIDRef());
// print("\nTacVersion = " . $installmentResults->getTacVersion());
// print("\nPlanAcceptanceId = " . $installmentResults->getPlanAcceptanceId());
// print("\nPlanStatus = " . $installmentResults->getPlanStatus());
// print("\nPlanResponse = " . $installmentResults->getPlanResponse());
?>

```

## 11.6 Cavv Pre-Authorization – Apple Pay & Google Pay™

The Cavv Pre-Authorization for Apple Pay and Google Pay™ transaction follows a 3-D Secure model but it does not require an MPI. Once the transaction payload has been decrypted, this transaction verifies funds on the customer's card, and holds the funds. To prepare the funds for deposit into the merchant's account please process a Pre-Authorization Completion transaction.

To perform the 3-D Secure authentication, the Moneris MPI or any third-party MPI may be used.

In addition to 3-D Secure transactions, this transaction can also be used to process Apple Pay and Google Pay™ transactions. This transaction is applicable only if choosing to integrate directly to Apple Wallet or Google Wallet (if not using the Moneris Apple Pay or Google Pay™ SDKs).

Refer to Apple or Google developer portals for details on integrating directly to their wallets to retrieve the payload data.

**WARNING:** Moneris strongly discourages the use of frames as part of a 3-D Secure implementation, and cannot guarantee their reliability when processing transactions in the production environment.

### Cavv Pre-Authorization for Apple Pay & Google Pay™ transaction object definition

```
$txnArray = array('type'=>'cavv_preauth', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for Cavv Pre-Authorization for Apple Pay & Google Pay™ transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

### Cavv Pre-Authorization for Apple Pay&Google Pay™ transaction request fields – Required

| Variable Name                                       | Type and Limits                                                                                                                                             | Set Method             |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| order ID                                            | String<br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                                                                                         | 'order_id'=>\$order_id |
| amount                                              | String<br>10-character decimal<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><div>EXAMPLE: 1234567.89</div> | 'amount'=>\$amount     |
| credit card number                                  | String<br>max 20-character alphanumeric                                                                                                                     | 'pan'=>\$pan           |
| Cardholder Authentication Verification Value (CAVV) | String<br>50-character alphanumeric                                                                                                                         | cavv=>\$cavv           |

| Variable Name                                                                                                                                                                                                                                                                                                                                                                                                       | Type and Limits                                   | Set Method                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|------------------------------|
| <b>NOTE:</b> For Apple Pay and Google Pay™ Cavv Purchase and Cavv Pre-Authorization transactions, CAVV field contains the decrypted cryptogram. For more, see Appendix A Definition of Request Fields.                                                                                                                                                                                                              |                                                   |                              |
| expiry date                                                                                                                                                                                                                                                                                                                                                                                                         | <i>String</i><br>4-character alphanumeric<br>YYMM | 'expiry_date'=>\$expiry_date |
| electronic commerce indicator                                                                                                                                                                                                                                                                                                                                                                                       | <i>String</i><br>1-character alphanumeric         | 'crypt_type'=>\$crypt        |
| <b>NOTE:</b> For Apple Pay and Google Pay™ Cavv Purchase and Cavv Pre-Authorization transactions, the e-commerce indicator is a mandatory field containing the value received from the decrypted payload or a default value of 5. If you get a 2-character value (e.g., 05 or 07) from the payload, remove the initial 0 and just send us the 2nd character. For more, see Appendix A Definition of Request Fields. |                                                   |                              |

### Cavv Pre-Authorization for Apple Pay & Google Pay™ transaction request fields – Optional

| Variable Name | Type and Limits                            | Set Method                                                                                                   |
|---------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| status check  | <i>Boolean</i><br>true/false               | <pre>\$mpgHttpPost =new<br/>mpgHttpPostStatus(\$store_<br/>id,\$api_<br/>token,\$status,\$mpgRequest);</pre> |
| customer ID   | <i>String</i><br>50-character alphanumeric | 'cust_id'=>\$cust_id                                                                                         |

| Variable Name      | Type and Limits                                                                                                                                                                                                                  | Set Method                                 |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
|                    | <b>NOTE:</b><br>Some special characters are not allowed:<br>< > \$ % = ? ^ { } [ ] \                                                                                                                                             |                                            |
| dynamic descriptor | <i>String</i><br>20-character alphanumeric<br>total of 22 characters including your merchant name and separator<br><b>NOTE:</b><br>Some special characters are not allowed:<br>< > \$ % = ? ^ { } [ ] \                          | 'dynamic_descriptor'=>\$dynamic_descriptor |
| card match ID      | <i>String</i><br>50-character alphanumeric<br><b>NOTE:</b> Applies to Offlinx™ only; must be unique value for each transaction                                                                                                   | 'cm_id' => \$transaction_id                |
| network            | <i>String</i><br>alphabetic<br><b>NOTE:</b> This request variable is mandatory for INTERAC® e-Commerce transactions conducted via Apple Pay or Google Pay™ only, and is not for use with credit card transactions.               |                                            |
| data type          | <i>String</i><br>3-character alphanumeric<br><b>NOTE:</b> This request variable is mandatory for INTERAC® e-Commerce transactions conducted via Apple Pay or Google Pay™ only, and is not for use with credit card transactions. |                                            |

| Variable Name | Type and Limits | Set Method |
|---------------|-----------------|------------|
|---------------|-----------------|------------|

### Sample Cavv Pre-Authorization for Apple Pay & Google Pay™

```
<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='monca03650';
$api_token='7Yw0MPTlhjBRcZiE6837';
/***** Transactional Variables *****/
$type='cavv_pauth';
$order_id='ord-' . date("dmy-G:i:s");
$cust_id='CUST887763';
$amount='4840.00';
$pan='5454545454545454';
$expiry_date='2212';
$cavv='AAABBJg0VhI0VniQEjRWAAAAA=';
$crypt_type = '7';
$wallet_indicator = "APP";
$dynamic_descriptor='123456';
// TrId and TokenCryptogram are optional, refer documentation for more details.
$tr_id = '50189815682';
$token_cryptogram = 'APmbM/41le0uAAH+s6xMAADFA==';
/***** Transaction Associative Array *****/
$txnArray=array(
    'type'=>$type,
    'order_id'=>$order_id,
    'cust_id'=>$cust_id,
    'amount'=>$amount,
    'pan'=>$pan,
    'expdate'=>$expiry_date,
    'cavv'=>$cavv,
    'crypt_type'=>$crypt_type, //mandatory for AMEX only
    //'wallet_indicator'=>$wallet_indicator, //set only for wallet transactions. e.g. APPLE PAY
    //'network'=> "Interac", //set only for Interac e-commerce
    //'data type'=> "3DSecure", //set only for Interac e-commerce
    'dynamic_descriptor'=>$dynamic_descriptor,
    'threeds_version' => '2', //Mandatory for 3DS Version 2.0+
    'threeds_server_trans_id' => 'e11d4985-8d25-40ed-99d6-c3803fe5e68f' //Mandatory for 3DS
    Version 2.0+ - obtained from MpiCavvLookup or MpiThreeDSAuthentication
    //'cm_id' => '8nAK8712sGaAkl56' //set only for usage with Offlinx - Unique max 50
    alphanumeric characters transaction id generated by merchant
    //'ds_trans_id' => '12345' //Optional - to be used only if you are using 3rd party 3ds 2.0
    service
    //'tr_id' => $tr_id
    //'token_cryptogram' => $token_cryptogram
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("U");
$cof->setPaymentInformation("2");
$cof->setIssuerId("139X3130ASCXAS9");
$mpgTxn->setCofInfo($cof);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false or comment out this line for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

```
/****** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nCavvResultCode = " . $mpgResponse->getCavvResultCode());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\nThreeDSVersion = " . $mpgResponse->getThreeDSVersion());
print("\nSourcePanLast4 = " . $mpgResponse->getSourcePanLast4());
print("\nAdviceCode = " . $mpgResponse->getAdviceCode());
?>
```

## 11.7 GooglePay Temporary Token Add – GooglePayTokenTempAdd

Creates a new temporary token credit card profile from an encrypted GooglePay payload. During the lifetime of this temporary token, it may be used to perform 3DS authentication and financial transactions via GooglePay Token Preauthorization or GooglePay Token Purchase.

The response field `GooglePaymentMethod` returned by this request will inform you if the underlying card within GooglePay is the funding card number ("FPAN") or a tokenized card number ("DPAN"). If a `GoogleTokenTempAdd` returns an FPAN, you may perform 3DS authentication with it; if it returns a DPAN, 3DS is not required.

Refer to Apple or Google developer portals for details on integrating directly to their wallets to retrieve the payload data.

### Things to Consider:

- The duration, or lifetime, of the temporary token can be set to be a maximum of 15 minutes.

### GooglePay Temporary Token Add transaction object definition

```
$txnArray = array('type'=>'googlePayTokenTempAdd', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for GooglePay Temporary Token Add transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits      |                          |
|---------------|----------------------|--------------------------|
| store ID      | <i>String</i><br>N/A | 'store_id'=>\$store_id   |
| API token     | <i>String</i><br>N/A | 'api_token'=>\$api_token |

### GooglePay Temporary Token Add transaction request fields – Required

For a full description of mandatory and optional values, see Appendix A Definition of Request Fields.

| Variable Name | Type and Limits             | Set Method                                                                                                                                                                                                                                                                                                      |
|---------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| network       | <i>String</i><br>alphabetic | <code>\$network = "MASTERCARD";</code><br><br>Card Brand name.<br><br>Field is case sensitive<br><br>Possible values:<br><br>Visa<br><br>Mastercard<br><br>American Express<br><br>Interac<br><br>Discover                                                                                                      |
| payment token | <i>Object</i><br>N/A        | <code>\$googlePayTokenTempAdd-&gt;setPaymentToken(\$signature,\$protocol_version,\$signed_message);</code><br><br>Payment details returned by Google in their <code>PaymentData</code> object for GooglePay transactions. See GooglePay Payment Token object request fields – Required below for field details. |

**GooglePay Payment Token object request fields – Required**

| Variable Name    | Type and Limits          | Set Method                                                                                                                                                                                                                                                                                                                                 |
|------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| signature        | <i>String</i><br><br>N/A | <code>\$signature = "ME...";</code><br><br>Verifies that the message came from Google. It's base64-encoded, and created with ECDSA by the intermediate signing key. Returned by Google in their <code>PaymentData</code> object for GooglePay transactions                                                                                 |
| protocol version | <i>String</i><br><br>N/A | <code>\$protocol_version = "ECv1";</code><br><br>Identifies the encryption or signing scheme under which the message is created. It allows the protocol to evolve over time, if needed. Returned by Google in their <code>PaymentData</code> object for GooglePay transactions                                                             |
| signed message   | <i>String</i><br><br>N/A | <code>\$signed_message = "{\\"encryptedMessage\\"}";</code><br><br>A JSON object serialized as an HTML-safe string that contains the encryptedMessage, ephemeralPublicKey, and tag. It's serialized to simplify the signature verification process. Returned by Google in their <code>PaymentData</code> object for GooglePay transactions |

**Sample GooglePay Temporary Token Add**

```

<?php
require "../mpgClasses.php";

/***** Request Variables *****/

$store_id='intuit_spd';
$sapi_token='spedguy';

/***** Transactional Variables *****/

$network = "MASTERCARD";

```



## Sample GooglePay Temporary Token Add

```

$signature =
"MEYCIQcdjfGZ1k/8h+eH9Ue5UxJsgDEFimp6YIrWhptpe+W3kAIhAOKikmz1B/C4WB5g3mXy139euOHHnSQ7bQWl2c
hgwole";
$protocol_version = "ECv1";
$signed_message = '
{"encryptedMessage": "nAEP5f0pzvU+cJHwxCwrCVPRr196NugevgfrdidPOB5B+WG7+yrsYoUVA7HopRD5y5GCld
QwrKnP2h2w/Qc2HBFn+G/g2IXqPBzMjguhpGitr6lV0tRLaYimxrgrbh/Xn8DxfW++pTHHoo+0xJiON6o3JC4vM6wMA
uhjjwEOgiDeKpgxJKEl8NULR2RK1OtvongkR80K8Et7CT+W0lXoMCoYrH3tJDKMtoVyFNfHPMAXLeV4NfVV+ZwhwD3F
+tGm7bQkPFMy2xUQxzdj7/H03vmyxwsblSKXhVG3hWKPMnY/+Gkb2K0pAicOHAB/SZuwaxHQ1130jaNafUIm96R9T2Y
c3p5gmnGiR03R9H5R8JqLL9Wb7LncvfiWuQppgbAKa6HdbuSjbehNOTw8S34VqxvpeSfQFUNYgkQ+fVEU/VaClE17Py
F8AMZKN10ZiZ1jj7jntqoD", "ephemeralPublicKey": "BD5snQM3HF2gdCyERaF9XBPdGOXL8fNyTM9QY/xNTi9Vk
WTtq5sg7dYgPXdLmQuwIhBN9OyLULAMsNcmsv2TT7k\u003d", "tag": "hyG7Ty/qQAZelt2INIMtDQPMAfoDVhUinW
451hJrcP4\u003d"}';

/***** Transactional Associative Array *****/

$googlePayTokenTempAdd = new GooglePayTokenTempAdd();
$googlePayTokenTempAdd->setPaymentToken($signature, $protocol_version, $signed_message);

/***** Transaction Object *****/

$mpgTxn = new mpgTransaction($googlePayTokenTempAdd);

/***** Request Object *****/

$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false or comment out this line for production
transactions

/***** HTTPS Post Object *****/

/* Status Check Example
$mpgHttpPost = new mpgHttpPostStatus($store_id, $api_token, $status_check, $mpgRequest);
*/
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);

/***** Response *****/

$mpgResponse = $mpgHttpPost->getMpgResponse();
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nMasked Pan = " . $mpgResponse->getResDataMaskedPan());
print("\nExp Date = " . $mpgResponse->getResDataExpDate());
print("\nPayment Type = " . $mpgResponse->getPaymentType());
print("\nGooglepayPaymentMethod = " . $mpgResponse->getGooglepayPaymentMethod());
?>

```

## 11.8 Google Pay™ Token Preauth

The Google Pay™ Token Preauth transaction is utilized after passing a GooglePay account into a temporary token using our GooglePay Token Temporary Add then performing 3DS authentication with the token. This transaction verifies funds on the customer's card and locks those funds for a time period specified by the card issuer.

To perform the 3-D Secure authentication, the Moneris MPI or any third-party MPI may be used.

Refer to Apple or Google developer portals for details on integrating directly to their wallets to retrieve the payload data.

### Google Pay™ Token Preauth for transaction object definition

```
$txnArray = array('type'=>'googlePayTokenPreauth', ...);  
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for Google Pay™ Token Preauth transaction

```
$mpgRequest = new mpgRequest($mpgTxn);  
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

### Google Pay™ Token Preauth transaction request fields – Required

| Variable Name | Type and Limits                                                     | Set Method                                                       |
|---------------|---------------------------------------------------------------------|------------------------------------------------------------------|
| order ID      | String<br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces | 'order_id'=>\$order_id                                           |
| data key      | String<br>25-character alphanumeric                                 | The temporary token returned by a GooglePayTokenTempAdd request. |
| amount        | String                                                              | 'amount'=>\$amount                                               |

| Variable Name                                                                                                                                                                                                                                                                                  | Type and Limits                                                                                                                                           | Set Method                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                | 10-character decimal<br><br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><br><div>EXAMPLE: 1234567.89</div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Cardholder Authentication Verification Value (CAVV)<br><br><div> <b>NOTE:</b> For Google Pay™ Token Pre-Authorization transactions, CAVV field contains the 3DS cryptogram only when 3DS is used prior. If you elected to skip 3DS Authentication, you may omit the CAVV field.         </div> | <i>String</i><br><br>50-character alphanumeric                                                                                                            | <code>cavv=&gt;\$cavv</code><br><br>The 3DS cryptogram.<br><br>Sent in all financial transactions with 3-D Secure, including Verified By Visa, MasterCard SecureCode, American Express SafeKey                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| electronic commerce indicator<br><br><div> <b>NOTE:</b> For Google Pay™ Token Purchase and Token Pre-Authorization transactions using 3DS Authentication, use the e-commerce indicator obtained from your 3DS Authentication.         </div>                                                   | <i>String</i><br><br>1-character alphanumeric                                                                                                             | <code>'crypt_type'=&gt;\$crypt</code><br><br>Describes the category of e-commerce transaction being processed. Allowable values are:<br><br>1 – Mail Order / Telephone Order—Single<br><br>2 – Mail Order / Telephone Order—Recurring<br><br>3 – Mail Order / Telephone Order—Instalment<br><br>4 – Mail Order / Telephone Order—Unknown classification<br><br>5 – Authenticated e-commerce transaction (3-D Secure)<br><br>6 – Non-authenticated e-commerce transaction (3-D Secure)<br><br>7 – SSL-enabled merchant<br><br><b>In Credential on File transactions where the request field e-commerce indicator is also being sent: the allow-</b> |

| Variable Name                                                                                                                                                                                                                                                                                  | Type and Limits                           | Set Method                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                |                                           | <p>able values for e-commerce indicator are dependent on the value sent for <b>payment indicator</b>, as follows:</p> <p>if payment indicator = R, then allowable values for e-commerce indicator: 2, 5 or 6</p> <p>if payment indicator = V, then allowable values for e-commerce indicator: 2, 5 or 6</p> <p>if payment indicator = C, then allowable values for e-commerce indicator: 1, 5, 6 or 7</p> <p>if payment indicator = U, then allowable values for e-commerce indicator: 1 or 7</p> <p>if payment indicator = Z, then allowable values for e-commerce indicator: 1, 5, 6 or 7</p> |
| 3DS server transaction ID <div> <b>NOTE:</b> Obtained from the Cavv Lookup request or MPI 3DS Authentication request. For Google Pay™ Token Purchase and Token Pre-Authorization transactions that do not use 3DS Authentication, you may omit the 3DS Server Transaction ID.           </div> | <i>String</i><br><br>36-character numeric | <pre>'threeds_server_trans_id'=&gt;\$threeds_server_trans_id</pre> <p>Data is obtained from a Cavv Lookup Request or MPI 3DS Authentication Request transaction</p>                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3DS version <div> <b>NOTE:</b> Mandatory for financial transactions using 3rd Party 3-D Secure services. If you elected to skip 3DS Authentication, you may omit the 3DS Version field.           </div>                                                                                       | <i>String</i><br><br>10-character numeric | <pre>'threeds_version'=&gt;\$threeds_version</pre> <p>Acceptable values:</p> <p>2.0.0 = 3DS protocol 2.0.0</p> <p>2.1.0 = 3DS protocol 2.1.0</p> <p>2.2.0 = 3DS protocol 2.2.0</p> <p>2.3.0 = 3DS protocol 2.3.0</p>                                                                                                                                                                                                                                                                                                                                                                            |
| network                                                                                                                                                                                                                                                                                        | <i>String</i><br><br>alphabetic           | Card Brand name.<br><br>Field is case sensitive                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Variable Name | Type and Limits | Set Method                                                                                            |
|---------------|-----------------|-------------------------------------------------------------------------------------------------------|
|               |                 | Possible values:<br><br>Visa<br><br>Mastercard<br><br>American Express<br><br>Interac<br><br>Discover |

### Google Pay™ Token preauth transaction request fields – Optional

| Variable Name      | Type and Limits                                                                                                                                                                                                                                             | Set Method                                 |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| customer ID        | <i>String</i><br><br>50-character alphanumeric<br><br><div> <b>NOTE:</b><br/>           Some special characters are not allowed:<br/>           &lt;&gt;\$%=?^{}[]\         </div>                                                                          | 'cust_id'=>\$cust_id                       |
| dynamic descriptor | <i>String</i><br><br>20-character alphanumeric<br><br>total of 22 characters including your merchant name and separator<br><br><div> <b>NOTE:</b><br/>           Some special characters are not allowed:<br/>           &lt;&gt;\$%=?^{}[]\         </div> | 'dynamic_descriptor'=>\$dynamic_descriptor |

### SampleGoogle Pay™ Token Preauth

```

<?php

require "../mpgClasses.php";

/***** Request Variables *****/

$store_id='intuit_spd';
$api_token='spedguy';

/***** Transactional Variables *****/

```

```
$order_id='ord-'.date("dmy-G:i:s");
$cust_id='nqa cust id';
$amount='1.00';
$script_type='2';
$network = "MASTERCARD";
$data_key = "ot-PLa88dGe5uSy8TwbI9BeMqurl";
$threads_server_trans_id = "delb97ee-c610-4877-b53f-c1c5ecd99bf0";
$threads_version = "2.2";
$cavv = "kAABApFSYyd4l2eQQFJjAAAAAAA=";
$dynamic_descriptor = "nqa-dd";

/***** Transactional Associative Array *****/

$googlePayTokenPreauth = new GooglePayTokenPreauth();
$googlePayTokenPreauth->setOrderId($order_id);
$googlePayTokenPreauth->setCustId($cust_id);
$googlePayTokenPreauth->setAmount($amount);
$googlePayTokenPreauth->setCryptType($script_type);
$googlePayTokenPreauth->setNetwork($network);
$googlePayTokenPreauth->setDataKey($data_key);
$googlePayTokenPreauth->setThreeDSSTransId($threads_server_trans_id);
$googlePayTokenPreauth->setThreeDSVersion($threads_version);
$googlePayTokenPreauth->setCavv($cavv);
$googlePayTokenPreauth->setDynamicDescriptor($dynamic_descriptor);

/***** Transaction Object *****/

$mpgTxn = new mpgTransaction($googlePayTokenPreauth);

/***** Request Object *****/

$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false or comment out this line for production transactions

/***** HTTPS Post Object *****/

/* Status Check Example
$mpgHttpPost =new mpgHttpPostStatus($store_id,$api_token,$status_check,$mpgRequest);
*/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);

/***** Response *****/

$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nStatusCode = " . $mpgResponse->getStatusCode());
print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
print("\nHostId = " . $mpgResponse->getHostId());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\nPayment Type = " . $mpgResponse->getPaymentType());
```

```

print("\nSourcePanLast4 = " . $mpgResponse->getSourcePanLast4());
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nCavvResultCode = " . $mpgResponse->getCavvResultCode());
print("\nPar = " . $mpgResponse->getPar());
print("\nThreeDSVersion = " . $mpgResponse->getThreeDSVersion());
print("\nGooglepayPaymentMethod = " . $mpgResponse->getGooglepayPaymentMethod());

?>

```

## 11.9 Google Pay™ Token Purchase

The Google Pay™ Token Purchase transaction is utilized after passing a GooglePay account into a temporary token using our GooglePay Token Temporary Add then performing 3DS authentication with the token. This transaction verifies funds on the customer's card, removes the funds and prepares them for deposit into the merchant's account.

To perform the 3-D Secure authentication, the Moneris MPI or any third-party MPI may be used.

Refer to Apple or Google developer portals for details on integrating directly to their wallets to retrieve the payload data.

### Google Pay™ Token Purchase for transaction object definition

```

$txnArray = array('type'=>'googlePayTokenPurchase', ...);

$mpgTxn = new mpgTransaction($txnArray);

```

### HttpPostRequest object for Google Pay™ Token Purchase transaction

```

$mpgRequest = new mpgRequest($mpgTxn);

$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);

```

### Core connection object fields (all API transactions)

| Variable Name | Type and Limits |                          |
|---------------|-----------------|--------------------------|
| store ID      | String<br>N/A   | 'store_id'=>\$store_id   |
| API token     | String<br>N/A   | 'api_token'=>\$api_token |

### Google Pay™ Token Purchase transaction request fields – Required

| Variable Name | Type and Limits                     | Set Method             |
|---------------|-------------------------------------|------------------------|
| order ID      | String<br>50-character alphanumeric | 'order_id'=>\$order_id |

| Variable Name                                                                                                                                                                                                                                                           | Type and Limits                                                                                                                                                                   | Set Method                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                         | a-Z A-Z 0-9 _ - : . @ spaces                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| data key                                                                                                                                                                                                                                                                | <p><i>String</i></p> <p>25-character alphanumeric</p>                                                                                                                             | <p><code>'data_key'=&gt;\$data_key</code></p> <p>The temporary token returned by a <code>GooglePayTokenTempAdd</code> request.</p>                                                                                                                                                                                                                                                                                                                                             |
| amount                                                                                                                                                                                                                                                                  | <p><i>String</i></p> <p>10-character decimal</p> <p>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point</p> <p><b>EXAMPLE:</b> 1234567.89</p> | <p><code>'amount'=&gt;\$amount</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p>Cardholder Authentication Verification Value (CAVV)</p> <p><b>NOTE:</b> For Google Pay™ Token Purchase transactions, CAVV field contains the 3DS cryptogram only when 3DS is used prior. If you elected to skip 3DS Authentication, you may omit the CAVV field.</p> | <p><i>String</i></p> <p>50-character alphanumeric</p>                                                                                                                             | <p><code>cavv=&gt;\$cavv</code></p> <p>The 3DS cryptogram.</p> <p>Sent in all financial transactions with 3-D Secure, including Verified By Visa, MasterCard SecureCode, American Express SafeKey</p>                                                                                                                                                                                                                                                                          |
| <p>electronic commerce indicator</p> <p><b>NOTE:</b> For Google Pay™ Token Purchase and Token Pre-Authorization transactions using 3DS Authentication, use the e-commerce indicator obtained from your 3DS Authentication.</p>                                          | <p><i>String</i></p> <p>1-character alphanumeric</p>                                                                                                                              | <p><code>'crypt_type'=&gt;\$crypt</code></p> <p>Describes the category of e-commerce transaction being processed. Allowable values are:</p> <ul style="list-style-type: none"> <li>1 – Mail Order / Telephone Order—Single</li> <li>2 – Mail Order / Telephone Order—Recurring</li> <li>3 – Mail Order / Telephone Order—Instalment</li> <li>4 – Mail Order / Telephone Order—Unknown classification</li> <li>5 – Authenticated e-commerce transaction (3-D Secure)</li> </ul> |



| Variable Name                                                                                                                                                                                                                                                                               | Type and Limits                       | Set Method                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                             |                                       | <p>6 – Non-authenticated e-commerce transaction (3-D Secure)</p> <p>7 – SSL-enabled merchant</p> <p><b>In Credential on File transactions where the request field e-commerce indicator is also being sent:</b> the allowable values for e-commerce indicator are dependent on the value sent for <b>payment indicator</b>, as follows:</p> <p>if payment indicator = R, then allowable values for e-commerce indicator: 2, 5 or 6</p> <p>if payment indicator = V, then allowable values for e-commerce indicator: 2, 5 or 6</p> <p>if payment indicator = C, then allowable values for e-commerce indicator: 1, 5, 6 or 7</p> <p>if payment indicator = U, then allowable values for e-commerce indicator: 1 or 7</p> <p>if payment indicator = Z, then allowable values for e-commerce indicator: 1, 5, 6 or 7</p> |
| 3DS server transaction ID <div> <p><b>NOTE:</b> Obtained from the Cavv Lookup request or MPI 3DS Authentication request. For Google Pay™ Token Purchase and Token Pre-Authorization transactions that do not use 3DS Authentication, you may omit the 3DS Server Transaction ID.</p> </div> | <i>String</i><br>36-character numeric | <pre>'threads_server_trans_id'=&gt;\$threads_server_trans_id</pre> <p>Data is obtained from a Cavv Lookup Request or MPI 3DS Authentication Request transaction</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 3DS version <div> <p><b>NOTE:</b> If you elected to skip 3DS Authentication, you may omit the 3DS Version field.</p> </div>                                                                                                                                                                 | <i>String</i><br>10-character numeric | <pre>'threads_version'=&gt;\$threads_version</pre> <p>Acceptable values:</p> <p>2.0.0 = 3DS protocol 2.0.0</p> <p>2.1.0 = 3DS protocol 2.1.0</p> <p>2.2.0 = 3DS protocol 2.2.0</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Variable Name | Type and Limits        | Set Method                                                                                                                                                                                    |
|---------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |                        | 2.3.0 = 3DS protocol 2.3.0                                                                                                                                                                    |
| network       | String<br>alphanumeric | \$network = "MASTERCARD";<br><br>Card Brand name.<br><br>Field is case sensitive<br><br>Possible values:<br><br>Visa<br><br>Mastercard<br><br>American Express<br><br>Interac<br><br>Discover |

### Google Pay™ Token Purchase transaction request fields – Optional

| Variable Name      | Type and Limits                                                                                                                                                                               | Set Method                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| customer ID        | String<br>50-character alphanumeric<br><br><b>NOTE:</b><br>Some special characters are not allowed:<br><>\$%=?^{}[]\                                                                          | 'cust_id'=>\$cust_id                       |
| dynamic descriptor | String<br>20-character alphanumeric<br><br>total of 22 characters including your merchant name and separator<br><br><b>NOTE:</b><br>Some special characters are not allowed:<br><>\$%=?^{}[]\ | 'dynamic_descriptor'=>\$dynamic_descriptor |

### Sample Google Pay™ Token Purchase

```
<?php
```

```

require "../mpgClasses.php";

/***** Request Variables *****/

$store_id='intuit_spd';
$api_token='spdguy';

/***** Transactional Variables *****/

$order_id='ord-'.date("dmy-G:i:s");
$cust_id='nqa cust id';
$amount='1.00';
$crypt_type='2';
$network = "MASTERCARD";
$data_key = "ot-eJ7JlFxbUrLkUow8Kko7Dj1e1";
$threeds_server_trans_id = "delb97ee-c610-4877-b53f-c1c5ecd99bf0";
$threeds_version = "2.2";
$cavv = "kAABApFSYd4l2eQQFJjAAAAA=";
$dynamic_descriptor = "nqa-dd";

/***** Transactional Associative Array *****/

$googlePayTokenPurchase = new googlePayTokenPurchase();
$googlePayTokenPurchase->setOrderId($order_id);
$googlePayTokenPurchase->setCustId($cust_id);
$googlePayTokenPurchase->setAmount($amount);
$googlePayTokenPurchase->setCryptType($crypt_type);
$googlePayTokenPurchase->setNetwork($network);
$googlePayTokenPurchase->setDataKey($data_key);
$googlePayTokenPurchase->setThreeDSSTransId($threeds_server_trans_id);
$googlePayTokenPurchase->setThreeDSVersion($threeds_version);
$googlePayTokenPurchase->setCavv($cavv);
$googlePayTokenPurchase->setDynamicDescriptor($dynamic_descriptor);

/***** Transaction Object *****/

$mpgTxn = new mpgTransaction($googlePayTokenPurchase);

/***** Request Object *****/

$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false or comment out this line for production transactions

/***** HTTPS Post Object *****/

/* Status Check Example
$mpgHttpPost = new mpgHttpPostStatus($store_id,$api_token,$status_check,$mpgRequest);
*/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);

/***** Response *****/

$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print("\nAuthCode = " . $mpgResponse->getAuthCode());

```

```
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nStatusCode = " . $mpgResponse->getStatusCode());
print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
print("\nHostId = " . $mpgResponse->getHostId());
print("\nIssuerId = " . $mpgResponse->getIssuerId());
print("\nPayment Type = " . $mpgResponse->getPaymentType());
print("\nSourcePanLast4 = " . $mpgResponse->getSourcePanLast4());
print("\nDataKey = " . $mpgResponse->getDataKey());
print("\nCavvResultCode = " . $mpgResponse->getCavvResultCode());
print("\nPar = " . $mpgResponse->getPar());
print("\nThreeDSVersion = " . $mpgResponse->getThreeDSVersion());
print("\nGooglepayPaymentMethod = " . $mpgResponse->getGooglepayPaymentMethod());

?>
```

## 12 Offlinx™

- What Is a Pixel Tag?
- Offlinx™ and API Transactions

### 12.1 What Is a Pixel Tag?

A pixel tag is a piece of code that goes on a web page and requests an image file (a tiny transparent image or pixel) when loaded, which, while not visible to the user, allows Offlinx™ to gather relevant information about the user.

The data collected by our pixel tag is:

- Anonymous (not personally identifiable) and compliant with privacy standards
- Secure — utilizes SSL communication to transmit the data securely
- Not shared with anyone

### 12.2 Offlinx™ and API Transactions

The Offlinx™ Card Match pixel tag feature can be implemented via the Unified API with the Card Match ID variable, which corresponds to the Transaction ID in Offlinx™. The Card Match ID must be a unique value for each transaction.

For more information about the Offlinx™ solution, consult the Offlinx™ Pixel Tag Setup Guide available from your account/service manager.

API transactions where this applies:

- Purchase
- Pre-Authorization
- Purchase with 3-D Secure – cavv\_purchase
- Pre-Authorization with 3-D Secure – cavv\_preauth

- Cavv Purchase – Apple Pay and Google Pay™
- Cavv Pre-Authorization – Apple Pay & Google Pay™

## 13 Convenience Fee

- 13.1 About Convenience Fee
- 13.3 Purchase with Convenience Fee
- 13.4 Purchase with Customer Info and Convenience Fee
- 13.5 Purchase with 3-D Secure and Convenience Fee

### 13.1 About Convenience Fee

The Convenience Fee program was designed to allow merchants to offer the convenience of an alternative payment channel to the cardholder at a charge. This applies only when providing a true "convenience" in the form of an alternative payment channel outside the merchant's customary face-to-face payment channels. The convenience fee will be a separate charge on top of what the consumer is paying for the goods and/or services they were given, and this charge will appear as a separate line item on the consumer's statement.

Convenience fee transactions do not support MCP or digital wallets.

**NOTE:** The Convenience Fee program is only offered to certain supported Merchant Category Codes (MCCs). Please speak to your account manager for further details.

### 13.2 Convenience Fee Information Object

Any transaction that supports Convenience Fee has an available set method for the Convenience Fee Information object.

The Convenience Fee Information object contains one request field, **convenience fee amount**.

#### Convenience Fee Info object definition

```
$mpgConvFee = new mpgConvFeeInfo($convFeeTemplate);
```

#### Convenience Fee Info object set method

```
$mpgTxn->setConvFeeInfo($mpgConvFee);
```

#### Convenience Fee Information object request fields

| Variable Name               | Type and Limits          | Description                                            |
|-----------------------------|--------------------------|--------------------------------------------------------|
| Convenience Fee Information | <i>Object</i><br><br>N/A | Contains fields related to the Convenience Fee feature |

| Variable Name          | Type and Limits                      | Description                                                |
|------------------------|--------------------------------------|------------------------------------------------------------|
| convenience fee amount | <i>String</i><br>9-character decimal | Dollar amount charged to the customer as a convenience fee |

### 13.3 Purchase with Convenience Fee

Information below describes a Purchase transaction request that also includes the Convenience Fee Info object.

#### Purchase with Convenience Fee transaction object definition

```
$txnArray = array('type'=>'purchase', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

#### HttpPostRequest object for Purchase with Convenience Fee transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
```

#### Purchase with Convenience Fee transaction request fields – Required

For a full description of mandatory and optional values, see Appendix A Definition of Request Fields.

| Variable Name               | Type and Limits                                                                                                                                                    | Set Method                                                         |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Convenience Fee Information | <i>Object</i><br>N/A                                                                                                                                               | <code>\$mpgConvFee = new mpgConvFeeInfo(\$convFeeTemplate);</code> |
| order ID                    | <i>String</i><br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                                                                                         | <code>'order_id'=&gt;\$order_id</code>                             |
| amount                      | <i>String</i><br>10-character decimal<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><div>EXAMPLE: 1234567.89</div> | <code>'amount'=&gt;\$amount</code>                                 |



| Variable Name                                                                                                                                                                                                                                                              | Type and Limits                                   | Set Method                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------|
| credit card number                                                                                                                                                                                                                                                         | <i>String</i><br>max 20-character alphanumeric    | 'pan'=>\$pan                                                      |
| expiry date                                                                                                                                                                                                                                                                | <i>String</i><br>4-character alphanumeric<br>YYMM | 'expiry_date'=>\$expiry_date                                      |
| electronic commerce indicator                                                                                                                                                                                                                                              | <i>String</i><br>1-character alphanumeric         | 'crypt_type'=>\$crypt                                             |
| <b>NOTE:</b> In Credential on File transactions where the request field e-commerce indicator is also being sent, the acceptable values for e-commerce indicator are dependent on the value sent for payment indicator; see request field definitions for more information. |                                                   |                                                                   |
| convenience fee amount                                                                                                                                                                                                                                                     | <i>String</i><br>9-character decimal              | \$convFeeTemplate = array<br>(convenience_fee=>\$convfee_amount); |

### Purchase with Convenience Fee transaction request fields – Optional

| Variable Name      | Type and Limits                                                                                                             | Set Method                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| customer ID        | <i>String</i><br>50-character alphanumeric<br><br><b>NOTE:</b><br>Some special characters are not allowed:<br><>\$%=?^{}[]\ | 'cust_id'=>\$cust_id                       |
| dynamic descriptor | <i>String</i><br>20-character alphanumeric                                                                                  | 'dynamic_descriptor'=>\$dynamic_descriptor |

| Variable Name   | Type and Limits                                                                                                              | Set Method                              |
|-----------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|                 | total of 22 characters including your merchant name and separator                                                            |                                         |
|                 | <div> <b>NOTE:</b><br/>           Some special characters are not allowed:<br/>           &lt;&gt;\$%=?^{}[]\         </div> |                                         |
| AVS Information | <i>Object</i><br><br>N/A                                                                                                     | \$mpgTxn->setAvsInfo<br>(\$mpgAvsInfo); |
| CVD Information | <i>Object</i><br><br>N/A                                                                                                     | \$mpgTxn->setCvdInfo<br>(\$mpgCvdInfo); |

### Sample Purchase with Convenience Fee

```

<?php
/* Moneris Gateway Canada Convenience Fee Account Required this transaction*/
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='monca00392';
$api_token='qYdISUhHiOdfTrlCLNpN';
// $status = 'false';
/***** Transaction Variables *****/
$orderid='ord-' . date("dmy-G:i:s");
$amount='10.00';
$pan='4242424242424242';
$expiry_date='1812';
$dynamic_descriptor='test';
/***** Transaction Array *****/
$txnArray=array(type=>'purchase',
order_id=>$orderid,
cust_id=>'cust',
amount=>$amount,
pan=>$pan,
expdate=>$expiry_date,
crypt_type=>'7',
dynamic_descriptor=>$dynamic_descriptor
);
/***** ConvFee Associative Array *****/
$convFeeTemplate = array(
convenience_fee=>'1.00'
);
/***** ConvFee Object *****/
$mpgConvFee = new mpgConvFeeInfo($convFeeTemplate);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Set ConvFee *****/
$mpgTxn->setConvFeeInfo($mpgConvFee);
/***** Request Object *****/

```

### Sample Purchase with Convenience Fee

```
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"CA" for sending transaction to Canadian
environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** mpgHttpPost Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
//Status check example
// $mpgHttpPost = new mpgHttpPostStatus($store_id,$api_token,$status,$mpgRequest);
/***** Response Object *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nISO = " . $mpgResponse->getISO());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nCardLevelResult = " . $mpgResponse->getCardLevelResult());
print("\nCfSuccess = " . $mpgResponse->getCfSuccess());
print("\nCfStatus = " . $mpgResponse->getCfStatus());
print("\nFeeAmount = " . $mpgResponse->getFeeAmount());
print("\nFeeRate = " . $mpgResponse->getFeeRate());
print("\nFeeType = " . $mpgResponse->getFeeType());
//print("\nStatusCode = " . $mpgResponse->getStatusCode());
//print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>
```

## 13.4 Purchase with Customer Info and Convenience Fee

Information below is a Purchase transaction request that also includes the Convenience Fee Info and Customer Information objects.

### Purchase with Customer Info and Convenience Fee transaction object definition

```
$txnArray = array('type'=>'purchase', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for Purchase with Customer Info and Convenience Fee transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Purchase with Customer Info and Convenience Fee transaction request fields – Required

For a full description of mandatory and optional values, see Appendix A Definition of Request Fields.

| Variable Name                 | Type and Limits                                                                                                                                                                     | Set Method                                                                  |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Convenience Fee Information   | <i>Object</i><br>N/A                                                                                                                                                                | <code>\$mpgConvFee = new<br/>mpgConvFeeInfo<br/>(\$convFeeTemplate);</code> |
| order ID                      | <i>String</i><br><br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                                                                                                      | <code>'order_id'=&gt;\$order_id</code>                                      |
| amount                        | <i>String</i><br><br>10-character decimal<br><br>Up to 7 digits (dollars) +<br>decimal point (.) + 2 digits<br>(cents) after the decimal<br>point<br><br><b>EXAMPLE:</b> 1234567.89 | <code>'amount'=&gt;\$amount</code>                                          |
| credit card number            | <i>String</i><br><br>max 20-character alpha-<br>numeric                                                                                                                             | <code>'pan'=&gt;\$pan</code>                                                |
| expiry date                   | <i>String</i><br><br>4-character alphanumeric<br><br>YYMM                                                                                                                           | <code>'expiry_date'=&gt;\$expiry_date</code>                                |
| electronic commerce indicator | <i>String</i><br><br>1-character alphanumeric                                                                                                                                       | <code>'crypt_type'=&gt;\$crypt</code>                                       |
| convenience fee amount        | <i>String</i><br><br>9-character decimal                                                                                                                                            | <code>\$mpgConvFee = new<br/>mpgConvFeeInfo<br/>(\$convFeeTemplate);</code> |

**Purchase with Customer Info and Convenience Fee transaction request fields – Optional**

| Variable Name | Type and Limits | Set Method                           |
|---------------|-----------------|--------------------------------------|
| customer ID   | <i>String</i>   | <code>'cust_id'=&gt;\$cust_id</code> |

| Variable Name                                                                                                                                                                                                                                                                                         | Type and Limits                                                                                                                                                                                                                                             | Set Method                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
|                                                                                                                                                                                                                                                                                                       | 50-character alphanumeric<br><br><div> <b>NOTE:</b><br/>           Some special characters are not allowed:<br/>           &lt;&gt;\$%=?^{}[]\         </div>                                                                                               |                                            |
| dynamic descriptor                                                                                                                                                                                                                                                                                    | <i>String</i><br><br>20-character alphanumeric<br><br>total of 22 characters including your merchant name and separator<br><br><div> <b>NOTE:</b><br/>           Some special characters are not allowed:<br/>           &lt;&gt;\$%=?^{}[]\         </div> | 'dynamic_descriptor'=>\$dynamic_descriptor |
| Customer Information                                                                                                                                                                                                                                                                                  | <i>Object</i><br><br>N/A                                                                                                                                                                                                                                    | \$mpgTxn->setCustInfo(\$mpgCustInfo);      |
| AVS Information                                                                                                                                                                                                                                                                                       | <i>Object</i><br><br>N/A                                                                                                                                                                                                                                    | \$mpgTxn->setAvsInfo(\$mpgAvsInfo);        |
| CVD Information<br><br><div> <b>NOTE:</b> When storing credentials on the initial transaction, the CVD object must be sent; for subsequent transactions using stored credentials, CVD can be sent with cardholder-initiated transactions only—<b>merchants must not store CVD information.</b> </div> | <i>Object</i><br><br>N/A                                                                                                                                                                                                                                    | \$mpgTxn->setCvdInfo(\$mpgCvdInfo);        |

## Sample Purchase with Customer Info and Convenience Fee

```

<?php
## Example php -q TestPurchase-CustInfo.php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='monca00392';
$api_token='qYdISUHiOdfTr1CLNpN';
/***** Transactional Variables *****/
$type='purchase';
$order_id='ord-' . date("dmy-G:i:s");
$cust_id='my cust id';
$amount='114.28';
$pan='4242424242424242';
$expiry_date='0812'; //December 2008
$crypt='7';
/***** Customer Information Variables *****/
$first_name = 'Cedric';
$last_name = 'Benson';
$company_name = 'Chicago Bears';
$address = '334 Michigan Ave';
$city = 'Chicago';
$province = 'Illinois';
$postal_code = 'M1M1M1';
$country = 'United States';
$phone_number = '453-989-9876';
$fax = '453-989-9877';
$tax1 = '1.01';
$tax2 = '1.02';
$tax3 = '1.03';
$shipping_cost = '9.95';
$email = 'Joe@widgets.com';
$instructions = "Make it fast";
/***** Line Item Variables *****/
$item_name = array();
$item_quantity = array();
$item_product_code = array();
$item_extended_amount = array();
$item_name[0] = 'Guy Lafleur Retro Jersey';
$item_quantity[0] = '1';
$item_product_code[0] = 'JRSCDA344';
$item_extended_amount[0] = '129.99';
$item_name[1] = 'Patrick Roy Signed Koho Stick';
$item_quantity[1] = '1';
$item_product_code[1] = 'JPREEA344';
$item_extended_amount[1] = '59.99';
/***** Customer Information Object *****/
$mpgCustInfo = new mpgCustInfo();
/***** Set Customer Information *****/
$billing = array(
    'first_name' => $first_name,
    'last_name' => $last_name,
    'company_name' => $company_name,
    'address' => $address,
    'city' => $city,
    'province' => $province,
    'postal_code' => $postal_code,
    'country' => $country,
    'phone_number' => $phone_number,
    'fax' => $fax,
    'tax1' => $tax1,
    'tax2' => $tax2,

```

## Sample Purchase with Customer Info and Convenience Fee

```

'tax3' => $tax3,
'shipping_cost' => $shipping_cost
);
$mpgCustInfo->setBilling($billing);
$shipping = array(
'first_name' => $first_name,
'last_name' => $last_name,
'company_name' => $company_name,
'address' => $address,
'city' => $city,
'province' => $province,
'postal_code' => $postal_code,
'country' => $country,
'phone_number' => $phone_number,
'fax' => $fax,
'tax1' => $tax1,
'tax2' => $tax2,
'tax3' => $tax3,
'shipping_cost' => $shipping_cost
);
$mpgCustInfo->setShipping($shipping);
$mpgCustInfo->setEmail($email);
$mpgCustInfo->setInstructions($instructions);
/***** Set Line Item Information *****/
$item[0] = array(
'name'=>$item_name[0],
'quantity'=>$item_quantity[0],
'product_code'=>$item_product_code[0],
'extended_amount'=>$item_extended_amount[0]
);
$item[1] = array(
'name'=>$item_name[1],
'quantity'=>$item_quantity[1],
'product_code'=>$item_product_code[1],
'extended_amount'=>$item_extended_amount[1]
);
$mpgCustInfo->setItems($item[0]);
$mpgCustInfo->setItems($item[1]);
/***** ConvFee Associative Array *****/
$convFeeTemplate = array(
'convenience_fee'=>'2.00'
);
/***** ConvFee Object *****/
$mpgConvFee = new mpgConvFeeInfo($convFeeTemplate);
/***** Transactional Associative Array *****/
$txnArray=array(
'type'=>$type,
'order_id'=>$order_id,
'cust_id'=>$cust_id,
'amount'=>$amount,
'pan'=>$pan,
'expdate'=>$expiry_date,
'crypt_type'=>$crypt
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Set Customer Information *****/
$mpgTxn->setCustInfo($mpgCustInfo);
/***** Set ConvFee *****/
$mpgTxn->setConvFeeInfo($mpgConvFee);

```

### Sample Purchase with Customer Info and Convenience Fee

```

/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
?>

```

## 13.5 Purchase with 3-D Secure and Convenience Fee

Information below describes a Purchase with 3-D Secure transaction request that also includes the Convenience Fee Info object.

### Convenience Fee Purchase with 3-D Secure transaction object definition

```
$txnArray = array('type'=>'cavv_purchase', ...);
```

```
$mpgTxn = new mpgTransaction($txnArray);
```

### HttpPostRequest object for Convenience Fee Purchase w/ 3-D Secure transaction

```
$mpgRequest = new mpgRequest($mpgTxn);
```

```
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
```

### Convenience Fee Purchase with 3-D Secure transaction request fields – Required

For a full description of mandatory and optional values, see Appendix A Definition of Request Fields.

| Variable Name               | Type and Limits | Set Method                                                   |
|-----------------------------|-----------------|--------------------------------------------------------------|
| Convenience Fee Information | Object<br>N/A   | \$mpgConvFee = new<br>mpgConvFeeInfo<br>(\$convFeeTemplate); |



| Variable Name                                       | Type and Limits                                                                                                                                                    | Set Method                                                                    |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| order ID                                            | <i>String</i><br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces                                                                                         | 'order_id'=>\$order_id                                                        |
| amount                                              | <i>String</i><br>10-character decimal<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><div>EXAMPLE: 1234567.89</div> | 'amount'=>\$amount                                                            |
| credit card number                                  | <i>String</i><br>max 20-character alphanumeric                                                                                                                     | 'pan'=>\$pan                                                                  |
| expiry date                                         | <i>String</i><br>4-character alphanumeric<br>YYMM                                                                                                                  | 'expiry_date'=>\$expiry_date                                                  |
| electronic commerce indicator                       | <i>String</i><br>1-character alphanumeric                                                                                                                          | 'crypt_type'=>\$crypt                                                         |
| Cardholder Authentication Verification Value (CAVV) | <i>String</i><br>50-character alphanumeric                                                                                                                         | cavv=>\$cavv                                                                  |
| convenience fee amount                              | <i>String</i><br>9-character decimal                                                                                                                               | <pre>\$convFeeTemplate = array (convenience_fee=&gt;\$convfee_ amount);</pre> |

### Purchase with 3-D Secure and Convenience Fee transaction request fields – Optional

| Variable Name | Type and Limits | Set Method         |
|---------------|-----------------|--------------------|
| status check  | <i>Boolean</i>  | \$mpgHttpPost =new |

| Variable Name                 | Type and Limits                                                                                                                                                                                                                                       | Set Method                                                                       |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|                               | true/false                                                                                                                                                                                                                                            | <code>mpgHttpPostStatus(\$store_id, \$api_token, \$status, \$mpgRequest);</code> |
| customer ID                   | <i>String</i><br>50-character alphanumeric<br><div> <b>NOTE:</b><br/>           Some special characters are not allowed:<br/> <code>&lt; &gt; \$ % = ? ^ { } [ ] \</code> </div>                                                                      | <code>'cust_id'=&gt;\$cust_id</code>                                             |
| dynamic descriptor            | <i>String</i><br>20-character alphanumeric<br>total of 22 characters including your merchant name and separator<br><div> <b>NOTE:</b><br/>           Some special characters are not allowed:<br/> <code>&lt; &gt; \$ % = ? ^ { } [ ] \</code> </div> | <code>'dynamic_descriptor'=&gt;\$dynamic_descriptor</code>                       |
| electronic commerce indicator | <i>String</i><br>1-character alphanumeric                                                                                                                                                                                                             | <code>'crypt_type'=&gt;\$crypt</code>                                            |
| Customer Information          | <i>Object</i><br>N/A                                                                                                                                                                                                                                  | <code>\$mpgTxn-&gt;setCustInfo(\$mpgCustInfo);</code>                            |
| AVS Information               | <i>Object</i><br>N/A                                                                                                                                                                                                                                  | <code>\$mpgTxn-&gt;setAvsInfo(\$mpgAvsInfo);</code>                              |
| CVD Information               | <i>Object</i><br>N/A<br><div> <b>NOTE:</b> When storing credentials on the initial transaction, the CVD object must be sent; for subsequent transactions using stored credentials, CVD can be sent         </div>                                     | <code>\$mpgTxn-&gt;setCvdInfo(\$mpgCvdInfo);</code>                              |

| Variable Name                                                                         | Type and Limits | Set Method |
|---------------------------------------------------------------------------------------|-----------------|------------|
| with cardholder-initiated transactions only—merchants must not store CVD information. |                 |            |

### Sample Purchase with 3-D Secure and Convenience Fee

```

<?php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='monca00392';
$api_token='qYdISUhHiOdfTrlCLNpN';
// $status = 'false';
/***** Transactional Variables *****/
$type='cavv_purchase';
$order_id="ord-".date("dmy-G:i:s");
$cust_id='customer1';
$amount='1.00';
$pan='4242424242424242';
$expiry_date='0912';
$cavv='AAABBJg0VhI0VniQEjRWAAAAAA';
// $cavv='AAABBJg0VhI0VniQEjRWAAAAAA=';
$commcard_invoice='Invoice 5757FRJ8';
$commcard_tax_amount='1.00';
$crypt_type = '7';
/***** Transaction Associative Array *****/
$txnArray=array(
    type=>$type,
    order_id=>$order_id,
    cust_id=>$cust_id,
    amount=>$amount,
    pan=>$pan,
    expdate=>$expiry_date,
    cavv=>$cavv,
    commcard_invoice=>$commcard_invoice,
    commcard_tax_amount=>$commcard_tax_amount,
    crypt_type=>$crypt_type, //mandatory for AMEX only
    dynamic_descriptor=>'test'
);
/***** ConvFee Associative Array *****/
$convFeeTemplate = array(
    convenience_fee=>'1.00'
);
/***** ConvFee Object *****/
$mpgConvFee = new mpgConvFeeInfo($convFeeTemplate);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Set ConvFee *****/
$mpgTxn->setConvFeeInfo($mpgConvFee);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "CA" for sending transaction to Canadian
environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/

```

**Sample Purchase with 3-D Secure and Convenience Fee**

```

$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
//Status check example
//$mpgHttpPost = new mpgHttpPostStatus($store_id,$api_token,$status,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTicket = " . $mpgResponse->getTicket());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nCardLevelResult = " . $mpgResponse->getCardLevelResult());
print("\nCavvResultCode = " . $mpgResponse->getCavvResultCode());
print("\nCfSuccess = " . $mpgResponse->getCfSuccess());
print("\nCfStatus = " . $mpgResponse->getCfStatus());
print("\nFeeAmount = " . $mpgResponse->getFeeAmount());
print("\nFeeRate = " . $mpgResponse->getFeeRate());
print("\nFeeType = " . $mpgResponse->getFeeType());
//print("\nStatusCode = " . $mpgResponse->getStatusCode());
//print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>

```

## 14 Recurring Billing

- 14.1 About Recurring Billing
- 14.2 Purchase with Recurring Billing
- 14.3 Recurring Billing Update
- 14.4 Recurring Billing Response Fields and Codes
- 14.5 Credential on File and Recurring Billing

### 14.1 About Recurring Billing

Recurring Billing allows you to set up payments whereby Moneris automatically processes the transactions and bills customers on your behalf based on the billing cycle information you provide.

Recurring Billing series are created by sending the Recurring Billing object in these transactions:

- Purchase
- Purchase with Vault
- Purchase with 3-D Secure (cavvPurchase)

You can modify a Recurring Billing series after it has been created by sending the Recurring Billing Update administrative transaction.

**NOTE:** Alternatively, if you prefer to manage recurring series on your own merchant system, you can send the periodic payments as basic Purchase transactions with the e-commerce indicator (`crypt_type`) value = 2 and with the Credential on File info object included.

### 14.2 Purchase with Recurring Billing

#### Recurring Billing Info Object Definition

```
$recurArray = array(  
    'recur_unit'=>$recurUnit, // (day | week | month)  
    'start_date'=>$startDate, //yyyy/mm/dd  
    'num_rekurs'=>$numRekurs,  
    'start_now'=>$startNow,  
    'period' => $recurInterval,
```

```
'recur_amount'=> $recurAmount
);

$mpgRecur = new mpgRecur($recurArray);
```

### Transaction object set method

```
$mpgTxn->setRecur($mpgRecur);
```

### Recurring Billing Info Object Request Fields

| Variable Name    | Type and Limits            | Description                                                                                                                                                                                                                   |
|------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| number of recurs | String<br>numeric<br>1-999 | The number of times that the transaction must recur                                                                                                                                                                           |
| period           | String<br>numeric<br>1-999 | Number of recur unit intervals that must pass between recurring billings                                                                                                                                                      |
| start date       | String<br>YYYYMMDD format  | Date of the first future recurring billing transaction; this must be a date in the future<br><br>If an additional charge will be made immediately, the start now variable must be set to true                                 |
| start now        | String<br>true/false       | Set to true if a charge will be made against the card immediately; otherwise set to false<br><br>When set to false, use Card Verification prior to sending the Purchase with Recurring Billing and Credential on File objects |
| recurring amount | String                     | Dollar amount of the recurring transaction                                                                                                                                                                                    |

**NOTE:** Amount to be billed immediately can differ from the subsequent recurring amounts

| Variable Name | Type and Limits                                                                                                                                                                 | Description                                                                                                                      |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
|               | 10-character decimal, minimum three digits<br><br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><br><div>EXAMPLE: 1234567.89</div> | This amount will be billed on the start date, and then billed repeatedly based on the interval defined by period and recur unit  |
| recur unit    | <i>String</i><br><br>day, week, month or eom                                                                                                                                    | Unit to be used as a basis for the interval<br><br>Works in conjunction with the period variable to define the billing frequency |

### Sample Purchase with Recurring Billing

```

<?php
##
## Example php -q TestPurchase-Recur.php store3 yesguy unique_order_id
##
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id = 'store5';
$api_token = 'yesguy';
/***** Recur Variables *****/
$recurUnit = 'eom';
$startDate = '2018/11/30';
$numRekurs = '4';
$recurInterval = '10';
$recurAmount = '31.00';
$startNow = 'true';
/***** Transactional Variables *****/
$orderId = 'ord-' . date("dmy-G:i:s");
$custId = 'student_number';
$creditCard = '5454545454545454';
$nowAmount = '10.00';
$expiryDate = '0912';
$scriptType = '7';
/***** Recur Associative Array *****/
$recurArray = array('recur_unit'=>$recurUnit, // (day | week | month)
'start_date'=>$startDate, // yyyy/mm/dd
'num_rekurs'=>$numRekurs,
'start_now'=>$startNow,
'period' => $recurInterval,
'recur_amount'=> $recurAmount
);
$mpgRecur = new mpgRecur($recurArray);
/***** Transactional Associative Array *****/
$txnArray=array('type'=>'purchase',
'order_id'=>$orderId,

```

## Sample Purchase with Recurring Billing

```

'cust_id'=>$custId,
'amount'=>$nowAmount,
'pan'=>$creditCard,
'expdate'=>$expiryDate,
'crypt_type'=>$cryptType
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Recur Object *****/
$mpgTxn->setRecur($mpgRecur);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setPaymentIndicator("R");
$cof->setPaymentInformation("2");
$cof->setIssuerId("168451306048014");
$mpgTxn->setCofInfo($cof);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id, $api_token, $mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print ("\nCardType = " . $mpgResponse->getCardType());
print ("\nTransAmount = " . $mpgResponse->getTransAmount());
print ("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print ("\nReceiptId = " . $mpgResponse->getReceiptId());
print ("\nTransType = " . $mpgResponse->getTransType());
print ("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print ("\nResponseCode = " . $mpgResponse->getResponseCode());
print ("\nISO = " . $mpgResponse->getISO());
print ("\nMessage = " . $mpgResponse->getMessage());
print ("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print ("\nAuthCode = " . $mpgResponse->getAuthCode());
print ("\nComplete = " . $mpgResponse->getComplete());
print ("\nTransDate = " . $mpgResponse->getTransDate());
print ("\nTransTime = " . $mpgResponse->getTransTime());
print ("\nTicket = " . $mpgResponse->getTicket());
print ("\nTimedOut = " . $mpgResponse->getTimedOut());
print ("\nRecurSuccess = " . $mpgResponse->getRecurSuccess());
print ("\nIssuerId = " . $mpgResponse->getIssuerId());
?>

```

## 14.3 Recurring Billing Update

After you have set up a Recurring Billing transaction series, you can change some of the details of the series as long as it has not yet completed the preset recurring duration (i.e., it hasn't terminated yet).

Before sending a Recurring Billing Update transaction that updates the credit card number, you must send a Card Verification request. This requirement does not apply if you are only updating the schedule or amount.



**Things to Consider:**

- When completing the update recurring billing portion please keep in mind that the recur bill dates cannot be changed to have an end date greater than 10 years from today and cannot be changed to have an end date end today or earlier.

**Recurring Billing Update transaction object definition**

```
$txnArray=array('type'=>'recur_update',... );
```

**HttpPostRequest object for Recurring Billing Update transaction**

```
$mpgTxn = new mpgTransaction($txnArray);
```

```
$mpgRequest = new mpgRequest($mpgTxn);
```

**Recurring Billing Update transaction values****Table 1 Recurring Billing Update – Basic Required Fields**

| Variable Name        | Type and Limits                            | Set Method           |
|----------------------|--------------------------------------------|----------------------|
| Order ID<br>order_id | <i>String</i><br>50-character alphanumeric | 'cust_id'=>\$cust_id |

**Table 2 Recurring Billing Update – Basic Optional Fields**

| Variable Name             | Type and Limits                            | Set Method               |
|---------------------------|--------------------------------------------|--------------------------|
| Customer ID<br>cust_id    | <i>String</i><br>50-character alphanumeric | 'cust_id'=>\$cust_id     |
| Credit card number<br>pan | <i>String</i><br>20-character alphanumeric | 'pan'=>\$pan             |
| Expiry date<br>expdate    | <i>String</i><br>YYMM                      | 'expdate'=>\$expiry_date |

**Table 3 Recurring Billing Update – Recurring Billing Required Fields**

| Variable Name                                | Type and Limits                 | Set Method                       | Description                                                                                                                                                                                                                                                                                          |
|----------------------------------------------|---------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recurring amount<br>recur_amount             | <i>String</i>                   | 'recur_amount'=>\$recur_amount   | Changes the amount that is billed recurrently<br><br>The change takes effect on the next charge                                                                                                                                                                                                      |
| Add number of recurs<br>add_num              | <i>String</i><br>numeric, 1-999 | 'add_num_rekurs'=> \$add_num     | <b>Adds</b> to the given number of recurring transactions to the current (remaining) number<br><br>This can be used if a customer decides to extend a membership or subscription<br><br>Cannot be used to decrease the current number of recurring transactions; use Change number of recurs instead |
| Change number of recurs<br>total_num         | <i>String</i><br>numeric, 1-999 | 'total_num_rekurs'=> \$total_num | <b>Replaces</b> the current (remaining) number of recurring transactions                                                                                                                                                                                                                             |
| Hold recurring billing<br>hold               | <i>String</i><br>true/false     | 'hold' => \$hold                 | Temporarily pauses recurring billing<br><br>While a transaction is on hold, it is not billed for the recurring amount; however, the number of remaining recurs continues to be decremented during that time                                                                                          |
| Terminate recurring transaction<br>terminate | <i>String</i><br>true/false     | 'terminate' => \$terminate       | Terminates recurring billing<br><br><b>NOTE:</b> After it has been terminated, a recurring transaction <b>cannot</b> be reactivated; a new purchase transaction with recurring billing must be submitted.                                                                                            |

### Sample Recurring Billing Update

```

<?php
##
## Example php -q TestRecurUpdate.php store1
##
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
/***** Transactional Variables *****/
$type='recur_update';
$cust_id='my cust id';
$order_id='ord-110515-10:45:21';
$recur_amount='1.00';
$pan='4242424242424242';
$expiry_date='1811';
$add_num='';
$total_num='7';
$hold = 'false';
$terminate = 'false';
/***** Transactional Associative Array *****/
$txnArray=array('type'=>$type,
'order_id'=>$order_id,
'cust_id'=>$cust_id,
'recur_amount'=>$recur_amount,
'pan'=>$pan,
'expdate'=>$expiry_date,
'add_num_rekurs' => $add_num,
'total_num_rekurs' => $total_num,
'hold' => $hold,
'terminate' => $terminate
);
/***** Credential on File *****/
$cof = new CofInfo();
$cof->setIssuerId("168451306048014");
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
$mpgTxn->setCofInfo($cof);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); // "US" for sending transaction to US environment
$mpgRequest->setTestMode(true); // false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost = new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());
print("\nTimedOut = " . $mpgResponse->getTimedOut());
print("\nRecurUpdateSuccess = " . $mpgResponse->getRecurUpdateSuccess());
print("\nNextRecurDate = " . $mpgResponse->getNextRecurDate());
print("\nRecurEndDate = " . $mpgResponse->getRecurEndDate());
?>

```

## 14.4 Recurring Billing Response Fields and Codes

Table 12 outlines the response fields that are part of recurring billing. Some are available when you set up recurring billing (such as with a Purchase transaction), and some are available when you update an existing transaction with the Recurring Billing transaction.

### Receipt object definition

```
$mpgResponse=$mpgHttpPost->getMpgResponse();
```

**Table 12: Recurring Billing response fields**

| Value                                                     | Type                                                        | Limits              | Get method                              |
|-----------------------------------------------------------|-------------------------------------------------------------|---------------------|-----------------------------------------|
|                                                           | Description                                                 |                     |                                         |
| Transaction object with Recurring Billing response fields |                                                             |                     |                                         |
| Response code                                             | String                                                      | 3-character numeric | \$mpgResponse->getResponseCode ()       |
|                                                           | See Table 13: for a description of possible response codes. |                     |                                         |
| Recur success                                             | String                                                      | TBD                 | \$mpgResponse->getRecurSuccess ()       |
|                                                           | Indicates whether the transaction successfully registered   |                     |                                         |
| Recur update object response fields                       |                                                             |                     |                                         |
| Recur update success                                      | String                                                      | true/false          | \$mpgResponse->getRecurUpdateSuccess () |
|                                                           | Indicates whether the transaction successfully updated.     |                     |                                         |
| Next recur date                                           | String                                                      | yyyy-mm-dd format   | \$mpgResponse->getNextRecurDate ()      |
|                                                           | Indicates when the transaction will be billed again.        |                     |                                         |
| Recur end date                                            | String                                                      | yyyy-mm-dd format   | \$mpgResponse->getRecurEndDate ()       |
|                                                           | Indicates when the Recurring Billing Transaction will end.  |                     |                                         |

The Recur Update response is a 3-digit numeric value. The following is a list of all possible responses after a Recur Update transaction has been sent.

**Table 13: Recur update response codes**

| Request Value | Definition                                                        |
|---------------|-------------------------------------------------------------------|
| 001           | Recurring transaction successfully updated (optional: terminated) |
| 983           | Cannot find the previous transaction                              |
| 984           | Data error: (optional: field name)                                |
| 985           | Invalid number of recurs                                          |
| 986           | Incomplete: timed out                                             |

Table 13: Recur update response codes (continued)

| Request Value | Definition           |
|---------------|----------------------|
| null          | Error: Malformed XML |

## 14.5 Credential on File and Recurring Billing

**NOTE:** The value of the **payment indicator** field must be **R** when sending Recurring Billing transactions.

For Recurring Billing transactions which are set to start **immediately**:

1. Send a Purchase transaction request with both the Recurring Billing and Credential on File info objects (with Recurring Billing object field **start now** = true)

For Recurring Billing transactions which are set to start on a **future** date:

1. Send Card Verification transaction request including the Credential on File info object to get the Issuer ID
2. Send Purchase transaction request with the Recur and Credential on File info objects included

For updating a Recurring Billing series where you are updating the card number (does not apply if you are only modifying the schedule or amount in a recurring series):

1. Send Card Verification request including the Credential on File info object to get the Issuer ID
2. Send a Recurring Billing Update transaction

For more information about the Recurring Billing object, see Definition of Request Fields – Recurring.



## 15 Customer Information

- 15.1 Using the Customer Information Object
- 15.2 Customer Information Sample Code

The Customer Information object offers a number of fields to be submitted as part of the financial transaction, and stored by Moneris. These details may be viewed in the future in the Merchant Resource Center.

The following transactions support the Customer Information object :

- Purchase (Basic, Interac Debit and Vault)
- Pre-Authorization (Basic and Vault)
- Re-Authorization (Basic)

The Customer Information object holds three types of information:

- Billing/Shipping information
- Miscellaneous customer information properties
- Item information

### Things to Consider:

- If you send characters that are not included in the allowed list, these extra transaction details may not be stored.
- All fields are alphanumeric and allow the following characters: a-z A-Z 0-9 \_ - : . @ \$ = /
- All French accents should be encoded as HTML entities, such as &acute;.
- The data sent in Billing and Shipping Address fields will not be used for any address verification.

### 15.1 Using the Customer Information Object

- 15.1.1 Customer Info Object – Miscellaneous Properties
- 15.1.2 Customer Info Object – Billing/Shipping Information
- 15.1.3 Customer Info Object – Item Information

In addition to instantiating a transaction object and a connection object (as you would for a normal transaction), you must instantiate a CustInfo object.

Any transaction that supports CustInfo has a setCustInfo method. This is used to write the customer information to the transaction object before writing the transaction object to the connection object.

### CustInfo object definition

```
$mpgCustInfo = new mpgCustInfo();
```

### Transaction object set method

```
$mpgTxn->setCustInfo($mpgCustInfo);
```

## 15.1.1 Customer Info Object – Miscellaneous Properties

While most of the Customer Information data is organized into objects, there are some values that are properties of the CustInfo object itself. They are explained in the table below.

**Table 14: CustInfo object miscellaneous properties**

| Value         | Type   | Limits                     | Set method                                              |
|---------------|--------|----------------------------|---------------------------------------------------------|
| Email Address | String | 60-character alphanumeric  | <code>\$mpgCustInfo-&gt;setEmail(\$email);</code>       |
| Instructions  | String | 100-character alphanumeric | <code>\$mpgCustInfo-&gt;setInstructions(\$note);</code> |

## 15.1.2 Customer Info Object – Billing/Shipping Information

Billing and shipping information is stored as part of the Customer Information object. They can be written to the object in one of two ways:

- Using set methods
- Using hash tables

Whichever method you use, you will be writing the information found in the table below for both the billing information and the shipping information.

All values are alphanumeric strings. Their maximum lengths are given in the Limit column.

**Table 15: Billing and shipping information values**

| Value      | Limit | Hash table key |
|------------|-------|----------------|
| First name | 30    | "first_name"   |



**Table 15: Billing and shipping information values (continued)**

| Value                      | Limit | Hash table key  |
|----------------------------|-------|-----------------|
| Last name                  | 30    | "last_name"     |
| Company name               | 50    | "company_name"  |
| Address                    | 70    | "address"       |
| City                       | 30    | "city"          |
| Province/State             | 30    | "province"      |
| Postal/Zip code            | 30    | "postal_code"   |
| Country                    | 30    | "country"       |
| Phone number (voice)       | 30    | "phone"         |
| Fax number                 | 30    | "fax"           |
| Federal tax                | 10    | "tax1"          |
| Provincial/State tax       | 10    | "tax2"          |
| County/Local/Specialty tax | 10    | "tax3"          |
| Shipping cost              | 10    | "shipping_cost" |

### 15.1.2.1 Set Methods for Billing and Shipping Info

The billing information and the shipping information for a given CustInfo object are written by using the `$mpgCustInfo->setBilling($billing)` ; and `$mpgCustInfo->setShipping($shipping)` ; methods respectively:

```
$billing = array(
    'first_name' => $first_name,
    'last_name' => $last_name,
    'company_name' => $company_name,
    'address' => $address,
    'city' => $city,
    'province' => $province,
```

```
'postal_code' => $postal_code,
'country' => $country,
'phone_number' => $phone_number,
'fax' => $fax,
'tax1' => $tax1,
'tax2' => $tax2,
'tax3' => $tax3,
'shipping_cost' => $shipping_cost
);
$mpgCustInfo->setBilling($billing);
$shipping = array(
'first_name' => $first_name,
'last_name' => $last_name,
'company_name' => $company_name,
'address' => $address,
'city' => $city,
'province' => $province,
'postal_code' => $postal_code,
'country' => $country,
'phone_number' => $phone_number,
'fax' => $fax,
'tax1' => $tax1,
'tax2' => $tax2,
'tax3' => $tax3,
'shipping_cost' => $shipping_cost
);
$mpgCustInfo->setShipping($shipping);
```

Both of these methods have the same set of mandatory arguments. They are described in the Billing and shipping information values table in 15.1.2.1 Set Methods for Billing and Shipping Info.

For sample code, see 15.2 Customer Information Sample Code.

### 15.1.2.2 Using Hash Tables for Billing and Shipping Info

Writing billing or shipping information using hash tables is done as follows:

1. Instantiate a CustInfo object.
2. Instantiate a hash table object. (The sample code uses a different hash table for billing and shipping for clarity purposes. However, the skillful developer can re-use the same one.)
3. Build the hash table using put methods with the hash table keys found in the Billing and shipping information values table in 15.1.2 Customer Info Object – Billing/Shipping Information.
4. Call the CustInfo object's setBilling/setShipping method to pass the hash table information to the CustInfo object
5. Call the transaction object's setCustInfo method to write the CustInfo object (with the billing/-shipping information to the transaction object.

For sample code, see 15.2 Customer Information Sample Code.

### 15.1.3 Customer Info Object – Item Information

The Customer Information object can hold information about multiple items. For each item, the values in the table below can be written.

All values are strings, but note the guidelines in the Limits column.

**Table 16: Item information values**

| Value                | Limits                                                                               | Hash table key    |
|----------------------|--------------------------------------------------------------------------------------|-------------------|
| Item name            | 45-character alphanumeric                                                            | "name"            |
| Item quantity        | 5-character numeric                                                                  | "quantity"        |
| Item product code    | 20-character alphanumeric                                                            | "product_code"    |
| Item extended amount | 9-character decimal with at least 3 digits and 2 penny values.<br><br>0.01-999999.99 | "extended_amount" |

One way of representing multiple items is with four arrays. This is the method used in the sample code. However, there are two ways to write the item information to the CustInfo object:

- Set methods
- Hash tables

### 15.1.3.1 Set Methods for Item Information

All the item information found in the Item information values table in 15.1.3 Customer Info Object – Item Information is written to the CustInfo object in one instruction for a given item. Such as:

```
customer.setItem(item_description, item_quantity, item_product_code, item_
extended_amount);
```

For sample code (showing how to use arrays to write information about two items), see 15.2 Customer Information Sample Code.

### 15.1.3.2 Using Hash Tables for Item Information

Writing item information using hash tables is done as follows:

1. Instantiate a CustInfo object.
2. Instantiate a hash table object. (The sample code uses a different hash table for each item for clarity purposes. However, the skillful developer can re-use the same one.)
3. Build the hash table using put methods with the hash table keys in the Item information values table in 15.1.3 Customer Info Object – Item Information.
4. Call the CustInfo object's setItem method to pass the hash table information to the CustInfo object
5. Call the transaction object's setCustInfo method to write the CustInfo object (with the item information to the transaction object.

For sample code that shows how to use arrays to write information about two items, see 15.2 Customer Information Sample Code.

## 15.2 Customer Information Sample Code

Below is an example of a Basic Purchase with Customer Information transaction.

Note that the two items ordered are represented by four arrays, and the billing and shipping details are the same.

### Sample Purchase with Customer Information

```
## Example php -q TestPurchase-CustInfo.php
require "../mpgClasses.php";
/***** Request Variables *****/
$store_id='store5';
$api_token='yesguy';
/***** Transactional Variables *****/
$type='purchase';
$order_id='ord-'.date("dmy-G:i:s");
$cust_id='my cust id';
$amount='1.00';
$pan='4242424242424242';
```

### Sample Purchase with Customer Information

```

$expiry_date='0812'; //December 2008
$crypt='7';
/***** Customer Information Variables *****/
$first_name = 'Cedric';
$last_name = 'Benson';
$company_name = 'Chicago Bears';
$address = '334 Michigan Ave';
$city = 'Chicago';
$province = 'Illinois';
$postal_code = 'M1M1M1';
$country = 'United States';
$phone_number = '453-989-9876';
$fax = '453-989-9877';
$tax1 = '1.01';
$tax2 = '1.02';
$tax3 = '1.03';
$shipping_cost = '9.95';
$email = 'Joe@widgets.com';
$instructions = "Make it fast";
/***** Line Item Variables *****/
$item_name[0] = 'Guy Lafleur Retro Jersey';
$item_quantity[0] = '1';
$item_product_code[0] = 'JRSCDA344';
$item_extended_amount[0] = '129.99';
$item_name[1] = 'Patrick Roy Signed Koho Stick';
$item_quantity[1] = '1';
$item_product_code[1] = 'JPREEA344';
$item_extended_amount[1] = '59.99';
/***** Customer Information Object *****/
$mpgCustInfo = new mpgCustInfo();
/***** Set Customer Information *****/
$billing = array(
    'first_name' => $first_name,
    'last_name' => $last_name,
    'company_name' => $company_name,
    'address' => $address,
    'city' => $city,
    'province' => $province,
    'postal_code' => $postal_code,
    'country' => $country,
    'phone_number' => $phone_number,
    'fax' => $fax,
    'tax1' => $tax1,
    'tax2' => $tax2,
    'tax3' => $tax3,
    'shipping_cost' => $shipping_cost
);
$mpgCustInfo->setBilling($billing);
$shipping = array(
    'first_name' => $first_name,
    'last_name' => $last_name,
    'company_name' => $company_name,
    'address' => $address,
    'city' => $city,
    'province' => $province,
    'postal_code' => $postal_code,
    'country' => $country,
    'phone_number' => $phone_number,
    'fax' => $fax,

```

## Sample Purchase with Customer Information

```

'tax1' => $tax1,
'tax2' => $tax2,
'tax3' => $tax3,
'shipping_cost' => $shipping_cost
);
$mpgCustInfo->setShipping($shipping);
$mpgCustInfo->setEmail($email);
$mpgCustInfo->setInstructions($instructions);
/***** Set Line Item Information *****/
$item[0] = array(
    'name'=>$item_name[0],
    'quantity'=>$item_quantity[0],
    'product_code'=>$item_product_code[0],
    'extended_amount'=>$item_extended_amount[0]
);
$item[1] = array(
    'name'=>$item_name[1],
    'quantity'=>$item_quantity[1],
    'product_code'=>$item_product_code[1],
    'extended_amount'=>$item_extended_amount[1]
);
$mpgCustInfo->setItems($item[0]);
$mpgCustInfo->setItems($item[1]);
/***** Transactional Associative Array *****/
$txnArray=array(
    'type'=>$type,
    'order_id'=>$order_id,
    'cust_id'=>$cust_id,
    'amount'=>$amount,
    'pan'=>$pan,
    'expdate'=>$expiry_date,
    'crypt_type'=>$crypt
);
/***** Transaction Object *****/
$mpgTxn = new mpgTransaction($txnArray);
/***** Set Customer Information *****/
$mpgTxn->setCustInfo($mpgCustInfo);
/***** Request Object *****/
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA"); //"US" for sending transaction to US environment
$mpgRequest->setTestMode(true); //false for production transactions
/***** HTTPS Post Object *****/
$mpgHttpPost =new mpgHttpPost($store_id,$api_token,$mpgRequest);
/***** Response *****/
$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nCardType = " . $mpgResponse->getCardType());
print("\nTransAmount = " . $mpgResponse->getTransAmount());
print("\nTxnNumber = " . $mpgResponse->getTxnNumber());
print("\nReceiptId = " . $mpgResponse->getReceiptId());
print("\nTransType = " . $mpgResponse->getTransType());
print("\nReferenceNum = " . $mpgResponse->getReferenceNum());
print("\nResponseCode = " . $mpgResponse->getResponseCode());
print("\nISO = " . $mpgResponse->getISO());
print("\nMessage = " . $mpgResponse->getMessage());
print("\nIsVisaDebit = " . $mpgResponse->getIsVisaDebit());
print("\nAuthCode = " . $mpgResponse->getAuthCode());
print("\nComplete = " . $mpgResponse->getComplete());
print("\nTransDate = " . $mpgResponse->getTransDate());
print("\nTransTime = " . $mpgResponse->getTransTime());

```

**Sample Purchase with Customer Information**

```
print("\nTicket = " . $mpgResponse->getTicket());  
print("\nTimedOut = " . $mpgResponse->getTimedOut());  
?>
```





## 16 Status Check

- 16.1 About Status Check
- 16.2 Using Status Check Response Fields
- 16.3 Sample Purchase with Status Check

### 16.1 About Status Check

Status Check is a connection object value that allows merchants to verify whether a previously sent transaction was processed successfully.

To submit a Status Check request, resend the original transaction with all the same parameter values, but set the status check value to either `true` or `false`.

Once set to “true”, the gateway will check the status of a transaction that has an `order_id` that matches the one passed.

- If the transaction is found, the gateway will respond with the specifics of that transaction.
- If the transaction is not found, the gateway will respond with a not found message.

Once it is set to “false”, the transaction will process as a new transaction.

For example, if you send a Purchase transaction with Status Check, include the same values as the original Purchase such as the order ID and the amount.

The feature must be enabled in your merchant profile. To have it enabled, contact Moneris.

#### Things to Consider:

- The Status Check request should only be used once and immediately (within 2 minutes) after the last transaction that had failed.
- The Status Check request should not be used to check `openTotals` & `batchClose` requests.
- Do not resend the Status Check request if it has timed out. Additional investigation is required.

### 16.2 Using Status Check Response Fields

After you have used the connection object to send a Status Check request, you can use the Receipt object to obtain the information you want regarding the success of the original transaction.

The status response fields related to the status check are Status Code and Status Message.

Possible Status Code response values:

- 0-49: successful transaction
- 50-999: unsuccessful transaction.

Possible Status Message response values:

- Found: Status code is 0-49
- Not found or Null: Status code is 50-999)

If the Status Message is `Found`, all other response fields are the same as those from the original transaction.

If the Status Message is `Not found`, all other response fields will be `Null`.

## 16.3 Sample Purchase with Status Check

### Sample Purchase transaction with Status Check

```
<?php
require "../mpgClasses.php";
$store_id='store5';
$api_token='yesguy';
$status_check = 'true';

$txnArray=array('type'=>'purchase',
    'order_id'=>'order',
    'cust_id'=>'cust',
    'amount'=>'1.00',
    'pan'=>'4242424242424242',
    'expdate'=>'2202',
    'crypt_type'=>'1',
    'dynamic_descriptor'=>'');

$mpgTxn = new mpgTransaction($txnArray);
$mpgRequest = new mpgRequest($mpgTxn);
$mpgRequest->setProcCountryCode("CA");
$mpgRequest->setTestMode(true); //false for production transactions
$mpgHttpPost =new mpgHttpPostStatus($store_id,$api_token,$status_check,$mpgRequest);

$mpgResponse=$mpgHttpPost->getMpgResponse();
print("\nStatusCode = " . $mpgResponse->getStatusCode());
print("\nStatusMessage = " . $mpgResponse->getStatusMessage());
?>
```

## 17 Testing a Solution

- 17.1 About the Merchant Resource Center
- 17.2 Logging In to the QA Merchant Resource Center
- 17.3 Test Credentials for Merchant Resource Center
- 17.4 Getting a Unique Test Store ID and API Token
- 17.5 Processing a Transaction
- 17.6 Testing MPI Solutions
- 1 Testing Visa Checkout
- 1 Test Cards
- 17.8 Simulator Host

### 17.1 About the Merchant Resource Center

The Merchant Resource Center is the user interface for Moneris Gateway services. There is also a QA version of the Merchant Resource Center site specifically allocated for you and other developers to use to test your API integrations with the gateway.

You can access the Merchant Resource Center in the test environment at:

<https://esqa.moneris.com/mpg> (Canada)

The test environment is generally available 24/7, but 100% availability is not guaranteed. Also, please be aware that other merchants are using the test environment in the Merchant Resource Center. Therefore, you may see transactions and user IDs that you did not create. As a courtesy to others who are testing, we ask that you use only the transactions/users that you created. This applies to processing Refund transactions, changing passwords or trying other functions.

### 17.2 Logging In to the QA Merchant Resource Center

To log in to the QA Merchant Resource Center for testing purposes:

1. Go to the Merchant Resource Center QA website at <https://esqa.moneris.com/mpg>
2. Enter your username and password, which are the same email address and password you use to log in to the Developer Portal
3. Enter your Store ID, which you obtained from the Developer Portal's My Testing Credentials as described in 17.3 Test Credentials for Merchant Resource Center

## 17.3 Test Credentials for Merchant Resource Center

For testing purposes, you can either use the pre-existing test stores in the Merchant Resource Center, or you can create your own unique test store where you will only see your own transactions. If you want to use the pre-existing stores, use the test credentials provided in the following tables with the corresponding lines of code, as in the examples below.

### Example of Corresponding Code For Canada:

```
$store_id='monca00392';

$api_token='qYdISUhHiOdfTr1CLNpN';

$mpgRequest->setProcCountryCode("CA");

$mpgRequest->setTestMode(true);
```

**Table 17: Test Server Credentials - Canada**

| store_id   | api_token   | Username | Password | Other Information                                   |
|------------|-------------|----------|----------|-----------------------------------------------------|
| store1     | yesguy      | demouser | password |                                                     |
| store2     | yesguy      | demouser | password |                                                     |
| store3     | yesguy      | demouser | password |                                                     |
| store4     | yesguy      | demouser | password |                                                     |
| store5     | yesguy      | demouser | password |                                                     |
| monca00392 | yesguy      | demouser | password | Use this store to test Convenience Fee transactions |
| moncaqagt1 | mgtokenguy1 | demouser | password | Use this store to test Token Sharing                |
| moncaqagt2 | mgtokenguy2 | demouser | password | Use this store to test Token Sharing                |
| moncaqagt3 | mgtokenguy3 | demouser | password | Use this store to test Token Sharing                |

| store_id   | api_token | Username | Password | Other Information                            |
|------------|-----------|----------|----------|----------------------------------------------|
| monca01428 | mcmpguy   | demouser | password | Use this store to test MasterCard MasterPass |

Alternatively, you can create and use a unique test store where you will only see your own transactions. For more on this, see [Getting a Unique Test Store ID and API Token](#) (page 521)

## 17.4 Getting a Unique Test Store ID and API Token

Transactions requests via the API will require you to have a Store ID and a corresponding API token. For testing purposes, you can either use the pre-existing test stores in the Merchant Resource Center, or you can create your own unique test store where you will only see your own transactions.

To get your unique Store ID and API token:

1. Log in to the Developer Portal at <https://developer.moneris.com>
2. In the My Profile dialog, click the Full Profile  button
3. Under My Testing Credentials, select Request Testing Credentials
4. Enter your Developer Portal password and select your country
5. Record the Store ID and API token that are given, as you will need them for logging in to the Merchant Resource Center (Store ID) and for API requests (API token).

Alternatively, you can use the pre-existing test stores already set up in the Merchant Resource Center as described in [Test Credentials for Merchant Resource Center](#) (page 520).



## 17.5 Processing a Transaction

- 1.1 Overview
- 1.2 HttpsPostRequest Object
- 1.3 Receipt Object

### 17.5.1 Overview

There are some common steps for every transaction that is processed.

1. Instantiate the transaction object (e.g., Purchase), and update it with object definitions that refer to the individual transaction.
2. Instantiate the HttpsPostRequest connection object and update it with connection information, host information and the transaction object that you created in step 17.5  
Section 17.5 (page 523) provides the HttpsPostRequest connection object definition. This object and its variables apply to **every** transaction request.
3. Invoke the HttpsPostRequest object's `send()` method.
4. Instantiate the Receipt object, by invoking the HttpsPostRequest object's get Receipt method.  
Use this object to retrieve the applicable response details.

Some transactions may require steps in addition to the ones listed here. Below is a sample Purchase transaction with each major step outlined. For extensive code samples of other transaction types, refer to the PHP API ZIP file.

**NOTE:** For illustrative purposes, the order in which lines of code appear below may differ slightly from the same sample code presented elsewhere in this document.

|                                                                                                                                                                                                                                                                |                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <pre>&lt;?php ## ## Example php -q TestPurchase.php store1 ## require "../mpgClasses.php";  \$type='purchase'; \$cust_id='cust id'; \$order_id='ord-' .date("dmy-G:i:s"); \$amount='1.00'; \$pan='4242424242424242'; \$expiry_date='1111'; \$script='7';</pre> | Include all necessary classes.                                     |
| <pre>\$store_id='store5'; \$api_token='yesguy';</pre>                                                                                                                                                                                                          | Define all mandatory values for the transaction object properties. |
|                                                                                                                                                                                                                                                                | Define all mandatory values for the connection object properties.  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <pre> \$txnArray=array('type'=&gt;\$type, 'order_id'=&gt;\$order_id, 'cust_id'=&gt;\$cust_id, 'amount'=&gt;\$amount, 'pan'=&gt;\$pan, 'expdate'=&gt;\$expiry_date, 'crypt_type'=&gt;\$crypt, 'dynamic_descriptor'=&gt;\$dynamic_descriptor );  \$mpgTxn = new mpgTransaction(\$txnArray);  \$mpgRequest = new mpgRequest(\$mpgTxn); \$mpgRequest-&gt;setProcCountryCode("CA"); //"US" for sending transaction to US environment \$mpgRequest-&gt;setTestMode(true); //false for production transactions </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>Instantiate the transaction object and assign values to properties.</p>                                               |
| <pre> /* Status Check Example \$mpgHttpPost =new mpgHttpsPostStatus(\$store_id,\$api_token,\$status_ check,\$mpgRequest); */ \$mpgHttpPost =new mpgHttpsPost(\$store_id,\$api_token,\$mpgRequest); </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Instantiate connection object and assign values to properties, including the transaction object you just created.</p> |
| <pre> \$mpgResponse=\$mpgHttpPost-&gt;getMpgResponse(); print("\nCardType = " . \$mpgResponse-&gt;getCardType()); print("\nTransAmount = " . \$mpgResponse-&gt;getTransAmount()); print("\nTxnNumber = " . \$mpgResponse-&gt;getTxnNumber()); print("\nReceiptId = " . \$mpgResponse-&gt;getReceiptId()); print("\nTransType = " . \$mpgResponse-&gt;getTransType()); print("\nReferenceNum = " . \$mpgResponse-&gt;getReferenceNum()); print("\nResponseCode = " . \$mpgResponse-&gt;getResponseCode()); print("\nISO = " . \$mpgResponse-&gt;getISO()); print("\nMessage = " . \$mpgResponse-&gt;getMessage()); print("\nIsVisaDebit = " . \$mpgResponse-&gt;getIsVisaDebit()); print("\nAuthCode = " . \$mpgResponse-&gt;getAuthCode()); print("\nComplete = " . \$mpgResponse-&gt;getComplete()); print("\nTransDate = " . \$mpgResponse-&gt;getTransDate()); print("\nTransTime = " . \$mpgResponse-&gt;getTransTime()); print("\nTicket = " . \$mpgResponse-&gt;getTicket()); print("\nTimedOut = " . \$mpgResponse-&gt;getTimedOut()); print("\nStatusCode = " . \$mpgResponse-&gt;getStatusCode()); print("\nStatusMessage = " . \$mpgResponse-&gt;getStatusMessage()); ?&gt; </pre> | <p>Instantiate the Receipt object and use its get methods to retrieve the desired response data.</p>                     |

## 17.5.2 HttpsPostRequest Object

The transaction object that you instantiate becomes a property of this object when you call its set transaction method.

### HttpsPostRequest Object Definition

```
HttpsPostRequest mpgReq = new HttpsPostRequest();
```

After instantiating the HttpsPostRequest object, update its mandatory and optional values as outlined in the following values tables.



**Table 18: HttpsPostRequest object mandatory values**

| Value                   | Type                                                                                                                                                                                                                                                           | Limits                    | Set method                                                                                              |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------|
|                         | Description                                                                                                                                                                                                                                                    |                           |                                                                                                         |
| Processing country code | String                                                                                                                                                                                                                                                         | 2-character alphabetic    | <code>\$mpgRequest-&gt;setProcCountryCode("CA");</code>                                                 |
|                         | CA for Canada, US for USA.                                                                                                                                                                                                                                     |                           |                                                                                                         |
| Test mode               | Boolean                                                                                                                                                                                                                                                        | true/false                | <code>\$mpgRequest-&gt;setTestMode(true);</code>                                                        |
|                         | Set to <code>true</code> when in test mode. Set to <code>false</code> (or comment out entire line) when in production mode.                                                                                                                                    |                           |                                                                                                         |
| Store ID                | String                                                                                                                                                                                                                                                         | 10-character alphanumeric | <code>\$mpgHttpPost = new mpgHttpPostStatus(\$store_id,\$api_token,\$status_check,\$mpgRequest);</code> |
|                         | Unique identifier provided by Moneris upon merchant account set up.<br>See 17.1 About the Merchant Resource Center for test environment details.                                                                                                               |                           |                                                                                                         |
| API Token               | String                                                                                                                                                                                                                                                         | 20-character alphanumeric | <code>\$mpgHttpPost = new mpgHttpPostStatus(\$store_id,\$api_token,\$status_check,\$mpgRequest);</code> |
|                         | Unique alphanumeric string assigned upon merchant account activation. To locate your production API token, refer to the Merchant Resource Center Admin Store Settings.<br>See 17.3 Test Credentials for Merchant Resource Center for test environment details. |                           |                                                                                                         |
| Transaction             | Object                                                                                                                                                                                                                                                         | Not applicable            | <code>\$mpgRequest = new mpgRequest(\$mpgTxn);</code>                                                   |
|                         | This argument is one of the numerous transaction types discussed in the rest of this manual. (Such as Purchase, Refund and so on.) This object is instantiated in step 1 above.                                                                                |                           |                                                                                                         |

Table 1 HttpsPostRequest object optional values

| Value        | Type                                                                                                                                                                                                                              | Limits     | Set method                                                                                                 |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------------------------|
|              | Description                                                                                                                                                                                                                       |            |                                                                                                            |
| Status Check | Boolean                                                                                                                                                                                                                           | true/false | <code>\$mpgHttpPost = new mpgHttpPostStatus(\$store_id, \$api_token, \$status_check, \$mpgRequest);</code> |
|              | See Appendix A Definition of Request Fields.                                                                                                                                                                                      |            |                                                                                                            |
|              | <div><b>NOTE:</b> while this value belongs to the <code>HttpPostRequest</code> object, it is only supported by some transactions. Check the individual transaction definition to find out whether Status Check can be used.</div> |            |                                                                                                            |

### 17.5.3 Receipt Object

After you send a transaction using the HttpsPostRequest object's send method, you can instantiate a receipt object.

#### Receipt Object Definition

```
$mpgResponse=$mpgHttpPost->getMpgResponse();
```

For an in-depth explanation of Receipt object methods and properties, see Appendix B Definitions of Response Fields.

## 17.6 Testing MPI Solutions

When testing your implementation of the Moneris MPI, you can use the Visa/MasterCard/Amex PIT (production integration testing) environment. The testing process is slightly different than a production environment in that when the inline window is generated, it does not contain any input boxes. Instead, it contains a window of data and a **Submit** button. Clicking **Submit** loads the response in the testing window. The response will not be displayed in production.

**NOTE:** MasterCard SecureCode and Amex SafeKey may not be directly tested within our current test environment. However, the process and behavior tested with the Visa test cards will be the same for MCSC and SafeKey.

When testing you may use the following test card numbers with any future expiry date. Use the appropriate test card information from the tables below: Visa and MasterCard use the same test card information, while Amex uses unique information.

**Table 19: MPI test card numbers (Visa and MasterCard only)**

| Card Number                          | VERes | PARes | Action                                                                                                                                                       |
|--------------------------------------|-------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4012001037141112<br>4242424242424242 | Y     | true  | TXN – Call function to create inLine window.<br>ACS – Send CAVV to Moneris Gateway using either the Cavv Purchase or the Cavv Pre-Authorization transaction. |
| 4012001038488884                     | U     | NA    | Send transaction to Moneris Gateway using either the basic Purchase or the basic Pre-Authorization transaction. Set crypt_type = 7.                          |
| 4012001038443335                     | N     | NA    | Send transaction to Moneris Gateway using either the basic Purchase or the basic Pre-Authorization transaction.<br>Set crypt_type = 6.                       |
| 4012001037461114                     | Y     | false | Card failed to authenticate. Merchant may chose to send transaction or decline transaction. If transaction is sent, use crypt type = 7.                      |

**Table 20: MPI test card numbers (Amex only)**

| Card Number     | Password |                 |       | Action                                                                                                                                                                           |
|-----------------|----------|-----------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | VERes    | Required?       | PARes |                                                                                                                                                                                  |
| 375987000000062 | U        | Not required    | N/A   | TXN – Call function to create inLine window.<br>ACS – Send CAVV to Moneris Gateway using either the Cavv Purchase or the Cavv Pre-Authorization transaction. Set crypt_type = 7. |
| 375987000000021 | Y        | Yes: test13fail | false | Card failed to authenticate. Merchant may chose to send transaction or decline transaction. If transaction is sent, use crypt type = 7.                                          |
| 375987000000013 | N        | Not required    | N/A   | Send transaction to Moneris Gateway using either the basic Purchase or the basic Pre-Authorization transaction. Set crypt_type = 6.                                              |
| 374500261001009 | Y        | Yes: test09     | true  | Card failed to authenticate. Merchant may choose to send transaction or decline transaction. Set crypt_type = 5.                                                                 |

**VERes**

The result U, Y or N is obtained by using getMessage().

**PARes**

The result “true” or “false” is obtained by using getSuccess().

To access the Merchant Resource Center in the test environment go to <https://esqa.moneris.com/mpg>.

Transactions in the test environment should not exceed \$11.00.

## 17.7 Test Card Numbers

Because of security and compliance reasons, the use of live credit and debit card numbers for testing is strictly prohibited. Only test credit and debit card numbers are to be used.

To test general transactions, use the following test card numbers:

| Card Plan  | Test Card Number                       |
|------------|----------------------------------------|
| Mastercard | 5454545454545454                       |
| Visa       | 4242424242424242                       |
| Amex       | 373599005095005                        |
| JCB        | 3566007770015365                       |
| Diners     | 36462462742008                         |
| Track2     | 5258968987035454=06061015454001060101? |
| Discover   | 6011000992927602                       |
| UnionPay   | 6250944000000771                       |

### 17.7.1 Test Card Numbers for Level 2/3

When testing Level 2/3 transactions, use the card numbers below.

| Card Brand | Test Card Number |
|------------|------------------|
| Mastercard | 5454545442424242 |
| Visa       | 4242424254545454 |
| Amex       | 373269005095005  |

---

## 17.7.2 Test Cards for Visa Checkout

| Card Plan        | Test Card Number                    |
|------------------|-------------------------------------|
| Visa             | 4005520201264821 (without card art) |
| Visa             | 4242424242424242 (with card art)    |
| MasterCard       | 5500005555555559                    |
| American Express | 340353278080900                     |
| Discover         | 6011003179988686                    |

## 17.8 Simulator Host

The test environment has been designed to replicate the production environment as closely as possible. One major difference is that Moneris is unable to send test transactions onto the production authorization network. Therefore, issuer responses are simulated. Additionally, the requirement to emulate approval, decline and error situations dictates that certain transaction variables initiate various response and error situations.

The test environment approves and declines transactions based on the penny value of the amount sent. For example, a transaction made for the amount of \$9.00 or \$1.00 is approved because of the .00 penny value.

Transactions in the test environment must not exceed \$11.00.

For a list of all current test environment responses for various penny values, please see the Test Environment Penny Response Table available at <https://developer.moneris.com>.

**NOTE:** These responses may change without notice. Check the Moneris Developer Portal (<https://developer.moneris.com>) regularly to access the latest documentation and downloads.



## 18 Moving to Production

- 18.1 Activating a Production Store Account
- 18.2 Configuring a Store for Production
- 18.3 Receipt Requirements
- 1 Getting Help

### 18.1 Activating a Production Store Account

The steps below outline how to activate your production account so that you can process production transactions.

1. Obtain your activation letter/fax from Moneris.
2. Go to <https://www.moneris.com/activate>.
3. Input your store ID and merchant ID from the letter/fax and click **Activate**.
4. Follow the on-screen instructions to create an administrator account. This account will grant you access to the Merchant Resource Center.
5. Log into the Merchant Resource Center at <https://www3.moneris.com/mpg> using the user credentials created in step 18.1.
6. Proceed to **ADMIN** and then **STORE SETTINGS**.
7. Locate the API token at the top of the page. You will use this API token along with the store ID that you received in your letter/fax and to send any production transactions through the API.

When your production store is activated, you need to configure your store so that it points to the production host. To learn how do to this, see Configuring a Store for Production (page 531)

**NOTE:** For more information about how to use the Merchant Resource Center, see the Moneris Gateway Merchant Resource Center User's Guide, which is available at <https://developer.moneris.com>.

### 18.2 Configuring a Store for Production

After you have completed your testing and have activated your production store, you are ready to point your store to the production host.

To configure a store for production:

1. Change the test mode set method from `true` to `false`.
2. Change the Store ID to reflect the production store ID that you received when you activated your production store. To review the steps for activating a production store, see Activating a Production Store Account (page 531).

3. Change the API token to the production token that you received during activation.
4. If you haven't done so already, change the code to reflect the correct processing country (Canada for most merchants). For more on this, see

The table below illustrates the steps above using the relevant code (and where **x** is an alphanumeric character).

| Step | Code in Testing                                                                                                      | Changes for Production                                                            |
|------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 1    | No string changes for this item, only set method is altered:<br><br><pre>\$mpgRequest-&gt;setTestMode(true);</pre>   | Set method for production:<br><br><pre>\$mpgRequest-&gt;setTestMode(false);</pre> |
| 2    | String:<br><br><pre>\$store_id='store5';</pre> Associated Set Method:<br><br><pre>'store_id'=&gt;\$store_id</pre>    | String for Production:<br><br><pre>\$store_id='monXXXXXXXX';</pre>                |
| 3    | String:<br><br><pre>\$api_token='yesguy';</pre> Associated Set Method:<br><br><pre>'api_token'=&gt;\$api_token</pre> | String for Production:<br><br><pre>\$api_token='xxxx';</pre>                      |

## 18.3 Receipt Requirements

Visa and MasterCard expect certain details to be provided to the cardholder and on the receipt when a transaction is approved.

Receipts must comply with the standards outlined within the Integration Receipts Requirements. For all the receipt requirements covering all transaction scenarios, visit the Moneris Developer Portal at <https://developer.moneris.com>.

Production of the receipt must begin when the appropriate response to the transaction request is received by the application. The transaction may be any of the following:

- **Sale** (Purchase)
- **Authorization** (PreAuth, Pre-Authorization)



- **Authorization Completion** (Completion, Capture)
- **Offline Sale** (Force Post)
- **Sale Void** (Purchase Correction, Void)
- **Refund.**

The boldface terms listed above are the names for transactions as they are to be displayed on receipts. Other terms used for the transaction are indicated in brackets.

### 18.3.1 Certification Requirements

Card-present transaction receipts are required to complete certification.

#### **Card-not-present integration**

Certification is optional but highly recommended.

#### **Card-present integration**

After you have completed the development and testing, your application must undergo a certification process where all the applicable transaction types must be demonstrated, and the corresponding receipts properly generated.

Contact a Client Integration Specialist for the Certification Test checklist that must be completed and returned for verification. (See "Getting Help" on page 1 for contact details.) Be sure to include the application version of your product. Any further changes to the product after certification requires re-certification.

After the certification requirements are met, Moneris will provide you with an official certification letter.

## Appendix A Definition of Request Fields

This section defines transaction request variables. Not all fields are required — refer to individual transaction type topics for information on whether a field is required or optional.

### A.1 Definition of Request Fields – Connection Fields

#### Core connection object fields (all API transactions)

| Variable Name          | Type and Limits | Description                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| store ID<br>store_id   | String<br>N/A   | Unique identifier provided by Moneris upon merchant account setup                                                                                                                                                                                                                                                                                                                                                            |
| API token<br>api_token | String<br>N/A   | <p>Unique alphanumeric string assigned by Moneris upon merchant account activation</p> <p>To find your API token, refer to your test or production store's Admin settings in the Merchant Resource Center, at the following URLs:</p> <p>Testing: <a href="https://esqa.moneris.com/mpg/">https://esqa.moneris.com/mpg/</a></p> <p>Production: <a href="https://www3.moneris.com/mpg/">https://www3.moneris.com/mpg/</a></p> |

### A.2 Definition of Request Fields – Core Fields

| Variable Name    | Type and Limits                                                                                                           | Description                                                                                                                                                                                |
|------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| amount<br>amount | String<br>10-character decimal<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point | <p>Transaction dollar amount</p> <p>This must contain at least 3 digits, two of which are penny values</p> <p>Minimum allowable value = \$0.01, maximum allowable value = \$9999999.99</p> |

| Variable Name                            | Type and Limits                                                                                                                                                                                                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <div>EXAMPLE: 1234567.89</div>                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| authorization code<br>auth_code          | String<br>8-character alphanumeric                                                                                                                                                                                   | An authorization code required to carry out a Force Post; provided in the transaction response from the issuing bank                                                                                                                                                                                                                                                                                                                                                                       |
| completion amount<br>comp_amount         | String<br>10-character decimal<br><br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><br><div>EXAMPLE: 1234567.89</div>                                                  | Dollar amount of a Pre-Authorization Completion transaction, which may differ from the original amount authorized in the Pre-Authorization                                                                                                                                                                                                                                                                                                                                                 |
| credit card number<br>pan                | String<br><br>max 20-character alphanumeric                                                                                                                                                                          | Credit card number, usually 16 digits —field can be maximum 20 digits in support of future expansion of card number ranges.<br><br>Carries the token for network tokenization transactions.                                                                                                                                                                                                                                                                                                |
| dynamic descriptor<br>dynamic_descriptor | String<br>20-character alphanumeric<br><br>total of 22 characters including your merchant name and separator<br><br><div>NOTE:<br/>Some special characters are not allowed:<br/>&lt; &gt; \$ % = ? ^ { } [ ] \</div> | Merchant-defined description sent on a per-transaction basis that will appear on the credit card statement appended to the merchant's business name<br><br>Dependent on the card issuer, the statement will typically show the dynamic descriptor appended to the merchant's existing business name separated by the "/" character; additional characters will be truncated<br><br><div>NOTE: The 22-character maximum limit must take the "/" into account as one of the characters</div> |

| Variable Name                 | Type and Limits          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| electronic commerce indicator | <i>String</i>            | Describes the category of e-commerce transaction being processed. Allowable values are:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| crypt_type                    | 1-character alphanumeric | <p>1 – Mail Order / Telephone Order—Single</p> <p>2 – Mail Order / Telephone Order—Recurring</p> <p>3 – Mail Order / Telephone Order—Instalment</p> <p>4 – Mail Order / Telephone Order—Unknown classification</p> <p>5 – Authenticated e-commerce transaction (3-D Secure)</p> <p>6 – Non-authenticated e-commerce transaction (3-D Secure)</p> <p>7 – SSL-enabled merchant</p> <p><b>In Credential on File transactions where the request field e-commerce indicator is also being sent:</b> the allowable values for e-commerce indicator are dependent on the value sent for <b>payment indicator</b>, as follows:</p> <p>if payment indicator = R, then allowable values for e-commerce indicator: 2, 5 or 6</p> <p>if payment indicator = V, then allowable values for e-commerce indicator: 2, 5 or 6</p> <p>if payment indicator = C, then allowable values for e-commerce indicator: 1, 5, 6 or 7</p> <p>if payment indicator = U, then allowable values for e-commerce indicator: 1 or 7</p> <p>if payment indicator = Z, then allowable values for e-commerce indicator: 1, 5, 6 or 7</p> <p><b>For Apple Pay or Google Pay™ transactions where you are doing decryption:</b> send the value of the <b>eciIndicator</b> or <b>3dsEciIndicator</b> field returned in the payload</p> <p>If the value is not present in the payload, send the value as 5; if you get a</p> |

| Variable Name                    | Type and Limits                                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  |                                                                  | <p>2-character value (e.g., 05 or 07), remove the initial 0 and just send us the 2nd character</p> <p>Allowable values for Apple Pay and Google Pay™ are:</p> <p>5: Authenticated e-commerce transaction</p> <p>7: SSL-enabled merchant</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| expiry date<br>expdate           | <p><i>String</i></p> <p>4-character alphanumeric</p> <p>YYMM</p> | <p>Expiry date of the credit card, in YYMM format.</p> <p><b>NOTE:</b> This is the reverse of the MMY date format that is presented on the card.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| is incremental<br>is_incremental | <p><i>Boolean</i></p> <p>true/false</p>                          | <p>'is_incremental'=&gt;\$is_incremental</p> <p>Indicates if this preauthorization is using an estimated amount. Estimations allow for incrementing the amount held via subsequent incrementalAuth requests. Defaults to false.</p> <p><b>NOTE:</b> Please note that if this field is true, the preauthorization is only eligible for a single Preauthorization Completion. Any completion sent for partial completion is treated as a full completion (ship_indicator= P is treated as = F when is_incremental= true on the original preauth).</p> <p><b>NOTE:</b> For preauthorizations using Account-Based Payments, as part of Pay By Bank, is_incremental = true indicates the transaction may be completed for up to 15% more than the original preauthorization amount.</p> |
| foreign indicator                | <i>Boolean</i>                                                   | 'foreign_indicator'=>\$foreign_indicator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Variable Name                        | Type and Limits                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | true or false                                                       | Used to identify domestic transactions processed by a marketplace merchant that is in a different country.                                                                                                                                                                                                                                                                                                                                                                           |
| order ID<br>order_id                 | String<br>50-character alphanumeric<br>a-Z A-Z 0-9 _ - : . @ spaces | Merchant-defined transaction identifier that must be unique for every Purchase, Pre-Authorization and Independent Refund transaction. No two transactions of these types may have the same order ID.<br><br>For Refund, Completion and Purchase Correction transactions, the order ID must be the same as that of the original transaction.                                                                                                                                          |
| original order ID<br>orig_order_id   | String                                                              | Order ID from the original Pre-Authorization transaction, used as a reference to retrieve the original payment details                                                                                                                                                                                                                                                                                                                                                               |
| shipping indicator<br>ship_indicator | String<br>1-character alphanumeric                                  | Used to identify completion transactions that require multiple shipments, also referred to as multiple completions<br><br>By default, if shipping indicator is not sent, the Pre-Authorization Completion is listed as final<br><br>To indicate that the Pre-Authorization Completion is to be left open by the issuer as supplemental shipments or completions are pending, submit shipping indicator with a value of P<br><br>Possible values:<br><br>P – Partial<br><br>F – Final |
| transaction number<br>txn_number     | String<br>255-character, alpha-                                     | Used to reference the original transaction when performing a follow-on transaction (i.e., Pre-Authorization                                                                                                                                                                                                                                                                                                                                                                          |

| Variable Name    | Type and Limits                                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | numeric, hyphens or under-scores<br><br>variable length | Completion, Purchase Correction or Refund)<br><br>This value is returned in the response of the original transaction<br><br>Pre-Authorization Completion: references a Pre-Authorization<br><br>Refund/Purchase Correction: references a Purchase or Pre-Authorization Completion                                                                                                                                                                                                                                                                                |
| wallet indicator | <i>String</i>                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| wallet_indicator | 3-character alphanumeric                                | Indicates when a card number has been collected via a digital wallet, such as in Apple Pay, Google Pay™, Visa Checkout and Mastercard MasterPass, or via network tokenization from the card brand.<br><br>Required for Apple Pay, Google Pay™ transactions whereby you are using your own API to decrypt the payload<br><br>Possible values:<br><br>APP – Apple Pay In-App<br><br>APW – Apple Pay on the Web<br><br>GPP – Google Pay™ In-App<br><br>GPW – Google Pay™ Web<br><br>VCO – Visa Checkout<br><br>MMP – Mastercard MasterPass<br><br>PBB - Pay By Bank |

**NOTE:** Please note that if this field is included to indicate Apple Pay or Google Pay™, then Convenience Fee is not supported.

| Variable Name | Type and Limits | Description |
|---------------|-----------------|-------------|
|---------------|-----------------|-------------|

**NOTE:** Network tokenization wallet indicators are not in the API call but are in the merchant resource centre (MRC).

### A.3 Definition of Request Fields – Credential on File

| Variable Name                                                                                                                                                                                                                                                              | Type and Limits                                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| issuer ID<br><br><b>NOTE:</b> This variable is required for all merchant-initiated transactions following the first one; upon sending the first transaction, the issuer ID value is received in the transaction response and then used in subsequent transaction requests. | <i>String</i><br><br>15-character alphanumeric<br><br>variable length | <p>Unique identifier for the cardholder's stored credentials</p> <p>Sent back in the response from the card brand when processing a Credential on File transaction</p> <p>If the cardholder's credentials are being stored for the first time, and the issuer ID was returned in the response, you must save the issuer ID on your system to use in subsequent Credential on File transactions (applies to merchant-initiated transactions only)</p> <p>The issuer ID must be saved to your systems when returned from Moneris Gateway in the response data, regardless if the value was received or not</p> <p>As a best practice, if the issuer ID is not returned and you received a value of NULL instead, store that value and send it in the subsequent transaction</p> |
| payment indicator                                                                                                                                                                                                                                                          | <i>String</i><br><br>1-character alphabetic                           | <p>Indicates the current or intended use of the credentials</p> <p>Possible values for first transactions:</p> <p>C - unscheduled Credential on File (first transactions only)</p> <p>R - recurring</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |



| Variable Name       | Type and Limits                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |                                                 | <p>V - recurring variable payment transaction</p> <p>Possible values for subsequent transactions:</p> <p>R - recurring</p> <p>V - recurring variable payment transaction</p> <p>U - unscheduled merchant-initiated transaction</p> <p>Z - unscheduled customer-initiated transaction</p> <p>In Credential on File transactions where the request field e-commerce indicator is also being sent, the acceptable values for e-commerce indicator are dependent on the value sent for payment indicator, as follows:</p> <p>if payment indicator = R, then allowable values for e-commerce indicator: 2, 5 or 6</p> <p>if payment indicator = V, then allowable values for e-commerce indicator: 2, 5 or 6</p> <p>if payment indicator = C, then allowable values for e-commerce indicator: 1, 5, 6 or 7</p> <p>if payment indicator = U, then allowable values for e-commerce indicator: 1 or 7</p> <p>if payment indicator = Z, then allowable values for e-commerce indicator: 1, 5, 6 or 7</p> |
| payment information | <p><i>String</i></p> <p>1-character numeric</p> | <p>Describes whether the transaction is the first or subsequent in the series</p> <p>Possible values:</p> <p>0 - first transaction in a series (storing payment details provided by the cardholder)</p> <p>2 - subsequent transactions (using previously stored payment details)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## A.4 Definition of Request Fields – GooglePay Token Temp Add

| Variable Name    | Type and Limits                 | Description                                                                                                                                                                                                                                                             |
|------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| payment token    | <i>Object</i><br><br><i>N/A</i> | Payment details returned by Google in their <code>PaymentData</code> object for GooglePay transactions. See Definition of Request Fields – GooglePay Token Temp Add below for field details.                                                                            |
| signature        | <i>String</i>                   | Verifies that the message came from Google. It's base64-encoded, and created with ECDSA by the intermediate signing key. Returned by Google in their <code>PaymentData</code> object for GooglePay transactions                                                         |
| protocol version | <i>String</i>                   | Identifies the encryption or signing scheme under which the message is created. It allows the protocol to evolve over time, if needed. Returned by Google in their <code>PaymentData</code> object for GooglePay transactions                                           |
| signed message   | <i>String</i>                   | A JSON object serialized as an HTML-safe string that contains the encryptedMessage, ephemeralPublicKey, and tag. It's serialized to simplify the signature verification process. Returned by Google in their <code>PaymentData</code> object for GooglePay transactions |

## A.5 Definition of Request Fields – Vault

| Variable Name | Type and Limits                                | Description                                                                                                                                                                                                                                                       |
|---------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data key      | <i>String</i><br><br>25-character alphanumeric | <p>Unique identifier for a Vault profile, and used in future Vault financial transactions to associate a transaction with that profile</p> <p>Data key is generated by Moneris and returned to you in the Receipt object when the profile is first registered</p> |

| Variable Name    | Type and Limits                                             | Description                                                                                                                                                                                                                                              |
|------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data key format  | <i>String</i><br>2-character alphanumeric                   | Specifies the data key format being returned<br><br>If left blank, data key format will default to 25-character alphanumeric<br><br>Possible values:<br><br>0 – 25 character alphanumeric data key<br><br>0U – unique 25-character alphanumeric data key |
| duration         | <i>String</i><br>3-character numeric<br>maximum 900 seconds | Amount of time the temporary token should be available                                                                                                                                                                                                   |
| email address    | <i>String</i><br>30-character alphanumeric                  | Customer's email address<br><br>Can be sent in when creating or updating a Vault profile                                                                                                                                                                 |
| note             | <i>String</i><br>30-character alphanumeric                  | Used for any supplementary information related to the customer<br><br>Can be sent in when creating or updating a Vault profile                                                                                                                           |
| phone number     | <i>String</i><br>30-character alphanumeric                  | Customer's phone number<br><br>Can be sent in when creating or updating a Vault profile                                                                                                                                                                  |
| return issuer ID | <i>Boolean</i><br>true/false                                | When true, Gateway returns the bank Issuer ID. Defaults to False.                                                                                                                                                                                        |

## A.6 Definition of Request Fields for Level 2/3 - Visa

Table 1 Visa - Corporate Card Common Data - Level 2 Request Fields

| Req* | Field Name   | Limits       | Set Method | Description  |
|------|--------------|--------------|------------|--------------|
| Y    | National Tax | 12-character | 'national_ | Must reflect |

| Req* | Field Name                                                 | Limits                    | Set Method                                           | Description                                                                                                                                                                                                                                 |
|------|------------------------------------------------------------|---------------------------|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                                            | decimal                   | <code>tax'=&gt;\$national_tax</code>                 | <p>the amount of National Tax (GST or HST) appearing on the invoice.</p> <p>Minimum - 0.01 Maximum - 999999.99. Must have 2 decimal places.</p>                                                                                             |
| Y    | Merchant VAT Registration/Single Business Reference Number | 20-character alphanumeric | <code>'merchant_vat_no'=&gt;\$merchant_vat_no</code> | <p>Merchant's Tax Registration Number</p> <p>must be provided if tax is included on the invoice</p> <div style="border: 1px solid black; padding: 5px; margin-top: 10px;"> <b>NOTE:</b> Must not be all spaces or all zeroes         </div> |
| C    | Local Tax                                                  | 12-character decimal      | <code>'local_tax'=&gt;\$local_tax</code>             | <p>Must reflect the amount of Local Tax (PST or QST) appearing on the invoice</p> <p>If Local Tax included then must not be all spaces or all zeroes; Must</p>                                                                              |

| Req* | Field Name                                 | Limits                    | Set Method                           | Description                                                                                                                                                                                                                                    |
|------|--------------------------------------------|---------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                            |                           |                                      | <p>be provided if Local Tax (PST or QST) applies</p> <p>Minimum = 0.01</p> <p>Maximum = 999999.99</p> <p>Must have 2 decimal places</p>                                                                                                        |
| C    | Local Tax (PST or QST) Registration Number | 15-character alphanumeric | 'local_tax_no'=>\$local_tax_no       | <p>Merchant's Local Tax (PST/QST) Registration Number</p> <p>Must be provided if tax is included on the invoice; If Local Tax included then must not be all spaces or all zeroes</p> <p>Must be provided if Local Tax (PST or QST) applies</p> |
| C    | Customer VAT Registration Number           | 13-character alphanumeric | 'customer_vat_no'=>\$customer_vat_no | <p>If the Customer's Tax Registration Number appears on the invoice to support tax exempt transactions it must</p>                                                                                                                             |

| Req* | Field Name                                        | Limits                    | Set Method                         | Description                                                                                                                     |
|------|---------------------------------------------------|---------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
|      |                                                   |                           |                                    | be provided here                                                                                                                |
| C    | Customer Code/Customer Reference Identifier (CRI) | 16-character alphanumeric | 'cri'=>\$cri                       | Value which the customer may choose to provide to the supplier at the point of sale – must be provided if given by the customer |
| N    | Customer Code                                     | 17-character alphanumeric | 'customer_code'=>\$customer_code   | Optional customer code field that will not be passed along to Visa, but will be included on Moneris reporting                   |
| N    | Invoice Number                                    | 17-character alphanumeric | 'invoice_number'=>\$invoice_number | Optional invoice number field that will not be passed along to Visa, but will be included on Moneris reporting                  |

\*Y = Required, N = Optional, C = Conditional

**Table 2 Visa - Corporate Card Common Data- Level 2 Request Fields (VSPurcha)**

| Req | Variable Name | Field Name | Size/Type                 | Description                                                                |
|-----|---------------|------------|---------------------------|----------------------------------------------------------------------------|
| C*  | Buyer Name    | buyer_name | 30-character alphanumeric | Buyer/Recipient Name<br><br>*only required by CRA if transaction is >\$150 |

| Req | Variable Name                 | Field Name         | Size/Type           | Description                                                                                                                                                                                                                                                                          |
|-----|-------------------------------|--------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C*  | Local tax rate                | local_tax_rate     | 4-character decimal | <p>Indicates the detailed tax rate applied in relationship to a local tax amount</p> <div> <b>EXAMPLE:</b> 8% PST should be 8.0. </div> <p>maximum 99.99</p> <p>*Must be provided if Local Tax (PST or QST) applies.</p>                                                             |
| N   | Duty Amount                   | duty_amount        | 9-character decimal | <p>Duty on total purchase amount</p> <p>A minus sign means 'amount is a credit', plus sign or no sign means 'amount is a debit'</p> <p>maximum without sign is 999999.99</p>                                                                                                         |
| N   | Invoice Discount Treatment    | discount_treatment | 1-character numeric | <p>Indicates how the merchant is managing discounts</p> <p>Must be one of the following values:</p> <p>0 - if no invoice level discounts apply for this invoice</p> <p>1 - if Tax was calculated on Post-Discount totals</p> <p>2 - if Tax was calculated on Pre-Discount totals</p> |
| N   | Invoice Level Discount Amount | discount_amt       | 9-character decimal | <p>Amount of discount (if provided at the invoice level accord-</p>                                                                                                                                                                                                                  |

| Req | Variable Name                    | Field Name         | Size/Type                  | Description                                                                                                                                                                                                                                     |
|-----|----------------------------------|--------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                                  |                    |                            | <p>ing to the Invoice Discount Treatment)</p> <p>Must be non-zero if Invoice Discount Treatment is 1 or 2</p> <p>Minimum amount is 0.00 and maximum is 999999.99</p>                                                                            |
| C*  | Ship To Postal Code / Zip Code   | ship_to_pos_code   | 10-character alpha-numeric | <p>The postal code or zip code for the destination where goods will be delivered</p> <p>*Required if shipment is involved</p> <p>Full alpha postal code - Valid ANA&lt;space&gt;NAN format required if shipping to an address within Canada</p> |
| C   | Ship From Postal Code / Zip Code | ship_from_pos_code | 10-character alpha-numeric | <p>The postal code or zip code from which items were shipped</p> <p>For Canadian addresses, requires full alpha postal code for the merchant with Valid ANA&lt;space&gt;NAN format</p>                                                          |
| C*  | Destination Country Code         | des_cou_code       | 2-character alpha-numeric  | <p>Code of country where purchased goods will be delivered</p>                                                                                                                                                                                  |



| Req | Variable Name                         | Field Name     | Size/Type                  | Description                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----|---------------------------------------|----------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                                       |                |                            | <p>Use ISO 3166-1 alpha-2 format</p> <div> <b>NOTE:</b> Required if it appears on the invoice for an international transaction </div>                                                                                                                                                                                                                                                                             |
| Y   | Unique VAT Invoice Reference Number   | vat_ref_num    | 25-character alpha-numeric | <p>Unique Value Added Tax Invoice Reference Number</p> <p>Must be populated with the invoice number and this cannot be all spaces or zeroes</p>                                                                                                                                                                                                                                                                   |
| Y   | Tax Treatment                         | tax_treatment  | 1-character numeric        | <p>Must be one of the following values:</p> <p>0 = Net Prices with tax calculated at line item level;</p> <p>1 = Net Prices with tax calculated at invoice level;</p> <p>2 = Gross prices given with tax information provided at line item level;</p> <p>3 = Gross prices given with tax information provided at invoice level;</p> <p>4 = No tax applies (small merchant) on the invoice for the transaction</p> |
| N   | Freight/Shipping Amount (Ship Amount) | freight_amount | 9-character decimal        | <p>Freight charges on total purchase</p> <p>If shipping is not provided as a line item it must be provided here, if applicable</p>                                                                                                                                                                                                                                                                                |

| Req | Variable Name          | Field Name             | Size/Type           | Description                                                                                                                                                                                                                                                                                |
|-----|------------------------|------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                        |                        |                     | Signed monetary amount: minus sign means 'amount is a credit', plus sign or no sign means 'amount is a debit', maximum without sign is 999999.99                                                                                                                                           |
| C   | GST HST Freight Rate   | gst_hst_freight_rate   | 4-character decimal | <p>Rate of GST (excludes PST) or HST charged on the shipping amount (in accordance with the Tax Treatment)</p> <p>If Freight/Shipping Amount is provided then this (National GST or HST) tax rate must be provided.</p> <p>Monetary amount, maximum is 99.99. Such as 13% HST is 13.00</p> |
| C   | GST HST Freight Amount | gst_hst_freight_amount | 9-character decimal | <p>Amount of GST (excludes PST) or HST charged on the shipping amount</p> <p>If Freight/Shipping Amount is provided then this (National GST or HST) tax amount must be provided if taxTreatment is 0 or 2</p> <p>Signed monetary amount: maximum without sign is 999999.99.</p>            |

**Table 3 Visa - Line Item Details - Level 3 Request Fields (VSPurchI)**

| Req | Variable Name       | Field Name       | Size/Type                  | Description                                                                                                                                                                                                                                                                                     |
|-----|---------------------|------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C   | Item Commodity Code | item_com_code    | 12-character alpha-numeric | Line item Commodity Code (if this field is not sent, then productCode must be sent)                                                                                                                                                                                                             |
| Y   | Product Code        | product_code     | 12-character alpha-numeric | <p>Product code for this line item – merchant’s product code, manufacturer’s product code or buyer’s product code</p> <p>Typically this will be the SKU or identifier by which the merchant tracks and prices the item or service</p> <p>This should always be provided for every line item</p> |
| Y   | Item Description    | item_description | 35-character alpha-numeric | Line item description                                                                                                                                                                                                                                                                           |
| Y   | Item Quantity       | item_quantity    | 12-character decimal       | <p>Quantity invoiced for this line item</p> <p>Up to 4 decimal places supported, whole numbers are accepted</p> <p>Minimum = 0.0001</p>                                                                                                                                                         |

| Req | Variable Name        | Field Name          | Size/Type                 | Description                                                                                                                      |
|-----|----------------------|---------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------|
|     |                      |                     |                           | Maximum = 999999999999                                                                                                           |
| Y   | Item Unit of Measure | item_uom            | 2-character alpha-numeric | Unit of Measure<br><br>Use ANSI X-12 EDI Allowable Units of Measure and Codes                                                    |
| Y   | Item Unit Cost       | unit_cost           | 12-character decimal      | Line item cost per unit<br><br>2-4 decimal places accepted<br><br>Minimum = 0.0001<br><br>Maximum = 999999.9999                  |
| N   | VAT Tax Amount       | vat_tax_amt         | 12-character decimal      | Any value-added tax or other sales tax amount<br><br>Must have 2 decimal places<br><br>Minimum = 0.01<br><br>Maximum = 999999.99 |
| N   | VAT Tax Rate         | vat_tax_rate        | 4-character decimal       | Sales tax rate<br><br><div>EXAMPLE: 8%<br/>PST should be 8.0</div><br>maximum 99.99                                              |
| Y   | Discount Treatment   | discount_treatmentL | 1-character numeric       | Must be one of the following values:                                                                                             |

| Req | Variable Name   | Field Name    | Size/Type            | Description                                                                                                                                                                                                                                                  |
|-----|-----------------|---------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                 |               |                      | <p>0 if no invoice level discounts apply for this invoice</p> <p>1 if Tax was calculated on Post-Discount totals</p> <p>2 if Tax was calculated on Pre-Discount totals.</p>                                                                                  |
| C   | Discount Amount | discount_amtL | 12-character decimal | <p>Amount of discount, if provided for this line item according to the Line Item Discount Treatment</p> <p>Must be non-zero if Line Item Discount Treatment is 1 or 2</p> <p>Must have 2 decimal places</p> <p>Minimum = 0.01</p> <p>Maximum = 999999.99</p> |

## A.7 Definition of Request Fields for Level 2/3 - Mastercard

Table 1 Objects - Level 2/3 MasterCard

| MCCorpais Objects | Description                |
|-------------------|----------------------------|
| MCCorpac          | Corporate Card Common data |
| MCCorpai          | Line Item Details          |

**Table 2 MasterCard - Corporate Card Common Data (MCCorpac) - Level 2 Request Fields**

| Req | Variable Name        | Field Name             | Size/Type                 | Description                                                                                                                                                                   |
|-----|----------------------|------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N   | AustinTetraNumber    | Austin-Tetra Number    | 15-character alphanumeric | Merchant's Austin-Tetra Number                                                                                                                                                |
| N   | NaicsCode            | NAICS Code             | 15-character alphanumeric | North American Industry Classification System (NAICS) code assigned to the merchant                                                                                           |
| N   | CustomerCode         | Customer Code          | 25-character alphanumeric | A control number, such as purchase order number, project number, department allocation number or name that the purchaser supplied the merchant. Left-justified; may be spaces |
| N   | UniqueInvoiceNumber  | Unique Invoice Number  | 17-character alphanumeric | Unique number associated with the individual transaction provided by the merchant                                                                                             |
| N   | CommodityCode        | Commodity Code         | 15-character alphanumeric | Code assigned by the merchant that best categorizes the item(s) being purchased                                                                                               |
| N   | OrderDate            | Order Date             | 6-character numeric       | The date the item was ordered. If present, must contain a valid date in the format YYMMDD.                                                                                    |
| N   | CorporationVatNumber | Corporation VAT Number | 20-character alphanumeric | Contains a corporation's value added tax (VAT) number                                                                                                                         |
| N   | CustomerVatNumber    | Customer VAT Number    | 20-character alphanumeric | Contains the VAT number for the customer/cardholder used to identify the customer when purchasing goods and services from the merchant                                        |
| N   | FreightAmount        | Freight                | 12-character              | The freight on the total                                                                                                                                                      |

| Req | Variable Name              | Field Name                        | Size/Type                 | Description                                                                                                                  |
|-----|----------------------------|-----------------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------|
|     |                            | Amount                            | decimal                   | purchase. Must have 2 decimals                                                                                               |
| N   | DutyAmount                 | Duty Amount                       | 12-character decimal      | The duty on the total purchase, Must have 2 decimals                                                                         |
| N   | DestinationProvinceCode    | Destination State / Province Code | 3-character alphanumeric  | State or Province of the country where the goods will be delivered. Left justified with trailing spaces. e.g., ONT - Ontario |
| N   | DestinationCountryCode     | Destination Country Code          | 3-character alphanumeric  | The country code where goods will be delivered. Left justified with trailing spaces. e.g., CAN - Canada                      |
| N   | ShipFromPosCode            | Ship From Postal Code             | 10-character alphanumeric | The postal code or zip code from which items were shipped                                                                    |
| N   | ShipToPosCode              | Destination Postal Code           | 10-character alphanumeric | The postal code or zip code where goods will be delivered                                                                    |
| N   | AuthorizedContactName      | Authorized Contact Name           | 36-character alphanumeric | Name of an individual or company contacted for company authorized purchases                                                  |
| N   | AuthorizedContactPhone     | Authorized Contact Phone          | 17-character alphanumeric | Phone number of an individual or company contacted for company authorized purchases                                          |
| N   | AdditionalCardAcceptordata | Additional Card Acceptor Data     | 40-character alphanumeric | Information pertaining to the card acceptor                                                                                  |
| N   | CardAcceptorType           | Card Acceptor Type                | 8-character alphanumeric  | Various classifications of business ownership characteristics                                                                |

| Req | Variable Name | Field Name | Size/Type | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----|---------------|------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |               |            |           | <p>This field takes 8 characters. Each character represents a different component, as follows:</p> <p>1st character represents 'Business Type' and contains a code to identify the specific classification or type of business:</p> <ul style="list-style-type: none"> <li>Corporation</li> <li>Not known</li> <li>Individual/Sole Proprietorship</li> <li>Partnership</li> <li>Association/Estate/Trust</li> <li>Tax Exempt Organizations (501C)</li> <li>International Organization</li> <li>Limited Liability Company (LLC)</li> <li>Government Agency</li> </ul> <p>2nd character represents 'Business Owner Type'. Contains a code to identify specific characteristics about the business owner.</p> <ul style="list-style-type: none"> <li>1 - No application classification</li> <li>2 - Female business owner</li> <li>3 - Physically handicapped female business owner</li> <li>4 - Physically handicapped male business owner</li> </ul> |



| Req | Variable Name | Field Name | Size/Type | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----|---------------|------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |               |            |           | <p>0 - Unknown</p> <p>3rd character represents 'Business Certification Type'. Contains a code to identify specific characteristics about the business certification type, such as small business, disadvantaged, or other certification type:</p> <p>1 - Not certified</p> <p>2 - Small Business Administration (SBA) certification small business</p> <p>3 - SBA certification as small disadvantaged business</p> <p>4 - Other government or agency-recognized certification (such as Minority Supplier Development Council)</p> <p>5 - Self-certified small business</p> <p>6 - SBA certification as small and other government or agency-recognized certification</p> <p>7 - SBA certification as small disadvantaged business and other government or agency-recognized certification</p> <p>8 - Other government or agency-recognized certification and self-certified small business</p> <p>A - SBA certification as 8(a)</p> <p>B - Self-certified small disadvantaged business</p> |

| Req | Variable Name | Field Name | Size/Type | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----|---------------|------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |               |            |           | <p>(SDB)</p> <p>C - SBA certification as HUBZone</p> <p>0 - Unknown</p> <p>4th character represents 'Business Racial/Ethnic Type'. Contains a code identifying the racial or ethnic type of the majority owner of the business.</p> <p>1 - African American</p> <p>2 - Asian Pacific American</p> <p>3 - Subcontinent Asian American</p> <p>4 - Hispanic American</p> <p>5 - Native American Indian</p> <p>6 - Native Hawaiian</p> <p>7 - Native Alaskan</p> <p>8 - Caucasian</p> <p>9 - Other</p> <p>0 - Unknown</p> <p>5th character represents 'Business Type Provided Code'</p> <p>Y - Business type is provided.</p> <p>N - Business type was not provided.</p> <p>R - Card acceptor refused to provide business type</p> <p>6th character represents 'Business Owner Type Provided Code'</p> |

| Req | Variable Name               | Field Name                     | Size/Type                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----|-----------------------------|--------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                             |                                |                           | <p>Y - Business owner type is provided.</p> <p>N - Business owner type was not provided.</p> <p>R - Card acceptor refused to provide business type</p> <p>7th character represents 'Business Certification Type Provided Code'</p> <p>Y - Business certification type is provided.</p> <p>N - Business certification type was not provided.</p> <p>R - Card acceptor refused to provide business type</p> <p>8th character represents 'Business Racial/Ethnic Type'</p> <p>Y - Business racial/ethnic type is provided.</p> <p>N - Business racial/ethnic type was not provided.</p> <p>R - Card acceptor refused to provide business racial/ethnic type</p> |
| N   | CardAcceptorTaxId           | Card Acceptor Tax ID           | 20-character alphanumeric | US Federal tax ID number for value added tax (VAT) ID.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| N   | CardAcceptorReferenceNumber | Card Acceptor Reference Number | 25-character alphanumeric | Code that facilitates card acceptor/corporation communication and record keeping                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Req | Variable Name         | Field Name               | Size/Type                 | Description                                                                                                                                                                                                                                                                  |
|-----|-----------------------|--------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N   | CardAcceptorVatNumber | Card Acceptor VAT Number | 20-character alphanumeric | Value added tax (VAT) number for the card acceptor location used to identify the card acceptor when collecting and reporting taxes                                                                                                                                           |
| C*  | Tax                   | Tax                      | up to 6 arrays            | <p>Can have up to 6 arrays contains different tax details. See Tax Array below for each field description.</p> <p>*This field is conditionally mandatory — if you use this array, you must fill in all tax array fields as listed in the Tax Array Request Fields below.</p> |

Table 3 MasterCard - Line Item Details (MCCorpal) - Level 3 Request Fields

| Req | Variable Name | Field Name     | Size/Type                 | Description                                                                                                                                                                   |
|-----|---------------|----------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N   | CustomerCode  | Customer Code  | 25-character alphanumeric | A control number, such as purchase order number, project number, department allocation number or name that the purchaser supplied the merchant. Left-justified; may be spaces |
| N   | LineItemDate  | Line Item Date | 6-character numeric       | <p>The purchase date of the line item referenced in the associated Corporate Card Line Item Detail.</p> <p>YYMMDD format</p>                                                  |

| Req | Variable Name   | Field Name       | Size/Type                 | Description                                                                                                                                                                                                                                                            |
|-----|-----------------|------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N   | ShipDate        | Ship Date        | 6-character numeric       | The date the merchandise was shipped to the destination.<br><br>YYMMDD format                                                                                                                                                                                          |
| N   | OrderDate       | Order Date       | 6-character numeric       | The date the item was ordered<br><br>YYMMDD format                                                                                                                                                                                                                     |
| Y   | ProductCode     | Product Code     | 12-character alphanumeric | Line item Product Code (if this field is not sent, then itemComCode)<br><br>If the order has a Freight/Shipping line item, the productCode value has to be "Freight/Shipping"<br><br>If the order has a Discount line item, the productCode value has to be "Discount" |
| Y   | ItemDescription | Item Description | 35-character alphanumeric | Line Item description                                                                                                                                                                                                                                                  |
| Y   | ItemQuantity    | Item Quantity    | 12-character alphanumeric | Quantity of line item                                                                                                                                                                                                                                                  |
| Y   | UnitCost        | Unit Cost        | 12-character decimal      | Line item cost per unit.<br><br>Must contain a minimum of 2 decimal places, up to 5 decimal places supported.<br><br>Minimum amount is                                                                                                                                 |

| Req | Variable Name   | Field Name           | Size/Type                 | Description                                                                                                                                                                                        |
|-----|-----------------|----------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                 |                      |                           | 0.00001 and maximum is 999999.99999                                                                                                                                                                |
| Y   | ItemUnitMeasure | Item Unit Measure    | 12-character alphanumeric | The line item unit of measurement code                                                                                                                                                             |
| Y   | ExtItemAmount   | Extended Item Amount | 9-character decimal       | <p>Contains the individual item amount that is normally calculated as price multiplied by quantity</p> <p>Must contain 2 decimal places</p> <p>Minimum amount is 0.00 and maximum is 999999.99</p> |
| N   | DiscountAmount  | Discount Amount      | 9-character decimal       | <p>Contains the item discount amount</p> <p>Must contain 2 decimal places</p> <p>Minimum amount is 0.00 and maximum is 999999.99</p>                                                               |
| N   | CommodityCode   | Commodity Code       | 15-character alphanumeric | Code assigned to the merchant that best categorizes the item(s) being purchased                                                                                                                    |
| C*  | Tax             | Tax                  | Up to 6 arrays            | <p>Can have up to 6 arrays contains different tax details. See Tax Array below for each field description.</p> <p>*This field is conditionally mandatory — if you use this</p>                     |

| Req | Variable Name | Field Name | Size/Type | Description                                                                                   |
|-----|---------------|------------|-----------|-----------------------------------------------------------------------------------------------|
|     |               |            |           | array, you must fill in all tax array fields as listed in the Tax Array Request Fields below. |

Table 4 Tax Array Request Fields - MasterCard Level 2/3 Transactions

| Req | Variable Name | Field Name | Size/Type                 | Description                                                                                                                                                                                                                                                                                                               |
|-----|---------------|------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M   | tax_amount    | Tax Amount | 12-character decimal      | <p>Contains detail tax amount for purchase of goods or service</p> <p>Must be 2 decimal places</p> <p>Maximum 999999.99</p>                                                                                                                                                                                               |
| M   | tax_rate      | Tax Rate   | 5-character decimal       | <p>Contains the detailed tax rate applied in relationship to a specific tax amount</p> <div style="border: 1px solid black; padding: 5px; margin: 10px 0;"> <p><b>EXAMPLE:</b> 5% GST should be '5.0' or 9.975% QST should be '9.975'</p> </div> <p>May contain up to 3 decimals, minimum 0.001, maximum up to 9999.9</p> |
| M   | tax_type      | Tax Type   | 4-character alpha-numeric | Contains tax type such as GST,QST,PST,HST                                                                                                                                                                                                                                                                                 |
| M   | tax_id        | Tax ID     | 20-character alpha-       | Provides an iden-                                                                                                                                                                                                                                                                                                         |

| Req | Variable Name         | Field Name                      | Size/Type                 | Description                                                                                                                                                                                                       |
|-----|-----------------------|---------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                       |                                 | numeric                   | tification number used by the card acceptor with the tax authority in relationship to a specific tax amount such as GST/HST number                                                                                |
| M   | tax_included_in_sales | Tax included in sales indicator | 1-character alpha-numeric | <p>This is the indicator used to reflect additional tax capture and reporting.</p> <p>Valid values are:</p> <p>Y = Tax included in total purchase amount</p> <p>N = Tax not included in total purchase amount</p> |

## A.8 Definition of Request Fields for Level 2/3 - Amex

Table 1 Amex- Level 2/3 Request Fields - Table 1 - Heading Fields

| Req | Variable Name | Field Name            | Size/Type                  | Description                                                                                                                                                                                                                                                                                            |
|-----|---------------|-----------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C   | big04         | Purchase Order Number | 22-character alpha-numeric | <p>The cardholder supplied Purchase Order Number, which is entered by the merchant at the point-of-sale</p> <p>This entry is used in the Statement/Reporting process and may include accounting information specific to the client</p> <p>Mandatory if the merchant's customer provides a Purchase</p> |



| Req       | Variable Name     | Field Name             | Size/Type                  | Description                                                                                                                                                                                                                                                                            |           |              |    |                |    |                   |    |                |    |              |    |               |
|-----------|-------------------|------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------|----|----------------|----|-------------------|----|----------------|----|--------------|----|---------------|
|           |                   |                        |                            | Order Number                                                                                                                                                                                                                                                                           |           |              |    |                |    |                   |    |                |    |              |    |               |
| N         | big05             | Release Number         | 30-character alpha-numeric | A number that identifies a release against a Purchase Order previously placed by the parties involved in the transaction                                                                                                                                                               |           |              |    |                |    |                   |    |                |    |              |    |               |
| N         | big10             | Invoice Number         | 8-character alpha-numeric  | Contains the Amex invoice/reference number                                                                                                                                                                                                                                             |           |              |    |                |    |                   |    |                |    |              |    |               |
| Y         | n101              | Entity Identifier Code | 2-character alpha-numeric  | Supported values:<br><br>‘R6’ - Requester (required)<br><br>‘BG’ - Buying Group (optional)<br><br>‘SF’ - Ship From (optional)<br><br>‘ST’ - Ship To (optional)<br><br>‘40’ - Receiver (optional)                                                                                       |           |              |    |                |    |                   |    |                |    |              |    |               |
| Y         | n102              | Name                   | 40-character alpha-numeric | <table><tr><th>n101 code</th><th>n102 meaning</th></tr><tr><td>R6</td><td>Requester Name</td></tr><tr><td>BG</td><td>Buying Group Name</td></tr><tr><td>SF</td><td>Ship From Name</td></tr><tr><td>ST</td><td>Ship To Name</td></tr><tr><td>40</td><td>Receiver Name</td></tr></table> | n101 code | n102 meaning | R6 | Requester Name | BG | Buying Group Name | SF | Ship From Name | ST | Ship To Name | 40 | Receiver Name |
| n101 code | n102 meaning      |                        |                            |                                                                                                                                                                                                                                                                                        |           |              |    |                |    |                   |    |                |    |              |    |               |
| R6        | Requester Name    |                        |                            |                                                                                                                                                                                                                                                                                        |           |              |    |                |    |                   |    |                |    |              |    |               |
| BG        | Buying Group Name |                        |                            |                                                                                                                                                                                                                                                                                        |           |              |    |                |    |                   |    |                |    |              |    |               |
| SF        | Ship From Name    |                        |                            |                                                                                                                                                                                                                                                                                        |           |              |    |                |    |                   |    |                |    |              |    |               |
| ST        | Ship To Name      |                        |                            |                                                                                                                                                                                                                                                                                        |           |              |    |                |    |                   |    |                |    |              |    |               |
| 40        | Receiver Name     |                        |                            |                                                                                                                                                                                                                                                                                        |           |              |    |                |    |                   |    |                |    |              |    |               |
| N         | n301              | Address                | 40-character alpha-numeric | Address                                                                                                                                                                                                                                                                                |           |              |    |                |    |                   |    |                |    |              |    |               |
| N         | n401              | City                   | 30-character alpha-numeric | City                                                                                                                                                                                                                                                                                   |           |              |    |                |    |                   |    |                |    |              |    |               |
| N         | n402              | State or Province      | 2-character alpha-numeric  | State or Province                                                                                                                                                                                                                                                                      |           |              |    |                |    |                   |    |                |    |              |    |               |
| N         | n403              | Postal Code            | 15-character alpha-        | Postal Code                                                                                                                                                                                                                                                                            |           |              |    |                |    |                   |    |                |    |              |    |               |

| Req        | Variable Name                                                                                                           | Field Name                         | Size/Type                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |                  |    |                                                                                                                         |    |                                        |    |     |    |     |    |     |
|------------|-------------------------------------------------------------------------------------------------------------------------|------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|----|-------------------------------------------------------------------------------------------------------------------------|----|----------------------------------------|----|-----|----|-----|----|-----|
|            |                                                                                                                         |                                    | numeric                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |                  |    |                                                                                                                         |    |                                        |    |     |    |     |    |     |
| Y          | ref01                                                                                                                   | Reference Identification Qualifier | 2-character alpha-numeric  | <div>This element may contain the following qualifiers for the corresponding occurrences of the N1Loop:</div> <table><thead><tr><th>n101 value</th><th>ref01 denotation</th></tr></thead><tbody><tr><td>R6</td><td>Supported values:<br/><br/>4C - Shipment Destination Code (mandatory)<br/><br/>CR - Customer Reference Number (conditional)</td></tr><tr><td>BG</td><td>n/a</td></tr><tr><td>SF</td><td>n/a</td></tr><tr><td>ST</td><td>n/a</td></tr><tr><td>40</td><td>n/a</td></tr></tbody></table> | n101 value | ref01 denotation | R6 | Supported values:<br><br>4C - Shipment Destination Code (mandatory)<br><br>CR - Customer Reference Number (conditional) | BG | n/a                                    | SF | n/a | ST | n/a | 40 | n/a |
| n101 value | ref01 denotation                                                                                                        |                                    |                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |                  |    |                                                                                                                         |    |                                        |    |     |    |     |    |     |
| R6         | Supported values:<br><br>4C - Shipment Destination Code (mandatory)<br><br>CR - Customer Reference Number (conditional) |                                    |                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |                  |    |                                                                                                                         |    |                                        |    |     |    |     |    |     |
| BG         | n/a                                                                                                                     |                                    |                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |                  |    |                                                                                                                         |    |                                        |    |     |    |     |    |     |
| SF         | n/a                                                                                                                     |                                    |                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |                  |    |                                                                                                                         |    |                                        |    |     |    |     |    |     |
| ST         | n/a                                                                                                                     |                                    |                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |                  |    |                                                                                                                         |    |                                        |    |     |    |     |    |     |
| 40         | n/a                                                                                                                     |                                    |                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |                  |    |                                                                                                                         |    |                                        |    |     |    |     |    |     |
| Y          | ref02                                                                                                                   | Reference Identification           | 15-character alpha-numeric | <div>VR is the Vendor ID Number, other codes describe the following:</div> <table><thead><tr><th>ref01 code</th><th>ref02 denotation</th></tr></thead><tbody><tr><td>4C</td><td>Ship to Zip or Canadian Postal Code (required)</td></tr><tr><td>CR</td><td>Cardmember Reference Number (optional)</td></tr></tbody></table>                                                                                                                                                                              | ref01 code | ref02 denotation | 4C | Ship to Zip or Canadian Postal Code (required)                                                                          | CR | Cardmember Reference Number (optional) |    |     |    |     |    |     |
| ref01 code | ref02 denotation                                                                                                        |                                    |                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |                  |    |                                                                                                                         |    |                                        |    |     |    |     |    |     |
| 4C         | Ship to Zip or Canadian Postal Code (required)                                                                          |                                    |                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |                  |    |                                                                                                                         |    |                                        |    |     |    |     |    |     |
| CR         | Cardmember Reference Number (optional)                                                                                  |                                    |                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |                  |    |                                                                                                                         |    |                                        |    |     |    |     |    |     |

Table 2 Amex - Level 2/3 Request Fields - Table 2 - Detail Fields

| Req | Variable Name | Field Name         | Size/Type            | Description      |
|-----|---------------|--------------------|----------------------|------------------|
| Y   | it102         | Line Item Quantity | 10-character decimal | Quantity of line |

| Req | Variable Name | Field Name                         | Size/Type                | Description                                                                                                                                                                                                                                                                                                                  |
|-----|---------------|------------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |               | Invoiced                           |                          | <p>item.</p> <p>Up to 2 decimal places supported.</p> <p>Minimum amount is 0.0 and maximum is 9999999999.</p>                                                                                                                                                                                                                |
| Y   | it103         | Unit or Basis for Measurement Code | 2-character alphanumeric | <p>The line item unit of measurement code</p> <p>Must contain a code that specifies the units in which the value is expressed or the manner in which a measurement is taken</p> <div> <p><b>EXAMPLE:</b> EA = each, E5=inches</p> </div> <p>See ANSI X-12 EDI Allowable Units of Measure and Codes for the list of codes</p> |
| Y   | it104         | Unit Price                         | 15-character decimal     | <p>Line item cost per unit</p> <p>Must contain 2 decimal places</p> <p>Minimum amount is 0.00 and maximum is 999999.99</p>                                                                                                                                                                                                   |
| N   | it105         | Basis or Unit Price Code           | 2-character alphanumeric | Code identifying the type of unit price for an item                                                                                                                                                                                                                                                                          |

| Req     | Variable Name             | Field Name                   | Size/Type                                                                                                                                                                                                                                 | Description                                                                                                                                                                                                                                                                                         |                     |    |                           |    |                           |       |                           |                                                                                                                                                    |
|---------|---------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----|---------------------------|----|---------------------------|-------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
|         |                           |                              |                                                                                                                                                                                                                                           | <div><b>EXAMPLE:</b> DR = dealer, AP = advise price</div><br>See ASC X12 004010 Element 639 for list of codes                                                                                                                                                                                       |                     |    |                           |    |                           |       |                           |                                                                                                                                                    |
| N       | it10618                   | Product/Service ID Qualifier | 2-character alphanumeric                                                                                                                                                                                                                  | Supported values:<br><br>'MG' - Manufacturer's Part Number<br><br>'VC' - Supplier Catalog Number<br><br>'SK' - Supplier Stock Keeping Unit Number<br><br>'UP' - Universal Product Code<br><br>'VP' – Vendor Part Number<br><br>'PO' – Purchase Order Number<br><br>'AN' – Client Defined Asset Code |                     |    |                           |    |                           |       |                           |                                                                                                                                                    |
| N       | it10719                   | Product/Service ID           | <table><tr><th>it10618</th><th>it10719 - size/type</th></tr><tr><td>VC</td><td>20-character alphanumeric</td></tr><tr><td>PO</td><td>22-character alphanumeric</td></tr><tr><td>Other</td><td>30-character alphanumeric</td></tr></table> | it10618                                                                                                                                                                                                                                                                                             | it10719 - size/type | VC | 20-character alphanumeric | PO | 22-character alphanumeric | Other | 30-character alphanumeric | Product/Service ID corresponds to the preceding qualifier defined in it10618<br><br>The maximum length depends on the qualifier defined in it10618 |
| it10618 | it10719 - size/type       |                              |                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                     |                     |    |                           |    |                           |       |                           |                                                                                                                                                    |
| VC      | 20-character alphanumeric |                              |                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                     |                     |    |                           |    |                           |       |                           |                                                                                                                                                    |
| PO      | 22-character alphanumeric |                              |                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                     |                     |    |                           |    |                           |       |                           |                                                                                                                                                    |
| Other   | 30-character alphanumeric |                              |                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                     |                     |    |                           |    |                           |       |                           |                                                                                                                                                    |
| C       | txi01                     | Tax Type code                | 2-character alphanumeric                                                                                                                                                                                                                  | Supported values:<br><br>'CA' – City Tax (optional)<br><br>'CT' – County/Tax (optional)                                                                                                                                                                                                             |                     |    |                           |    |                           |       |                           |                                                                                                                                                    |

| Req | Variable Name | Field Name      | Size/Type           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----|---------------|-----------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |               |                 |                     | <p>'EV' – Environmental Tax (optional)</p> <p>'GS' – Good and Services Tax (GST) (optional)</p> <p>'LS' – State and Local Sales Tax (optional)</p> <p>'LT' – Local Sales Tax (optional)</p> <p>'PG' – Provincial Sales Tax (PST) (optional)</p> <p>'SP' – State/Provincial Tax a.k.a. Quebec Sales Tax (QST) (optional)</p> <p>'ST' – State Sales Tax (optional)</p> <p>'TX' – All Taxes (required)</p> <p>'VA' – Value-Added Tax a.k.a. Canadian Harmonized Sales Tax (HST) (optional)</p>                                                      |
| C   | txi02         | Monetary Amount | 6-character decimal | <p>This element may contain the monetary tax amount that corresponds to the Tax Type Code in txi01</p> <div data-bbox="1177 1455 1411 1791" style="border: 1px solid black; padding: 5px; margin-top: 10px;"> <p><b>NOTE:</b><br/>           If txi02 is used in mandatory occurrence txi01=TX, txi02 must contain the total tax amount applicable to the entire invoice (transaction)<br/><br/>           If taxes are not applicable for the entire invoice (transaction), txi02 must be 0.00.</p> </div> <p>The maximum value that can be</p> |

| Req | Variable Name | Field Name      | Size/Type                | Description                                                                                                                                                                                                                                                                                                                                                                                |
|-----|---------------|-----------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |               |                 |                          | <p>entered in this field is "9999.99", which is \$9,999.99 (CAD)</p> <p>A debit is entered as: 9999.99</p> <p>A credit is entered as: -9999.99</p>                                                                                                                                                                                                                                         |
| C   | txi03         | Percent         | 10-character decimal     | <p>Contains the tax percentage (in decimal format) that corresponds to the tax type code defined in txi01</p> <p>Up to 2 decimal places supported</p>                                                                                                                                                                                                                                      |
| C   | txi06         | Tax Exempt Code | 1-character alphanumeric | <p>This element may contain the Tax Exempt Code that identifies the exemption status from sales and tax that corresponds to the Tax Type Code in txi01</p> <p>Supported values:</p> <p>1 – Yes (Tax Exempt)</p> <p>2 – No (Not Tax Exempt)</p> <p>4 – Not Exempt/For Resale</p> <p>A – Labor Taxable, Material Exempt</p> <p>B – Material Taxable, Labor Exempt</p> <p>C – Not Taxable</p> |

| Req | Variable Name | Field Name                | Size/Type                 | Description                                                                                                                                                                                       |
|-----|---------------|---------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |               |                           |                           | <p>F – Exempt (Goods / Services Tax)</p> <p>G – Exempt (Provincial Sales Tax)</p> <p>L – Exempt Local Service</p> <p>R – Recurring Exempt</p> <p>U – Usage Exempt</p>                             |
| Y   | pam05         | Line Item Extended Amount | 8-character decimal       | <p>Contains the individual item amount that is normally calculated as price multiplied by quantity</p> <p>Must contain 2 decimal places</p> <p>Minimum amount is 0.00 and maximum is 99999.99</p> |
| Y   | pid05         | Line Item Description     | 80-character alphanumeric | <p>Line Item description</p> <p>Contains the description of the individual item purchased</p> <p>This field pertain to each line item in the transaction</p>                                      |

Table 3 Amex - Level 2/3 Request Fields - Table 3 - Summary Fields

| Req | Variable Name | Field Name    | Size/Type                | Description       |
|-----|---------------|---------------|--------------------------|-------------------|
| C   | txi01         | Tax Type code | 2-character alphanumeric | Supported values: |

| Req | Variable Name | Field Name      | Size/Type           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----|---------------|-----------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |               |                 |                     | <p>'CA' – City Tax (optional)</p> <p>'CT' – County/Tax (optional)</p> <p>'EV' – Environmental Tax (optional)</p> <p>'GS' – Good and Services Tax (GST) (optional)</p> <p>'LS' – State and Local Sales Tax (optional)</p> <p>'LT' – Local Sales Tax (optional)</p> <p>'PG' – Provincial Sales Tax (PST) (optional)</p> <p>'SP' – State/Provincial Tax a.k.a. Quebec Sales Tax (QST) (optional)</p> <p>'ST' – State Sales Tax (optional)</p> <p>'TX' – All Taxes (required)</p> <p>'VA' – Value-Added Tax a.k.a. Canadian Harmonized Sales Tax (HST) (optional)</p> |
| C   | txi02         | Monetary Amount | 6-character decimal | <p>This element may contain the monetary tax amount that corresponds to the Tax Type Code in txi01</p> <div> <p><b>NOTE:</b><br/>If txi02 is used in mandatory occurrence txi01=TX, txi02 must contain the total tax amount applicable</p> </div>                                                                                                                                                                                                                                                                                                                 |



| Req | Variable Name | Field Name      | Size/Type                 | Description                                                                                                                                                                                                                                                                                                               |
|-----|---------------|-----------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |               |                 |                           | <div>to the entire invoice (transaction)<br/>If taxes are not applicable for the entire invoice (transaction), txi02 must be 0.00.</div> <p>The maximum value that can be entered in this field is “9999.99”, which is \$9,999.99 (CAD)</p> <p>A debit is entered as: 9999.99</p> <p>A credit is entered as: –9999.99</p> |
| C   | txi03         | Percent         | 10-character decimal      | <p>Contains the tax percentage (in decimal format) that corresponds to the tax type code defined in txi01</p> <p>Up to 2 decimal places supported</p>                                                                                                                                                                     |
| C   | txi06         | Tax Exempt Code | 1-character alpha-numeric | <p>Supported values:</p> <p>1 – Yes (Tax Exempt)</p> <p>2 – No (Not Tax Exempt)</p> <p>4 – Not Exempt/For Resale</p> <p>A – Labor Taxable, Material Exempt</p> <p>B – Material Taxable, Labor Exempt</p> <p>C – Not Taxable</p>                                                                                           |

| Req | Variable Name | Field Name | Size/Type | Description                                                                                                                                                    |
|-----|---------------|------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |               |            |           | F – Exempt (Goods / Services Tax)<br><br>G – Exempt (Provincial Sales Tax)<br><br>L – Exempt Local Service<br><br>R – Recurring Exempt<br><br>U – Usage Exempt |

## A.9 Definition of Request Fields – 3-D Secure 2.2

| Variable Name                              | Type and Limits                             | Description                                                                                                |
|--------------------------------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------|
| billing address<br>BillAddress1            | <i>String</i><br>50-character alphanumeric  | Cardholder billing address                                                                                 |
| billing city<br>BillCity                   | <i>String</i><br>50-character alphanumeric  | Cardholder billing city                                                                                    |
| billing country<br>BillCountry             | <i>String</i><br>3-character alphanumeric   | Defined as 3 digit country code ISO 3166-1                                                                 |
| billing postal code<br>BillPostalCode      | <i>String</i><br>16-character alphanumeric  | Cardholder billing postal code                                                                             |
| billing province<br>BillProvince           | <i>String</i><br>3-character alphanumeric   | Cardholder province or state<br><br>Defined in country subdivision ISO 3166-2                              |
| browser java enabled<br>BrowserJavaEnabled | <i>String</i><br><br>1-character alphabetic | Indicates whether Java is enabled in the browser<br><br>Allowable values:<br><br>T = True<br><br>F = False |

| Variable Name                                | Type and Limits                                                                                             | Description                                                                                                                                                                                                |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| browser language<br>BrowserLanguage          | <i>String</i><br>8-character alphanumeric                                                                   | As defined in IETF BCP47                                                                                                                                                                                   |
| browser screen height<br>BrowserScreenHeight | <i>String</i><br>6-character numeric                                                                        | Pixel height of cardholder screen                                                                                                                                                                          |
| browser screen width<br>BrowserScreenWidth   | <i>String</i><br>6-character numeric                                                                        | Pixel width of cardholder screen                                                                                                                                                                           |
| browser user agent<br>BrowserUserAgent       | <i>String</i><br>2048-character alpha-numeric                                                               | Browser User Agent                                                                                                                                                                                         |
| cardholder name<br>CardholderName            | <i>String</i><br>45-character alphanumeric<br><div><b>NOTE:</b> Accented characters are not allowable</div> | Name of the cardholder                                                                                                                                                                                     |
| challenge window size<br>ChallengeWindowSize | <i>String</i><br>2-character alphanumeric                                                                   | Relates to the rendering of the ACS challenge within the browser.<br><br>Allowable values:<br><br>01 = 250 x 400<br><br>02 = 390 x 400<br><br>03 = 500 x 600<br><br>04 = 600 x 400<br><br>05 = Full screen |
| cres<br>CRes                                 | <i>String</i><br>200-character alpha-numeric                                                                | Response data from the challenge                                                                                                                                                                           |
| currency<br>currency                         | <i>String</i><br>3-character numeric                                                                        | ISO 4217 3 digit currency code (CAD = 124, USD = 840)                                                                                                                                                      |

| Variable Name                                                                                                                        | Type and Limits                      | Description                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div> <b>NOTE:</b> This field should not be sent unless Multi Currency Pricing is enabled on your merchant account </div>            |                                      |                                                                                                                                                                                                                                              |
| DS transaction ID<br>DSTransId<br><div> <b>NOTE:</b> Only used in financial transactions using 3rd Party 3-D Secure services. </div> | String<br>36-character alphanumeric  | Refers to the DSTransID in the response of the previous 3DS authentication.                                                                                                                                                                  |
| decoupled request async URL<br>DecoupledRequestAsyncUrl                                                                              | String<br>256-character alphanumeric | Your URL where Moneris will POST the response back from ACS. Moneris reattempts 3 times to POST the response.                                                                                                                                |
| decoupled request indicator<br>DecoupledRequestIndicator                                                                             | String<br>1-character alphabetic     | Whether the request utilizes Decoupled Authentication or not, if the ACS confirms its use.<br><br>Y = Decoupled Authentication is supported and preferred if challenge is necessary<br><br>N = Do not use Decoupled Authentication (Default) |
| decoupled request max time<br>DecoupledRequestMaxTime                                                                                | String<br>5-character numeric        | The maximum minutes that Moneris waits for an ACS to provide results.<br><br>Numeric values between 1 and 10080. The max is equivalent to 7 days.                                                                                            |
| device channel<br>DeviceChannel                                                                                                      | String<br>2-character numeric        | The interface used to initiate the authentication:<br><br>02 = Browser (BRW)<br><br>03 = 3DS Requestor Initiated (3RI)                                                                                                                       |
| email                                                                                                                                | String                               | Cardholder email address                                                                                                                                                                                                                     |

| Variable Name                                                | Type and Limits                       | Description                                                                                                                                                                                                                        |
|--------------------------------------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Email                                                        | 254-character alpha-numeric           | <div> <b>NOTE:</b> This field is not mandatory, but it is required. It is highly recommended to provide the cardholder's email address. Lack of providing the cardholder's address, might increase the risk of rejects. </div>     |
| message category<br>MessageCategory                          | String<br>2-character numeric         | <p>Whether the authentication request is for a payment or non-payment use:</p> <p>01 = payment authentication (PA)</p> <p>02 = non-payment authentication (NPA)</p>                                                                |
| notification URL<br>NotificationURL                          | String<br>256-character alpha-numeric | Notification URL for receiving the 3DS Method POST response from the issuer ACS.                                                                                                                                                   |
| prior request ref<br>prior_request_auth_ref                  | String<br>36-character alphanumeric   | Refers to the 3DS ACS Transaction ID in the response of the previous 3DS authentication.                                                                                                                                           |
| prior request auth method<br>prior_request_auth_method       | String<br>2-character numeric         | <p>Mechanism used by the cardholder to authenticate in the previous 3DS authentication:</p> <p>01 = Frictionless authentication</p> <p>02 = Challenge authentication</p> <p>03 = AVS verified</p> <p>04 = Other issuer methods</p> |
| prior request auth timestamp<br>prior_request_auth_timestamp | String<br>12-character numeric        | <p>Date and time in UTC of the prior cardholder authentication. Found in the previous 3DS authentication response as 3DS Auth TimeStamp.</p> <p>Format is YYYYMMDDHHMM.</p>                                                        |
| recurring expiry                                             | String                                | End date after which no further                                                                                                                                                                                                    |

| Variable Name       | Type and Limits           | Description                                                                                                                                                                                                              |
|---------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RecurringExpiry     | 8-character numeric       | recurring transactions shall be performed. Format is YYYYMMDD.                                                                                                                                                           |
| recurring frequency | <i>String</i>             | The minimum number of days between recurring transactions. Numeric values between 1 and 9999, leading zeroes accepted.                                                                                                   |
| RecurringFrequency  | 4-character numeric       |                                                                                                                                                                                                                          |
| request challenge   | <i>String</i>             | Indicates whether a browser-based challenge is requested for this transaction. Standard is "01"                                                                                                                          |
| RequestChallenge    | 2-character numeric       |                                                                                                                                                                                                                          |
|                     |                           | <ul style="list-style-type: none"> <li>• 01 = No preference</li> <li>• 02 = No challenge requested</li> <li>• 03 = Challenge requested: 3DS Requestor Preference</li> <li>• 04 = Challenge requested: Mandate</li> </ul> |
| request type        | <i>String</i>             | Indicates the type of browser-based authentication request:                                                                                                                                                              |
| RequestType         | 2-character alphanumeric  |                                                                                                                                                                                                                          |
|                     |                           | 01 = cardholder initiated payment<br>02 = recurring transaction<br>03 = installment transaction<br>04 = add card<br>05 = maintain card<br>06 = cardholder verification as part of EMV token ID & V                       |
| shipping address    | <i>String</i>             | Shipping destination address                                                                                                                                                                                             |
| ShipAddress1        | 50-character alphanumeric |                                                                                                                                                                                                                          |
| ri indicator        | <i>String</i>             | The type of 3DS Requestor Initiated (3RI) request:                                                                                                                                                                       |

| Variable Name                                                                                                                                                                                                                                                                                         | Type and Limits                     | Description                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RiIndicator<br><br><div> <b>NOTE:</b> Visa Secure only support<br/>           ri_Indicator = 01, 02, 06, 08, 09,<br/>           or 11 for Payment Transactions<br/>           and ri Indicator = 03, 04, 05, 08,<br/>           09 or 10 for Non-Payment Trans-<br/>           actions         </div> | 2-character numeric                 | 01 = Recurring<br><br>02 = Installment<br><br>03 = Add Card<br><br>04 = Maintain Card Information<br><br>05 = Account verification<br><br>06 = Split/Delayed Shipment<br><br>07 = Top-up<br><br>08 = Mail Order<br><br>09 = Telephone Order<br><br>10 = Whitelist<br><br>11 = Other Payment |
| shipping city<br>ShipCity                                                                                                                                                                                                                                                                             | String<br>50-character alphanumeric | Shipping destination city                                                                                                                                                                                                                                                                   |
| shipping country<br>ShipCountry                                                                                                                                                                                                                                                                       | String<br>3-character alphanumeric  | Shipping destination country<br><br>Defined as 3-digit country code in<br>ISO 3166-1                                                                                                                                                                                                        |
| shipping postal code<br>ShipPostalCode                                                                                                                                                                                                                                                                | String<br>16-character alphanumeric | Shipping destination postal or ZIP code                                                                                                                                                                                                                                                     |
| shipping province<br>ShipProvince                                                                                                                                                                                                                                                                     | String<br>3-character alphanumeric  | Shipping destination province<br><br>Defined in country subdivision ISO<br>3166-2                                                                                                                                                                                                           |
| 3DS completion indicator<br>ThreeDSCompletionInd                                                                                                                                                                                                                                                      | String<br>1-character alphanumeric  | indicates whether 3ds method<br>MpiCardLookup was successfully<br>completed<br><br>Allowable values:<br><br>Y = Successfully completed<br><br>N = Did not successfully complete                                                                                                             |

| Variable Name                                | Type and Limits                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| U = Unavailable                              |                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| browser IP Address<br><BrowserIP>            | String<br><br>Allows '.' and ':'<br><br>45-character alphanumeric | IP address of the browser as returned by the HTTP headers to the 3DS Requestor.<br><br><b>NOTE:</b> This field is not mandatory, but it is required. It is highly recommended to provide. Lack of providing this field, might increase the risk of rejects.                                                                                                                                                                                                                               |
| cardholder work phone number<br><WorkPhone>  | Object<br><br>N/A                                                 | Cardholder work phone number<br><br><b>NOTE:</b> This field is not mandatory, but it is required. It is highly recommended to provide at least one of the Cardholder Phone Number. Lack of providing at least one of the Cardholder Phone Number, might increase the risk of rejects.<br><br><b>NOTE:</b> This is a nested object within the transaction. For information about fields in the Cardholder Phone Number Info object, see Cardholder Phone Number Info Object and Variables. |
| cardholder home phone number<br><home_phone> | Object<br><br>N/A                                                 | Cardholder home phone number<br><br><b>NOTE:</b> This field is not mandatory, but it is required. It is highly recommended to provide at least one of the Cardholder Phone Number. Lack of providing at least one of the Cardholder Phone Number, might increase the risk of rejects.<br><br><b>NOTE:</b> This is a nested object within the transaction. For information about fields in the Cardholder Phone Number Info object, see Cardholder Phone Number Info Object and Variables. |
| cardholder mobile phone number               | Object                                                            | Cardholder mobile phone number                                                                                                                                                                                                                                                                                                                                                                                                                                                            |



| Variable Name  | Type and Limits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <mobile_phone> | N/A             | <p><b>NOTE:</b> This field is not mandatory, but it is required. It is highly recommended to provide at least one of the Cardholder Phone Number. Lack of providing at least one of the Cardholder Phone Number, might increase the risk of rejects.</p> <p><b>NOTE:</b> This is a nested object within the transaction. For information about fields in the Cardholder Phone Number Info object, see Cardholder Phone Number Info Object and Variables.</p> |

### MPI 3DS Cardholder Phone Number

| Variable Name                  | Type and Limits                       | Description                                              |
|--------------------------------|---------------------------------------|----------------------------------------------------------|
| country code<br><country_code> | <i>String</i><br>3-character numeric  | Country Code of phone number provided by the Cardholder. |
| phone number<br><phone_number> | <i>String</i><br>15-character numeric | The phone number provided by the Cardholder.             |

## A.10 Definition of Request Fields – PBB Info Object

This table includes fields for the Pay by Bank object pbb\_info included in financial authorization transactions, such as a Purchase or Preauthorization. This object contains the necessary fields to link the authorization to the consent

For the fields used Pay By Bank authentication calls (Create Consent, Get Consent ID, and Get Payment Transaction Data), refer to "Definition of Request Fields – Pay By Bank" on page 583.

| Variable Name            | Type and Limits                                      | Description                                                                                                                                                    |
|--------------------------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Consent ID<br>consent_id | <i>String</i><br>36-character alphanumeric, variable | <p><code>\$pbb_info-&gt;setConsentId(\$consent_id);</code></p> <p>Consent token obtained from the Interac Konek SDK. Used to authenticate PBB connections.</p> |

| Variable Name                          | Type and Limits                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Payment Method<br>payment_method       | <i>String</i><br><br>27-character alphanumeric, variable  | <pre>\$pbb_info-&gt;setPaymentMethod(\$payment_method);</pre> <p>The type of account for the payment</p> <p>BANK_ACCOUNT_SAVINGS, BANK_ACCOUNT_CHEQUING, BANK_ACCOUNT_LINE_OF_CREDIT, VISA_CREDIT, VISA_DEBIT, MASTERCARD_CREDIT, MASTERCARD_DEBIT, AMEX_CREDIT</p>                                                                                                                                                                                                                                                                                                                       |
| PBB Cryptogram<br>cryptogram           | <i>String</i><br><br>256-character alphanumeric, variable | <pre>\$pbb_info-&gt;setCryptogram(\$cryptogram);</pre> <p>Payment cryptogram used with payment token in subsequent financial authorization request.</p> <p>This field is conditional; including it is required if a Cryptogram was returned in the TransactionData object. Otherwise, omit this field.</p> <div> <p><b>NOTE:</b> Payment cryptograms are only required for Interac Konek Direct transactions involving a tokenized credit card product. If the underlying payment method is a bank account, a cryptogram is not returned for use in the financial transaction.</p> </div> |
| Cryptogram Expiry<br>cryptogram_expiry | <i>String</i><br><br>24-character numeric, fixed          | <pre>\$pbb_info-&gt;setCryptogramExpiry(\$cryptogram_expiry);</pre> <p>UTC Date &amp; time at which cryptogram expires in ISO 8601 standard.</p> <p>This field is conditional; including it is required if a CryptogramExpiry was returned in the TransactionData object. Otherwise, omit this field.</p>                                                                                                                                                                                                                                                                                 |
| Channel<br>channel                     | <i>String</i><br><br>256-character alpha-                 | <pre>\$pbb_info-&gt;setChannel(\$channel);</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Variable Name | Type and Limits   | Description                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | numeric, variable | <p>The channel used by the cardholder for initiating the request. Supports the values:</p> <p>MOBILE_APP : For applications on mobile devices</p> <p>MOBILE_WEB : For web browsers on mobile devices</p> <p>DESTOP_WEB : For desktop web browsers</p> <div> <p><b>NOTE:</b> This field is required for Interac Direct Transactions. When missing, the default value is DESTOP_WEB</p> </div> |

## A.11 Definition of Request Fields – Pay By Bank

This table includes fields for the Create Consent, Get Consent ID, and Get Payment Transaction Data API calls used in the authentication flow for a Pay By Bank transaction.

For the fields used in the pbb\_info object of a financial transaction, such as a Purchase or Preauthorization, refer instead to "Definition of Request Fields – PBB Info Object" on page 581

| Variable Name            | Type and Limits                                                                                             | Description                                                                                                        |
|--------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Amount<br>amount         | <p><i>Number</i></p> <p>10-character numeric, variable</p> <p>minimum: 0.01</p> <p>maximum: 99999999.99</p> | <p>placeholder</p> <p>Amount for use in subsequent financial transaction.</p>                                      |
| Consent ID<br>consent_id | <p><i>String</i></p> <p>36-character alphanumeric, variable</p>                                             | <p>placeholder</p> <p>Consent token obtained from the Interac Konek SDK. Used to authenticate PBB connections.</p> |
| Currency<br>currency     | <p><i>String</i></p> <p>3-character alphanumeric</p>                                                        | <p>placeholder</p> <p>ISO 4217 currency code used for the</p>                                                      |

| Variable Name                        | Type and Limits                                     | Description                                                                                                                                                                                                                                                                                                     |
|--------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      |                                                     | payment. Possible values<br>CAD                                                                                                                                                                                                                                                                                 |
| Required Contact Fields              | <i>Array of strings</i><br>N/A                      | List of cardholder contact fields flagged as required by the merchant. Array contains contact_field strings for each field flagged. Possible values are:<br><br>NAME, EMAIL, PHONE, SHIPPING_ADDRESS                                                                                                            |
| Pay By Bank Merchant ID              | <i>String</i><br>10-50 alphanumeric                 | Unique Merchant identifier assigned by Interac Konek during onboarding for Pay By Bank                                                                                                                                                                                                                          |
| Supported Payment Methods            | <i>Array of strings</i><br>N/A                      | List of payment methods (payment_method) the merchant supports for the transaction. Values are of the type AccountType, with the possible values as:<br><br>BANK_ACCOUNT_SAVINGS, BANK_ACCOUNT_CHEQUING, BANK_ACCOUNT_LINE_OF_CREDIT, VISA_CREDIT, VISA_DEBIT, MASTERCARD_CREDIT, MASTERCARD_DEBIT, AMEX_CREDIT |
| Transaction Type<br>transaction_type | <i>String</i><br>9-character alphanumeric, variable | placeholder<br><br>transaction type.<br><br>Enums: PURCHASE, PRE_AUTH, CAPTURE                                                                                                                                                                                                                                  |

### A.11.1 Pay By Bank - Payment Object

| Variable Name        | Type and Limits                           | Description                                                                               |
|----------------------|-------------------------------------------|-------------------------------------------------------------------------------------------|
| Currency<br>currency | <i>String</i><br>3-character alphanumeric | \$pbbCreateConsent->setPbbPayment(\$currency);<br><br>ISO 4217 currency code used for the |

| Variable Name                  | Type and Limits                                                                              | Description                                                                                                                                                                                                                                                             |
|--------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                |                                                                                              | payment. Possible values<br>CAD                                                                                                                                                                                                                                         |
| Amount<br>amount               | Number<br><br>10-character numeric, variable<br><br>minimum: 0.01<br><br>maximum: 9999999.99 | \$pbbCreateConsent->setPbbPayment(\$amount);<br><br>First payment amount of the order.<br><br>For RECURRING payments where other payment methods such as Gift Cards are used for the initial payment, the Amount can be 0. Otherwise it must be a value greater than 0. |
| Sub Total<br>sub_total         | Number<br><br>10-character numeric, variable                                                 | \$pbbCreateConsent->setPbbPayment(\$sub_total);<br><br>Subtotal of the order                                                                                                                                                                                            |
| Tax<br>tax                     | Number<br><br>10-character numeric, variable                                                 | \$pbbCreateConsent->setPbbPayment(\$tax);<br><br>Total tax calculated for the order                                                                                                                                                                                     |
| Shipping Cost<br>shipping_cost | Number<br><br>10-character numeric                                                           | \$pbbCreateConsent->setPbbPayment(\$shipping_cost);<br><br>Shipping cost of the order                                                                                                                                                                                   |

### A.11.2 Pay By Bank - Order Object

| Variable Name                  | Type and Limits                          | Description                                                                                                                                    |
|--------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Shipment Type<br>shipment_type | String<br><br>6-character alphanumeric   | \$shipment_type = "NONE";<br><br>Indicates how the merchandise will be shipped to the customer. Possible values are<br><br>SINGLE, SPLIT, NONE |
| Items<br>items                 | Array object<br><br>Max 100 item objects | \$item1 = array(                                                                                                                               |

| Variable Name                                      | Type and Limits                                            | Description                                                                                                                                                             |
|----------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    |                                                            | A list of Items objects for the line items within the order, with their recurring setups if available. See "Pay By Bank - Item Object" below below.                     |
| Merchant Order Reference<br>merchant_order_ref     | <i>String</i><br><br>100-character alpha-numeric, variable | <code>\$merchant_order_ref = "1695401825L1CLwAJ98p7JwAc";</code><br><br>Unique identifier for the order or cart                                                         |
| Placement Mode<br>placement_mode                   | <i>String</i><br><br>8-character alphanumeric, variable    | <code>\$placement_mode = "STANDARD";</code><br><br>Placement mode<br><br>Enums: STANDARD, EXPRESS                                                                       |
| Merchant Customer Reference<br>MerchantCustomerRef | <i>String</i><br><br>50-character alpha-numeric, variable  | Merchant assigned internal value uniquely identifying a merchant's customer. Useful to associate transactions to a specific customer for reporting and troubleshooting. |

### A.11.3 Pay By Bank - Item Object

| Variable Name                          | Type and Limits                                        | Description                                                                                                                              |
|----------------------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Item Payment Type<br>item_payment_type | <i>String</i><br><br>9-character alphaumeric, variable | <code>"item_payment_type" =&gt; "SINGLE",</code><br><br>Payment type.<br><br>Enums: SINGLE, RECURRING                                    |
| Amount<br>Amount                       | <i>Number</i><br><br>10-character numeric, variable    | <code>"amount" =&gt; 1.00,</code><br><br>Amount of the transaction.<br><br>For RECURRING payments, this is the recurring payment amount. |
| Item Quantity<br>qty                   | <i>String</i><br><br>5-digits                          | <code>"qty" =&gt; 1,</code><br><br>Quantity of the item                                                                                  |

| Variable Name                   | Type and Limits                                        | Description                                                                       |
|---------------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------|
| Item Description<br>description | <i>String</i><br>100-character alpha-numeric, variable | "description" => "Item1",<br>Short description of the line item.                  |
| Item Type<br>item_type          | <i>String</i><br>15-character numeric, variable        | Type of item in the card.<br><br>Enums: GIFT_CARD, CRYPTO_CURRENCY, TOP_UP, OTHER |
| Item Reference<br>item_ref      | <i>String</i><br>100-character numeric, variable       | Unique value to identify the transaction                                          |
| Item Name<br>item_name          | <i>String</i><br>255-character alpha-numeric, variable | "item_ref" => "Testing"<br><br>Name of the item.                                  |

## A.12 Definition of Request Fields – MCP

| Variable Name            | Type and Limits                                                                     | Description                                                                             |
|--------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| MCP version number       | <i>String</i><br>numeric<br>current version is 1.0                                  | Release version number for MCP                                                          |
| cardholder amount        | <i>String</i><br>12-character numeric<br>smallest discrete unit of foreign currency | Amount, in units of foreign currency, the cardholder will be charged on the transaction |
| cardholder currency code | <i>String</i><br>3-character numeric                                                | ISO code representing the foreign currency of the cardholder                            |

**Optional MCP fields**

| Variable Name  | Type and Limits          | Description                                                                                                                                                                                                       |
|----------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MCP rate token | <i>String</i><br><br>N/A | Token representing a temporarily locked-in foreign exchange rate, obtained in the response of the MCP Get Rate transaction and used in subsequent MCP financial transaction requests in order to redeem that rate |

**MCP Get Rate transaction request fields**

| Variable Name         | Type and Limits                                            | Description                                                                                                                                                        |
|-----------------------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MCP version number    | <i>String</i><br><br>numeric<br><br>current version is 1.0 | Release version number for MCP                                                                                                                                     |
| rate transaction type | <i>String</i><br><br>1-character alphabetic                | Value representing the type of subsequent transaction request that the rate token will be used for.<br><br>Allowable values:<br><br>P – Purchase<br><br>R – Refund |
| MCP Rate Info         | <i>Object</i><br><br>N/A                                   | Nested object in the MCP Get Rate transaction containing the <b>add cardholder amount</b> and <b>add merchant settlement</b> fields                                |

**MCP Rate Info object request fields**

At least one of the following variables must be sent:

| Variable Name         | Type and Limits                                                                                        | Description                                                                                                                                     |
|-----------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| add cardholder amount | <i>String array</i><br><br>12-character numeric, 3-character numeric<br><br>(smallest discrete unit of | A string array representing: <ul style="list-style-type: none"> <li>the amount, in units of foreign currency, the cardholder will be</li> </ul> |



| Variable Name                  | Type and Limits                                                                                                           | Description                                                                                                                                                                                                                                               |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | foreign currency, currency code)                                                                                          | <p>charged, and</p> <ul style="list-style-type: none"> <li>the ISO currency code corresponding to the foreign currency of the cardholder</li> </ul>                                                                                                       |
| add merchant settlement amount | <p><i>String array</i></p> <p>12-character numeric, 3-character numeric</p> <p>(amount in CAD pennies, currency code)</p> | <p>A string array representing:</p> <ul style="list-style-type: none"> <li>the amount the merchant will receive in the transaction, in Canadian dollars</li> <li>the ISO currency code corresponding to the foreign currency of the cardholder</li> </ul> |

### A.13 Definition of Request Fields – Offlinx™

Applies to Offlinx™ integration only

| Variable Name | Type and Limits                                       | Description                                                                                                                                                                       |
|---------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| card match ID | <p><i>String</i></p> <p>50-character alphanumeric</p> | <p>Corresponds to the Transaction ID used for the Offlinx™ Card Match Pixel Tag, a unique identifier created by the merchant</p> <p>Must be unique value for each transaction</p> |

### A.14 Definition of Request Fields – Convenience Fee

| Variable Name               | Type and Limits                 | Description                                            |
|-----------------------------|---------------------------------|--------------------------------------------------------|
| Convenience Fee Information | <p><i>Object</i></p> <p>N/A</p> | Contains fields related to the Convenience Fee feature |

| Variable Name          | Type and Limits                      | Description                                                |
|------------------------|--------------------------------------|------------------------------------------------------------|
| convenience fee amount | <i>String</i><br>9-character decimal | Dollar amount charged to the customer as a convenience fee |

## A.15 Definition of Request Fields – Recurring

### Recurring Billing Info Object Request Fields

| Variable Name    | Type and Limits                   | Description                                                                                                                                                                                                                   |
|------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| number of recurs | <i>String</i><br>numeric<br>1-999 | The number of times that the transaction must recur                                                                                                                                                                           |
| period           | <i>String</i><br>numeric<br>1-999 | Number of recur unit intervals that must pass between recurring billings                                                                                                                                                      |
| start date       | <i>String</i><br>YYYYMMDD format  | Date of the first future recurring billing transaction; this must be a date in the future<br><br>If an additional charge will be made immediately, the start now variable must be set to true                                 |
| start now        | <i>String</i><br>true/false       | Set to true if a charge will be made against the card immediately; otherwise set to false<br><br>When set to false, use Card Verification prior to sending the Purchase with Recurring Billing and Credential on File objects |

**NOTE:** Amount to be billed immediately can differ from the subsequent recurring amounts

| Variable Name    | Type and Limits                                                                                                                                                                          | Description                                                                                                                                                                   |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| recurring amount | <i>String</i><br>10-character decimal, minimum three digits<br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><div>EXAMPLE: 1234567.89</div> | Dollar amount of the recurring transaction<br>This amount will be billed on the start date, and then billed repeatedly based on the interval defined by period and recur unit |
| recur unit       | <i>String</i><br>day, week, month or eom                                                                                                                                                 | Unit to be used as a basis for the interval<br>Works in conjunction with the period variable to define the billing frequency                                                  |

## A.16 Definition of Request Fields – Installments by Visa

| Variable Name                | Type and Limits                                                                 | Description                                                                                                                                                           |
|------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Installment Info             | <i>Object</i><br>N/A                                                            | Contains request fields related to installments                                                                                                                       |
| installment plan ID          | <i>String</i><br>36-character alphanumeric<br>fixed length                      | Card brand-generated identifier for an installment plan                                                                                                               |
| installment plan reference   | <i>String</i><br>10-character alphanumeric<br>fixed length                      | Unique, human friendly name for the installment plan                                                                                                                  |
| terms and conditions version | <i>String</i><br>10-character alphanumeric<br>variable length (1-10 characters) | Version of the terms and conditions of the installment plan accepted by the cardholder<br>The version is auto-incremented every time an update is made to the plan by |

| Variable Name | Type and Limits | Description |
|---------------|-----------------|-------------|
|               |                 | the issuer  |

## A.17 Definition of Request Fields – Account Name Verification Object

Request fields within the Account Name Verification object. The object can only be included in Card Verification transactions. Account name verification is only applicable to Visa credit cards.

| Variable Name                | Type and Limits                            | Description            |
|------------------------------|--------------------------------------------|------------------------|
| First Name<br><first_name>   | <i>String</i><br>32-character alphanumeric | Cardholder last name   |
| Middle Name<br><middle_name> | <i>String</i><br>32-character alphanumeric | Cardholder middle name |
| Last Name<br><last_name>     | <i>String</i><br>32-character alphanumeric | Cardholder last name   |

## Appendix B Definitions of Response Fields

**Table 21: Receipt object response values**

| Value                   | Type                                                                                                                                                                                                                                                                                                                                  | Limits                                                                                                                                                        | Get Method                        |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
|                         | Description                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                               |                                   |
| General response fields |                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                               |                                   |
| Card type               | String                                                                                                                                                                                                                                                                                                                                | 2-character alphabetic (min. 1)                                                                                                                               | \$mpgResponse->getCardType() ;    |
|                         | Represents the type of card in the transaction, e.g., Visa, Mastercard.<br>Possible values: <ul style="list-style-type: none"><li>• V = Visa</li><li>• M = Mastercard</li><li>• AX = American Express</li><li>• DC = Diner's Card</li><li>• NO = Novus/Discover</li><li>• SE = Sears</li><li>• D = Debit</li><li>• C1 = JCB</li></ul> |                                                                                                                                                               |                                   |
| Transaction amount      | String                                                                                                                                                                                                                                                                                                                                | 10-character decimal<br><br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><br><div>EXAMPLE:<br/>1234567.89</div> | \$mpgResponse->getTransAmount() ; |
|                         | Transaction amount that was processed.                                                                                                                                                                                                                                                                                                |                                                                                                                                                               |                                   |

**Table 21: Receipt object response values (continued)**

| Value              | Type                                                                                                                                                                   | Limits                     | Get Method                                      |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------|
|                    | Description                                                                                                                                                            |                            |                                                 |
| Transaction number | String                                                                                                                                                                 | 255-character alphanumeric | <code>\$mpgResponse-&gt;getTxnNumber()</code> ; |
|                    | Gateway Transaction identifier often needed for follow-on transactions (such as Refund and Purchase Correction) to reference the originally processed transaction.     |                            |                                                 |
| Receipt ID         | String                                                                                                                                                                 | 50-character alphanumeric  | <code>\$mpgResponse-&gt;getReceiptId()</code> ; |
|                    | Order ID that was specified in the transaction request.                                                                                                                |                            |                                                 |
| Transaction type   | String                                                                                                                                                                 | 2-character alphanumeric   | <code>\$mpgResponse-&gt;getTransType()</code> ; |
|                    | <ul style="list-style-type: none"> <li>• 0 = Purchase</li> <li>• 1 = Pre-Authorization</li> <li>• 2 = Completion</li> <li>• 4 = Refund</li> <li>• 11 = Void</li> </ul> |                            |                                                 |

Table 21: Receipt object response values (continued)

| Value            | Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Limits               | Get Method                        |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------------------|
|                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                      |                                   |
| Reference number | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 18-character numeric | \$mpgResponse->getReferenceNum(); |
|                  | <p>Terminal used to process the transaction as well as the shift, batch and sequence number. This data is typically used to reference transactions on the host systems, and must be displayed on any receipt presented to the customer.</p> <p>This information is to be stored by the merchant.</p> <p>Example: 660123450010690030</p> <ul style="list-style-type: none"> <li>• 66012345: Terminal ID</li> <li>• 001: Shift number</li> <li>• 069: Batch number</li> <li>• 003: Transaction number within the batch.</li> </ul> |                      |                                   |
| Response code    | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3-character numeric  | \$mpgResponse->getResponseCode(); |
|                  | <ul style="list-style-type: none"> <li>• &lt; 50: Transaction approved</li> <li>• ≥ 50: Transaction declined</li> <li>• Null: Transaction incomplete.</li> </ul> <p>For further details on the response codes that are returned, see the Response Codes document at <a href="https://developer.moneris.com">https://developer.moneris.com</a>.</p>                                                                                                                                                                               |                      |                                   |
| ISO              | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2-character numeric  | \$mpgResponse->getISO();          |
|                  | ISO response code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                                   |
| Bank totals      | Object                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                      | code to come                      |
|                  | Response data returned in a Batch Close and Open Totals request. See "Definitions of Response Fields" on page 593.                                                                                                                                                                                                                                                                                                                                                                                                               |                      |                                   |

Table 21: Receipt object response values (continued)

| Value               | Type Limits                                                                                                                                                                     |                            | Get Method                       |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------|
|                     | Description                                                                                                                                                                     |                            |                                  |
| Message             | String                                                                                                                                                                          | 100-character alphanumeric | \$mpgResponse->getMessage();     |
|                     | Response description returned from issuer.<br>The message returned from the issuer is intended for merchant information only, and is <b>not</b> intended for customer receipts. |                            |                                  |
| Authorization code  | String                                                                                                                                                                          | 8-character alphanumeric   | \$mpgResponse->getAuthCode();    |
|                     | Authorization code returned from the issuing institution.                                                                                                                       |                            |                                  |
| Complete            | String                                                                                                                                                                          | true/false                 | \$mpgResponse->getComplete();    |
|                     | Transaction was sent to authorization host and a response was received                                                                                                          |                            |                                  |
| Transaction date    | String                                                                                                                                                                          | Format: yyyy-mm-dd         | \$mpgResponse->getTransDate();   |
|                     | Processing host date stamp                                                                                                                                                      |                            |                                  |
| Transaction time    | String                                                                                                                                                                          | Format: ##:##:##           | \$mpgResponse->getTransTime();   |
|                     | Processing host time stamp                                                                                                                                                      |                            |                                  |
| Ticket              | String                                                                                                                                                                          | N/A                        | \$mpgResponse->getTicket();      |
|                     | Reserved field.                                                                                                                                                                 |                            |                                  |
| Timed out           | String                                                                                                                                                                          | true/false                 | \$mpgResponse->getTimedOut();    |
|                     | Transaction failed due to a process timing out.                                                                                                                                 |                            |                                  |
| Is Visa Debit       | String                                                                                                                                                                          | true/false                 | \$mpgResponse->getIsVisaDebit(); |
|                     | Indicates whether the card processed is a Visa Debit.                                                                                                                           |                            |                                  |
| PBBLifeCycleTraceID | String                                                                                                                                                                          | 15-alphanumeric            |                                  |
|                     | Unique transaction identifier from Interac Direct systems. Applies to Interac Direct transactions only and is used to link follow-on transactions.                              |                            |                                  |



**Table 21: Receipt object response values (continued)**

| Value                                   | Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Limits                    | Get Method                             |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------|
|                                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                           |                                        |
| Account Name Verification Result        | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 10-character alphanumeric | \$mpgResponse->getAccountNameResult(); |
|                                         | Code indicating the results of Visa Account Name Verification.<br>Position 1 and 2: Overall inquiry status.<br><br>Position 3 and 4: Full name match status<br><br>Position 5 and 6: last name match status<br><br>Position 7 and 8: middle name match status<br><br>Position 9 to 10: first name match status<br><br><br>Inquiry status values:<br>00 = Name match performed<br>01 = Name match no performed<br>02 = Name match not supported<br><br><br>Values for full name match and last/middle/first name match:<br>01 = Match<br>50 = Partial Match<br>99 = No Match |                           |                                        |
| Batch Close/Open Totals response fields |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                           |                                        |
| Processed card types                    | String Array                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | N/A                       |                                        |
|                                         | Returns all of the processed card types in the current batch for the terminal ID/ECR Number from the request.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                           |                                        |
| Terminal IDs                            | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 8-character alphanumeric  | code to come                           |
|                                         | Returns the terminal ID/ECR Number from the request.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                           |                                        |

Table 21: Receipt object response values (continued)

| Value            | Type        |                              | Limits | Get Method                                                                                                                                                                                                                                                                                                         |
|------------------|-------------|------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | Description |                              |        |                                                                                                                                                                                                                                                                                                                    |
| Purchase count   | String      | 4-character<br>numeric       |        | <code>\$mpgResponse-&gt;getPurchaseCount(\$secr_number,\$creditCards[\$i]);</code>                                                                                                                                                                                                                                 |
|                  |             |                              |        | Indicates the # of Purchase, Pre-Authorization Completion and Force Post transactions processed. If none were processed in the batch, then the value returned will be 0000.                                                                                                                                        |
| Purchase amount  | String      | 11-character<br>alphanumeric |        | <code>\$mpgResponse-&gt;getPurchaseAmount(\$secr_number,\$creditCards[\$i]);</code>                                                                                                                                                                                                                                |
|                  |             |                              |        | Indicates the dollar amount processed for Purchase, Pre-Authorization Completion or Force Post transactions. This field begins with a + and is followed by 10 numbers, the first 8 indicate the amount and the last 2 indicate the penny value.<br><br><b>EXAMPLE:</b> +0000000000 = 0.00 and +0000041625 = 416.25 |
| Refund count     | String      | 4-character<br>numeric       |        | <code>\$mpgResponse-&gt;getRefundAmount(\$secr_number,\$creditCards[\$i]);</code>                                                                                                                                                                                                                                  |
|                  |             |                              |        | Indicates the # of Refund or Independent Refund transactions processed. If none were processed in the batch, then the value returned will be 0000.                                                                                                                                                                 |
| Refund amount    | String      | 11-character<br>alphanumeric |        | <code>\$mpgResponse-&gt;getRefundAmount(\$secr_number,\$creditCards[\$i]);</code>                                                                                                                                                                                                                                  |
|                  |             |                              |        | Indicates the dollar amount processed for Refund, Independent Refund or ACH Credit transactions. This field begins with a + and is followed by 10 numbers, the first 8 indicate the amount and the last 2 indicate the penny value.<br><br>Example, +0000000000 = 0.00 and +0000041625 = 416.25                    |
| Correction count | String      | 4-character<br>numeric       |        | <code>\$mpgResponse-&gt;getCorrectionCount(\$secr_number,\$creditCards[\$i]);</code>                                                                                                                                                                                                                               |
|                  |             |                              |        | Indicates the # of Purchase Correction transactions processed. If none were processed in the batch, then the value returned will be 0000.                                                                                                                                                                          |

Table 21: Receipt object response values (continued)

| Value                                                      | Type                                                                                                                                                                                                           | Limits                    | Get Method                                                            |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----------------------------------------------------------------------|
|                                                            | Description                                                                                                                                                                                                    |                           |                                                                       |
| Correction amount                                          | String                                                                                                                                                                                                         | 11-character alphanumeric | \$mpgResponse->getCorrectionAmount(\$secr_number,\$creditCards[\$i]); |
|                                                            | Indicates the dollar amount processed for Purchase Correction transactions. This field begins with a + and is followed by 10 numbers, the first 8 indicate the amount and the last 2 indicate the penny value. |                           |                                                                       |
|                                                            | EXAMPLE: +0000000000 = 0.00 and +0000041625 = 416.25                                                                                                                                                           |                           |                                                                       |
| Recurring Billing Response Fields (see Appendix A, page 1) |                                                                                                                                                                                                                |                           |                                                                       |
| Recurring billing success                                  | String                                                                                                                                                                                                         | true/false                | \$mpgResponse->getRecurSuccess();                                     |
|                                                            | Indicates whether the recurring billing transaction has been successfully set up for future billing.                                                                                                           |                           |                                                                       |
| Recur update success                                       | String                                                                                                                                                                                                         | true/false                | \$mpgResponse->getRecurUpdateSuccess();                               |
|                                                            | Indicates recur update success.                                                                                                                                                                                |                           |                                                                       |
| Next recur date                                            | String                                                                                                                                                                                                         | yyyy-mm-dd                | \$mpgResponse->getNextRecurDate();                                    |
|                                                            | Indicates next recur billing date.                                                                                                                                                                             |                           |                                                                       |
| Recur end date                                             | String                                                                                                                                                                                                         | yyyy-mm-dd                | \$mpgResponse->getRecurEndDate();                                     |
|                                                            | Indicates final recur billing date.                                                                                                                                                                            |                           |                                                                       |
| Status Check response fields (see )                        |                                                                                                                                                                                                                |                           |                                                                       |
| Status code                                                | String                                                                                                                                                                                                         | 3-character alphanumeric  | \$mpgResponse->getStatusCode();                                       |
|                                                            | <ul style="list-style-type: none"><li>&lt; 50: Transaction found and successful</li><li>≥ 50: Transaction not found and not successful</li></ul>                                                               |                           |                                                                       |
|                                                            | NOTE: the status code is only populated if the connection object's Status Check property is set to true.                                                                                                       |                           |                                                                       |

Table 21: Receipt object response values (continued)

| Value                                     | Type                                                                                                                                                                                                                                                                           | Limits                    | Get Method                                        |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------------------------------|
|                                           | Description                                                                                                                                                                                                                                                                    |                           |                                                   |
| Status message                            | String                                                                                                                                                                                                                                                                         | found/not found           | \$mpgResponse->getStatusMessage()<br>();          |
|                                           | <ul style="list-style-type: none"><li>Found: <math>0 \leq \text{Status Code} \leq 49</math></li><li>Not Found or null: <math>50 \leq \text{Status Code} \leq 999</math>.</li></ul>                                                                                             |                           |                                                   |
|                                           | <div><b>NOTE:</b> The status message is only populated if the connection object's Status Check property is set to <b>true</b>.</div>                                                                                                                                           |                           |                                                   |
| AVS response fields (see 10.1, page 408)  |                                                                                                                                                                                                                                                                                |                           |                                                   |
| AVS result code                           | String                                                                                                                                                                                                                                                                         | 1-character alpha-numeric | \$mpgResponse->getAvsResultCode()<br>();          |
|                                           | Indicates the address verification result. For a full list of possible response codes refer to Section Appendix B.                                                                                                                                                             |                           |                                                   |
| CVD response fields (see )                |                                                                                                                                                                                                                                                                                |                           |                                                   |
| CVD result code                           | String                                                                                                                                                                                                                                                                         | 2-character alpha-numeric | \$mpgResponse->getCvdResultCode()<br>();          |
|                                           | Indicates the CVD validation result. The first byte is the numeric CVD indicator sent in the request; the second byte is the response code. Possible response codes are shown in Appendix B                                                                                    |                           |                                                   |
| GooglePay Token response fields           |                                                                                                                                                                                                                                                                                |                           |                                                   |
| GooglePay Payment Method                  | String                                                                                                                                                                                                                                                                         | 4-character alpha-numeric | \$mpgResponse->GetGooglepayPaymentMethod()<br>(); |
|                                           | Indicates if the underlying card used in the GooglePay digital wallet is the funding card number ("FPAN") or a tokenized card number ("DPAN"). If a GoogleTokenTempAdd returns an FPAN, you may perform 3DS authentication with it; if it returns a DPAN, 3DS is not required. |                           |                                                   |
| MPI response fields (see "MPI" on page 1) |                                                                                                                                                                                                                                                                                |                           |                                                   |
| Type                                      | String                                                                                                                                                                                                                                                                         | 99-character alphanumeric |                                                   |
|                                           | VERes, PAREs or error defines what type of response you are receiving .                                                                                                                                                                                                        |                           |                                                   |

Table 21: Receipt object response values (continued)

| Value    | Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Limits                      | Get Method                      |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|---------------------------------|
|          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                             |                                 |
| Success  | Boolean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | true/false                  | \$mpgResponse->getMpiSuccess(); |
|          | True if attempt was successful, false if attempt was unsuccessful.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                             |                                 |
| Message  | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 100-character alphabetic    | \$mpgResponse->getMpiMessage(); |
|          | <p>MPI TXN transactions can produce the following values:</p> <ul style="list-style-type: none"> <li>Y: Create VBV verification form popup window.</li> <li>N: Send purchase or preauth with crypt type 6</li> <li>U: Send purchase or preauth with crypt type 7.</li> </ul> <p>MPI ACS transactions can produce the following values:</p> <ul style="list-style-type: none"> <li>Y or A: (Also <code>receipt.getMpiSuccess()=true</code>) Proceed with cavv purchase or cavv preauth.</li> <li>N: Authentication failed or high-risk transaction. It is recommended that you do not to proceed with the transaction.</li> </ul> <p>Depending on a merchant's risk tolerance and results from other methods of fraud detection, transaction may proceed with crypt type 7.</p> <ul style="list-style-type: none"> <li>U or time out: Send purchase or preauth as crypt type 7.</li> </ul> |                             |                                 |
| Term URL | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 255-character alphanumeric  |                                 |
|          | URL to which the PAREs is returned                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                             |                                 |
| MD       | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1024-character alphanumeric |                                 |
|          | Merchant-defined data that was echoed back                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                             |                                 |
| ACS URL  | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 255-character alphanumeric  |                                 |
|          | URL that will be for the generated pop-up                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                             |                                 |
| MPI CAVV | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 28-character alphanumeric   | \$mpgResponse->getMpiCavv();    |
|          | VbV/MCSC/American Express SafeKey authentication data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                             |                                 |

**Table 21: Receipt object response values (continued)**

| Value                                    | Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Limits                    | Get Method                                |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-------------------------------------------|
|                                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                           |                                           |
| MPI E-Commerce Indicator                 | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1-character alpha-numeric |                                           |
| CAVV result code                         | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1-character alpha-numeric | \$mpgResponse->getCavvResultCode()<br>(); |
|                                          | Indicates the Visa CAVV result. For more information, see 1 Cavv Result Codes for Verified by Visa.<br><br><ul style="list-style-type: none"><li>• 0 = CAVV authentication results invalid</li><li>• 1 = CAVV failed validation; authentication</li><li>• 2 = CAVV passed validation; authentication</li><li>• 3 = CAVV passed validation; attempt</li><li>• 4 = CAVV failed validation; attempt</li><li>• 7 = CAVV failed validation; attempt (US issued cards only)</li><li>• 8 = CAVV passed validation; attempt (US issued cards only)</li><li>• The CAVV result code indicates the result of the CAVV validation.</li></ul> |                           |                                           |
| MPI inline form                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                           | \$mpgResponse->getMpiInLineForm()<br>();  |
|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                           |                                           |
| Vault response fields (see 4.1, page 68) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                           |                                           |
| Data key                                 | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 28-character alphanumeric | \$mpgResponse->getDataKey()<br>();        |
|                                          | The data key response field is populated when you send a Vault Add Credit Card – ResAddCC (page 71), Vault Encrypted Add Credit Card – EncResAddCC (page 75), Vault Tokenize Credit Card – ResTokenizeCC (page 99), Vault Temporary Token Add – ResTempAdd (page 78) or Vault Add Token – ResAddToken (page 96) transaction. It is the profile identifier that all future financial Vault transactions will use to associate with the saved information.                                                                                                                                                                         |                           |                                           |

Table 21: Receipt object response values (continued)

| Value                        | Type                                                                                                                          | Limits                    | Get Method                            |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------------------|
|                              | Description                                                                                                                   |                           |                                       |
| Vault payment type           | String                                                                                                                        | cc                        | \$mpgResponse->getPaymentType();      |
|                              | Indicates the payment type associated with a Vault profile                                                                    |                           |                                       |
| Expiring card's Payment type | String                                                                                                                        | cc                        | \$mpgResponse->getExpPaymentType();   |
|                              | Indicates the payment type associated with a Vault profile. Applicable to Vault Get Expiring transaction type.                |                           |                                       |
| Vault masked PAN             | String                                                                                                                        | 20-character numeric      | \$mpgResponse->getResDataMaskedPan(); |
|                              | Returns the first 4 and/or last 4 of the card number saved in the profile.                                                    |                           |                                       |
| Expiring card's Masked PAN   | String                                                                                                                        | 20-character numeric      | \$mpgResponse->getResDataMaskedPan(); |
|                              | Returns the first 4 and/or last 4 of the card number saved in the profile. Applicable to Vault Get Expiring transaction type. |                           |                                       |
| Vault success                | String                                                                                                                        | true/false                | \$mpgResponse->getResSuccess();       |
|                              | Indicates whether Vault transaction was successful.                                                                           |                           |                                       |
| Vault customer ID            | String                                                                                                                        | 30-character alphanumeric | \$mpgResponse->getResDataCustId();    |
|                              | Returns the customer ID saved in the profile.                                                                                 |                           |                                       |
| Expiring card's customer ID  | String                                                                                                                        | 30-character alphanumeric | \$mpgResponse->getResDataCustId();    |
|                              | Returns the customer ID saved in the profile. Applicable to Vault Get Expiring transaction type.                              |                           |                                       |
| Vault phone number           | String                                                                                                                        | 30-character alphanumeric | \$mpgResponse->getResDataPhone();     |
|                              | Returns the phone number saved in the profile.                                                                                |                           |                                       |

**Table 21: Receipt object response values (continued)**

| Value                         | Type                                                                                                                             | Limits                    | Get Method                                             |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------------------------------------|
|                               | Description                                                                                                                      |                           |                                                        |
| Expiring card's phone number  | String                                                                                                                           | 30-character alphanumeric | <code>\$mpgResponse-&gt;getResDataPhone()</code> ;     |
|                               | Returns the phone number saved in the profile. Applicable to Vault Get Expiring transaction type.                                |                           |                                                        |
| Vault email address           | String                                                                                                                           | 30-character alphanumeric | <code>\$mpgResponse-&gt;getResDataEmail()</code> ;     |
|                               | Returns the email address saved in the profile.                                                                                  |                           |                                                        |
| Expiring card's email address | String                                                                                                                           | 30-character alphanumeric | <code>\$mpgResponse-&gt;getResDataEmail()</code> ;     |
|                               | Returns the email address saved in the profile. Applicable to Vault Get Expiring transaction type.                               |                           |                                                        |
| Vault note                    | String                                                                                                                           | 30-character alphanumeric | <code>\$mpgResponse-&gt;getResDataNote()</code> ;      |
|                               | Returns the note saved in the profile.                                                                                           |                           |                                                        |
| Expiring card's note          | String                                                                                                                           | 30-character alphanumeric | <code>\$mpgResponse-&gt;getResDataNote()</code> ;      |
|                               | Returns the note saved in the profile. Applicable to Vault Get Expiring transaction type.                                        |                           |                                                        |
| Vault expiry date             | String                                                                                                                           | 4-character numeric       | <code>\$mpgResponse-&gt;getResDataExpDate()</code> ;   |
|                               | Returns the expiry date of the card number saved in the profile. YYMM format.                                                    |                           |                                                        |
| Expiring card's expiry date   | String                                                                                                                           | 4-character numeric       | <code>\$mpgResponse-&gt;getResDataExpDate()</code> ;   |
|                               | Returns the expiry date of the card number saved in the profile. YYMM format. Applicable to Vault Get Expiring transaction type. |                           |                                                        |
| Vault E-commerce indicator    | String                                                                                                                           | 1-character numeric       | <code>\$mpgResponse-&gt;getResDataCryptType()</code> ; |
|                               | Returns the e-commerce indicator saved in the profile.                                                                           |                           |                                                        |



Table 21: Receipt object response values (continued)

| Value                                | Type Limits                                                                                                                                                                                                                                                 |                           | Get Method                                  |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------------------------|
|                                      | Description                                                                                                                                                                                                                                                 |                           |                                             |
| Expiring card's E-commerce indicator | String                                                                                                                                                                                                                                                      | 1-character numeric       | \$mpgResponse->getResDataCryptType();       |
|                                      | Returns the e-commerce indicator saved in the profile. Applicable to Vault Get Expiring transaction type.                                                                                                                                                   |                           |                                             |
| Vault AVS street number              | String                                                                                                                                                                                                                                                      | 19-character alphanumeric | \$mpgResponse->getResDataAvsStreetNumber(); |
|                                      | Returns the AVS street number saved in the profile. If no other AVS street number is passed in the transaction request, this value will be submitted along with the financial transaction to the issuer.                                                    |                           |                                             |
| Expiring card's AVS street number    | String                                                                                                                                                                                                                                                      | 19-character alphanumeric | \$mpgResponse->getResDataAvsStreetNumber(); |
|                                      | Returns the AVS street number saved in the profile. If no other AVS street number is passed in the transaction request, this value will be submitted along with the financial transaction to the issuer. Applicable to Vault Get Expiring transaction type. |                           |                                             |
| Vault AVS street name                | String                                                                                                                                                                                                                                                      | 19-character alphanumeric | \$mpgResponse->getResDataAvsStreetName();   |
|                                      | Returns the AVS street name saved in the profile. If no other AVS street number is passed in the transaction request, this value will be submitted along with the financial transaction to the issuer.                                                      |                           |                                             |
| Expiring card's AVS street name      | String                                                                                                                                                                                                                                                      | 19-character alphanumeric | \$mpgResponse->getResDataAvsStreetName();   |
|                                      | Returns the AVS street name saved in the profile. If no other AVS street number is passed in the transaction request, this value will be submitted along with the financial transaction to the issuer. Applicable to Vault Get Expiring transaction type.   |                           |                                             |
| Vault AVS ZIP code                   | String                                                                                                                                                                                                                                                      | 9-character alphanumeric  | \$mpgResponse->getResDataAvsZipcode();      |
|                                      | Returns the AVS zip/postal code saved in the profile. If no other AVS street number is passed in the transaction request, this value will be submitted along with the financial transaction to the issuer.                                                  |                           |                                             |

**Table 21: Receipt object response values (continued)**

| Value                                                       | Type                                                                                                                                                                                                                                                          | Limits                    | Get Method                              |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----------------------------------------|
|                                                             | Description                                                                                                                                                                                                                                                   |                           |                                         |
| Expiring card's AVS ZIP code                                | String                                                                                                                                                                                                                                                        | 9-character alpha-numeric | \$mpgResponse->getResDataAvsZipcode (); |
|                                                             | Returns the AVS zip/postal code saved in the profile. If no other AVS street number is passed in the transaction request, this value will be submitted along with the financial transaction to the issuer. Applicable to Vault Get Expiring transaction type. |                           |                                         |
| Vault credit card number                                    | String                                                                                                                                                                                                                                                        | 20-character numeric      | \$mpgResponse->getResDataPan ();        |
|                                                             | Returns the full credit card number saved in the Vault profile. Applicable to Vault Lookup Full transaction only.                                                                                                                                             |                           |                                         |
| Corporate card                                              | String                                                                                                                                                                                                                                                        | true/false                | \$mpgResponse->getCorporateCard ();     |
|                                                             | Indicates whether the card associated with the Vault profile is a corporate card.                                                                                                                                                                             |                           |                                         |
| Encrypted Mag Swipe response fields (see Section 1, page 1) |                                                                                                                                                                                                                                                               |                           |                                         |
| Masked credit card number                                   | String                                                                                                                                                                                                                                                        | 20-character alphanumeric | \$mpgResponse->getMaskedPan ();         |
|                                                             |                                                                                                                                                                                                                                                               |                           |                                         |
| Convenience Fee response fields (see Appendix A, page 1)    |                                                                                                                                                                                                                                                               |                           |                                         |
| Convenience fee success                                     | String                                                                                                                                                                                                                                                        | true/false                | \$mpgResponse->getCfSuccess ();         |
|                                                             | Indicates whether the Convenience Fee transaction processed successfully.                                                                                                                                                                                     |                           |                                         |

Table 21: Receipt object response values (continued)

| Value                  | Type Limits Get Method                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                           |                                                |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|------------------------------------------------|
|                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                           |                                                |
| Convenience fee status | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3-character alpha-numeric | <code>\$mpgResponse-&gt;getCfStatus();</code>  |
|                        | <p>Indicates the status of the merchant and convenience fee transactions. The CfStatus field provides details about the transaction behavior and should be referenced when contacting Moneris Customer Support.</p> <p>Possible values are:</p> <ul style="list-style-type: none"> <li>• 1 or 1F – Completed 1st purchase transaction</li> <li>• 2 or 2F – Completed 2nd purchase transaction</li> <li>• 3 – Completed void transaction</li> <li>• 4A or 4D – Completed refund transaction</li> <li>• 7 or 7F – Completed merchant independent refund transaction</li> <li>• 8 or 8F – Completed merchant refund transaction</li> <li>• 9 or 9F – Completed 1st void transaction</li> <li>• 10 or 10F – Completed 2nd void transaction</li> <li>• 11A or 11D – Completed refund transaction</li> </ul> |                           |                                                |
| Convenience fee amount | String                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 9-character decimal       | <code>\$mpgResponse-&gt;getFeeAmount();</code> |
|                        | <p>The expected Convenience Fee amount. This field will return the amount submitted by the merchant for a successful transaction. For an unsuccessful transaction, it will return the expected convenience fee amount</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                           |                                                |

**Table 21: Receipt object response values (continued)**

| Value                               | Type                                                                                                                                                                                                                                                          | Limits                    | Get Method                        |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----------------------------------|
|                                     | Description                                                                                                                                                                                                                                                   |                           |                                   |
| Convenience fee rate                | String                                                                                                                                                                                                                                                        | 9-character decimal       | \$mpgResponse->getFeeRate () ;    |
|                                     | The convenience fee rate that has been defined on the merchant's profile.<br>For example:<br><br>1.00 – a fixed amount or<br><br>10.0 - a percentage amount                                                                                                   |                           |                                   |
| Convenience fee type                | String                                                                                                                                                                                                                                                        | AMT/PCT                   | \$mpgResponse->getFeeType () ;    |
|                                     | The type of convenience fee that has been defined on the merchant's profile.<br><br>Available options are:<br><br>AMT – fixed amount<br><br>PCT – percentage                                                                                                  |                           |                                   |
| Merchant Advice Code response field |                                                                                                                                                                                                                                                               |                           |                                   |
| Advice Code                         | String                                                                                                                                                                                                                                                        | 2-character alpha-numeric | \$mpgResponse->getAdviceCode () ; |
|                                     | The message returned from the issuer is intended for merchant information only, and is not intended for customer receipts.<br><br>For further details on the response codes that are returned, see the Advice Code document at https://developer.moneris.com. |                           |                                   |

**Table 22: Financial transaction response codes**

| Code | Description                                |
|------|--------------------------------------------|
| < 50 | Transaction approved                       |
| ≥ 50 | Transaction declined                       |
| NULL | Transaction was not sent for authorization |

For more details on the response codes that are returned, see the Response Codes document available at <https://developer.moneris.com>

Table 23: Vault Admin Responses

| Code | Description                                                                                                                                                                                                                          |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 001  | Successfully registered CC details.<br>Successfully updated CC details.<br>Successfully deleted CC details.<br>Successfully located CC details.<br>Successfully located # expiring cards.<br>(NOTE: # = the number of cards located) |
| 983  | Cannot find previous                                                                                                                                                                                                                 |
| 986  | Incomplete: timed out                                                                                                                                                                                                                |
| 987  | Invalid transaction                                                                                                                                                                                                                  |
| 988  | Cannot find expiring cards                                                                                                                                                                                                           |
| Null | Error: Malformed XML                                                                                                                                                                                                                 |

## B.1 Definition of Response Fields – 3-D Secure

The following response fields are specific to 3-D Secure transactions

| Variable Name                          | Type and Limits                                | Description                                                                                                                                                                                       |
|----------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cardholder Authentication Value (CAVV) | <i>String</i><br><br>50-character alphanumeric | <code>\$mpgResponse-&gt;getMpiCavv()</code><br><br>Cardholder Authentication Value, to be provided in the financial request                                                                       |
| challenge completion indicator         | <i>String</i><br><br>1-character alphabetic    | <code>\$mpgResponse-&gt;getMpiChallengeCompletionIndicator()</code><br><br>Indicates the result of the challenge request <ul style="list-style-type: none"> <li>Y = challenge complete</li> </ul> |

| Variable Name             | Type and Limits                                   | Description                                                                                                                                                                                                                                 |
|---------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           |                                                   | <ul style="list-style-type: none"> <li>N = challenge not complete</li> <li>null = field not present</li> </ul>                                                                                                                              |
| challenge URL             | <i>String</i><br>2048-character alphanumeric      | <code>\$mpgResponse-&gt;getMpiChallengeURL()</code><br>If the transaction status is "C" this field will be populated with the URL to POST the challenge data to create the cardholder challenge screen.                                     |
| Message Type              | <i>String</i><br>4-character alpha-betic          | <code>\$mpgResponse-&gt;getMpiMessageType()</code><br>Denotes the request message<br>EMV nomenclature "ARES"                                                                                                                                |
| 3DS method URL            | <i>String</i><br>alphanumeric                     | <code>\$mpgResponse-&gt;getMpiThreeDSMethodURL()</code><br>Device fingerprinting endpoint                                                                                                                                                   |
| 3DS method data           | <i>String</i><br>alphanumeric                     | <code>\$mpgResponse-&gt;getMpiThreeDSMethodData()</code><br>Data that must be posted to 3DS Method URL                                                                                                                                      |
| 3DS server transaction ID | <i>String</i><br><b>36-character alphanumeric</b> | <code>\$mpgResponse-&gt;getMpiThreeDSServerTransId()</code><br>3-D Secure unique transaction identifier                                                                                                                                     |
| 3DS version               | <i>String</i><br>10-character numeric             | <code>\$mpgResponse-&gt;getThreeDSVersion();</code><br>The 3DS version number.                                                                                                                                                              |
| transaction status        | <i>String</i><br>1-character alpha-betic          | <code>\$mpgResponse-&gt;getMpiTransStatus()</code><br>Indicates the result of the authentication. See 3DS TransStatus in the appendix for a full list: <ul style="list-style-type: none"> <li>Y or A = successful authentication</li> </ul> |

| Variable Name             | Type and Limits           | Description                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           |                           | <ul style="list-style-type: none"> <li>• C = proceed with challenge</li> <li>• D = wait for decoupled authentication</li> </ul>                                                                                                                                                                                                                                                                                                        |
| challenge data            | alphanumeric              | If the transStatus is “C” this field will be populated with the data to post the challengeURL. The data must be posted in a field with the name “creq”.                                                                                                                                                                                                                                                                                |
| ECI                       | 1-character alphabetic    | <p>Crypt Type to be provided in the financial request.</p> <p>If transaction status is “Y” or “A” this value will return in the response. The ECI is used by subsequent financial transactions.</p>                                                                                                                                                                                                                                    |
| DS trans ID               | 36-character alphanumeric | Universally unique transaction identifier assigned by the 3DS Directory Server (DS) to identify a single transaction.                                                                                                                                                                                                                                                                                                                  |
| transaction status reason | 2-character numeric       | <p>Provides additional information on why the TransStatus field has the specified value.</p> <p>Possible values:</p> <ul style="list-style-type: none"> <li>• 01 = Card authentication failed</li> <li>• 02 = Unknown Device</li> <li>• 03 = Unsupported Device</li> <li>• 04 = Exceeds authentication frequency limit</li> <li>• 05 = Expired card</li> <li>• 06 = Invalid card number</li> <li>• 07 = Invalid transaction</li> </ul> |

| Variable Name          | Type and Limits            | Description                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        |                            | <ul style="list-style-type: none"> <li>• 08 = No Card record</li> <li>• 09 = Security failure</li> <li>• 10 = Stolen card</li> <li>• 11 = Suspected fraud</li> <li>• 12 = Transaction not permitted to cardholder</li> <li>• 13 = Cardholder not enrolled in service</li> <li>• 14 = Transaction timed out at the ACS</li> <li>• 15 = Low confidence</li> <li>• 16 = Medium confidence</li> </ul> |
| cardholder information | 128-character alphanumeric | Text provided by the ACS/Issuer to Cardholder during a Frictionless or Decoupled transaction. This provides information to the cardholder. For example, "Additional authentication is needed for this transaction, please contact (Issuer Name) at xxx-xxx-xxxx."                                                                                                                                 |
| Authentication-Type    | 2-character numeric        | <p>Authentication method the issuer will use to challenge the cardholder.</p> <ul style="list-style-type: none"> <li>• 01 = Static</li> <li>• 02 = Dynamic</li> <li>• 03 = OOB</li> <li>• 04 = Decoupled</li> </ul>                                                                                                                                                                               |
| 3DS ACS Transaction ID | 36-character alphanumeric  | Universally Unique transaction identifier assigned by the issuer Access Control Server (ACS) to                                                                                                                                                                                                                                                                                                   |



| Variable Name      | Type and Limits      | Description                                                                              |
|--------------------|----------------------|------------------------------------------------------------------------------------------|
|                    |                      | identify a single transaction.                                                           |
| 3DS Auth TimeStamp | 12-character numeric | Date and time in UTC of the cardholder authentication.<br><br>Date format = YYYYMMDDHHMM |

## B.2 Definition of Response Fields – MCP

### MCP response fields

| Variable Name                | Type and Limits                                                                                                                                                                | Get Method and Description                                                                                                                                                                                                                                                                                                                               |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MCP rate                     | <i>String</i><br><br>9-character decimal<br><br>variable length                                                                                                                | <code>\$mpgResponse-&gt;getMCPRate();</code><br><br>The foreign exchange rate (foreign currency to CAD) that will be used for the transaction<br><br>If a <b>MCP rate token</b> was used, it will reflect the rate secured by the MCP Get Rate transaction; if no token was used, the rate is the current exchange rate retrieved by the Moneris Gateway |
| merchant settlement currency | <i>String</i><br><br>3-character numeric                                                                                                                                       | <code>\$mpgResponse-&gt;getMerchantSettlementCurrency();</code><br><br>Currency that the merchant is settling in                                                                                                                                                                                                                                         |
| merchant settlement amount   | <i>String</i><br><br>10-character decimal<br><br>Up to 7 digits (dollars) + decimal point (.) + 2 digits (cents) after the decimal point<br><br><div>EXAMPLE: 1234567.89</div> | <code>\$mpgResponse-&gt;getMerchantSettlementAmount();</code><br><br>Amount that will be paid to the merchant, in Canadian dollars                                                                                                                                                                                                                       |

| Variable Name            | Type and Limits                                                 | Get Method and Description                                                                                                                    |
|--------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| cardholder currency code | <i>String</i><br>3-character numeric                            | <code>\$mpgResponse-&gt;getCardholderCurrencyCode();</code><br><br>ISO code for the foreign currency the cardholder is using to pay           |
| cardholder amount        | <i>String</i><br>12-character numeric<br>variable length        | <code>\$mpgResponse-&gt;getCardholderAmount();</code><br><br>Amount, in units of foreign currency, the cardholder will pay on the transaction |
| MCP error status code    | <i>String</i><br>4-character numeric<br>variable length         | <code>\$mpgResponse-&gt;getMCPErrorStatusCode();</code><br><br>A number representing a MCP error code response                                |
| MCP error message        | <i>String</i><br>250-character alpha-numeric<br>variable length | <code>\$mpgResponse-&gt;getMCPErrorMessage();</code><br><br>Message corresponding with an MCP error code                                      |
| host ID                  | <i>String</i><br>15-character alpha-numeric                     | <code>\$mpgResponse-&gt;getHostId();</code><br><br>Unique identifier used across the Moneris platform                                         |

**Response fields specific to MCP Get Rate**

| Variable Name         | Type and Limits                                                   | Get Method and Description                                                                                                                                               |
|-----------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rate transaction type | <i>String</i><br>max 8-character alphabetic<br>PURCHASE or REFUND | <code>\$mpgResponse-&gt;getRateTxnType();</code><br><br>Reflects the transaction type being sent in the request                                                          |
| MCP rate token        | <i>String</i><br>17-character alphanumeric                        | <code>\$mpgResponse-&gt;getMCPRateToken();</code><br><br>Time-limited token representing a temporarily locked in foreign exchange rate for use in financial transactions |

| Variable Name            | Type and Limits                                         | Get Method and Description                                                                                           |
|--------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
|                          |                                                         | This field is returned in the response to a MCP Get Rate request                                                     |
| rate inquiry start time  | <i>String</i><br>24-character alphanumeric              | <code>\$mpgResponse-&gt;getRateInqStartTime();</code><br><br>The local time (ISO 8601) when the rate is requested    |
| rate inquiry end time    | <i>String</i><br>24-character alphanumeric              | <code>\$mpgResponse-&gt;getRateInqEndTime();</code><br><br>The local time (ISO 8601) when the rate is returned       |
| rate validity start time | <i>String</i><br>10-character numeric                   | <code>\$mpgResponse-&gt;getRateValidityStartTime();</code><br><br>The time (unix UTC) of when the rate is valid from |
| rate validity end time   | <i>String</i><br>10-character numeric                   | <code>\$mpgResponse-&gt;getRateValidityEndTime();</code><br><br>The time (unix UTC) of when the rate is valid until  |
| rate validity period     | <i>String</i><br>3-character numeric<br>variable length | <code>\$mpgResponse-&gt;getRateValidityPeriod();</code><br><br>The time in minutes this rate is valid for            |

### B.3 Definition of Response Fields –Installments by Visa

#### Response fields appearing in the Installment Plan Lookup transaction

| Variable Name              | Type and Limits      | Description                                                             |
|----------------------------|----------------------|-------------------------------------------------------------------------|
| Eligible Installment Plans | <i>Object</i><br>N/A | Contains fields related to the installment plan                         |
| plan count                 | <i>String</i>        | Total number of installment plans available for offer to the cardholder |

| Variable Name                | Type and Limits                                     | Description                                                                                                                                                                                                                                                          |
|------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | numeric                                             |                                                                                                                                                                                                                                                                      |
| Plan Details                 | Array object<br>N/A                                 | Contains fields related to the particular installment plan<br><br>Each installment plan on offer to the cardholder is represented by a distinct Plan Details object                                                                                                  |
| annual percentage rate (APR) | String<br>numeric                                   | Annual percentage rate (APR) attached to the installment plan payments; for display purposes only and not used for calculations<br><br>Allowable values: 0-10000<br><br>Percentage rate is represented with two implicit decimals<br><br><b>EXAMPLE:</b> 320 is 3.2% |
| installment frequency        | String<br>max 10-character alphabetic               | Frequency of installments for the plan<br><br>Potential values:<br>WEEKLY<br>BIWEEKLY<br>MONTHLY<br>BIMONTHLY                                                                                                                                                        |
| installment plan ID          | String<br>36-character alphanumeric<br>fixed length | Card brand-generated identifier for an installment plan<br><br>Used as a request field in the Installment Info object                                                                                                                                                |
| installment plan name        | String<br>max 255-character alphanumeric            | Name of the installment plan; may not be unique                                                                                                                                                                                                                      |
| installment plan reference   | String<br>10-character alphanumeric                 | Unique, human friendly name for the installment plan                                                                                                                                                                                                                 |

| Variable Name            | Type and Limits                                         | Description                                                                                                                          |
|--------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
|                          | fixed length                                            | Used as a request field in the Installment Info object                                                                               |
| installment plan type    | <i>String</i><br>max 20 character alpha-numeric         | Type of installment plans<br><br>Potential values:<br>ISSUER_PROMOTION<br>BI_LATERAL<br>ISSUER_DEFAULT<br>MARKET                     |
| number of installments   | <i>String</i><br>4-character numeric<br>min 1, max 1000 | Maximum number of installments in the plan                                                                                           |
| First Installment        | <i>Object</i><br>N/A                                    | Contains cost details for the first installment                                                                                      |
| first installment amount | <i>String</i><br>max 9-character numeric                | Amount of the first installment payment<br><br>Final two digits represent penny values<br><br><div>EXAMPLE: 123112 = \$1231.12</div> |
| first installment fee    | <i>String</i><br>max 9-character numeric                | Fee charged on the first installment<br><br>Final two digits represent penny values<br><br><div>EXAMPLE: 123112 = \$1231.12</div>    |
| upfront fee              | <i>String</i><br>numeric                                | The up-front fee charged to the cardholder for the installment plan; only charged on the first installment                           |

| Variable Name           | Type and Limits                               | Description                                                                                                                      |
|-------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| total amount            | <i>String</i><br>numeric                      | Sum of upfront fee and installment fee                                                                                           |
| Last Installment        | <i>Object</i><br>N/A                          | Contains cost details for the last installment                                                                                   |
| last installment amount | <i>String</i><br>max 9-character numeric      | Amount of the final installment payment<br><br>Final two digits represent penny values<br><br><b>EXAMPLE:</b> 123112 = \$1231.12 |
| last installment fee    | <i>String</i><br>max 9-character numeric      | Fee charged on the last installment<br><br>Final two digits represent penny values<br><br><b>EXAMPLE:</b> 123112 = \$1231.12     |
| total amount            | <i>String</i><br>numeric                      | Sum of upfront fee and installment fee                                                                                           |
| Promotion Info          | <i>Object</i><br>N/A                          | Contains promotion information shared between the issuer and the merchant                                                        |
| promotion code          | <i>String</i><br>2-character alphanumeric     | An external identifier for the plan provided by the issuer                                                                       |
| promotion ID            | <i>String</i><br>max 8-character alphanumeric | An external identifier provided by the issuer that identifies a program or promotion                                             |
| Terms and Conditions    | <i>Array object</i>                           | Contains fields related to terms and                                                                                             |

| Variable Name                | Type and Limits                                                                     | Description                                                                                                                                                                          |
|------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | N/A                                                                                 | conditions presented to the cardholder                                                                                                                                               |
| terms and conditions count   | <i>String</i><br>numeric                                                            | Number of instances of the set of terms and conditions attached to a particular installment plan, representing the number of languages they are offered in                           |
| Terms and Conditions Details | <i>Object</i><br>N/A                                                                | Contains details related to a particular language set (English, French, etc.) of terms and conditions being offered<br><br>Each language set has its own object                      |
| language code                | <i>String</i><br>3-character alphanumeric                                           | Language code for the terms and conditions text                                                                                                                                      |
| text                         | <i>String</i><br>max 2000-character alphanumeric                                    | Text of the terms and conditions for the installment plan                                                                                                                            |
| terms and conditions URL     | <i>String</i><br>max 1000 character-alphanumeric                                    | A terms and conditions HTTPS URL hosted by the issuer for displaying to the cardholder                                                                                               |
| terms and conditions version | <i>String</i><br>10-character alphanumeric<br><br>variable length (1-10 characters) | Version of the terms and conditions of the installment plan accepted by the cardholder<br><br>The version is auto-incremented every time an update is made to the plan by the issuer |
| total fees                   | <i>String</i><br><br>max 9-character numeric                                        | Total fees charged by the plan<br><br>Final two digits represent penny values                                                                                                        |

EXAMPLE: 123112 = \$1231.12

| Variable Name   | Type and Limits   | Description                                                                                                                                                                                                                          |
|-----------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| total plan cost | String<br>numeric | Represents the total amount the selected installment plan will cost<br><br>The right-most digits represent minor units (e.g., cents in CAD); no fractional minor units<br><br><b>EXAMPLE:</b> 123112 in CAD represents CAD \$1231.12 |

### Response fields appearing in financial transactions

| Variable Name                | Type and Limits                                                          | Description                                                                                                                                                                            |
|------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Installment Results          | Object<br>N/A                                                            | Contains fields related to the installment plan in financial transactions                                                                                                              |
| installment plan ID          | String<br>36-character alphanumeric<br>fixed length                      | Card brand-generated identifier for an installment plan                                                                                                                                |
| installment plan reference   | String<br>10-character alphanumeric<br>fixed length                      | Unique, human friendly name for the installment plan                                                                                                                                   |
| terms and conditions version | String<br>10-character alphanumeric<br>variable length (1-10 characters) | Version of the terms and conditions of the installment plan accepted by the cardholder.<br><br>The version is auto-incremented every time an update is made to the plan by the issuer. |
| plan acceptance ID           | String<br>36-character alphanumeric<br>fixed length                      | Visa-generated, alphanumeric, unique and short human-readable name for the installment plan                                                                                            |
| installment plan status      | String                                                                   | Potential values:                                                                                                                                                                      |



| Variable Name | Type and Limits                           | Description                                                                                                                                                                                                                     |
|---------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | 1-character alphabetic<br>fixed length    | N – new plan, not accepted yet<br>A – accepted plan<br>C – cancelled plan                                                                                                                                                       |
| plan response | <i>String</i><br>max 50-character numeric | Response code for the installment plan<br>Potential values:<br>00 – processed and approved<br>If not 00, indicates installment plan processing failure; a verbose error response message as received from from Visa is returned |

## Appendix C Response Codes

### Approved Response Codes

| Response Code | Messages                                                                                                                                                                                                    |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 000           | <p>Approved, Account Balances</p> <p>Included (Balance Inquiry), No Reason to Decline</p> <p>Approved (Balances)</p> <p>File Processed/Successful transaction with fault</p>                                |
| 001           | <p>Approved, Account Balances Not Included</p> <p>Approved – No Balances/Approved or completed successfully</p> <p>VIP</p> <p>Approved (No Balances)/Advice Acknowledged – Financial Liability Accepted</p> |
| 002           | Approved, country club                                                                                                                                                                                      |
| 003           | Approved, maybe more ID                                                                                                                                                                                     |
| 004           | Approved, pending ID (sign paper draft)                                                                                                                                                                     |
| 005           | Approved, blind                                                                                                                                                                                             |
| 006           | Approved, VIP                                                                                                                                                                                               |
| 007           | Approved, administrative transaction                                                                                                                                                                        |
| 008           | Approved, national NEG file hit OK                                                                                                                                                                          |
| 009           | Approved, commercial                                                                                                                                                                                        |
| 010           | Approved for partial amount                                                                                                                                                                                 |
| 023           | Amex - credit approval                                                                                                                                                                                      |
| 024           | Amex 77 - credit approval                                                                                                                                                                                   |

| Response Code | Messages                        |
|---------------|---------------------------------|
| 027           | Transaction already reversed    |
| 028           | VIP Credit Approved             |
| 029           | Credit Response Acknowledgement |
| 900           | Global Error                    |
| 901           | Invalid URL                     |
| 902           | Malformed XML                   |

**Declined Response Codes**

| Response Code | Messages                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 050           | Do Not Honor<br>Decline<br>Refer to card issuer<br>ID certification fails<br>Deny – Do not Honour<br>Card not initialized<br>Declined:<br>Deny – Unacceptable Fee<br>Unable to locate original transaction<br>Suspected Fraud<br>Deny – Card Acceptor Call Acquirer's Security Dep<br>Amount Not Reconciled – Totals Provided<br>ATM/POS terminal number cannot be located<br>MAC failed<br>Declined:<br>MAC failed |

| Response Code | Messages                                                                                                                                                                                                                                                                               |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | Reserved<br>Security processing failure<br>No arrears (transaction receipt not printed)<br>Invalid File Type<br>No such File<br>File Locked<br>Unsuccessful<br>Incorrect File Length<br>File Decompression Error<br>File Name Error<br>File cannot be received<br>Deny – Do Not Honour |
| 051           | Expired Card                                                                                                                                                                                                                                                                           |
| 052           | PIN retries exceeded<br>PIN try limit exceeded<br>Allowable number of PIN tries exceeded                                                                                                                                                                                               |
| 053           | No sharing                                                                                                                                                                                                                                                                             |
| 054           | No security module                                                                                                                                                                                                                                                                     |
| 055           | Invalid transaction                                                                                                                                                                                                                                                                    |
| 056           | No Support/Transaction Not Permitted to Acquirer<br>Tran Not Supported by FI/Not Supported by Receiver                                                                                                                                                                                 |
| 057           | Lost or stolen card                                                                                                                                                                                                                                                                    |
| 058           | Invalid status                                                                                                                                                                                                                                                                         |
| 059           | Deny (Keep Card) – Restricted Card                                                                                                                                                                                                                                                     |

| Response Code | Messages                                                                                                                                                                 |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | Restricted Card                                                                                                                                                          |
| 060           | No Chequing account<br>No Savings Account                                                                                                                                |
| 061           | No PBF                                                                                                                                                                   |
| 062           | PBF update error                                                                                                                                                         |
| 063           | Invalid authorization type                                                                                                                                               |
| 064           | Bad Track 2                                                                                                                                                              |
| 065           | Adjustment not allowed                                                                                                                                                   |
| 066           | Invalid credit card advance increment                                                                                                                                    |
| 067           | Invalid transaction date                                                                                                                                                 |
| 068           | PTLF error                                                                                                                                                               |
| 069           | Bad Message Error/No CVM Results<br>Bad message – edit error/Format error                                                                                                |
| 070           | No IDF<br>Invalid Issuer<br>Invalid Issuer/Deny – Issuer/Bank Not Found                                                                                                  |
| 071           | Invalid route authorization<br>Unable to route/Financial institution or intermediate network facility cannot be found for routing<br>Invalid Rout to Auth /Incorrect IIN |
| 072           | Card on National NEG file                                                                                                                                                |
| 073           | Invalid route service (destination)                                                                                                                                      |
| 074           | Unable to authorize                                                                                                                                                      |

| Response Code | Messages                                                                                                                                                                                                                                              |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | Re-enter Transaction<br>Transaction Cannot be Completed<br>Deny – Security Violation Deny – Violation of Law<br>System problem - ask cardholder to insert card in chip card reader<br>Merchant Link not logged on (Network Management Logon required) |
| 075           | Invalid PAN length                                                                                                                                                                                                                                    |
| 076           | Low funds                                                                                                                                                                                                                                             |
| 077           | Pre-auth full                                                                                                                                                                                                                                         |
| 078           | Duplicate transaction<br>Duplicate transaction/Request in progress                                                                                                                                                                                    |
| 079           | Maximum online refund reached                                                                                                                                                                                                                         |
| 080           | Maximum offline refund reached                                                                                                                                                                                                                        |
| 081           | Maximum credit per refund reached                                                                                                                                                                                                                     |
| 082           | Number of times used exceeded                                                                                                                                                                                                                         |
| 083           | Maximum refund credit reached                                                                                                                                                                                                                         |
| 084           | Duplicate transaction - authorization number has already been corrected by host                                                                                                                                                                       |
| 085           | Inquiry not allowed                                                                                                                                                                                                                                   |
| 086           | Over floor limit                                                                                                                                                                                                                                      |
| 087           | Maximum number of refund credit by retailer                                                                                                                                                                                                           |
| 088           | Place call                                                                                                                                                                                                                                            |
| 089           | CAF status inactive or closed                                                                                                                                                                                                                         |
| 090           | Referral file full                                                                                                                                                                                                                                    |

| Response Code | Messages                                                              |
|---------------|-----------------------------------------------------------------------|
| 091           | NEG file problem                                                      |
| 092           | Advance less than minimum                                             |
| 093           | Delinquent                                                            |
| 094           | Over table limit                                                      |
| 095           | Amount over maximum<br>Amt Over Max/Transaction amount limit exceeded |
| 096           | PIN required                                                          |
| 097           | Mod 10 check failure                                                  |
| 098           | Force Post                                                            |
| 099           | Bad PBF                                                               |

### Referral Response Codes

| Response Code | Messages                                                                                                                                                                 |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 100           | Unable to process transaction<br><br>Invalid Request. Contact Moneris Client POS Certification for repeat declines.<br><br>Network Unavailable<br><br>System Malfunction |
| 101           | Place call                                                                                                                                                               |
| 102           | Refer – Call<br><br>Expired Card<br><br>Card Acceptor Contact<br><br>Call Card Accpt Acq Secur                                                                           |
| 103           | NEG file problem                                                                                                                                                         |

| Response Code | Messages                                     |
|---------------|----------------------------------------------|
| 104           | CAF problem                                  |
| 105           | Card not supported                           |
| 106           | Amount over maximum                          |
| 107           | Over daily limit                             |
| 108           | CAF Problem                                  |
| 109           | Advance less than minimum                    |
| 110           | Number of times used exceeded                |
| 111           | Delinquent                                   |
| 112           | Over table limit                             |
| 113           | Timeout                                      |
| 115           | PTLF error                                   |
| 121           | Administration file problem                  |
| 122           | Unable to validate PIN: security module down |

**System Error Response Codes**

| Response Code | Messages                                                                                       |
|---------------|------------------------------------------------------------------------------------------------|
| 150           | Invalid Service Code/Merchant<br>Merchant Not On File<br>Merchant Not on File/Invalid Merchant |
| 200           | Invalid account<br>Invalid Card Number<br>Invalid Account/Deny – No Account Type Requested     |
| 201           | Incorrect PIN                                                                                  |



| Response Code | Messages                                                                                                             |
|---------------|----------------------------------------------------------------------------------------------------------------------|
|               | Invalid PIN/Incorrect personal identification number<br>PIN Block Error                                              |
| 202           | Advance less than minimum                                                                                            |
| 203           | Administrative card needed                                                                                           |
| 204           | Amount over maximum                                                                                                  |
| 205           | Invalid Advance Amount<br>Original Amnt Incorrect<br>Bad message/Invalid Amount<br>Original transaction amount error |
| 206           | CAF not found<br>Invalid “to” account<br>Invalid “from” account<br>Invalid account                                   |
| 207           | Invalid transaction date                                                                                             |
| 208           | Invalid expiration date                                                                                              |
| 209           | Invalid transaction code                                                                                             |
| 210           | PIN key sync error                                                                                                   |
| 212           | Destination not available                                                                                            |
| 251           | Error on cash amount                                                                                                 |
| 252           | Debit not supported                                                                                                  |

**American Express Response Codes (Declines)**

| Response Code | Messages                                  |
|---------------|-------------------------------------------|
| 426           | AMEX - Denial 12                          |
| 427           | AMEX - Invalid merchant                   |
| 429           | AMEX - Account error                      |
| 430           | AMEX - Expired card                       |
| 431           | AMEX - Call Amex                          |
| 434           | AMEX - Call 03<br>Note: Invalid CVD (CID) |
| 435           | AMEX - System down                        |
| 436           | AMEX - Call 05                            |
| 437           | AMEX - Declined                           |
| 438           | AMEX - Declined                           |
| 439           | AMEX - Service error                      |
| 440           | AMEX - Call Amex                          |
| 441           | AMEX - Amount error                       |

**Credit Card Response Codes (Declines)**

| Response Code | Messages                                                                                                             |
|---------------|----------------------------------------------------------------------------------------------------------------------|
| 408           | CREDIT CARD - Card use limited - Refer to branch                                                                     |
| 475           | CREDIT CARD - Invalid expiration date                                                                                |
| 476           | CREDIT CARD - Invalid transaction, rejected<br>No Credit Account<br>Invalid transaction/Invalid related transactions |

| Response Code | Messages                                                                                                                                                                                                                                                                    |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | Unable to process/Suspected malfunction; related transaction error<br>Unable to Authorize:<br>Cut off is in process<br>Issuer not capable to process<br>Switch system malfunction<br>Issuer response not received by CUPS<br>Unable to Authorize/Illegal Status of Acquirer |
| 477           | CREDIT CARD - Refer Call/Invalid Card Number<br>Invalid card number (no such account)<br>Deny – Card Not Found<br>Items not on Bankbook beyond limit, declined/Invalid card number                                                                                          |
| 478           | CREDIT CARD - Decline, Pick up card, Call                                                                                                                                                                                                                                   |
| 479           | CREDIT CARD - Decline, Pick up card                                                                                                                                                                                                                                         |
| 480           | CREDIT CARD - Decline, Pick up card                                                                                                                                                                                                                                         |
| 481           | CREDIT CARD - Decline<br>Transaction not allowed to be processed by cardholder<br>Low funds/Insufficient Balance<br>Invalid Transaction<br>Transaction not allowed to be processed by merchant                                                                              |
| 482           | CREDIT CARD - Expired Card                                                                                                                                                                                                                                                  |
| 483           | CREDIT CARD – Refer/Refer to Issuer<br>Deny – Card Acceptor Contact Acquirer                                                                                                                                                                                                |
| 484           | CREDIT CARD - Expired card - refer                                                                                                                                                                                                                                          |
| 485           | CREDIT CARD - Not authorized                                                                                                                                                                                                                                                |

| Response Code | Messages                                                                                        |
|---------------|-------------------------------------------------------------------------------------------------|
| 486           | CREDIT CARD - CVV Cryptographic error                                                           |
| 487           | CREDIT CARD - Invalid CVV                                                                       |
| 489           | CREDIT CARD - Invalid CVV                                                                       |
| 490           | CREDIT CARD - Invalid CVV                                                                       |
| 492           | System problem - ask cardholder to insert card in chip card reader<br>Withdrawal count exceeded |

**System Decline Response Codes**

| Response Code | Messages                                          |
|---------------|---------------------------------------------------|
| 800           | Bad format                                        |
| 801           | Bad data                                          |
| 802           | Invalid Clerk ID                                  |
| 809           | Bad close                                         |
| 810           | System timeout                                    |
| 811           | System error                                      |
| 821           | Bad response length                               |
| 842           | Installment plan lookup failure                   |
| 877           | Invalid PIN block                                 |
| 878           | PIN length error                                  |
| 880           | Final packet of a multi-packet transaction        |
| 881           | Intermediate packet of a multi-packet transaction |
| 889           | MAC key sync error                                |

| Response Code | Messages                                                                                                                                                                                                                    |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 898           | Bad MAC value                                                                                                                                                                                                               |
| 899           | Bad sequence number - resend transaction                                                                                                                                                                                    |
| 900           | Capture - PIN Tries Exceeded                                                                                                                                                                                                |
| 901           | Capture - Expired Card                                                                                                                                                                                                      |
| 902           | Capture - NEG Capture                                                                                                                                                                                                       |
| 903           | Capture - CAF Status 3                                                                                                                                                                                                      |
| 904           | Capture - Advance < Minimum                                                                                                                                                                                                 |
| 905           | Capture - Num Times Used                                                                                                                                                                                                    |
| 906           | Capture - Delinquent                                                                                                                                                                                                        |
| 907           | Capture - Over Limit Table                                                                                                                                                                                                  |
| 908           | Capture - Amount Over Maximum<br>Capture - Capture<br>Pick up Card<br>Suspected Fraud<br>Hard Capture<br>Deny – Keep Card:<br>Special Conditions<br>Expired Card<br>Fraud<br>Card Acceptor Call Acquirer's<br>Do Not Honour |
| 950           | Admin card is not enabled on Merchant profile                                                                                                                                                                               |

**Other Response Codes**

| Response Code | Message |
|---------------|---------|
| 599           | Decline |

**Admin Response Codes**

| Response Code | Messages                                                                                                                                                    |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 960           | Initialization Failure - No Match on Merchant ID                                                                                                            |
| 961           | Initialization Failure - No Match on PED ID                                                                                                                 |
| 962           | Initialization Failure - No match on Printer ID                                                                                                             |
| 963           | No match on Poll code                                                                                                                                       |
| 964           | Initialization Failure - No match on Concentrator ID                                                                                                        |
| 965           | Invalid software version number                                                                                                                             |
| 966           | Duplicate terminal name                                                                                                                                     |
| 970           | Terminal/Clerk table full                                                                                                                                   |
| 983           | Clerk Totals Unavailable: selected Clerk IDs do not exist or have zero totals                                                                               |
| 989           | MAC Error on Transaction 95 (Initialization and Handshake), most often, this indicates that the wrong keys have been injected into a device/KMAC Sync Error |

**EMV Reversal Request Codes**

| Response Code | Messages                                                  |
|---------------|-----------------------------------------------------------|
| 990           | Chip card declines a host approved transaction            |
| 991           | Chip card removed before ICC communications are completed |

**Surcharge Response Codes**

| Response Code | Messages                      |
|---------------|-------------------------------|
| 976           | BIN not found - DO NOT RETRY  |
| 977           | Prepaid Card BIN              |
| 978           | Debit Card BIN                |
| 979           | Internal System Error - RETRY |
| 980           | Surcharge Limit Exceeded      |





## Appendix D Error Messages

### Error messages that are returned if the gateway is unreachable

#### Global Error Receipt

You are not connecting to our servers. This can be caused by a firewall or your internet connection.

#### Response Code = NULL

The response code can be returned as null for a variety of reasons. The majority of the time, the explanation is contained within the Message field.

When a 'NULL' response is returned, it can indicate that the issuer, the credit card host, or the gateway is unavailable. This may be because they are offline or because you are unable to connect to the internet.

A 'NULL' can also be returned when a transaction message is improperly formatted.

### Error messages that are returned in the Message field of the response

#### XML Parse Error in Request: <System specific detail>

An improper XML document was sent from the API to the servlet.

#### XML Parse Error in Response: <System specific detail>

An improper XML document was sent back from the servlet.

#### Transaction Not Completed Timed Out

Transaction timed out before the host responds to the gateway.

#### Request was not allowed at this time

The host is disconnected.

#### Could not establish connection with the gateway: <System specific detail>

Gateway is not accepting transactions or server does not have proper access to internet.

#### Input/Output Error: <System specific detail>

Servlet is not running.

#### The transaction was not sent to the host because of a duplicate order id

Tried to use an order id which was already in use.

#### The transaction was not sent to the host because of a duplicate order id

Expiry Date was sent in the wrong format.

### Vault error messages

#### Can not find previous

Data key provided was not found in our records or profile is no longer active.

#### Invalid Transaction

Transaction cannot be performed because improper data was sent.

or

Mandatory field is missing or an invalid SEC code was sent.

#### Malformed XML

Parse error.

**Incomplete**

Timed out.

**or**

Cannot find expiring cards.

## 19 Konek Error Codes

Delete this text and replace it with your own content.

| CODE | TYPE                               | MESSAGE                                                   |
|------|------------------------------------|-----------------------------------------------------------|
| 101  | PHOENIX                            | Error with the request                                    |
| 102  | PHOENIX                            | Error with the field                                      |
| 4000 | Bad Request                        | Request cannot be processed                               |
| 4001 | Request Header Validation Error    | Request Header is missing or invalid:<br>{header name}    |
| 4002 | Request Parameter Validation Error | Request Parameter is missing or invalid:<br>{param name}  |
| 4003 | Request Body Validation Error      | Request Body field is missing or invalid:<br>{field name} |
| 4004 | Request Body Validation Error      | Request Body field is not expected: {field<br>name}       |
| 4005 | Request Body Validation Error      | Request Body cannot be parsed                             |
| 4098 |                                    | Minor aged user logged into the System.                   |
| 4099 |                                    | Business Error codes defined by FI for<br>KONEK           |
| 4100 | Not Found                          | The requested resource cannot be found.<br>{type}: {id}   |
| 4101 | Not Found                          | The requested path does not exist                         |
| 4200 | Decryption Error                   | Failed to decrypt encrypted data                          |
| 4201 | Decryption Error                   | Decrypted data cannot be parsed                           |
| 4300 | Consent Error                      | Consent status is invalid                                 |
| 4400 | Invalid Requester                  | Requester ID is invalid or does not exist                 |
| 5000 | Internal Server Error              | The server failed to process this request                 |

| CODE | TYPE                  | MESSAGE                                                            |
|------|-----------------------|--------------------------------------------------------------------|
| 8000 | Token Requester Error | Token Requester Error: {error message from TR}                     |
| 8001 | Token Requester Error | Token Requester Error: {downstream system}-{error message from TR} |
| 8002 | Token Requester Error | An Error has occurred while trying to access the Token Requester   |

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